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NMEA 2000®

Appendices A & B - Parameter Groups (PGNs)
NMEA Network Messages

Standard for Serial-Data Networking of Marine Electronic Devices

National Marine Electronics Association

International Marine Electronics Association

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NMEA 2000 ®

STANDARD FOR SERIAL-DATA NETWORKING OF MARINE ELECTRONIC DEVICES

Appendix A

Application Layer
(Parameter Group Definitions)

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A-1. Parameter Group Definition

Data transmitted on the NMEA 2000 network consist of Parameter Groups of related information identified by a Parameter Group Number. The manner in which relevant data is organized into parameter groups determines the utility and the efficiency of the network. The basic philosophy applied when organizing Parameter Groups of associated data is as follows:

a) Instantaneous and real-time data are transmitted rapidly using the minimum number of frames, preferably single-frame. This data is often provided without qualifying status fields or other information that is needed less frequently.
    i. *Example*: Latitude and longitude is provided as a real-time position fix in a single frame with no additional information, the Source Address continuously identifies the source of the data.

b) Data that changes less frequently are transmitted periodically along with associated qualifiers and a time stamp or sequence identifier (SID) to link data from the same source address occurring at the same time, this is usually a fast-packet message.
    i. *Example*: Position is formatted as absolute position in multiple frames containing: SID, latitude, longitude, altitude, datum, geoidal offset, operating mode, and time. Source Address links this information with the instantaneous data.

c) Additional qualifying data is contained in additional fast-packet messages.
    i. *Example:* A status message is formatted with: SID, number of satellites in view, number of satellites used, HDOP, VDOP, PDOP, Mode, DGPS reference station ID, age of data, and time. SID and Source Address link this data to the absolute position message.

The Parameter Group Number, which is a part of the CAN Identifier field, only influences access to the network when there is contention for the network.

The primary information necessary to describe a Parameter Group is shown in Table A-1 below:

Table A-1 Parameter Group Definition

Name	Unique descriptive name or acronym for the Parameter Group
Parameter Group Number	PGNs are assigned by the NMEA 2000 Committee according to the nature of the data to be transmitted. Blocks of numbers are available for manufacturer's proprietary Parameter Groups and for globally and specifically addressed Parameter Groups. PGNs set network access priority but are subordinate to the Priority bits described below; low-number PGNs have higher priority. Fast-packet messages may be assigned PGNs over the full range of values.
Destination (Global or Addressed)	Determines if a specific device address may be included in the CAN Identifier field or if the message is intended for global use.

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Table A-1 Parameter Group Definition	
Description	Free-form text description of the Parameter Group, special conditions that apply, relationships with other Parameter Groups, etc. The section may contain requirements and constraints on the use of the message but shall not modify the definition of specific fields.
Default Priority Value	The default values within the 3-bit (0 – 7) priority range. General categories are: Alarms, safety, and network critical messages: default = 2 Commands, real-time data, and mission critical proprietary messages : default = 3 Non-critical messages (status and data): default = 5 and 6 Multi-packet messages: default = 7 Proprietary messages: default = 7
Default Update Rate	Specifies the required default value of how often this PGN should be broadcast in units of milliseconds. There may be situations where the default priority and or default broadcast rates are not adequate to properly support safety or mission critical data. See Appendix D § 4.10 "*Meeting Changing Demands.*"
Frequency	Provides the Default Update Rate in units of Cycles per Second (Hertz).
Query Support **- Required** **- Optional** **- Prohibited**	Query Support specifies whether or not nodes that transmit this message shall respond to the Request Group Function (PGN 126208) with the requested data. • **Required-** Node must respond with the requested data. • **Optional-** Node must respond with requested data or a NACK. • **Prohibited-** A node must not respond. Notes: 1. See the section titled "Request Parameter" below. 2. The Prohibited state does not preclude the ability to alter transmission interval and offset.

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Table A-1 Parameter Group Definition

Command Support **- Required** **- Optional** **- Prohibited**	Command Support specifies whether or not nodes that transmit this message shall accept Command Group Function (PGN 126208) commands that change the message's Data Fields' values. Changing the Data Fields' values may change the configuration and/or behavior of the node. • **Required-** The node must support commanding one or more Data Fields within this PGN. • **Optional-** The node may support commanding one or more Data Fields within this PGN. There are no Data Fields with mandatory Command Parameter support within this PGN. There are one or more Data Fields with optional command parameter support within this PGN. • **Prohibited-** The node shall not support commanding Data Fields within this PGN. Notes: 1. Refer to the Parameter Group's Data Fields' Command Parameter values to determine which Data Fields must or may be supported. 2. The Prohibited state does not preclude the ability to alter the Priority Setting. 3. See the section titled "Command Parameter" below.
Single Frame (Yes or No)	Specifies whether or not this message is defined as a single-frame message. Once defined as single frame, the Parameter Group cannot become a multi-frame. Generally single-frame messages are defined using all 8-bytes in the data field with unused bits set aside as reserved bits.
Acknowledgment Requirements	Specifies the reply required in response to this Parameter Group message. See NMEA 2000 Main Document Section 3.2.3.

Data Field Definitions

Field No.	Field Name	Field Size Bytes	Field Size Bits	Request Parameter	Command Parameter	Data Dictionary Item No.	Notes
see note 1	see note 2	see note 3	see note 3	see note 4	see note 5	see note 6	see note 7
see note 8	⇓	⇓	⇓	⇓	⇓	⇓	⇓

Notes**:**

1. **Field Number** is a unique sequence number incrementing from 1, for each Data Field which contains an identifiable parameter in a Parameter Group. A Data Field does not have to start or end on a byte boundary. The Field Number may be used by other messages and in applications to specify or identify a parameter (e.g., by a request or a command message) and shall not be changed once assigned.

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The NMEA 2000 committee may, in the future, add additional Data Fields to the end of previously defined messages and may partition a reserved-bit field if it is the last Data Field in a message. With this exception, and also that of defining the meaning but not changing the size of reserved bit fields within a message, the message content or structure shall not change once defined.

2. **Field Name-** Each Data Field shall have a local name. A local name is assigned to a parameter and may be used to further define data identified by the data dictionary item (e.g., field name “Waypoint Number” may be used with data dictionary item “Generic Numeric ID” to define a Waypoint Number field).

3. **Field Size** defines the quantity of the bytes and bits in each Data Field. The following subsections define the conventions for handling Data Fields. See Appendix B, Section 1, "Data placement within the CAN frame", for a thorough explanation of these fundamental rules, and several illustrations.
   - The data portion of a message shall consist of byte-fields and bit-fields. Multiple bit-fields are allowed within bytes and bit-fields may cross byte boundaries.
   - Single or multiple bit-fields shall be filled with reserved bits (logical “1”) if necessary to provide an integer number of bytes per frame.
   - Data Fields other than bit-fields shall always start on a byte boundary.
   - Bit-fields are expressed by their size from 1 to n-bits. Bit-fields, single or multiple, not separated by byte-fields shall be sent collectively as byte-fields to the CAN controller. The least significant bit (LSb) of the first bit-field is the LSB of the byte(s) sent to CAN and the most significant bit (MSb) of the last bit-field is the MSB of the last byte sent to CAN.
   - Character byte-fields may be variable length; the length is defined in the data dictionary item for that field or may be defined in the Notes field.
   - Data Fields are transmitted sequentially starting at field No. 1. Data Fields are transmitted in this order regardless of byte and frame boundaries.
   - Multi-byte data shall be sent to the CAN controller least significant byte (LSB) first (Intel or little endian byte order).
   - Character strings are sent to the CAN controller first character first (e.g., in the string “ABC<string terminator>” character <A> is sent first).

4. **Request Parameter** specifies whether or not the Data Field is intended as a qualifier when requesting specific data with the Request Group Function (PGN 126208). NMEA Reserved fields do not have a Request Parameter setting. Request Parameter can have the following values:

      - ***Required -*** A “Required” value in the Request Parameter means that manufacturers must support the Data Field’s Field Number as a requested parameter value in the Request Group Function (PGN 126208). This ensures that every NMEA certified product provides the same level of request support for a specific Parameter Group.

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        - Example- PGN 126996 (Product Information) contains Field 2 “NMEA Manufacturers Product Code” which states “Required” in the Request Parameter. By specifying a value for Field 2 in a Request PGN (126208) which is requesting PGN 126996, only nodes that have a match for this parameter would respond with their Product Information (PGN 126996). This example illustrates how a node may interrogate the NMEA network to determine the number of nodes with the same make and model (i.e. multiple GPS devices from the same manufacturer) is on the network.

    - ***Optional -*** An “Optional” value in the Request Parameter means the manufacturer may support the Data Field’s Field Number as a requested parameter value in the Request Group Function (PGN 126208). The NMEA will be phasing out the “No” value in NMEA Network Message Database versions after 1.301 and replacing the “No” value with “Optional”.

    - ***Note -*** A “Note_” value in the Request Parameter means the manufacturer must read the additional explanation found in the Parameter Group description to determine necessary support. The note describes how to implement the Data Field’s Field Number as a requested parameter value, and will indicate if this is an “Optional” or “Required” behavior.

  It is recommended that manufacturers use this facility to provide request capabilities beyond the minimum specified by this standard.

5. **Command Parameter** specifies whether or not the Data Field is configurable with the Command Group Function (PGN 126208). NMEA Reserved fields do not have a Command Parameter setting. Command Parameter can have the following values:

    - ***Required -*** A “Required” value in the Command Parameter means that manufacturers must support the Data Field’s Field Number as a commanded parameter value in the Command Group Function (PGN 126208). This ensures that every NMEA certified product provides the same level of command support for a specific Parameter Group.

    - ***Optional -*** An “Optional” value in the Command Parameter means the manufacturer may support the Data Field’s Field Number as a commanded parameter value in the Command Group Function (PGN 126208).

    - ***Prohibited -*** A “Prohibited” value in the Command Parameter means the manufacturer shall not support the Data Field’s Field Number as a commanded parameter value in the Command Group Function (PGN 126208).

    - ***Note -*** A “Note_” value in the Command Parameter means the manufacturer must read the additional explanation found in the Parameter Group description to determine necessary support. The note describes how to implement the Data

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Field's Field Number as a commanded parameter value, and will indicate if this is an "Optional" or "Required" behavior.

6. **Data Dictionary**- Except for qualifying information contained in the Field Name, the referenced Data Dictionary Item (see A-14.) is the only definition for the data being transmitted. A Data Dictionary item shall define a Data Field, other parts of the Parameter Group shall not be used to modify or make exception to the Data Dictionary definition. See A-3. – Parameter Group Field Definitions.

   In addition to describing the application of the Data Field, the Data Dictionary item defines the data-type and size of the data transmitted, physical parameter units, and range and resolution.
   All data types defined and supported by NMEA 2000 contain two or three special states or values. Every data field may thus indicate "Data not available" or "Out of range" by using the specific values. This allows devices to provide partial information within a Parameter Group when the value for all the Data Fields within the Parameter Group are not known, not available, or not yet measured. See section A.10 Data Type for these special data states and values.

7. The **"Notes"** field is only used to qualify the use of the Data Field within the message (e.g., specify the maximum size of a variable size field) or to define relationships between Data Fields (e.g., reset a process if two Data Fields have value = xx). The "Notes" field shall not be used to define the meaning of the Data Field. Data Field definitions are found only in the Data Dictionary.

8. Certain messages may have a variable number of additional Data Fields beyond the fixed fields that are defined. In these cases, the description of these Data Fields is found in the Field Name and the use is defined in the Notes field.

A-2. Parameter Group Data Field Definitions

All Data Fields in all Parameter Groups are defined in a Data Dictionary common to all messages. Local definitions, other than implied by the Field Name, in a Parameter Group are not allowed. The development of Data Field definitions follows the sequence:

Data Type
The method of representing numbers, characters, and individual bit-fields. Makes use of standard character representation and numerical types for integers and floating point numbers.
⇓
Data Formats
Uses the standard Data Types to describe the range

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and resolution of all physical parameters and special entities such as strings and numerical counters. .

⇓

Data Dictionary

Using only the defined Data Formats, the Data Dictionary provides specific definitions for each field in all NMEA Network messages. This common Data Dictionary serves all messages; data items are re-used by different messages whenever possible.

⇓

Parameter Group Definition

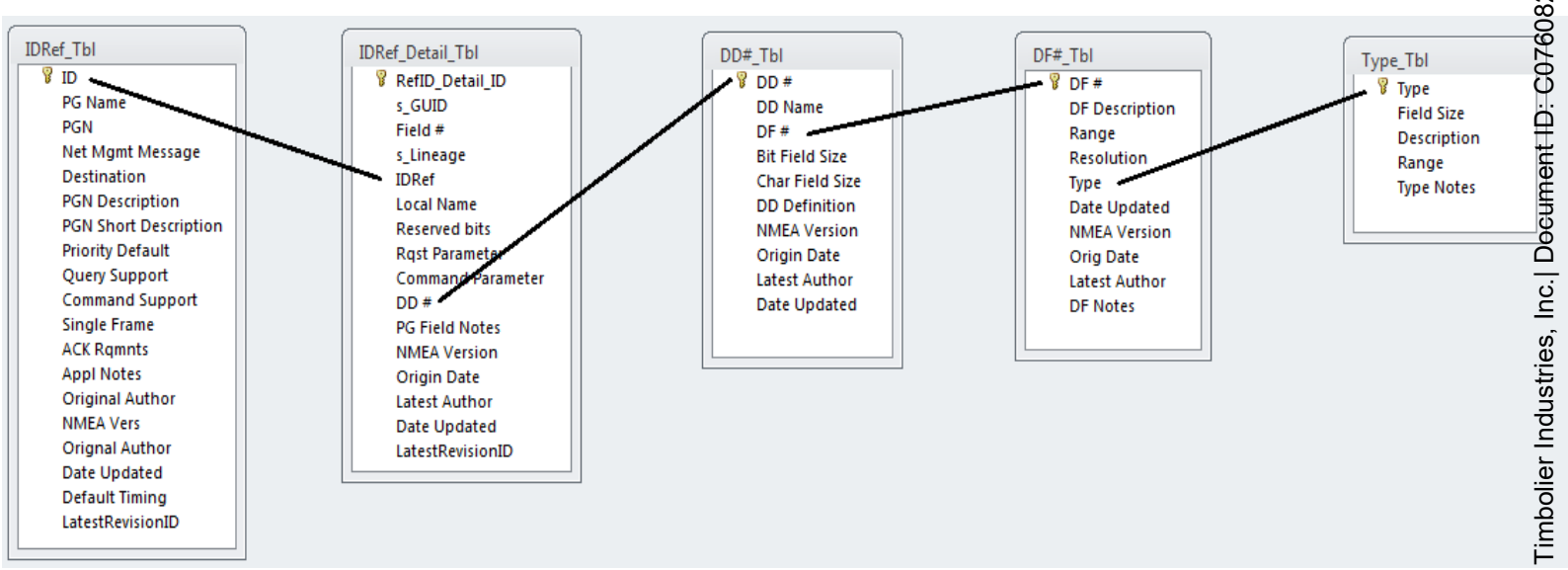
Locates the Data Dictionary items within the Data Fields of each message. Assigns a Field Number and a Field Name for each Data Dictionary item used in a message.

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A-3. Parameter Group Database Structure

NMEA Network Messages are constructed from tables as shown in Figure A-1 and are maintained in the Microsoft Access database file “NMEA Network Message.mdb”. Individual Parameter Groups are developed in two parts in compliance with section A.1. The first part, Parameter Group Definition (Table “IDRef_Tbl”), provides the Parameter Group Name, Parameter Group Number, Destination Type, Description, Default Priority, Default Timing, Query Support, Command Support, Acknowledgment Requirements, and Single-Frame status. The second part is the Data Field portion of the Parameter Group, table “IDRef_Detail_Tbl”. This table is linked to the Parameter Group definition table “IDRef_Tbl” by the ID reference field. Individual Data Fields are constructed by associating Field Names with Data Dictionary items and by providing Data Field Notes and reserved-bit counts as necessary. Data Dictionary items, defined in table “DD#_Tbl”, are linked to Data Format and Data Type tables “DF#_Tbl” and “Type_Tbl” respectively to automatically provide all of the additional Parameter Group Data Field information.

Figure A-1 – Data Definition Structure



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A-4. Parameter Group Example

The Parameter Group definition for Distance Log, PGN 128275 is shown below in Figure A-2.

Distance Log **PGN: 128275** **hex: 1F513**

This PGN provides two distance values recorded from one log which measures speed through water. The UTC time of the last distance increment is captured as Measurement Date & Time (which will be near current time). The distance values are stored during power down and resume counting after power up.

Total Cumulative Distance is normally set to zero when the log is installed and never reset after that.

Note 1: Distance Since Last Reset may manually be set to zero at any suitable time.
The "Distance Since Last Reset" is reset by setting its value to 0 with the "Command Group Function" PGN 126208.

Single Frame:	No	Priority Default:	6	Default Update Rate:	1000 milliseconds	Frequency:	1. cycles per second
Destination:	Global	Query Support:	Optional	Command Support:	Optional	ACK Rqmnts:	

Original Reference ID # 39

Field #	Field Name		Byte Field Size / Bit Field Size	Request Parameter / Command Parameter	Range	Resolution	
1	**Measurement Date**		2 /	Optional / Optional			
	DD039 Generic date				Days since January 1, 1970, Date is relative to UTC Time.		
	DF41 Date, day count	**uint16**			0 to 65,532 days	1 day	0 = January 1, 1970, max = ~179 years
2	**Measurement Time**		4 /	Optional / Optional			
	DD158 Generic time of day				24 hour clock, 0 = midnight, time is in UTC		
	DF06 Time of day	**uint32**			0 to 86,401 s	1x10E-4 s	~24 hours, 0 = midnight, range allows for up to two leap seconds per day
3	**Total Cumulative Distance**		4 /	Optional / Optional			
	DD120 Distance, voyage						
	DF11 Distance, long	**uint32**			0 to ~4.295x10E+9 m	1 m	
4	**Distance Since Last Reset**		4 /	Optional / Note 1			
	DD120 Distance, voyage						
	DF11 Distance, long	**uint32**			0 to ~4.295x10E+9 m	1 m	

Figure A-2 – PGN Example

The data provided by this Parameter Group are the cumulative distance traveled (Total Cumulative Distance = field 3), distance from a specified point from reset (Distance Since Last Reset = field 4), and the date and time (fields 1 and 2) of the measurements. This NMEA Network message is a global broadcast message (Destination = Global) with a default priority level of “6”. The default update rate for this message is 1 Hz or once every second

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The Distance Log message is defined as a fast-packet message requiring 3 CAN frames. The message consists of 4 Data Fields spread over the 2 frames. The boundaries of the Data Fields with respect to the CAN frames have no significance and it is a task of the receiving node to reassemble the Data Fields from the received frames.

The time and date may be initialized or modified by setting Data Fields 1 and 2 with the Command Group Function message. If the device does not know the time and date when this Parameter Group is generated, the time and date fields should indicate this with the appropriate values. For a uint16 Data Type, as in the Date field, the value that indicates "Data not available" is 65,535. For a unit32 Data Type, as in the Time field, the value that indicates "Data not available" is 4,294,967,295. Definitions for the information in all 4 Data Fields are provided in the Data Dictionary.

A-5. Requesting New PGNs to NMEA

The NMEA welcomes manufacturers to develop and propose new NMEA Network Message PGNs. All new PGNs are based on consensus and approval of the NMEA 2000 Technical Committee and specific working groups. To propose a new PGN, please email the NMEA Director of Standards at sspitzer@nmea.org so that specific forms can be sent to you.

A-6. Proprietary Parameter Groups

Proprietary messages provide a means for manufacturers to use the parameter group structure and definitions of this standard to transfer data which is beyond the scope of approved NMEA Network Parameter Groups.

These are the only conditions for the use of proprietary messages:

- Condition1: The message is device specific such as, unique calibration or special configuration.
- Condition 2: Data is being used for test purposes.
- Condition 3: Data PGN is not available in the current NMEA Network Message Database.

In the case of condition 3, proprietary PGNs shall only be used until the necessary data fields are supported by a NMEA published Network Message.

The NMEA 2000 standard strongly encourages manufacturers to calibrate and configure their respective devices over the network.

NMEA encourages manufacturers to publish their proprietary PGNs, PGNs field descriptions and PGNs transmission rates.

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Proprietary Format

Different Parameter Group Numbers are used in order to distinguish between single-frame and fast-packet or ISO multi-packet proprietary messages.

NMEA Network Message Proprietary PGNs are required to contain 3 fields: a Manufacturer Code (11-bit), a Reserved field (2-bit), and the Industry Group (3-bit). These fields are necessary so that the format and structure of a proprietary PGN may be known. Every PGN contains the source or sender's address. This alone is not always sufficient in identifying the contents of a proprietary PGN.

A Definer Tag is a 16-bit entity, composed of 3 fields: a Manufacturer Code, a Reserved 2-bit field, and the Industry Group. See Figure A-3 below for an illustration of the fields that comprise the Definer Tag. NMEA 2000 proprietary PGNs are required to contain a Definer Tag in the first 2 data bytes of each message. The Definer Tag is necessary so that the format and structure of a proprietary PGN may be known. Every PGN contains the source or sender's address. This alone is not always sufficient in identifying the contents of a proprietary PGN.

For example: Device 1 may wish to have device 2 send a proprietary PGN of device 1's design. The source address indicates that this proprietary PGN is originating from device 2. The Definer Tag indicates that this proprietary PGN from device 2 adheres to a proprietary format defined by device 1. Additional identification fields may be necessary, depending upon how a manufacturer implements proprietary PGN formats.

Proprietary Message Certification

Data that is defined within standardized NMEA Network Messages shall be conveyed using the standardized messages. For the benefit of the maritime community as a whole, the NMEA strongly encourages and supports all manufacturers to propose the standardization of their respective proprietary messages.

All proprietary messages shall be disclosed to NMEA for certification purposes. NMEA will not disclose this information. Manufacturers shall disclose the following confidential information:

- PGN number
- PGN title and or purpose
- PGN priority
- PGN update rate
- PGN acknowledgement requirements
- PGN data field descriptions

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Single-Frame Proprietary

PGNs 061184 (Destination Addressable) and 065280 through 065535 (Destination Global) are reserved for single-frame proprietary usage only.

These PGNs shall not be used for multi-frame messages, and cannot utilize the ISO 11783 multi-packet or NMEA Network Message fast-packet protocols.

Fast-Packet Proprietary

PGNs 126720 (Destination Addressable) and 130816 through 131071 (Destination Global) are reserved for non-single-frame proprietary usage only.

These PGNS shall be sent using the ISO multi-packet or NMEA Network Message fast-packet protocols.

An example of a proprietary PGN message is shown below.

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Proprietary **PGN: XXXXX**

This PGN provides the manufacturer with the ability to communicate data, commands, or any other proprietary information over the NMEA 2000 network. The first three fields are mandatory for all NMEA 2000 proprietary PGNS. The manufacture may define as many additional fields as necessary, beginning with field 4. See section 3.4 in the main document for more information on proprietary PGNs

Single Frame:	N	*Priority Default:*	6	*Default Update Rate:*	2500 milliseconds	*Frequency:*	.4	cycles per second	
Destination:	Global	*Query Support:*	Required	*Command Support:*	Required	*ACK Rqmnts:*	None		

Original Reference ID # 225

Field #	*Field Name*							
1	**Manufacturer Code**		*Byte Field Size:*		*Request Parameter*	Required		
			Bit Field Size:	11	*Command Parameter:*	Required		
	DD172 NMEA Manufacturer Code		Assigned by NMEA Committee					
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields	
2	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*			
			Bit Field Size:	resv 2	*Command Parameter:*			
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"					
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields	
3	**Industry Group**		*Byte Field Size:*		*Request Parameter*	Required		
			Bit Field Size:	3	*Command Parameter:*	Required		
	DD168 Industry Group		0 = Global; 1 = On-Highway; 2 = Agricultural and Forestry; 3 = Construction; 4 = Marine; 5 = Industrial - Process Control - Stationary (Gen-Sets) 6 = Reserved for SAE 7 = Reserved for SAE					
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields	
4	**Manufacturer Defined Fields**		*Byte Field Size:*		*Request Parameter*	Optional		
			Bit Field Size:		*Command Parameter:*	Optional		
	DD							
	DF	**bit(n)**	*Range:*		*Resolution:*			

Figure A-3 – Proprietary PGN Example

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A-7. Deprecation Definition and Policy of PGNs

Within NMEA standards or documentation, the term “deprecation” means:
When applied to a PGN it is an indication that the respective PGN and its functionality has been superseded by or included in other PGN(s). This is intended to raise awareness that alternative methods have been published.

Deprecation Does NOT Mean Obsolescence.

Policy:

Manufacturers are required to certify their products to the latest version of the NMEA Network Message Database. Currently and in general, NMEA grants approximately 18 months for manufacturers to implement the new Network Message database version for new products. NMEA publishes technical bulletins with implementation dates regarding specific PGN(s) and Network Message database versions.

Deprecated PGNs will remain in the NMEA Network Message database. Future product designs that use functionality supported by deprecated PGN(s) shall support the new PGN(s). In order to provide backward compatibility with legacy equipment, the NMEA strongly recommends that manufacturers also continue to support the deprecated PGN(s).

With each Deprecated PGN, NMEA will add the following language in the header of the affected PGN:

> ***PGN XXX- DEPRECATED***
>
> *This PGN has been deprecated (as of NMEA Network Message Database version x.x) and is not recommended for new designs. However, support of the deprecated PGN xxxxxx is strongly recommended to ensure compatibility with legacy equipment.*
>
> *PGN xxxxxx shall be used for new designs.*

NMEA will list all Deprecated PGNs within the revision history of Appendix B.

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A-8. Revision History

Revision	Date	Description
1.000	Sept 2001	Initial Release
1.001	Oct 2001	Section A-5 corrected section reference in example Prop PGN from X.X to 3.4
1.002	Nov 2002	Distance Log Example corrected text form @CAN frames to # CAN frames
1.003	May 2003	Version number changed
2.000	January 2013	Added Command and Request parameter text. Updated screen images Added New PGN request section-A5 Added Proprietary Parameter Group Section-A6 Added Deprecation Definition & Policy-A7
2.100	February 2015	Added note 2 to Query Support Table A-1 Added note 2 to Command Support Table A-1 Added reference to Appendix D in Table A-1 *Default Update Rates* Added Definer Tag text in Section A-6 Proprietary Format

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PGN Report (sorted by Number)

National Marine Electronics Association



NMEA 2000 ®

Appendix B.1 -- Parameter Groups

STANDARD FOR SERIAL-DATA NETWORKING OF MARINE ELECTRONIC DEVICES

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EFFECTIVE DATE AUGUST 1, 2012

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ISO Acknowledgment

PGN: 059392
hex: E800

This message is provided by ISO 11783 for a handshake mechanism between transmitting and receiving devices. This message is the possible response to acknowledge the reception of a “normal broadcast” message or the response to a specific command to indicate compliance or failure. The application layer is responsible for determining when this message is desired, outside of network management requirements specified by this standard (e.g. response to ISO Request message). The destination address of this PGN shall always contain a destination specific address.

Note 1: Version 1.000 of the NMEA 2000 Standard required the destination address to be the global address of 255.

Note 2: It is prohibited to request this PGN.

Single Frame: Yes *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Address *Query Support:* Prohibited *Command Support:* Prohibited *ACK Rqmnts:* None

Field #	*Field Name*						Original Reference ID # 64
1	**Control Byte**		*Byte Field Size:*		*Request Parameter*	Prohibited	
			Bit Field Size:	8	*Command Parameter:*	Prohibited	
	DD177 ISO 11783 ACK Status			0x00 = Positive Acknowledgment; 0x01 = Negative Acknowledgment; 0x02 = PGN supported but access denied; 0x03 to 0xFF = Reserved			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
2	**Group Function Value**		*Byte Field Size:*		*Request Parameter*	Prohibited	
			Bit Field Size:	8	*Command Parameter:*	Prohibited	
	DD178 Group Function Value			Group Function of PGN being acknowledged. This field identifies for a device the specific group function of a PGN being acknowledged or declined. This field is not used if the PGN being acknowledged or declined is not a group function PGN.			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
3	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*		
			Bit Field Size:	resv 24	*Command Parameter:*		
	DD001 Reserved field			Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.						
4	**PGN of Requested Information**		*Byte Field Size:*		*Request Parameter*	Prohibited	
			Bit Field Size:	24	*Command Parameter:*	Prohibited	
	DD009 PGN			24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields

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ISO Request

PGN: 059904
hex: EA00

ISO 11783 defines this message to provide a method for requesting the transmission of a PGN from a network device or devices. As defined by ISO, this message has a data length of 3 bytes with no padding added to complete the single frame. The appropriate response to this message is based on the PGN being requested, and whether the receiver supports the requested PGN. See section 3.4.2 of 11783-3 for the rules governing the response to this PGN.

Single Frame:	Yes	*Priority Default:*	6	*Default Update Rate:*		milliseconds	*Frequency:*	NA	cycles per second
Destination:	Address	*Query Support:*	Optional	*Command Support:*	Optional	*ACK Rqmnts:*	Requested data or 59392 ISO Acknowledge with error code.		

Field # *Field Name* Original Reference ID # 65

1	**PGN being requested**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	24	*Command Parameter:*	Optional	
	DD009 PGN		24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	

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ISO Transport Protocol, Data Transfer

PGN: 060160
hex: EB00

ISO 11783 defines this PGN as part of the transport protocol method used for transmitting messages that have 9 or more data bytes. This PGN represents a single packet of a multipacket message and is used in conjunction with PGN 60416. Once a connection has been established or a broadcast announcement has been made, this message is transmitted using the timing and handshake requirements in section 3.10 of 11783-3 until all the message's packets are transmitted or the transmission is aborted. Although this PGN is addressable, when used with the Broadcast Announce Message (BAM) method, the destination shall be 255.

Single Frame: Yes *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Address *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 71

Field #	*Field Name*					
1	**Sequence number of multi-packet frame**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	8	*Command Parameter:*	Optional	
	DD180 Multi-packet frame counter		Valid range 0x01 to 0xFF			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
2	**Multi-packet packetized data**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	56	*Command Parameter:*	Optional	
	DD181 Multi-packet packetized data		56 bits organized as seven 8-bit bytes, each of the bytes is transmitted in the same order as it would in a standard (i.e., single frame) message. Unused bytes shall be filled with 0xFF.			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	

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ISO Transport Protocol, Connection Management - RTS group function

PGN: 060416
hex: EC00

ISO 11783 defines this group function PGN as part of the transport protocol method used for transmitting messages that have 9 or more data bytes. This PGN's role in the transport process is determined by the group function value found in the first data byte of the PGN.

RTS - When the group function is Request To Send (RTS), the PGN is asking a specific node on the network for permission to transmit a larger than 8 byte message to the node. This process is referred to as opening a connection.

For a complete description of this PGN's usage and timing requirements see section 3.10 of 11783-3.

Single Frame:	Yes	*Priority Default:*	6	*Default Update Rate:*		*Frequency:*	NA cycles per second
Destination:	Address	*Query Support:*	Optional	*Command Support:*	Optional	*ACK Rqmnts:*	Refer to Section 3.10 of ISO `11783-3

Field # *Field Name* — Original Reference ID # 66

1 RTS Group Function Code

Byte Field Size: *Bit Field Size:* 8 — *Request Parameter* Optional — *Command Parameter:* Optional

DD179 Group Function, Connection Management

0x10 = Request to Send;
0x11 = Clear to Send;
0x13 = End of Message;
0x20 = Broadcast Announce Message;
0xFF = Abort;
0x00 to 0xF, 0x12, 0x14 to 0x1F, 0x21 to 0xFE = Reserved

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

This is the RTS message, set = 0x10

2 Total message size, bytes

Byte Field Size: 2 — *Request Parameter* Optional — *Command Parameter:* Optional

DD007 Generic numeric ID, medium

Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

Only values in the range of 9 to 1785 are allowed.

3 Total number of frames to be transmitted

Byte Field Size: *Bit Field Size:* 8 — *Request Parameter* Optional — *Command Parameter:* Optional

DD180 Multi-packet frame counter

Valid range 0x01 to 0xFF

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

4 NMEA Reserved

Byte Field Size: *Bit Field Size:* resv 8 — *Request Parameter* — *Command Parameter:*

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

5 PGN of multi-packet message

Byte Field Size: *Bit Field Size:* 24 — *Request Parameter* Optional — *Command Parameter:* Optional

DD009 PGN

24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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ISO Transport Protocol, Connection Management - EOM group function

PGN: 060416
hex: EC00

ISO 11783 defines this group function PGN as part of the transport protocol method used for transmitting messages that have 9 or more data bytes. This PGN's role in the transport process is determined by the group function value found in the first data byte of the PGN.

EOM - When the group function is End Of Message (EOM), the PGN is notifying the transmitter of a multipacket message that all packets were received successfully. This group function signals the successful conclusion of a transmission started with the RTS group function thus closing the connection.

For a complete description of this PGN's usage and timing requirements see section 3.10 of 11783-3.

Single Frame: Yes *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Address *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 68

Field #	Field Name					
1	**EOM Group Function Code**	Byte Field Size:		Request Parameter	Optional	
		Bit Field Size:	8	Command Parameter:	Optional	
	DD179 Group Function, Connection Management	0x10 = Request to Send; 0x11 = Clear to Send; 0x13 = End of Message; 0x20 = Broadcast Announce Message; 0xFF = Abort; 0x00 to 0xF, 0x12, 0x14 to 0x1F, 0x21 to 0xFE = Reserved				
	DF52 Bit field	**bit(n)**	Range: Variable	Resolution: 1	Used to construct bit fields	
	This is the EOM message, set = 0x13					
2	**Total message size, bytes**	Byte Field Size:	2	Request Parameter	Optional	
		Bit Field Size:		Command Parameter:	Optional	
	DD007 Generic numeric ID, medium	Number of route, waypoint, event, mark, etc.				
	DF54 Integer, 16 bit unsigned	**uint16**	Range: 0 to 65,532	Resolution: 1 bit	Unit-less number	
	Only values in the range of 9 to 1785 are allowed.					
3	**Total number of frames received**	Byte Field Size:		Request Parameter	Optional	
		Bit Field Size:	8	Command Parameter:	Optional	
	DD180 Multi-packet frame counter	Valid range 0x01 to 0xFF				
	DF52 Bit field	**bit(n)**	Range: Variable	Resolution: 1	Used to construct bit fields	
4	**NMEA Reserved**	Byte Field Size:		Request Parameter		
		Bit Field Size:	resv 8	Command Parameter:		
	DD001 Reserved field	Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	Range: Variable	Resolution: 1	Used to construct bit fields	
	Used to align subsequent data on a byte boundary.					
5	**PGN of multi-packet message**	Byte Field Size:		Request Parameter	Optional	
		Bit Field Size:	24	Command Parameter:	Optional	
	DD009 PGN	24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first				
	DF52 Bit field	**bit(n)**	Range: Variable	Resolution: 1	Used to construct bit fields	

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ISO Transport Protocol, Connection Management - Abort group function

PGN: 060416
hex: EC00

ISO 11783 defines this group function PGN as part of the transport protocol method used for transmitting messages that have 9 or more data bytes. This PGN's role in the transport process is determined by the group function value found in the first data byte of the PGN.

ABORT - When the group function is Abort, the PGN is notifying the transmitter or receiver of the multipacket message that the other partner is terminating the connection without completing the transfer of the message. This group function can also be used to refuse the connection when the RTS group function is initially received.

For a complete description of this PGN's usage and timing requirements see section 3.10 of 11783-3.

Single Frame: Yes | *Priority Default:* 7 | *Default Update Rate:* | milliseconds | *Frequency:* NA cycles per second
Destination: Address | *Query Support:* Optional | *Command Support:* Optional | *ACK Rqmnts:* None

Original Reference ID # 70

Field #	Field Name				
1	**Abort Group Function Code**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	8	*Command Parameter:*	Optional

DD179 Group Function, Connection Management

0x10 = Request to Send;
0x11 = Clear to Send;
0x13 = End of Message;
0x20 = Broadcast Announce Message;
0xFF = Abort;
0x00 to 0xF, 0x12, 0x14 to 0x1F, 0x21 to 0xFE = Reserved

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

This is the Abort message, set = 0xFF

2	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 32	*Command Parameter:*	

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

3	**PGN of multi-packet message**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	24	*Command Parameter:*	Optional

DD009 PGN

24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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ISO Transport Protocol, Connection Management - BAM group function

PGN: 060416
hex: EC00

ISO 11783 defines this group function PGN as part of the transport protocol method used for transmitting messages that have 9 or more data bytes. This PGN's role in the transport process is determined by the group function value found in the first data byte of the PGN.

BAM - When the group function is Broadcast Announce Message (BAM), the PGN is notifying all network nodes that a multipacket message is about to be transmitted on the network. Because the BAM group function is broadcast, no further handshake is required before the multipacket message is transmitted.

For a complete description of this PGN's usage and timing requirements see section 3.10 of 11783-3.

Single Frame: Yes *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Address *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 69

1 BAM Group Function Code
Byte Field Size: *Bit Field Size:* 8
Request Parameter Optional
Command Parameter: Optional

DD179 Group Function, Connection Management
0x10 = Request to Send;
0x11 = Clear to Send;
0x13 = End of Message;
0x20 = Broadcast Announce Message;
0xFF = Abort;
0x00 to 0xF, 0x12, 0x14 to 0x1F, 0x21 to 0xFE = Reserved

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

This is the BAM message, set = 0x20

2 Total message size, bytes
Byte Field Size: 2 *Bit Field Size:*
Request Parameter Optional
Command Parameter: Optional

DD007 Generic numeric ID, medium
Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

Maximum value = 1785

3 Total number of frames to be transmitted
Byte Field Size: *Bit Field Size:* 8
Request Parameter Optional
Command Parameter: Optional

DD180 Multi-packet frame counter
Valid range 0x01 to 0xFF

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

4 NMEA Reserved
Byte Field Size: *Bit Field Size:* resv 8
Request Parameter
Command Parameter:

DD001 Reserved field
Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

5 PGN of multi-packet message
Byte Field Size: *Bit Field Size:* 24
Request Parameter Optional
Command Parameter: Optional

DD009 PGN
24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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ISO Transport Protocol, Connection Management - CTS group function

PGN: 060416
hex: EC00

ISO 11783 defines this group function PGN as part of the transport protocol method used for transmitting messages that have 9 or more data bytes. This PGN's role in the transport process is determined by the group function value found in the first data byte of the PGN.

CTS - If the group function is Clear To Send (CTS), the PGN is notifying the transmitter of a multipacket message how many packets the node is ready to receive and which packets have been received successfully. This group function is the proper response to an RTS group function to start the transmission of the multipacket data and allows the receiver to control the flow of data throughout the multipacket transmission.

For a complete description of this PGN's usage and timing requirements see section 3.10 of 11783-3.

Single Frame: Yes | *Priority Default:* 7 | *Default Update Rate:* milliseconds | *Frequency:* NA cycles per second
Destination: Address | *Query Support:* Optional | *Command Support:* Optional | *ACK Rqmnts:* None

Field # *Field Name* — Original Reference ID # 67

1 CTS Group Function Code — *Byte Field Size:* | *Bit Field Size:* 8 | *Request Parameter* Optional | *Command Parameter:* Optional

DD179 Group Function, Connection Management

0x10 = Request to Send;
0x11 = Clear to Send;
0x13 = End of Message;
0x20 = Broadcast Announce Message;
0xFF = Abort;
0x00 to 0xF, 0x12, 0x14 to 0x1F, 0x21 to 0xFE = Reserved

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

This is the CTS message, set = 0x11

2 Number of frames that can be sent — *Byte Field Size:* | *Bit Field Size:* 8 | *Request Parameter* Optional | *Command Parameter:* Optional

DD180 Multi-packet frame counter

Valid range 0x01 to 0xFF

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 Number of next frame to be transmitted — *Byte Field Size:* | *Bit Field Size:* 8 | *Request Parameter* Optional | *Command Parameter:* Optional

DD180 Multi-packet frame counter

Valid range 0x01 to 0xFF

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

4 NMEA Reserved — *Byte Field Size:* | *Bit Field Size:* resv 16 | *Request Parameter* | *Command Parameter:*

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

5 PGN of multi-packet message — *Byte Field Size:* | *Bit Field Size:* 24 | *Request Parameter* Optional | *Command Parameter:* Optional

DD009 PGN

24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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ISO Address Claim

PGN: 060928
hex: EE00

This network management message is used to claim a network address and to respond with device information (NAME) requested by the ISO Request (PGN 059904) or Complex Request Group Function (PGN 126208). This parameter group is always transmitted with a destination address global.

This PGN contains several fields that are Request Parameters. These Request Parameters can be used independently or in any combination.

A node receiving an ISO Request (PGN 059904) for this PGN shall respond by providing this PGN.

If a Complex Request Group Function (PGN 126208) requesting this PGN is received, the receiving node shall respond in the following manner:

If no Request Parameter fields have been specified with the Complex Request, then the response is to return this PGN, just like responding to the ISO Request (PGN 59904) described above.

If the Complex Request (PGN 126208) specifies one or more Request Parameter fields, then the response shall be filtered by the one or more fields and field values contained within the request. For example, if the Complex Request for this PGN contained a value for field 2, the Manufacturer's Code, then the device would respond with this PGN, if and only if the device's Manufacturer Code matched the value requested. If the device's Manufacturer code did not match the value requested, then the response would depend on whether the request was global or addressed. A global request containing one or more requested field values that do not match requires no response, while an addressed request containing requested field values, in which one or more do not match, requires a response with the Acknowledge Group PGN (126208), containing the appropriate error codes for each of the requested fields, such as "0x3 = Request or command parameter out-of-range", for the fields that did not match.

ISO Requests for Address Claim parameter group are restricted in address mode and request frequency by the Main NMEA 2000 document, section 8.3.2, "Address to NAME Association Tables".

The ISO Address Claim parameter group is not to be used as a heartbeat message.

Single Frame: Yes *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Address *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* Refer to Section 6.4 of ISO 11783-5

Field # *Field Name* Original Reference ID # 63

1	**Unique Number (ISO Identity Number)**		*Byte Field Size:* *Bit Field Size:* 21		*Request Parameter* Optional *Command Parameter:* Optional	
	DD173 NMEA Unique Number		Binary number assigned by manufacturer to ensure that the NAME field for each manufactured device is unique, reference NMEA Network Management Section 8.			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
2	**Manufacturer Code**		*Byte Field Size:* *Bit Field Size:* 11		*Request Parameter* Required *Command Parameter:* Optional	
	DD172 NMEA Manufacturer Code		Assigned by NMEA Committee			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
3	**Device Instance Lower (ISO ECU Instance)**		*Byte Field Size:* *Bit Field Size:* 3		*Request Parameter* Required *Command Parameter:* Optional	
	DD201 Generic instance 2 (3-bit)		0x0 to 0x7 = instance 0 to 7			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	

The combination of fields 3 & 4 make up the 8 bit NMEA Device Instance. NMEA Device Instance values are not intended to be unique on the network. When NMEA Device Instances are configured, they should be unique within all devices of the same Class & Function code.

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ISO Address Claim

PGN: 060928
hex: EE00

4	**Device Instance Upper (ISO Function Instance)**		*Byte Field Size:*		*Request Parameter*	Required
			Bit Field Size:	5	*Command Parameter:*	Optional
	DD174 Generic instance (5-bit)		0x00 to 0x1F = Instance 0 to 31;			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	The combination of fields 3 & 4 make up the 8 bit NMEA Device Instance. NMEA Device Instance values are not intended to be unique on the network. When NMEA Device Instances are configured, they should be unique within all devices of the same Class & Function code.					
5	**Device Function (ISO Function)**		*Byte Field Size:*		*Request Parameter*	Required
			Bit Field Size:	8	*Command Parameter:*	Optional
	DD171 NMEA Function Code		Dependent on NMEA Device Class DD170, reference NMEA Class & Function Codes, Appendix B6.			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
6	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*	
			Bit Field Size:	1	*Command Parameter:*	
	DD175 Dominant Bit		Set = 0			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
7	**Device Class**		*Byte Field Size:*		*Request Parameter*	Required
			Bit Field Size:	7	*Command Parameter:*	Optional
	DD170 NMEA Device Class		Dependent on Industry Group DD 168, reference NMEA Class & Function Codes, Appendix B6.			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
8	**System Instance (ISO Device Class Instance)**		*Byte Field Size:*		*Request Parameter*	Required
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD169 Generic instance (4-bit)		0x0 to 0xF = Instance number 0 to 15;			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
9	**Industry Group**		*Byte Field Size:*		*Request Parameter*	Required
			Bit Field Size:	3	*Command Parameter:*	Optional
	DD168 Industry Group		0 = Global; 1 = On-Highway; 2 = Agricultural and Forestry; 3 = Construction; 4 = Marine; 5 = Industrial - Process Control - Stationary (Gen-Sets) 6 = Reserved for SAE 7 = Reserved for SAE			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	Marine Industry Group, set = 4					
10	**NMEA Reserved (ISO Self Configurable)**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	resv 1	*Command Parameter:*	Optional
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields

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ISO Commanded Address

PGN: 065240
hex: FED8

ISO 11783 defined this message to provide a mechanism for assigning a network address to a node. The NAME information in the data portion of the message must match the name information of the node whose network address is to be set. ISO 11783-5 requires this message to be sent using the BAM Transport Protocol method. The appropriate response to this message is defined in section 5.2.3 of 11783-5.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* Refer to Section 6.4 of ISO 11783-5

Field # *Field Name* Original Reference ID # 83

1	**Unique Number (ISO Identity Number)**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	21	*Command Parameter:*	Optional	
	DD173 NMEA Unique Number		Binary number assigned by manufacturer to ensure that the NAME field for each manufactured device is unique, reference NMEA Network Management Section 8.				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
2	**Manufacturer Code**		*Byte Field Size:*		*Request Parameter*	Required	
			Bit Field Size:	11	*Command Parameter:*	Optional	
	DD172 NMEA Manufacturer Code		Assigned by NMEA Committee				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
3	**Device Instance Lower (ISO ECU Instance)**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	3	*Command Parameter:*	Optional	
	DD201 Generic instance 2 (3-bit)		0x0 to 0x7 = instance 0 to 7				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
4	**Device Instance Upper (ISO Function Instance)**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	5	*Command Parameter:*	Optional	
	DD174 Generic instance (5-bit)		0x00 to 0x1F = Instance 0 to 31;				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
5	**Device Function (ISO Function)**		*Byte Field Size:*		*Request Parameter*	Required	
			Bit Field Size:	8	*Command Parameter:*	Optional	
	DD171 NMEA Function Code		Dependent on NMEA Device Class DD170, reference NMEA Class & Function Codes, Appendix B6.				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
6	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*		
			Bit Field Size:	1	*Command Parameter:*		
	DD175 Dominant Bit		Set = 0				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
7	**Device Class**		*Byte Field Size:*		*Request Parameter*	Required	
			Bit Field Size:	7	*Command Parameter:*	Optional	
	DD170 NMEA Device Class		Dependent on Industry Group DD 168, reference NMEA Class & Function Codes, Appendix B6.				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields

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ISO Commanded Address

PGN: 065240
hex: FED8

8	**System Instance (ISO Device Class Instance)**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	4	*Command Parameter:*	Optional	
	DD169 Generic instance (4-bit)		0x0 to 0xF = Instance number 0 to 15;				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
9	**Industry Group**		*Byte Field Size:*		*Request Parameter*	Required	
			Bit Field Size:	3	*Command Parameter:*	Optional	
	DD168 Industry Group		0 = Global; 1 = On-Highway; 2 = Agricultural and Forestry; 3 = Construction; 4 = Marine; 5 = Industrial - Process Control - Stationary (Gen-Sets) 6 = Reserved for SAE 7 = Reserved for SAE				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
	Marine Group, set = 4						
10	**Reserved (ISO Self Configurable)**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	resv 1	*Command Parameter:*	Optional	
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
11	**New Source Address**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	8	*Command Parameter:*	Optional	
	DD176 Network Address		0x00 to 0xFB (0 to 251) = Claimable NMEA network address space; 0xFC (252) = Reserved; 0xFD (253) = Reserved; 0xFE (254) = Null address; 0xFF (255) = Global address				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
	Only values less than 252 shall be used.						

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NMEA - Request group function

PGN: 126208
hex: 1ED00

The Request / Command / Acknowledge Group type of function is defined by first field. The message will be a Request, Command, or Acknowledge Group Function.

The Request Group Function is defined as follows:

This message requests the transmission of a specific set of data in a Parameter Group by setting variable parameters within the Parameter Group specified by the field number. Field number and parameter value may appear in any order in this message. When multiple fields and parameters are specified the request is treated as an "AND" function. This PGN may be used to set the transmission interval and the delay before the first transmission.

NMEA encourages manufacturers to provide the capability of altering the values of transmission time for NMEA certified products on the network. When the values of transmission time or delay before the first transmission are altered, they should be maintained after power cycling. All parameter value fields of this request must be padded if necessary to ensure byte boundaries are adhered to.

Note 1: Acknowledge Requirements - Requested data as scheduled or Acknowledge Group Function with acknowledgement error codes.

Single Frame: No *Priority Default:* 3 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Address *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* Note 1

Field # *Field Name* Original Reference ID # 1

1 **Request Group Function Code** *Byte Field Size:* 1 *Bit Field Size:* *Request Parameter* Required *Command Parameter:* Required

DD144 Group Function, Request/Command/Acknowledge
0 = Request Message,
1 = Command Message,
2 = Acknowledge Message,
3 = Read Fields,
4 = Read Fields Reply,
5 = Write Fields,
6 = Write Fields Reply

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number
This is the Request message, set = 0x00.

2 **Requested PGN** *Byte Field Size:* *Bit Field Size:* 24 *Request Parameter* Required *Command Parameter:* Required

DD009 PGN
24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 **Transmission interval** *Byte Field Size:* 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD035 Data transmit interval
Time interval between data transmissions Where:
0x0000 0000 = Turn-off transmission,
0xFFFF FFFE = Restore Default Interval
0xFFFF FFFF = Do not change interval

DF65 Time interval, standard **uint32** *Range:* 0 to ~4.295x10E+6 s *Resolution:* 1x10E-3 s
0xFFFF FFFF in this field and 0xFFFF in field 4: Transmit now without changing timing variables.

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NMEA - Request group function

PGN: 126208
hex: 1ED00

4 Transmission interval offset
Byte Field Size: 2 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD036 Data transmit offset
Offset in transmit time from time of request command:
0x0 = transmit immediately
0xFFFF = Do not change offset.

DF66 Time interval, .01sec **uint16** *Range:* 0 to 655.32s *Resolution:* 1x10E-2sec
0xFFFF in this field and 0xFFFF FFFF in field 3: Transmit now without changing timing variables.

---

5 Number of Pairs of Request Parameters to follow
Byte Field Size: 1 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD006 Generic counter, short
Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

---

6 Field number of first requested parameter
Byte Field Size: 1 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD005 Generic numeric ID, short
Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

---

7 Value of first requested parameter
Byte Field Size: ? — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD000 Undefined
Application specific, defined at time of use

DF00 Undefined **Undefined** *Range:* undefined *Resolution:* undefined Application specific, defined at time of use.
Requested parameter size and type is dependent on the PGN and the specific request parameter field.

---

8 Variable Number of fields, Field number 6 repeated
Byte Field Size: 1 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD005 Generic numeric ID, short
Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

---

9 Variable Number of fields, Field number 7 repeated
Byte Field Size: ? — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD000 Undefined
Application specific, defined at time of use

DF00 Undefined **Undefined** *Range:* undefined *Resolution:* undefined Application specific, defined at time of use.
Requested parameter size and type is dependent on the PGN and the specific request parameter field.

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NMEA - Command group function

PGN: 126208
hex: 1ED00

The Request / Command / Acknowledge Group type of function is defined by first field. The message will be a Request, Command, or Acknowledge Group Function.

The Command Group Function is defined as follows:

The Command Group Function message is directed to a specific address, the Global Address (255) shall not be used. This command sets the value of one, some or all parameters in a Parameter Group. The number of parameters to set is in field 5, then follows the field number and the new value repeated for each of them. A Parameter Group may contain one group of parameters out of multiple instances where the instance number of the group is given in one field. A command to set any parameter of such a group must contain the field number and value of the group instance number.

Example is setting the name of a Tracked Target in PGN 128520, where the Target ID must be specified with the new Name. The field number and value of the group instance is included in the number of parameters to set (field 5) and is listed together with them. Field number and parameter value may appear in any order in this message, also all parameter value fields will be padded if necessary to ensure byte boundaries are adhered to.

When sending a command to a device, when accepted if the PGN referenced has command capability, then fields 6 and 7 shall identify the parameters being commanded, and the commanded values.

Note 1: Acknowledge Requirements- Acknowledgment Group Function Message (PGN 126208) shall be sent in response to all Commands.

Single Frame: No *Priority Default:* 3 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Address *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* Note 1

Field # *Field Name* Original Reference ID # 2

1 **Command Group Function Code** *Byte Field Size:* 1 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Required

DD144 Group Function, Request/Command/Acknowledge
0 = Request Message,
1 = Command Message,
2 = Acknowledge Message,
3 = Read Fields,
4 = Read Fields Reply,
5 = Write Fields,
6 = Write Fields Reply

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

This is the Command message, set = 0x01.

2 **Commanded PGN** *Byte Field Size:* *Request Parameter* Required
Bit Field Size: 24 *Command Parameter:* Required

DD009 PGN
24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 **Priority Setting** *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: 4 *Command Parameter:* Optional

DD182 Priority, Set
0x0 to 0x7 = commanded priority value;
0x8 = do not change priority;
0x9 = return priority to default;
0xA to 0xF = reserved

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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4	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*		
		Bit Field Size:	resv 4	*Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field **bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.					
5	**Number of Pairs of Commanded Parameters to follow**	*Byte Field Size:*	1	*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	
	DD006 Generic counter, short		Numeric count, event counter, sequence counter			
	DF53 Integer, 8 bit unsigned **uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number
6	**Field number of first commanded parameter**	*Byte Field Size:*	1	*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned **uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number
7	**Value of first command parameter**	*Byte Field Size:*	?	*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	
	DD000 Undefined		Application specific, defined at time of use			
	DF00 Undefined **Undefined**	*Range:*	undefined	*Resolution:*	undefined	Application specific, defined at time of use.
	Commanded parameter size and type is dependent on the PGN and the specific command parameter field.					
8	**Variable Number of fields, Field number 6 repeated**	*Byte Field Size:*	1	*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned **uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number
9	**Variable Number of fields, Field number 7 repeated**	*Byte Field Size:*	?	*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	
	DD000 Undefined		Application specific, defined at time of use			
	DF00 Undefined **Undefined**	*Range:*	undefined	*Resolution:*	undefined	Application specific, defined at time of use.
	Commanded parameter size and type is dependent on the PGN and the specific command parameter field.					

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NMEA - Acknowledge group function

PGN: 126208
hex: 1ED00

The Request / Command / Acknowledge Group type of function is defined by first field. The message will be a Request, Command, or Acknowledge Group Function.

The Acknowledge Group Function is defined as follows:

The Acknowledgement reply is transmitted in response to a Request, Command, Read or Write Group Function Message. The Acknowledge Group Function shall be transmitted in response to every Command Message received, indicating acknowledgement or containing the appropriate Error Code. The Acknowledge Group Function is only required in response to a Request, Read or Write Group Function that cannot be satisfied. The Acknowledge Group function shall transmit all fields applicable to the Group Function being acknowledged, fields where the error does not exist are set to 0x0 (No Error/Acknowledge).

Single Frame: No *Priority Default:* 3 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Address *Query Support:* Required *Command Support:* *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 3

1 **Acknowledgment Group Function Code** *Byte Field Size:* 1 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Required

DD144 Group Function, Request/Command/Acknowledge
0 = Request Message,
1 = Command Message,
2 = Acknowledge Message,
3 = Read Fields,
4 = Read Fields Reply,
5 = Write Fields,
6 = Write Fields Reply

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

This is the Acknowledgement message, set = 0x02.

2 **Requested or Commanded PGN # being acknowledged** *Byte Field Size:* *Request Parameter* Required
Bit Field Size: 24 *Command Parameter:* Required

DD009 PGN
24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 **PGN error code** *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: 4 *Command Parameter:* Optional

DD037 Error codes, Acknowledgement
0x0 = No Error/Acknowledge,
0x1 = PGN not supported,
0x2 = PGN temporarily not available,
0x3 = Access denied,
0x4 = Request or Command is not supported,
0x5 = Definer Tag is not supported,
0x6 = Read or Write is not supported

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

4 **Transmission Interval / Priority error code** *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: 4 *Command Parameter:* Optional

DD139 Error codesTransmit interval acknowledgement
0x0 = No Error/Acknowledge,
0x1 = Transmit Interval /Priority not supported,
0x2 = Transmit interval is less than measurement/calculation interval,
0x3 = Access denied.
0x4 = Request is not supported.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

For read or write, group function acknowledgement, this field shall be set to 0x0.

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5	**Number of Requested or Commanded Parameters**	*Byte Field Size:* 1	*Request Parameter*	Optional
		Bit Field Size:	*Command Parameter:*	Optional

DD006 Generic counter, short — Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

If Field 3 is 0x4 or you are replying to a read or write group function, this field shall be set to 0xFF. Value 0xFF indicates no fields follow.

6	**First parameter error code**	*Byte Field Size:*	*Request Parameter*	Optional
		Bit Field Size: 4	*Command Parameter:*	Optional

DD141 Error Codes Command Acknowledgement

0x0 = No Error/Acknowledge;
0x1 = Invalid request or command parameter field;
0x2 = Temporarily unable to comply;
0x3 = Request or command parameter out-of-range;
0x4 = Access denied;
0x5 = Request or Command is not supported;
0x6 = Read or Write is not supported;

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

7	**Variable Number of fields, Field number 6 repeated**	*Byte Field Size:*	*Request Parameter*	Optional
		Bit Field Size: 4	*Command Parameter:*	Optional

DD141 Error Codes Command Acknowledgement

0x0 = No Error/Acknowledge;
0x1 = Invalid request or command parameter field;
0x2 = Temporarily unable to comply;
0x3 = Request or command parameter out-of-range;
0x4 = Access denied;
0x5 = Request or Command is not supported;
0x6 = Read or Write is not supported;

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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NMEA - Read Fields - group function

PGN: 126208
hex: 1ED00

This Read Fields Group Function provides a means to read specific fields in a PGN.

When Field 2 (PGN number) contains a non-proprietary PGN number, field 3, field 4 and field 5 are not included in this message.

If the receiver of this message can comply, the receiver will send to the transmitter a Read Fields Reply Group Function. If the receiver cannot comply, an Acknowledge Group Function shall be sent.

Single Frame:	No	*Priority Default:*	3	*Default Update Rate:*	0	milliseconds	*Frequency:*	NA	cycles per second
Destination:	Address	*Query Support:*	Optional	*Command Support:*	Optional		*ACK Rqmnts:*	None	

Original Reference ID # 191

Field #	Field Name							
1	**Complex Request Group Function Code**		*Byte Field Size:*	1		*Request Parameter*	Optional	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD144 Group Function, Request/Command/Acknowledge			0 = Request Message, 1 = Command Message, 2 = Acknowledge Message, 3 = Read Fields, 4 = Read Fields Reply, 5 = Write Fields, 6 = Write Fields Reply				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number	
	This is the Read Fields message, set = 0x03							
2	**PGN Number**		*Byte Field Size:*			*Request Parameter*	Optional	
			Bit Field Size:	24		*Command Parameter:*	Optional	
	DD009 PGN			24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields	
3	**Manufacturer's Code**		*Byte Field Size:*			*Request Parameter*	Optional	
			Bit Field Size:	11		*Command Parameter:*	Optional	
	DD172 NMEA Manufacturer Code			Assigned by NMEA Committee				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields	
	When Field 2 (PGN number) contains a non-proprietary PGN number, this field is not included in the message.							
4	**NMEA Reserved**		*Byte Field Size:*			*Request Parameter*		
			Bit Field Size:	resv 2		*Command Parameter:*		
	DD001 Reserved field			Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields	
	When Field 2 (PGN number) contains a non-proprietary PGN number, this field is not included in the message.							

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5	**Industry Group**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	3	*Command Parameter:*	Optional	
	DD168 Industry Group		0 = Global; 1 = On-Highway; 2 = Agricultural and Forestry; 3 = Construction; 4 = Marine; 5 = Industrial - Process Control - Stationary (Gen-Sets) 6 = Reserved for SAE 7 = Reserved for SAE				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields

When Field 2 (PGN number) contains a non-proprietary PGN number, this field is not included in the message.

6	**Unique ID**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number

This field links the read fields reply group function with the read fields group function.

7	**Number of Selection Pairs**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD006 Generic counter, short		Numeric count, event counter, sequence counter				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number
8	**Number of Parameter Pairs to be Read**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD006 Generic counter, short		Numeric count, event counter, sequence counter				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number
9	**Field Number of First Selection Pair**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number
10	**Field Value of First Selection Pair**		*Byte Field Size:*	?	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD000 Undefined		Application specific, defined at time of use				
	DF00 Undefined	**Undefined**	*Range:*	undefined	*Resolution:*	undefined	Application specific, defined at time of use.
11	**Variable Number of fields, field 9 repeated**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number

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12	**Variable Number of Fields, field 10 repeated**		*Byte Field Size:*	?	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD000 Undefined		Application specific, defined at time of use				
	DF00 Undefined	**Undefined**	*Range:*	undefined	*Resolution:*	undefined	Application specific, defined at time of use.
13	**Field Number of First Parameter Pair to be Read**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number
14	**Variable Number of Fields, field 13 repeated**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number

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NMEA - Read Fields Reply - group function

PGN: 126208
hex: 1ED00

The Read Fields Reply Group Function is a reply to the Read Fields Group Function.

When the Read Fields Group Function is received, if the receiver can comply, the Read Fields Reply Group Function will be transmitted with the resulting read values.

Single Frame:	No	*Priority Default:*	3	*Default Update Rate:*	0 milliseconds	*Frequency:*	NA cycles per second		
Destination:	Address	*Query Support:*	Optional	*Command Support:*	Optional	*ACK Rqmnts:*	None		

Field # *Field Name* Original Reference ID # 192

1 **Complex Request Group Function Code** — *Byte Field Size:* 1 — *Bit Field Size:* — *Request Parameter* Optional — *Command Parameter:* Optional

DD144 Group Function, Request/Command/Acknowledge

0 = Request Message,
1 = Command Message,
2 = Acknowledge Message,
3 = Read Fields,
4 = Read Fields Reply,
5 = Write Fields,
6 = Write Fields Reply

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

This is the Read Fields Reply message, set = 0x04

2 **PGN Number** — *Byte Field Size:* — *Bit Field Size:* 24 — *Request Parameter* Optional — *Command Parameter:* Optional

DD009 PGN — 24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 **Manufacturer's Code** — *Byte Field Size:* — *Bit Field Size:* 11 — *Request Parameter* Optional — *Command Parameter:* Optional

DD172 NMEA Manufacturer Code — Assigned by NMEA Committee

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

When Field 2 (PGN number) contains a non-proprietary PGN number, this field is not included in the message.

4 **NMEA Reserved** — *Byte Field Size:* — *Bit Field Size:* resv 2 — *Request Parameter* — *Command Parameter:*

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

When Field 2 (PGN number) contains a non-proprietary PGN number, this field is not included in the message.

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5 **Industry Group** — *Byte Field Size:* — *Bit Field Size:* 3 — *Request Parameter* Optional — *Command Parameter:* Optional

DD168 Industry Group

0 = Global;
1 = On-Highway;
2 = Agricultural and Forestry;
3 = Construction;
4 = Marine;
5 = Industrial - Process Control - Stationary (Gen-Sets)
6 = Reserved for SAE
7 = Reserved for SAE

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

When Field 2 (PGN number) contains a non-proprietary PGN number, this field is not included in the message.

6 **Unique ID** — *Byte Field Size:* 1 — *Bit Field Size:* — *Request Parameter* Optional — *Command Parameter:* Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

This field links the read fields group function with the read fields reply group function.

7 **Number of Selection Pairs** — *Byte Field Size:* 1 — *Bit Field Size:* — *Request Parameter* Optional — *Command Parameter:* Optional

DD006 Generic counter, short — Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

8 **Number of Parameter Pairs to be Read** — *Byte Field Size:* 1 — *Bit Field Size:* — *Request Parameter* Optional — *Command Parameter:* Optional

DD006 Generic counter, short — Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

9 **Field Number of First Selection Pair** — *Byte Field Size:* 1 — *Bit Field Size:* — *Request Parameter* Optional — *Command Parameter:* Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

10 **Field Value of First Selection Pair** — *Byte Field Size:* ? — *Bit Field Size:* — *Request Parameter* Optional — *Command Parameter:* Optional

DD000 Undefined — Application specific, defined at time of use

DF00 Undefined **Undefined** *Range:* undefined *Resolution:* undefined Application specific, defined at time of use.

11 **Variable Number of fields, field 9 repeated** — *Byte Field Size:* 1 — *Bit Field Size:* — *Request Parameter* Optional — *Command Parameter:* Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

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12 **Variable Number of Fields, field 10 repeated**
Byte Field Size: ? *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD000 Undefined — Application specific, defined at time of use

DF00 Undefined **Undefined** *Range:* undefined *Resolution:* undefined Application specific, defined at time of use.

---

13 **Field Number of First Parameter Pair to be Read**
Byte Field Size: 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

---

14 **Field Value of First Parameter Pair to be Read**
Byte Field Size: ? *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD000 Undefined — Application specific, defined at time of use

DF00 Undefined **Undefined** *Range:* undefined *Resolution:* undefined Application specific, defined at time of use.

---

15 **Variable Number of Fields, field 13 repeated**
Byte Field Size: 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

---

16 **Variable Number of Fields, field 14 repeated**
Byte Field Size: ? *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD000 Undefined — Application specific, defined at time of use

DF00 Undefined **Undefined** *Range:* undefined *Resolution:* undefined Application specific, defined at time of use.

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NMEA - Write Fields - group function

PGN: 126208
hex: 1ED00

This Write Fields Group Function provides a means to write specific fields in a PGN.

This Group Function is especially useful when configuring an instance or reference to be used by a node when transmitting a PGN such as PGN 127508. Use Fields 7, 9 and 10 to identify the currently assigned instance value. Use Fields 8, 13 and 14 to identify the new assigned instance value.

When Field 2 (PGN number) contains a non-proprietary PGN number, field 3, field 4 and field 5 are not included in this message.

If the receiver of this message can comply, the receiver will send to the transmitter a Write Fields Reply Group Function. If the receiver cannot comply, an Acknowledge Group Function shall be sent.

Single Frame: No *Priority Default:* 3 *Default Update Rate:* 0 milliseconds *Frequency:* NA cycles per second

Destination: Address *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 193

Field #	*Field Name*					
1	**Complex Request Group Function Code**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD144 Group Function, Request/Command/Acknowledge		0 = Request Message, 1 = Command Message, 2 = Acknowledge Message, 3 = Read Fields, 4 = Read Fields Reply, 5 = Write Fields, 6 = Write Fields Reply			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
	This is the Write Fields message, set = 0x05					
2	**PGN Number**		*Byte Field Size:* *Bit Field Size:* 24		*Request Parameter* *Command Parameter:*	Optional Optional
	DD009 PGN		24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
3	**Manufacturer's Code**		*Byte Field Size:* *Bit Field Size:* 11		*Request Parameter* *Command Parameter:*	Optional Optional
	DD172 NMEA Manufacturer Code		Assigned by NMEA Committee			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
	When Field 2 (PGN number) contains a non-proprietary PGN number, this field is not included in the message.					
4	**NMEA Reserved**		*Byte Field Size:* *Bit Field Size:* resv 2		*Request Parameter* *Command Parameter:*	
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
	When Field 2 (PGN number) contains a non-proprietary PGN number, this field is not included in the message.					

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5	**Industry Group**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	3	*Command Parameter:*	Optional
	DD168 Industry Group		0 = Global; 1 = On-Highway; 2 = Agricultural and Forestry; 3 = Construction; 4 = Marine; 5 = Industrial - Process Control - Stationary (Gen-Sets) 6 = Reserved for SAE 7 = Reserved for SAE			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields

When Field 2 (PGN number) contains a non-proprietary PGN number, this field is not included in the message.

6	**Unique ID**		*Byte Field Size:*	1	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:* 1 bit	Unit-less number

This field links the write fields reply group function with the write fields group function.

7	**Number of Selection Pairs**		*Byte Field Size:*	1	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD006 Generic counter, short		Numeric count, event counter, sequence counter			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:* 1 bit	Unit-less number
8	**Number of Parameter Pairs to be Written**		*Byte Field Size:*	1	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD006 Generic counter, short		Numeric count, event counter, sequence counter			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:* 1 bit	Unit-less number
9	**Field Number of First Selection Pair**		*Byte Field Size:*	1	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:* 1 bit	Unit-less number
10	**Field Value of First Selection Pair**		*Byte Field Size:*	?	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD000 Undefined		Application specific, defined at time of use			
	DF00 Undefined	**Undefined**	*Range:*	undefined	*Resolution:* undefined	Application specific, defined at time of use.
11	**Variable Number of fields, field 9 repeated**		*Byte Field Size:*	1	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:* 1 bit	Unit-less number

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12 **Variable Number of Fields, field 10 repeated**
Byte Field Size: ? *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD000 Undefined — Application specific, defined at time of use

DF00 Undefined **Undefined** *Range:* undefined *Resolution:* undefined Application specific, defined at time of use.

---

13 **Field Number of First Parameter Pair to be Written**
Byte Field Size: 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

---

14 **Field Value of First Parameter Pair to be Written**
Byte Field Size: ? *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD000 Undefined — Application specific, defined at time of use

DF00 Undefined **Undefined** *Range:* undefined *Resolution:* undefined Application specific, defined at time of use.

---

15 **Variable Number of Fields, field 13 repeated**
Byte Field Size: 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

---

16 **Variable Number of Fields, field 14 repeated**
Byte Field Size: ? *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD000 Undefined — Application specific, defined at time of use

DF00 Undefined **Undefined** *Range:* undefined *Resolution:* undefined Application specific, defined at time of use.

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NMEA - Write Fields Reply - group function

PGN: 126208
hex: 1ED00

The Write Fields Reply Group Function is a reply to the Write Fields Group Function.

When the Write Fields Group Function is received, if the receiver can comply, the Write Fields Reply Group Function will be transmitted. If a parameter value is accepted then the modified value will be returned. If a parameter value is not accepted, the original value will be returned

Single Frame:	No	*Priority Default:*	3	*Default Update Rate:*	0 milliseconds	*Frequency:*	NA cycles per second
Destination:	Address	*Query Support:*	Optional	*Command Support:*	Optional	*ACK Rqmnts:*	None

Field # *Field Name* — Original Reference ID # 194

1 **Complex Request Group Function Code** — *Byte Field Size:* 1 — *Request Parameter* Optional — *Command Parameter:* Optional

DD144 Group Function, Request/Command/Acknowledge

0 = Request Message,
1 = Command Message,
2 = Acknowledge Message,
3 = Read Fields,
4 = Read Fields Reply,
5 = Write Fields,
6 = Write Fields Reply

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

This is the Write Fields Reply message, set = 0x06

2 **PGN Number** — *Bit Field Size:* 24 — *Request Parameter* Optional — *Command Parameter:* Optional

DD009 PGN — 24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 **Manufacturer's Code** — *Bit Field Size:* 11 — *Request Parameter* Optional — *Command Parameter:* Optional

DD172 NMEA Manufacturer Code — Assigned by NMEA Committee

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

When Field 2 (PGN number) contains a non-proprietary PGN number, this field is not included in the message.

4 **NMEA Reserved** — *Bit Field Size:* resv 2

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

When Field 2 (PGN number) contains a non-proprietary PGN number, this field is not included in the message.

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5	**Industry Group**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	3	*Command Parameter:*	Optional	
	DD168 Industry Group		0 = Global; 1 = On-Highway; 2 = Agricultural and Forestry; 3 = Construction; 4 = Marine; 5 = Industrial - Process Control - Stationary (Gen-Sets) 6 = Reserved for SAE 7 = Reserved for SAE				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields

When Field 2 (PGN number) contains a non-proprietary PGN number, this field is not included in the message.

6	**Unique ID**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number

This field links the write fields group function with the write fields reply group function.

7	**Number of Selection Pairs**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD006 Generic counter, short		Numeric count, event counter, sequence counter				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number
8	**Number of Parameter Pairs to be Written**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD006 Generic counter, short		Numeric count, event counter, sequence counter				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number
9	**Field Number of First Selection Pair**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number
10	**Field Value of First Selection Pair**		*Byte Field Size:*	?	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD000 Undefined		Application specific, defined at time of use				
	DF00 Undefined	**Undefined**	*Range:*	undefined	*Resolution:*	undefined	Application specific, defined at time of use.
11	**Variable Number of fields, field 9 repeated**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number

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12 **Variable Number of Fields, field 10 repeated**
Byte Field Size: ? *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD000 Undefined — Application specific, defined at time of use

DF00 Undefined **Undefined** *Range:* undefined *Resolution:* undefined Application specific, defined at time of use.

13 **Field Number of First Parameter Pair to be Written**
Byte Field Size: 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

14 **Field Value of First Parameter Pair to be Written**
Byte Field Size: ? *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD000 Undefined — Application specific, defined at time of use

DF00 Undefined **Undefined** *Range:* undefined *Resolution:* undefined Application specific, defined at time of use.

15 **Variable Number of Fields, field 13 repeated**
Byte Field Size: 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

16 **Variable Number of Fields, field 14 repeated**
Byte Field Size: ? *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD000 Undefined — Application specific, defined at time of use

DF00 Undefined **Undefined** *Range:* undefined *Resolution:* undefined Application specific, defined at time of use.

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PGN List - Received PGNs group function

PGN: 126464
hex: 1EE00

The PGN List group function type is defined by the first field.

Receive PGNs group function: This message contains a list of the Receive PGNs that are supported by a node.

If the Group Function Code (Field #1) is not specified in the "Command Request" or an ISO Request is made of this PGN, the response will be with both the Transmitted PGNs and Received PGNs group functions. (This PGN will be transmitted twice.)

All ISO Request (PGN 059904) requests for the PGN List parameter group shall be destination address specific (as opposed to global). Once a node issues an ISO Request for PGN List, the node shall not issue a subsequent ISO Request for PGN List until the full response has been received for the first request or the response has timed out in accordance with ISO 11783-3, “Controller response time and time out defaults”.

All NMEA Request group function (PGN 126208) requests for the PGN List parameter group shall be destination address specific (as opposed to global). Once a node issues an NMEA Request for PGN List, the node shall not issue a subsequent NMEA Request for PGN List until the full response has been received for the first request or the response has timed out in accordance with ISO 11783-3, “Controller response time and time out defaults”.

Single Frame:	No	*Priority Default:*	6	*Default Update Rate:*	milliseconds	*Frequency:*	NA cycles per second
Destination:	Address	*Query Support:*	Required	*Command Support:*	Required	*ACK Rqmnts:*	Requested data or AcknowledgementGroup Function containing error codes of reasons for non-compliance.

Field # *Field Name* Original Reference ID # 7

1 **Received PGN Group Function Code**
Byte Field Size: 1 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Required

DD145 Group Function, Transmit and Receive PGN List — 0 = Transmit PGN List Message, 1 = Receive PGN List Message

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

This is the Receive Group List, set = 0x01

2 **First PGN supported**
Byte Field Size: *Request Parameter* Optional
Bit Field Size: 24 *Command Parameter:* Optional

DD009 PGN — 24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 **Variable Number of fields, Field number 2 repeated**
Byte Field Size: *Request Parameter* Optional
Bit Field Size: 24 *Command Parameter:* Optional

DD009 PGN — 24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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PGN List - Transmit PGNs group function

PGN: 126464
hex: 1EE00

The PGN List group function type is defined by the first field.

Transmit PGNs group function: This message contains a list of the Transmitted PGNs that are supported by a node.

If the Group Function Code (Field #1) is not specified in the "Command Request" or an ISO Request is made of this PGN, the response will be with both the Transmitted PGNs and Received PGNs group functions. (This PGN will be transmitted twice.)

All ISO Request (PGN 059904) requests for the PGN List parameter group shall be destination address specific (as opposed to global). Once a node issues an ISO Request for PGN List, the node shall not issue a subsequent ISO Request for PGN List until the full response has been received for the first request or the response has timed out in accordance with ISO 11783-3, “Controller response time and time out defaults”.

All NMEA Request group function (PGN 126208) requests for the PGN List parameter group shall be destination address specific (as opposed to global). Once a node issues an NMEA Request for PGN List, the node shall not issue a subsequent NMEA Request for PGN List until the full response has been received for the first request or the response has timed out in accordance with ISO 11783-3, “Controller response time and time out defaults”.

This PGN will be requested as needed.

Single Frame:	No	*Priority Default:*	6	*Default Update Rate:*		milliseconds	*Frequency:*	NA	cycles per second
Destination:	Address	*Query Support:*	Required	*Command Support:*	Required		*ACK Rqmnts:*	Requested data or Acknowledge mentGroup Function containing error codes of reasons for non-compliance.	

Field # *Field Name* Original Reference ID # 6

1 **Transmitted PGN Group Function Code** *Byte Field Size:* 1 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Required

DD145 Group Function, Transmit and Receive PGN List 0 = Transmit PGN List Message, 1 = Receive PGN List Message

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

This is the Transmitted Group List, set = 0x00

2 **First PGN supported** *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: 24 *Command Parameter:* Optional

DD009 PGN 24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 **Variable Number of fields, Field number 2 repeated** *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: 24 *Command Parameter:* Optional

DD009 PGN 24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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Alert

PGN: 126983
hex: 1F007

This PGN is used to report the status of an alert and is required for all Alert Service implementations. The PGN is transmitted according to the update rate when the Alert is active or upon request.

The Command Group function PGN (126208) is used to request or command the Alert PGN.

Query Support for this PGN is required.

ISO request is not allowed.

Note 1: When requested or commanded using PGN 126208, if any field 1 thru to 5 is included within the parameter list, it represents a qualifier identifying the target alert or alerts. Any or all of the fields not included as qualifiers are treated as wild cards and the request or command is applied to the remaining fields.

Note 2: Field 18, Test mode can only be commanded, if the existing trigger condition is disabled. Once in test mode it is only possible to command back to disabled.

Note 3: Field 19 can only be commanded if the current trigger condition is test mode.

Single Frame: No *Priority Default:* 2 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 206

1 **Alert Type** *Byte Field Size:* *Request Parameter* Note 1
Bit Field Size: 4 *Command Parameter:* Note 1

DD315 Alert Type

0 = Reserved
1 = Emergency Alarm
2 = Alarm
3 = Reserved
4 = Reserved
5 = Warning
6 = Reserved
7 = Reserved
8 = Caution
13 = Reserved
14 = Data out of range
15 = Data not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Alert Service shall convey the Alert type as a means of identification

2 **Alert Category** *Byte Field Size:* *Request Parameter* Note 1
Bit Field Size: 4 *Command Parameter:* Note 1

DD316 Alert Category

0 = Navigational
1 = Technical
13 = Reserved
14 = Data out of range
15 = Data not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Alert Service shall convey the Alert Category as a means of identification. Dependent on NMEA Alert Code Definition Table.

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Alert

PGN: 126983
hex: 1F007

3 Alert System

Byte Field Size: 1 — *Request Parameter* Note 1
Bit Field Size: — *Command Parameter:* Note 1

DD317 Alert System — Values defined by the Alert Codes in Appendix B

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The Alert Service shall convey the Alert System as a means of identification, which will describe the main system where the cause of the Alert has occurred. Dependent on NMEA Alert Code Definition Table.

4 Alert Sub-System

Byte Field Size: 1 — *Request Parameter* Note 1
Bit Field Size: — *Command Parameter:* Note 1

DD318 Alert Sub-System — Values defined by the Alert Codes in Appendix B

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The Alert Service shall convey the Alert Sub-system as a means of identification, which will describe the sub system where the cause of the Alert has occurred. Dependent on NMEA Alert Code Definition Table.

5 Alert ID

Byte Field Size: 2 — *Request Parameter* Note 1
Bit Field Size: — *Command Parameter:* Note 1

DD319 Alert ID — Values defined by the Alert Codes in Appendix B

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

The Alert Service shall convey the Alert ID as a means of describing the actual cause of an Alert. Dependent on NMEA Alert Code Definition Table.

6 Data Source Network ID NAME

Byte Field Size: 8 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Prohibited

DD320 Network ID NAME — This field is defined by fields 1- 10 of PGN 60928

DF56 Integer, 64 bit unsigned **uint64** *Range:* 0 to (2E+64)-4 *Resolution:* 1 bit Unit-less number

The Data Source 64 bit network NAME field found within PGN 060928. This is the NAME of the device providing the data that is monitored by this alert.

7 Data Source Instance

Byte Field Size: — *Request Parameter* Optional
Bit Field Size: 8 — *Command Parameter:* Prohibited

DD128 Generic instance

0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Data Source Instance identifies either the Device Instance or the Data Instance (as applicable) of the source of data monitored by this Alert. This parameter is used in conjunction with the Data Source Network ID and the Data Source Index to uniquely identify Alerts of the same type, from the same source but from using a different input or sensor to generate the Alert.
If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.

Note: Refer to Alert Codes(Appendix B) & Alert Application Notes (Appendix D).

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Alert

PGN: 126983
hex: 1F007

8 **Data Source Index / Source**

Byte Field Size:
Bit Field Size: 8

Request Parameter Optional
Command Parameter: Prohibited

DD128 Generic instance

0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Data Source Index is an index into a set of data associated with the same Data Source Instance.
If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.
Note: Refer to Alert Codes(Appendix B) & Alert Application Notes (Appendix D).

---

9 **Alert Occurrence Number**

Byte Field Size: 1
Bit Field Size:

Request Parameter Optional
Command Parameter: Prohibited

DD006 Generic counter, short

Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

This parameter is a number starting at zero and incremented by one for each occurrence of the same Alert.

---

10 **Temporary Silence Status**

Byte Field Size:
Bit Field Size: 1

Request Parameter Optional
Command Parameter: Prohibited

DD321 Temporary Silence Status

0 = Not Temporary Silence
1 = Temporary Silence

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

This single bit status denotes whether a temporary silence period is active for the current alert.

---

11 **Acknowledge Status**

Byte Field Size:
Bit Field Size: 1

Request Parameter Optional
Command Parameter: Prohibited

DD322 Acknowledge Status

0 = Not Acknowledged
1 = Acknowledged

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

This single bit status denotes whether the current alert has been acknowledged.

---

12 **Escalation Status**

Byte Field Size:
Bit Field Size: 1

Request Parameter Optional
Command Parameter: Prohibited

DD323 Escalation Status

0 = Not Escalated
1 = Escalated

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

This single bit status denotes whether the current alert has been escalated.

---

13 **Temporary Silence Support**

Byte Field Size:
Bit Field Size: 1

Request Parameter Optional
Command Parameter: Optional

DD324 Temporary Silence Support

0 = Not Supported
1 = Supported

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

This single bit status denotes whether the current Alert supports the temporary silence feature.

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14 **Acknowledge Support**
Byte Field Size: *Bit Field Size:* 1
Request Parameter Optional
Command Parameter: Optional

DD325 Acknowledge Support
0 = Not Supported
1 = Supported

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
This single bit status denotes whether the current Alert supports the acknowledge feature.

---

15 **Escalation Support**
Byte Field Size: *Bit Field Size:* 1
Request Parameter Optional
Command Parameter: Optional

DD326 Escalation Support
0 = Not Supported
1 = Supported

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
This single bit status denotes whether the current Alert supports escalation.

---

16 **NMEA Reserved**
Byte Field Size: *Bit Field Size:* resv 2
Request Parameter
Command Parameter:

DD001 Reserved field
Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
Used to align subsequent data on byte boundary.

---

17 **Acknowledge Source Network ID NAME**
Byte Field Size: 8 *Bit Field Size:*
Request Parameter Optional
Command Parameter: Prohibited

DD320 Network ID NAME
This field is defined by fields 1- 10 of PGN 60928

DF56 Integer, 64 bit unsigned **uint64** *Range:* 0 to (2E+64)-4 *Resolution:* 1 bit Unit-less number
The Acknowledge Source 64 bit network NAME field found within PGN 060928. This is the NAME of the device that acknowledged this alert.

---

18 **Trigger Condition**
Byte Field Size: *Bit Field Size:* 4
Request Parameter Optional
Command Parameter: Note 2

DD328 Trigger Condition
0 = Manual
1 = Auto
2 = Test
3 = Disabled
13 = Reserved
14 = Data out of range
15 = Data not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
0 = Manual (triggered by human)
1 = Auto (triggered autonomously)
2 = Test (alert is in test mode)
3 = Disabled (alert is not operational)

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19 **Threshold Status** *Byte Field Size:* *Bit Field Size:* 4 *Request Parameter* Optional *Command Parameter:* Note 3

DD329 Threshold Status

0 = Normal
1 = Threshold Exceeded
2 = Extreme Threshold Exceeded, field added to support 0183
3 = Low Threshold Exceeded, field added to support 0183
4 = Extreme Low Threshold Exceeded, field added to support 0183
13 = Reserved
14 = Data out of range
15 = Data not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

20 **Alert Priority** *Byte Field Size:* 1 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD006 Generic counter, short

Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The Alert priority is a numeric value in the range of 0 to 252, where 0 is the highest priority.

---

21 **Alert State** *Byte Field Size:* 1 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Prohibited

DD330 Alert State

0 = Disabled
1 = Normal
2 = Active
3 = Silenced
4 = Acknowledged
5 = Awaiting Acknowledge
253 = Reserved
254 = Data out of range
255 = Data not available

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

---

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Alert Response

PGN: 126984
hex: 1F008

This PGN is used to control an active Alert. The function of this PGN is to send an acknowledge command or a temporary silence command to the Alert Source. Since this is a broadcast PGN, a destination Alert Source name field is included. The PGN also includes data source instance to enable the product providing an alert to distinguish from multiple sensors.

Note 1: It is not possible to request or command this PGN.

Single Frame: No *Priority Default:* 2 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Prohibited *Command Support:* Prohibited *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 207

1 Alert Type
Byte Field Size: *Bit Field Size:* 4 *Request Parameter* Note 1 *Command Parameter:* Prohibited

DD315 Alert Type

0 = Reserved
1 = Emergency Alarm
2 = Alarm
3 = Reserved
4 = Reserved
5 = Warning
6 = Reserved
7 = Reserved
8 = Caution
13 = Reserved
14 = Data out of range
15 = Data not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Alert Service shall convey the Alert type as a means of identification.

2 Alert Category
Byte Field Size: *Bit Field Size:* 4 *Request Parameter* Note 1 *Command Parameter:* Prohibited

DD316 Alert Category

0 = Navigational
1 = Technical
13 = Reserved
14 = Data out of range
15 = Data not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Alert Service shall convey the Alert Category as a means of identification. Dependent on NMEA Alert Code Definition Table.

3 Alert System
Byte Field Size: 1 *Bit Field Size:* *Request Parameter* Note 1 *Command Parameter:* Prohibited

DD317 Alert System

Values defined by the Alert Codes in Appendix B

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The Alert Service shall convey the Alert System as a means of identification, which will describe the main system where the cause of the Alert has occurred. Dependent on NMEA Alert Code Definition Table

4 Alert Sub-System
Byte Field Size: 1 *Bit Field Size:* *Request Parameter* Note 1 *Command Parameter:* Prohibited

DD318 Alert Sub-System

Values defined by the Alert Codes in Appendix B

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The Alert Service shall convey the Alert Sub-system as a means of identification, which will describe the sub system where the cause of the Alert has occurred. Dependent on NMEA Alert Code Definition Table.

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Alert Response

PGN: 126984
hex: 1F008

5 **Alert ID**

Byte Field Size: 2
Bit Field Size:

Request Parameter Note 1
Command Parameter: Prohibited

DD319 Alert ID — Values defined by the Alert Codes in Appendix B

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

The Alert Service shall convey the Alert ID as a means of describing the actual cause of an Alert. Dependent on NMEA Alert Code Definition Table.

---

6 **Data Source Network ID NAME**

Byte Field Size: 8
Bit Field Size:

Request Parameter Note 1
Command Parameter: Prohibited

DD320 Network ID NAME — This field is defined by fields 1- 10 of PGN 60928

DF56 Integer, 64 bit unsigned **uint64** *Range:* 0 to (2E+64)-4 *Resolution:* 1 bit Unit-less number

The Data Source 64 bit network NAME field found within PGN 060928. This is the NAME of the device providing the data that is monitored by this alert.

---

7 **Data Source Instance**

Byte Field Size:
Bit Field Size: 8

Request Parameter Note 1
Command Parameter: Prohibited

DD128 Generic instance

0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Data Source Instance identifies either the Device Instance or the Data Instance (as applicable) of the source of data monitored by this Alert. This parameter is used in conjunction with the Data Source Network ID and the Data Source Index to uniquely identify Alerts of the same type, from the same source but from using a different input or sensor to generate the Alert.

Note: Refer to Alert Codes(Appendix B) & Alert Application Notes (Appendix D).

---

8 **Data Source Index / Source**

Byte Field Size:
Bit Field Size: 8

Request Parameter Note 1
Command Parameter: Prohibited

DD128 Generic instance

0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Data Source Index is an index into a set of data associated with the same Data Source Instance.
Note: Refer to Alert Codes(Appendix B) & Alert Application Notes (Appendix D).

---

9 **Alert Occurrence Number**

Byte Field Size: 1
Bit Field Size:

Request Parameter Note 1
Command Parameter: Prohibited

DD006 Generic counter, short — Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

This parameter is a number starting at zero and incremented by one for each occurrence of the same Alert.

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Alert Response

PGN: 126984
hex: 1F008

10	**Acknowledge Source Network ID NAME**	*Byte Field Size:*	8	*Request Parameter*	Note 1	
		Bit Field Size:		*Command Parameter:*	Prohibited	
	DD320 Network ID NAME	This field is defined by fields 1- 10 of PGN 60928				
	DF56 Integer, 64 bit unsigned **uint64**	*Range:*	0 to (2E+64)-4	*Resolution:*	1 bit	Unit-less number
	The Acknowledge Source 64 bit network NAME field found within PGN 060928. This is the NAME of the device that acknowledged this alert.					
11	**Response Command**	*Byte Field Size:*		*Request Parameter*	Note 1	
		Bit Field Size:	2	*Command Parameter:*	Prohibited	
	DD332 Response Command	0 = Acknowledge 1 = Temporary Silence 2 = Test Command off 3 = Test Command on				
	DF52 Bit field **bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
12	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*		
		Bit Field Size:	resv 6	*Command Parameter:*		
	DD001 Reserved field	Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field **bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
	Used to align subsequent data on byte boundary.					

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Alert Text

PGN: 126985
hex: 1F009

The Alert text PGN is used to convey identification and location text strings associated with source of an Alert. This PGN is not a mandatory requirement to implement the Alert Service.

This PGN would either be transmitted cyclically once an alert is active or upon request. If the PGN is transmitted cyclically then the frequency shall not be any more than once every 10 seconds.

This PGN shall be transmitted upon request regardless of alert status.

The Command Group function PGN (126208) is used to request or command the Alert Text PGN.

Note 1: When requested or commanded using PGN 126208, if any field 1 thru to 5 is included within the parameter list, it represents a qualifier identifying the target alert or alerts. Any or all of the fields not included as qualifiers are not considered or used in the selection process of applying the request or command to the remaining fields (These are considered as wild cards).

Note 2: Field 10 can be used to request text strings by a defined language.

Single Frame: No *Priority Default:* 2 *Default Update Rate:* 10000 milliseconds *Frequency:* .1 cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 208

1 Alert Type
Byte Field Size: *Bit Field Size:* 4
Request Parameter Note 1
Command Parameter: Note 1

DD315 Alert Type

0 = Reserved
1 = Emergency Alarm
2 = Alarm
3 = Reserved
4 = Reserved
5 = Warning
6 = Reserved
7 = Reserved
8 = Caution
13 = Reserved
14 = Data out of range
15 = Data not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Alert Service shall convey the Alert type as a means of identification.

2 Alert Category
Byte Field Size: *Bit Field Size:* 4
Request Parameter Note 1
Command Parameter: Note 1

DD316 Alert Category

0 = Navigational
1 = Technical
13 = Reserved
14 = Data out of range
15 = Data not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Alert Service shall convey the Alert Category as a means of identification. Dependent on NMEA Alert Code Definition Table

3 Alert System
Byte Field Size: 1 *Bit Field Size:*
Request Parameter Note 1
Command Parameter: Note 1

DD317 Alert System

Values defined by the Alert Codes in Appendix B

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The Alert Service shall convey the Alert System as a means of identification, which will describe the main system where the cause of the Alert has occurred. Dependent on NMEA Alert Code Definition Table.

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Alert Text

PGN: 126985
hex: 1F009

4 **Alert Sub-System** — *Byte Field Size:* 1 — *Bit Field Size:* — *Request Parameter* Note 1 — *Command Parameter:* Note 1

DD318 Alert Sub-System — Values defined by the Alert Codes in Appendix B

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The Alert Service shall convey the Alert Sub-system as a means of identification, which will describe the sub system where the cause of the Alert has occurred. Dependent on NMEA Alert Code Definition Table.

---

5 **Alert ID** — *Byte Field Size:* 2 — *Bit Field Size:* — *Request Parameter* Note 1 — *Command Parameter:* Note 1

DD319 Alert ID — Values defined by the Alert Codes in Appendix B

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

The Alert Service shall convey the Alert ID as a means of describing the actual cause of an Alert. Dependent on NMEA Alert Code Definition Table.

---

6 **Data Source Network ID NAME** — *Byte Field Size:* 8 — *Bit Field Size:* — *Request Parameter* Note 2 — *Command Parameter:* Note 2

DD320 Network ID NAME — This field is defined by fields 1- 10 of PGN 60928

DF56 Integer, 64 bit unsigned **uint64** *Range:* 0 to (2E+64)-4 *Resolution:* 1 bit Unit-less number

The Data Source 64 bit network NAME field found within PGN 060928. This is the NAME of the device providing the data that is monitored by this alert.

---

7 **Data Source Instance** — *Byte Field Size:* — *Bit Field Size:* 8 — *Request Parameter* Note 2 — *Command Parameter:* Note 2

DD128 Generic instance

0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Data Source Instance identifies either the Device Instance or the Data Instance (as applicable) of the source of data monitored by this Alert. This parameter is used in conjunction with the Data Source Network ID and the Data Source Index to uniquely identify Alerts of the same type, from the same source but from using a different input or sensor to generate the Alert.
If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.

Note: Refer to Alert Codes(Appendix B) & Alert Application Notes (Appendix D).

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Alert Text

PGN: 126985
hex: 1F009

8	**Data Source Index / Source**	*Byte Field Size:*		*Request Parameter*	Note 2
		Bit Field Size:	8	*Command Parameter:*	Note 2

DD128 Generic instance

0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Data Source Index is an index into a set of data associated with the same Data Source Instance.
If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.

Note: Refer to Alert Codes(Appendix B) & Alert Application Notes (Appendix D).

9	**Alert Occurrence Number**	*Byte Field Size:*	1	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Prohibited

DD006 Generic counter, short

Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The parameter is a number starting at zero and incremented by one for each occurrence of the same Alert.

10	**Language ID**	*Byte Field Size:*	1	*Request Parameter*	Note 2
		Bit Field Size:		*Command Parameter:*	Optional

DD355 Language ID

0 = English (US)
1 = English (UK)
2 = Arabic
3 = Chinese (simplified)
4 = Croatian
5 = Danish
6 = Dutch
7 = Finnish
8 = French
9 = German
10 = Greek
11 = Italian
12 = Japanese
13 = Korean
14 = Norwegian
15 = Polish
16 = Portuguese(Brazilian)
17 = Russian
18 = Spanish
19 = Sweedish

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

This is a numeric number ranged 0 to 252 denoting the language used within both of the following text strings.

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Alert Text

PGN: 126985
hex: 1F009

11 **Alert Text Description**

Byte Field Size: 8 or 16 | n
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character

2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

This is a text string describing the Alert.

---

12 **Alert Location Text Description**

Byte Field Size: 8 or 16 | n
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character

2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

This is a text string describing the location of the Alert.

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Alert Configuration

PGN: 126986
hex: 1F00A

This PGN shall be required for all Alert Service implementations.

This PGN is used to report the configuration of an alert. When reporting the configuration, the “User Defined Alert Assignment” Field 11 shall be set to a value of “2” meaning “Not Applicable”

This PGN can also be used in conjunction with the Command Group Function PGN (126208) to change the configuration of an alert. This PGN can also be used in conjunction with the Command Group Function PGN (126208) to create a new “User Defined” Alert that does not exist within the current network and is not already defined within the NMEA 2000 Alert Codes. (See NMEA 2000 Appendix B, Alert Codes). This PGN can also be used in conjunction with the Command Group Function PGN (126208) to remove a single “User Defined” Alert that was previously created within the current network. This PGN shall be broadcast after the acknowledgement required by the Command Group Function (126208).

Note 1: When commanded using PGN 126208, if any field 1 thru 5 is included in the commanded parameter list, it represents a qualifier identifying the specific alert or alerts that the command will apply to. Any field 1 thru 5 that is not included is treated as a wild card and the command is applied to all alerts matching the remaining fields.

Note 2: Fields 6, 7 & 8, can also optionally be requested or commanded. Fields 7 & 8 are dependent on field 6. For more information, see Alert Application Notes in Appendix D.

Note 3: The ability to command Field 10 shall be required.

Note 4: Commanding the “User Defined Alert Assignment” Field 11 is optional. When commanded this field is used for two purposes:

1 - To create a new (user defined) Alert that does not exist within the current network and that is not already defined within the NMEA 2000 Alert Codes. (See NMEA 2000 Appendix B). When commanded using PGN 126208, this field is set to a value of “0 = Instantiate (new) User Defined Alert”.

When commanded using PGN 126208, fields 2 thru 5 shall be included in the commanded parameter list, identifying the specific alert that the command will apply to. The fields shall be interpreted as follows:

For Field 2 – “Alert Category” shall be used as a qualifier to determine which NMEA defined category (Navigational or Technical) the new “User Defined Alert” should be assigned to.

For Field 3 – “Alert System”, the values determine the action taken. If the provided value is within the NMEA “User Defined Range for this field” and does not exist within the current network, a new “User Defined” Alert System entry is created; If the provided value is already defined by NMEA or previous “User Defined” actions, the value is treated as qualifier as to what Alert System the new “User Defined Alert” is assigned to.

For Field 4 – “Alert Subsystem”, the values determine the action taken. If the provided value is within the NMEA “User Defined Range for this field” and does not exist within the current network, a new “User Defined” Alert Subsystem entry is created; If the provided value is already defined by NMEA or previous “User Defined” actions, the value is treated as qualifier as to what Alert Subsystem the new “User Defined Alert” is assigned to.

For Field 5 – “Alert ID”: The provided value must be within the NMEA “User Defined” Range for this field and does not exist within the current network, then a new “User Defined” Alert ID entry is created; If the provided value is already defined by NMEA or previous “User Defined” actions, this is an invalid command and should be Acknowledged as such.

2 - To remove an existing “User Defined” Alert that exists within the current network. When commanded using PGN 126208, this field is set to a value of “1 = Remove (existing) User Defined Alert”.

When commanded using PGN 126208, fields 2 thru 5 shall be included in the as necessary commanded parameter list, identifying the specific alert that the command will apply to. For instance if there is only a “User Defined” Alert System with no associated Alert Subsystem, the Alert Subsystem Field 4 does not need to be commanded. The fields shall be interpreted as follows;

For Field 2 – “Alert Category” shall be used as a qualifier to determine which NMEA defined category (Navigational or Technical) the existing “User Defined Alert” should be removed from.

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Alert Configuration

PGN: 126986
hex: 1F00A

For Field 3 – “Alert System”, the values determine the action taken. If the provided value is within the NMEA “User Defined Range for this field”, and exists within the current network, and there are no “User Defined” Alert Subsystems associated with the Alert System value, the “User Defined” Alert System is removed; If the provided value is already defined by NMEA, or defined by previous “User Defined” actions and is associated with one or more “User Defined” Alert Subsystems, the value is treated as qualifier as to what Alert System the “User Defined Alert” is to be removed from.

For Field 4 – “Alert Subsystem”, the values determine the action taken. If the provided value is within the NMEA “User Defined Range for this field”, exists within the current network, and there are no “User Defined” Alert IDs associated with the Alert Subsystem value, the “User Defined” Alert Subsystem is removed; If the provided value is already defined by NMEA, or defined by previous “User Defined” actions and is associated with one or more “User Defined” Alert IDs, the value is treated as qualifier as to what Alert Subsystem the “User Defined Alert” is to be removed from.

For Field 5 – “Alert ID”: The provided value must be within the NMEA “User Defined” Range for this field and exists within the current network, only then can the “User Defined” Alert ID is removed; If the provided value is already defined by NMEA or does not exist from previous “User Defined” actions, this is an invalid command and should be Acknowledged as such.

NMEA defined Alerts shall not be affected by the “User Defined Alert Assignment” Field 11.

Single Frame: No *Priority Default:* 2 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Required *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 210

1 **Alert Type** *Byte Field Size:* *Bit Field Size:* 4 *Request Parameter* Required *Command Parameter:* Optional

DD315 Alert Type

0 = Reserved
1 = Emergency Alarm
2 = Alarm
3 = Reserved
4 = Reserved
5 = Warning
6 = Reserved
7 = Reserved
8 = Caution
13 = Reserved
14 = Data out of range
15 = Data not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Alert Service shall convey the Alert type as a means of identification.

2 **Alert Category** *Byte Field Size:* *Bit Field Size:* 4 *Request Parameter* Required *Command Parameter:* Note 1

DD316 Alert Category

0 = Navigational
1 = Technical
13 = Reserved
14 = Data out of range
15 = Data not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Alert Service shall convey the Alert Category as a means of identification. Dependent on NMEA Alert Code Definition Table.

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Alert Configuration

PGN: 126986
hex: 1F00A

3 Alert System

Byte Field Size: 1 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Note 1

DD317 Alert System — Values defined by the Alert Codes in Appendix B

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The Alert Service shall convey the Alert System as a means of identification, which will describe the main system where the cause of the Alert has occurred. Dependent on NMEA Alert Code Definition Table.

4 Alert Sub-System

Byte Field Size: 1 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Note 1

DD318 Alert Sub-System — Values defined by the Alert Codes in Appendix B

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The Alert Service shall convey the Alert Sub-system as a means of identification, which will describe the sub system where the cause of the Alert has occurred. Dependent on NMEA Alert Code Definition Table.

5 Alert ID

Byte Field Size: 2 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Note 1

DD319 Alert ID — Values defined by the Alert Codes in Appendix B

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

The Alert Service shall convey the Alert ID as a means of describing the actual cause of an Alert. Dependent on NMEA Alert Code Definition Table.

6 Data Source Network ID NAME

Byte Field Size: 8 *Request Parameter* Note 2
Bit Field Size: *Command Parameter:* Note 2

DD320 Network ID NAME — This field is defined by fields 1- 10 of PGN 60928

DF56 Integer, 64 bit unsigned **uint64** *Range:* 0 to (2E+64)-4 *Resolution:* 1 bit Unit-less number

The Data Source 64 bit network NAME field found within PGN 060928. This is the NAME of the device providing the data that is monitored by this alert.

7 Data Source Instance

Byte Field Size: *Request Parameter* Note 2
Bit Field Size: 8 *Command Parameter:* Note 2

DD128 Generic instance

0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Data Source Instance identifies either the Device Instance or the Data Instance (as applicable) of the source of data monitored by this Alert. This parameter is used in conjunction with the Data Source Network ID and the Data Source Index to uniquely identify Alerts of the same type, from the same source but from using a different input or sensor to generate the Alert. If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.

Note: Refer to Alert Codes(Appendix B) & Alert Application Notes (Appendix D).

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Alert Configuration

PGN: 126986
hex: 1F00A

8 **Data Source Index / Source**

Byte Field Size: *Bit Field Size:* 8

Request Parameter Note 2
Command Parameter: Note 2

DD128 Generic instance

0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Data Source Index is an index into a set of data associated with the same Data Source Instance. If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.
Note: Refer to Alert Codes(Appendix B) & Alert Application Notes (Appendix D).

---

9 **Alert Occurrence Number**

Byte Field Size: 1 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Prohibited

DD006 Generic counter, short

Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The parameter is a number starting at zero and incremented by one for each occurrence of the same Alert.

---

10 **Alert Control**

Byte Field Size: *Bit Field Size:* 2

Request Parameter Optional
Command Parameter: Required

DD333 Alert Control

0 = Disabled
1 = Enabled

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

0 = Alert Disabled
1 = Alert Enabled

---

11 **User Defined Alert Assignment**

Byte Field Size: *Bit Field Size:* 2

Request Parameter Optional
Command Parameter: Note 4

DD411 User Defined Alert Assignment

0 = Instantiate (new) User Defined Alert
1 = Remove (existing) User Defined Alert
2 = Null – not applicable

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

12 **NMEA Reserved**

Byte Field Size: *Bit Field Size:* resv 4

Request Parameter
Command Parameter:

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

---

13 **Reactivation Period**

Byte Field Size: 4 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Optional

DD334 Reactivation Period

Range: 0 to ~4.295x10E+9 s

DF67 Time interval, large **uint32** *Range:* 0 to ~4.295x10E+9 s *Resolution:* 1 sec

This field contains the reactivation period associated with the target Alert. The reactivation period is used to change the state of an Alert from acknowledged back to Active. This value is in seconds

---

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14	**Temporary Silence Period**	*Byte Field Size:*	4	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD335 Temporary Silence Period — Range: 0 to ~4.295x10E+9 s

DF67 Time interval, large **uint32** *Range:* 0 to ~4.295x10E+9 s *Resolution:* 1 sec

This field contains the temporary silence period associated with the target Alert. The temporary Silence period is used to determine how long an audible device remains muted after a temporary silence command has been issued. This value is in seconds.

15	**Escalation Period**	*Byte Field Size:*	4	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD336 Escalation Period — Range: 0 to ~4.295x10E+9 s

DF67 Time interval, large **uint32** *Range:* 0 to ~4.295x10E+9 s *Resolution:* 1 sec

This field contains the escalation period associated with the target Alert. The Escalation period is used as an elapsed time before the status of an Alert is escalated. The value is defined in seconds.

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Alert Threshold

PGN: 126987
hex: 1F00B

The Alert Threshold PGN is used to convey or program the trigger method and threshold level associated with an Alert. This PGN is not a mandatory requirement to implement the Alert Service.

The Command Group function PGN (126208) is used to request or command the Alert Threshold PGN.

Note 1: When requested or commanded fields 1 thru to 5 represent a qualifier indentifying the target alert or alerts. Any or all of the fields 1 thru to 5 not included as qualifiers are treated as wild cards and the request or command is applied to the remaining fields.

Note 2: Fields 6, 7 and 8 can also be used to qualify a specific request or command of this PGN. Fields 7 and 8 are dependent on field 6.

Note 3: Fields 10 thru 15 shall be specified when this PGN is commanded. When programming threshold parameters, fields 11 thru 14 for each parameter must be specified each time this Alert is commanded.
Each issuance of commanding the Alert Threshold PGN replaces all previously assigned parameters.

Single Frame: No *Priority Default:* 2 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 209

1 **Alert Type** *Byte Field Size:* *Request Parameter* Note 1
Bit Field Size: 4 *Command Parameter:* Note 1

DD315 Alert Type

0 = Reserved
1 = Emergency Alarm
2 = Alarm
3 = Reserved
4 = Reserved
5 = Warning
6 = Reserved
7 = Reserved
8 = Caution
13 = Reserved
14 = Data out of range
15 = Data not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Alert Service shall convey the Alert type as a means of identification.

2 **Alert Category** *Byte Field Size:* *Request Parameter* Note 1
Bit Field Size: 4 *Command Parameter:* Note 1

DD316 Alert Category

0 = Navigational
1 = Technical
13 = Reserved
14 = Data out of range
15 = Data not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Alert Service shall convey the Alert Category as a means of identification. Dependent on NMEA Alert Code Definition Table

3 **Alert System** *Byte Field Size:* 1 *Request Parameter* Note 1
Bit Field Size: *Command Parameter:* Note 1

DD317 Alert System

Values defined by the Alert Codes in Appendix B

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The Alert Service shall convey the Alert System as a means of identification, which will describe the main system where the cause of the Alert has occurred. Dependent on NMEA Alert Code Definition Table.

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Alert Threshold

PGN: 126987
hex: 1F00B

4 **Alert Sub-System**

Byte Field Size: 1 *Request Parameter* Note 1
Bit Field Size: *Command Parameter:* Note 1

DD318 Alert Sub-System — Values defined by the Alert Codes in Appendix B

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The Alert Service shall convey the Alert Sub-system as a means of identification, which will describe the sub system where the cause of the Alert has occurred. Dependent on NMEA Alert Code Definition Table.

---

5 **Alert ID**

Byte Field Size: 2 *Request Parameter* Note 1
Bit Field Size: *Command Parameter:* Note 1

DD319 Alert ID — Values defined by the Alert Codes in Appendix B

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

The Alert Service shall convey the Alert ID as a means of describing the actual cause of an Alert. Dependent on NMEA Alert Code Definition Table.

---

6 **Data Source Network ID NAME**

Byte Field Size: 8 *Request Parameter* Note 2
Bit Field Size: *Command Parameter:* Note 2

DD320 Network ID NAME — This field is defined by fields 1- 10 of PGN 60928

DF56 Integer, 64 bit unsigned **uint64** *Range:* 0 to (2E+64)-4 *Resolution:* 1 bit Unit-less number

The Data Source 64 bit network NAME field found within PGN 060928. This is the NAME of the device providing the data that is monitored by this alert.

---

7 **Data Source Instance**

Byte Field Size: *Request Parameter* Note 2
Bit Field Size: 8 *Command Parameter:* Note 2

DD128 Generic instance

0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Data Source Instance identifies either the Device Instance or the Data Instance (as applicable) of the source of data monitored by this Alert. This parameter is used in conjunction with the Data Source Network ID and the Data Source Index to uniquely identify Alerts of the same type, from the same source but from using a different input or sensor to generate the Alert.
If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.

Note: Refer to Alert Codes(Appendix B) & Alert Application Notes (Appendix D).

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Alert Threshold

PGN: 126987
hex: 1F00B

8 **Data Source Index / Source**
Byte Field Size: *Bit Field Size:* 8
Request Parameter Note 2
Command Parameter: Note 2

DD128 Generic instance

0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Data Source Index is an index into a set of data associated with the same Data Source Instance. If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.
Note: Refer to Alert Codes(Appendix B) & Alert Application Notes (Appendix D).

---

9 **Alert Occurrence Number**
Byte Field Size: 1 *Bit Field Size:*
Request Parameter Optional
Command Parameter: Prohibited

DD006 Generic counter, short

Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The parameter is a number starting at zero and incremented by one for each occurrence of the same Alert.

---

10 **Total Number of Threshold Parameters**
Byte Field Size: 1 *Bit Field Size:*
Request Parameter Optional
Command Parameter: Note 3

DD006 Generic counter, short

Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

This field contains a parameter that defines the total number of threshold parameters used to trigger the alert.

---

11 **Parameter Number**
Byte Field Size: 1 *Bit Field Size:*
Request Parameter Optional
Command Parameter: Note 3

DD005 Generic numeric ID, short

Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

This field defines numerically which parameter is to follow up to a maximum of the total threshold parameters.

---

12 **Trigger Method**
Byte Field Size: *Bit Field Size:* 8
Request Parameter Optional
Command Parameter: Note 3

DD337 Trigger Method

0 = Manual
1 = Automatic
2 = Threshold Reached Ascending
3 = Threshold Reached Descending
253 = Reserved
254 = Data out of range
255 = Data not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

13 **Threshold Data Format**
Byte Field Size: 1 *Bit Field Size:*
Request Parameter Optional
Command Parameter: Note 3

DD338 Threshold Data Format

As an example to reference an unsigned high voltage (DF96), the Threshold Data format would be decimal 96. Range 0 to 252.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

This field contains the NMEA Network Message data format code used to define the threshold level below.

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Alert Threshold

PGN: 126987
hex: 1F00B

14	**Threshold Level**	*Byte Field Size:*	?	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Note 3

DD339 Threshold Level — This parameter is defined by the Threshold Data Format and is a preset value used to trigger an Alert.

DF00 Undefined — **Undefined** *Range:* undefined *Resolution:* undefined — Application specific, defined at time of use.

This field contains the threshold level that is used as a comparison against the data value to trigger the associated Alert.
This field can contain a variable sized parameter defined by the Threshold Data Format code defined above.

15	**Fields 11 to 14 Repeat as necessary**	*Byte Field Size:*	?	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Note 3

DD000 Undefined — Application specific, defined at time of use

DF00 Undefined — **Undefined** *Range:* undefined *Resolution:* undefined — Application specific, defined at time of use.

Application Specific defined at time of use

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Alert Value

PGN: 126988
hex: 1F00C

The Alert Value PGN is used to convey the instantaneous value parameter directly linked with an associated Alert. This PGN is not a mandatory requirement to implement the Alert Service.

The Request Group function PGN (126208) is used to request the Alert Value PGN. The Command Group Function PGN (126208) shall not be used with this PGN.

Note 1: When requested or commanded fields 1 thru to 5 represent a qualifier identifying the target alert or alerts. Any or all of the fields 1 thru to 5 not included as qualifiers are treated as wild cards and the request or command is applied to the remaining fields.

Note 2: Fields 6, 7 and 8 can also be used to qualify a specific request of this PGN. Fields 7 and 8 are dependent on field 6.

Single Frame: No *Priority Default:* 2 *Default Update Rate:* 10000 milliseconds *Frequency:* .1 cycles per second
Destination: Global *Query Support:* Required *Command Support:* Prohibited *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 211

1 Alert Type
Byte Field Size: *Bit Field Size:* 4 *Request Parameter* Note 1 *Command Parameter:* Prohibited

DD315 Alert Type

0 = Reserved
1 = Emergency Alarm
2 = Alarm
3 = Reserved
4 = Reserved
5 = Warning
6 = Reserved
7 = Reserved
8 = Caution
13 = Reserved
14 = Data out of range
15 = Data not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Alert Service shall convey the Alert type as a means of identification.

2 Alert Category
Byte Field Size: *Bit Field Size:* 4 *Request Parameter* Note 1 *Command Parameter:* Prohibited

DD316 Alert Category

0 = Navigational
1 = Technical
13 = Reserved
14 = Data out of range
15 = Data not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Alert Service shall convey the Alert Category as a means of identification. Dependent on NMEA Alert Code Definition Table.

3 Alert System
Byte Field Size: 1 *Bit Field Size:* *Request Parameter* Note 1 *Command Parameter:* Prohibited

DD317 Alert System

Values defined by the Alert Codes in Appendix B

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The Alert Service shall convey the Alert System as a means of identification, which will describe the main system where the cause of the Alert has occurred. Dependent on NMEA Alert Code Definition Table.

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Alert Value

PGN: 126988
hex: 1F00C

4	**Alert Sub-System**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Note 1 Prohibited

DD318 Alert Sub-System — Values defined by the Alert Codes in Appendix B

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The Alert Service shall convey the Alert Sub-system as a means of identification, which will describe the sub system where the cause of the Alert has occurred. Dependent on NMEA Alert Code Definition Table.

5	**Alert ID**	*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Note 1 Prohibited

DD319 Alert ID — Values defined by the Alert Codes in Appendix B

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

The Alert Service shall convey the Alert ID as a means of describing the actual cause of an Alert. Dependent on NMEA Alert Code Definition Table.

6	**Data Source Network ID NAME**	*Byte Field Size:* 8 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Note 2 Prohibited

DD320 Network ID NAME — This field is defined by fields 1- 10 of PGN 60928

DF56 Integer, 64 bit unsigned **uint64** *Range:* 0 to (2E+64)-4 *Resolution:* 1 bit Unit-less number

The Data Source 64 bit network NAME field found within PGN 060928. This is the NAME of the device providing the data that is monitored by this alert.

7	**Data Source Instance**	*Byte Field Size:* *Bit Field Size:* 8		*Request Parameter* *Command Parameter:*	Note 2 Prohibited

DD128 Generic instance

0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Data Source Instance identifies either the Device Instance or the Data Instance (as applicable) of the source of data monitored by this Alert. This parameter is used in conjunction with the Data Source Network ID and the Data Source Index to uniquely identify Alerts of the same type, from the same source but from using a different input or sensor to generate the Alert. If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.

Note: Refer to Alert Codes(Appendix B) & Alert Application Notes (Appendix D).

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Alert Value

PGN: 126988
hex: 1F00C

8 **Data Source Index / Source** — Byte Field Size: — Bit Field Size: 8 — Request Parameter: Note 2 — Command Parameter: Prohibited

DD128 Generic instance

0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

The Data Source Index is an index into a set of data associated with the same Data Source Instance. If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.
Note: Refer to Alert Codes(Appendix B) & Alert Application Notes (Appendix D).

---

9 **Alert Occurrence Number** — Byte Field Size: 1 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Prohibited

DD006 Generic counter, short — Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** Range: 0 to 252 Resolution: 1 bit Unit-less number

The parameter is a number starting at zero and incremented by one for each occurrence of the same Alert.

---

10 **Total Number of Value Parameters** — Byte Field Size: 1 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Prohibited

DD006 Generic counter, short — Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** Range: 0 to 252 Resolution: 1 bit Unit-less number

This field contains a parameter that defines the total number of value parameters used to trigger the alert.

---

11 **Value Parameter Number** — Byte Field Size: 1 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Prohibited

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** Range: 0 to 252 Resolution: 1 bit Unit-less number

This field defines numerically which parameter is to follow up to a maximum of the total value parameters.

---

12 **Value Data Format** — Byte Field Size: 1 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Prohibited

DD340 Value Data Format — As an example to reference an unsigned high voltage (DF96), the Threshold Data format would be decimal 96. Range 0 to 252.

DF53 Integer, 8 bit unsigned **uint8** Range: 0 to 252 Resolution: 1 bit Unit-less number

This field contains the NMEA Network Message data format code used to define the value data below.

---

13 **Value Data** — Byte Field Size: ? — Bit Field Size: — Request Parameter: Optional — Command Parameter: Prohibited

DD341 Value Data — The value data range is defined by the value data format. The Value data is the dynamic parameter used to trigger the alert.

DF00 Undefined **Undefined** Range: undefined Resolution: undefined Application specific, defined at time of use.

This field contains the value data that compared against the threshold level to trigger the associated Alert. This field can contain a variable sized parameter defined by the Value Data Format code defined above.

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Alert Value

PGN: 126988
hex: 1F00C

14 **Fields 11 to 13 Repeat as necessary** *Byte Field Size:* ? *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Prohibited

DD000 Undefined — Application specific, defined at time of use

DF00 Undefined **Undefined** *Range:* undefined *Resolution:* undefined Application specific, defined at time of use.
Application Specific defined at time of use

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System Time

PGN: 126992
hex: 1F010

The purpose of this parameter group is to provide a regular transmission of UTC time and date, with minimal latency. Optionally, the transmission of this parameter group may use the SID to provide synchronism for measurement data in other parameter groups from the same source address.

The default update rate for this PGN is only applicable when sending the current UTC time and date independantly, otherwise the update rate will be determined by the update rate of the PGN(s) related by the SID.

Single Frame: Yes *Priority Default:* 3 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Prohibited *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 29

1 Sequence ID
Byte Field Size: 1 *Bit Field Size:* *Request Parameter* Note 1 *Command Parameter:* Prohibited

DD056 Sequence ID
An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

2 Source
Byte Field Size: *Bit Field Size:* 4 *Request Parameter* Note 1 *Command Parameter:* Prohibited

DD066 Time Source
0x0 = GPS,
0x1 =GLONASS,
0x2 = WWV or equivalent Radio Station Time Sync,
0x3 = Local Cesium clock,
0x4 = Local Rubidium clock,
0x5 = Local Crystal clock,
0x6 - 0xE = reserved,
0xF = Null

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 NMEA Reserved
Byte Field Size: *Bit Field Size:* resv 4 *Request Parameter* *Command Parameter:*

DD001 Reserved field
Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
Used to align subsequent data on a byte boundary.

4 Date
Byte Field Size: 2 *Bit Field Size:* *Request Parameter* Note 1 *Command Parameter:* Optional

DD039 Generic date
Days since January 1, 1970, Date is relative to UTC Time.

DF41 Date, day count **uint16** *Range:* 0 to 65,532 days *Resolution:* 1 day 0 = January 1, 1970, max = ~179 years

5 Time
Byte Field Size: 4 *Bit Field Size:* *Request Parameter* Note 1 *Command Parameter:* Optional

DD158 Generic time of day
24 hour clock, 0 = midnight, time is in UTC

DF06 Time of day **uint32** *Range:* 0 to 86,401 s *Resolution:* 1x10E-4 s ~24 hours, 0 = midnight, range allows for up to two leap seconds per day

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Heartbeat

PGN: 126993
hex: 1F011

This PGN shall be transmitted by all NMEA devices.

Reception of this PGN confirms that a device is still present on the network.

Reception of this PGN may also be used to maintain an address to NAME association table within the receiving device.

The transmission interval may be used by the receiving unit to determine the time-out value for the connection supervision.

The value contained in Field 1 of this PGN reflects the PGN's current Transmission Interval. Changes to this PGN's Transmission Interval shall be reflected in Field 1.

The transmission interval can only be changed by using the Request Group Function PGN 126208 with no pairs of request parameters provided. Field 3 of the Request Group Function PGN 126208 may contain values between 1,000ms and 60,000ms.

This PGN cannot be requested by the ISO Request PGN 059904 or Request Group Function PGN 126208. In Request Group Function PGN 126208, setting Field 3 to a value of 0xFFFF FFFF and Field 4 to a value of 0xFFFF: "Transmit now without changing timing variables." is prohibited.

The Command Group Function PGN 126208 shall not be used with this PGN.

Fields 3 and 4 of this PGN provide information which can be used to distinguish short duration disturbances from permanent failures. See ISO 11898 -1 Sections 6.12, 6.13, 6.14, 13.1.1, 13.1.4 , 13.1.4.3 and Figure 16 (node status transition diagram) for additional context.

Single Frame: Y *Priority Default:* 7 *Default Update Rate:* 60,000 milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Prohibited *Command Support:* Prohibited *ACK Rqmnts:* None

Original Reference ID # 228

Field #	*Field Name*					
1	**Update Rate**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Prohibited *Command Parameter:* Prohibited	
	DD383 Data Interval		Reporting interval in seconds. 0xFFFD=Reserved 0xFFFE=Error 0xFFFF=Data not available			
	DF116 Time Interval, .001 sec.	**uint16**	*Range:* 0 - 65.532s	*Resolution:* 1x10E-3 sec.		
	Valid data range is between 1.000 sec and 60.000 sec.					
2	**Heartbeat Sequence Counter**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Prohibited *Command Parameter:* Prohibited	
	DD006 Generic counter, short		Numeric count, event counter, sequence counter			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
3	**Class 1 CAN Controller State**		*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter* Prohibited *Command Parameter:* Prohibited	
	DD384 CAN State		00= error active 01= error passive 10= bus off 11= not available Represents the worst state detected since the last transmission of this information.			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
	Reflects the node state , as defined in ISO 11898-1 section 13.1.4, of the Class 1 CAN controller or the Class 2 System Instance 0 CAN controller.					

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Heartbeat

PGN: 126993
hex: 1F011

4	**Class 2 Second CAN Controller State**	*Byte Field Size:*		*Request Parameter*	Prohibited
		Bit Field Size:	2	*Command Parameter:*	Prohibited

DD384 CAN State

00= error active
01= error passive
10= bus off
11= not available

Represents the worst state detected since the last transmission of this information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Reflects the node state, as defined in ISO 11898-1 section 13.1.4, of the Class 2. System Instance 1 CAN controller.

5	**Equipment Status**	*Byte Field Size:*		*Request Parameter*	Prohibited
		Bit Field Size:	2	*Command Parameter:*	Prohibited

DD385 Device Status

00= operational
01= fault
10= reserved
11= not available

Operational = device has no active error conditions
Fault = active error conditions exist

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

6	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 34	*Command Parameter:*	

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

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Product Information

PGN: 126996
hex: 1F014

Provides product information onto the network that could be important for determining quality of data coming from this product. Each field must be provided if the data is typically available from such a product. At minimum the NMEA Network Database Version, NMEA Manufacturer's Product Code, and NMEA Certification Level must be provided.

All ISO Request (PGN 059904) requests for the Product Information parameter group shall be destination address specific (as opposed to global). Once a node issues an ISO Request for Product Information, the node shall not issue a subsequent ISO Request for Product Information until the full response has been received for the first request or the response has timed out in accordance with ISO 11783-3, "Controller response time and time out defaults".

All NMEA Request group function (PGN 126208) requests for the Product Information parameter group without specifying request parameters shall be destination address specific (as opposed to global). Once a node issues an NMEA Request for Product Information, the node shall not issue a subsequent NMEA Request for Product Information until the full response has been received for the first request or the response has timed out in accordance with ISO 11783-3, "Controller response time and time out defaults".

Note 1: NMEA encourages manufacturers to provide the capability to support multiple NMEA Network Message database versions within their product. This provides a means for manufacturers products to retain backwards compatability with previous versions of the database and existing legacy products.

This parameter group contains static information and therefore should not be requested periodically, but may be requested as needed.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field #	*Field Name*			Original Reference ID # 4
1	**NMEA Network Message Database Version**	*Byte Field Size:* 2 *Bit Field Size:*	*Request Parameter*	Optional Note 1
	DD007 Generic numeric ID, medium	Number of route, waypoint, event, mark, etc.		
	DF54 Integer, 16 bit unsigned **uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number
	Defined by NMEA: Decimal Number of the format AA.BBB where AA is the major release and BBB is a minor release. The decimal point position is assumed. i.e. The first release of this standard is value 1000, which is to be refered to as Version 1.000			
2	**NMEA Manufacturer's Product Code**	*Byte Field Size:* 2 *Bit Field Size:*	*Request Parameter*	Required Prohibited
	DD007 Generic numeric ID, medium	Number of route, waypoint, event, mark, etc.		
	DF54 Integer, 16 bit unsigned **uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number
	Assigned by NMEA: Decimal Number assigned numerically to a manufacturer's product. If this field is not specified in the "Command Request" or an ISO Request is made of this PGN, the response will with the unit's product code. Otherwise if this field is specified only the units with a matching product code will respond with this PGN.			

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3 **Manufacturer's Model ID**
Byte Field Size: char n
Bit Field Size:
Request Parameter Optional
Command Parameter: Prohibited

DD192 Generic String, ASCII, Fixed length — Length specified by PGN field definition.

DF63 String, fixed **char8(n)** *Range:* 0 to 1,785 characters *Resolution:* 1 char

This is a n=32 character string, Format defined and documented by manufacturer. The beginning of the field should clearly define Manufacturer's Product.

0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.

---

4 **Manufacturer's Software Version Code**
Byte Field Size: char n
Bit Field Size:
Request Parameter Optional
Command Parameter: Optional

DD192 Generic String, ASCII, Fixed length — Length specified by PGN field definition.

DF63 String, fixed **char8(n)** *Range:* 0 to 1,785 characters *Resolution:* 1 char

This is a n=32 character string, Format defined and documented by manufacturer.

0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.

---

5 **Manufacturer's Model Version**
Byte Field Size: char n
Bit Field Size:
Request Parameter Optional
Command Parameter: Prohibited

DD192 Generic String, ASCII, Fixed length — Length specified by PGN field definition.

DF63 String, fixed **char8(n)** *Range:* 0 to 1,785 characters *Resolution:* 1 char

This is a n=32 character string, Format defined and documented by manufacturer.

0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.

---

6 **Manufacturer's Model Serial Code**
Byte Field Size: char n
Bit Field Size:
Request Parameter Optional
Command Parameter: Prohibited

DD192 Generic String, ASCII, Fixed length — Length specified by PGN field definition.

DF63 String, fixed **char8(n)** *Range:* 0 to 1,785 characters *Resolution:* 1 char

This is a n=32 character string, Format defined and documented by manufacturer.

0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.

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7	**NMEA 2000 Certification Level**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Prohibited

DD226 NMEA Certification Level

0 = Level A
1 = Level B
2 = Not Applicable

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

As of February 2015 and Version 2.000 of the NMEA 2000 Main Document, Certification Level Field (7) shall be set to 2 meaning “Not Applicable”.

8	**Load Equivalency**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Prohibited

DD257 Load Equivalency Number

The "Load Equivalency Number" is defined in the NMEA 2000 Standard's Main Document in section 2.4.7 Interface Power. Please consult this for details.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

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Configuration Information

PGN: 126998
hex: 1F016

Free-form alphanumeric fields describing the installation (e.g., starboard engine room location) of the device and installation notes (e.g., calibration data).

All ISO Request (PGN 059904) requests for the Configuration Information parameter group shall be destination address specific (as opposed to global). Once a node issues an ISO Request for Configuration Information, the node shall not issue a subsequent ISO Request for Configuration Information until the full response has been received for the first request or the response has timed out in accordance with ISO 11783-3, "Controller response time and time out defaults".

All NMEA Request group function (PGN 126208) requests for the Configuration Information parameter group without specifying request parameters shall be destination address specific (as opposed to global). Once a node issues an NMEA Request for Configuration Information, the node shall not issue a subsequent NMEA Request for Configuration Information until the full response has been received for the first request or the response has timed out in accordance with ISO 11783-3, "Controller response time and time out defaults".

This parameter group contains static information and therefore should not be requested periodically, but may be requested as needed.

Note 1: Using this field as a request parameter is prohibited.

Single Frame:	No	*Priority Default:*	6	*Default Update Rate:*		milliseconds	*Frequency:*	NA	cycles per second
Destination:	Global	*Query Support:*	Optional	*Command Support:*	Optional	*ACK Rqmnts:*	None		

Field #	*Field Name*					Original Reference ID # 5
1	**Installation Description, Field 1**	*Byte Field Size:*	8 or 16	n	*Request Parameter*	Note 1
		Bit Field Size:			*Command Parameter:*	Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character

2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

70 ASCII or 35 Unicode characters maximum

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Configuration Information

PGN: 126998
hex: 1F016

2 **Installation Description, Field 2**

Byte Field Size: 8 or 16 n
Bit Field Size:

Request Parameter Note 1
Command Parameter: Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character

2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

70 ASCII or 35 Unicode characters maximum

---

3 **Manufacturer Information, Field 3**

Byte Field Size: 8 or 16 n
Bit Field Size:

Request Parameter Note 1
Command Parameter: Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character

2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

70 ASCII or 35 Unicode characters maximum

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Man Overboard Notification(MOB)

PGN: 127233
hex: 1F101

The MOB PGN is intended to provide notification from a MOB monitoring system. The included position information may be that of the vessel or the MOB device itself as identified in field “X”, position source. Additional information may include the current state of the MOB device, time of activation, and MOB device battery status.

This PGN may be used to set a MOB waypoint, or to initiate an alert process.

This PGN may be used to command or register a MOB device emitter Ids or other applicable fields in the message with an MOB System or other equipment. If the fields in this PGN are configured over the network, the Command Group Function (PGN 126208) shall be used.

Queries for this PGN shall be requested using either the ISO Request (PGN 059904) or the NMEA Request Group Function (PGN 126208).

A device receiving an ISO (PGN 059904) for this PGN (127233), shall respond by providing as many of these PGNs (127233) as necessary for every MOB Emitter ID that has associated data fields.

If a Request Group Function (PGN 126208) requesting this PGN (127233) is received, the receiving device shall respond in the following manner:
•If no requested fields have been included with the Request Group Function then the response is to return one or more PGNs, just like responding to the ISO Request (PGN 055904) described above.
•If the Request Group Function (PGN 126208) includes the MOB Emitter ID field or MOB Status field, then the response shall be filtered by these fields contained within this request resulting in one or more PGN (127233) responses.

If the MOB Emitter ID requested is not considered a valid MOB Emitter ID by the receiving device, then the appropriate response would be the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied.” And the requested MOB Emitter ID field parameter error code (Field 6) of “0x3 = Requested or command parameter out-of-range;”.

The Default update rate of this PGN is autonomous, as it is dependant upon notification rates of MOB devices.

Single Frame: N *Priority Default:* 3 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 227

1 Sequence ID *Byte Field Size:* 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD056 Sequence ID — An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

2 MOB Emitter ID *Byte Field Size:* 4 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Optional

DD010 Generic numeric ID, large — Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number
This provides a unique Identifier for each MOB emitter (MOB emitter is unique to the vessel)

3 Man Overboard (MOB) Status *Byte Field Size:* *Request Parameter* Required
Bit Field Size: 3 *Command Parameter:* Optional

DD369 Man Over Board (MOB) Status

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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Man Overboard Notification(MOB)

PGN: 127233
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4 **NMEA Reserved**
Byte Field Size: *Bit Field Size:* resv 5
Request Parameter *Command Parameter:*

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

---

5 **UTC Time of MOB Activation**
Byte Field Size: 4 *Bit Field Size:*
Request Parameter Optional *Command Parameter:* Optional

DD158 Generic time of day — 24 hour clock, 0 = midnight, time is in UTC

DF06 Time of day **uint32** *Range:* 0 to 86,401 s *Resolution:* 1x10E-4 s ~24 hours, 0 = midnight, range allows for up to two leap seconds per day

UTC of MOB activation provides the time of the initial MOB device activation.

---

6 **Position Source**
Byte Field Size: *Bit Field Size:* 3
Request Parameter Optional *Command Parameter:* Optional

DD370 Position Source — Identifies the source of the position information
0=MOB Position estimated by the Vessel
1=MOB position reported by MOB emitter
2-5= Reserved
6=Error
7=Unknown
/ Unavailable

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

7 **NMEA Reserved**
Byte Field Size: *Bit Field Size:* resv 5
Request Parameter *Command Parameter:*

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

---

8 **UTC Date of Position**
Byte Field Size: 2 *Bit Field Size:*
Request Parameter Optional *Command Parameter:* Optional

DD039 Generic date — Days since January 1, 1970, Date is relative to UTC Time.

DF41 Date, day count **uint16** *Range:* 0 to 65,532 days *Resolution:* 1 day 0 = January 1, 1970, max = ~179 years

---

9 **UTC Time of Position**
Byte Field Size: 4 *Bit Field Size:*
Request Parameter Optional *Command Parameter:* Optional

DD158 Generic time of day — 24 hour clock, 0 = midnight, time is in UTC

DF06 Time of day **uint32** *Range:* 0 to 86,401 s *Resolution:* 1x10E-4 s ~24 hours, 0 = midnight, range allows for up to two leap seconds per day

The UTC of position provides the time of the position information.

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Man Overboard Notification(MOB)

PGN: 127233
hex: 1F101

10 Latitude

Byte Field Size: 4 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD022 Latitude, WGS-84

Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23 Latitude — **int32** — *Range:* +/- 90 deg — *Resolution:* 1x10E-7 deg — "-" = South, resolution ~1.1 cm

---

11 Longitude

Byte Field Size: 4 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD023 Longitude, WGS-84

Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF25 Longitude — **int32** — *Range:* +/- 180 deg — *Resolution:* 1x10E-7 deg — "-" = West, resolution ~1.1 cm

---

12 Course over ground Reference

Byte Field Size: — *Request Parameter* Optional
Bit Field Size: 2 — *Command Parameter:* Optional

DD117 Direction reference

0 = True,
1 = Magnetic,
2 = Error,
3 = Null

DF52 Bit field — **bit(n)** — *Range:* Variable — *Resolution:* 1 — Used to construct bit fields

---

13 NMEA Reserved

Byte Field Size: — *Request Parameter* Optional
Bit Field Size: resv 6 — *Command Parameter:* Optional

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field — **bit(n)** — *Range:* Variable — *Resolution:* 1 — Used to construct bit fields

---

14 Course over ground

Byte Field Size: 2 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD165 Course-Over-Ground (COG)

The direction of the path over ground actually followed by a vessel.

DF02 Angle — **uint16** — *Range:* 0 to 2Pi rad — *Resolution:* 1x10E-4 rad — Resolution ~0.0057deg, 1 deg = .01745 rad

---

15 Speed over ground

Byte Field Size: 2 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD044 Generic Speed

DF35 Speed — **uint16** — *Range:* 0 to 655.32 m/s — *Resolution:* 1x10E-2 m/s — 1 Knot = 0.5144 m/s

---

16 MMSI of vessel of Origin

Byte Field Size: 4 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD010 Generic numeric ID, large

Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned — **uint32** — *Range:* 0 to 4,294,967,292 — *Resolution:* 1 bit — Unit-less number

The MMSI number of the ship of origin may be set to 2,147,483,647 if unknown.

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Man Overboard Notification(MOB)

PGN: 127233
hex: 1F101

17	**MOB Emitter Battery Status**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	3	*Command Parameter:*	Optional
	DD371 Battery Status		0=Good 1=Low 2–5=Reserved 6=Error 7=Unavailable			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
18	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*	
			Bit Field Size:	resv 5	*Command Parameter:*	
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.					

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Heading/Track Control

PGN: 127237
hex: 1F105

Sends Commands to, and receives data from, heading control systems. Allows for navigational (remote) control of a heading control system and direct rudder control. When used as a command, the Commanded Rudder Direction field and the Commanded Rudder Angle should never contain order values at the same time.

Single Frame: No *Priority Default:* 2 *Default Update Rate:* 250 milliseconds *Frequency:* 4. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 54

1 **Rudder Limit Exceeded** *Byte Field Size:* *Bit Field Size:* 2 *Request Parameter* Optional *Command Parameter:* Prohibited

DD002 Generic status pair
MSB/LSB:
00 = [No, Off, Disabled, Reset, "0"],
01 = [Yes, On, Enabled, Set, "1"],
10 = Error,
11= [Unavailable, Unknown]

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

2 **Off-Heading Limit Exceeded** *Byte Field Size:* *Bit Field Size:* 2 *Request Parameter* Optional *Command Parameter:* Prohibited

DD002 Generic status pair
MSB/LSB:
00 = [No, Off, Disabled, Reset, "0"],
01 = [Yes, On, Enabled, Set, "1"],
10 = Error,
11= [Unavailable, Unknown]

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Off-Heading Limit field can be generated if the Steering Mode is Heading Control Standalone, Heading Control, or Track Control.

3 **Off-Track Limit Exceeded** *Byte Field Size:* *Bit Field Size:* 2 *Request Parameter* Optional *Command Parameter:* Prohibited

DD002 Generic status pair
MSB/LSB:
00 = [No, Off, Disabled, Reset, "0"],
01 = [Yes, On, Enabled, Set, "1"],
10 = Error,
11= [Unavailable, Unknown]

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Off-Track Limit field can be generated if the Steering Mode is Track Control.

4 **Override** *Byte Field Size:* *Bit Field Size:* 2 *Request Parameter* Optional *Command Parameter:* Prohibited

DD163 Autopilot Override
1 = Yes, 0 = No. Yes means a temporary interruption of the selected steering mode. As long as this field is Yes, Steering Mode and Turn Mode shall be ignored by the heading/track controller and its computing parts shall operate as if manual steering was selected.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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5 **Steering Mode**

Byte Field Size:
Bit Field Size: 3

Request Parameter Optional
Command Parameter: Prohibited

DD153 Steering Mode

MSB/LSB:
000 = Main Steering,
001 = Non-Follow-up Device,
010 = Follow-up Device,
011 = Heading Control Standalone,
100 = Heading Control,
101 = Track Control.
Definitions: Main Steering/Outside System – The main steering system is in use.
Non-Follow-up Device – The system provides non-follow-up control. Rudder is moved in the commanded direction but a specific angle is not maintained.
Follow-up – The system provides follow-up control. Rudder is moved to the commanded angle and maintained at that angle.
Heading Control Standalone – The system works as a standalone heading controller.
Heading Control – The system works as a remotely controlled heading controller with the commanded course input from an external device.
Track Control – The system works as a track controller by correcting a commanded course. Corrections are based on additionally received track errors.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

6 **Turn Mode**

Byte Field Size:
Bit Field Size: 3

Request Parameter Optional
Command Parameter: Prohibited

DD152 Turn Method

MSB/LSB:
000 = Rudder Limit controlled,
001 = turn rate controlled,
010 = radius controlled.
Definitions: Rudder – An operator set rudder limit determines the maximum turning capability available, in which case the system outputs rudder orders up to and limited by the rudder limit.
Turn Rate – An operator set turn rate limit determines the maximum turning capability available, in which case the system controls the rate of turn using whatever value of rudder order is necessary to control the rate.
Radius – An operator set turn radius determines the maximum turning capability available which in this case is described as the minimum turn radius, in which case the system controls the rate of turn corresponding to the set turn radius.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

7 **Heading Reference**

Byte Field Size:
Bit Field Size: 2

Request Parameter Optional
Command Parameter: Prohibited

DD117 Direction reference

0 = True,
1 = Magnetic,
2 = Error,
3 = Null

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

8 **NMEA Reserved**

Byte Field Size:
Bit Field Size: resv 5

Request Parameter
Command Parameter:

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

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Heading/Track Control

PGN: 127237
hex: 1F105

9	**Commanded Rudder Direction**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	3	*Command Parameter:*	Optional	
	DD147 Directional Command		MSB/LSB: 000 = No Order, 001 = Move to starboard, 010 = Move to port.				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
10	**Commanded Rudder Angle**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD146 Rudder Angle Signed		Rudder angle where positive values are starboard and negative values are port				
	DF04 Angle, signed	**int16**	*Range:*	+/-Pi rad	*Resolution:*	1x10E-4 rad	Resolution ~0.0057deg
11	**Heading-To-Steer (Course)**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD113 Course		The horizontal direction in which a vessel is steered or intended to be steered, expressed as angular distance 000 north, clockwise through 359 degrees. Not to be confused with Course-over-ground, Track, or Heading.				
	DF02 Angle	**uint16**	*Range:*	0 to 2Pi rad	*Resolution:*	1x10E-4 rad	Resolution ~0.0057deg, 1 deg = .01745 rad
12	**Track**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD166 Track		The intended or desired horizontal direction of travel with respect to the earth. The track expressed in degrees of the compass may differ from the course due to allowances made in the course for such factors as sea and weather conditions in order to achieve the desired track. This field represents the course line (leg) between two waypoints. It may be altered dynamically in a track-controlled turn along a pre-planned radius.				
	DF02 Angle	**uint16**	*Range:*	0 to 2Pi rad	*Resolution:*	1x10E-4 rad	Resolution ~0.0057deg, 1 deg = .01745 rad
13	**Rudder Limit**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Prohibited	
	DD148 Angular Limit						
	DF02 Angle	**uint16**	*Range:*	0 to 2Pi rad	*Resolution:*	1x10E-4 rad	Resolution ~0.0057deg, 1 deg = .01745 rad
14	**Off-Heading Limit**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Prohibited	
	DD148 Angular Limit						
	DF02 Angle	**uint16**	*Range:*	0 to 2Pi rad	*Resolution:*	1x10E-4 rad	Resolution ~0.0057deg, 1 deg = .01745 rad
	Off-Heading Limit field can be generated if the Steering Mode is Heading Control Standalone, Heading Control, or Track Control.						
15	**Radius of Turn Order**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD149 Distance ordered		A commanded distance like radius order, off-track limit, etc.				
	DF74 Distance, rough	**int16**	*Range:*	+/-32,764 m	*Resolution:*	1 m	

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16	**Rate of Turn Order**		*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* *Command Parameter:*	Optional Optional
	DD150 Rate of Turn		+ = Bow turning to starboard, 1 deg/min = .00029 rad/sec				
	DF73 Angular rate, signed	**int16**	*Range:* +/-1.0 rad/s	*Resolution:*	1/32 x 10E-3 rad/s	Resolution 0.1 deg/min	
17	**Off-Track Limit**		*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* *Command Parameter:*	Optional Prohibited
	DD149 Distance ordered		A commanded distance like radius order, off-track limit, etc.				
	DF74 Distance, rough	**int16**	*Range:* +/-32,764 m	*Resolution:*	1 m		
	Off-Track field can be generated if the Steering Mode is Track Control.						
18	**Vessel Heading**		*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* *Command Parameter:*	Optional Prohibited
	DD167 Heading		The horizontal direction in which a ship actually points or heads at any instant, expressed in angular units from a reference direction, usually from 000 at the reference direction clockwise through 359 degrees.				
	DF02 Angle	**uint16**	*Range:* 0 to 2Pi rad	*Resolution:*	1x10E-4 rad	Resolution ~0.0057deg, 1 deg = .01745 rad	

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Rudder

PGN: 127245
hex: 1F10D

Rudder order command in direction or angle with current rudder angle reading. The Direction Order field is for non-follow-up rudder orders and the Angle Order field is for follow-up rudder orders. When used as a command, the Direction Order field and the Angle Order field shall never contain order values at the same time. The Position should be set to 'Data Not Available' unless the unit which sources the command also sources the current angle.

When used as a feedback from the rudder, the commanded rudder may be returned together with the current rudder angle reading.

It is recommended to send both the commands and the current rudder angle messages at fixed intervals. The unit which controls the rudder should monitor the reception of rudder angle messages.

Single Frame: Yes *Priority Default:* 2 *Default Update Rate:* 100 milliseconds *Frequency:* 10. cycles per second
Destination: Global *Query Support:* Required *Command Support:* Required *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 53

1 **Rudder Instance** *Byte Field Size:* *Bit Field Size:* 8 *Request Parameter* Required *Command Parameter:* Required

DD128 Generic instance

0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

If this field is not specified in the "Request Group Function" or an ISO Request is made of this PGN, the response will be with all defined Rudder Instances. (This PGN will be transmitted for each instance.) If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.

2 **Direction Order** *Byte Field Size:* *Bit Field Size:* 3 *Request Parameter* Optional *Command Parameter:* Optional

DD147 Directional Command

MSB/LSB:
000 = No Order,
001 = Move to starboard,
010 = Move to port.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Value should be "Not Available" when Angle Order is provided.

3 **NMEA Reserved** *Byte Field Size:* *Bit Field Size:* resv 5 *Request Parameter* *Command Parameter:*

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

4 **Angle Order** *Byte Field Size:* 2 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD146 Rudder Angle Signed

Rudder angle where positive values are starboard and negative values are port

DF04 Angle, signed **int16** *Range:* +/-Pi rad *Resolution:* 1x10E-4 rad Resolution ~0.0057deg

Value should be "Not Available" when Direction Order is provided.

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Rudder

PGN: 127245
hex: 1F10D

5	**Position**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Prohibited	
	DD146 Rudder Angle Signed		Rudder angle where positive values are starboard and negative values are port				
	DF04 Angle, signed	**int16**	*Range:*	+/-Pi rad	*Resolution:*	1x10E-4 rad	Resolution ~0.0057deg
6	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*		
			Bit Field Size:	resv 16	*Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.						

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Vessel Heading

PGN: 127250
hex: 1F112

Heading sensor value with a flag for True or Magnetic. If the sensor value is Magnetic, the deviation field can be used to produce a Magnetic heading, and the variation field can be used to correct the Magnetic heading to produce a True heading.

To obtain Magnetic Heading from the Heading Sensor Reading: Add Deviation to Heading Sensor Reading.

To obtain True Heading: Add Variation to Magnetic Heading.

The Heading Sensor Reading may or may not be corrected for Deviation and the Deviation field set to 'DataNotAvailable'. (A fluxgate compass may be compensated for Deviation without being able to produce the Deviation corresponding to every Heading)

If the Heading sensor does not provide Variation by itself, it shall set the value to 'Data Not Available'.
A source which provides Magnetic Variation only, should use PGN 127258 Magnetic Variation.

Variation provided in this PGN is the one currently in use by this device. If this device is also a source of variation, it should also transmit Magnetic Variation PGN 127258.

A steering compass must send rapidly, a second backup compass may send at a slower rate, a Variation only source may send this at a slow rate.

A deviation table may be programmed into the compass using the Command Group Function (PGN 126208) message and sending Heading Sensor Reading and Deviation for each entry of the table.

Single Frame: Yes | *Priority Default:* 2 | *Default Update Rate:* 100 milliseconds | *Frequency:* 10. cycles per second
Destination: Global | *Query Support:* Optional | *Command Support:* Optional | *ACK Rqmnts:* None

Original Reference ID # 56

Field #	Field Name						
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD056 Sequence ID			An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252		*Resolution:* 1 bit		Unit-less number
2	**Heading Sensor Reading**		*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD118 Heading Sensor Reading			Primary output of heading as indicated by the heading sensor.			
	DF02 Angle	**uint16**	*Range:* 0 to 2Pi rad		*Resolution:* 1x10E-4 rad		Resolution ~0.0057deg, 1 deg = .01745 rad
3	**Deviation**		*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD151 Magnetic Heading Correction			Positive values are Easterly and negative values are Westerly.			
	DF04 Angle, signed	**int16**	*Range:* +/-Pi rad		*Resolution:* 1x10E-4 rad		Resolution ~0.0057deg
4	**Variation**		*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD151 Magnetic Heading Correction			Positive values are Easterly and negative values are Westerly.			
	DF04 Angle, signed	**int16**	*Range:* +/-Pi rad		*Resolution:* 1x10E-4 rad		Resolution ~0.0057deg

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Vessel Heading

PGN: 127250
hex: 1F112

5	**Heading Sensor Reference**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD117 Direction reference		0 = True, 1 = Magnetic, 2 = Error, 3 = Null			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
6	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*	
			Bit Field Size:	resv 6	*Command Parameter:*	
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.					

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Rate of Turn

PGN: 127251
hex: 1F113

Rate of Turn is the rate of change of the Heading.

Heading is defined as the direction of the vertical projection of the fore-and-aft line of the ship onto the horizontal plane.

Single Frame: Yes *Priority Default:* 2 *Default Update Rate:* 100 milliseconds *Frequency:* 10. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 90

1 Sequence ID
Byte Field Size: 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD056 Sequence ID
An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

2 Rate of Turn
Byte Field Size: 4 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD224 Rate of Turn
+ = Bow turning to starboard, 1 deg/min = .00029 rad/sec

DF85 Angular rate, signed - Pre **int32** *Range:* +/-67.0 rad/s (approx. 230703 deg/min *Resolution:* 1/32 x 10E-6 rad/s

3 NMEA Reserved
Byte Field Size: *Request Parameter*
Bit Field Size: resv 24 *Command Parameter:*

DD001 Reserved field
Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
Used to align subsequent data on a byte boundary.

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Heave

PGN: 127252
hex: 1F114

The Heave PGN reports the vertical distance perpendicular to (smooth, wave-free water on) the earth's surface. The vertical distance is the height that a vessel is lifted by the crest of a wave (-), or the depth that a vessel drops in the trough of a wave (+). This convention conforms to the North-East-Down(NED) coordinate system, which is also referred to as positive down. Heave mesurements can be used by a sounder to remove the effects of waves (heave) from the data presented on its display.

Field 3 may be configured over the network. The Command Group Function (PGN126208) shall be used for any configuration of this PGN.

Note 1: Field 3 command support is optional. If it is supported, the sensor may accept a command to change the source of the delay value:
1) Change the delay to one of the manufacturer's preset values, or
2) Change the delay to a user defined value

Single Frame: Y *Priority Default:* 3 *Default Update Rate:* 100 milliseconds *Frequency:* 10. cycles per second
Destination: G *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 229

Field #	Field Name						
1	**Sequence ID**			*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD056 Sequence ID			An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53	Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
2	**Heave**			*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD372 Heave			Vertical displacement perpendicular to (smooth, wave-free water on) the earth's surface. The height that a vessel is lifted by the crest of a wave (-), or the depth that a vessel drops in the trough of a wave (+). This convention conforms to the North-East-Down(NED) coordinate system, which is also referred to as positive down.			
	DF14	Distance, short, signed	**int16**	*Range:* +/-327.64 m	*Resolution:* 1x10E-2 m		
3	**Delay**			*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Note 1
	DD373 Delay			Delay added by calculations. When commanded: 0x0000 - 0xfff9 = Value within valid uint16 range 0xfffa = Set minimum value 0xfffb = Set maximum value 0xfffc = Restore Factory Defaults			
	DF66	Time interval, .01sec	**uint16**	*Range:* 0 to 655.32s	*Resolution:* 1x10E-2sec		
	The delay parameter is optional. If not supported, it must be transmitted as Data Not Available. Commanding this field may allow the sensor to fine tune its calculations given more or less time than its default setting.						

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4	**Delay Source**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	4	*Command Parameter:*	Optional

DD374 Measurement Delay Source

0 = User Defined
1 = Minimum Supported Value
2 = Factory Default Value
3 = Maximum Supported Value
4 thru 13 = Reserved
14 = Data out of range
15 = Data not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

This field is used to report the source of the time delay value in field 3 in this PGN.

5	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 20	*Command Parameter:*	

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

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Attitude

PGN: 127257
hex: 1F119

This PGN provides a single transmission that describes the position of a vessel relative to both horizontal and vertical planes. This would typically be used for vessel stabilization, vessel control and onboard platform stabilization.

Products that directly interface to pitch, roll and yaw transducers would transmit this PGN

Single Frame:	Yes	*Priority Default:*	3	*Default Update Rate:*	1000 milliseconds	*Frequency:*	1.	cycles per second	
Destination:	Global	*Query Support:*	Optional	*Command Support:*	Optional	*ACK Rqmnts:*	None		

Field # *Field Name* — Original Reference ID # 35

#	Field Name						
1	**Sequence ID**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD056 Sequence ID			An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number
2	**Yaw**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD063 Yaw			Oscillation of ship about it's vertical axis. Bow moving to starboard is positive.			
	DF04 Angle, signed	**int16**	*Range:*	+/-Pi rad	*Resolution:*	1x10E-4 rad	Resolution ~0.0057deg
3	**Pitch**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD062 Pitch			Oscillation of ship about it's latitudinal axis. Bow moving up is positive.			
	DF04 Angle, signed	**int16**	*Range:*	+/-Pi rad	*Resolution:*	1x10E-4 rad	Resolution ~0.0057deg
4	**Roll**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD061 Roll			Oscillation of ship about it's longitudinal axis. Roll to the starboard is positive.			
	DF04 Angle, signed	**int16**	*Range:*	+/-Pi rad	*Resolution:*	1x10E-4 rad	Resolution ~0.0057deg
	Roll to starboard is positive +/- 180 degrees						
5	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*		
			Bit Field Size:	resv 8	*Command Parameter:*		
	DD001 Reserved field			Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.						

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Magnetic Variation

PGN: 127258
hex: 1F11A

Message for transmitting variation. The message contains a sequence number to allow synchronization of other messages such as Heading or Course over Ground. The quality of service and age of service are provided to enable recipients to determine an appropriate level of service if multiple transmissions exist.

Single Frame: Yes *Priority Default:* 6 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 92

1 Sequence ID
Byte Field Size: 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD056 Sequence ID
An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

2 Variation Source
Byte Field Size: *Request Parameter* Optional
Bit Field Size: 4 *Command Parameter:* Optional

DD232 Variation Source
0x00 = Manual Entry, "Variation has been entered via key entry"
0x01 = Automatic - Chart, "Variation is derived from cartograhy present in system"
0x02 = Automatic Table, "Variation has been computed from tabular based system"
0x03 = Automatic Calculation, "Variation has been derived via calculation"
0x04 = WMM 2000, "Variation is calc via World Magnetic Model for 2000"
0x05 = WMM 2005, "Variation is calc via World Magnetic Model for 2005"
0x06 = WMM 2010, "Variation is calc via World Magnetic Model for 2010"
0x07 = WMM 2015, "Variation is calc via World Magnetic Model for 2015"
0x08 = WMM 2020, "Variation is calc via World Magnetic Model for 2020"
0x09 = Reserved,
thru
0x0E = Reserved,
0x0F = Data Not Available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 NMEA Reserved
Byte Field Size: *Request Parameter*
Bit Field Size: resv 4 *Command Parameter:*

DD001 Reserved field
Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
Used to align subsequent data on a byte boundary.

4 Age of Service (Date)
Byte Field Size: 2 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD039 Generic date
Days since January 1, 1970, Date is relative to UTC Time.

DF41 Date, day count **uint16** *Range:* 0 to 65,532 days *Resolution:* 1 day 0 = January 1, 1970, max = ~179 years

5 Variation
Byte Field Size: 2 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD151 Magnetic Heading Correction
Positive values are Easterly and negative values are Westerly.

DF04 Angle, signed **int16** *Range:* +/-Pi rad *Resolution:* 1x10E-4 rad Resolution ~0.0057deg

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Magnetic Variation

PGN: 127258
hex: 1F11A

6	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 16	*Command Parameter:*	

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

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Engine Parameters, Rapid Update

PGN: 127488
hex: 1F200

Provides data with a high update rate for a specific engine in a single frame message. The first field provides information as to which engine. This PGN if used with PGN 127489 will provide most Engine data.

Single Frame: Yes | *Priority Default:* 2 | *Default Update Rate:* 100 milliseconds | *Frequency:* 10. cycles per second
Destination: Global | *Query Support:* Required | *Command Support:* Optional | *ACK Rqmnts:* None

Original Reference ID # 88

Field #	Field Name	Byte Field Size	Bit Field Size	Request Parameter	Command Parameter	Description
1	**Engine Instance**		8	Required	Optional	
	DD410 Engine / Transmission Instance					This data instance field shall be enumerated on a globally unique basis beginning on the Port side with instance number 0 and incrementing towards Starboard. Devices in Propulsion Class 50 and Function Code 140 shall be the first enumeration sequence (i.e. 0, 1, 2, 3, ...n) prior to moving to a different Class and Function Code. Devices of the next Class and Function Code shall continue the enumeration sequence (i.e. n+1, n+2,...).
	DF52 Bit field **bit(n)**			Range: Variable	Resolution: 1	Used to construct bit fields
	If this field is not specified in the "Command Request" or an ISO Request is made of this PGN, the response will be with all defined Engine Instances. (This PGN will be transmitted for each instance.) If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.					
2	**Engine Speed**	2		Optional	Optional	
	DD129 Rate of rotation					
	DF72 Rotational rate, unsigned **uint16**			Range: 0-16,383 RPM	Resolution: 1/4 RPM	
3	**Engine Boost Pressure**	2		Optional	Optional	
	DD049 Generic Pressure					
	DF47 Pressure, medium **uint16**			Range: 0 to 6,553,200 Pa	Resolution: 1x10E+2 Pa	
4	**Engine tilt/trim**	1		Optional	Optional	
	DD138 Generic percent of range					
	DF30 Percent, Relative measure **int8**			Range: +/- 124%	Resolution: 1%	
	Range 0 - 100%, where 0% =Full Down (trim) and 100% = Full Up (tilt) Positions					
5	**NMEA Reserved**		resv 16			
	DD001 Reserved field					Variable number of reserved bits, all set to logic "1"
	DF52 Bit field **bit(n)**			Range: Variable	Resolution: 1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.					

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Engine Parameters, Dynamic

PGN: 127489
hex: 1F201

Used to provide real-time operational data and status relevant to a specific engine, indicated by the engine instance field. This message would normally be broadcasted periodically to provide information for instrumentation or control functions.

The Engine Discrete Status 1 and Engine Discrete Status 2 fields (11 and 12) are required and may not be transmitted as unavailable. Reserved bits in these two fields shall be set to 0.

For example, if a sensor is not provided to detect water in the fuel, a 0 shall be placed in the corresponding bit position; If a sensor is provided to detect water in the fuel and water in the fuel is not indicated a 0 shall be placed in the bit position; The only time that a 1 is placed in the bit position is when both a sensor is provided to detect water in the fuel, and water in the fuel is indicated.

Single Frame: No *Priority Default:* 2 *Default Update Rate:* 500 milliseconds *Frequency:* 2. cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 47

Field #	Field Name					
1	**Engine instance**		*Byte Field Size:* *Bit Field Size:* 8		*Request Parameter* Required *Command Parameter:* Optional	
	DD410 Engine / Transmission Instance			This data instance field shall be enumerated on a globally unique basis beginning on the Port side with instance number 0 and incrementing towards Starboard. Devices in Propulsion Class 50 and Function Code 140 shall be the first enumeration sequence (i.e. 0, 1, 2, 3, ...n) prior to moving to a different Class and Function Code. Devices of the next Class and Function Code shall continue the enumeration sequence (i.e. n+1, n+2,...).		
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
	If this field is not specified in the "Command Request" or an ISO Request is made of this PGN, the response will be with all defined Engine Instances. (This PGN will be transmitted for each instance.) If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.					
2	**Engine oil pressure**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Prohibited	
	DD049 Generic Pressure					
	DF47 Pressure, medium	**uint16**	*Range:* 0 to 6,553,200 Pa	*Resolution:* 1x10E+2 Pa		
3	**Engine oil temp.**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Prohibited	
	DD130 Temperature, high					
	DF38 Temperature, high	**uint16**	*Range:* 0 to 6,553.2 deg K	*Resolution:* 1x10E-1 deg K		
4	**Engine temp.**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Prohibited	
	DD043 Generic Temperature					
	DF39 Temperature, low	**uint16**	*Range:* 0 to 655.32 deg K	*Resolution:* 1x10E-2 deg K		
5	**Alternator potential**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Prohibited	
	DD136 Voltage, DC					
	DF42 Voltage, high	**int16**	*Range:* +/- 327.64 V	*Resolution:* 1x10E-2 V		

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Engine Parameters, Dynamic

PGN: 127489
hex: 1F201

6 Fuel rate

Byte Field Size: 2 | Bit Field Size: | Request Parameter: Optional | Command Parameter: Prohibited

DD131 Flow rate, low

DF18	Flow rate, low	**int16**	Range:	+/-3.2764 cu-m/hr	Resolution:	1x10E-4 cu-m/hr

7 Total engine hours

Byte Field Size: 4 | Bit Field Size: | Request Parameter: Optional | Command Parameter: Optional

DD132 Run time, Engine

DF67	Time interval, large	**uint32**	Range:	0 to ~4.295x10E+9 s	Resolution:	1 sec

8 Engine coolant pressure

Byte Field Size: 2 | Bit Field Size: | Request Parameter: Optional | Command Parameter: Prohibited

DD049 Generic Pressure

DF47	Pressure, medium	**uint16**	Range:	0 to 6,553,200 Pa	Resolution:	1x10E+2 Pa

9 Fuel Pressure

Byte Field Size: 2 | Bit Field Size: | Request Parameter: Optional | Command Parameter: Prohibited

DD225 Generic Pressure High

DF29	Pressure	**uint16**	Range:	0 to 65,532,000 Pa	Resolution:	1x10E+3 Pa

10 Not Available

Byte Field Size: | Bit Field Size: resv 8 | Request Parameter: | Command Parameter:

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52	Bit field	**bit(n)**	Range:	Variable	Resolution:	1

Used to construct bit fields

Previously asssigned Eng tilt/trim, moved to PGN127488 field 4 for faster update rate. This value to be always set to +127 (Not Available) until future reuse.

11 Engine Discrete Status 1

Byte Field Size: | Bit Field Size: 16 | Request Parameter: Optional | Command Parameter: Prohibited

DD206 Engine Discrete Warning Status

xxxx xxxx xxxx xxx1 = Check Engine,
xxxx xxxx xxxx xx1x = Over Temperature,
xxxx xxxx xxxx x1xx = Low Oil Pressure,
xxxx xxxx xxxx 1xxx = Low Oil Level,
xxxx xxxx xxx1 xxxx = Low Fuel Pressure,
xxxx xxxx xx1x xxxx = Low System Voltage,
xxxx xxxx x1xx xxxx = Low Coolant Level,
xxxx xxxx 1xxx xxxx = Water Flow,
xxxx xxx1 xxxx xxxx = Water in Fuel,
xxxx xx1x xxxx xxxx = Charge Indicator,
xxxx x1xx xxxx xxxx = Preheat Indicator,
xxxx 1xxx xxxx xxxx = High Boost Pressure,
xxx1 xxxx xxxx xxxx = Rev Limit Exceeded,
xx1x xxxx xxxx xxxx = EGR System,
x1xx xxxx xxxx xxxx = Throttle Position Sensor,
1xxx xxxx xxxx xxxx = Engine Emergency Stop Mode
where x = don't care

Inactive states shall be transmitted as 0. Reserved and unsupported bits within the field shall be transmitted as 0.

DF52	Bit field	**bit(n)**	Range:	Variable	Resolution:	1

Used to construct bit fields

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12 **Engine Discrete Status 2**

Byte Field Size:
Bit Field Size: 16

Request Parameter Optional
Command Parameter: Prohibited

DD223 Engine Discrete Warning Status

xxxx xxxx xxxx xxx1 = Warning Level 1,
xxxx xxxx xxxx xx1x = Warning Level 2,
xxxx xxxx xxxx x1xx = Power Reduction,
xxxx xxxx xxxx 1xxx = Maintenance Needed,
xxxx xxxx xxx1 xxxx = Engine Comm Error,
xxxx xxxx xx1x xxxx = Sub or Secondary Throttle,
xxxx xxxx x1xx xxxx = Neutral Start Protect,
xxxx xxxx 1xxx xxxx = Engine Shutting Down,
xxxx xxx1 xxxx xxxx = reserved,
xxxx xx1x xxxx xxxx = reserved,
xxxx x1xx xxxx xxxx = reserved,
xxxx 1xxx xxxx xxxx = reserved,
xxx1 xxxx xxxx xxxx = reserved,
xx1x xxxx xxxx xxxx = reserved,
x1xx xxxx xxxx xxxx = reserved,
1xxx xxxx xxxx xxxx = reserved,
where x = don't care

Inactive states shall be transmitted as 0. Reserved and unsupported bits within the field shall be transmitted as 0.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

13 **Percent Engine Load**

Byte Field Size: 1
Bit Field Size:

Request Parameter Optional
Command Parameter: Prohibited

DD138 Generic percent of range

DF30 Percent, Relative measure **int8** *Range:* +/- 124% *Resolution:* 1%
Range 0 - 124%

---

14 **Percent Engine Torque**

Byte Field Size: 1
Bit Field Size:

Request Parameter Optional
Command Parameter: Prohibited

DD138 Generic percent of range

DF30 Percent, Relative measure **int8** *Range:* +/- 124% *Resolution:* 1%
Range 0 - 124%

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Transmission Parameters, Dynamic

PGN: 127493
hex: 1F205

Used to provide the operational state and internal operating parameters of a specific transmission, indicated by the transmission instance field. This message would normally be broadcasted periodically to provide information for instrumentation or control functions.

Single Frame: Yes | *Priority Default:* 2 | *Default Update Rate:* 100 milliseconds | *Frequency:* 10. cycles per second
Destination: Global | *Query Support:* Required | *Command Support:* Required | *ACK Rqmnts:* None

Original Reference ID # 43

Field #	Field Name					
1	**Transmission instance**	Byte Field Size: / Bit Field Size: 8		Request Parameter: Required / Command Parameter: Required		
	DD410 Engine / Transmission Instance	This data instance field shall be enumerated on a globally unique basis beginning on the Port side with instance number 0 and incrementing towards Starboard. Devices in Propulsion Class 50 and Function Code 140 shall be the first enumeration sequence (i.e. 0, 1, 2, 3, ...n) prior to moving to a different Class and Function Code. Devices of the next Class and Function Code shall continue the enumeration sequence (i.e. n+1, n+2,...).				
	DF52 Bit field	**bit(n)**	Range: Variable	Resolution: 1	Used to construct bit fields	
	If this field is not specified in the "Request Group Function" or an ISO Request is made of this PGN, the response will be with all defined Transmission Instances. (This PGN will be transmitted for each instance.) If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.					
2	**Transmission Gear**	Byte Field Size: / Bit Field Size: 2		Request Parameter: Optional / Command Parameter: Optional		
	DD222 Transmission Gear	0 = Forward. 1 = Neutral, 2 = Reverse, 3 = [Unavailable, Unknown]				
	DF52 Bit field	**bit(n)**	Range: Variable	Resolution: 1	Used to construct bit fields	
3	**NMEA Reserved**	Byte Field Size: / Bit Field Size: resv 6		Request Parameter: / Command Parameter:		
	DD001 Reserved field	Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	Range: Variable	Resolution: 1	Used to construct bit fields	
	Used to align subsequent data on a byte boundary.					
4	**Transmission oil pressure**	Byte Field Size: 2 / Bit Field Size:		Request Parameter: Optional / Command Parameter: Prohibited		
	DD049 Generic Pressure					
	DF47 Pressure, medium	**uint16**	Range: 0 to 6,553,200 Pa	Resolution: 1x10E+2 Pa		
5	**Transmission oil temperature**	Byte Field Size: 2 / Bit Field Size:		Request Parameter: Optional / Command Parameter: Prohibited		
	DD130 Temperature, high					
	DF38 Temperature, high	**uint16**	Range: 0 to 6,553.2 deg K	Resolution: 1x10E-1 deg K		

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6	**Transmission Discrete Status**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	8	*Command Parameter:*	Prohibited

DD221 Transmission Discrete Warning Status

xxxx xxx1 = Check Tranmission,
xxxx xx1x = Over Temperature,
xxxx x1xx = Low Oil Pressure,
xxxx 1xxx = Low Oil Level,
xxx1 xxxx = Sail Drive,
xx1x xxxx = reserved,
x1xx xxxx = reserved,
1xxx xxxx = reserved,
where x = don't care

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

7	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 8	*Command Parameter:*	

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

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Trip Fuel Consumption, Vessel

PGN: 127496
hex: 1F208

Trip fuel parameters relative to Vessel

Single Frame: No *Priority Default:* 5 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 95

Field #	Field Name					
1	**Time to Empty**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD134 Run time, Trip					
	DF65 Time interval, standard	**uint32**	*Range:* 0 to ~4.295x10E+6 s	*Resolution:* 1x10E-3 s		
2	**Distance to Empty /Fuel Range**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD199 Distance, Unsigned					
	DF09 Distance	**uint32**	*Range:* 0 to ~4.295x10E+7 m	*Resolution:* 1x10E-2 m		
3	**Estimated Fuel Remaining**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD135 Volume					
	DF44 Volume	**uint16**	*Range:* 0 to 65.532 cu m	*Resolution:* 1x10E-3 cu m		
4	**Trip Run Time**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD134 Run time, Trip					
	DF65 Time interval, standard	**uint32**	*Range:* 0 to ~4.295x10E+6 s	*Resolution:* 1x10E-3 s		

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Trip Fuel Consumption, Engine

PGN: 127497
hex: 1F209

Engine related trip / fuel information.

This PGN will be requested as needed.

Single Frame: No *Priority Default:* 5 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Required *Command Support:* Required *ACK Rqmnts:* None

Original Reference ID # 48

Field #	Field Name			Byte Field Size / Bit Field Size	Request Parameter / Command Parameter	Range / Resolution
1	**Engine instance**			Bit Field Size: 8	Request Parameter: Required Command Parameter: Required	
	DD410 Engine / Transmission Instance				This data instance field shall be enumerated on a globally unique basis beginning on the Port side with instance number 0 and incrementing towards Starboard. Devices in Propulsion Class 50 and Function Code 140 shall be the first enumeration sequence (i.e. 0, 1, 2, 3, ...n) prior to moving to a different Class and Function Code. Devices of the next Class and Function Code shall continue the enumeration sequence (i.e. n+1, n+2,...).	
	DF52 Bit field	**bit(n)**				Range: Variable; Resolution: 1; Used to construct bit fields
	If this field is not specified in the "Request Group Function" or an ISO Request is made of this PGN, the response will be with all defined Engine Instances. (This PGN will be transmitted for each instance.) If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.					
2	**Trip fuel used**			Byte Field Size: 2 Bit Field Size:	Request Parameter: Optional Command Parameter: Optional	
	DD135 Volume					
	DF44 Volume	**uint16**				Range: 0 to 65.532 cu m; Resolution: 1x10E-3 cu m
3	**Fuel Rate, Average**			Byte Field Size: 2 Bit Field Size:	Request Parameter: Optional Command Parameter: Optional	
	DD131 Flow rate, low					
	DF18 Flow rate, low	**int16**				Range: +/-3.2764 cu-m/hr; Resolution: 1x10E-4 cu-m/hr
4	**Fuel Rate, Economy**			Byte Field Size: 2 Bit Field Size:	Request Parameter: Optional Command Parameter: Optional	
	DD131 Flow rate, low					
	DF18 Flow rate, low	**int16**				Range: +/-3.2764 cu-m/hr; Resolution: 1x10E-4 cu-m/hr
5	**Instantaneous Fuel Economy**			Byte Field Size: 2 Bit Field Size:	Request Parameter: Optional Command Parameter: Optional	
	DD131 Flow rate, low					
	DF18 Flow rate, low	**int16**				Range: +/-3.2764 cu-m/hr; Resolution: 1x10E-4 cu-m/hr

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Engine Parameters, Static

PGN: 127498
hex: 1F20A

Provides identification information and rated engine speed for the engine indicated by the engine instance field. Used primarily by display devices.

This PGN will be requested as needed.

Single Frame: No | Priority Default: 5 | Default Update Rate: | milliseconds | Frequency: NA cycles per second
Destination: Global | Query Support: Required | Command Support: Required | ACK Rqmnts: None

Original Reference ID # 49

Field #	Field Name						
1	**Engine instance**		Byte Field Size:		Request Parameter	Required	
			Bit Field Size:	8	Command Parameter:	Required	
	DD410 Engine / Transmission Instance						This data instance field shall be enumerated on a globally unique basis beginning on the Port side with instance number 0 and incrementing towards Starboard. Devices in Propulsion Class 50 and Function Code 140 shall be the first enumeration sequence (i.e. 0, 1, 2, 3, ...n) prior to moving to a different Class and Function Code. Devices of the next Class and Function Code shall continue the enumeration sequence (i.e. n+1, n+2,...).
	DF52 Bit field	**bit(n)**	Range:	Variable	Resolution:	1	Used to construct bit fields
	If this field is not specified in the "Request Group Function" or an ISO Request is made of this PGN, the response will be with all defined Engine Instances. (This PGN will be transmitted for each instance.) If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.						
2	**Rated engine speed**		Byte Field Size:	2	Request Parameter	Optional	
			Bit Field Size:		Command Parameter:	Optional	
	DD129 Rate of rotation						
	DF72 Rotational rate, unsigned	**uint16**	Range:	0-16,383 RPM	Resolution:	1/4 RPM	
3	**VIN**		Byte Field Size:	8 or 16 n	Request Parameter	Optional	
			Bit Field Size:		Command Parameter:	Prohibited	
	DD004 Generic name string, short						Name of place, route, waypoint, destination, vessel, vehicle, etc.
	DF50 String, variable, short	**ch8or16(n)**	Range:	0 to 250 ASCII or 0 to 125 Unicode Characters	Resolution:	1 ASCII or 1 Unicode Character	2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.
	200 characters maximum						

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4	**Software ID**		*Byte Field Size:*	8 or 16	n	*Request Parameter*	Optional
			Bit Field Size:			*Command Parameter:*	Optional
	DD004 Generic name string, short					Name of place, route, waypoint, destination, vessel, vehicle, etc.	
	DF50 String, variable, short	**ch8or16(n)**	*Range:*	0 to 250 ASCII or 0 to 125 Unicode Characters	*Resolution:*	1 ASCII or 1 Unicode Character	2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

200 characters maximum

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Load Controller Connection State / Control

PGN: 127500
hex: 1F20C

Broadcast the state and status of a Load Controller output/connection & control of the output/connection with PGN 126208 Command Group Function.

This PGN has two purposes. First, it provides a mechanism to broadcast the current state and status of a Load Controller output electrical connection/pin/channel. The values provided include the connection state, status, pulse width modulation (PWM) duty cycle, and timing parameters to control one-shot pulses. Depending on the state, the PWM and timing status parameters may not be defined and are set to 0xFF.

Second, the PGN provides the mechanism to command the state and status of a Load Controller output electrical connection/pin/channel. Commands are transmitted to a specific addressed Load Controller using the NMEA Command Group Function (PGN 126208, Function 0x01). When used in this manner, values must be provided for all fields defined in this parameter group to ensure unambiguous understanding.

Global ISO requests (PGN 059904) and Global NMEA Requests (PGN 126208) for this Parameter Group shall be ignored.

Response to an addressed ISO request for this Parameter Group shall be the transmission of this PGN once for each defined connection/pin/channel.

Response to an addressed NMEA Request for this Parameter Group without specifying a value for PGN 127500, Field #2 (Connection ID) shall be the transmission of this PGN once for each defined connection/pin/channel.

Response to an addressed NMEA Request for this Parameter Group with a specified value for PGN 127500, Field #2 (Connection ID) shall be the transmission of this PGN for the specified connection/pin/channel.

The recommended/default frequency is every 15,000ms.

The Sequence ID field (SID) is used to link this PGN to other related PGN's from the same source address. When no linkage exists, the value of the SID shall be set to 255.

Application specific examples can be found in Appendix D.

This PGN is part of the Distributed Power System / Architecture.

Single Frame: Y *Priority Default:* 3 *Default Update Rate:* 1500 milliseconds *Frequency:* .7 cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 212

1 Sequence ID
Byte Field Size: 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD056 Sequence ID
An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number
The Sequence ID field (SID) is used to link this PGN to other related PGN's from the same source address. When no linkage exists, the value of the SID shall be set to 255.

2 Connection ID
Byte Field Size: 1 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Optional

DD005 Generic numeric ID, short
Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number
This field identifies the connection/pin/channel. The combination of the Connection ID and Source Address of the device sending the message is globally unique.

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Load Controller Connection State / Control

PGN: 127500
hex: 1F20C

3 **State**

Byte Field Size:
Bit Field Size: 8

Request Parameter Optional
Command Parameter: Optional

DD351 Connection State

Commandable States- Codes that may be reported as states and commanded using PGN 126208
0 = OFF Connection is OFF - Hi Impedance
1 = LOW Connected to Ground
2 = HIGH Connected to Supply
3 = TRIPPED Max. Device capabilities Exceeded - Connection is OFF
4 = PWM Connection is Pulse Width Modulated - Dimmer
5 = TIMED_OFF Connection is OFF for TimeOFF, then HIGH
6 = TIMED_ON Connection is HIGH for TimeON, then OFF
7 = CYCLE_OFF Connection is OFF for TimeOFF, then CYCLE_ON
8 = CYCLE_ON Connection is HIGH for TimeON, then CYCLE_OFF
9-127 = Reserved

State Only- Codes that can only be reported as states
128 = FAULT Internal Failure - Connection is OFF
129-191 = Reserved

Command Only- Codes that may be commanded using PGN 126208, but are never reported as states
192 = RESET Reset TRIPPED to OFF
193 = DESTROY Command to Destroy Fuse - Disable Output - set FAULT state
194-253 = Reserved
254 = Error
255 = Data not Available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The physical State of the Connection.

---

4 **Status**

Byte Field Size:
Bit Field Size: 16

Request Parameter Optional
Command Parameter: Optional

DD352 Connection Status

x xxxx xxxx xxx1 = SENSED_HIGH-HIGH voltage level detected
x xxxx xxxx xx1x = SENSED_LOW- voltage level detected
x xxxx xxxx x1xx = SHORT- Current draw Exceeds device capability
x xxxx xxxx 1xxx = NO_LOAD- When HIGH, No Load detected - Amps=0
x xxxx xxx1 xxxx = FUSE_FAIL- Blown or removed
x xxxx xx1x xxxx = FUSE_DESTRUCT - Fuse was Commanded to destroy
x xxxx x1xx xxxx = GFCI_FAULT- Ground fault leakage detected
x xxxx 1xxx xxxx = OVER_VOLTAGE- Voltage Greater than Over Voltage Value
x xxx1 xxxx xxxx = OVER_CURRENT- Current Greater than Over Current Value
x xx1x xxxx xxxx =UNDER_VOLTAGE- Voltage Less than Under Voltage Value
x x1xx xxxx xxxx = UNDER_CURRENT- Current Less than Under Current Value
x 0xxx xxxx xxxx = Reserved
0 xxxx xxxx xxxx = Reserved
where x = don't care

Inactive states shall be transmitted as 0. Reserved and unsupported bits within the field shall be transmitted as 0.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The Status of the Connection.

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Load Controller Connection State / Control

PGN: 127500
hex: 1F20C

5	**Operational Status & Control**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	3	*Command Parameter:*	Optional

DD362 Operational Status & Control

xx1 = Locked- Disable/Enable ability to Change State
x1x = Security-Disable/Enable ability to Lock/Unlock
0xx = Reserved
where x = don't care

Inactive states shall be transmitted as 0.
Reserved and unsupported bits within the field shall be transmitted as 0.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

6	**PWM Duty Cycle**	*Byte Field Size:*	1	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD263 Generic Absolute Percentage 0-252%

DF93 Percent, Absolute **uint8** *Range:* 0 - 252% *Resolution:* 1%

The Duty Cycle Percentage value of Pulse Width Modulation (Only valid from 0% to 100%)

7	**TimeON**	*Byte Field Size:*	1	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD361 Event Time- Short — Interval of time remaining before event change

DF111 Time Interval Medium- S **uint8** *Range:* 0.00 - 2.52 seconds *Resolution:* 0.01 seconds

Time remaining before the connection is changed to the OFF or CYCLE_OFF State

8	**TimeOFF**	*Byte Field Size:*	1	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD361 Event Time- Short — Interval of time remaining before event change

DF111 Time Interval Medium- S **uint8** *Range:* 0.00 - 2.52 seconds *Resolution:* 0.01 seconds

Time remaining before the connection is changed to the HIGH or CYCLE_ON State

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Binary Status Report

PGN: 127501
hex: 1F20D

Universal status report for multiple banks of two-state indicators.

Single Frame: Yes *Priority Default:* 3 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Prohibited *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 44

1	**Binary Device Bank Instance**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Required Prohibited
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
	If this field is not specified in the "Command Request" or an ISO Request is made of this PGN, the response will be with all defined Binary Device Bank Instances. (This PGN will be transmitted for each instance.)					
2	**Status 1**		*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter* *Command Parameter:*	Optional Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
3	**Status 2**		*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter* *Command Parameter:*	Optional Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
4	**Status 3**		*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter* *Command Parameter:*	Optional Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
5	**Status 4**		*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter* *Command Parameter:*	Optional Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	

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6	**Status 5**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
7	**Status 6**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
8	**Status 7**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
9	**Status 8**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
10	**Status 9**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
11	**Status 10**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields

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12	**Status 11**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
13	**Status 12**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
14	**Status 13**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
15	**Status 14**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
16	**Status 15**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
17	**Status 16**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields

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18	**Status 17**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
19	**Status 18**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
20	**Status 19**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
21	**Status 20**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
22	**Status 21**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
23	**Status 22**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Prohibited
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields

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24	**Status 23**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	2	*Command Parameter:*	Prohibited	
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
25	**Status 24**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	2	*Command Parameter:*	Prohibited	
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
26	**Status 25**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	2	*Command Parameter:*	Prohibited	
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
27	**Status 26**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	2	*Command Parameter:*	Prohibited	
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
28	**Status 27**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	2	*Command Parameter:*	Prohibited	
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
29	**Status 28**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	2	*Command Parameter:*	Prohibited	
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields

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Switch Bank Control

PGN: 127502
hex: 1F20E

Universal commands to multiple banks of two-state devices.

Single Frame: Yes *Priority Default:* 3 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Required *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 45

1 Switch Bank Instance *Byte Field Size:* 1 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Required

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

If this field is not specified in the "Request Group Function" or an ISO Request is made of this PGN, the response will be with all defined Switch Bank Instances. (This PGN will be transmitted for each instance.) If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.

2 Switch 1 *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: 2 *Command Parameter:* Required

DD003 Generic command pair

00 = [Turn Off, Disable, Reset, Make "0"],
01 = [Turn On, Enable, Set, Make "1"],
02 = Reserved,
03 = No action

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 Switch 2 *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: 2 *Command Parameter:* Required

DD003 Generic command pair

00 = [Turn Off, Disable, Reset, Make "0"],
01 = [Turn On, Enable, Set, Make "1"],
02 = Reserved,
03 = No action

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

4 Switch 3 *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: 2 *Command Parameter:* Required

DD003 Generic command pair

00 = [Turn Off, Disable, Reset, Make "0"],
01 = [Turn On, Enable, Set, Make "1"],
02 = Reserved,
03 = No action

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

5 Switch 4 *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: 2 *Command Parameter:* Required

DD003 Generic command pair

00 = [Turn Off, Disable, Reset, Make "0"],
01 = [Turn On, Enable, Set, Make "1"],
02 = Reserved,
03 = No action

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

6 Switch 5 *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: 2 *Command Parameter:* Required

DD003 Generic command pair

00 = [Turn Off, Disable, Reset, Make "0"],
01 = [Turn On, Enable, Set, Make "1"],
02 = Reserved,
03 = No action

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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7	**Switch 6**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
8	**Switch 7**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
9	**Switch 8**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
10	**Switch 9**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
11	**Switch 10**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
12	**Switch 11**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields

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13	**Switch 12**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
14	**Switch 13**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
15	**Switch 14**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
16	**Switch 15**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
17	**Switch 16**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
18	**Switch 17**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields

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19	**Switch 18**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
20	**Switch 19**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
21	**Switch 20**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
22	**Switch 21**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
23	**Switch 22**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
24	**Switch 23**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields

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25	**Switch 24**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
26	**Switch 25**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
27	**Switch 26**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
28	**Switch 27**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
29	**Switch 28**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields

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AC Input Status -DEPRECATED

PGN: 127503
hex: 1F20F

This PGN has been deprecated (as of NMEA Network Message Database version 1.300) and is not recommended for new designs. However, support of the deprecated PGN 127503 is strongly recommended to ensure compatibility with legacy equipment.

Manufacturers shall use PGNs defined by J1939-75 for AC parameters related to Generators and Utility Connections. J1939-75 PGNs shall be implemented using the status bits as defined in J1939-71.

The PGN description at the time of deprecation was as follows:

Any device with an AC Input may transmit this message. Fields 3 through 12 may repeat as indicated by the Number of Lines.

If requested via the ISO Request, a separate message will be returned for each AC Instance connected to the device.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* 1500 milliseconds *Frequency:* .7 cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 128

Field #	*Field Name*					
1	**AC Instance**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter*	Required Optional	*Command Parameter:*
	DD005 Generic numeric ID, short	Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned **uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
	The subsequent parameters pertain to this AC source.					
2	**Number of Lines**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter*	Optional Optional	*Command Parameter:*
	DD006 Generic counter, short	Numeric count, event counter, sequence counter				
	DF53 Integer, 8 bit unsigned **uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
	This is the number of lines (tuples) being reported.					
3	**Line**	*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter*	Optional Optional	*Command Parameter:*
	DD270 AC Line	0x00 = Line 1, 0x01 = Line 2, 0x02 = Line 3 0x03 = Reserved				
	DF52 Bit field **bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
	This is the physical connector that is supplying power. In the case of split phase there are two lines.					
4	**Acceptability**	*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter*	Optional Optional	*Command Parameter:*
	DD259 AC Acceptability	0x00 = Bad Level, 0x01 = Bad Frequency, 0x02 = Being Qualified, 0x03 = Good				
	DF52 Bit field **bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
5	**NMEA Reserved**	*Byte Field Size:* *Bit Field Size:* resv 4		*Request Parameter*		*Command Parameter:*
	DD001 Reserved field	Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field **bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
	Used to align subsequent data on a byte boundary.					

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AC Input Status -DEPRECATED

PGN: 127503
hex: 1F20F

6	**Voltage**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD260 Voltage, AC RMS					
	DF96 Voltage, high, unsigned	**uint16**	*Range:* +/- 655.32 V	*Resolution:*	1x10E-2 V	
7	**Current**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD269 Current, Electric, Unsigned					
	DF95 Current, electric, high	**uint16**	*Range:* 0 - 6553.2 A	*Resolution:*	1x10E-1 A	
8	**Frequency**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD267 Frequency					
	DF22 Frequency, low	**uint16**	*Range:* 0 to 655.32 Hz	*Resolution:*	1x10E-2 Hz	
9	**Breaker Size**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD269 Current, Electric, Unsigned					
	DF95 Current, electric, high	**uint16**	*Range:* 0 - 6553.2 A	*Resolution:*	1x10E-1 A	
10	**Real Power**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD261 Power (watts)					
	DF94 Power	**uint32**	*Range:* 0 - 4,294,967,292 W	*Resolution:*	1 W	
11	**Reactive Power**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD262 Volt Amps Reactive Power (VAR)					
	DF92 Power - VAR	**uint32**	*Range:* 0 - 4,294,967,292 VAR	*Resolution:*	1 VAR	
12	**Power Factor**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD271 Power Factor		Cosine of the angle between the AC voltage and current			
	DF97 Power Factor	**int8**	*Range:* +/- 1.00	*Resolution:*	1x10E-2	

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AC Output Status -DEPRECATED

PGN: 127504
hex: 1F210

This PGN has been deprecated (as of NMEA Network Message Database version 1.300) and is not recommended for new designs. However, support of the deprecated PGN 127504 is strongly recommended to ensure compatibility with legacy equipment.

Manufacturers shall use PGNs defined by J1939-75 for AC parameters related to Generators and Utility Connections. J1939-75 PGNs shall be implemented using the status bits as defined in J1939-71.

The PGN description at the time of deprecation was as follows:

Any device with an AC Output may transmit this message. Fileds 3 though 12 may repeat as indicated by the Number of Lines.

If requested via the ISO Request, a separate message will be returned for each AC Instance connected to the device.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* 1500 milliseconds *Frequency:* .7 cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 129

Field #	*Field Name*					
1	**AC Instance**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter*	Required Optional	*Command Parameter:*
	DD005 Generic numeric ID, short	Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned **uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
	The subsequent parameters pertain to this AC source.					
2	**Number of lines**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter*	Optional Optional	*Command Parameter:*
	DD006 Generic counter, short	Numeric count, event counter, sequence counter				
	DF53 Integer, 8 bit unsigned **uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
	This is the number of lines (tuples) being reported.					
3	**Line**	*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter*	Optional Optional	*Command Parameter:*
	DD270 AC Line	0x00 = Line 1, 0x01 = Line 2, 0x02 = Line 3 0x03 = Reserved				
	DF52 Bit field **bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
	This is the physical connector that is supplying power. In the case of split phase there are two lines.					
4	**Waveform**	*Byte Field Size:* *Bit Field Size:* 3		*Request Parameter*	Optional Optional	*Command Parameter:*
	DD273 Waveform	0x00 = Sine Wave 0x01 = Modified Sine Wave 0x02 = Reserved thru 0x05 = Reserved 0x06 = Error 0x07 = Data Not Available				
	DF52 Bit field **bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		

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AC Output Status -DEPRECATED

PGN: 127504
hex: 1F210

5	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*		
			Bit Field Size:	resv 3	*Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.						
6	**Voltage**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD260 Voltage, AC RMS						
	DF96 Voltage, high, unsigned	**uint16**	*Range:*	+/- 655.32 V	*Resolution:*	1x10E-2 V	
7	**Current**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD269 Current, Electric, Unsigned						
	DF95 Current, electric, high	**uint16**	*Range:*	0 - 6553.2 A	*Resolution:*	1x10E-1 A	
8	**Frequency**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD267 Frequency						
	DF22 Frequency, low	**uint16**	*Range:*	0 to 655.32 Hz	*Resolution:*	1x10E-2 Hz	
9	**Breaker Size**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD269 Current, Electric, Unsigned						
	DF95 Current, electric, high	**uint16**	*Range:*	0 - 6553.2 A	*Resolution:*	1x10E-1 A	
10	**Real Power**		*Byte Field Size:*	4	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD261 Power (watts)						
	DF94 Power	**uint32**	*Range:*	0 - 4,294,967,292 W	*Resolution:*	1 W	
11	**Reactive Power**		*Byte Field Size:*	4	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD262 Volt Amps Reactive Power (VAR)						
	DF92 Power - VAR	**uint32**	*Range:*	0 - 4,294,967,292 VAR	*Resolution:*	1 VAR	
12	**Power Factor**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD271 Power Factor		Cosine of the angle between the AC voltage and current				
	DF97 Power Factor	**int8**	*Range:*	+/- 1.00	*Resolution:*	1x10E-2	

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Fluid Level

PGN: 127505
hex: 1F211

Fluid Level contains an instance number, type of fluid, level of fluid, and tank capacity. For example the fluid instance may be the level of fuel in a tank or the level of water in the bilge. Used primarily by display or instrumentation devices.

Single Frame: Yes *Priority Default:* 6 *Default Update Rate:* 2500 milliseconds *Frequency:* .4 cycles per second
Destination: Global *Query Support:* Required *Command Support:* Required *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 51

1 Fluid Instance — *Byte Field Size:* *Bit Field Size:* 4 — *Request Parameter* Required; *Command Parameter:* Required

DD169 Generic instance (4-bit) — 0x0 to 0xF = Instance number 0 to 15;

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

If this field is not specified in the "Request Group Function" or an ISO Request is made of this PGN, the response will be with all defined Fluid Instances. (This PGN will be transmitted for each instance.) If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.

2 Fluid Type — *Byte Field Size:* *Bit Field Size:* 4 — *Request Parameter* Optional; *Command Parameter:* Optional

DD208 Type of Fluid

0x00 = Fuel,
0x01 = Fresh Water,
0x02 = Waste Water,
0x03 = Live Well,
0x04 = Oil,
0x05 = Black Water (Sewage)
0x06 = Fuel (Gasoline)
0x07 = Reserved,
thru
0x0D = Reserved,
0x0E = Error,
0x0F = Data not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 Fluid Level — *Byte Field Size:* 2 *Bit Field Size:* — *Request Parameter* Optional; *Command Parameter:* Optional

DD215 Generic Percent of Range, Medium

DF84 Percent, Relative Measur **int16** *Range:* -131.072% to 131.056% *Resolution:* 4x10E-3 %

Range 0 - 100%, where 0% =Empty and 100% = Full

4 Tank Capacity — *Byte Field Size:* 4 *Bit Field Size:* — *Request Parameter* Optional; *Command Parameter:* Optional

DD227 Volume

DF86 Volume. Large **uint32** *Range:* 0 to~4.296x10E+5 cu m *Resolution:* 1x10E-4 cu m

5 NMEA Reserved — *Byte Field Size:* *Bit Field Size:* resv 8 — *Request Parameter* *Command Parameter:*

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

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DC Detailed Status

PGN: 127506
hex: 1F212

Provides parameteric data for a specific battery, indicated by the battery instance field. Used primarily by display or instrumentation devices, but may also be used by battery management controls.

This PGN is an extension, via the SID, to the DC Status PGN.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* 1500 milliseconds *Frequency:* .7 cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 130

1 Sequence ID
Byte Field Size: 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD056 Sequence ID
An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

2 DC Instance
Byte Field Size: 1 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Optional

DD005 Generic numeric ID, short
Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The subsequent parameters pertain to this DC source.
Note: This maps to the Battery Instance field.

3 DC Type
Byte Field Size: *Request Parameter* Optional
Bit Field Size: 8 *Command Parameter:* Optional

DD288 DC Type
0x00 = Battery,
0x01 = Alternator,
0x02 = Convertor,
0x03 = Solar Cell,
0x04 = Wind Generator,
0x05 = Reserved,
thru
0xFD = Reserved
0xFE = Error
0xFF = Data Not Available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

4 State of Charge
Byte Field Size: 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD263 Generic Absolute Percentage 0-252%

DF93 Percent, Absolute **uint8** *Range:* 0 - 252% *Resolution:* 1%

% of total charge remaining

5 State of Health
Byte Field Size: 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD263 Generic Absolute Percentage 0-252%

DF93 Percent, Absolute **uint8** *Range:* 0 - 252% *Resolution:* 1%

% of total life remaining

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6	**Time Remaining**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD268 Time					
	DF98 Time interval, medium, u	**uint16**	*Range:* 0 - 65,532 minutes	*Resolution:* 1 minute		
	Time remaing at current rate of discharge					
7	**Ripple Voltage**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD287 AC Ripple					
	DF102 AC Vrms	**uint16**	*Range:* 0 - 65.532	*Resolution:* 1x10E-3 V		
	Resolution is 1mV.					
8	**Amp Hours**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD283 Battery Capacity (Coulombs)					
	DF100 Battery Capacity	**uint16**	*Range:* 0 - 235,915,200 Coulombs	*Resolution:* 3600 C	This maps directly into Ampere Hours (AH) where 3600C = 1AH.	
	The Amp Hours is the cumulative amp hours drawn since a battery was full. When a battery is fully charged again the Amp Hours should be zero. The Amp Hours differ from the State Of Charge since it does not depend on the effective capacity of the battery (Which is influenced by the Peukert exponent and Battery Temperature. e.g.) The Amp Hours is defined as unsigned to keep the same datatype as the Battery Capacity [0 – 65000 Ah in steps of 1Ah]. The value is typically displayed as a (strictly) negative number to indicate that is the charge removed from the					

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Charger Status- DEPRECATED

PGN: 127507
hex: 1F213

This PGN has been deprecated (as of NMEA Network Message Database version 2.000) and is not recommended for new designs. However, support of the deprecated PGN 127507 is strongly recommended to ensure compatibility with legacy equipment.

PGN 127750 shall be used for new designs.

The PGN description at the time of deprecation was as follows:

Any device capable of charging a battery may transmit this message as a status report.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* 1500 milliseconds *Frequency:* .7 cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 131

1 **Charger Instance** *Byte Field Size:* 1 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Required

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

2 **Battery Instance** *Byte Field Size:* 1 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Required

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The subsequent parameters pertain to this DC source.

3 **Operating State** *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: 4 *Command Parameter:* Optional

DD264 Charger Operating State

0x00 = Not Charging,
0x01 = Bulk,
0x02 = Absorption,
0x03 = Overcharge,
0x04 = Equalize,
0x05 = Float,
0x06 = No Float
0x07 = Constant VI,
0x08 = Disabled,
0x09 = Fault,
thru
0x0D = Reserved
0x0E = Error
0x0F = Data Not Available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

4 **Charge Mode** *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: 4 *Command Parameter:* Optional

DD265 Charger Mode

0x00 = Standalone,
0x01 = Primary,
0x02 = Secondary,
0x03 = Echo
0x04 = Reserved
thru
0x0D = Reserved
0x0E = Error
0x0F = Data Not Available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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Charger Status- DEPRECATED

PGN: 127507
hex: 1F213

5	**Charger Enable/Disable**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
6	**Equalization Pending**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
7	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*	
			Bit Field Size:	resv 4	*Command Parameter:*	
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	Used to align subsequent data on byte boundary.					
8	**Equalization Time Remaining**		*Byte Field Size:*	2	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD268 Time					
	DF98 Time interval, medium, u	**uint16**	*Range:*	0 - 65,532 minutes	*Resolution:* 1 minute	

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Battery Status

PGN: 127508
hex: 1F214

Battery, Solar Cell, etc
Provides parametric data for a specific DC Source, indicated by the instance field. The type of DC Source can be identified from the DC Detailed Status PGN. Used primarily by display or instrumentation devices, but may also be used by power management controls.

Single Frame: Yes *Priority Default:* 6 *Default Update Rate:* 1500 milliseconds *Frequency:* .7 cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 50

1 Battery Instance
Byte Field Size: 1 *Bit Field Size:* *Request Parameter* Required *Command Parameter:* Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

If this field is not specified in the "Request Group Function" or an ISO Request is made of this PGN, the response will be with all defined Battery Instances. (This PGN will be transmitted for each instance.)

2 Battery Voltage
Byte Field Size: 2 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD136 Voltage, DC

DF42 Voltage, high **int16** *Range:* +/- 327.64 V *Resolution:* 1x10E-2 V

3 Battery Current
Byte Field Size: 2 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD140 Current, Electric — + = current consumed, - = current supplied

DF07 Current, electric, high **int16** *Range:* +/- 3,276.4 A *Resolution:* 1x10E-1 A

4 Battery Case Temperature
Byte Field Size: 2 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD043 Generic Temperature

DF39 Temperature, low **uint16** *Range:* 0 to 655.32 deg K *Resolution:* 1x10E-2 deg K

5 Sequence ID
Byte Field Size: 1 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD056 Sequence ID — An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

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Inverter Status- DEPRECATED

PGN: 127509
hex: 1F215

This PGN has been deprecated (as of version 2.000) and is not recommended for new designs. However, support of PGN 127509 may be necessary to ensure compatibility with legacy equipment.

PGN 127751 shall be used for new designs.

The PGN description at the time of deprecation was as follows:

Any device capable of inverting a DC source to an AC output may transmit this message.

If requested via the ISO Request, a separate message will be returned for each AC Instance connected to the device.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* 1500 milliseconds *Frequency:* .7 cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 132

Field #	Field Name						
1	**Inverter Instance**		*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* Required *Command Parameter:* Optional	
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
2	**AC Instance**		*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* Required *Command Parameter:* Optional	
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
	The subsequent parameters pertain to this AC source.						
3	**DC Instance**		*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
	The subsequent parameters pertain to this DC source. Note: This maps to the Battery Instance field.						
4	**Operating State**		*Byte Field Size:* *Bit Field Size:* 4			*Request Parameter* Optional *Command Parameter:* Optional	
	DD266 Invertor Operating State		0x00 = Invert, 0x01 = AC Passthru, 0x02 = Load Sense, 0x03 = Fault, 0x04 = Disabled 0x05 = Reserved thru 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		

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Inverter Status- DEPRECATED

PGN: 127509
hex: 1F215

5	**Inverter Enable/Disable**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
6	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*	
			Bit Field Size:	resv 2	*Command Parameter:*	
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.					

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Charger Configuration Status

PGN: 127510
hex: 1F216

Any device capable of charging a battery may transmit this message.

If requested via the ISO Request, a separate message will be returned for each Battery Instance connected to the device.

The Request and Command group function messages (PGN 126208) can be used to set parameters within this PGN.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Required *ACK Rqmnts:* None

Original Reference ID # 135

Field #	Field Name						
1	**Charger Instance**		*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* Required *Command Parameter:* Required	
	DD005 Generic numeric ID, short			Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
2	**Battery Instance**		*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* Required *Command Parameter:* Required	
	DD005 Generic numeric ID, short			Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
	The subsequent parameters pertain to this DC source.						
3	**Charger Enable/Disable**		*Byte Field Size:* *Bit Field Size:* 2			*Request Parameter* Optional *Command Parameter:* Optional	
	DD003 Generic command pair			00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
4	**NMEA Reserved**		*Byte Field Size:* *Bit Field Size:* resv 6			*Request Parameter* *Command Parameter:*	
	DD001 Reserved field			Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
	Used to align subsequent data on a byte boundary.						
5	**Charge Current Limit**		*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD263 Generic Absolute Percentage 0-252%						
	DF93 Percent, Absolute	**uint8**	*Range:* 0 - 252%	*Resolution:* 1%			
	Limits charger output current to a percentage (0-100%) of the designed maximum.						

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Charger Configuration Status

PGN: 127510
hex: 1F216

6 **Charging Algorithm**

Byte Field Size:
Bit Field Size: 4

Request Parameter Optional
Command Parameter: Optional

DD272 Charging Algorithm

0x00 = Trickle
0x01 = CVCC (Constant Voltage Constant Current)
0x02 = 2 Stage (No float)
0x03 = 3 Stage
0x04 = Reserved
thru
0x0D = Reserved
0x0E = Error
0x0F = Data Not Available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

7 **Charger Mode**

Byte Field Size:
Bit Field Size: 4

Request Parameter Optional
Command Parameter: Optional

DD265 Charger Mode

0x00 = Standalone,
0x01 = Primary,
0x02 = Secondary,
0x03 = Echo
0x04 = Reserved
thru
0x0D = Reserved
0x0E = Error
0x0F = Data Not Available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Default is standalone. For installations where two or more chargers are charging the same battery bank, one charger is the primary and the others are secondary. Some chargers include a second, lower power echo charger for maintaining a second battery bank (e.g., start battery).

---

8 **Estimated Battery Temp - When No Sensor Present**

Byte Field Size:
Bit Field Size: 4

Request Parameter Optional
Command Parameter: Optional

DD274 Batt Temp - No Sensor

0x00 = Cold
0x01 = Warm
0x02 = Hot
0x03 = Reserved
thru
0x0D = Reserved
0x0E = Error
0x0F = Data Not Available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

If there is no battery temperature sensor or it is defective then the charger will use this field to determine the battery temperature.

---

9 **Equalize One Time Enable/Disable**

Byte Field Size:
Bit Field Size: 2

Request Parameter Optional
Command Parameter: Optional

DD003 Generic command pair

00 = [Turn Off, Disable, Reset, Make "0"],
01 = [Turn On, Enable, Set, Make "1"],
02 = Reserved,
03 = No action

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Equalizing over charges the battery in an attempt to bring the battery's cells up to the same level of charge.

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Charger Configuration Status

PGN: 127510
hex: 1F216

10 **Over Charge Enable/Disable**

Byte Field Size:
Bit Field Size: 2

Request Parameter Optional
Command Parameter: Optional

DD003 Generic command pair

00 = [Turn Off, Disable, Reset, Make "0"],
01 = [Turn On, Enable, Set, Make "1"],
02 = Reserved,
03 = No action

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

For chargers that support regular over charging, this field enables the feature.

---

11 **Equalize Time**

Byte Field Size: 2
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD268 Time

DF98 Time interval, medium, u **uint16** *Range:* 0 - 65,532 minutes *Resolution:* 1 minute

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Inverter Configuration Status

PGN: 127511
hex: 1F217

Any device capable of inverting DC to AC may transmit this message.

If requested via the ISO Request, a separate message will be returned for each AC Instance connected to the device.

The Complex Request/Command/Acknowledgement group function message can be used to set the following parameters.

Note: While less than 8 bytes… we are anticipating that this message is expected to grow.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Required *ACK Rqmnts:* None

Original Reference ID # 136

Field #	*Field Name*							
1	**Inverter Instance**		*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* *Command Parameter:*	Required Optional	
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.					
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number			
2	**AC Instance**		*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* *Command Parameter:*	Required Optional	
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.					
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number			
	The subsequent parameters pertain to this AC source.							
3	**DC Instance**		*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* *Command Parameter:*	Optional Optional	
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.					
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number			
	The subsequent parameters pertain to this DC source. Note: This maps to the Battery Instance field.							
4	**Inverter Enable/Disable**		*Byte Field Size:* *Bit Field Size:* 2			*Request Parameter* *Command Parameter:*	Optional Optional	
	DD003 Generic command pair		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action					
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields			
5	**Inverter Mode**		*Byte Field Size:* *Bit Field Size:* 4			*Request Parameter* *Command Parameter:*	Optional Optional	
	DD275 Inverter Mode		0x00 = Standalone 0x01 = Series Master 0x02 = Series Slave 0x03 = Parallel Master 0x04 = Parallel Slave 0x05 = Reserved thru 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available					
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields			

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6	**Load Sense Enable/Disable**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional

DD003 Generic command pair

00 = [Turn Off, Disable, Reset, Make "0"],
01 = [Turn On, Enable, Set, Make "1"],
02 = Reserved,
03 = No action

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

7	**Load Sense Power Threshold**		*Byte Field Size:*	2	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional

DD276 Power, medium (watts)

DF28 Power **uint16** *Range:* 0 to 65,532 W *Resolution:* 1 W

When load sensing is enabled and the inverter is in the standby state, a load requiring at least this amount of power must be applied to enter the inverting state.

8	**Load Sense Interval**		*Byte Field Size:*	2	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional

DD036 Data transmit offset

Offset in transmit time from time of request command:
0x0 = transmit immediately
0xFFFF = Do not change offset.

DF66 Time interval, .01sec **uint16** *Range:* 0 to 655.32s *Resolution:* 1x10E-2sec

When load sensing is enabled and the inverter is in the standby state, the inverter will periodically check for a load on this interval.

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AGS Configuration Status

PGN: 127512
hex: 1F218

Automatic Generator Starter

Any device that is capable of starting/stopping a generator may transmit this message.

The Complex Request/Command/Acknowledgement group function message can be used to set the following parameters.

Note: While less than 8 bytes… we are anticipating that this message is expected to grow.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 137

Field #	*Field Name*						
1	**AGS Instance**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Required *Command Parameter:* Optional		
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
2	**Generator Instance**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Required *Command Parameter:* Optional		
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
3	**AGS Mode**		*Byte Field Size:* *Bit Field Size:* 4		*Request Parameter* Optional *Command Parameter:* Optional		
	DD277 AGS Mode		0x00 = Off 0x01 = On 0x02 = Automatic 0x03 = Reserved thru 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
4	**NMEA Reserved**		*Byte Field Size:* *Bit Field Size:* resv 44		*Request Parameter* Optional *Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
	Used to align subsequent data on a byte boundary.						

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Battery Configuration Status

PGN: 127513
hex: 1F219

Any device connected to a battery may transmit this message.

If requested via the ISO Request, a separate message will be returned for each Battery Instance connected to the device.

The Complex Request/Command/Acknowledgement group function message can be used to set the following parameters.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 138

Field #	Field Name						
1	**Battery Instance**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Required *Command Parameter:* Optional		
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
	The subsequent parameters pertain to this DC source.						
2	**Battery Type**		*Byte Field Size:* *Bit Field Size:* 4		*Request Parameter* Optional *Command Parameter:* Optional		
	DD282 Battery Type		0x00 = Flooded 0x01 = GEL 0x02 = AGM 0x03 = Reserved thru 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
3	**Supports Equalization**		*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter* Optional *Command Parameter:* Optional		
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
	Indicates if the battery supports equalization.						
4	**NMEA Reserved**		*Byte Field Size:* *Bit Field Size:* resv 2		*Request Parameter* *Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
	Used to align subsequent data on a byte boundary.						

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Battery Configuration Status

PGN: 127513
hex: 1F219

5	**Nominal Voltage**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	4	*Command Parameter:*	Optional	
	DD284 Nominal Voltage		0x00 = 6 Volts 0x01 = 12 Volts 0x02 = 24 Volts 0x03 = 32 Volts 0x04 = 36 Volts 0x05 = 42 Volts 0x06 = 48 Volts 0x07 = Reserved thru 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
6	**Battery Chemistry**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	4	*Command Parameter:*	Optional	
	DD285 Battery Chemistry		0x00 = Lead Acid 0x01 = LiIon 0x02 = NiCad 0x03 = ZnO 0x04 = NiMH 0x05 = Reserved thru 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
7	**Battery Capacity**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD283 Battery Capacity (Coulombs)						
	DF100 Battery Capacity	**uint16**	*Range:*	0 - 235,915,200 Coulombs	*Resolution:*	3600 C	This maps directly into Ampere Hours (AH) where 3600C = 1AH.
8	**Battery Temperature Coefficient**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD138 Generic percent of range						
	DF30 Percent, Relative measure	**int8**	*Range:*	+/- 124%	*Resolution:*	1%	
9	**Peukert Exponent**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD286 Peukert Exponent						
	DF101 Peukert Exponent	**uint8**	*Range:*	1 - 1.5	*Resolution:*	2x10E-3	
10	**Charge Efficiency Factor**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD138 Generic percent of range						
	DF30 Percent, Relative measure	**int8**	*Range:*	+/- 124%	*Resolution:*	1%	

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AGS Status

PGN: 127514
hex: 1F21A

Automatic Generator Starter

Any device capable of starting/stopping a generator may transmit this message.

Note: While less than 8 bytes… we are anticipating that this message is expected to grow.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* 1500 milliseconds *Frequency:* .7 cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field #	*Field Name*					Original Reference ID # 133
1	**AGS Instance**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Required Optional
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
2	**Generator Instance**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Required Optional
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
3	**AGS Operating State**		*Byte Field Size:* *Bit Field Size:* 4		*Request Parameter* *Command Parameter:*	Optional Optional
	DD278 AGS Operating State		0x00 = Quiet Time 0x01 = Auto On 0x02 = Auto Off 0x03 = Manual On 0x04 = Manual Off 0x05 = Generator Shutdown 0x06 = External Shutdown 0x07 = Fault 0x08 = Suspend 0x09 = Not Operating 0x0A = Reserved thru 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	

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4	**Generator State**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD279 Generator State		0x00 = Preheating 0x01 = Start Delay 0x02 = Cranking 0x03 = Starter Cooling 0x04 = Warming Up 0x05 = Cooling Down 0x06 = Spinning Down 0x07 = Shutdown Bypass 0x08 = Stopping 0x09 = Running 0x0A = Stopped 0x0B = Crank Delay 0x0C = Reserved 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
5	**Generator On Reason**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	8	*Command Parameter:*	Optional
	DD280 Generator On Reason		0x00 = Not On 0x01 = DC Voltage Low 0x02 = Battery State Of Charge Low 0x03 = AC Current High 0x04 = Contact Closed 0x05 = Manual On 0x06 = Exercise 0x07 = Non Quiet Time 0x08 = External On Via AGS 0x09 = External On Via Generator 0x0A = Unable to Stop 0x0B = Reserved thru 0xFD = Reserved 0xFE = Error 0xFF = Data Not Available			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields

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AGS Status

PGN: 127514
hex: 1F21A

6	**Generator Off Reason**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	8	*Command Parameter:*	Optional

DD281 Generator Off Reason

0x00 = Not Off
0x01 = DC Voltage High
0x02 = Battery State Of Charge High
0x03 = AC Current Low
0x04 = Contact Opened
0x05 = Reached Absorption
0x06 = Reached Float
0x07 = Manual Off
0x08 = Max Run Time
0x09 = Max Auto Cycle
0x0A = Exercise Done
0x0B = Quiet Time
0x0C = External Off Via AGS
0x0D = Safe Mode
0x0E = External Off Via Generator
0x0F = External Shutdown
0x10 = Auto Off
0x11 = Fault
0x12 = Unable to Start
0x13 = Reserved
thru
0xFD = Reserved
0xFE = Error
0xFF = Data Not Available

DF52	Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields

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AC Power / Current- Phase A

PGN: 127744
hex: 1F300

The purpose of this PGN is to provide a common way to report Phase A AC Current / Power status. It provides a mechanism to report on a specified connection address and instance with precision to 0.1 amperes and a range from 0 - 6553.2 Amperes. It can be requested by another device such as a logic controller (Command Source) or display and can be broadcast periodically or upon a status change. Using the SID, this status information can be combined with the current and voltage information of phase B and C to provide the total power of a three phase power connection (input or output).

This PGN is part of the Distributed Power System / Architecture.

Note 1: If Query Support for this PGN is implemented, then manufacturers shall include support for queries that specify a value for Field 2.

Note 2: If command support for this PGN is implemented, then each command shall include a value for Field 2, which shall identify the connection being commanded.

Single Frame: Y *Priority Default:* 6 *Default Update Rate:* 1500 milliseconds *Frequency:* .7 cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 213

Field #	Field Name						
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD056 Sequence ID			An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
	For example, an identical Sequence ID in this field indicates simultaneous measurements for phases A, B and C.						
2	**Connection Number**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Note 1 *Command Parameter:* Note 2		
	DD005 Generic numeric ID, short			Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
	Fixed instance within the source of this message, identifying the connection; the combination of the Connection Number and device sending the message is bus wide unique						
3	**AC RMS Current**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD269 Current, Electric, Unsigned						
	DF95 Current, electric, high	**uint16**	*Range:* 0 - 6553.2 A	*Resolution:* 1x10E-1 A			
4	**Power**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD349 Power- Extended Range			+2,147,483,645 = Reserved +2,147,483,646 = Out of Range +2,147,483,647 = Data not available or Do not change			
	DF108 Power- Extended Range	**int32**	*Range:* -2,147,483,648 to +2,147,483,644 Watts	*Resolution:* 1W			

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AC Power / Current- Phase B

PGN: 127745
hex: 1F301

The purpose of this PGN is to provide a common way to report Phase B AC Current / Power status. It provides a mechanism to report on a specified connection address and instance with precision to 0.1 amperes and a range from 0 - 6553.2 Amperes. It can be requested by another device such as a logic controller (Command Source) or display and can be broadcast periodically or upon a status change. Using the SID, this status information can be combined with the current and voltage information of phase A and C to provide the total power of a three phase power connection (input or output).

This PGN is part of the Distributed Power System / Architecture.

Note 1: If Query Support for this PGN is implemented, then manufacturers shall include support for queries that specify a value for Field 2.

Note 2: If command support for this PGN is implemented, then each command shall include a value for Field 2, which shall identify the connection being commanded.

Single Frame: Y *Priority Default:* 6 *Default Update Rate:* 1500 milliseconds *Frequency:* .7 cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 214

Field #	Field Name						
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD056 Sequence ID			An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
	For example, an identical Sequence ID in this field indicates simultaneous measurements for phases A, B and C.						
2	**Connection Number**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Note 1 *Command Parameter:* Note 2		
	DD005 Generic numeric ID, short			Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
	Fixed instance within the source of this message, identifying the connection; the combination of the Connection Number and device sending the message is bus wide unique						
3	**AC RMS Current**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD269 Current, Electric, Unsigned						
	DF95 Current, electric, high	**uint16**	*Range:* 0 - 6553.2 A	*Resolution:* 1x10E-1 A			
4	**Power**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD349 Power- Extended Range			+2,147,483,645 = Reserved +2,147,483,646 = Out of Range +2,147,483,647 = Data not available or Do not change			
	DF108 Power- Extended Range	**int32**	*Range:* -2,147,483,648 to +2,147,483,644 Watts	*Resolution:* 1W			

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AC Power / Current- Phase C

PGN: 127746
hex: 1F302

The purpose of this PGN is to provide a common way to report Phase C AC Current / Power status. It provides a mechanism to report on a specified connection address and instance with precision to 0.1 amperes and a range from 0 - 6553.2 Amperes. It can be requested by another device such as a logic controller (Command Source) or display and can be broadcast periodically or upon a status change. Using the SID, this status information can be combined with the current and voltage information of phase A and B to provide the total power of a three phase power connection (input or output).

This PGN is part of the Distributed Power System / Architecture.

Note 1: If Query Support for this PGN is implemented, then manufacturers shall include support for queries that specify a value for Field 2.

Note 2: If command support for this PGN is implemented, then each command shall include a value for Field 2, which shall identify the connection being commanded.

Single Frame: Y *Priority Default:* 6 *Default Update Rate:* 1500 milliseconds *Frequency:* .7 cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 215

Field #	Field Name						
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD056 Sequence ID			An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
	For example, an identical Sequence ID in this field indicates simultaneous measurements for phases A, B and C.						
2	**Connection Number**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Note 1 *Command Parameter:* Note 2		
	DD005 Generic numeric ID, short			Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
	Fixed instance within the source of this message, identifying the connection; the combination of the Connection Number and device sending the message is bus wide unique						
3	**AC RMS Current**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD269 Current, Electric, Unsigned						
	DF95 Current, electric, high	**uint16**	*Range:* 0 - 6553.2 A	*Resolution:* 1x10E-1 A			
4	**Power**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD349 Power- Extended Range			+2,147,483,645 = Reserved +2,147,483,646 = Out of Range +2,147,483,647 = Data not available or Do not change			
	DF108 Power- Extended Range	**int32**	*Range:* -2,147,483,648 to +2,147,483,644 Watts	*Resolution:* 1W			

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AC Voltage / Frequency-Phase A

PGN: 127747
hex: 1F303

The purpose of this PGN is to provide a common way to report Phase A AC Voltage/ Frequency status. It provides a mechanism to report on a specified connection address and instance with precision to 0.1 Volts and a range from 0.1 to 6553.2 Volts. It can be requested by another device such as a logic controller (Command Source) or display and can be broadcast periodically or upon a status change. Using the SID, this status information can be combined with the current and voltage information of phase B and C to provide the total power of a three phase power connection (input or output).

This PGN is part of the Distributed Power System / Architecture.

Note 1: If Query Support for this PGN is implemented, then manufacturers shall include support for queries that specify a value for Field 2.

Note 2: If command support for this PGN is implemented, then each command shall include a value for Field 2, which shall identify the connection being commanded.

Single Frame: Y *Priority Default:* 6 *Default Update Rate:* 1500 milliseconds *Frequency:* .7 cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 216

Field #	Field Name						
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD056 Sequence ID		An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
	For example, an identical Sequence ID in this field indicates simultaneous measurements for phases A, B and C.						
2	**Connection Number**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Note 1 *Command Parameter:* Note 2		
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
	Fixed instance within the source of this message, identifying the connection; the combination of the Connection Number and device sending the message is bus wide unique						
3	**AC RMS Voltage Line to Neutral**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD347 Voltage, AC, High Range		6553.3 = Reserved 6553.4 = Out of Range 6553.5 = Not Available or Do Not Change				
	DF106 Voltage AC- High Range	**uint16**	*Range:* 0 - 6553.2 V	*Resolution:* 0.1V			
4	**AC RMS Voltage Line to Line**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD347 Voltage, AC, High Range		6553.3 = Reserved 6553.4 = Out of Range 6553.5 = Not Available or Do Not Change				
	DF106 Voltage AC- High Range	**uint16**	*Range:* 0 - 6553.2 V	*Resolution:* 0.1V			

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AC Voltage / Frequency-Phase A

PGN: 127747
hex: 1F303

5	**AC Frequency**	*Byte Field Size:* 2	*Request Parameter*	Optional
		Bit Field Size:	*Command Parameter:*	Optional

DD348 Frequency-Medium

6553.3 = Reserved
6553.4 = Out of Range
6553.5 = Not Available or Do Not Change

DF107 Frequency-Medium **uint16** *Range:* 0-6553.2 Hz *Resolution:* 0.1 Hz

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AC Voltage / Frequency-Phase B

PGN: 127748
hex: 1F304

The purpose of this PGN is to provide a common way to report Phase B AC Voltage/ Frequency status. It provides a mechanism to report on a specified connection address and instance with precision to 0.1 Volts and a range from 0.1 to 6553.2 Volts. It can be requested by another device such as a logic controller (Command Source) or display and can be broadcast periodically or upon a status change. Using the SID, this status information can be combined with the current and voltage information of phase A and C to provide the total power of a three phase power connection (input or output).

This PGN is part of the Distributed Power System / Architecture.

Note 1: If Query Support for this PGN is implemented, then manufacturers shall include support for queries that specify a value for Field 2.

Note 2: If command support for this PGN is implemented, then each command shall include a value for Field 2, which shall identify the connection being commanded.

Single Frame: Y *Priority Default:* 6 *Default Update Rate:* 1500 milliseconds *Frequency:* .7 cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 217

Field #	Field Name					
1	**Sequence ID**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter*	Optional Optional	*Command Parameter:*
	DD056 Sequence ID	An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
	For example, an identical Sequence ID in this field indicates simultaneous measurements for phases A, B and C.					
2	**Connection Number**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter*	Note 1 Note 2	*Command Parameter:*
	DD005 Generic numeric ID, short	Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
	Fixed instance within the source of this message, identifying the connection; the combination of the Connection Number and device sending the message is bus wide unique					
3	**AC RMS Voltage Line to Neutral**	*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter*	Optional Optional	*Command Parameter:*
	DD347 Voltage, AC, High Range	6553.3 = Reserved 6553.4 = Out of Range 6553.5 = Not Available or Do Not Change				
	DF106 Voltage AC- High Range	**uint16**	*Range:* 0 - 6553.2 V	*Resolution:* 0.1V		
4	**AC RMS Voltage Line to Line**	*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter*	Optional Optional	*Command Parameter:*
	DD347 Voltage, AC, High Range	6553.3 = Reserved 6553.4 = Out of Range 6553.5 = Not Available or Do Not Change				
	DF106 Voltage AC- High Range	**uint16**	*Range:* 0 - 6553.2 V	*Resolution:* 0.1V		

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AC Voltage / Frequency-Phase B

PGN: 127748
hex: 1F304

5	**AC Frequency**	*Byte Field Size:* 2	*Request Parameter*	Optional
		Bit Field Size:	*Command Parameter:*	Optional

DD348 Frequency-Medium

6553.3 = Reserved
6553.4 = Out of Range
6553.5 = Not Available or Do Not Change

DF107 Frequency-Medium **uint16** *Range:* 0-6553.2 Hz *Resolution:* 0.1 Hz

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AC Voltage / Frequency-Phase C

PGN: 127749
hex: 1F305

The purpose of this PGN is to provide a common way to report Phase C AC Voltage/ Frequency status. It provides a mechanism to report on a specified connection address and instance with precision to 0.1 Volts and a range from 0.1 to 6553.2 Volts. It can be requested by another device such as a logic controller (Command Source) or display and can be broadcast periodically or upon a status change. Using the SID, this status information can be combined with the current and voltage information of phase A and B to provide the total power of a three phase power connection (input or output).

This PGN is part of the Distributed Power System / Architecture.

Note 1: If Query Support for this PGN is implemented, then manufacturers shall include support for queries that specify a value for Field 2.

Note 2: If command support for this PGN is implemented, then each command shall include a value for Field 2, which shall identify the connection being commanded.

Single Frame: Y *Priority Default:* 6 *Default Update Rate:* 1500 milliseconds *Frequency:* .7 cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 218

Field #	Field Name					
1	**Sequence ID**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter*	Optional Optional	*Command Parameter:*
	DD056 Sequence ID	An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
	For example, an identical Sequence ID in this field indicates simultaneous measurements for phases A, B and C.					
2	**Connection Number**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter*	Note 1 Note 2	*Command Parameter:*
	DD005 Generic numeric ID, short	Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
	Fixed instance within the source of this message, identifying the connection; the combination of the Connection Number and device sending the message is bus wide unique					
3	**AC RMS Voltage Line to Neutral**	*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter*	Optional Optional	*Command Parameter:*
	DD347 Voltage, AC, High Range	6553.3 = Reserved 6553.4 = Out of Range 6553.5 = Not Available or Do Not Change				
	DF106 Voltage AC- High Range	**uint16**	*Range:* 0 - 6553.2 V	*Resolution:* 0.1V		
4	**AC RMS Voltage Line to Line**	*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter*	Optional Optional	*Command Parameter:*
	DD347 Voltage, AC, High Range	6553.3 = Reserved 6553.4 = Out of Range 6553.5 = Not Available or Do Not Change				
	DF106 Voltage AC- High Range	**uint16**	*Range:* 0 - 6553.2 V	*Resolution:* 0.1V		

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AC Voltage / Frequency-Phase C

PGN: 127749
hex: 1F305

5	**AC Frequency**	*Byte Field Size:* 2	*Request Parameter*	Optional
		Bit Field Size:	*Command Parameter:*	Optional

DD348 Frequency-Medium

6553.3 = Reserved
6553.4 = Out of Range
6553.5 = Not Available or Do Not Change

DF107 Frequency-Medium **uint16** *Range:* 0-6553.2 Hz *Resolution:* 0.1 Hz

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Converter (Inverter/Charger) Status

PGN: 127750
hex: 1F306

Provides both state and status information about a Charger, Inverter or combined Inverter / Charger. It provides a mechanism to broadcast the current state of a power Converter with single or multiple inputs and outputs/connections/pins/channels. The values provided include the connection state and temperature.

This PGN also provides detailed status of the various characteristics of the connection(s) including: Low Power Mode, Fault, Bulk, Absorption, Float, Storage, Equalize, Pass-thru, Inverting, and Assisting

This PGN does not add 'On / Off' functionality for these devices/connection since they can be handled with other On/Off commands.

This PGN replaces PGN 127507 (Charger Status), which has been deprecated. This PGN can be used in conjunction with the DC Connection Status PGN to provide detailed information regarding the parameters of a Convertor device and all of its connections from/to the power grid. The use of the SID can assist in combining this PGN with similar PGN broadcasts to control or report power and other parameters for multiple connections of the Converter.

The recommended/default frequency is every 1500 ms.

This PGN is part of the Distributed Power System / Architecture.

Note 1: If Query Support for this PGN is implemented, then manufacturers shall include support for queries that specify a value for Field 2.

Note 2: If command support for this PGN is implemented, then each command shall include a value for Field 2, which shall identify the connection being commanded.

Single Frame: Y *Priority Default:* 6 *Default Update Rate:* 1500 milliseconds *Frequency:* .7 cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 219

Field #	*Field Name*					
1	**Sequence ID**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional	
	DD056 Sequence ID	An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
	Combine with other PGN's to provide detailed state and status of a Converter Device					
2	**Connection Number**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Note 1 Note 2	
	DD005 Generic numeric ID, short	Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
	Fixed instance within the source of this message, identifying the DC connection; the combination of the Connection Number and node sending is bus wide unique. Remark: This field references to a (DC) connection when the device is capable of independently charging several batteries. When a device only has a single state this field shall be set to 0xFF. The state then applies to all connections					

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Converter (Inverter/Charger) Status

PGN: 127750
hex: 1F306

3	**Operating State**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	8	*Command Parameter:*	Optional	
	DD342 Converter Operating State	0x0=Off, 0x1=Low Power Mode, 0x2=Fault, 0x3=Bulk, 0x4=Absorption, 0x5=Float, 0x6=Storage, 0x7=Equalize, 0x8=Pass thru, 0x9=Inverting, 0xA=Assisting, 0xB-0xFE=Reserved, 0xFF=Not Available				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
4	**Temperature State**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	2	*Command Parameter:*	Optional	
	DD343 Converter Temperature State	0x0=Ok, 0x1=Warning, 0x2=Over Temperature, 0x3=Not Available				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
5	**Overload State**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	2	*Command Parameter:*	Optional	
	DD344 Converter Overload State	0x0=Ok, 0x1=Warning, 0x2=Overload, 0x3=Not Available				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
6	**Low DC Voltage State**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	2	*Command Parameter:*	Optional	
	DD345 Converter Low DC Voltage State	0x0=Ok, 0x1=Warning, 0x2=DC voltage too low, 0x3=Not Available				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
7	**Ripple State**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	2	*Command Parameter:*	Optional	
	DD346 Converter Ripple State	0x0=Ok, 0x1=Ripple Warning, 0x2=Ripple Too High, 0x3=Not Available				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	

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Converter (Inverter/Charger) Status

PGN: 127750
hex: 1F306

8	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*	
			Bit Field Size:	resv 32	*Command Parameter:*	
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	Used to align subsequent data on byte boundary.					

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DC Voltage / Current

PGN: 127751
hex: 1F307

The purpose of this PGN is to provide a common way to report and extended range of DC Voltage and DC Current status including high voltage / high power systems.

The use of the SID can assist in combining the Voltage and Current of multiple connections with similar PGN broadcasts to control or report power and other parameters.

The message shall be broadcasted periodically or upon request. The recommended/default frequency is every 1500 ms. We do not recommend a period lower than 500 ms or beyond 5000 ms to maintain system integrity and optimal performance.

This PGN can be used in conjunction with 127506, 127508, and 127510 to provide precision voltage and current status. This PGN replaces 127509 (Inverter Status), which has been deprecated.

The connection identification is system wide unique and includes the device and instance number allowing multiple connections within a single device.

This PGN is part of the Distributed Power System / Architecture.

Note 1: If Query Support for this PGN is implemented, then manufacturers shall include support for queries that specify a value for Field 2.

Note 2: If command support for this PGN is implemented, then each command shall include a value for Field 2, which shall identify the connection being commanded.

Single Frame: Y *Priority Default:* 6 *Default Update Rate:* 1500 milliseconds *Frequency:* .7 cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 226

1 Sequence ID *Byte Field Size:* 1 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD056 Sequence ID

An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

To Report multiple connections at a single point in time

2 Connection Number *Byte Field Size:* 1 *Bit Field Size:* *Request Parameter* Note 1 *Command Parameter:* Note 2

DD005 Generic numeric ID, short

Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

Fixed instance within the source of this message, identifying the connection; the combination of the Connection Number and device sending the message is bus wide unique

3 DC Voltage *Byte Field Size:* 2 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD367 Voltage, DC, High Range/Resolution

6553.3 = Reserved
6553.4 = Out of Range
6553.5 = Not Available or Do Not Change

DF106 Voltage AC- High Range **uint16** *Range:* 0 - 6553.2 V *Resolution:* 0.1V

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DC Voltage / Current

PGN: 127751
hex: 1F307

4	**DC Current**	*Byte Field Size:* 3 *Bit Field Size:*		*Request Parameter*	Optional
				Command Parameter:	Optional

DD368 Current, DC, High Range/Resolution

83886.06 = Reserved
83886.07 = Out of Range
83886.08 = Not Available or Do Not Change

DF114 Current, DC, high range/r **int24** *Range:* -83886.08 A to 83886.05 A *Resolution:* 0.01A

5	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 16	*Command Parameter:*	

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

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Speed, Water Referenced

PGN: 128259
hex: 1F503

The purpose of this PGN is to provide a single transmission that describes the motion of a vessel over water.

As of version 1.210 of this standard, the name of this PGN was changed from "Speed", field 4 "Speed Water Reference Type" was added, and field 3 "Speed Ground Referenced" was noted not for new designs.

As of version 1.310 of this standard, field 5 "Speed Direction" was added to provide direction information. This reduced the reserved field (now field 6) from 16 bits to 12 bits.

Single Frame: Yes *Priority Default:* 2 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 32

Field #	Field Name					
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD056 Sequence ID		An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
2	**Speed Water Referenced**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD044 Generic Speed					
	DF35 Speed	**uint16**	*Range:* 0 to 655.32 m/s	*Resolution:* 1x10E-2 m/s	1 Knot = 0.5144 m/s	
3	**Speed Ground Referenced**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD044 Generic Speed					
	DF35 Speed	**uint16**	*Range:* 0 to 655.32 m/s	*Resolution:* 1x10E-2 m/s	1 Knot = 0.5144 m/s	
	Not for new designs. Value to be obtained from PGN 129026. Value to be set to Not Available.					
4	**Speed Water Referenced Type**		*Byte Field Size:* *Bit Field Size:* 8		*Request Parameter* *Command Parameter:*	Optional Optional
	DD293 Speed Water Reference Type		00 = Paddle Wheel 01 = Pitot Tube 02 = Doppler Log 03 = Correlation Log (Ultra-Sound) 04 = EM Log (Electro - Magnetic) 05 through 128 Reserved 129 through 252 Generic Speed Sources other than those defined 253 = Not Supported 254 = Error 255 = Do Not Change/Not Available			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
	Added as of version 1.210. Previously was reserved					

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5	**Speed Direction**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	4	*Command Parameter:*	Optional

DD356 Speed Direction

00 = Forward (Ahead)
01 = Astern (Backward)
02 through 14 Reserved
15 = Data Not Available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

When provided, this field identifies the direction of the "Speed Water Referenced" value in field 2.

6	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 12	*Command Parameter:*	

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

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Water Depth

PGN: 128267
hex: 1F50B

Water depth relative to the transducer and offset of the measuring transducer. Positive offset numbers provide the distance from the transducer to the waterline. Negative offset numbers provide the distance from the transducer to the part of the keel of interest.

Single Frame: Yes *Priority Default:* 3 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 60

Field #	Field Name					
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD056 Sequence ID		An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
2	**Water Depth, Transducer**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD162 Water Depth At Transducer		Depth relative to the transducer location. Range of value specified in "Maximum Depth Range" (field 4).			
	DF09 Distance	**uint32**	*Range:* 0 to ~4.295x10E+7 m	*Resolution:* 1x10E-2 m		
3	**Offset**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD161 Transducer Offset		Positive values represent distance from transducer to water line and negative values represent distance from the transducer to the keel.			
	DF46 Distance, signed, medium	**int16**	*Range:* +/- 32.764 m	*Resolution:* 1x10E-3 m		
4	**Maximum Depth Range**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD350 Maximum Depth Range		Device classification of the Maximum Range over which water depth can be measured. 253 = Deeper than 2,520 meters 254 = Error 255 = Data Not Available			
	DF109 Distance, Rough Approxi	**uint8**	*Range:* 0 - 2,520 meters	*Resolution:* 10 meters		

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Distance Log

PGN: 128275
hex: 1F513

This PGN provides two distance values recorded from one log which measures speed through water. The UTC time of the last distance increment is captured as Measurement Date & Time (which will be near current time). The distance values are stored during power down and resume counting after power up.

Total Cumulative Distance is normally set to zero when the log is installed and never reset after that.

Note 1: Distance Since Last Reset may manually be set to zero at any suitable time.
The "Distance Since Last Reset" is reset by setting its value to 0 with the "Command Group Function" PGN 126208.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 39

Field #	Field Name						
1	**Measurement Date**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD039 Generic date		Days since January 1, 1970, Date is relative to UTC Time.				
	DF41 Date, day count	**uint16**	*Range:* 0 to 65,532 days	*Resolution:* 1 day	0 = January 1, 1970, max = ~179 years		
2	**Measurement Time**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD158 Generic time of day		24 hour clock, 0 = midnight, time is in UTC				
	DF06 Time of day	**uint32**	*Range:* 0 to 86,401 s	*Resolution:* 1x10E-4 s	~24 hours, 0 = midnight, range allows for up to two leap seconds per day		
3	**Total Cumulative Distance**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD120 Distance, voyage						
	DF11 Distance, long	**uint32**	*Range:* 0 to ~4.295x10E+9 m	*Resolution:* 1 m			
4	**Distance Since Last Reset**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Note 1		
	DD120 Distance, voyage						
	DF11 Distance, long	**uint32**	*Range:* 0 to ~4.295x10E+9 m	*Resolution:* 1 m			

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Tracked Target Data

PGN: 128520
hex: 1F608

Message for reporting status and target data from tracking radar external devices. The reporting interval will vary by the values for target status and quantity of track data files.

Single Frame: No *Priority Default:* 2 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 87

1 Sequence ID
Byte Field Size: 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD056 Sequence ID
An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

2 Target ID #
Byte Field Size: 2 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Optional

DD007 Generic numeric ID, medium
Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

3 Track Status
Byte Field Size: *Request Parameter* Required
Bit Field Size: 4 *Command Parameter:* Optional

DD216 Track Status
xx00 = Cancelled or Not Available,
xx01 = Initial Acquisition Target,
xx10 = Tracking Target,
xx11 = Lost Target,
x0xx = Reported Target No,
x1xx = Reported Target Yes,
0xxx = Acquision Manual,
1xxx = Acquision Auto,
where x = don't care

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

4 Bearing Reference
Byte Field Size: *Request Parameter* Optional
Bit Field Size: 2 *Command Parameter:* Optional

DD218 Direction reference for target
0 = True,
1 = Magnetic,
2 = Error,
3 = Releative

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

5 NMEA Reserved
Byte Field Size: *Request Parameter*
Bit Field Size: resv 2 *Command Parameter:*

DD001 Reserved field
Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
Used to align subsequent data on a byte boundary.

6 Bearing
Byte Field Size: 2 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD127 Generic Direction -True
Degrees clockwise relative to True North.

DF02 Angle **uint16** *Range:* 0 to 2Pi rad *Resolution:* 1x10E-4 rad Resolution ~0.0057deg, 1 deg = .01745 rad

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Tracked Target Data

PGN: 128520
hex: 1F608

7	**Distance**		*Byte Field Size:*	4	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD115 Distance						
	DF15 Distance, signed	**int32**	*Range:*	+/-~2.147x10E+7 m	*Resolution:*	1x10E-2 m	
8	**Course**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD127 Generic Direction -True		Degrees clockwise relative to True North.				
	DF02 Angle	**uint16**	*Range:*	0 to 2Pi rad	*Resolution:*	1x10E-4 rad	Resolution ~0.0057deg, 1 deg = .01745 rad
9	**Speed**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD044 Generic Speed						
	DF35 Speed	**uint16**	*Range:*	0 to 655.32 m/s	*Resolution:*	1x10E-2 m/s	1 Knot = 0.5144 m/s
10	**CPA**		*Byte Field Size:*	4	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD115 Distance						
	DF15 Distance, signed	**int32**	*Range:*	+/-~2.147x10E+7 m	*Resolution:*	1x10E-2 m	
11	**TCPA**		*Byte Field Size:*	4	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD034 Time-elapsed/Time-to-go		Time interval in milli-sec. "-" = time elapsed since event, "+" = time to go before event.				
	DF40 Time interval, signed, sta	**int32**	*Range:*	+/- ~2.148x10E+6 s	*Resolution:*	1x10E-3 s	
12	**UTC of Fix**		*Byte Field Size:*	4	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD158 Generic time of day		24 hour clock, 0 = midnight, time is in UTC				
	DF06 Time of day	**uint32**	*Range:*	0 to 86,401 s	*Resolution:*	1x10E-4 s	~24 hours, 0 = midnight, range allows for up to two leap seconds per day

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Tracked Target Data

PGN: 128520
hex: 1F608

13	**Name**		*Byte Field Size:*	8 or 16 n	*Request Parameter*	Optional		
			Bit Field Size:		*Command Parameter:*	Optional		
	DD004 Generic name string, short		Name of place, route, waypoint, destination, vessel, vehicle, etc.					
	DF50 String, variable, short	**ch8or16(n)**	*Range:*	0 to 250 ASCII or 0 to 125 Unicode Characters	*Resolution:*	1 ASCII or 1 Unicode Character	2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.	
14	**Reference Target**		*Byte Field Size:*		*Request Parameter*	Optional		
			Bit Field Size:	2	*Command Parameter:*	Optional		
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]					
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields	

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Position, Rapid Update

PGN: 129025
hex: 1F801

This PGN provides latitude and longitude referenced to WGS84. Being defined as single frame message, as opposed to other PGNs that include latitude and longitude and are defined as fast or multi-packet, this PGN lends itself to being transmitted more frequently without using up excessive bandwidth on the bus for the benefit of receiving equipment that may require rapid position updates.

Single Frame:	Yes	*Priority Default:*	2	*Default Update Rate:*	100 milliseconds	*Frequency:*	10.	cycles per second	
Destination:	Global	*Query Support:*	Optional	*Command Support:*	Optional	*ACK Rqmnts:*	None		

Field # *Field Name* Original Reference ID # 18

1 Latitude

Byte Field Size: 4 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD022 Latitude, WGS-84

Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23 Latitude **int32** *Range:* +/- 90 deg *Resolution:* 1x10E-7 deg "-" = South, resolution ~1.1 cm

2 Longitude

Byte Field Size: 4 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD023 Longitude, WGS-84

Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF25 Longitude **int32** *Range:* +/- 180 deg *Resolution:* 1x10E-7 deg "-" = West, resolution ~1.1 cm

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COG & SOG, Rapid Update

PGN: 129026
hex: 1F802

This PGN is a single frame PGN that provides Course Over Ground (COG) and Speed Over Ground (SOG).

Being a single frame message, as opposed to other PGNs that include COG and SOG and are defined as multi-packet, this PGN lends itself to being transmitted more frequently, without using up excessive bandwidth on the bus. This may be of benefit to receiving equipment requiring rapid COG and SOG updates.

Single Frame: Yes *Priority Default:* 2 *Default Update Rate:* 250 milliseconds *Frequency:* 4. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 31

Field #	Field Name						
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD056 Sequence ID		An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
2	**COG Reference**		*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter* Optional *Command Parameter:* Optional		
	DD117 Direction reference		0 = True, 1 = Magnetic, 2 = Error, 3 = Null				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
3	**NMEA Reserved**		*Byte Field Size:* *Bit Field Size:* resv 6		*Request Parameter* *Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
	Used to align subsequent data on a byte boundary.						
4	**Course Over Ground**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD165 Course-Over-Ground (COG)		The direction of the path over ground actually followed by a vessel.				
	DF02 Angle	**uint16**	*Range:* 0 to 2Pi rad	*Resolution:* 1x10E-4 rad	Resolution ~0.0057deg, 1 deg = .01745 rad		
5	**Speed Over Ground**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD044 Generic Speed						
	DF35 Speed	**uint16**	*Range:* 0 to 655.32 m/s	*Resolution:* 1x10E-2 m/s	1 Knot = 0.5144 m/s		
6	**NMEA Reserved**		*Byte Field Size:* *Bit Field Size:* resv 16		*Request Parameter* *Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
	Used to align subsequent data on a byte boundary.						

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Position Delta, High Precision Rapid Update

PGN: 129027
hex: 1F803

The "Position Delta, High Precision Rapid Update" Parameter Group is intended for applications where very high precision and very fast update rates are needed for position data. This PGN can provide delta position changes down to 1 millimeter with a delta time period accurate to 5 milliseconds. One example application for this PGN is high precision positioning and guidance of automated machinery such as tractor implements utilized in the agriculture industry. Similar high precision positioning needs in the marine industry may be satisfied with this parameter Group. This PGN only has meaning when it is associated (via the Sequence ID field) with another PGN such as the GNSS Position Data Parameter Group. Association with the "Altitude Delta, High Precision Rapid Update" PGN is needed for a complete 3D position update.

Single Frame: Yes *Priority Default:* 2 *Default Update Rate:* 100 milliseconds *Frequency:* 10. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 93

Field #	Field Name					
1	**Sequence ID**	*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD056 Sequence ID	An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
2	**Time Delta**	*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD233 Time Value (Short resolution 5 msec)					
	DF88 Time Interval, short	**uint8**	*Range:* 0 to 1.26 sec	*Resolution:* 5x10E-3 sec		
3	**Latitude Delta**	*Byte Field Size:* 3 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD234 Latitude					
	DF89 Latitude, 24 bit	**int24**	*Range:* +/-83 sec (")	*Resolution:* 1x10E-5 sec(")		
4	**Longitude Delta**	*Byte Field Size:* 3 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD235 Longitude					
	DF90 Longitude, 24 bit	**int24**	*Range:* +/-83 sec (")	*Resolution:* 1x10E-5 sec(")		

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Altitude Delta, High Precision Rapid Update

PGN: 129028
hex: 1F804

The "Altitude Delta, High Precision Rapid Update" Parameter Group is intended for applications where very high precision and very fast update rates are needed for altitude and course over ground data. This PGN can provide delta altitude changes down to 1 millimeter, a change in direction as small as 0.0057 degrees, and with a delta time period accurate to 5 milliseconds. One example application for this PGN is high precision positioning and guidance of automated machinery such as tractor implements utilized in the agriculture industry. Similar high precision positioning needs in the marine industry may be satisfied with this parameter Group. This PGN only has meaning when it is associated (via the Sequence ID field) with another PGN such as the GNSS Position Data Parameter Group. Association with the "Position Delta, High Precision Rapid Update" PGN is needed for a complete 3D position update.

Single Frame: Yes | *Priority Default:* 2 | *Default Update Rate:* 100 milliseconds | *Frequency:* 10. cycles per second
Destination: Global | *Query Support:* Optional | *Command Support:* Optional | *ACK Rqmnts:* None

Original Reference ID # 94

Field #	*Field Name*					
1	**Sequence ID**	*Byte Field Size:* 1		*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	
	DD056 Sequence ID	An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
2	**Time Delta**	*Byte Field Size:* 1		*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	
	DD233 Time Value (Short resolution 5 msec)					
	DF88 Time Interval, short	**uint8**	*Range:* 0 to 1.26 sec	*Resolution:* 5x10E-3 sec		
3	**GNSS Quality**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size: 4		*Command Parameter:*	Optional	
	DD067 Quality, GNSS	0 = no GPS, 1 = GNSS fix, 2 = DGNSS fix, 3 = Precise GNSS*, 4 = RTK Fixed Integer, 5 = RTK Float, 6 = Estimated (DR) mode, 7 = Manual Input, 8 = Simulate mode, 9-13 = Reserved, 14 = Error, 15 = Null. *Precise GNSS means no deliberate degradation (such as SA) and higher resolution code (P-code), and 2 frequencies are used to correct atmospheric delays.				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
4	**Direction**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size: 2		*Command Parameter:*	Optional	
	DD117 Direction reference	0 = True, 1 = Magnetic, 2 = Error, 3 = Null				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	

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Altitude Delta, High Precision Rapid Update

PGN: 129028
hex: 1F804

5	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*		
			Bit Field Size:	resv 2	*Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.						
6	**Course Over Ground**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD165 Course-Over-Ground (COG)		The direction of the path over ground actually followed by a vessel.				
	DF02 Angle	**uint16**	*Range:*	0 to 2Pi rad	*Resolution:*	1x10E-4 rad	Resolution ~0.0057deg, 1 deg = .01745 rad
7	**Altitude Delta**		*Byte Field Size:*	3	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD236 Altitude						
	DF91 Altitude, 24 bit	**int24**	*Range:*	+/-8,388m	*Resolution:*	1x10E-3m	

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GNSS Position Data

PGN: 129029
hex: 1F805

This PGN conveys a comprehensive set of Global Navigation Satellite System (GNSS) parameters, including position information. Equipment transmitting this PGN would typically also transmit PGN 129025 (Position – Rapid Update).

The Sequence ID may be used to synchronize the data with data from other PGNs originating from the same source. For example, a GPS chart plotter with an integrated depth finder might output both PGN 129025 (Position, Rapid Update) and PGN 128267 (Water Depth) for each position.

Single Frame: No *Priority Default:* 3 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 19

Field #	Field Name						
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD056 Sequence ID			An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252		*Resolution:* 1 bit		Unit-less number
2	**Position date**		*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD039 Generic date			Days since January 1, 1970, Date is relative to UTC Time.			
	DF41 Date, day count	**uint16**	*Range:* 0 to 65,532 days		*Resolution:* 1 day		0 = January 1, 1970, max = ~179 years
3	**Position time**		*Byte Field Size:* 4 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD158 Generic time of day			24 hour clock, 0 = midnight, time is in UTC			
	DF06 Time of day	**uint32**	*Range:* 0 to 86,401 s		*Resolution:* 1x10E-4 s		~24 hours, 0 = midnight, range allows for up to two leap seconds per day
4	**Latitude**		*Byte Field Size:* 8 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD202 Latitude (Extended Resolution)			Latitude referenced to WGS-84			
	DF76 Latitude (Extended)	**int64**	*Range:* +/- 90 deg		*Resolution:* 1x10E-16 deg		"-" = South, resolution ~.01 nanometer
5	**Longitude**		*Byte Field Size:* 8 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD203 Longitude (Extended Resolution)			Longitude referenced to WGS-84			
	DF77 Longitude (Extended)	**int64**	*Range:* +/- 180 deg		*Resolution:* 1x10E-16 deg		"-" = West, resolution ~.01 nanometer
6	**Altitude**		*Byte Field Size:* 8 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD204 Altitude (Extended Resolution)			Altitude referenced to WGS-84			
	DF78 Distance (Extended)	**int64**	*Range:* +/-~9.223x10E+12 m		*Resolution:* 1x10E-6 m		

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7	**Type of System**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	4	*Command Parameter:*	Optional	
	DD207 Type of System		0x0 = GPS 0x1 = GLONASS 0x2 = GPS+GLONASS 0x3 = GPS+SBAS(WAAS) 0x4 = GPS+SBAS+GLONASS 0x5 - 0xF = Reserved for future combinations				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
8	**Method, GNSS**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	4	*Command Parameter:*	Optional	
	DD067 Quality, GNSS		0 = no GPS, 1 = GNSS fix, 2 = DGNSS fix, 3 = Precise GNSS*, 4 = RTK Fixed Integer, 5 = RTK Float, 6 = Estimated (DR) mode, 7 = Manual Input, 8 = Simulate mode, 9-13 = Reserved, 14 = Error, 15 = Null. *Precise GNSS means no deliberate degradation (such as SA) and higher resolution code (P-code), and 2 frequencies are used to correct atmospheric delays.				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
	Position Fixed Method						
9	**Integrity**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	2	*Command Parameter:*	Optional	
	DD209 GNSS Integrity		0 = No Integrity checking,* 1 = Safe, 2 = Caution, 3 = Unsafe * means the receiver does not have this capability				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
10	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*		
			Bit Field Size:	resv 6	*Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.						
11	**Number of SVs**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD006 Generic counter, short		Numeric count, event counter, sequence counter				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number

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12 **HDOP** *Byte Field Size:* 2 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD055 DOP — Dilution of Precision (DOP) indicates the contribution of satellite configuration geometry to positioning error. A lower DOP value is preferred because less error is being introduced. Reported as components: HDOP (Horizontal), VDOP (Vertical), TDOP (Time). Minimum DOP value is 1.0 (no error introduced).

DF69 Ratio, Relative measure, **int16** *Range:* +/-327.64 *Resolution:* 1x10E-2 Unit-less number

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13 **PDOP** *Byte Field Size:* 2 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD055 DOP — Dilution of Precision (DOP) indicates the contribution of satellite configuration geometry to positioning error. A lower DOP value is preferred because less error is being introduced. Reported as components: HDOP (Horizontal), VDOP (Vertical), TDOP (Time). Minimum DOP value is 1.0 (no error introduced).

DF69 Ratio, Relative measure, **int16** *Range:* +/-327.64 *Resolution:* 1x10E-2 Unit-less number

---

14 **Geoidal Separation** *Byte Field Size:* 4 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD069 Geoidal Separation — The difference between the earth ellipsoid and mean-sea-level (geoid) defined by the reference datum used in the position solution, "-" = mean-sea-level below ellipsoid. Reference datum is defined in this packet.

DF15 Distance, signed **int32** *Range:* +/-~2.147x10E+7 m *Resolution:* 1x10E-2 m

---

15 **Number of Reference Stations** *Byte Field Size:* 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD006 Generic counter, short — Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

---

16 **Reference Station Type"1"** *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: 4 *Command Parameter:* Optional

DD070 Ref Station Type — Reference Station Type.
0x0=GPS;
0x1=GLONASS;
0x2 to 0xD=Reserved;
0XE=Error;
0XF=Null

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

17 **Reference Station ID"1"** *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: 12 *Command Parameter:* Optional

DD071 Ref Station — Reference Station ID. Reference Station number as provided by the Service Provider.[Reference document required]

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

18 **Age of DGNSS Corrections "1"** *Byte Field Size:* 2 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD060 Differential Age — Age of Differential corrections

DF66 Time interval, .01sec **uint16** *Range:* 0 to 655.32s *Resolution:* 1x10E-2sec

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19	**Reference Station Type "n"**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	4	*Command Parameter:*	Optional

DD070 Ref Station Type

Reference Station Type.
0x0=GPS;
0x1=GLONASS;
0x2 to 0xD=Reserved;
0XE=Error;
0XF=Null

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Variable Number of fields, Field number 18 repeated

20	**Reference Station ID "n"**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	12	*Command Parameter:*	Optional

DD071 Ref Station

Reference Station ID. Reference Station number as provided by the Service Provider.[Reference document required]

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Variable Number of fields, Field number 19 repeated

21	**Age of DGNSS Reference Station "n"**	*Byte Field Size:*	2	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD060 Differential Age

Age of Differential corrections

DF66 Time interval, .01sec **uint16** *Range:* 0 to 655.32s *Resolution:* 1x10E-2sec

Variable Number of fields, Field number 20 repeated

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Local Time Offset

PGN: 129033
hex: 1F809

This PGN is used for the purpose of identifying the local time offset from UTC. It shall be transmitted on request or when the local offset is changed. The Time and Date Fields represent the current time.

Current time is transmitted periodically using System Time PGN (126992).

In Versions 1.301 and prior, the title of this PGN was "Time & Date". The default update rate was set at 1 second. The 1 second interval is not recommended for new designs. The description of this PGN was changed to clarify how and when it should be used.

Single Frame: Yes *Priority Default:* 3 *Default Update Rate:* n/a milliseconds *Frequency:* #Type! cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 40

Field #	Field Name						
1	**Date**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD039 Generic date		Days since January 1, 1970, Date is relative to UTC Time.				
	DF41 Date, day count	**uint16**	*Range:* 0 to 65,532 days	*Resolution:* 1 day	0 = January 1, 1970, max = ~179 years		
2	**Time**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD158 Generic time of day		24 hour clock, 0 = midnight, time is in UTC				
	DF06 Time of day	**uint32**	*Range:* 0 to 86,401 s	*Resolution:* 1x10E-4 s	~24 hours, 0 = midnight, range allows for up to two leap seconds per day		
3	**Local Offset, Minutes**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD121 Time, Local Offset		Local offset from UTC to obtain Local Time. This value includes Time Zone, daylight Savings Time, etc.				
	DF71 Time interval, medium	**int16**	*Range:* +/-32,764 minutes	*Resolution:* 1.0 minute			

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AIS Class A Position Report

PGN: 129038
hex: 1F80E

This parameter group provides data associated with the ITU-R M.1371 Messages 1, 2, and 3 Position Reports, autonomous, assigned, and response to interrogation, respectively. An AIS device may generate this parameter group either upon receiving a VHF data link message 1,2 or 3 (see ITU-R M.1371 for additional information), or upon a device's own transmissionof a Message 1, 2, or 3.

Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

Single Frame: No *Priority Default:* 4 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 109

1 Message ID
Byte Field Size: *Bit Field Size:* 6 *Request Parameter* Required *Command Parameter:* Optional

DD188 AIS Message Identifier
Message Identifier (range of 0 to 63).

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
1 = Autonomously Scheduled Position Report Message,
2 = Assigned Scheduled Position Report Message,
3 = Special (response to interrogation) Position Report Message.

2 Repeat Indicator
Byte Field Size: *Bit Field Size:* 2 *Request Parameter* Optional *Command Parameter:* Optional

DD185 AIS Repeater Indicator
Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3).

0 = Default
1 = First retransmission
2 = Second retransmission
3 = Final retransmission

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 User ID
Byte Field Size: 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD010 Generic numeric ID, large
Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number
MMSI number of mobile station reporting its position.

4 Longitude
Byte Field Size: 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD023 Longitude, WGS-84
Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF25 Longitude **int32** *Range:* +/- 180 deg *Resolution:* 1x10E-7 deg "-" = West, resolution ~1.1 cm
Longitude of mobile station reporting its position.

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AIS Class A Position Report

PGN: 129038
hex: 1F80E

5 Latitude

Byte Field Size: 4
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD022 Latitude, WGS-84

Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23 Latitude **int32** Range: +/- 90 deg Resolution: 1x10E-7 deg "-" = South, resolution ~1.1 cm

Latitude of mobile station reporting its position.

6 Position Accuracy

Byte Field Size:
Bit Field Size: 1
Request Parameter: Optional
Command Parameter: Optional

DD184 AIS Position Accuracy

0 = low accuracy > 10m such as nondifferential GNSS (default),
1 = high accuracy < 10m such as DGNSS

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

7 RAIM-flag

Byte Field Size:
Bit Field Size: 1
Request Parameter: Optional
Command Parameter: Optional

DD189 AIS RAIM-flag

0 = RAIM not in use (default),
1 = RAIM in use

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

8 Time Stamp

Byte Field Size:
Bit Field Size: 6
Request Parameter: Optional
Command Parameter: Optional

DD186 AIS Time Stamp

0-59 = UTC second when the report was generated,
60 = time stamp not available (default),
61 = positioning system is in manual input mode,
62 = Electronic position fixing system operates in estimated (dead reckoning) mode,
63 = positioning system is inoperative

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

9 COG

Byte Field Size: 2
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD165 Course-Over-Ground (COG)

The direction of the path over ground actually followed by a vessel.

DF02 Angle **uint16** Range: 0 to 2Pi rad Resolution: 1x10E-4 rad Resolution ~0.0057deg, 1 deg = .01745 rad

COG of mobile station reporting its position.

10 SOG

Byte Field Size: 2
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD044 Generic Speed

DF35 Speed **uint16** Range: 0 to 655.32 m/s Resolution: 1x10E-2 m/s 1 Knot = 0.5144 m/s

SOG of mobile station reporting its position.

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AIS Class A Position Report

PGN: 129038
hex: 1F80E

11 Communication State
Byte Field Size:
Bit Field Size: 19
Request Parameter: Optional
Command Parameter: Optional

DD187 AIS Communication State

The Communication State contains information used by the various TDMA slot allocation algorithms and synchronization information

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

12 AIS Transceiver Information
Byte Field Size:
Bit Field Size: 5
Request Parameter: Optional
Command Parameter: Optional

DD246 AIS Transceiver Information

0 = Channel A VDL reception,
1 = Channel B VDL reception,
2 = Channel A VDL transmission,
3 = Channel B VDL transmission,
4 = Own information not broadcast,
5-31 = Reserved.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

13 True Heading
Byte Field Size: 2
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD127 Generic Direction -True

Degrees clockwise relative to True North.

DF02 Angle **uint16** Range: 0 to 2Pi rad Resolution: 1x10E-4 rad Resolution ~0.0057deg, 1 deg = .01745 rad

True Heading of mobile station reporting its position.

14 Rate of Turn
Byte Field Size: 2
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD150 Rate of Turn

+ = Bow turning to starboard, 1 deg/min = .00029 rad/sec

DF73 Angular rate, signed **int16** Range: +/-1.0 rad/s Resolution: 1/32 x 10E-3 rad/s Resolution 0.1 deg/min

Rate of turn of mobile station reporting its position.

15 Navigational Status
Byte Field Size:
Bit Field Size: 4
Request Parameter: Optional
Command Parameter: Optional

DD183 AIS Navigational Status

0 = under way using engine,
1 = at anchor,
2 = not under command,
3 = restricted manoeuvrability,
4 = constrained by her draught,
5 = moored,
6 = aground,
7 = engaged in fishing,
8 = under way sailing,
9 = reserved for future amendment of navigational status for ships carrying DG, HS, or MP, or IMO hazard or pollutant category C (HSC),
10 = reserved for future amendment of navigational status for ships carrying DG, HS or MP, or IMO hazard or pollutant category A (WIG),
11-14 = reserved for future use,
15 = not defined (default)

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

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16 **Special Maneuver Indicator**

Byte Field Size:
Bit Field Size: 2

Request Parameter: Optional
Command Parameter: Optional

DD310 Special Maneuver Indicator

0 = not available (default)
1 = not engaged in special Maneuver
2 = engaged in special Maneuver (i.e.: regional passing arrangement on Inland Waterway)
3 = reserved

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

17 **NMEA Reserved**

Byte Field Size:
Bit Field Size: resv 2

Request Parameter:
Command Parameter:

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

---

18 **AIS Spare**

Byte Field Size:
Bit Field Size: resv 3

Request Parameter:
Command Parameter:

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** Range: N2KUnitless Resolution: 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field.

---

19 **NMEA Reserved**

Byte Field Size:
Bit Field Size: resv 5

Request Parameter:
Command Parameter:

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

---

20 **Sequence ID**

Byte Field Size: 1
Bit Field Size:

Request Parameter: Optional
Command Parameter: Optional

DD056 Sequence ID

An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** Range: 0 to 252 Resolution: 1 bit Unit-less number

---

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AIS Class B Position Report

PGN: 129039
hex: 1F80F

This parameter group provides data associated with the ITU-R M.1371 Message 18 Standard Class B Equipment Position Report. An AIS device may generate this parameter group either upon receiving a VHF data link message 18, or upon receipt of an ISO or NMEA request PGN (see ITU-R M.1371-1 for additional information). Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

Single Frame: No *Priority Default:* 4 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 122

1 **Message ID** *Byte Field Size:* *Bit Field Size:* 6 *Request Parameter* Optional *Command Parameter:* Optional

DD188 AIS Message Identifier

Message Identifier (range of 0 to 63).

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

18 = Standard Class B Equipment Position Report Message

2 **Repeat Indicator** *Byte Field Size:* *Bit Field Size:* 2 *Request Parameter* Optional *Command Parameter:* Optional

DD185 AIS Repeater Indicator

Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3).

0 = Default
1 = First retransmission
2 = Second retransmission
3 = Final retransmission

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 **User ID** *Byte Field Size:* 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD010 Generic numeric ID, large

Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number

MMSI number of mobile station reporting position.

4 **Longitude** *Byte Field Size:* 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD023 Longitude, WGS-84

Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF25 Longitude **int32** *Range:* +/- 180 deg *Resolution:* 1x10E-7 deg "-" = West, resolution ~1.1 cm

Longitude of mobile station reporting position.

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AIS Class B Position Report

PGN: 129039
hex: 1F80F

5 Latitude

Byte Field Size: 4 | Request Parameter: Optional
Bit Field Size: | Command Parameter: Optional

DD022 Latitude, WGS-84

Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23 Latitude — **int32** — Range: +/- 90 deg — Resolution: 1x10E-7 deg — "-" = South, resolution ~1.1 cm

Latitude of mobile station reporting position.

---

6 Position Accuracy

Byte Field Size: | Request Parameter: Optional
Bit Field Size: 1 | Command Parameter: Optional

DD184 AIS Position Accuracy

0 = low accuracy > 10m such as nondifferential GNSS (default),
1 = high accuracy < 10m such as DGNSS

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field — **bit(n)** — Range: Variable — Resolution: 1 — Used to construct bit fields

---

7 RAIM-flag

Byte Field Size: | Request Parameter: Optional
Bit Field Size: 1 | Command Parameter: Optional

DD189 AIS RAIM-flag

0 = RAIM not in use (default),
1 = RAIM in use

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field — **bit(n)** — Range: Variable — Resolution: 1 — Used to construct bit fields

---

8 Time Stamp

Byte Field Size: | Request Parameter: Optional
Bit Field Size: 6 | Command Parameter: Optional

DD186 AIS Time Stamp

0-59 = UTC second when the report was generated,
60 = time stamp not available (default),
61 = positioning system is in manual input mode,
62 = Electronic position fixing system operates in estimated (dead reckoning) mode,
63 = positioning system is inoperative

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field — **bit(n)** — Range: Variable — Resolution: 1 — Used to construct bit fields

---

9 COG

Byte Field Size: 2 | Request Parameter: Optional
Bit Field Size: | Command Parameter: Optional

DD165 Course-Over-Ground (COG)

The direction of the path over ground actually followed by a vessel.

DF02 Angle — **uint16** — Range: 0 to 2Pi rad — Resolution: 1x10E-4 rad — Resolution ~0.0057deg, 1 deg = .01745 rad

COG of mobile station reporting position.

---

10 SOG

Byte Field Size: 2 | Request Parameter: Optional
Bit Field Size: | Command Parameter: Optional

DD044 Generic Speed

DF35 Speed — **uint16** — Range: 0 to 655.32 m/s — Resolution: 1x10E-2 m/s — 1 Knot = 0.5144 m/s

SOG of mobile station reporting position.

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11 Communication State

Byte Field Size:
Bit Field Size: 19

Request Parameter: Optional
Command Parameter: Optional

DD187 AIS Communication State

The Communication State contains information used by the various TDMA slot allocation algorithms and synchronization information

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

12 AIS Transceiver Information

Byte Field Size:
Bit Field Size: 5

Request Parameter: Optional
Command Parameter: Optional

DD246 AIS Transceiver Information

0 = Channel A VDL reception,
1 = Channel B VDL reception,
2 = Channel A VDL transmission,
3 = Channel B VDL transmission,
4 = Own information not broadcast,
5-31 = Reserved.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

13 True Heading

Byte Field Size: 2
Bit Field Size:

Request Parameter: Optional
Command Parameter: Optional

DD167 Heading

The horizontal direction in which a ship actually points or heads at any instant, expressed in angular units from a reference direction, usually from 000 at the reference direction clockwise through 359 degrees.

DF02 Angle **uint16** Range: 0 to 2Pi rad Resolution: 1x10E-4 rad Resolution ~0.0057deg, 1 deg = .01745 rad

True heading of mobile station reporting position. A value of 65535 indicates that data is not available.

14 Reserved for Regional Applications

Byte Field Size:
Bit Field Size: resv 8

Request Parameter: Optional
Command Parameter: Optional

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** Range: N2KUnitless Resolution: 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Reserved for Regional Applications" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

15 Reserved for Regional Applications

Byte Field Size:
Bit Field Size: resv 2

Request Parameter: Optional
Command Parameter: Optional

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** Range: N2KUnitless Resolution: 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Reserved for Regional Applications" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

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16 **Class B unit flag**

Byte Field Size:
Bit Field Size: 1

Request Parameter: Optional
Command Parameter: Optional

DD294 AIS ClassB Unit Flag

0 = Class B SOTDMA unit
1 = Class B "CS" unit

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

17 **Class B Display Flag**

Byte Field Size:
Bit Field Size: 1

Request Parameter: Optional
Command Parameter: Optional

DD295 AIS Class B Display Flag

0 = No display available; not capable of displaying ITU-R M.1371 Messages12 and 14
1 = Equipped with integrated display displaying ITU-R M.1371 Messages 12 and 14

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

18 **Class B DSC Flag**

Byte Field Size:
Bit Field Size: 1

Request Parameter: Optional
Command Parameter: Optional

DD296 AIS Class B DSC Flag

0 = Not equipped with DSC function
1 = Equipped with DSC function (dedicated or time-shared)

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

19 **Class B Band Flag**

Byte Field Size:
Bit Field Size: 1

Request Parameter: Optional
Command Parameter: Optional

DD297 AIS Class B Band Flag

0 = Capable of operating over the upper 525 kHz band of the marine band
1 = Capable of operating over the whole marine band (irrelevant if "Class B Message 22 flag" is 0)

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

20 **Class B Msg 22 Flag**

Byte Field Size:
Bit Field Size: 1

Request Parameter: Optional
Command Parameter: Optional

DD298 AIS Class B Msg 22 Flag

0 = No frequency management via Message 22 , operating on AIS1 and AIS2 only
1 = Frequency management via Message 22

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

21 **Mode Flag**

Byte Field Size:
Bit Field Size: 1

Request Parameter: Optional
Command Parameter: Optional

DD299 AIS Mode Flag

0 = Station operating in autonomous mode (default)
1 = Station operating in assigned mode

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

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22	**Communication State Selector Flag**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	1	*Command Parameter:*	Optional
	DD245 AIS Communication State Selctor Flag		0=SOTDMA communication state, 1=ITDMA communication state follows. See the latest version of ITU-R M.1371 for more information.			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
23	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	resv 7	*Command Parameter:*	Optional
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	Used to align subsequent data on byte boundary.					
24	**Sequence ID**		*Byte Field Size:*	1	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD056 Sequence ID		An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:* 1 bit	Unit-less number

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AIS Class B Extended Position Report

PGN: 129040
hex: 1F810

This parameter group provides data associated with the ITU-R M.1371 Message 19 Extended Class B Equipment Position Report containing position and static information. An AIS device may generate this parameter group either upon receiving a VHF data link message 19, or upon receipt of an ISO or NMEA request PGN. The Command Group Function PGN 126208 may be used with this PGN to configure static parameters such as ship dimensions, antenna location, and type of electronic position fixing device (see ITU-R M.1371-1 for additional information). Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

Single Frame: No *Priority Default:* 4 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 123

1 Message ID
Byte Field Size: *Bit Field Size:* 6 *Request Parameter* Optional *Command Parameter:* Optional

DD188 AIS Message Identifier
Message Identifier (range of 0 to 63).

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
19 = Extended Class B Equipment Position Report

2 Repeat Indicator
Byte Field Size: *Bit Field Size:* 2 *Request Parameter* Optional *Command Parameter:* Optional

DD185 AIS Repeater Indicator
Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3).

0 = Default
1 = First retransmission
2 = Second retransmission
3 = Final retransmission

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 User ID
Byte Field Size: 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD010 Generic numeric ID, large
Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number
MMSI number of mobile station reporting position.

4 Longitude
Byte Field Size: 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD023 Longitude, WGS-84
Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF25 Longitude **int32** *Range:* +/- 180 deg *Resolution:* 1x10E-7 deg "-" = West, resolution ~1.1 cm
Longitude of mobile station reporting position.

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AIS Class B Extended Position Report

PGN: 129040
hex: 1F810

5 Latitude

Byte Field Size: 4
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD022 Latitude, WGS-84

Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23 Latitude — **int32** — Range: +/- 90 deg — Resolution: 1x10E-7 deg — "-" = South, resolution ~1.1 cm

Latitude of mobile station reporting position.

---

6 Position Accuracy

Byte Field Size:
Bit Field Size: 1
Request Parameter: Optional
Command Parameter: Optional

DD184 AIS Position Accuracy

0 = low accuracy > 10m such as nondifferential GNSS (default),
1 = high accuracy < 10m such as DGNSS

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field — **bit(n)** — Range: Variable — Resolution: 1 — Used to construct bit fields

---

7 RAIM-flag

Byte Field Size:
Bit Field Size: 1
Request Parameter: Optional
Command Parameter: Optional

DD189 AIS RAIM-flag

0 = RAIM not in use (default),
1 = RAIM in use

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field — **bit(n)** — Range: Variable — Resolution: 1 — Used to construct bit fields

---

8 Time Stamp

Byte Field Size:
Bit Field Size: 6
Request Parameter: Optional
Command Parameter: Optional

DD186 AIS Time Stamp

0-59 = UTC second when the report was generated,
60 = time stamp not available (default),
61 = positioning system is in manual input mode,
62 = Electronic position fixing system operates in estimated (dead reckoning) mode,
63 = positioning system is inoperative

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field — **bit(n)** — Range: Variable — Resolution: 1 — Used to construct bit fields

---

9 COG

Byte Field Size: 2
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD165 Course-Over-Ground (COG)

The direction of the path over ground actually followed by a vessel.

DF02 Angle — **uint16** — Range: 0 to 2Pi rad — Resolution: 1x10E-4 rad — Resolution ~0.0057deg, 1 deg = .01745 rad

COG of mobile station reporting position.

---

10 SOG

Byte Field Size: 2
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD044 Generic Speed

DF35 Speed — **uint16** — Range: 0 to 655.32 m/s — Resolution: 1x10E-2 m/s — 1 Knot = 0.5144 m/s

SOG of mobile station reporting position.

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AIS Class B Extended Position Report

PGN: 129040
hex: 1F810

11 **Reserved for Regional Applications**

Byte Field Size:
Bit Field Size: resv | 8

Request Parameter
Command Parameter:

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1

Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Reserved for Regional Applications" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

---

12 **Reserved for Regional Applications**

Byte Field Size:
Bit Field Size: resv | 4

Request Parameter
Command Parameter:

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1

Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Reserved for Regional Applications" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

---

13 **NMEA Reserved**

Byte Field Size:
Bit Field Size: resv | 4

Request Parameter
Command Parameter:

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1

Used to construct bit fields

Used to align subsequent data on byte boundary.

---

14 **Ship/Cargo Type**

Byte Field Size:
Bit Field Size: 8

Request Parameter Optional
Command Parameter: Optional

DD193 Ship/Cargo Type

0=Not Available or no ship (default),
1-99= (See the latest version of ITU-R M.1371 Section 3.3.8.2.3.2 Table 18),
100-199=Reserved for Regional (See the latest version of ITU-R M.1371),
200-255=Reserved for future (See the latest version of ITU-R M.1371).

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1

Used to construct bit fields

---

15 **True Heading**

Byte Field Size: 2
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD165 Course-Over-Ground (COG)

The direction of the path over ground actually followed by a vessel.

DF02 Angle **uint16** *Range:* 0 to 2Pi rad *Resolution:* 1x10E-4 rad

Resolution ~0.0057deg, 1 deg = .01745 rad

True Heading of mobile station reporting its position.

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16 **NMEA Reserved**
Byte Field Size:
Bit Field Size: resv 4
Request Parameter
Command Parameter:

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

---

17 **Type of Electronic Positioning Device**
Byte Field Size:
Bit Field Size: 4
Request Parameter: Optional
Command Parameter: Optional

DD191 AIS Electronic Positioning Device Type

0 =Undefined (default)
1 = GPS
2 = GLONASS
3 = Combined GPS/GLONASS
4 = Loran-C
5 = Chayka
6 = Integrated Navigation System
7 = Surveyed; For fixed AtoN and virtual AtoN, the charted position should be used. The accurate position enhances its function as a radar reference target
8 = Galilieo
9-15 = Reserved for future use.

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

18 **Ship Length**
Byte Field Size: 2
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD194 Distance, medium — Dependent upon PG Field definition.

DF75 Distance, Medium **uint16** Range: 0 to 6553.2 m Resolution: 1x10E-1 m

Length of mobile station reporting its position. A value of 65535 indicates that data is not available.

---

19 **Ship Beam**
Byte Field Size: 2
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD194 Distance, medium — Dependent upon PG Field definition.

DF75 Distance, Medium **uint16** Range: 0 to 6553.2 m Resolution: 1x10E-1 m

Beam of mobile station reporting its position. A value of 65535 indicates that data is not available.

---

20 **Position Reference Point from Starboard**
Byte Field Size: 2
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD194 Distance, medium — Dependent upon PG Field definition.

DF75 Distance, Medium **uint16** Range: 0 to 6553.2 m Resolution: 1x10E-1 m

Position reference point from starboard side of mobile station reporting its position. A value of 65535 indicates that data is not available.

---

21 **Position Reference Point aft of Ship's Bow**
Byte Field Size: 2
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD194 Distance, medium — Dependent upon PG Field definition.

DF75 Distance, Medium **uint16** Range: 0 to 6553.2 m Resolution: 1x10E-1 m

Position reference point from aft of ship's bow of mobile station reporting its position. A value of 65535 indicates that data is not available.

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22 **Name**

Byte Field Size: char n
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD192 Generic String, ASCII, Fixed length — Length specified by PGN field definition.

DF63 String, fixed **char8(n)** *Range:* 0 to 1,785 characters *Resolution:* 1 char

0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.

This is a 20 character string, see ITU-R M.1371-1 for more information.

---

23 **Data Terminal Equipment (DTE)**

Byte Field Size:
Bit Field Size: 1

Request Parameter Optional
Command Parameter: Optional

DD242 Data Terminal Equipment (DTE)

0=Available,
1=not available.

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

24 **Mode Flag**

Byte Field Size:
Bit Field Size: 1

Request Parameter Optional
Command Parameter: Optional

DD299 AIS Mode Flag

0 = Station operating in autonomous mode (default)
1 = Station operating in assigned mode

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

25 **AIS Spare**

Byte Field Size:
Bit Field Size: resv 4

Request Parameter
Command Parameter:

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1

Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

---

26 **AIS Transceiver Information**

Byte Field Size:
Bit Field Size: 5

Request Parameter Optional
Command Parameter: Optional

DD246 AIS Transceiver Information

0 = Channel A VDL reception,
1 = Channel B VDL reception,
2 = Channel A VDL transmission,
3 = Channel B VDL transmission,
4 = Own information not broadcast,
5-31 = Reserved.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

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AIS Class B Extended Position Report

PGN: 129040
hex: 1F810

27	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*	
			Bit Field Size:	resv 5	*Command Parameter:*	
	DD001 Reserved field			Variable number of reserved bits, all set to logic "1"		
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	Used to align subsequent data on byte boundary.					
28	**Sequence ID**		*Byte Field Size:*	1	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD056 Sequence ID			An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.		
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:* 1 bit	Unit-less number

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AIS Aids to Navigation (AtoN) Report

PGN: 129041
hex: 1F811

This PGN provides information received from an AtoN AIS station conforming to ITU-R M.1371-4 Message 21. The AtoN station may be mounted on an aid-to-navigation or this message may be transmitted by a fixed station when the functionality of an AtoN station is integrated into the fixed station. This message is typically transmitted autonomously at a rate of once every three (3) min. Other reporting rates are possible when the AtoN device has received an assigned mode command (Message 16) via the VHF data link, or by an external command such as PGN 129804 - AIS Assignment Mode Command.

Application of all data fields is dependant upon the latest version of ITU-R M.1371.

Single Frame: N | *Priority Default:* 4 | *Default Update Rate:* N/A milliseconds | *Frequency:* #Type! cycles per second
Destination: Global | *Query Support:* Optional | *Command Support:* Optional | *ACK Rqmnts:* None

Field # *Field Name* — Original Reference ID # 203

1 Message ID — *Byte Field Size:* | *Bit Field Size:* 6 | *Request Parameter* Optional | *Command Parameter:* Optional

DD188 AIS Message Identifier

Message Identifier (range of 0 to 63).

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
21 = AIS Aids to Navigation (AtoN) Report

2 Repeat Indicator — *Byte Field Size:* | *Bit Field Size:* 2 | *Request Parameter* Optional | *Command Parameter:* Optional

DD185 AIS Repeater Indicator

Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3).

0 = Default
1 = First retransmission
2 = Second retransmission
3 = Final retransmission

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 ID — *Byte Field Size:* 4 | *Bit Field Size:* | *Request Parameter* Optional | *Command Parameter:* Optional

DD010 Generic numeric ID, large

Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number
MMSI number of AtoN station reporting position

4 Longitude — *Byte Field Size:* 4 | *Bit Field Size:* | *Request Parameter* Optional | *Command Parameter:* Optional

DD023 Longitude, WGS-84

Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF25 Longitude **int32** *Range:* +/- 180 deg *Resolution:* 1x10E-7 deg "-" = West, resolution ~1.1 cm

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AIS Aids to Navigation (AtoN) Report

PGN: 129041
hex: 1F811

5 **Latitude**
Byte Field Size: 4 | Bit Field Size: | Request Parameter: Optional | Command Parameter: Optional

DD022 Latitude, WGS-84
Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23 Latitude **int32** Range: +/- 90 deg Resolution: 1x10E-7 deg "-" = South, resolution ~1.1 cm

---

6 **Position Accuracy**
Byte Field Size: | Bit Field Size: 1 | Request Parameter: Optional | Command Parameter: Optional

DD184 AIS Position Accuracy
0 = low accuracy > 10m such as nondifferential GNSS (default),
1 = high accuracy < 10m such as DGNSS

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

7 **RAIM Flag**
Byte Field Size: | Bit Field Size: 1 | Request Parameter: Optional | Command Parameter: Optional

DD189 AIS RAIM-flag
0 = RAIM not in use (default),
1 = RAIM in use

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

8 **Time Stamp**
Byte Field Size: | Bit Field Size: 6 | Request Parameter: Optional | Command Parameter: Optional

DD186 AIS Time Stamp
0-59 = UTC second when the report was generated,
60 = time stamp not available (default),
61 = positioning system is in manual input mode,
62 = Electronic position fixing system operates in estimated (dead reckoning) mode,
63 = positioning system is inoperative

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

9 **AtoN Structure Length/Diameter**
Byte Field Size: 2 | Bit Field Size: | Request Parameter: Optional | Command Parameter: Optional

DD194 Distance, medium
Dependent upon PG Field definition.

DF75 Distance, Medium **uint16** Range: 0 to 6553.2 m Resolution: 1x10E-1 m
See Message 21 Dimension/reference for position data field and notes for interpretation of these values in the latest version of ITU-R M.1371.

---

10 **AtoN Structure Beam/Diameter**
Byte Field Size: 2 | Bit Field Size: | Request Parameter: Optional | Command Parameter: Optional

dd194 Distance, medium
Dependent upon PG Field definition.

DF75 Distance, Medium **uint16** Range: 0 to 6553.2 m Resolution: 1x10E-1 m
See Message 21 Dimension/reference for position data field and notes for interpretation of these values in the latest version of ITU-R M.1371.

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11	**Position Reference Point from Starboard Structure Edge/Radius**	*Byte Field Size:*	2	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD194 Distance, medium — Dependent upon PG Field definition.

DF75 Distance, Medium **uint16** *Range:* 0 to 6553.2 m *Resolution:* 1x10E-1 m

See Message 21 Dimension/reference for position data field and notes for interpretation of these values in the latest version of ITU-R M.1371.

12	**Position Reference Point from True North facing Structure Edge/Radius**	*Byte Field Size:*	2	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD194 Distance, medium — Dependent upon PG Field definition.

DF75 Distance, Medium **uint16** *Range:* 0 to 6553.2 m *Resolution:* 1x10E-1 m

See Message 21 Dimension/reference for position data field and notes for interpretation of these values in the latest version of ITU-R M.1371.

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13 **Aid to Navigation (AtoN) Type**

Byte Field Size:
Bit Field Size: 5

Request Parameter Optional
Command Parameter: Optional

DD305 AIS Aids to Navigation Type

0 = Type of AtoN not specified (default)
1 = Reference point
2 = RACON
3 = Fixed structures off-shore, such as oil platforms, wind farms; This code identifies an obstruction that is fitted with an AtoN AIS station
4 = Spare, Reserved for future use

Fixed AtoN:
5 = Light, without sectors
6 = Light, with sectors
7 = Leading Light Front
8 = Leading Light Rear
9 = Beacon, Cardinal N
10 = Beacon, Cardinal E
11 = Beacon, Cardinal S
12 = Beacon, Cardinal W
13 = Beacon, Port hand
14 = Beacon, Starboard hand
15 = Beacon, Preferred Channel port hand
16 = Beacon, Preferred Channel starboard hand
17 = Beacon, Isolated danger
18 = Beacon, Safe water
19 = Beacon, Special mark

Floating AtoN:
20 = Cardinal Mark N
21 = Cardinal Mark E
22 = Cardinal Mark S
23 = Cardinal Mark W
24 = Port hand Mark
25 = Starboard hand Mark
26 = Preferred Channel Port hand
27 = Preferred Channel Starboard hand
28 = Isolated danger
29 = Safe Water
30 = Special Mark
31 = Light Vessel/LANBY/Rigs

NOTE 1: The types of aids to navigation listed above are based on the IALA Maritime Buoyage System, where applicable.
NOTE 2: There is potential for confusion when deciding whether an aid is lighted or unlighted. Competent authorities may wish to use the regional/local section of the message to indicate this.

See Latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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14 **Off Position Indicator**

Byte Field Size:
Bit Field Size: 1

Request Parameter Optional
Command Parameter: Optional

DD306 Off position indicator

For floating AtoN, only:

0 = on position
1 = off position

NOTE 1: This flag should only be considered valid by receiving station, if the AtoN is a floating aid, and if time stamp is equal to or below 59. For floating AtoN the guard zone parameters should be set on installation.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

15 **Virtual AtoN Flag**

Byte Field Size:
Bit Field Size: 1

Request Parameter Optional
Command Parameter: Optional

DD307 Virtual AtoN Flag

0 = real AtoN at indicated position (default)
1 = virtual AtoN, does not physically exist

See note 2 of MSG 21 in ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

16 **Assigned Mode Flag**

Byte Field Size:
Bit Field Size: 1

Request Parameter Optional
Command Parameter: Optional

DD308 AIS Assigned Mode Flag

0 = Station operating in autonomous and continuous mode (default)
1 = Station operating in assigned mode

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

17 **AIS Spare**

Byte Field Size:
Bit Field Size: resv 1

Request Parameter
Command Parameter:

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

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18	**Electronic Fixing Position Fixing Device Type**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	4	*Command Parameter:*	Optional

DD191 AIS Electronic Positioning Device Type

0 =Undefined (default)
1 = GPS
2 = GLONASS
3 = Combined GPS/GLONASS
4 = Loran-C
5 = Chayka
6 = Integrated Navigation System
7 = Surveyed; For fixed AtoN and virtual AtoN, the charted position should be used. The accurate position enhances its function as a radar reference target
8 = Galilieo
9-15 = Reserved for future use.

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

19	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 3	*Command Parameter:*	

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

20	**AtoN Status**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	8	*Command Parameter:*	Optional

DD309 AtoN Status

Reserved for indicating AtoN Status; Only defined value is 00000000 (default).

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

21	**AIS Transceiver Information**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	5	*Command Parameter:*	

DD246 AIS Transceiver Information

0 = Channel A VDL reception,
1 = Channel B VDL reception,
2 = Channel A VDL transmission,
3 = Channel B VDL transmission,
4 = Own information not broadcast,
5-31 = Reserved.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

22	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 3	*Command Parameter:*	

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary

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AIS Aids to Navigation (AtoN) Report

PGN: 129041
hex: 1F811

23	**Aid to Navigation (AtoN) Name**	*Byte Field Size:*	8 or 16 n	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character

2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

This field is always transmitted using ASCII and combines two fields from ITU-R M.1371 message 21. 'Name of Aids-to-Navigation' is a fixed 20 ASCII character field that is always present, and 'Name of Aid-to-Navigation Extension' is an optional field of up to 14 ASCII characters. The total length of this vield is variable and will be from 20 to 34 ASCII characters.

See ITU-R M.1371 for more information.

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Datum

PGN: 129044
hex: 1F814

Local geodetic datum and datum offsets from a reference datum. This PGN is used to define the datum to which a position location output by the same device in other PGNs is referenced.

This PGN will normally be requested as needed.

Single Frame:	No	*Priority Default:*	6	*Default Update Rate:*	10000 milliseconds	*Frequency:*	.1 cycles per second
Destination:	Global	*Query Support:*	Optional	*Command Support:*	Optional	*ACK Rqmnts:*	None

Field # *Field Name* — Original Reference ID # 20

1 Local Datum

Byte Field Size: char 4 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD068 Datum

4-character code for the datum currently being output for the position solution. The datum's are defined in the IHO Publication S-60, Appendices B and C. The first three characters are the datum ID as per the IHO tables. The fourth character is the local datum subdivision code. A null character indicates the datum or subdivision code is unknown or not used.

DF63 String, fixed — **char8(n)** — *Range:* 0 to 1,785 characters — *Resolution:* 1 char

0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.

2 Delta Latitude

Byte Field Size: 4 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD106 Latitude, Offset

Offsets change with position; position in the local datum is offset from the position in the reference datum in the directions indicated: Plocal datum = Pref datum + offset.

DF23 Latitude — **int32** — *Range:* +/- 90 deg — *Resolution:* 1x10E-7 deg

"-" = South, resolution ~1.1 cm

'+' is North

3 Delta Longitude

Byte Field Size: 4 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD107 Longitude, Offset

Offsets change with position; position in the local datum is offset from the position in the reference datum in the directions indicated: Plocal datum = Pref datum + offset.

DF25 Longitude — **int32** — *Range:* +/- 180 deg — *Resolution:* 1x10E-7 deg

"-" = West, resolution ~1.1 cm

'+' is East

4 Delta Altitude

Byte Field Size: 4 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD115 Distance

DF15 Distance, signed — **int32** — *Range:* +/-~2.147x10E+7 m — *Resolution:* 1x10E-2 m

'+' is Up

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Datum

PGN: 129044
hex: 1F814

5	**Reference Datum**	*Byte Field Size:*	char 4	*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	

DD068 Datum

4-character code for the datum currently being output for the position solution. The datum's are defined in the IHO Publication S-60, Appendices B and C. The first three characters are the datum ID as per the IHO tables. The fourth character is the local datum subdivision code. A null character indicates the datum or subdivision code is unknown or not used.

DF63	String, fixed	**char8(n)**	*Range:* 0 to 1,785 characters	*Resolution:* 1 char		0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.

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User Datum Settings

PGN: 129045
hex: 1F815

Transformation parameters for converting from WGS-84 to other Datums.

This PGN will normally be requested as needed.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 21

Field #	Field Name					
1	**Delta X**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD108 Axis Delta shift		Delta Shift in X, Y, or Z axis from WGS 84.			
	DF15 Distance, signed	**int32**	*Range:* +/-~2.147x10E+7 m	*Resolution:* 1x10E-2 m		
2	**Delta Y**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD108 Axis Delta shift		Delta Shift in X, Y, or Z axis from WGS 84.			
	DF15 Distance, signed	**int32**	*Range:* +/-~2.147x10E+7 m	*Resolution:* 1x10E-2 m		
3	**Delta Z**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD108 Axis Delta shift		Delta Shift in X, Y, or Z axis from WGS 84.			
	DF15 Distance, signed	**int32**	*Range:* +/-~2.147x10E+7 m	*Resolution:* 1x10E-2 m		
4	**Rotation in X**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD109 Axis Rotational shift		Rotational shift in X, Y, or Z axis from WGS 84. Rotations presented use the geodetic sign convention. When looking along the positive axis towards the origin, counter-clockwise rotations are positive.			
	DF70 Angle, tiny	**float32**	*Range:* Variable radians	*Resolution:* Floats, radian		
5	**Rotation in Y**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD109 Axis Rotational shift		Rotational shift in X, Y, or Z axis from WGS 84. Rotations presented use the geodetic sign convention. When looking along the positive axis towards the origin, counter-clockwise rotations are positive.			
	DF70 Angle, tiny	**float32**	*Range:* Variable radians	*Resolution:* Floats, radian		
6	**Rotation in Z**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD109 Axis Rotational shift		Rotational shift in X, Y, or Z axis from WGS 84. Rotations presented use the geodetic sign convention. When looking along the positive axis towards the origin, counter-clockwise rotations are positive.			
	DF70 Angle, tiny	**float32**	*Range:* Variable radians	*Resolution:* Floats, radian		
7	**Scale**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD110 Scale		Scale factor expressed in parts-per-million			
	DF49 Ratio, Relative measure	**float32**	*Range:* Variable	*Resolution:* Floats	Unit-less number	

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User Datum Settings

PGN: 129045
hex: 1F815

8	**Ellipsoid Semi-major Axis**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD111 Ellipsoid Semi-major Axis		Semi-major axis (a) of the User Datum ellipsoid.			
	DF15 Distance, signed	**int32**	*Range:*	+/-~2.147x10E+7 m	*Resolution:*	1x10E-2 m
9	**Ellipsoid Flattening Inverse**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD112 Ellipsoid Flattening Inverse		Flattening (1/f) of the user Datum ellipsoid.			
	DF49 Ratio, Relative measure	**float32**	*Range:*	Variable	*Resolution:*	Floats Unit-less number
10	**Datum Name**		*Byte Field Size:*	char 4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD068 Datum		4-character code for the datum currently being output for the position solution. The datum's are defined in the IHO Publication S-60, Appendices B and C. The first three characters are the datum ID as per the IHO tables. The fourth character is the local datum subdivision code. A null character indicates the datum or subdivision code is unknown or not used.			
	DF63 String, fixed	**char8(n)**	*Range:*	0 to 1,785 characters	*Resolution:*	1 char 0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.

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Cross Track Error

PGN: 129283
hex: 1F903

This PGN provides the magnitude of position error perpendicular to the desired course.

Single Frame: Yes | Priority Default: 3 | Default Update Rate: 1000 milliseconds | Frequency: 1. cycles per second
Destination: Global | Query Support: Optional | Command Support: Optional | ACK Rqmnts: None

Original Reference ID # 34

Field #	Field Name		Byte Field Size	Bit Field Size	Request Parameter	Command Parameter	Description
1	**Sequence ID**		1		Optional	Optional	
	DD056 Sequence ID						An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.
	DF53 Integer, 8 bit unsigned	**uint8**			Range: 0 to 252	Resolution: 1 bit	Unit-less number
2	**XTE Mode**			4	Optional	Optional	
	DD025 Mode, Data						0x0 = Autonomous mode, 0x1 = Differential, enhanced mode, 0x2 = Estimated mode, 0x3 = Simulator mode, 0x4 = Manual mode, 0x5 to 0xD = Reserved 0xE = Error, 0xF = Data not available
	DF52 Bit field	**bit(n)**			Range: Variable	Resolution: 1	Used to construct bit fields
3	**NMEA Reserved**			resv 2			Variable number of reserved bits, all set to logic "1"
	DD001 Reserved field						
	DF52 Bit field	**bit(n)**			Range: Variable	Resolution: 1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.						
4	**Navigation Terminated**			2	Optional	Optional	
	DD002 Generic status pair						MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]
	DF52 Bit field	**bit(n)**			Range: Variable	Resolution: 1	Used to construct bit fields
	Flag should be set as follows: NO - when Navigation is running normally, YES - In the last PGN when Navigation was Terminated (manually or automatically), ERROR - in case of a navigation error, UNAVAIL - if flag is not supported.						
5	**XTE**		4		Optional	Optional	
	DD114 XTE						Cross-track-error of a route. "-" = Left of track, need to steer right
	DF15 Distance, signed	**int32**			Range: +/-~2.147x10E+7 m	Resolution: 1x10E-2 m	

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Cross Track Error

PGN: 129283
hex: 1F903

6	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 16	*Command Parameter:*	

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

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Navigation Data

PGN: 129284
hex: 1F904

This PGN provides essential navigation data for following a route.

Transmissions will originate from products that can create and manage routes using waypoints. This information is intended for navigational repeaters.

Transmission intervals should be aligned to latitude and longitude update rates

Single Frame: No *Priority Default:* 3 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 37

Field #	Field Name					
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD056 Sequence ID		An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
2	**Distance to Destination Waypoint**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD199 Distance, Unsigned					
	DF09 Distance	**uint32**	*Range:* 0 to ~4.295x10E+7 m	*Resolution:* 1x10E-2 m		
3	**Course/Bearing Ref.**		*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter* Optional *Command Parameter:* Optional	
	DD117 Direction reference		0 = True, 1 = Magnetic, 2 = Error, 3 = Null			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
4	**Perpendicular Crossed**		*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter* Optional *Command Parameter:* Optional	
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
5	**Arrival Circle Entered**		*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter* Optional *Command Parameter:* Optional	
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	

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6 Calculation Type
Byte Field Size:
Bit Field Size: 2
Request Parameter: Optional
Command Parameter: Optional

DD119 Calculation Type
0 = Great Circle calculations,
1 = Rhumb Line calculations,
2 = Error,
3 = Null

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

7 ETA Time
Byte Field Size: 4
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD158 Generic time of day
24 hour clock, 0 = midnight, time is in UTC

DF06 Time of day **uint32** Range: 0 to 86,401 s Resolution: 1x10E-4 s ~24 hours, 0 = midnight, range allows for up to two leap seconds per day

---

8 ETA Date
Byte Field Size: 2
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD039 Generic date
Days since January 1, 1970, Date is relative to UTC Time.

DF41 Date, day count **uint16** Range: 0 to 65,532 days Resolution: 1 day 0 = January 1, 1970, max = ~179 years

---

9 Bearing, Origin To Destination Waypoint
Byte Field Size: 2
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD164 Bearing
The horizontal direction of one terrestrial point from another, expressed as the angular distance from a reference direction, measured from 000 at the reference direction clockwise through 359 degrees.

DF02 Angle **uint16** Range: 0 to 2Pi rad Resolution: 1x10E-4 rad Resolution ~0.0057deg, 1 deg = .01745 rad

---

10 Bearing, Position To Destination Waypoint
Byte Field Size: 2
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD164 Bearing
The horizontal direction of one terrestrial point from another, expressed as the angular distance from a reference direction, measured from 000 at the reference direction clockwise through 359 degrees.

DF02 Angle **uint16** Range: 0 to 2Pi rad Resolution: 1x10E-4 rad Resolution ~0.0057deg, 1 deg = .01745 rad

---

11 Origin Waypoint Number
Byte Field Size: 4
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD010 Generic numeric ID, large
Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** Range: 0 to 4,294,967,292 Resolution: 1 bit Unit-less number
Applies to current route and at this time is limited to 16 bits

---

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Navigation Data

PGN: 129284
hex: 1F904

12 **Destination Waypoint Number** — *Byte Field Size:* 4 — *Bit Field Size:* — *Request Parameter* Optional — *Command Parameter:* Optional

DD010 Generic numeric ID, large — Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number

Applies to current route and at this time is limited to 16 bits

---

13 **Destination Wpt Latitude** — *Byte Field Size:* 4 — *Bit Field Size:* — *Request Parameter* Optional — *Command Parameter:* Optional

DD022 Latitude, WGS-84 — Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23 Latitude **int32** *Range:* +/- 90 deg *Resolution:* 1x10E-7 deg "-" = South, resolution ~1.1 cm

---

14 **Destination Wpt Longitude** — *Byte Field Size:* 4 — *Bit Field Size:* — *Request Parameter* Optional — *Command Parameter:* Optional

DD023 Longitude, WGS-84 — Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF25 Longitude **int32** *Range:* +/- 180 deg *Resolution:* 1x10E-7 deg "-" = West, resolution ~1.1 cm

---

15 **Waypoint Closing Velocity** — *Byte Field Size:* 2 — *Bit Field Size:* — *Request Parameter* Optional — *Command Parameter:* Optional

DD228 Generic speed, signed - large — Positive values represent ahead or starboard transverse speed and negative values represent astern or port transverse speed.

DF87 Speed, signed - large **int16** *Range:* +/- 327.66 m/s (+/- 636 knots) *Resolution:* 1x10E-2 m/s

Positive value indicates approaching Wpt, negative indicates moving away from Wpt

---

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Navigation - Route/WP information

PGN: 129285
hex: 1F905

This PGN shall return Route and WP data ahead in the Active Route.
It can be requested or may be transmitted without a request, typically at each Waypoint advance.

When navigating the Route in Forward direction, the Waypoints shall be included in the order of increasing RPS#s. When navigating in Reverse direction the order shall be decreasing RPS#s.

The first Waypoint shall be the origin WP. When navigating towards a single WP or when the first WP of the Route is not yet passed, the origin RPS# shall be 65535 (NA). The WP Name and Position may optionally be filled with the name and position where the navigation started, or it shall be set to NA.

ISO request for this PGN shall return origin and destination WP, next WP may be added - but it is not required.
For a complete description of the Route and WP PGNs, see the application note in Appendix D.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 105

Field #	Field Name					
1	**Start RPS#**	*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD007 Generic numeric ID, medium		Number of route, waypoint, event, mark, etc.			
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number	
	Route Point Sequence Number (RPS#) of the Origin Waypoint. Set to 65535 (NA) if the origin WP is not defined.					
2	**nItems**	*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Required *Command Parameter:* Optional		
	DD007 Generic numeric ID, medium		Number of route, waypoint, event, mark, etc.			
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number	
	n RPS#'s requested/sent. If not specified in the request, the default is n = 2.					
3	**Database ID**	*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD007 Generic numeric ID, medium		Number of route, waypoint, event, mark, etc.			
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number	
4	**Route ID**	*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD007 Generic numeric ID, medium		Number of route, waypoint, event, mark, etc.			
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number	
5	**Navigation direction in route**	*Byte Field Size:* *Bit Field Size:* 3		*Request Parameter* Optional *Command Parameter:* Optional		
	DD241 Navigation Direction		0=Forward, 1=Reverse, 2-5 Reserved, 6= Error, 7=Null (info not available)			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
	Forward=increasing Route Point Sequence Number (RPS#)					

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6	**Supplementary Route/WP data available**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	2	*Command Parameter:*	Optional

DD002 Generic status pair

MSB/LSB:
00 = [No, Off, Disabled, Reset, "0"],
01 = [Yes, On, Enabled, Set, "1"],
10 = Error,
11= [Unavailable, Unknown]

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

YES=there are supplementary data available
The Database ID, Route ID, WPID/RPS# may be used to request other "Route and WP Service" PGNs with supplementary data.

7	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 3	*Command Parameter:*	

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

8	**Route Name**	*Byte Field Size:*	8 or 16 n	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character

2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

Max 30 ASCII or Unicode Characters

9	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 8	*Command Parameter:*	

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

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10 **WPID**

Byte Field Size: 2 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

Shall have valid data if the Waypoint exists in the WP-List.

The waypoints shall be included in the order of appearance in the Navigation Direction.

---

11 **WP Name**

Byte Field Size: 8 or 16 n *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character 2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

Max. 30 ASCII or Unicode Characters

---

12 **WP Latitude**

Byte Field Size: 4 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD022 Latitude, WGS-84 — Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23 Latitude **int32** *Range:* +/- 90 deg *Resolution:* 1x10E-7 deg "-" = South, resolution ~1.1 cm

---

13 **WP Longitude**

Byte Field Size: 4 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD023 Longitude, WGS-84 — Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF25 Longitude **int32** *Range:* +/- 180 deg *Resolution:* 1x10E-7 deg "-" = West, resolution ~1.1 cm

---

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Navigation - Route/WP information

PGN: 129285
hex: 1F905

14	**Fields 10 thru 13 repeat as needed**	*Byte Field Size:* ?		*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD000 Undefined — Application specific, defined at time of use

DF00 Undefined — **Undefined** *Range:* undefined *Resolution:* undefined — Application specific, defined at time of use.

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Set & Drift, Rapid Update

PGN: 129291
hex: 1F90B

The Set and Drift effect on the Vessel is the direction and the speed of a current. The Course & Speed (through water) vector added to the Set & Drift vector is the COG & SOG vector. The bearings may be True or Magnetic referenced.
When Set & Drift is calculated from data from a GPS, a compass and a speed log, the Set & Drift estimate will be influenced by current, weather and anything that sets the ship off from the intended Course.
The Sequence ID may be used to tie the data to time, position, sample number.

Single Frame: Yes | *Priority Default:* 3 | *Default Update Rate:* 1000 milliseconds | *Frequency:* 1. cycles per second
Destination: Global | *Query Support:* Optional | *Command Support:* Optional | *ACK Rqmnts:* None

Original Reference ID # 33

Field #	Field Name						
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional	
	DD056 Sequence ID			An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252		*Resolution:* 1 bit		Unit-less number
2	**Set Reference**		*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter* *Command Parameter:*	Optional Optional	
	DD117 Direction reference			0 = True, 1 = Magnetic, 2 = Error, 3 = Null			
	DF52 Bit field	**bit(n)**	*Range:* Variable		*Resolution:* 1		Used to construct bit fields
3	**NMEA Reserved**		*Byte Field Size:* *Bit Field Size:* resv 6		*Request Parameter* *Command Parameter:*		
	DD001 Reserved field			Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:* Variable		*Resolution:* 1		Used to construct bit fields
	Used to align subsequent data on a byte boundary.						
4	**Set**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional	
	DD048 Current flow direction			Direction towards which current flows. Degrees relative to True North.			
	DF02 Angle	**uint16**	*Range:* 0 to 2Pi rad		*Resolution:* 1x10E-4 rad		Resolution ~0.0057deg, 1 deg = .01745 rad
5	**Drift**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional	
	DD044 Generic Speed						
	DF35 Speed	**uint16**	*Range:* 0 to 655.32 m/s		*Resolution:* 1x10E-2 m/s		1 Knot = 0.5144 m/s
6	**NMEA Reserved**		*Byte Field Size:* *Bit Field Size:* resv 16		*Request Parameter* *Command Parameter:*		
	DD001 Reserved field			Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:* Variable		*Resolution:* 1		Used to construct bit fields
	Used to align subsequent data on a byte boundary.						

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Time to/from Mark

PGN: 129301
hex: 1F915

Time to go to or elapsed from a generic mark, that may be non-fixed. The mark is not generally a specific geographic point but may vary continuously and is most often determined by calculation (the recommended turning or tacking point for sailing vessels, the wheel-over point for vessels making turns, a predicted collision point, etc.)

Single Frame: No *Priority Default:* 3 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 41

Field #	Field Name						
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD056 Sequence ID			An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252		*Resolution:* 1 bit		Unit-less number
2	**Time elapsed (from) or to-go to mark**		*Byte Field Size:* 4 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD034 Time-elapsed/Time-to-go			Time interval in milli-sec. "-" = time elapsed since event, "+" = time to go before event.			
	DF40 Time interval, signed, sta	**int32**	*Range:* +/- ~2.148x10E+6 s		*Resolution:* 1x10E-3 s		
3	**Mark Type**		*Byte Field Size:* *Bit Field Size:* 4			*Request Parameter* Optional *Command Parameter:* Optional	
	DD122 Mark Type			0 = Collision, 1 = Turning Point, 2 = Reference (general), 3 = Wheelover, 4 = Waypoint, 5-13 = Reserved, 14 = Error, 15 = Null			
	DF52 Bit field	**bit(n)**	*Range:* Variable		*Resolution:* 1		Used to construct bit fields
4	**NMEA Reserved**		*Byte Field Size:* *Bit Field Size:* resv 4			*Request Parameter* *Command Parameter:*	
	DD001 Reserved field			Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:* Variable		*Resolution:* 1		Used to construct bit fields
	Used to align subsequent data on a byte boundary.						
5	**Mark ID**		*Byte Field Size:* 4 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD010 Generic numeric ID, large			Number of route, waypoint, event, mark, etc.			
	DF55 Integer, 32 bit unsigned	**uint32**	*Range:* 0 to 4,294,967,292		*Resolution:* 1 bit		Unit-less number

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Bearing and Distance between two Marks

PGN: 129302
hex: 1F916

Bearing and distance from the origin mark to the destination mark, calculated at the origin mark, for any two arbitrary generic marks. The calculation type (Rhumb Line, Great Circle) is specified, as well as the bearing reference (Mag, True).

This PGN will normally be requested as needed.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 38

#	Field Name							
1	**Sequence ID**		*Byte Field Size:* 1			*Request Parameter*	Optional	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD056 Sequence ID			An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252		*Resolution:* 1 bit	Unit-less number		
2	**Bearing Ref.**		*Byte Field Size:*			*Request Parameter*	Optional	
			Bit Field Size: 2			*Command Parameter:*	Optional	
	DD117 Direction reference			0 = True, 1 = Magnetic, 2 = Error, 3 = Null				
	DF52 Bit field	**bit(n)**	*Range:* Variable		*Resolution:* 1	Used to construct bit fields		
3	**Calculation Type**		*Byte Field Size:*			*Request Parameter*	Optional	
			Bit Field Size: 2			*Command Parameter:*	Optional	
	DD119 Calculation Type			0 = Great Circle calculations, 1 = Rhumb Line calculations, 2 = Error, 3 = Null				
	DF52 Bit field	**bit(n)**	*Range:* Variable		*Resolution:* 1	Used to construct bit fields		
4	**NMEA Reserved**		*Byte Field Size:*			*Request Parameter*		
			Bit Field Size: resv 4			*Command Parameter:*		
	DD001 Reserved field			Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	*Range:* Variable		*Resolution:* 1	Used to construct bit fields		
	Used to align subsequent data on a byte boundary.							
5	**Bearing, Origin To Destination**		*Byte Field Size:* 2			*Request Parameter*	Optional	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD164 Bearing			The horizontal direction of one terrestrial point from another, expressed as the angular distance from a reference direction, measured from 000 at the reference direction clockwise through 359 degrees.				
	DF02 Angle	**uint16**	*Range:* 0 to 2Pi rad		*Resolution:* 1x10E-4 rad	Resolution ~0.0057deg, 1 deg = .01745 rad		
6	**Distance**		*Byte Field Size:* 4			*Request Parameter*	Optional	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD199 Distance, Unsigned							
	DF09 Distance	**uint32**	*Range:* 0 to ~4.295x10E+7 m		*Resolution:* 1x10E-2 m			

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7	**Origin Mark Type**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD122 Mark Type		0 = Collision, 1 = Turning Point, 2 = Reference (general), 3 = Wheelover, 4 = Waypoint, 5-13 = Reserved, 14 = Error, 15 = Null			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
8	**Destination Mark Type**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD122 Mark Type		0 = Collision, 1 = Turning Point, 2 = Reference (general), 3 = Wheelover, 4 = Waypoint, 5-13 = Reserved, 14 = Error, 15 = Null			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
9	**Origin Mark Id**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD010 Generic numeric ID, large		Number of route, waypoint, event, mark, etc.			
	DF55 Integer, 32 bit unsigned	**uint32**	*Range:*	0 to 4,294,967,292	*Resolution:* 1 bit	Unit-less number
10	**Destination Mark ID**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD010 Generic numeric ID, large		Number of route, waypoint, event, mark, etc.			
	DF55 Integer, 32 bit unsigned	**uint32**	*Range:*	0 to 4,294,967,292	*Resolution:* 1 bit	Unit-less number

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GNSS Control Status

PGN: 129538
hex: 1FA02

GNSS common satellite receiver parameter status.

This PGN will be requested as needed.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 17

Field #	Field Name						
1	**SV Elevation Mask**		*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD054 Elevation			Angle above or below the horizon. -90° to +90°; negative below the horizon			
	DF04 Angle, signed	**int16**	*Range:* +/-Pi rad		*Resolution:* 1x10E-4 rad		Resolution ~0.0057deg
	Do not use satellites below this value.						
2	**PDOP Mask**		*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD055 DOP			Dilution of Precision (DOP) indicates the contribution of satellite configuration geometry to positioning error. A lower DOP value is preferred because less error is being introduced. Reported as components: HDOP (Horizontal), VDOP (Vertical), TDOP (Time). Minimum DOP value is 1.0 (no error introduced).			
	DF69 Ratio, Relative measure,	**int16**	*Range:* +/-327.64		*Resolution:* 1x10E-2		Unit-less number
	When exceeded, GNSS Receiver shall indicate No GNSS fix or DR Mode in PGN 129029						
3	**PDOP Switch**		*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD055 DOP			Dilution of Precision (DOP) indicates the contribution of satellite configuration geometry to positioning error. A lower DOP value is preferred because less error is being introduced. Reported as components: HDOP (Horizontal), VDOP (Vertical), TDOP (Time). Minimum DOP value is 1.0 (no error introduced).			
	DF69 Ratio, Relative measure,	**int16**	*Range:* +/-327.64		*Resolution:* 1x10E-2		Unit-less number
	When exceeded GNSS Receiver shall switch from 3D to 2D mode						
4	**SNR Mask**		*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD057 SNR Value			SNR expressed in C/No			
	DF31 dB, relative measure	**int16**	*Range:* +/- 327.64 dB		*Resolution:* 1x10E-2 dB		
	Do not use satellites below this value.						
5	**GNSS Mode**		*Byte Field Size:* *Bit Field Size:* 3			*Request Parameter* Optional *Command Parameter:* Optional	
	DD058 Mode, GNSS			0 = 1D, 1 = 2D, 2 = 3D, 3 = Auto, 4-5 = Reserved, 6 = Error, 7 = Null			
	DF52 Bit field	**bit(n)**	*Range:* Variable		*Resolution:* 1		Used to construct bit fields

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GNSS Control Status

PGN: 129538
hex: 1FA02

6	**DGNSS Mode**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	3	*Command Parameter:*	Optional
	DD059 Mode, DGNSS		0 = Off, 1 = Auto, 2 = RTCM SC104 Pseudorange Corrections, 3 = RTCA SC159 Network Corrections, 4 = RTK Differential Corrections, 5 = Reserved, 6 = Error; 7 = Null			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
7	**Position / Velocity Filter**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
8	**Max Correction Age**		*Byte Field Size:*	2	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD060 Differential Age		Age of Differential corrections			
	DF66 Time interval, .01sec	**uint16**	*Range:*	0 to 655.32s	*Resolution:*	1x10E-2sec
9	**Antenna Altitude for 2D Mode**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD024 Altitude, WGS-84		Altitude referenced to WGS-84			
	DF15 Distance, signed	**int32**	*Range:*	+/-~2.147x10E+7 m	*Resolution:*	1x10E-2 m
10	**Use Antenna Altitude for 2D Mode**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
	00 = Use last good calculated Altitude for 2D mode.					
11	**Reserved**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	resv 6	*Command Parameter:*	Optional
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields

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GNSS DOPs

PGN: 129539
hex: 1FA03

This PGN provides a single transmission containing GNSS status and dilution of precision components (DOP) that indicate the contribution of satellite geometry to the overall positioning error. There are three DOP parameters reported, horizontal (HDOP), Vertical (VDOP) and time (TDOP).

Single Frame: Yes *Priority Default:* 6 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 22

Field #	Field Name						
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD056 Sequence ID		An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
2	**Set Mode**		*Byte Field Size:* *Bit Field Size:* 3		*Request Parameter* Optional *Command Parameter:* Optional		
	DD058 Mode, GNSS		0 = 1D, 1 = 2D, 2 = 3D, 3 = Auto, 4-5 = Reserved, 6 = Error, 7 = Null				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
3	**Op Mode**		*Byte Field Size:* *Bit Field Size:* 3		*Request Parameter* Optional *Command Parameter:* Optional		
	DD058 Mode, GNSS		0 = 1D, 1 = 2D, 2 = 3D, 3 = Auto, 4-5 = Reserved, 6 = Error, 7 = Null				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
4	**NMEA Reserved**		*Byte Field Size:* *Bit Field Size:* resv 2		*Request Parameter* *Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
	Used to align subsequent data on a byte boundary.						
5	**HDOP**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD055 DOP		Dilution of Precision (DOP) indicates the contribution of satellite configuration geometry to positioning error. A lower DOP value is preferred because less error is being introduced. Reported as components: HDOP (Horizontal), VDOP (Vertical), TDOP (Time). Minimum DOP value is 1.0 (no error introduced).				
	DF69 Ratio, Relative measure,	**int16**	*Range:* +/-327.64	*Resolution:* 1x10E-2	Unit-less number		

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GNSS DOPs

PGN: 129539
hex: 1FA03

6	**VDOP**		*Byte Field Size:*	2	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD055 DOP			Dilution of Precision (DOP) indicates the contribution of satellite configuration geometry to positioning error. A lower DOP value is preferred because less error is being introduced. Reported as components: HDOP (Horizontal), VDOP (Vertical), TDOP (Time). Minimum DOP value is 1.0 (no error introduced).		
	DF69	Ratio, Relative measure,	**int16**	*Range:* +/-327.64	*Resolution:* 1x10E-2	Unit-less number
7	**TDOP**		*Byte Field Size:*	2	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD055 DOP			Dilution of Precision (DOP) indicates the contribution of satellite configuration geometry to positioning error. A lower DOP value is preferred because less error is being introduced. Reported as components: HDOP (Horizontal), VDOP (Vertical), TDOP (Time). Minimum DOP value is 1.0 (no error introduced).		
	DF69	Ratio, Relative measure,	**int16**	*Range:* +/-327.64	*Resolution:* 1x10E-2	Unit-less number

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GNSS Sats in View

PGN: 129540
hex: 1FA04

GNSS information on current satellites in view tagged by sequence ID. Information includes PRN, elevation, azimuth, and SNR. Field 4 defines the number of satellites. Fields 5 thru 11 define the satellite number and the information. These fields sequentially repeated for each satellite to be transmitted as indicated by "n" in fields 12 thru 18.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 23

1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD056 Sequence ID			An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252		*Resolution:* 1 bit	Unit-less number	
2	**Mode**		*Byte Field Size:* *Bit Field Size:* 2			*Request Parameter* Optional *Command Parameter:* Optional	
	DD072 Range Residual Mode			Range Residual used in position calculation or range residuals were calculated after the position. 0=range residuals were used to calculate data; 1=range residuals were calculated after the position., 2=Error, 3=Null			
	DF52 Bit field	**bit(n)**	*Range:* Variable		*Resolution:* 1	Used to construct bit fields	
3	**NMEA Reserved**		*Byte Field Size:* *Bit Field Size:* resv 6			*Request Parameter* *Command Parameter:*	
	DD001 Reserved field			Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:* Variable		*Resolution:* 1	Used to construct bit fields	
	Used to align subsequent data on a byte boundary.						
4	**Number of SVs**		*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD006 Generic counter, short			Numeric count, event counter, sequence counter			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252		*Resolution:* 1 bit	Unit-less number	
5	**PRN "1"**		*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD074 Satellite ID Number			0 = value not used, 1-32 = GPS, 33-64 = SBAS, Satellite, Based Augmentation System (ie WAAS) 65-96 = GLONASS. For GLONASS, satellites are identified by 64+satellite slot number. The slot numbers are 1 through 24 for the full GLONASS constellation of 24 satellites, this gives a range of 65 through 88. The numbers 89 through 96 are available if slot numbers above 24 are allocated to on-orbit spares.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252		*Resolution:* 1 bit	Unit-less number	

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GNSS Sats in View

PGN: 129540
hex: 1FA04

6	**Elevation "1"**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD054 Elevation		Angle above or below the horizon. -90° to +90°; negative below the horizon				
	DF04 Angle, signed	**int16**	*Range:*	+/-Pi rad	*Resolution:*	1x10E-4 rad	Resolution ~0.0057deg
7	**Azimuth "1"**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD127 Generic Direction -True		Degrees clockwise relative to True North.				
	DF02 Angle	**uint16**	*Range:*	0 to 2Pi rad	*Resolution:*	1x10E-4 rad	Resolution ~0.0057deg, 1 deg = .01745 rad
8	**SNR "1"**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD057 SNR Value		SNR expressed in C/No				
	DF31 dB, relative measure	**int16**	*Range:*	+/- 327.64 dB	*Resolution:*	1x10E-2 dB	
9	**Range Residuals 1**		*Byte Field Size:*	4	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD073 Range Residuals		Range Residual value in meters				
	DF79 Distance signed fine	**int32**	*Range:*	+/-~2.147x10E+4 m	*Resolution:*	1x10E-5 m	
10	**PRN Status "1"**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	4	*Command Parameter:*	Optional	
	DD124 PRN Usage Status		0 = Not Tracked, 1 = Tracked but not used in solution, 2 = Used in solution without Differential corrections, 3 = Differential Corrections available, 4 = Tracked with Differential Corrections, 5 = used with Differential Corrections, 6-13 =Reserved, 14 = Error, 15 = No Selection				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
11	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*		
			Bit Field Size:	resv 4	*Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.						

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GNSS Sats in View

PGN: 129540
hex: 1FA04

12 **PRN "n"** — Byte Field Size: 1 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Optional

DD074 Satellite ID Number

0 = value not used,
1-32 = GPS,
33-64 = SBAS, Satellite, Based Augmentation System (ie WAAS)
65-96 = GLONASS.
For GLONASS, satellites are identified by 64+satellite slot number. The slot numbers are 1 through 24 for the full GLONASS constellation of 24 satellites, this gives a range of 65 through 88. The numbers 89 through 96 are available if slot numbers above 24 are allocated to on-orbit spares.

DF53 Integer, 8 bit unsigned **uint8** — Range: 0 to 252 — Resolution: 1 bit — Unit-less number

Variable Number of fields, Field number 5 repeated

---

13 **Elevation "n"** — Byte Field Size: 2 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Optional

DD054 Elevation

Angle above or below the horizon. -90° to +90°; negative below the horizon

DF04 Angle, signed **int16** — Range: +/-Pi rad — Resolution: 1x10E-4 rad — Resolution ~0.0057deg

Variable Number of fields, Field number 6 repeated

---

14 **Azimuth "n"** — Byte Field Size: 2 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Optional

DD127 Generic Direction -True

Degrees clockwise relative to True North.

DF02 Angle **uint16** — Range: 0 to 2Pi rad — Resolution: 1x10E-4 rad — Resolution ~0.0057deg, 1 deg = .01745 rad

Variable Number of fields, Field number 7 repeated

---

15 **SNR "n"** — Byte Field Size: 2 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Optional

DD057 SNR Value

SNR expressed in C/No

DF31 dB, relative measure **int16** — Range: +/- 327.64 dB — Resolution: 1x10E-2 dB

Variable Number of fields, Field number 8 repeated

---

16 **Range Residuals "n"** — Byte Field Size: 4 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Optional

DD073 Range Residuals

Range Residual value in meters

DF79 Distance signed fine **int32** — Range: +/-~2.147x10E+4 m — Resolution: 1x10E-5 m

Variable number of fields, Filed Number 9 repeated

---

17 **PRN Status "n"** — Byte Field Size: — Bit Field Size: 4 — Request Parameter: Optional — Command Parameter: Optional

DD124 PRN Usage Status

0 = Not Tracked,
1 = Tracked but not used in solution,
2 = Used in solution without Differential corrections,
3 = Differential Corrections available,
4 = Tracked with Differential Corrections,
5 = used with Differential Corrections,
6-13 =Reserved,
14 = Error,
15 = No Selection

DF52 Bit field **bit(n)** — Range: Variable — Resolution: 1 — Used to construct bit fields

Variable Number of fields, Field number 10 repeated

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GNSS Sats in View

PGN: 129540
hex: 1FA04

18	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 4	*Command Parameter:*	

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Variable Number of fields, Field number 11 repeated

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GPS Almanac Data

PGN: 129541
hex: 1FA05

This PGN provides a single transmission that contains relevant almanac data for GPS products. The almanac contains satellite vehicle course orbital parameters. This information is not considered precise and is only valid for several months at a time. GPS products receive almanac data directly from the satellites.

This information would either be transmitted to and from GPS products for update, or system interrogation.

This information would generally be transmitted upon request, during calibration or installation, but not at regular intervals.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 24

Field #	Field Name					
1	**PRN**		Byte Field Size: 1 Bit Field Size:		Request Parameter: Optional Command Parameter: Optional	
	DD074 Satellite ID Number		0 = value not used, 1-32 = GPS, 33-64 = SBAS, Satellite, Based Augmentation System (ie WAAS) 65-96 = GLONASS. For GLONASS, satellites are identified by 64+satellite slot number. The slot numbers are 1 through 24 for the full GLONASS constellation of 24 satellites, this gives a range of 65 through 88. The numbers 89 through 96 are available if slot numbers above 24 are allocated to on-orbit spares.			
	DF53 Integer, 8 bit unsigned	**uint8**	Range: 0 to 252	Resolution: 1 bit	Unit-less number	
2	**GPS Week number**		Byte Field Size: 2 Bit Field Size:		Request Parameter: Optional Command Parameter: Optional	
	DD082 GPS Week Number		GPS week number. Starting on 6 Jan., 1980. The GPS week number roll over will not affect this value, i.e., it will continue to count up 1022, 1023, 1024, 1025.			
	DF54 Integer, 16 bit unsigned	**uint16**	Range: 0 to 65,532	Resolution: 1 bit	Unit-less number	
3	**SV Health Bits**		Byte Field Size: Bit Field Size: 8		Request Parameter: Optional Command Parameter: Optional	
	DD083 SV Health Bits		SV health, bits 17-24 of each almanac page. Reference ICD-GPS-200 paragraph 20.3.3.5.1.3, Table 20-VII and Table 20-VIII			
	DF52 Bit field	**bit(n)**	Range: Variable	Resolution: 1	Used to construct bit fields	
4	**Eccentricity**		Byte Field Size: Bit Field Size: 16		Request Parameter: Optional Command Parameter: Optional	
	DD084 Eccentricity, e		Eccentricity, e. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.			
	DF52 Bit field	**bit(n)**	Range: Variable	Resolution: 1	Used to construct bit fields	
5	**Almanac Reference Time**		Byte Field Size: Bit Field Size: 8		Request Parameter: Optional Command Parameter: Optional	
	DD085 Almanac Reference Time, toa		Almanac reference time. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.			
	DF52 Bit field	**bit(n)**	Range: Variable	Resolution: 1	Used to construct bit fields	
6	**Inclination Angle**		Byte Field Size: Bit Field Size: 16		Request Parameter: Optional Command Parameter: Optional	
	DD086 Almanac parameter, (sigma)I		Inclination angle. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.			
	DF52 Bit field	**bit(n)**	Range: Variable	Resolution: 1	Used to construct bit fields	

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7	**Rate of Right Ascension**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	16	*Command Parameter:*	Optional
	DD087 Almanac parameter, OMEGADOT	Rate of right ascension, OMEGADOT. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.			
	DF52 Bit field **bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
8	**Root of Semi-major Axis**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	24	*Command Parameter:*	Optional
	DD088 Almanac parameter, (A)1/2	Root of semi-major axis. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.			
	DF52 Bit field **bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
9	**Argument of Perigee**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	24	*Command Parameter:*	Optional
	DD089 Almanac parameter, (omega)	Argument of Perigee. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.			
	DF52 Bit field **bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
10	**Longitude of Ascension Node**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	24	*Command Parameter:*	Optional
	DD090 Almanac parameter, (omega)0	Longitude of ascension node. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.			
	DF52 Bit field **bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
11	**Mean Anomaly**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	24	*Command Parameter:*	Optional
	DD091 Almanac parameter, M0	Mean anomaly. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.			
	DF52 Bit field **bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
12	**Clock Parameter 1**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	11	*Command Parameter:*	Optional
	DD092 Almanac parameter, af0	Clock Parameter 1. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.			
	DF52 Bit field **bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
13	**Clock Parameter 2**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	11	*Command Parameter:*	Optional
	DD093 Almanac parameter, af1	Clock Parameter 2. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.			
	DF52 Bit field **bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
14	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 2	*Command Parameter:*	
	DD001 Reserved field	Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field **bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
	Used to align subsequent data on a byte boundary.				

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GNSS Pseudorange Noise Statistics

PGN: 129542
hex: 1FA06

GNSS pseudorange measurement noise statistics can be translated in the position domain in order to give statistical measures of the quality of the position solution. Intended for use with a Receiver Autonomous Integrity Monitoring (RAIM) application.

Single Frame: No | *Priority Default:* 6 | *Default Update Rate:* 1000 milliseconds | *Frequency:* 1. cycles per second
Destination: Global | *Query Support:* Optional | *Command Support:* Optional | *ACK Rqmnts:* None

Original Reference ID # 26

Field #	Field Name						
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD056 Sequence ID				An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.		
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
2	**RMS of Position Uncertainty**		*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD075 Error Distances				Error distances expressed in meters.		
	DF13 Distance, short	**uint16**	*Range:* 0 to 655.32 m	*Resolution:* 1x10E-2 m			
3	**STD of Major axis**		*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD075 Error Distances				Error distances expressed in meters.		
	DF13 Distance, short	**uint16**	*Range:* 0 to 655.32 m	*Resolution:* 1x10E-2 m			
4	**STD of Minor axis**		*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD075 Error Distances				Error distances expressed in meters.		
	DF13 Distance, short	**uint16**	*Range:* 0 to 655.32 m	*Resolution:* 1x10E-2 m			
5	**Orientation of Major axis**		*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD127 Generic Direction -True				Degrees clockwise relative to True North.		
	DF02 Angle	**uint16**	*Range:* 0 to 2Pi rad	*Resolution:* 1x10E-4 rad	Resolution ~0.0057deg, 1 deg = .01745 rad		
6	**STD of Lat Error**		*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD075 Error Distances				Error distances expressed in meters.		
	DF13 Distance, short	**uint16**	*Range:* 0 to 655.32 m	*Resolution:* 1x10E-2 m			
7	**STD of Lon Error**		*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD075 Error Distances				Error distances expressed in meters.		
	DF13 Distance, short	**uint16**	*Range:* 0 to 655.32 m	*Resolution:* 1x10E-2 m			

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GNSS Pseudorange Noise Statistics

PGN: 129542
hex: 1FA06

8	**STD of Alt Error**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD075 Error Distances		Error distances expressed in meters.			
	DF13 Distance, short	**uint16**	*Range:* 0 to 655.32 m		*Resolution:*	1x10E-2 m

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GNSS RAIM Output

PGN: 129545
hex: 1FA09

This PGN is used to provide the output from a GNSS Receiver's Receiver Autonomous Integrity Monitoring (RAIM) process. The Integrity field value is based upon the parameters set in PGN 130059 GNS RAIM Settings.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 84

Field #	Field Name						
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD056 Sequence ID		An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
2	**Integrity Flag**		*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter* Optional *Command Parameter:* Optional		
	DD209 GNSS Integrity		0 = No Integrity checking,* 1 = Safe, 2 = Caution, 3 = Unsafe * means the receiver does not have this capability				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
3	**NMEA Reserved**		*Byte Field Size:* *Bit Field Size:* resv 6		*Request Parameter* *Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
	Used to align subsequent data on a byte boundary.						
4	**Latitude expected error**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD220 Measure						
	DF14 Distance, short, signed	**int16**	*Range:* +/-327.64 m	*Resolution:* 1x10E-2 m			
5	**Longitude expected error**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD220 Measure						
	DF14 Distance, short, signed	**int16**	*Range:* +/-327.64 m	*Resolution:* 1x10E-2 m			
6	**Altitude expected error**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD220 Measure						
	DF14 Distance, short, signed	**int16**	*Range:* +/-327.64 m	*Resolution:* 1x10E-2 m			

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GNSS RAIM Output

PGN: 129545
hex: 1FA09

7	**SV ID of most likely failed sat**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD074 Satellite ID Number		0 = value not used, 1-32 = GPS, 33-64 = SBAS, Satellite, Based Augmentation System (ie WAAS) 65-96 = GLONASS. For GLONASS, satellites are identified by 64+satellite slot number. The slot numbers are 1 through 24 for the full GLONASS constellation of 24 satellites, this gives a range of 65 through 88. The numbers 89 through 96 are available if slot numbers above 24 are allocated to on-orbit spares.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
8	**Probability of missed detection**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD220 Measure					
	DF14 Distance, short, signed	**int16**	*Range:* +/-327.64 m	*Resolution:* 1x10E-2 m		
9	**Estimate of pseudorange bias**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD220 Measure					
	DF14 Distance, short, signed	**int16**	*Range:* +/-327.64 m	*Resolution:* 1x10E-2 m		
10	**Std Deviation of bias**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD220 Measure					
	DF14 Distance, short, signed	**int16**	*Range:* +/-327.64 m	*Resolution:* 1x10E-2 m		

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GNSS RAIM Settings

PGN: 129546
hex: 1FA0A

This PGN is used to report the control parameters for a GNSS Receiver Autonomous Integrity Monitoring (RAIM) process. The Command Group Function PGN 126208 provides the means to set these values over the network.

Single Frame: Yes *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 85

#	Field Name						
1	**Radial Position Error Maximum threshold**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional	
	DD075 Error Distances		Error distances expressed in meters.				
	DF13 Distance, short	**uint16**	*Range:* 0 to 655.32 m	*Resolution:* 1x10E-2 m			
2	**Probability of False Alarm**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional	
	DD138 Generic percent of range						
	DF30 Percent, Relative measure	**int8**	*Range:* +/- 124%	*Resolution:* 1%			
3	**Probability of Missed Detection**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional	
	DD138 Generic percent of range						
	DF30 Percent, Relative measure	**int8**	*Range:* +/- 124%	*Resolution:* 1%			
4	**Pseudorange Residual Filtering Time Constant**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional	
	DD210 Time Value, resolution 1 sec		Time in seconds				
	DF80 Time, 1sec	**uint16**	*Range:* 0 to 65532 seconds	*Resolution:* 1 second			
5	**NMEA Reserved**		*Byte Field Size:* *Bit Field Size:* resv 16		*Request Parameter* *Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
	Used to align subsequent data on a byte boundary.						

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GNSS Pseudorange Error Statistics

PGN: 129547
hex: 1FA0B

This parameter group is used to support Receiver Autononmous Integrity Monitoring (RAIM). Pseudorange measurement error statistics can be translated in the position domain in order to give statistical measures of the quality of the position solution.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 89

Field #	Field Name						
1	**Sequence ID**		Byte Field Size: 1 Bit Field Size:			Request Parameter: Optional Command Parameter: Optional	
	DD056 Sequence ID				An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.		
	DF53 Integer, 8 bit unsigned	**uint8**	Range: 0 to 252	Resolution: 1 bit	Unit-less number		
2	**RMS Std Dev of Range Inputs**		Byte Field Size: 2 Bit Field Size:			Request Parameter: Optional Command Parameter: Optional	
	DD219 Standard Deviation						
	DF13 Distance, short	**uint16**	Range: 0 to 655.32 m	Resolution: 1x10E-2 m			
	RMS value of the standard deviation of the range inputs to the navigation process. Range inputs include pseudoranges & DGNSS corrections.						
3	**Std Dev major error ellipse**		Byte Field Size: 2 Bit Field Size:			Request Parameter: Optional Command Parameter: Optional	
	DD219 Standard Deviation						
	DF13 Distance, short	**uint16**	Range: 0 to 655.32 m	Resolution: 1x10E-2 m			
	Standard deviation of the semi-major axis of error ellipse (meters)						
4	**Std Dev minor error ellipse**		Byte Field Size: 2 Bit Field Size:			Request Parameter: Optional Command Parameter: Optional	
	DD219 Standard Deviation						
	DF13 Distance, short	**uint16**	Range: 0 to 655.32 m	Resolution: 1x10E-2 m			
	Standard deviation of the semi-minor axis of error ellipse (meters).						
5	**Orientation of error ellipse**		Byte Field Size: 2 Bit Field Size:			Request Parameter: Optional Command Parameter: Optional	
	DD127 Generic Direction -True				Degrees clockwise relative to True North.		
	DF02 Angle	**uint16**	Range: 0 to 2Pi rad	Resolution: 1x10E-4 rad	Resolution ~0.0057deg, 1 deg = .01745 rad		
	Orientation of semi-major axis of error ellipse (from true north)						
6	**Std Dev Latitude error**		Byte Field Size: 2 Bit Field Size:			Request Parameter: Optional Command Parameter: Optional	
	DD219 Standard Deviation						
	DF13 Distance, short	**uint16**	Range: 0 to 655.32 m	Resolution: 1x10E-2 m			
	Standard deviation of Latitude error (meters)						

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GNSS Pseudorange Error Statistics

PGN: 129547
hex: 1FA0B

7	**Std Dev Longitude error**	*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* *Command Parameter:*	Optional Optional
	DD219 Standard Deviation					
	DF13 Distance, short	**uint16**	*Range:* 0 to 655.32 m	*Resolution:* 1x10E-2 m		
	Standard deviation of Longitude error (meters)					
8	**Std Dev Altitude error**	*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* *Command Parameter:*	Optional Optional
	DD219 Standard Deviation					
	DF13 Distance, short	**uint16**	*Range:* 0 to 655.32 m	*Resolution:* 1x10E-2 m		
	Standard deviation of altitude error (meters)					

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DGNSS Corrections

PGN: 129549
hex: 1FA0D

This PGN provides a means to pass differential GNSS corrections between NMEA devices. Passing DGNSS data this way allows for more flexibility than traditional methods. One differential correction receiver could supply multiple GNSS receivers. Multiple differential correction receivers or data streams could be connected to a GNSS receiver allowing for network DGNSS approaches. This PGN can accommodate DGPS and DGLONASS corrections. Future systems can be indicated by allocation of the reserved states in field 3. These corrections can be related to the position solution and to time through proper application of the sequence ID field.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 86

Field #	Field Name					
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD056 Sequence ID		An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
2	**Reference Station ID**		*Byte Field Size:* *Bit Field Size:* 12		*Request Parameter* Optional *Command Parameter:* Optional	
	DD071 Ref Station		Reference Station ID. Reference Station number as provided by the Service Provider.[Reference document required]			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
3	**Reference Station Type**		*Byte Field Size:* *Bit Field Size:* 4		*Request Parameter* Optional *Command Parameter:* Optional	
	DD070 Ref Station Type		Reference Station Type. 0x0=GPS; 0x1=GLONASS; 0x2 to 0xD=Reserved; 0XE=Error; 0XF=Null			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
4	**Time of corrections**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD211 Time Value, resolution 0.1 sec		Time in seconds			
	DF81 Time, 0.1sec	**uint16**	*Range:* 0 to 6553.2 seconds	*Resolution:* 1x10E-1 sec		
5	**Station Health**		*Byte Field Size:* *Bit Field Size:* 4		*Request Parameter* Optional *Command Parameter:* Optional	
	DD212 Station Heath		0x00 = Not Working, 0x01 = Unmonitored, 0x02 = Healthy & Operational, 0x03 = Healthy & in Test Mode, 0x04 = In Test Mode - DO NOT USE, 0x05 - 0x15 = Reserved			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	

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6	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*		
			Bit Field Size:	resv 4	*Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.						
7	**Satellite ID**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD074 Satellite ID Number		0 = value not used, 1-32 = GPS, 33-64 = SBAS, Satellite, Based Augmentation System (ie WAAS) 65-96 = GLONASS. For GLONASS, satellites are identified by 64+satellite slot number. The slot numbers are 1 through 24 for the full GLONASS constellation of 24 satellites, this gives a range of 65 through 88. The numbers 89 through 96 are available if slot numbers above 24 are allocated to on-orbit spares.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number
8	**PRC**		*Byte Field Size:*	4	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD213 Distance, int 32 4dp						
	DF83 Distance, signed 4dp	**int32**	*Range:*	+/-~2.147x10E+5 m	*Resolution:*	1x10E-4 m	
9	**RRC**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD214 Generic Speed						
	DF82 Speed, signed 4dpt	**int16**	*Range:*	+/-3.2764 m/s	*Resolution:*	1x10E-4 m/s	
10	**UDRE**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD195 Distance, short		Dependent upon PG Field definition.				
	DF13 Distance, short	**uint16**	*Range:*	0 to 655.32 m	*Resolution:*	1x10E-2 m	
	value 655.32 (all 1's) indicates satellite invalid do not use or stop using immediately.						
11	**IOD**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number

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GNSS Differential Correction Receiver Interface

PGN: 129550
hex: 1FA0E

GNSS common differential correction receiver parameter status.

Single Frame: Yes *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 27

1 Channel
Byte Field Size: 1 *Bit Field Size:*
Request Parameter Optional *Command Parameter:* Optional

DD076 Receiver channel number
The channel number of the correction receiver. If the receiver only has one beacon input channel, this value shall be 1; 0 is undefined.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

2 Frequency
Byte Field Size: 4 *Bit Field Size:*
Request Parameter Optional *Command Parameter:* Optional

DD077 Differential Correction Receiver frequency
This is the input frequency of the correction receiver.

DF21 Frequency **uint32** *Range:* 0 to ~4.295x10E+10 Hz *Resolution:* 10 Hz

3 Serial Interface Bit Rate
Byte Field Size: *Bit Field Size:* 5
Request Parameter Optional *Command Parameter:* Optional

DD078 Differential Correction Broadcast Bit Rate
This is the bit rate of the correction receiver.
0 = 25bps,
1 = 50bps,
2 = 100bps,
3 = 200bps,
4 = 300bps,
5 = 500bps,
6 = 1200bps,
7 = 2400bps,
8 = 4800bps,
9 = 9600bps
10 = 19200bps,
11 = 38400bps,
12 = 57600bps,
13-29 = Reserved,
30 = Error,
31 = Null

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

4 Serial Interface Detection Mode
Byte Field Size: *Bit Field Size:* 3
Request Parameter Optional *Command Parameter:* Optional

DD079 Mode, Bit Rate
This is the mode of operation for the correction receiver.
0 = Auto bit rate set,
1 = Manual bit rate set,
2-5 = Reserved
6 = Error,
7 = Null.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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5	**Differential Source**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD125 Differential Source		0 = Auto Select, 1 = Loran Communications 2 = MSK Beacon, 3 = FM Subcarrier, 4 = AIS (Automatic Identification System), 5 = Other Ground-based Radio, 6 = SBAS (Satellite Based Augmentation System, 7 = Other Satellite, 8-13 = Reserved, 14 = Error, 15 = No Selection			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
6	**Differential Operation Mode**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD126 Differential Operating Mode		0 = Manual, 1 = Auto Power, 2 = Auto Range, 3-13 = Reserved, 14 = Error, 15 = No Selection			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
7	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*	
			Bit Field Size:	resv 8	*Command Parameter:*	
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
	Used to align subsequent data on a byte boundary.					

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GNSS Differential Correction Receiver Signal

PGN: 129551
hex: 1FA0F

GNSS differential correction receiver status tagged by sequence ID. Status information includes frequency, SNR, and use as a correction source.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 28

Field #	Field Name						
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD056 Sequence ID			An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252		*Resolution:* 1 bit		Unit-less number
2	**Channel**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD076 Receiver channel number			The channel number of the correction receiver. If the receiver only has one beacon input channel, this value shall be 1; 0 is undefined.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252		*Resolution:* 1 bit		Unit-less number
3	**Signal Strength**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD080 Correction Receiver Signal Strength			This is the signal strength expressed in dB with respect to 1uV/m.			
	DF16 Electric field	**int32**	*Range:* +/-327.64 dB re: uV/m		*Resolution:* 1x10E-2 dB re: uV/m		
4	**Signal SNR**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD081 SNR Value			SNR expressed in dB.			
	DF31 dB, relative measure	**int16**	*Range:* +/- 327.64 dB		*Resolution:* 1x10E-2 dB		
5	**Frequency**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD077 Differential Correction Receiver frequency			This is the input frequency of the correction receiver.			
	DF21 Frequency	**uint32**	*Range:* 0 to ~4.295x10E+10 Hz		*Resolution:* 10 Hz		
6	**Station Type**		*Byte Field Size:* *Bit Field Size:* 4		*Request Parameter* Optional *Command Parameter:* Optional		
	DD070 Ref Station Type			Reference Station Type. 0x0=GPS; 0x1=GLONASS; 0x2 to 0xD=Reserved; 0XE=Error; 0XF=Null			
	DF52 Bit field	**bit(n)**	*Range:* Variable		*Resolution:* 1		Used to construct bit fields

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GNSS Differential Correction Receiver Signal

PGN: 129551
hex: 1FA0F

7	**Station ID**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	12	*Command Parameter:*	Optional
	DD071 Ref Station		Reference Station ID. Reference Station number as provided by the Service Provider.[Reference document required]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
8	**Differential Signal Bit Rate**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	5	*Command Parameter:*	Optional
	DD078 Differential Correction Broadcast Bit Rate		This is the bit rate of the correction receiver. 0 = 25bps, 1 = 50bps, 2 = 100bps, 3 = 200bps, 4 = 300bps, 5 = 500bps, 6 = 1200bps, 7 = 2400bps, 8 = 4800bps, 9 = 9600bps 10 = 19200bps, 11 = 38400bps, 12 = 57600bps, 13-29 = Reserved, 30 = Error, 31 = Null			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
9	**Differential Signal Detection Mode**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	3	*Command Parameter:*	Optional
	DD079 Mode, Bit Rate		This is the mode of operation for the correction receiver. 0 = Auto bit rate set, 1 = Manual bit rate set, 2-5 = Reserved 6 = Error, 7 = Null.			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
10	**Used as Correction Source**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
11	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*	
			Bit Field Size:	resv 2	*Command Parameter:*	
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.					

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GNSS Differential Correction Receiver Signal

PGN: 129551
hex: 1FA0F

12	**Differential Source**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD125 Differential Source		0 = Auto Select, 1 = Loran Communications 2 = MSK Beacon, 3 = FM Subcarrier, 4 = AIS (Automatic Identification System), 5 = Other Ground-based Radio, 6 = SBAS (Satellite Based Augmentation System, 7 = Other Satellite, 8-13 = Reserved, 14 = Error, 15 = No Selection			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
13	**Time Since Last Sat Differential Sync**		*Byte Field Size:*	2	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD060 Differential Age		Age of Differential corrections			
	DF66 Time interval, .01sec	**uint16**	*Range:*	0 to 655.32s	*Resolution:*	1x10E-2sec
14	**Satellite Service ID No.**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	16	*Command Parameter:*	Optional
	DD143 Satellite Service ID		Satellite Service ID number as provided by the Service Provider.[Reference document required]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields

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GLONASS Almanac Data

PGN: 129556
hex: 1FA14

This PGN provides a single transmission that contains relevant almanac data for Glonass products. The almanac contains satellite vehicle course orbital parameters. This information is not considered precise and is only valid for several months at a time. Glonass products receive almanac data directly from the satellites.

This information would either be transmitted to and from Glonass products for update, or system interrogation.

This information would generally be transmitted upon request, during calibration or installation, but not at regular intervals.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field #	Field Name					
						Original Reference ID # 30
1	**PRN**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD074 Satellite ID Number		0 = value not used, 1-32 = GPS, 33-64 = SBAS, Satellite, Based Augmentation System (ie WAAS) 65-96 = GLONASS. For GLONASS, satellites are identified by 64+satellite slot number. The slot numbers are 1 through 24 for the full GLONASS constellation of 24 satellites, this gives a range of 65 through 88. The numbers 89 through 96 are available if slot numbers above 24 are allocated to on-orbit spares.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
2	**NA**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD094 Almanac parameter, NA		Calendar day count within the four year period beginning with the previous leap year			
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number	
3	**NMEA Reserved**		*Byte Field Size:* *Bit Field Size:* resv 2		*Request Parameter* *Command Parameter:*	
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
	Used to align subsequent data on a byte boundary.					
4	**CnA**		*Byte Field Size:* *Bit Field Size:* 1		*Request Parameter* Optional *Command Parameter:* Optional	
	DD095 Almanac parameter, CnA		Generalized health of the Satellite, reference GLONASS ICD.			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
5	**HnA**		*Byte Field Size:* *Bit Field Size:* 5		*Request Parameter* Optional *Command Parameter:* Optional	
	DD096 Almanac parameter, HnA		Carrier frequency number respectively, reference GLONASS ICD.			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
6	**(epsilon)nA**		*Byte Field Size:* *Bit Field Size:* 16		*Request Parameter* Optional *Command Parameter:* Optional	
	DD097 Almanac parameter, (epsilon)nA		Eccentricity, reference GLONASS ICD Section 4.5 Table 4.3 (fill unused bits with zeros).			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	

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GLONASS Almanac Data

PGN: 129556
hex: 1FA14

7	**(deltaTnA)DOT**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	8	*Command Parameter:*	Optional	
	DD098 Almanac parameter, (deltaTnA)DOT		Rate of change of the draconitic circling time, reference GLONASS ICD Section 4.5 Table 4.3 (fill unused bits with zeros).			
	DF52 Bit field **bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
8	**(omega)nA**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	16	*Command Parameter:*	Optional	
	DD099 Almanac parameter, (omega)nA		Argument of Perigee, reference GLONASS ICD Section 4.5 Table 4.3 (fill unused bits with zeros).			
	DF52 Bit field **bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
9	**(delta)TnA**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	24	*Command Parameter:*	Optional	
	DD100 Almanac parameter, (delta)TnA		Correction to the average value of the draconitic circling time, reference GLONASS ICD Section 4.5 Table 4.3 (fill unused bits with zeros).			
	DF52 Bit field **bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
10	**tnA**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	24	*Command Parameter:*	Optional	
	DD101 Almanac parameter, tnA		Time of the ascension node, almanac reference time, reference GLONASS ICD Section 4.5 Table 4.3 (fill unused bits with zeros).			
	DF52 Bit field **bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
11	**(lambda)nA**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	24	*Command Parameter:*	Optional	
	DD102 Almanac parameter, (lambda)nA		Greenwich longitude of the ascension node, reference GLONASS ICD Section 4.5 Table 4.3 (fill unused bits with zeros).			
	DF52 Bit field **bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
12	**(delta)inA**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	24	*Command Parameter:*	Optional	
	DD103 Almanac parameter, (delta)inA		Correction to the average value of the inclination angle, reference GLONASS ICD Section 4.5 Table 4.3 (fill unused bits with zeros).			
	DF52 Bit field **bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
13	**tcA**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	28	*Command Parameter:*	Optional	
	DD104 Almanac parameter, (tau)cA		System time scale correction, reference GLONASS ICD Section 4.5 Table 4.3 (fill unused bits with zeros).			
	DF52 Bit field **bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
14	**tnA**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	12	*Command Parameter:*	Optional	
	DD105 Almanac parameter, (tau)nA		Course value of the time scale shift, reference GLONASS ICD Section 4.5 Table 4.3 (fill unused bits with zeros).			
	DF52 Bit field **bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields

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AIS DGNSS Broadcast Binary Message

PGN: 129792
hex: 1FB00

This parameter group provides data associated with the ITU-R M.1371 Message 17 GNSS Broadcast Binary Message containing DGNSS corrections from a base station. An AIS device may generate this parameter group either upon receiving a VHF data link message 17, or upon receipt of an ISO or NMEA request PGN (see ITU-R M.1371-1 for additional information). Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 121

Field #	Field Name						
1	**Message ID**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	6	*Command Parameter:*	Optional	
	DD188 AIS Message Identifier		Message Identifier (range of 0 to 63). See the latest version of ITU-R M.1371 for more information.				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
	17 = GNSS Broadcast Binary Message						
2	**Repeat Indicator**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	2	*Command Parameter:*	Optional	
	DD185 AIS Repeater Indicator		Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3). 0 = Default 1 = First retransmission 2 = Second retransmission 3 = Final retransmission See the latest version of ITU-R M.1371 for more information.				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
3	**Source ID**		*Byte Field Size:*	4	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD010 Generic numeric ID, large		Number of route, waypoint, event, mark, etc.				
	DF55 Integer, 32 bit unsigned	**uint32**	*Range:*	0 to 4,294,967,292	*Resolution:*	1 bit	Unit-less number
	MMSI number of base station reporting DGNSS information.						
4	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*		
			Bit Field Size:	resv 1	*Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
	Used to align subsequent data on byte boundary.						
5	**AIS Transceiver Information**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	5	*Command Parameter:*	Optional	
	DD246 AIS Transceiver Information		0 = Channel A VDL reception, 1 = Channel B VDL reception, 2 = Channel A VDL transmission, 3 = Channel B VDL transmission, 4 = Own information not broadcast, 5-31 = Reserved.				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields

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6 AIS Spare

Byte Field Size:
Bit Field Size: resv 2

Request Parameter
Command Parameter:

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** Range: N2KUnitless Resolution: 1

Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

7 Longitude

Byte Field Size: 4
Bit Field Size:

Request Parameter: Optional
Command Parameter: Optional

DD023 Longitude, WGS-84

Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF25 Longitude **int32** Range: +/- 180 deg Resolution: 1x10E-7 deg

"-" = West, resolution ~1.1 cm

Longitude of base station reporting DGNSS information.

8 Latitude

Byte Field Size: 4
Bit Field Size:

Request Parameter: Optional
Command Parameter: Optional

DD022 Latitude, WGS-84

Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23 Latitude **int32** Range: +/- 90 deg Resolution: 1x10E-7 deg

"-" = South, resolution ~1.1 cm

Latitude of base station reporting DGNSS information.

9 NMEA Reserved

Byte Field Size:
Bit Field Size: resv 3

Request Parameter
Command Parameter:

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** Range: Variable Resolution: 1

Used to construct bit fields

Used to align subsequent data on byte boundary.

10 AIS Spare

Byte Field Size:
Bit Field Size: resv 5

Request Parameter
Command Parameter:

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** Range: N2KUnitless Resolution: 1

Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

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AIS DGNSS Broadcast Binary Message

PGN: 129792
hex: 1FB00

11 **Number of Bits in Binary Data Field**
Byte Field Size: 2 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

Indicates the number of binary data bits that are contained within the Binary Data field.

12 **Binary Data**
Byte Field Size: *Request Parameter* Optional
Bit Field Size: n *Command Parameter:* Optional

DD142 Binary Bit Field — Binary data bit field.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Differential correction data. See ITU-R M.1371-1. The size of this field is provided in field 11, Number of Bits in Binary Data Field.

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AIS UTC and Date Report

PGN: 129793
hex: 1FB01

This parameter group provides data from ITU-R M.1371 message 4 Base Station Report providing position, time, date, and current slot number of a base station, and 11 UTC and date response message providing current UTC and date if available. An AIS device may generate this parameter group either upon receiving a VHF data link message 4 or 11, or upon receipt of an ISO or NMEA request PGN. The Command Group Function PGN 126208 may be used with this PGN to configure base station parameters (see ITU-R M.1371-1 for additional information). Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

Single Frame: No *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 110

1 Message ID
Byte Field Size: *Bit Field Size:* 6 *Request Parameter* Required *Command Parameter:* Optional

DD188 AIS Message Identifier
Message Identifier (range of 0 to 63).

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
4 = Base Station Report
11 = UTC and Date Response

2 Repeat Indicator
Byte Field Size: *Bit Field Size:* 2 *Request Parameter* Optional *Command Parameter:* Optional

DD185 AIS Repeater Indicator
Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3).

0 = Default
1 = First retransmission
2 = Second retransmission
3 = Final retransmission

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 User ID
Byte Field Size: 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD010 Generic numeric ID, large
Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number
MMSI number of station reporting its UTC and date.

4 Longitude
Byte Field Size: 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD023 Longitude, WGS-84
Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF25 Longitude **int32** *Range:* +/- 180 deg *Resolution:* 1x10E-7 deg "-" = West, resolution ~1.1 cm
Longitude of station reporting its UTC and date.

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AIS UTC and Date Report

PGN: 129793
hex: 1FB01

5 Latitude

Byte Field Size: 4 | Bit Field Size:

Request Parameter: Optional | Command Parameter: Optional

DD022 Latitude, WGS-84

Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23 Latitude **int32** Range: +/- 90 deg Resolution: 1x10E-7 deg "-" = South, resolution ~1.1 cm

Latitude of station reporting its UTC and date.

---

6 Position accuracy

Byte Field Size: | Bit Field Size: 1

Request Parameter: Optional | Command Parameter: Optional

DD184 AIS Position Accuracy

0 = low accuracy > 10m such as nondifferential GNSS (default),
1 = high accuracy < 10m such as DGNSS

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

7 RAIM-flag

Byte Field Size: | Bit Field Size: 1

Request Parameter: Optional | Command Parameter: Optional

DD189 AIS RAIM-flag

0 = RAIM not in use (default),
1 = RAIM in use

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

8 NMEA Reserved

Byte Field Size: | Bit Field Size: resv 6

Request Parameter: Optional | Command Parameter: Optional

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

---

9 Position time

Byte Field Size: 4 | Bit Field Size:

Request Parameter: Optional | Command Parameter: Optional

DD158 Generic time of day

24 hour clock, 0 = midnight, time is in UTC

DF06 Time of day **uint32** Range: 0 to 86,401 s Resolution: 1x10E-4 s ~24 hours, 0 = midnight, range allows for up to two leap seconds per day

Time of position of station reporting its UTC and date.

---

10 Communication State

Byte Field Size: | Bit Field Size: 19

Request Parameter: Optional | Command Parameter: Optional

DD187 AIS Communication State

The Communication State contains information used by the various TDMA slot allocation algorithms and synchronization information

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

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AIS UTC and Date Report

PGN: 129793
hex: 1FB01

11 **AIS Transceiver Information**

Byte Field Size: | *Bit Field Size:* 5 | *Request Parameter* Optional | *Command Parameter:* Optional

DD246 AIS Transceiver Information

0 = Channel A VDL reception,
1 = Channel B VDL reception,
2 = Channel A VDL transmission,
3 = Channel B VDL transmission,
4 = Own information not broadcast,
5-31 = Reserved.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

12 **Position Date**

Byte Field Size: 2 | *Bit Field Size:* | *Request Parameter* Optional | *Command Parameter:* Optional

DD039 Generic date

Days since January 1, 1970, Date is relative to UTC Time.

DF41 Date, day count **uint16** *Range:* 0 to 65,532 days *Resolution:* 1 day 0 = January 1, 1970, max = ~179 years

---

13 **NMEA Reserved**

Byte Field Size: | *Bit Field Size:* resv 4 | *Request Parameter* Optional | *Command Parameter:*

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

---

14 **Type of Electronic Positioning Device**

Byte Field Size: | *Bit Field Size:* 4 | *Request Parameter* Optional | *Command Parameter:* Optional

DD191 AIS Electronic Positioning Device Type

0 =Undefined (default)
1 = GPS
2 = GLONASS
3 = Combined GPS/GLONASS
4 = Loran-C
5 = Chayka
6 = Integrated Navigation System
7 = Surveyed; For fixed AtoN and virtual AtoN, the charted position should be used. The accurate position enhances its function as a radar reference target
8 = Galilieo
9-15 = Reserved for future use.

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

15 **AIS Spare**

Byte Field Size: | *Bit Field Size:* resv 10 | *Request Parameter* | *Command Parameter:*

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Netowrk Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

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AIS UTC and Date Report

PGN: 129793
hex: 1FB01

16	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*	
			Bit Field Size:	resv 6	*Command Parameter:*	
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	Used to align subsequent data on byte boundary.					
17	**Sequence ID**		*Byte Field Size:*	1	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD056 Sequence ID		An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:* 1 bit	Unit-less number

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AIS Class A Static and Voyage Related Data

PGN: 129794
hex: 1FB02

This parameter group provides data associated with the ITU-R M.1371 Message 5 Ship Static and Voyage Related Data Message. An AIS device may generate this parameter group either upon receiving a VHF data link message 5, or upon receipt of an ISO or NMEA request PGN. The Command Group Function PGN 126208 may be used with this PGN to configure static and voyage related parameters (see ITU-R M.1371-1 for additional information). Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

The Sequence ID field (SID) is used to link this PGN to other related PGN's from the same source address. When no linkage exists, the value of the SID shall be set to 255.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 111

Field #	Field Name					
1	**Message ID**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	6	*Command Parameter:*	Optional
	DD188 AIS Message Identifier		Message Identifier (range of 0 to 63). See the latest version of ITU-R M.1371 for more information.			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	5 = Ship Static and Voyage Related Data Message.					
2	**Repeat Indicator**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD185 AIS Repeater Indicator		Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3). 0 = Default 1 = First retransmission 2 = Second retransmission 3 = Final retransmission See the latest version of ITU-R M.1371 for more information.			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
3	**User ID**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD010 Generic numeric ID, large		Number of route, waypoint, event, mark, etc.			
	DF55 Integer, 32 bit unsigned	**uint32**	*Range:*	0 to 4,294,967,292	*Resolution:* 1 bit	Unit-less number
	MMSI number of mobile station reporting its static and voyage related data.					
4	**IMO**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD010 Generic numeric ID, large		Number of route, waypoint, event, mark, etc.			
	DF55 Integer, 32 bit unsigned	**uint32**	*Range:*	0 to 4,294,967,292	*Resolution:* 1 bit	Unit-less number
	IMO number of mobile station reporting its static and voyage related data.					

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AIS Class A Static and Voyage Related Data

PGN: 129794
hex: 1FB02

5 Call Sign

Byte Field Size: char | n
Bit Field Size:
Request Parameter Optional
Command Parameter: Optional

DD192 Generic String, ASCII, Fixed length — Length specified by PGN field definition.

DF63 String, fixed **char8(n)** *Range:* 0 to 1,785 characters *Resolution:* 1 char — 0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.

This is a 7 character string, see ITU-R M.1371-1 for more information.

---

6 Name

Byte Field Size: char | n
Bit Field Size:
Request Parameter Optional
Command Parameter: Optional

DD192 Generic String, ASCII, Fixed length — Length specified by PGN field definition.

DF63 String, fixed **char8(n)** *Range:* 0 to 1,785 characters *Resolution:* 1 char — 0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.

This is a 20 character string, see ITU-R M.1371-1 for more information.

---

7 Ship/Cargo Type

Byte Field Size:
Bit Field Size: 8
Request Parameter Optional
Command Parameter: Optional

DD193 Ship/Cargo Type — 0=Not Available or no ship (default), 1-99= (See the latest version of ITU-R M.1371 Section 3.3.8.2.3.2 Table 18), 100-199=Reserved for Regional (See the latest version of ITU-R M.1371), 200-255=Reserved for future (See the latest version of ITU-R M.1371).

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 — Used to construct bit fields

---

8 Ship Length

Byte Field Size: 2
Bit Field Size:
Request Parameter Optional
Command Parameter: Optional

DD194 Distance, medium — Dependent upon PG Field definition.

DF75 Distance, Medium **uint16** *Range:* 0 to 6553.2 m *Resolution:* 1x10E-1 m

Length of mobile station reporting its static and voyage related data. A value of 65535 indicates that data is not available.

---

9 Ship Beam

Byte Field Size: 2
Bit Field Size:
Request Parameter Optional
Command Parameter: Optional

DD194 Distance, medium — Dependent upon PG Field definition.

DF75 Distance, Medium **uint16** *Range:* 0 to 6553.2 m *Resolution:* 1x10E-1 m

Beam of mobile station reporting its static and voyage related data. A value of 65535 indicates that data is not available.

---

10 Position Reference Point from Starboard

Byte Field Size: 2
Bit Field Size:
Request Parameter Optional
Command Parameter: Optional

DD194 Distance, medium — Dependent upon PG Field definition.

DF75 Distance, Medium **uint16** *Range:* 0 to 6553.2 m *Resolution:* 1x10E-1 m

Position reference point from starboard side of mobile station reporting its static and voyage related data. A value of 65535 indicates that data is not available.

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AIS Class A Static and Voyage Related Data

PGN: 129794
hex: 1FB02

11 **Position Reference Point aft of Ship's Bow** — Byte Field Size: 2 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Optional

DD194 Distance, medium — Dependent upon PG Field definition.

DF75 Distance, Medium **uint16** Range: 0 to 6553.2 m Resolution: 1x10E-1 m

Position reference point from aft of ship's bow of mobile station reporting its static and voyage related data. A value of 65535 indicates that data is not available.

---

12 **Estimated Date of Arrival** — Byte Field Size: 2 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Optional

DD039 Generic date — Days since January 1, 1970, Date is relative to UTC Time.

DF41 Date, day count **uint16** Range: 0 to 65,532 days Resolution: 1 day — 0 = January 1, 1970, max = ~179 years

EDA of mobile station reporting its static and voyage related data.

---

13 **Estimated Time of Arrival** — Byte Field Size: 4 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Optional

DD158 Generic time of day — 24 hour clock, 0 = midnight, time is in UTC

DF06 Time of day **uint32** Range: 0 to 86,401 s Resolution: 1x10E-4 s — ~24 hours, 0 = midnight, range allows for up to two leap seconds per day

ETA of mobile station reporting its static and voyage related data.

---

14 **Draft** — Byte Field Size: 2 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Optional

DD196 Draft — The depth of a ship in the water. The vertical distance between the waterline and the keel.

DF13 Distance, short **uint16** Range: 0 to 655.32 m Resolution: 1x10E-2 m

---

15 **Destination** — Byte Field Size: char n — Bit Field Size: — Request Parameter: Optional — Command Parameter: Optional

DD192 Generic String, ASCII, Fixed length — Length specified by PGN field definition.

DF63 String, fixed **char8(n)** Range: 0 to 1,785 characters Resolution: 1 char — 0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.

This is a 20 character string, see ITU-R M.1371-1 for more information.

---

16 **AIS Version** — Byte Field Size: — Bit Field Size: 2 — Request Parameter: Optional — Command Parameter: Optional

DD304 AIS Version Indicator

0 = Station compliant with AIS edition 0
1-3 = Station compliant with AIS editions 1, 2, and 3

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 — Used to construct bit fields

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17 **Type of Electronic Positioning Device**

Byte Field Size:
Bit Field Size: 4
Request Parameter: Optional
Command Parameter: Optional

DD191 AIS Electronic Positioning Device Type

0 =Undefined (default)
1 = GPS
2 = GLONASS
3 = Combined GPS/GLONASS
4 = Loran-C
5 = Chayka
6 = Integrated Navigation System
7 = Surveyed; For fixed AtoN and virtual AtoN, the charted position should be used. The accurate position enhances its function as a radar reference target
8 = Galilieo
9-15 = Reserved for future use.

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

18 **Data Terminal Equipment (DTE)**

Byte Field Size:
Bit Field Size: 1
Request Parameter: Optional
Command Parameter: Optional

DD242 Data Terminal Equipment (DTE)

0=Available,
1=not available.

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

19 **AIS Spare**

Byte Field Size:
Bit Field Size: resv 1
Request Parameter:
Command Parameter:

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** Range: N2KUnitless Resolution: 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

---

20 **AIS Transceiver Information**

Byte Field Size:
Bit Field Size: 5
Request Parameter: Optional
Command Parameter: Optional

DD246 AIS Transceiver Information

0 = Channel A VDL reception,
1 = Channel B VDL reception,
2 = Channel A VDL transmission,
3 = Channel B VDL transmission,
4 = Own information not broadcast,
5-31 = Reserved.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

21 **NMEA Reserved**

Byte Field Size:
Bit Field Size: resv 3
Request Parameter:
Command Parameter:

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

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AIS Class A Static and Voyage Related Data

PGN: 129794
hex: 1FB02

22	**Sequence ID**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional

DD056 Sequence ID

An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53	Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number

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AIS Addressed Binary Message

PGN: 129795
hex: 1FB03

This parameter group provides data associated with the ITU-R M.1371 Message 6 Addressed Binary Message supporting address communication of binary data. An AIS device may generate this parameter group either upon receiving a VHF data link message 6, or upon receipt of an ISO or NMEA request PGN. The Command Group Function PGN 126208 may be used with this PGN to configure parameters such as the Destination ID and Binary Data (see ITU-R M.1371-1 for additional information). Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

Single Frame: No *Priority Default:* 5 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 112

1 **Message ID** *Byte Field Size:* *Bit Field Size:* 6 *Request Parameter* Optional *Command Parameter:* Optional

DD188 AIS Message Identifier

Message Identifier (range of 0 to 63).

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

6 = Addressed Binary Message.

2 **Repeat Indicator** *Byte Field Size:* *Bit Field Size:* 2 *Request Parameter* Optional *Command Parameter:* Optional

DD185 AIS Repeater Indicator

Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3).

0 = Default
1 = First retransmission
2 = Second retransmission
3 = Final retransmission

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 **Source ID** *Byte Field Size:* 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD010 Generic numeric ID, large

Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number

MMSI number of source station.

4 **NMEA Reserved** *Byte Field Size:* *Bit Field Size:* resv 1 *Request Parameter* *Command Parameter:*

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

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5 **AIS Transceiver Information**

Byte Field Size: *Bit Field Size:* 5

Request Parameter Optional
Command Parameter: Optional

DD246 AIS Transceiver Information

0 = Channel A VDL reception,
1 = Channel B VDL reception,
2 = Channel A VDL transmission,
3 = Channel B VDL transmission,
4 = Own information not broadcast,
5-31 = Reserved.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

6 **Sequence Number**

Byte Field Size: *Bit Field Size:* 2

Request Parameter Optional
Command Parameter: Optional

DD243 AIS Sequence Number

Range 0-3

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

7 **Destination ID**

Byte Field Size: 4 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Optional

DD010 Generic numeric ID, large

Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number

MMSI Number of destination station.

---

8 **NMEA Reserved**

Byte Field Size: *Bit Field Size:* resv 6

Request Parameter
Command Parameter:

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

---

9 **Retransmit Flag**

Byte Field Size: *Bit Field Size:* 1

Request Parameter Optional
Command Parameter: Optional

DD244 AIS Retransmit Flag

0=No retransmission,
1=retransmitted.

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

10 **AIS Spare**

Byte Field Size: *Bit Field Size:* resv 1

Request Parameter
Command Parameter:

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

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AIS Addressed Binary Message

PGN: 129795
hex: 1FB03

11	**Number of Bits in Binary Data Field**	*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

Indicates the number of binary data bits that are contained within the Binary Data field.

12	**Binary Data**	*Byte Field Size:* *Bit Field Size:* n		*Request Parameter* *Command Parameter:*	Optional Optional

DD142 Binary Bit Field — Binary data bit field.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Application specific data.

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AIS Acknowledge

PGN: 129796
hex: 1FB04

This parameter group provides data associated with the ITU-R M.1371 Messages 7 Binary Acknowledge Message and 13 Safety Related Acknowledge Message. Message 7 acknowledges receipt of message 6 while message 13 acknowledges receipt of message 14. An AIS device may generate this parameter group either upon receiving a VHF data link message 7 or 13, or upon receipt of an ISO or NMEA request PGN (see ITU-R M.1371-1 for additional information). Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

Single Frame: No *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 113

Field #	Field Name					
1	**Message ID**	*Byte Field Size:*		*Request Parameter*	Required	
		Bit Field Size:	6	*Command Parameter:*	Optional	
	DD188 AIS Message Identifier	Message Identifier (range of 0 to 63). See the latest version of ITU-R M.1371 for more information.				
	DF52 Bit field **bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
	7 = Binary Acknowledge Message, 13 = Safety Related Acknowledge Message.					
2	**Repeat Indicator**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	2	*Command Parameter:*	Optional	
	DD185 AIS Repeater Indicator	Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3). 0 = Default 1 = First retransmission 2 = Second retransmission 3 = Final retransmission See the latest version of ITU-R M.1371 for more information.				
	DF52 Bit field **bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
3	**Source ID**	*Byte Field Size:*	4	*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	
	DD010 Generic numeric ID, large	Number of route, waypoint, event, mark, etc.				
	DF55 Integer, 32 bit unsigned **uint32**	*Range:* 0 to 4,294,967,292	*Resolution:* 1 bit	Unit-less number		
	MMSI number of source station for this acknowledge.					
4	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*		
		Bit Field Size:	resv 1	*Command Parameter:*		
	DD001 Reserved field	Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field **bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
	Used to align subsequent data on byte boundary.					

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AIS Acknowledge

PGN: 129796
hex: 1FB04

5 AIS Transceiver Information

Byte Field Size:
Bit Field Size: 5
Request Parameter: Optional
Command Parameter: Optional

DD246 AIS Transceiver Information

0 = Channel A VDL reception,
1 = Channel B VDL reception,
2 = Channel A VDL transmission,
3 = Channel B VDL transmission,
4 = Own information not broadcast,
5-31 = Reserved.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

6 AIS Spare

Byte Field Size:
Bit Field Size: resv 2
Request Parameter:
Command Parameter:

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** Range: N2KUnitless Resolution: 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

7 Destination ID"1"

Byte Field Size: 4
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD010 Generic numeric ID, large

Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** Range: 0 to 4,294,967,292 Resolution: 1 bit Unit-less number

MMSI number of first destination for this acknowledge.

8 NMEA Reserved

Byte Field Size:
Bit Field Size: resv 6
Request Parameter:
Command Parameter:

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

9 Sequence Number for ID"1"

Byte Field Size:
Bit Field Size: 2
Request Parameter: Optional
Command Parameter: Optional

DD243 AIS Sequence Number

Range 0-3

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Sequence number of message to be acknowledged, range 0-3.

10 Destination ID"n"

Byte Field Size: 4
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD010 Generic numeric ID, large

Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** Range: 0 to 4,294,967,292 Resolution: 1 bit Unit-less number

Variable Number of fields, Field number 7 repeated.

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AIS Acknowledge

PGN: 129796
hex: 1FB04

11	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*	
			Bit Field Size:	resv 6	*Command Parameter:*	
	DD001 Reserved field			Variable number of reserved bits, all set to logic "1"		
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	Variable Number of fields, Field number 8 repeated.					
12	**Sequence Number for ID"n"**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD243 AIS Sequence Number			Range 0-3 See the latest version of ITU-R M.1371 for more information.		
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	Variable Number of fields, Field number 9 repeated.					
13	**Sequence ID**		*Byte Field Size:*	1	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD056 Sequence ID			An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.		
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:* 1 bit	Unit-less number

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AIS Binary Broadcast Message

PGN: 129797
hex: 1FB05

This parameter group provides data associated with the ITU-R M.1371 Message 8 Binary Broadcast Message supporting broadcast communication of binary data. An AIS device may generate this parameter group either upon receiving a VHF data link message 8, or upon receipt of an ISO or NMEA request PGN. The Command Group Function PGN 126208 may be used with this PGN to configure parameters such as the Binary Data (see ITU-R M.1371-1 for additional information). Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

Single Frame: No *Priority Default:* 5 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 114

1 Message ID — *Byte Field Size:* *Bit Field Size:* 6 — *Request Parameter* Optional — *Command Parameter:* Optional

DD188 AIS Message Identifier — Message Identifier (range of 0 to 63).

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

8 = Binary Broadcast Message.

2 Repeat Indicator — *Byte Field Size:* *Bit Field Size:* 2 — *Request Parameter* Optional — *Command Parameter:* Optional

DD185 AIS Repeater Indicator — Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3).

0 = Default
1 = First retransmission
2 = Second retransmission
3 = Final retransmission

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 Source ID — *Byte Field Size:* 4 *Bit Field Size:* — *Request Parameter* Optional — *Command Parameter:* Optional

DD010 Generic numeric ID, large — Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number

MMSI number of source station.

4 NMEA Reserved — *Byte Field Size:* *Bit Field Size:* resv 1 — *Request Parameter* — *Command Parameter:*

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

5 AIS Transceiver Information — *Byte Field Size:* *Bit Field Size:* 5 — *Request Parameter* Optional — *Command Parameter:* Optional

DD246 AIS Transceiver Information — 0 = Channel A VDL reception,
1 = Channel B VDL reception,
2 = Channel A VDL transmission,
3 = Channel B VDL transmission,
4 = Own information not broadcast,
5-31 = Reserved.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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6	**AIS Spare**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 2	*Command Parameter:*	

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

7	**Number of Bits in Binary Data Field**	*Byte Field Size:*	2	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD007 Generic numeric ID, medium

Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

Indicates the number of binary data bits that are contained within the Binary Data field.

8	**Binary Data**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	n	*Command Parameter:*	Optional

DD142 Binary Bit Field

Binary data bit field.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Application specific data.

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AIS SAR Aircraft Position Report

PGN: 129798
hex: 1FB06

This parameter group provides data associated with the ITU-R M.1371 Message 9 SAR Aircraft Position Report Message for Airborne AIS units conducting Search and Rescue operations. An AIS device may generate this parameter group either upon receiving a VHF data link message 9, or upon receipt of an ISO or NMEA request PGN (see ITU-R M.1371-1 for additional information). Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

Single Frame: No *Priority Default:* 4 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 115

Field #	Field Name					
1	**Message ID**		*Byte Field Size:* *Bit Field Size:* 6		*Request Parameter* Optional *Command Parameter:* Optional	
	DD188 AIS Message Identifier			Message Identifier (range of 0 to 63). See the latest version of ITU-R M.1371 for more information.		
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
	9 = SAR Aircraft Position Report Message					
2	**Repeat Indicator**		*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter* Optional *Command Parameter:* Optional	
	DD185 AIS Repeater Indicator			Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3). 0 = Default 1 = First retransmission 2 = Second retransmission 3 = Final retransmission See the latest version of ITU-R M.1371 for more information.		
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
3	**User ID**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD010 Generic numeric ID, large			Number of route, waypoint, event, mark, etc.		
	DF55 Integer, 32 bit unsigned	**uint32**	*Range:* 0 to 4,294,967,292	*Resolution:* 1 bit	Unit-less number	
	MMSI number of SAR aircraft reporting position.					
4	**Longitude**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD023 Longitude, WGS-84			Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.		
	DF25 Longitude	**int32**	*Range:* +/- 180 deg	*Resolution:* 1x10E-7 deg	"-" = West, resolution ~1.1 cm	
	Longitude of SAR aircraft reporting position.					

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AIS SAR Aircraft Position Report

PGN: 129798
hex: 1FB06

5 Latitude

Byte Field Size: 4 | Bit Field Size: | Request Parameter: Optional | Command Parameter: Optional

DD022 Latitude, WGS-84

Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23 Latitude — **int32** — Range: +/- 90 deg — Resolution: 1x10E-7 deg — "-" = South, resolution ~1.1 cm

Latitude of SAR aircraft reporting position.

---

6 Position Accuracy

Byte Field Size: | Bit Field Size: 1 | Request Parameter: Optional | Command Parameter: Optional

DD184 AIS Position Accuracy

0 = low accuracy > 10m such as nondifferential GNSS (default),
1 = high accuracy < 10m such as DGNSS

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field — **bit(n)** — Range: Variable — Resolution: 1 — Used to construct bit fields

---

7 RAIM-Flag

Byte Field Size: | Bit Field Size: 1 | Request Parameter: Optional | Command Parameter: Optional

DD189 AIS RAIM-flag

0 = RAIM not in use (default),
1 = RAIM in use

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field — **bit(n)** — Range: Variable — Resolution: 1 — Used to construct bit fields

---

8 Time Stamp

Byte Field Size: | Bit Field Size: 6 | Request Parameter: Optional | Command Parameter: Optional

DD186 AIS Time Stamp

0-59 = UTC second when the report was generated,
60 = time stamp not available (default),
61 = positioning system is in manual input mode,
62 = Electronic position fixing system operates in estimated (dead reckoning) mode,
63 = positioning system is inoperative

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field — **bit(n)** — Range: Variable — Resolution: 1 — Used to construct bit fields

---

9 COG

Byte Field Size: 2 | Bit Field Size: | Request Parameter: Optional | Command Parameter: Optional

DD165 Course-Over-Ground (COG)

The direction of the path over ground actually followed by a vessel.

DF02 Angle — **uint16** — Range: 0 to 2Pi rad — Resolution: 1x10E-4 rad — Resolution ~0.0057deg, 1 deg = .01745 rad

COG of SAR aircraft reporting position.

---

10 SOG

Byte Field Size: 2 | Bit Field Size: | Request Parameter: Optional | Command Parameter: Optional

DD044 Generic Speed

DF35 Speed — **uint16** — Range: 0 to 655.32 m/s — Resolution: 1x10E-2 m/s — 1 Knot = 0.5144 m/s

SOG of SAR aircraft reporting position.

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AIS SAR Aircraft Position Report

PGN: 129798
hex: 1FB06

11 Communication State

Byte Field Size: | *Bit Field Size:* 19 | *Request Parameter* Optional | *Command Parameter:* Optional

DD187 AIS Communication State

The Communication State contains information used by the various TDMA slot allocation algorithms and synchronization information

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

12 AIS Transceiver Information

Byte Field Size: | *Bit Field Size:* 5 | *Request Parameter* Optional | *Command Parameter:* Optional

DD246 AIS Transceiver Information

0 = Channel A VDL reception,
1 = Channel B VDL reception,
2 = Channel A VDL transmission,
3 = Channel B VDL transmission,
4 = Own information not broadcast,
5-31 = Reserved.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

13 Altitude

Byte Field Size: 4 | *Bit Field Size:* | *Request Parameter* Optional | *Command Parameter:* Optional

DD115 Distance

DF15 Distance, signed **int32** *Range:* +/-~2.147x10E+7 m *Resolution:* 1x10E-2 m

Altitude of SAR aircraft reporting position.

---

14 Reserved for Regional Applications

Byte Field Size: | *Bit Field Size:* resv 8 | *Request Parameter* | *Command Parameter:*

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Reserved for Regional Applications" bit field found within the corresponding AIS such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

---

15 Data Terminal Equipment (DTE)

Byte Field Size: | *Bit Field Size:* 1 | *Request Parameter* Optional | *Command Parameter:* Optional

DD242 Data Terminal Equipment (DTE)

0=Available,
1=not available.

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

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16 **AIS Spare**

Byte Field Size:
Bit Field Size: resv 5

Request Parameter
Command Parameter:

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1

Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

---

17 **NMEA Reserved**

Byte Field Size:
Bit Field Size: resv 2

Request Parameter
Command Parameter:

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

---

18 **Sequence ID**

Byte Field Size: 1
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD056 Sequence ID

An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

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Radio Frequency/Mode/Power

PGN: 129799
hex: 1FB07

This PGN provides status and control for a Radiotelephone, connected to a NMEA network.

The Radiotelephone will transmit and receive status along with remote control and repeater products.

This information will either need to be transmitted on change, by request or on a low duty cycle.

Single Frame: No *Priority Default:* 3 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 62

Field #	Field Name					
1	**Rx Frequency**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD016 Radio Tx or Rx Frequency					
	DF21 Frequency	**uint32**	*Range:* 0 to ~4.295x10E+10 Hz	*Resolution:* 10 Hz		
2	**Tx Frequency**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD016 Radio Tx or Rx Frequency					
	DF21 Frequency	**uint32**	*Range:* 0 to ~4.295x10E+10 Hz	*Resolution:* 10 Hz		
3	**Radio Channel**		*Byte Field Size:* char 6 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD017 Radio Tx or Rx Channel				MF/HF telephone channels to have first digit 3 followed by ITU channel numbers with leading zeros as required. MF/HF teletype channels to have first digit 4; the second and third digit give the frequency bands; and the fourth to sixth digits ITU channel numbers; each with leading zeros as required. VHF channels to have he first digit 9 followed by zero. The next number is "1" indicating the ship station's transmit frequency is being used as a simplex channel frequency, or "2" indicating the coast station's transmit frequency is being used as a simplex channel frequency, "0" otherwise. The remaining three numbers are the VHF channel numbers with leading zeros as required.	
	DF63 String, fixed It is not necessary to supply both RX/TX frequency and channel but if both are present, frequency takes priority in case of mismatch.	**char8(n)**	*Range:* 0 to 1,785 characters	*Resolution:* 1 char	0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.	
4	**Tx Power**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD018 Radio Tx Power					
	DF28 Power If equipment has insufficient resolution to apply a commanded power, the next lower available power setting should be used.	**uint16**	*Range:* 0 to 65,532 W	*Resolution:* 1 W		

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5	**Mode**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	8	*Command Parameter:*	Optional

DD019 Mode, Radiotelephone

Radiotelephone mode settings:
0 = F3E/G3E simplex, telephone;
1 = F3E/G3E duplex, telephone;
2 = J3E, telephone;
3 = H3E, telephone;
4 = F1B/J2B FEC NBDP, telex/teleprinter;
5 = F1B/J2B ARQ NBDP, telex/teleprinter;
6 = F1B/J2B receive only, teleprinter/DSC;
7 = F1B/J2B, teleprinter/DSC;
8 = A1A Morse, tape recorder;
9 = A1A Morse, Morse key/head set;
10 =F1C/F2C/F3C, FAX-machine;
11-253 = reserved;
254 = error;
255 = unavailable/do not change

DF52	Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields

6	**Channel Bandwidth**	*Byte Field Size:*	2	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD020 Radio Channel Bandwidth

DF26	Frequency, mid	**uint16**	*Range:*	0 to 65,532 Hz	*Resolution:*	1 Hz

If equipment has insufficient resolution to apply a commanded bandwidth, the closest available setting should be used.

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AIS UTC/Date Inquiry

PGN: 129800
hex: 1FB08

This parameter group provides data associated with the ITU-R M.1371 Message 10 UTC and Date Inquiry Message used to request current UTC and date. An AIS device may generate this parameter group either upon receiving a VHF data link message 10, or upon receipt of an ISO or NMEA request PGN. The Command Group Function PGN 126208 may be used with this PGN to configure parameters such as the Destination ID (see ITU-R M.1371-1 for additional information). Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

Single Frame: No *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 116

Field #	Field Name					
1	**Message ID**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	6	*Command Parameter:*	Optional
	DD188 AIS Message Identifier		Message Identifier (range of 0 to 63). See the latest version of ITU-R M.1371 for more information.			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	10 = AIS UTC and Date Inquiry Message					
2	**Repeat Indicator**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD185 AIS Repeater Indicator		Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3). 0 = Default 1 = First retransmission 2 = Second retransmission 3 = Final retransmission See the latest version of ITU-R M.1371 for more information.			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
3	**Source ID**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD010 Generic numeric ID, large		Number of route, waypoint, event, mark, etc.			
	DF55 Integer, 32 bit unsigned	**uint32**	*Range:*	0 to 4,294,967,292	*Resolution:* 1 bit	Unit-less number
	MMSI number of station which inquires UTC.					
4	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*	
			Bit Field Size:	resv 1	*Command Parameter:*	
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	Used to align subsequent data on byte boundary.					
5	**AIS Transceiver Information**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	5	*Command Parameter:*	Optional
	DD246 AIS Transceiver Information		0 = Channel A VDL reception, 1 = Channel B VDL reception, 2 = Channel A VDL transmission, 3 = Channel B VDL transmission, 4 = Own information not broadcast, 5-31 = Reserved.			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields

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6	**AIS Spare**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 2	*Command Parameter:*	

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

7	**Destination ID**	*Byte Field Size:*	4	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD010 Generic numeric ID, large

Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number

MMSI number of station which is inquired.

8	**AIS Spare**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 2	*Command Parameter:*	

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

9	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 6	*Command Parameter:*	

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

10	**Sequence ID**	*Byte Field Size:*	1	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD056 Sequence ID

An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

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AIS Addressed Safety Related Message

PGN: 129801
hex: 1FB09

This parameter group provides data associated with the ITU-R M.1371 Message 12 Addressed Safety Related Message supporting addressed communication of safety related data. An AIS device may generate this parameter group either upon receiving a VHF data link message 12, or upon receipt of an ISO or NMEA request PGN. The Command Group Function PGN 126208 may be used with this PGN to configure safety related message parameters (see ITU-R M.1371-1 for additional information). Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

Single Frame: No *Priority Default:* 5 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 117

1 Message ID
Byte Field Size: *Bit Field Size:* 6 *Request Parameter* Optional *Command Parameter:* Optional

DD188 AIS Message Identifier
Message Identifier (range of 0 to 63).

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
12 = Addressed Safety Related Message

2 Repeat Indicator
Byte Field Size: *Bit Field Size:* 2 *Request Parameter* Optional *Command Parameter:* Optional

DD185 AIS Repeater Indicator
Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3).

0 = Default
1 = First retransmission
2 = Second retransmission
3 = Final retransmission

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 Source ID
Byte Field Size: 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD010 Generic numeric ID, large
Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number
MMSI number of station which is the source of the message.

4 NMEA Reserved
Byte Field Size: *Bit Field Size:* resv 1 *Request Parameter* *Command Parameter:*

DD001 Reserved field
Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
Used to align subsequent data on byte boundary.

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AIS Addressed Safety Related Message

PGN: 129801
hex: 1FB09

5 AIS Transceiver Information

Byte Field Size: *Bit Field Size:* 5

Request Parameter Optional
Command Parameter: Optional

DD246 AIS Transceiver Information

0 = Channel A VDL reception,
1 = Channel B VDL reception,
2 = Channel A VDL transmission,
3 = Channel B VDL transmission,
4 = Own information not broadcast,
5-31 = Reserved.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

6 Sequence Number

Byte Field Size: *Bit Field Size:* 2

Request Parameter Optional
Command Parameter: Optional

DD243 AIS Sequence Number

Range 0-3

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

7 Destination ID

Byte Field Size: 4 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Optional

DD010 Generic numeric ID, large

Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number

MMSI number of station which is the destination of this message.

---

8 NMEA Reserved

Byte Field Size: *Bit Field Size:* resv 6

Request Parameter
Command Parameter:

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

---

9 Retransmit Flag

Byte Field Size: *Bit Field Size:* 1

Request Parameter Optional
Command Parameter: Optional

DD244 AIS Retransmit Flag

0=No retransmission,
1=retransmitted.

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

10 AIS Spare

Byte Field Size: *Bit Field Size:* resv 1

Request Parameter
Command Parameter:

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

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AIS Addressed Safety Related Message

PGN: 129801
hex: 1FB09

11 **Safety Related Text**

Byte Field Size: 8 or 16 n
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD004 Generic name string, short

Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)**

Range: 0 to 250 ASCII or 0 to 125 Unicode Characters

Resolution: 1 ASCII or 1 Unicode Character

2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

Maximum size is 156 8-bit ASCII characters.

12 **Sequence ID**

Byte Field Size: 1
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD056 Sequence ID

An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8**

Range: 0 to 252

Resolution: 1 bit

Unit-less number

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AIS Safety Related Broadcast Message

PGN: 129802
hex: 1FB0A

This parameter group provides data associated with the ITU-R M.1371 Message 14 Safety Related Broadcast Message supporting broadcast communication of safety related data. An AIS device may generate this parameter group either upon receiving a VHF data link message 14, or upon receipt of an ISO or NMEA request PGN. The Command Group Function PGN 126208 may be used with this PGN to configure parameters such as the Safety Related Text (see ITU-R M.1371-1 for additional information). Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

Single Frame: No *Priority Default:* 5 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 118

Field #	Field Name					
1	**Message ID**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	6	*Command Parameter:*	Optional	
	DD188 AIS Message Identifier	Message Identifier (range of 0 to 63). See the latest version of ITU-R M.1371 for more information.				
	DF52 Bit field **bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
	14 = Safety Related Broadcast Message.					
2	**Repeat Indicator**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	2	*Command Parameter:*	Optional	
	DD185 AIS Repeater Indicator	Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3). 0 = Default 1 = First retransmission 2 = Second retransmission 3 = Final retransmission See the latest version of ITU-R M.1371 for more information.				
	DF52 Bit field **bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
3	**Source ID**	*Byte Field Size:*	4	*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	
	DD010 Generic numeric ID, large	Number of route, waypoint, event, mark, etc.				
	DF55 Integer, 32 bit unsigned **uint32**	*Range:* 0 to 4,294,967,292	*Resolution:* 1 bit	Unit-less number		
	MMSI number of station which is the source of the message.					
4	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*		
		Bit Field Size:	resv 1	*Command Parameter:*		
	DD001 Reserved field	Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field **bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
	Used to align subsequent data on byte boundary.					

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AIS Safety Related Broadcast Message

PGN: 129802
hex: 1FB0A

5 AIS Transceiver Information

Byte Field Size:
Bit Field Size: 5
Request Parameter: Optional
Command Parameter: Optional

DD246 AIS Transceiver Information

0 = Channel A VDL reception,
1 = Channel B VDL reception,
2 = Channel A VDL transmission,
3 = Channel B VDL transmission,
4 = Own information not broadcast,
5-31 = Reserved.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

6 AIS Spare

Byte Field Size:
Bit Field Size: resv 2
Request Parameter:
Command Parameter:

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** Range: N2KUnitless Resolution: 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

7 Safety Related Text

Byte Field Size: 8 or 16 n
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD004 Generic name string, short

Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** Range: 0 to 250 ASCII or 0 to 125 Unicode Characters Resolution: 1 ASCII or 1 Unicode Character 2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

Maximum size is 163 8-bit ASCII characters.

8 Sequence ID

Byte Field Size: 1
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD056 Sequence ID

An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** Range: 0 to 252 Resolution: 1 bit Unit-less number

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AIS Interrogation

PGN: 129803
hex: 1FB0B

This parameter group provides data associated with the ITU-R M.1371 Message 15 Interrogation Message used to request a specific ITU-R M.1371 message resulting in responses from one or more AIS mobile units. An AIS device may generate this parameter group either upon receiving a VHF data link message 15, or upon receipt of an ISO or NMEA request PGN. The Command Group Function PGN 126208 shall be used with this PGN to configure base station interrogation parameters (see ITU-R M.1371-1 for additional information). Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

Single Frame: No *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 119

1 Message ID
Byte Field Size: *Bit Field Size:* 6 *Request Parameter* Optional *Command Parameter:* Optional

DD188 AIS Message Identifier
Message Identifier (range of 0 to 63).

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
15 = Interrogation Message

2 Repeat Indicator
Byte Field Size: *Bit Field Size:* 2 *Request Parameter* Optional *Command Parameter:* Optional

DD185 AIS Repeater Indicator
Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3).

0 = Default
1 = First retransmission
2 = Second retransmission
3 = Final retransmission

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 Source ID
Byte Field Size: 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD010 Generic numeric ID, large
Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number
MMSI number of interrogating station.

4 NMEA Reserved
Byte Field Size: *Bit Field Size:* resv 1 *Request Parameter* *Command Parameter:*

DD001 Reserved field
Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
Used to align subsequent data on byte boundary.

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AIS Interrogation

PGN: 129803
hex: 1FB0B

5 AIS Transceiver Information

Byte Field Size:
Bit Field Size: 5
Request Parameter: Optional
Command Parameter: Optional

DD246 AIS Transceiver Information

0 = Channel A VDL reception,
1 = Channel B VDL reception,
2 = Channel A VDL transmission,
3 = Channel B VDL transmission,
4 = Own information not broadcast,
5-31 = Reserved.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

6 AIS Spare

Byte Field Size:
Bit Field Size: resv 2
Request Parameter:
Command Parameter:

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** Range: N2KUnitless Resolution: 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

7 Destination ID 1

Byte Field Size: 4
Bit Field Size:
Request Parameter: Optional
Command Parameter: Required

DD010 Generic numeric ID, large

Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** Range: 0 to 4,294,967,292 Resolution: 1 bit Unit-less number

MMSI number of first interrogated station.

8 NMEA Reserved

Byte Field Size:
Bit Field Size: resv 2
Request Parameter:
Command Parameter:

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

9 Message ID 1.1

Byte Field Size:
Bit Field Size: 6
Request Parameter: Optional
Command Parameter: Required

DD188 AIS Message Identifier

Message Identifier (range of 0 to 63).

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

First Requested message type from first interrogated station.

10 Slot Offset 1.1

Byte Field Size: 2
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD007 Generic numeric ID, medium

Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** Range: 0 to 65,532 Resolution: 1 bit Unit-less number

Response Slot offset for first requested message from first interrogated station.

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11 **AIS Spare** — Byte Field Size: — Bit Field Size: resv 2 — Request Parameter — Command Parameter:

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** Range: N2KUnitless Resolution: 1

Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

---

12 **Message ID 1.2** — Byte Field Size: — Bit Field Size: 6 — Request Parameter: Optional — Command Parameter: Required

DD188 AIS Message Identifier

Message Identifier (range of 0 to 63).

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1

Used to construct bit fields

Second requested message type from first interrogated station.

---

13 **Slot Offset 1.2** — Byte Field Size: 2 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Optional

DD007 Generic numeric ID, medium

Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** Range: 0 to 65,532 Resolution: 1 bit

Unit-less number

Response Slot offset for second requested message from second interrogated station.

---

14 **NMEA Reserved** — Byte Field Size: — Bit Field Size: resv 6 — Request Parameter — Command Parameter:

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** Range: Variable Resolution: 1

Used to construct bit fields

Used to align subsequent data on byte boundary.

---

15 **AIS Spare** — Byte Field Size: — Bit Field Size: resv 2 — Request Parameter — Command Parameter:

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** Range: N2KUnitless Resolution: 1

Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

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16 **Destination ID 2** — Byte Field Size: 4 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Required

DD010 Generic numeric ID, large — Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** Range: 0 to 4,294,967,292 Resolution: 1 bit Unit-less number

MMSI number of second interrogated station.

---

17 **NMEA Reserved** — Byte Field Size: — Bit Field Size: resv 2 — Request Parameter: — Command Parameter:

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

---

18 **Message ID 2.1** — Byte Field Size: — Bit Field Size: 6 — Request Parameter: Optional — Command Parameter: Required

DD188 AIS Message Identifier — Message Identifier (range of 0 to 63).

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Requested message type from second interrogated station.

---

19 **Slot Offset 2.1** — Byte Field Size: 2 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** Range: 0 to 65,532 Resolution: 1 bit Unit-less number

Response slot offset for requested message from second interrogated station.

---

20 **AIS Spare** — Byte Field Size: — Bit Field Size: resv 2 — Request Parameter: — Command Parameter:

DD311 AIS Spare Field — Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** Range: N2KUnitless Resolution: 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

---

21 **NMEA Reserved** — Byte Field Size: — Bit Field Size: resv 6 — Request Parameter: — Command Parameter:

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

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AIS Interrogation

PGN: 129803
hex: 1FB0B

22	**Sequence ID**	*Byte Field Size:* 1		*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD056 Sequence ID

An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

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AIS Assignment Mode Command

PGN: 129804
hex: 1FB0C

This parameter group provides data associated with the ITU-R M.1371 Message 16 Assigned Mode Command Message for assigning specific behavior by a competent authority. An AIS device may generate this parameter group either upon receiving a VHF data link message 16, or upon receipt of an ISO or NMEA request PGN. The Command Group Function PGN 126208 shall be used with this PGN to configure base station assigned mode parameters (see ITU-R M.1371-1 for additional information). Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

Single Frame: No *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Required *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 120

1 Message ID
Byte Field Size: *Bit Field Size:* 6 *Request Parameter* Optional *Command Parameter:* Optional

DD188 AIS Message Identifier
Message Identifier (range of 0 to 63).

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
16 = Assigned Mode Command Message

2 Repeat Indicator
Byte Field Size: *Bit Field Size:* 2 *Request Parameter* Optional *Command Parameter:* Optional

DD185 AIS Repeater Indicator
Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3).

0 = Default
1 = First retransmission
2 = Second retransmission
3 = Final retransmission

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 Source ID
Byte Field Size: 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD010 Generic numeric ID, large
Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number
MMSI number of assigning station.

4 NMEA Reserved
Byte Field Size: *Bit Field Size:* resv 1 *Request Parameter* *Command Parameter:*

DD001 Reserved field
Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
Used to align subsequent data on byte boundary.

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AIS Assignment Mode Command

PGN: 129804
hex: 1FB0C

5	**AIS Transceiver Information**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	5	*Command Parameter:*	Required
	DD246 AIS Transceiver Information		0 = Channel A VDL reception, 1 = Channel B VDL reception, 2 = Channel A VDL transmission, 3 = Channel B VDL transmission, 4 = Own information not broadcast, 5-31 = Reserved.			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
6	**AIS Spare**		*Byte Field Size:*		*Request Parameter*	
			Bit Field Size:	resv 2	*Command Parameter:*	
	DD311 AIS Spare Field		Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.			
	DF115 Bit field defaulting to zer	**bit0(n)**	*Range:*	N2KUnitless	*Resolution:* 1	Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use
	This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.					
7	**Destination ID A**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Required
	DD010 Generic numeric ID, large		Number of route, waypoint, event, mark, etc.			
	DF55 Integer, 32 bit unsigned	**uint32**	*Range:*	0 to 4,294,967,292	*Resolution:* 1 bit	Unit-less number
	MMSI number of destination station A.					
8	**Offset A**		*Byte Field Size:*	2	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Required
	DD007 Generic numeric ID, medium		Number of route, waypoint, event, mark, etc.			
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:*	0 to 65,532	*Resolution:* 1 bit	Unit-less number
	Offset from Current slot to first assigned slot.					
9	**Increment A**		*Byte Field Size:*	2	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Required
	DD007 Generic numeric ID, medium		Number of route, waypoint, event, mark, etc.			
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:*	0 to 65,532	*Resolution:* 1 bit	Unit-less number
	Increment to next assigned slot.					
10	**Destination ID B**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD010 Generic numeric ID, large		Number of route, waypoint, event, mark, etc.			
	DF55 Integer, 32 bit unsigned	**uint32**	*Range:*	0 to 4,294,967,292	*Resolution:* 1 bit	Unit-less number
	MMSI number of destination station B.					

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11 **Offset B**

Byte Field Size: 2 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

Offset from Current slot to first assigned slot.

---

12 **Increment B**

Byte Field Size: 2 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

Increment to next assigned slot.

---

13 **AIS Spare**

Byte Field Size: *Bit Field Size:* resv 4

Request Parameter *Command Parameter:*

DD311 AIS Spare Field — Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

---

14 **NMEA Reserved**

Byte Field Size: *Bit Field Size:* resv 4

Request Parameter *Command Parameter:*

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

---

15 **Sequence ID**

Byte Field Size: 1 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Optional

DD056 Sequence ID — An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

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AIS Data Link Management Message

PGN: 129805
hex: 1FB0D

This parameter group provides data associated with the ITU-R M.1371 Message 20 Data Link Management Message for reserving slots for base stations. An AIS device may generate this parameter group either upon receiving a VHF data link message 20, or upon receipt of an ISO or NMEA request PGN. The Command Group Function PGN 126208 shall be used with this PGN to configure base station data link management parameters (see ITU-R M.1371-1 for additional information). Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

Single Frame: No *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 124

Field #	Field Name					
1	**Message ID**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	6	*Command Parameter:*	Optional
	DD188 AIS Message Identifier			Message Identifier (range of 0 to 63). See the latest version of ITU-R M.1371 for more information.		
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	20 = Data Link Management Message					
2	**Repeat Indicator**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD185 AIS Repeater Indicator			Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3). 0 = Default 1 = First retransmission 2 = Second retransmission 3 = Final retransmission See the latest version of ITU-R M.1371 for more information.		
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
3	**Source Station ID**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD010 Generic numeric ID, large			Number of route, waypoint, event, mark, etc.		
	DF55 Integer, 32 bit unsigned	**uint32**	*Range:*	0 to 4,294,967,292	*Resolution:* 1 bit	Unit-less number
	MMSI number of base station transmitting management message.					
4	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*	
			Bit Field Size:	resv 1	*Command Parameter:*	
	DD001 Reserved field			Variable number of reserved bits, all set to logic "1"		
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	Used to align subsequent data on byte boundary.					

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AIS Data Link Management Message

PGN: 129805
hex: 1FB0D

5 AIS Transceiver Information

Byte Field Size: *Bit Field Size:* 5

Request Parameter Optional
Command Parameter: Optional

DD246 AIS Transceiver Information

0 = Channel A VDL reception,
1 = Channel B VDL reception,
2 = Channel A VDL transmission,
3 = Channel B VDL transmission,
4 = Own information not broadcast,
5-31 = Reserved.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

6 AIS Spare

Byte Field Size: *Bit Field Size:* resv 2

Request Parameter *Command Parameter:*

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

7 Offset Number 1

Byte Field Size: 2 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Required

DD007 Generic numeric ID, medium

Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

0=Not available,
1-4095=respective reserved offset number.

See ITU-R M.1371-1 for more information.

8 Number of Slots 1

Byte Field Size: 1 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Required

DD005 Generic numeric ID, short

Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

0=Not available,
1-15=respective number of reserved consecutive slots.

See ITU-R M.1371-1 for more information.

9 Time Out 1

Byte Field Size: 1 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Required

DD005 Generic numeric ID, short

Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

0=Not available,
1-7=respective time-out value in minutes.

See ITU-R M.1371-1 for more information.

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10 **Increment 1**

Byte Field Size: 2 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Required

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

0=Not available,
1-2047=respective increment to repeat reservation block.

See ITU-R M.1371-1 for more information.

---

11 **Offset Number 2**

Byte Field Size: 2 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

0=Not available,
1-4095=respective reserved offset number.

See ITU-R M.1371-1 for more information.

---

12 **Number of Slots 2**

Byte Field Size: 1 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

0=Not available,
1-15=respective number of reserved consecutive slots.

See ITU-R M.1371-1 for more information.

---

13 **Time Out 2**

Byte Field Size: 1 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

0=Not available,
1-7=respective time-out value in minutes.

See ITU-R M.1371-1 for more information.

---

14 **Increment 2**

Byte Field Size: 2 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

0=Not available,
1-2047=respective increment to repeat reservation block.

See ITU-R M.1371-1 for more information.

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15 **Offset Number 3**

Byte Field Size: 2 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

0=Not available,
1-4095=respective reserved offset number.

See ITU-R M.1371-1 for more information.

---

16 **Number of Slots 3**

Byte Field Size: 1 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

0=Not available,
1-15=respective number of reserved consecutive slots.

See ITU-R M.1371-1 for more information.

---

17 **Time Out 3**

Byte Field Size: 1 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

0=Not available,
1-7=respective time-out value in minutes.

See ITU-R M.1371-1 for more information.

---

18 **Increment 3**

Byte Field Size: 2 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

0=Not available,
1-2047=respective increment to repeat reservation block.

See ITU-R M.1371-1 for more information.

---

19 **Offset Number 4**

Byte Field Size: 2 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

0=Not available,
1-4095=respective reserved offset number.

See ITU-R M.1371-1 for more information.

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AIS Data Link Management Message

PGN: 129805
hex: 1FB0D

20 **Number of Slots 4**

Byte Field Size: 1 | *Request Parameter* Optional
Bit Field Size: | *Command Parameter:* Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

0=Not available,
1-15=respective number of reserved consecutive slots.

See ITU-R M.1371-1 for more information.

---

21 **Time Out 4**

Byte Field Size: 1 | *Request Parameter* Optional
Bit Field Size: | *Command Parameter:* Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

0=Not available,
1-7=respective time-out value in minutes.

See ITU-R M.1371-1 for more information.

---

22 **Increment 4**

Byte Field Size: 2 | *Request Parameter* Optional
Bit Field Size: | *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

0=Not available,
1-2047=respective increment to repeat reservation block.

See ITU-R M.1371-1 for more information.

---

23 **AIS Spare**

Byte Field Size: | *Request Parameter*
Bit Field Size: resv 6 | *Command Parameter:*

DD311 AIS Spare Field — Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

---

24 **NMEA Reserved**

Byte Field Size: | *Request Parameter*
Bit Field Size: resv 2 | *Command Parameter:*

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

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AIS Data Link Management Message

PGN: 129805
hex: 1FB0D

25	**Sequence ID**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter*	Optional
				Command Parameter:	Optional

DD056 Sequence ID

An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

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AIS Channel Management

PGN: 129806
hex: 1FB0E

This parameter group provides data associated with the ITU-R M.1371 Message 22 Channel Management Message supporting management of transceiver modes and channels by a base station. An AIS device may generate this parameter group either upon receiving a VHF data link message 5, or upon receipt of an ISO or NMEA request PGN. The Command Group Function PGN126208 shall be used with this PGN to configure channel management parameters (see ITU-R M.1371 for additional information). Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

Response to Request:
The response to either an ISO request PGN 059904 or NMEA Request Group Function 126208 for this PGN 129806 shall be the same. The AIS unit shall provide this PGN 129806 for each and every channel management geographical region currently stored in memory.

The latitude and longitude fields 11, 12, 13, 14 in this PGN are only used at a truncated precision to 1/10 of a minute.

The Sequence ID field (SID) is used to link this PGN to other related PGN's from the same source address. When no linkage exists, the value of the SID shall be set to 255.

Single Frame: No *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Required *ACK Rqmnts:* None

Original Reference ID # 202

Field #	*Field Name*					
1	**Message ID**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	6	*Command Parameter:*	Optional
	DD188 AIS Message Identifier		Message Identifier (range of 0 to 63). See the latest version of ITU-R M.1371 for more information.			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
	22 = Channel Management Message.					
2	**Repeat Indicator**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD185 AIS Repeater Indicator		Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3). 0 = Default 1 = First retransmission 2 = Second retransmission 3 = Final retransmission See the latest version of ITU-R M.1371 for more information.			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
3	**Station ID**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD010 Generic numeric ID, large		Number of route, waypoint, event, mark, etc.			
	DF55 Integer, 32 bit unsigned	**uint32**	*Range:* 0 to 4,294,967,292	*Resolution:* 1 bit	Unit-less number	
	MMSI number of base station.					
4	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*	
			Bit Field Size:	resv 1	*Command Parameter:*	
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
	Used to align subsequent data on byte boundary.					

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AIS Channel Management

PGN: 129806
hex: 1FB0E

5 AIS Transceiver Information

Byte Field Size: *Bit Field Size:* 5

Request Parameter Optional
Command Parameter: Required

DD246 AIS Transceiver Information

0 = Channel A VDL reception,
1 = Channel B VDL reception,
2 = Channel A VDL transmission,
3 = Channel B VDL transmission,
4 = Own information not broadcast,
5-31 = Reserved.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

6 AIS Spare

Byte Field Size: *Bit Field Size:* resv 2

Request Parameter *Command Parameter:*

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

---

7 Channel A

Byte Field Size: 2 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Required

DD007 Generic numeric ID, medium

Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

Channel number according to recommendation ITU-R M.1084, Annex 4.

---

8 Channel B

Byte Field Size: 2 *Bit Field Size:*

Request Parameter Optional
Command Parameter: Required

DD007 Generic numeric ID, medium

Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

Channel number according to recommendation ITU-R M.1084, Annex 4.

---

9 Source Identifier

Byte Field Size: *Bit Field Size:* 3

Request Parameter Optional
Command Parameter: Required

DD353 AIS Channel Management Source

Source identifiers:
0 = When this PGN is sent to an AIS unit
1 = ITU-R M.1371 message 22: Channel Management addressed message
2 = ITU-R M.1371 message 22: Channel Management broadcast geographical area message
3 = NMEA 0183 ACA Sentence or NMEA Network PGN # 129806
4 = DSC Channel 70 Telecommand
5 = Operator manual input
6 thru 7 = Reserved for future use

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

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10	**Power**		Byte Field Size:		Request Parameter	Optional
			Bit Field Size:	1	Command Parameter:	Required
	DD252 AIS Power		0=High (default), 1=low. See the latest version of ITU-R M.1371 for more information.			
	DF52 Bit field	**bit(n)**	Range:	Variable	Resolution: 1	Used to construct bit fields
11	**Tx/Rx Mode**		Byte Field Size:		Request Parameter	Optional
			Bit Field Size:	4	Command Parameter:	Required
	DD253 AIS Tx/Rx Mode		0=Tx A/Tx B, Rx A/Rx B (default), 1=Tx A, Rx A/Rx B, 2=Tx B, Rx A/Rx B, 3-15=not used. See the latest version of ITU-R M.1371 for more information.			
	DF52 Bit field	**bit(n)**	Range:	Variable	Resolution: 1	Used to construct bit fields
12	**North East Longitude Corner 1**		Byte Field Size:	4	Request Parameter	Optional
			Bit Field Size:		Command Parameter:	Required
	DD023 Longitude, WGS-84		Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.			
	DF25 Longitude	**int32**	Range:	+/- 180 deg	Resolution: 1x10E-7 deg	"-" = West, resolution ~1.1 cm
	North East longitude corner of geographic area designated in this message. The precision of the latitude and longitude fields shall be limited to 1 decimal place of minutes (1/10 of a minute). If a higher precision is provided to an AIS unit, the receiving AIS unit shall truncate the value to a precision of 1/10 of a minute.					
13	**North East Latitude Corner 1**		Byte Field Size:	4	Request Parameter	Optional
			Bit Field Size:		Command Parameter:	Required
	DD022 Latitude, WGS-84		Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.			
	DF23 Latitude	**int32**	Range:	+/- 90 deg	Resolution: 1x10E-7 deg	"-" = South, resolution ~1.1 cm
	North East latitude corner of geographic area designated in this message. The precision of the latitude and longitude fields shall be limited to 1 decimal place of minutes (1/10 of a minute). If a higher precision is provided to an AIS unit, the receiving AIS unit shall truncate the value to a precision of 1/10 of a minute.					
14	**South West Longitude Corner 2**		Byte Field Size:	4	Request Parameter	Optional
			Bit Field Size:		Command Parameter:	Required
	DD023 Longitude, WGS-84		Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.			
	DF25 Longitude	**int32**	Range:	+/- 180 deg	Resolution: 1x10E-7 deg	"-" = West, resolution ~1.1 cm
	South West longitude corner of geographic area designated in this message. The precision of the latitude and longitude fields shall be limited to 1 decimal place of minutes (1/10 of a minute). If a higher precision is provided to an AIS unit, the receiving AIS unit shall truncate the value to a precision of 1/10 of a minute.					

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15 **South West Latitude Corner 2**

Byte Field Size: 4 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Required

DD022 Latitude, WGS-84

Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23 Latitude **int32** *Range:* +/- 90 deg *Resolution:* 1x10E-7 deg "-" = South, resolution ~1.1 cm

South West latitude cornerof geographic area designated in this message. The precision of the latitude and longitude fields shall be limited to 1 decimal place of minutes (1/10 of a minute). If a higher precision is provided to an AIS unit, the receiving AIS unit shall truncate the value to a precision of 1/10 of a minute.

---

16 **NMEA Reserved**

Byte Field Size: *Request Parameter*
Bit Field Size: resv 1 *Command Parameter:*

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

---

17 **Addressed or Broadcast Message Indicator**

Byte Field Size: *Request Parameter* Optional
Bit Field Size: 1 *Command Parameter:* Required

DD254 AIS Addressed or Broadcast Message Indicator

0=Broadcast geographical area message (default),
1=addressed message (to individual station(s)).

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

18 **Channel A Bandwidth**

Byte Field Size: *Request Parameter* Optional
Bit Field Size: 1 *Command Parameter:* Required

DD255 AIS Channel Bandwidth

0=default (as specified by channel number),
1=12.5 kHz bandwidth.

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

When sent to an AIS unit:
A value of 0 corresponds to the bandwidth as specified by channel number, see ITU-R M.1084, Annex 4 For AIS equipment designed to ITU-R M.1371-1 and 1371-2, a value of 1, bandwidth is 12.5 kHz .
For AIS equipment designed to ITU-R M.1371-3 and 1371-4, this value is ignored.
When sent from an AIS unit:
A value of 0 indicates that bandwidth is specified by channel number, see ITU-R M.1084, Annex 4 For AIS equipment designed to ITU-R M.1371-1 and 1371-2, a value of 1, bandwidth is 12.5 kHz .
For AIS equipment designed to ITU-R M.1371-3 and 1371-4, this value is always set to 0.

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19 **Channel B Bandwidth**

Byte Field Size: | *Bit Field Size:* 1

Request Parameter | *Command Parameter:* Optional / Required

DD255 AIS Channel Bandwidth

0=default (as specified by channel number),
1=12.5 kHz bandwidth.

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

When sent to an AIS unit:
A value of 0 corresponds to the bandwidth as specified by channel number, see ITU-R M.1084, Annex 4 For AIS equipment designed to ITU-R M.1371-1 and 1371-2, a value of 1, bandwidth is 12.5 kHz .
For AIS equipment designed to ITU-R M.1371-3 and 1371-4, this value is ignored.
When sent from an AIS unit:
A value of 0 indicates that bandwidth is specified by channel number, see ITU-R M.1084, Annex 4 For AIS equipment designed to ITU-R M.1371-1 and 1371-2, a value of 1, bandwidth is 12.5 kHz .
For AIS equipment designed to ITU-R M.1371-3 and 1371-4, this value is always set to 0.

---

20 **NMEA Reserved**

Byte Field Size: | *Bit Field Size:* resv 1

Request Parameter | *Command Parameter:*

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

---

21 **Transitional Zone Size**

Byte Field Size: | *Bit Field Size:* 3

Request Parameter | *Command Parameter:* Optional / Required

DD256 AIS Transitional Zone Size

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

22 **AIS Spare**

Byte Field Size: | *Bit Field Size:* resv 23

Request Parameter | *Command Parameter:*

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

---

23 **NMEA Reserved**

Byte Field Size: | *Bit Field Size:* resv 1

Request Parameter | *Command Parameter:*

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

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24	**In-Use Flag**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	2	*Command Parameter:*	Required

DD354 AIS Channel Management Parameters Usage

0 = “Not in use” when this PGN is sent from an AIS unit or when commanded with PGN 126208 to a AIS Base Station
1 = “In use” when this PGN is sent from an AIS unit or when commanded with PGN 126208 to an AIS Base Station
2 - 3 = Reserved for future use

AIS Mobile units shall not accept commands on this field with PGN 126208.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

This value is set to indicate that the other parameters in this PGN are "in-use" by an AIS unit at the time that the AIS unit sends this PGN.

25	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 6	*Command Parameter:*	

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

26	**Time of in-use Flag Change**	*Byte Field Size:*	4	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD158 Generic time of day

24 hour clock, 0 = midnight, time is in UTC

DF06 Time of day **uint32** *Range:* 0 to 86,401 s *Resolution:* 1x10E-4 s ~24 hours, 0 = midnight, range allows for up to two leap seconds per day

This is the UTC time that the "in-use-flag" field changed to the indicated state. AIS units shall not accept commands on this field with PGN 126208.

27	**Sequence ID**	*Byte Field Size:*	1	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD056 Sequence ID

An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

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AIS Group Assignment

PGN: 129807
hex: 1FB0F

The Group Assignment Command is transmitted by a base station when operating as a controlling entity for AIS Stations. ITU-R M.1371 Message 23 contains three criteria (position, ship and cargo type, and station type) that areused by each station that receives the message to determine if the message content applies to that station.

Application of all data fields is dependant upon the latest version of ITU-R M.1371.

Use of the NMEA Command Group Function (PGN 126208) with this PGN is only applicable to AIS base stations.

Single Frame: No *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Required *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 201

1 Message ID
Byte Field Size: *Bit Field Size:* 6 *Request Parameter* Optional *Command Parameter:* Optional

DD188 AIS Message Identifier
Message Identifier (range of 0 to 63).

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
23 = Group Assignment Command for AIS

2 Repeat Indicator
Byte Field Size: *Bit Field Size:* 2 *Request Parameter* Optional *Command Parameter:* Optional

DD185 AIS Repeater Indicator
Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3).

0 = Default
1 = First retransmission
2 = Second retransmission
3 = Final retransmission

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 Source ID
Byte Field Size: 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD010 Generic numeric ID, large
Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number
MMSI number of base station

4 AIS Spare
Byte Field Size: *Bit Field Size:* resv 2 *Request Parameter* *Command Parameter:*

DD311 AIS Spare Field
Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use
This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within theAIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

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5 **Tx/Rx Mode** — Byte Field Size: — Bit Field Size: 4 — Request Parameter: Optional — Command Parameter: Required

DD253 AIS Tx/Rx Mode

0=Tx A/Tx B, Rx A/Rx B (default),
1=Tx A, Rx A/Rx B,
2=Tx B, Rx A/Rx B,
3-15=not used.

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

6 **NMEA Reserved** — Byte Field Size: — Bit Field Size: resv 2 — Request Parameter: — Command Parameter:

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Used to align subsequent data on byte boundary

---

7 **North East Longitude Corner 1** — Byte Field Size: 4 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Required

DD023 Longitude, WGS-84

Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF25 Longitude **int32** Range: +/- 180 deg Resolution: 1x10E-7 deg "-" = West, resolution ~1.1 cm

North East Longitude corner of geographic area designated in this message

---

8 **North East Latitude Corner 1** — Byte Field Size: 4 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Required

DD022 Latitude, WGS-84

Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23 Latitude **int32** Range: +/- 90 deg Resolution: 1x10E-7 deg "-" = South, resolution ~1.1 cm

North East Latitude corner of geographic area designated in this message.

---

9 **South West Longitude Corner 2** — Byte Field Size: 4 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Required

DD023 Longitude, WGS-84

Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF25 Longitude **int32** Range: +/- 180 deg Resolution: 1x10E-7 deg "-" = West, resolution ~1.1 cm

South West Longitude corner of geographic area designated in this message.

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AIS Group Assignment

PGN: 129807
hex: 1FB0F

10 **South West Latitude Corner 2** — *Byte Field Size:* 4 — *Bit Field Size:* — *Request Parameter* / *Command Parameter:* Optional / Required

DD022 Latitude, WGS-84

Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23 Latitude **int32** *Range:* +/- 90 deg *Resolution:* 1x10E-7 deg "-" = South, resolution ~1.1 cm

South West Latitude corner of geographic area designated in this message.

---

11 **Station Type** — *Byte Field Size:* — *Bit Field Size:* 4 — *Request Parameter* / *Command Parameter:* Optional / Required

DD301 AIS Station Type

0 = All types of mobiles (default)
1 = Reserved for future use
2 = All types of Class B mobile stations
3 = SAR airborne mobile station
4 = AtoN station
5 = Class B "CS" shipborne mobile station only
6 = Inland waterways
7 to 9 = Reserved for regional use
10 to 15 = Reserved for future use

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Specifies type of AIS Station this is intended for.

---

12 **NMEA Reserved** — *Byte Field Size:* — *Bit Field Size:* resv 4 — *Request Parameter* / *Command Parameter:*

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary

---

13 **Ship and Cargo Filter** — *Byte Field Size:* — *Bit Field Size:* 8 — *Request Parameter* / *Command Parameter:* Optional / Required

DD300 Ship/Cargo Filter

0 = all types (default)
1 - 99 - See ITU-R M.1371 Table "Identifiers to be used by ships to report their type"
100 - 199 = Reserved for regional use
200 - 255 = Reserved for future use

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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14 **AIS Spare**

Byte Field Size:
Bit Field Size: resv 22

Request Parameter
Command Parameter:

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within theAIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network Messages are encodedwith logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's

---

15 **NMEA Reserved**

Byte Field Size:
Bit Field Size: resv 2

Request Parameter
Command Parameter:

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary

---

16 **Reporting Interval**

Byte Field Size:
Bit Field Size: 4

Request Parameter Optional
Command Parameter: Required

DD302 AIS Reporting Interval for Class B

0 = As given by the autonomous mode
1 = 10 min
2 = 6 min
3 = 3 min
4 = 1 min
5 = 30 sec
6 = 15 sec
7 = 10 sec
8 = 5 sec
9 = Next shorter reporting interval
10 = Next longer reporting interval
11 = 2 sec (not applicable to the Class B “CS”)
12 - 15 = Reserved for future use

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Specifies how often the position report is transmitted. When in dual channel mode (see field 5) the transmission rate is maintained by alternating transmissions between channels, each channel transmitting half the required reports. When in single channel mode the single selected channel transmits all the required reports.

---

17 **Quiet Time**

Byte Field Size:
Bit Field Size: 4

Request Parameter Optional
Command Parameter: Required

DD303 AIS Quiet Time

0 = No quiet time commanded
1-15 = Quiet time of 1 to 15 min

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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18 **AIS Spare** *Byte Field Size:* *Bit Field Size:* resv 6 *Request Parameter* *Command Parameter:*

DD311 AIS Spare Field — Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1 — Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this field. Normally, spare or reserved bits in NMEA Network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's

19 **NMEA Reserved** *Byte Field Size:* *Bit Field Size:* resv 2 *Request Parameter* *Command Parameter:*

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 — Used to construct bit fields

Used to align subsequent data on byte boundary.

20 **AIS Transceiver Information** *Byte Field Size:* *Bit Field Size:* 5 *Request Parameter:* Optional *Command Parameter:* Required

DD246 AIS Transceiver Information

0 = Channel A VDL reception,
1 = Channel B VDL reception,
2 = Channel A VDL transmission,
3 = Channel B VDL transmission,
4 = Own information not broadcast,
5-31 = Reserved.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 — Used to construct bit fields

21 **NMEA Reserved** *Byte Field Size:* *Bit Field Size:* resv 3 *Request Parameter* *Command Parameter:*

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 — Used to construct bit fields

Used to align subsequent data on byte boundary.

22 **Sequence ID** *Byte Field Size:* 1 *Bit Field Size:* *Request Parameter:* Optional *Command Parameter:* Optional

DD056 Sequence ID — An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit — Unit-less number

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This PGN provides Digital Selective Calling (DSC) data according to ITU M.493-9 with optional expansion according to ITU M.821-1. DSC is a paging system that is used to automate distress alerts sent over terrestrial communication systems such as VHF, MF and HF marine radio systems. DSC provides a mechanism to report significantly more information regarding a distress call rather than just the distress itself.
Products equipped with DSC will transmit and receive this information.

This PGN will be transmitted as and when required.

Calls to be transmitted should generally use the command Group Function Message (PGN 126208) in conjunction with this parameter group to ensure that the correct transmitter is selected.

This PGN will be transmitted as and when required.

Single Frame:	No	*Priority Default:*	4	*Default Update Rate:*		milliseconds	*Frequency:*	NA	cycles per second
Destination:	Global	*Query Support:*	Required	*Command Support:*	Optional	*ACK Rqmnts:*			

Original Reference ID # 61

Field #	*Field Name*							
1	**DSC Format Symbol**		*Byte Field Size:*	1		*Request Parameter*	Optional	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD011 DSC Symbol Definitions				Integer numbers within the range 000 to 127 representing DSC Symbols defined by ITU-R M.493 Table 3 for: Phasing and Unique Functions; Format Specifier; Category; Nature of Distress; First Telecommand; Second Telecommand			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number	
2	**DSC Category Symbol**		*Byte Field Size:*	1		*Request Parameter*	Optional	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD011 DSC Symbol Definitions				Integer numbers within the range 000 to 127 representing DSC Symbols defined by ITU-R M.493 Table 3 for: Phasing and Unique Functions; Format Specifier; Category; Nature of Distress; First Telecommand; Second Telecommand			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number	
3	**DSC Message Address**		*Byte Field Size:*	char 5		*Request Parameter*	Optional	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD012 DSC Address or Geographic Area				Individual characters having only decimal values in the range 0 to 127 for the DSC symbols defined by ITU-R M.493 are used to code the address or geographic area as defined in ITU-R M.493-9 Section 5. This may represent an individual MMSI, a group MMSI, or a geographic area.			
	DF63 String, fixed	**char8(n)**	*Range:*	0 to 1,785 characters	*Resolution:*	1 char	0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.	
4	**Nature Of Distress or 1st Telecommand**		*Byte Field Size:*	1		*Request Parameter*	Optional	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD011 DSC Symbol Definitions				Integer numbers within the range 000 to 127 representing DSC Symbols defined by ITU-R M.493 Table 3 for: Phasing and Unique Functions; Format Specifier; Category; Nature of Distress; First Telecommand; Second Telecommand			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number	

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5 **Subsequent Communication Mode or 2nd Telecommand**

Byte Field Size: 1 *Request Parameter* Optional

Bit Field Size: *Command Parameter:* Optional

DD011 DSC Symbol Definitions

Integer numbers within the range 000 to 127 representing DSC Symbols defined by ITU-R M.493 Table 3 for: Phasing and Unique Functions; Format Specifier; Category; Nature of Distress; First Telecommand; Second Telecommand

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

---

6 **Proposed Rx Frequency/Channel**

Byte Field Size: char 6 *Request Parameter* Optional

Bit Field Size: *Command Parameter:* Optional

DD017 Radio Tx or Rx Channel

MF/HF telephone channels to have first digit 3 followed by ITU channel numbers with leading zeros as required. MF/HF teletype channels to have first digit 4; the second and third digit give the frequency bands; and the fourth to sixth digits ITU channel numbers; each with leading zeros as required. VHF channels to have he first digit 9 followed by zero. The next number is "1" indicating the ship station's transmit frequency is being used as a simplex channel frequency, or "2" indicating the coast station's transmit frequency is being used as a simplex channel frequency, "0" otherwise. The remaining three numbers are the VHF channel numbers with leading zeros as required.

DF63 String, fixed **char8(n)** *Range:* 0 to 1,785 characters *Resolution:* 1 char 0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.

---

7 **Proposed Tx Frequency/Channel**

Byte Field Size: char 6 *Request Parameter* Optional

Bit Field Size: *Command Parameter:* Optional

DD017 Radio Tx or Rx Channel

MF/HF telephone channels to have first digit 3 followed by ITU channel numbers with leading zeros as required. MF/HF teletype channels to have first digit 4; the second and third digit give the frequency bands; and the fourth to sixth digits ITU channel numbers; each with leading zeros as required. VHF channels to have he first digit 9 followed by zero. The next number is "1" indicating the ship station's transmit frequency is being used as a simplex channel frequency, or "2" indicating the coast station's transmit frequency is being used as a simplex channel frequency, "0" otherwise. The remaining three numbers are the VHF channel numbers with leading zeros as required.

DF63 String, fixed **char8(n)** *Range:* 0 to 1,785 characters *Resolution:* 1 char 0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.

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8	**Telephone Number**		*Byte Field Size:*	8 or 16 n	*Request Parameter*	Required
			Bit Field Size:		*Command Parameter:*	Optional

DD015 DSC Symbol String

Individual characters having only decimal values in the range 0 to 127 for the DSC symbols defined by ITU-R M.493 are used to code: Telephone number as defined by ITU-R M.493 Section 8.2.3; DSC Expansion Data as defined by ITU-R M.821 Section 2.

DF50	String, variable, short	**ch8or16(n)**	*Range:*	0 to 250 ASCII or 0 to 125 Unicode Characters	*Resolution:*	1 ASCII or 1 Unicode Character	2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

16 ASCII characters maximum, no Unicode

9	**Latitude of Vessel Reported**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional

DD022 Latitude, WGS-84

Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23	Latitude	**int32**	*Range:*	+/- 90 deg	*Resolution:*	1x10E-7 deg	"-" = South, resolution ~1.1 cm

10	**Longitude of Vessel Reported**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional

DD023 Longitude, WGS-84

Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF25	Longitude	**int32**	*Range:*	+/- 180 deg	*Resolution:*	1x10E-7 deg	"-" = West, resolution ~1.1 cm

11	**Time of Position**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional

DD158 Generic time of day

24 hour clock, 0 = midnight, time is in UTC

DF06	Time of day	**uint32**	*Range:*	0 to 86,401 s	*Resolution:*	1x10E-4 s	~24 hours, 0 = midnight, range allows for up to two leap seconds per day

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12 **MMSI Of Ship In Distress**

Byte Field Size: char 5 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD012 DSC Address or Geographic Area

Individual characters having only decimal values in the range 0 to 127 for the DSC symbols defined by ITU-R M.493 are used to code the address or geographic area as defined in ITU-R M.493-9 Section 5. This may represent an individual MMSI, a group MMSI, or a geographic area.

DF63 String, fixed **char8(n)** *Range:* 0 to 1,785 characters *Resolution:* 1 char

0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.

---

13 **DSC EOS Symbol**

Byte Field Size: 1 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD011 DSC Symbol Definitions

Integer numbers within the range 000 to 127 representing DSC Symbols defined by ITU-R M.493 Table 3 for: Phasing and Unique Functions; Format Specifier; Category; Nature of Distress; First Telecommand; Second Telecommand

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit

Unit-less number

---

14 **Expansion Enabled**

Byte Field Size: — *Request Parameter* Optional
Bit Field Size: 2 — *Command Parameter:* Optional

DD002 Generic status pair

MSB/LSB:
00 = [No, Off, Disabled, Reset, "0"],
01 = [Yes, On, Enabled, Set, "1"],
10 = Error,
11= [Unavailable, Unknown]

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1

Used to construct bit fields

---

15 **NMEA Reserved**

Byte Field Size: — *Request Parameter*
Bit Field Size: resv 6 — *Command Parameter:*

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1

Used to construct bit fields

Used to align subsequent data on a byte boundary.

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16 **Calling Rx Frequency/Channel**

Byte Field Size: char 6 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD017 Radio Tx or Rx Channel

MF/HF telephone channels to have first digit 3 followed by ITU channel numbers with leading zeros as required. MF/HF teletype channels to have first digit 4; the second and third digit give the frequency bands; and the fourth to sixth digits ITU channel numbers; each with leading zeros as required. VHF channels to have he first digit 9 followed by zero. The next number is "1" indicating the ship station's transmit frequency is being used as a simplex channel frequency, or "2" indicating the coast station's transmit frequency is being used as a simplex channel frequency, "0" otherwise. The remaining three numbers are the VHF channel numbers with leading zeros as required.

DF63 String, fixed **char8(n)** *Range:* 0 to 1,785 characters *Resolution:* 1 char — 0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.

17 **Calling Tx Frequency/Channel**

Byte Field Size: char 6 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD017 Radio Tx or Rx Channel

MF/HF telephone channels to have first digit 3 followed by ITU channel numbers with leading zeros as required. MF/HF teletype channels to have first digit 4; the second and third digit give the frequency bands; and the fourth to sixth digits ITU channel numbers; each with leading zeros as required. VHF channels to have he first digit 9 followed by zero. The next number is "1" indicating the ship station's transmit frequency is being used as a simplex channel frequency, or "2" indicating the coast station's transmit frequency is being used as a simplex channel frequency, "0" otherwise. The remaining three numbers are the VHF channel numbers with leading zeros as required.

DF63 String, fixed **char8(n)** *Range:* 0 to 1,785 characters *Resolution:* 1 char — 0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.

18 **Time of Receipt/Transmission**

Byte Field Size: 4 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD158 Generic time of day

24 hour clock, 0 = midnight, time is in UTC

DF06 Time of day **uint32** *Range:* 0 to 86,401 s *Resolution:* 1x10E-4 s — ~24 hours, 0 = midnight, range allows for up to two leap seconds per day

19 **Date of Receipt/Transmission**

Byte Field Size: 2 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD039 Generic date

Days since January 1, 1970, Date is relative to UTC Time.

DF41 Date, day count **uint16** *Range:* 0 to 65,532 days *Resolution:* 1 day — 0 = January 1, 1970, max = ~179 years

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20 **DSC Equipment Assigned Message ID**

Byte Field Size: 2 — *Request Parameter* Required
Bit Field Size: — *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned — **uint16** — *Range:* 0 to 65,532 — *Resolution:* 1 bit — Unit-less number

If this field is not specified in the "Command Request" or an ISO Request is made of this PGN, the response will with the current ID. Otherwise if this field is specified only the units with a matching ID will respond with this PGN.

21 **DSC Expansion Field Symbol**

Byte Field Size: 1 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD014 DSC Expansion Symbol Definitions — Integer numbers within the range 000 to 127 representing DSC Symbols defined by ITU-R M.821 Table 1.

DF53 Integer, 8 bit unsigned — **uint8** — *Range:* 0 to 252 — *Resolution:* 1 bit — Unit-less number

22 **DSC Expansion Field Data**

Byte Field Size: 8 or 16 n — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD015 DSC Symbol String — Individual characters having only decimal values in the range 0 to 127 for the DSC symbols defined by ITU-R M.493 are used to code: Telephone number as defined by ITU-R M.493 Section 8.2.3; DSC Expansion Data as defined by ITU-R M.821 Section 2.

DF50 String, variable, short — **ch8or16(n)** — *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters — *Resolution:* 1 ASCII or 1 Unicode Character — 2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

The meaning and size of this field is determined by the DSC Expansion symbol in field 21.
String length 38 ASCII characters maximum, no Unicode.

23 **Variable Number Of Fields, Field 21 Repeated, Expansion Field Type**

Byte Field Size: 1 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD014 DSC Expansion Symbol Definitions — Integer numbers within the range 000 to 127 representing DSC Symbols defined by ITU-R M.821 Table 1.

DF53 Integer, 8 bit unsigned — **uint8** — *Range:* 0 to 252 — *Resolution:* 1 bit — Unit-less number

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24 **Variable Number Of Fields, Field 22 Repeated, Expansion Field Data**

Byte Field Size: 8 or 16 n

Request Parameter Optional

Bit Field Size:

Command Parameter: Optional

DD015 DSC Symbol String

Individual characters having only decimal values in the range 0 to 127 for the DSC symbols defined by ITU-R M.493 are used to code: Telephone number as defined by ITU-R M.493 Section 8.2.3; DSC Expansion Data as defined by ITU-R M.821 Section 2.

DF50	String, variable, short	**ch8or16(n)**	*Range:*	0 to 250 ASCII or 0 to 125 Unicode Characters	*Resolution:* 1 ASCII or 1 Unicode Character	2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

The meaning and size of this field is determined by the DSC Expansion symbol in field 23.
String length 38 ASCII characters maximum, no Unicode.

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AIS Class B "CS" Static Data Report, Part A

PGN: 129809
hex: 1FB11

This parameter group is used by Class B "CS" shipborne mobile equipment each time Part A of ITU-R M.1372 Message 24 is received. The parameter group is the first of two parts, the second being transmitted in PGN 129810. This is Part A of ITU-R M.1372 Message 24. Message 24 part B is normally transmitted within 1 minute following Message 24 part A, and these parameter groups follow accordingly with PGN 129810 following 1 minute after PGN129809.
Reception of Part A followed immediately by Part B will occur in response to an AIS interrogation for Message 24. In that case the parameter groups will follow accordingly with PGN 129809 followed immediately by PGN 129810.

Application of all data fields is dependant upon the latest version of ITU-R M.1371.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 199

Field #	Field Name					
1	**Message ID**		*Byte Field Size:* *Bit Field Size:* 6		*Request Parameter* *Command Parameter:*	Optional Optional
	DD188 AIS Message Identifier		Message Identifier (range of 0 to 63). See the latest version of ITU-R M.1371 for more information.			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
	24 = AIS Class B Static Data Part A					
2	**Repeat Indicator**		*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter* *Command Parameter:*	Optional Optional
	DD185 AIS Repeater Indicator		Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3). 0 = Default 1 = First retransmission 2 = Second retransmission 3 = Final retransmission See the latest version of ITU-R M.1371 for more information.			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
3	**User ID**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD010 Generic numeric ID, large		Number of route, waypoint, event, mark, etc.			
	DF55 Integer, 32 bit unsigned	**uint32**	*Range:* 0 to 4,294,967,292	*Resolution:* 1 bit	Unit-less number	
	MMSI number of mobile station reporting its static information					
4	**Name**		*Byte Field Size:* char n *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD192 Generic String, ASCII, Fixed length		Length specified by PGN field definition.			
	DF63 String, fixed	**char8(n)**	*Range:* 0 to 1,785 characters	*Resolution:* 1 char	0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.	
	20 character string, default value is "@@@@@@@@@@@@@@@@@@@@" = not available					

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AIS Class B "CS" Static Data Report, Part A

PGN: 129809
hex: 1FB11

5	**AIS Transceiver Information**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	5	*Command Parameter:*	Optional	
	DD246 AIS Transceiver Information		0 = Channel A VDL reception, 1 = Channel B VDL reception, 2 = Channel A VDL transmission, 3 = Channel B VDL transmission, 4 = Own information not broadcast, 5-31 = Reserved.				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
6	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*		
			Bit Field Size:	resv 3	*Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
	Used to align subsequent data on byte boundary.						
7	**Sequence ID**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD056 Sequence ID		An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:*	0 to 252	*Resolution:*	1 bit	Unit-less number

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AIS Class B "CS" Static Data Report, Part B

PGN: 129810
hex: 1FB12

This parameter group is used by Class B “CS” shipborne mobile equipment each time Part B of ITU-R M.1372 Message 24 is received. The parameter group is the second of two parts, the first part being transmitted in PGN 129809. This is Part B of ITU-R M.1372 Message 24. Message 24 part B is normally transmitted within 1 minute following Message 24 part A, and these parameter groups follow accordingly with PGN 129810 following 1 minute after PGN129809.
Reception of Part A followed immediately by Part B will occur in response to an AIS interrogation for Message 24. In that case the parameter groups will follow accordingly with PGN 129809 followed immediately by PGN 129810.

Application of all data fields is dependant upon the latest version of ITU-R M.1371.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 200

1 Message ID
Byte Field Size: *Bit Field Size:* 6 *Request Parameter* Optional *Command Parameter:* Optional

DD188 AIS Message Identifier
Message Identifier (range of 0 to 63).

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
24 = AIS Class B static data Part B

2 Repeat Indicator
Byte Field Size: *Bit Field Size:* 2 *Request Parameter* Optional *Command Parameter:* Optional

DD185 AIS Repeater Indicator
Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3).

0 = Default
1 = First retransmission
2 = Second retransmission
3 = Final retransmission

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 User ID
Byte Field Size: 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD010 Generic numeric ID, large
Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number
MMSI number of mobile station reporting its static information

4 Type of Ship and Cargo
Byte Field Size: *Bit Field Size:* 8 *Request Parameter* Optional *Command Parameter:* Optional

DD193 Ship/Cargo Type
0=Not Available or no ship (default),
1-99= (See the latest version of ITU-R M.1371 Section 3.3.8.2.3.2 Table 18),
100-199=Reserved for Regional (See the latest version of ITU-R M.1371),
200-255=Reserved for future (See the latest version of ITU-R M.1371).

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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AIS Class B "CS" Static Data Report, Part B

PGN: 129810
hex: 1FB12

5 **Vendor ID**

Byte Field Size: char | n
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD192 Generic String, ASCII, Fixed length — Length specified by PGN field definition.

DF63 String, fixed **char8(n)** *Range:* 0 to 1,785 characters *Resolution:* 1 char

7 character string - Unique identification of the unit by a number as defined by the manufacturer; ("@@@@@@@" = not available = default)

0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.

6 **Call Sign**

Byte Field Size: char | n
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD192 Generic String, ASCII, Fixed length — Length specified by PGN field definition.

DF63 String, fixed **char8(n)** *Range:* 0 to 1,785 characters *Resolution:* 1 char

7 character string - See the latest version of ITU-R M.1371 for more information; ("@@@@@@@" = not available = default)

0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.

7 **Ship Length**

Byte Field Size: 2
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD194 Distance, medium — Dependent upon PG Field definition.

DF75 Distance, Medium **uint16** *Range:* 0 to 6553.2 m *Resolution:* 1x10E-1 m

Length of mobile station reporting its static data; Valid range = 0-1022, and must be greater than or equal to field 10 "Reference Point Position Aft of Bow", a value of 65535 indicates that data is not available. For more information, see IMO Circular 227 Section 5.3 Ships Dimensions or NMEA 0400 Section 19.3.2 Vessel Reference Point (This field is valid only if field 3 "User ID" contains a value <= 999999999)

8 **Ship Beam**

Byte Field Size: 2
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD194 Distance, medium — Dependent upon PG Field definition.

DF75 Distance, Medium **uint16** *Range:* 0 to 6553.2 m *Resolution:* 1x10E-1 m

Beam of mobile station reporting its statis data; A value of 65535 indicates that data is not available (This field is valid only if field 3 "User ID" contains a value <= 999999999)

9 **Reference Point Position from Starboard**

Byte Field Size: 2
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD194 Distance, medium — Dependent upon PG Field definition.

DF75 Distance, Medium **uint16** *Range:* 0 to 6553.2 m *Resolution:* 1x10E-1 m

Distance to reference point measured from the starboard side of mobile station reporting its static data; Valid range = 0-63, a value of 65535 indicates that data is not available. For more information, see IMO Circular 227 Section 5.3 Ships Dimensions or NMEA 0400 Section 19.3.2 Vessel Reference Point (This field is valid only if field 3 "User ID" contains a value <= 999999999)

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AIS Class B "CS" Static Data Report, Part B

PGN: 129810
hex: 1FB12

10 **Reference Point Position Aft of Bow** — Byte Field Size: 2 — Request Parameter: Optional — Bit Field Size: — Command Parameter: Optional

DD194 Distance, medium — Dependent upon PG Field definition.

DF75 Distance, Medium **uint16** Range: 0 to 6553.2 m Resolution: 1x10E-1 m

Distance to reference point measured aft from the bow of mobile station reporting its static position; Valid range = 0-511, a value of 65535 indicates that data is not available. For more information, see IMO Circular 227 Section 5.3 Ships Dimensions or NMEA 0400 Section 19.3.2 Vessel Reference Point (This field is valid only if field 3 "User ID" contains a value <= 999999999)

11 **Mother Ship MMSI** — Byte Field Size: 4 — Request Parameter: Optional — Bit Field Size: — Command Parameter: Optional

DD010 Generic numeric ID, large — Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** Range: 0 to 4,294,967,292 Resolution: 1 bit Unit-less number

For unregistered daughter vessels, this is the MMSI assigned to the mother ship; (This field is valid only if field 3 "User ID" contains a value > 999999999)

12 **NMEA Reserved** — Byte Field Size: — Request Parameter: — Bit Field Size: resv 2 — Command Parameter:

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

13 **AIS Spare** — Byte Field Size: — Request Parameter: — Bit Field Size: resv 6 — Command Parameter:

DD311 AIS Spare Field — Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** Range: N2KUnitless Resolution: 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "Spare" bit field found within the corresponding AIS message such that future definition within the AIS message can also be accommodated within this parameter group while maintaining field order; Normally, spare or reserved bits in NMEA Network Messages are encoded with logic 1's, however for AIS parameter groups the spare or reserved bits are encoded as logic 0's.

14 **AIS Transceiver Information** — Byte Field Size: — Request Parameter: Optional — Bit Field Size: 5 — Command Parameter: Optional

DD246 AIS Transceiver Information — 0 = Channel A VDL reception, 1 = Channel B VDL reception, 2 = Channel A VDL transmission, 3 = Channel B VDL transmission, 4 = Own information not broadcast, 5-31 = Reserved.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

15 **NMEA Reserved** — Byte Field Size: — Request Parameter: — Bit Field Size: resv 3 — Command Parameter:

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

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AIS Class B "CS" Static Data Report, Part B

PGN: 129810
hex: 1FB12

16	**Sequence ID**	*Byte Field Size:*	1	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD056 Sequence ID

An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

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AIS Single Slot Binary Message

PGN: 129811
hex: 1FB13

This parameter group provides data associated with the ITU-R M.1371 Message 25 Single Slot Binary Message supporting both addressed and broadcast communication of binary data. An AIS device may generate this parameter group either upon receiving a VHF Data Link Message 25, or upon receipt of ISO or NMEA request PGN. The Command Group Function PGN 126208 may be used with this PGN to configure parameters such as the binary data (see ITU-R M.1371-5 Message 25 for additional information). Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

Single Frame: N *Priority Default:* 5 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 231

1 Sequence ID
Byte Field Size: 1 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD056 Sequence ID
An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

2 Message ID
Byte Field Size: *Bit Field Size:* 6 *Request Parameter* Optional *Command Parameter:* Optional

DD188 AIS Message Identifier
Message Identifier (range of 0 to 63).

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
25 equals Single Slot Binary Message

3 Repeat Indicator
Byte Field Size: *Bit Field Size:* 2 *Request Parameter* Optional *Command Parameter:* Optional

DD185 AIS Repeater Indicator
Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3).

0 = Default
1 = First retransmission
2 = Second retransmission
3 = Final retransmission

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

4 Source ID
Byte Field Size: 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD010 Generic numeric ID, large
Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number
MMSI Number of source station.

5 Destination Indicator
Byte Field Size: *Bit Field Size:* 1 *Request Parameter* Optional *Command Parameter:* Optional

DD387 AIS Destination Indicator
0 = Broadcast (no Destination ID field used)
1 = Addressed (Destination ID field uses 30 data bits for MMSI

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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AIS Single Slot Binary Message

PGN: 129811
hex: 1FB13

6 Binary data flag

Byte Field Size:
Bit Field Size: 1
Request Parameter: Optional
Command Parameter: Optional

DD386 AIS Binary Structure

0 = unstructured binary data (no Application Identifier bits used)
1 = binary data coded as defined by using the 16-bit Application Identifier

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

7 AIS Spare

Byte Field Size:
Bit Field Size: resv 2
Request Parameter:
Command Parameter:

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** Range: N2KUnitless Resolution: 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "spare" bit field found within the corresponding AIS message, such that future definition within the AIS message can also be accommodated within this field.

8 NMEA Reserved

Byte Field Size:
Bit Field Size: resv 4
Request Parameter:
Command Parameter:

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

9 Destination ID

Byte Field Size: 4
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD010 Generic numeric ID, large

Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** Range: 0 to 4,294,967,292 Resolution: 1 bit Unit-less number

MMSI Number of destination station.

10 Number of bits in Binary Data Field

Byte Field Size: 1
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD006 Generic counter, short

Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** Range: 0 to 252 Resolution: 1 bit Unit-less number

Indicates the number of binary of data bits that are contained within the binary data field

11 Binary Data

Byte Field Size:
Bit Field Size: n
Request Parameter: Optional
Command Parameter: Optional

DD142 Binary Bit Field

Binary data bit field.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Application specific data

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AIS Multi Slot Binary Message

PGN: 129812
hex: 1FB14

This parameter group provides data associated with the ITU-R M.1371 Message 26 Multi Slot Binary Message with Communication State supporting both addressed and broadcast communication of binary data. This message is primarily intended for scheduled binary transmissions by application of the SOTDMA or ITDMA access scheme. An AIS device may generate this parameter group either upon receiving a VHF Data Link Message 26, or upon receipt of ISO or NMEA request PGN. The Command Group Function PGN 126208 may be used with this PGN to configure parameters such as the binary data (see ITU-R M.1371-5 Message 26 for additional information). Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

This PGN is variable length and provides binary data from 1 to 5 slots. Fields 21 through 24 only exist when there is binary data in the related slot. For example; with a 3 slot binary message, the last field of this PGN is Field 22.

Due to variable length fields at the end of the message, all unused bits in the last byte (of the last field) shall be set to 1.

Single Frame: N *Priority Default:* 5 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 232

1 Sequence ID
Byte Field Size: 1 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD056 Sequence ID
An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

2 Message ID
Byte Field Size: *Bit Field Size:* 6 *Request Parameter* Optional *Command Parameter:* Optional

DD188 AIS Message Identifier
Message Identifier (range of 0 to 63).

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
26 equals Multi Slot Binary Message with communications state.

3 Repeat Indicator
Byte Field Size: *Bit Field Size:* 2 *Request Parameter* Optional *Command Parameter:* Optional

DD185 AIS Repeater Indicator
Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3).

0 = Default
1 = First retransmission
2 = Second retransmission
3 = Final retransmission

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

4 Source ID
Byte Field Size: 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD010 Generic numeric ID, large
Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number
MMSI Number of source station.

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5 **Destination Indicator**
Byte Field Size: *Bit Field Size:* 1
Request Parameter Optional
Command Parameter: Optional

DD387 AIS Destination Indicator
0 = Broadcast (no Destination ID field used)
1 = Addressed (Destination ID field uses 30 data bits for MMSI

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

6 **Binary data flag**
Byte Field Size: *Bit Field Size:* 1
Request Parameter Optional
Command Parameter: Optional

DD386 AIS Binary Structure
0 = unstructured binary data (no Application Identifier bits used)
1 = binary data coded as defined by using the 16-bit Application Identifier

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

7 **NMEA Reserved**
Byte Field Size: *Bit Field Size:* resv 6
Request Parameter *Command Parameter:*

DD001 Reserved field
Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
Used to align subsequent data on a byte boundary.

---

8 **Destination ID**
Byte Field Size: 4 *Bit Field Size:*
Request Parameter Optional
Command Parameter: Optional

DD010 Generic numeric ID, large
Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number
MMSI Number of destination station.

---

9 **AIS Spare**
Byte Field Size: *Bit Field Size:* resv 2
Request Parameter *Command Parameter:*

DD311 AIS Spare Field
Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use
This field mirrors the "spare" bit field found within the corresponding AIS message, such that future definition within the AIS message can also be accommodated within this field.

---

10 **Communication state selector flag**
Byte Field Size: *Bit Field Size:* 1
Request Parameter Optional
Command Parameter: Optional

DD245 AIS Communication State Selctor Flag
0=SOTDMA communication state,
1=ITDMA communication state follows.

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

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AIS Multi Slot Binary Message

PGN: 129812
hex: 1FB14

11 Communication state

Byte Field Size:
Bit Field Size: 19
Request Parameter: Optional
Command Parameter: Optional

DD187 AIS Communication State

The Communication State contains information used by the various TDMA slot allocation algorithms and synchronization information

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

12 AIS Spare

Byte Field Size:
Bit Field Size: resv 4
Request Parameter:
Command Parameter:

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** Range: N2KUnitless Resolution: 1 Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "spare" bit field found within the corresponding AIS message, such that future definition within the AIS message can also be accommodated within this field.

---

13 NMEA Reserved

Byte Field Size:
Bit Field Size: resv 6
Request Parameter:
Command Parameter:

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

---

14 Number of Bits in Binary Data Field – 1st slot

Byte Field Size: 1
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD006 Generic counter, short

Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** Range: 0 to 252 Resolution: 1 bit Unit-less number

Indicates the number of data bits that are contained within the binary data field for 1st slot

---

15 Number of Bits in Binary Data Field – 2nd slot

Byte Field Size: 1
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD006 Generic counter, short

Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** Range: 0 to 252 Resolution: 1 bit Unit-less number

Indicates the number of data bits that are contained within the binary data field for 2nd slot

---

16 Number of Bits in Binary Data Field – 3rd slot

Byte Field Size: 1
Bit Field Size:
Request Parameter: Optional
Command Parameter: Optional

DD006 Generic counter, short

Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** Range: 0 to 252 Resolution: 1 bit Unit-less number

Indicates the number of data bits that are contained within the binary data field for 3rd slot

---

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17 **Number of Bits in Binary Data Field – 4th slot** *Byte Field Size:* 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD006 Generic counter, short — Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

Indicates the number of data bits that are contained within the binary data field for 4th slot

---

18 **Number of Bits in Binary Data Field – 5th slot** *Byte Field Size:* 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD006 Generic counter, short — Numeric count, event counter, sequence counter

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

Indicates the number of data bits that are contained within the binary data field for 5th slot

---

19 **Binary Data – 1st slot** *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: n *Command Parameter:* Optional

DD142 Binary Bit Field — Binary data bit field.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Application specific data for the 1st slot

---

20 **Binary Data – 2nd slot** *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: n *Command Parameter:* Optional

DD142 Binary Bit Field — Binary data bit field.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Application specific data for the 2nd slot.
When the "Number of Bits in Binary Data Field – 2nd slot" equals 0 this field and subsequent binary data fields are not included in the PGN.

---

21 **Binary Data – 3rd slot** *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: n *Command Parameter:* Optional

DD142 Binary Bit Field — Binary data bit field.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Application specific data for the 3rd slot.
When the "Number of Bits in Binary Data Field – 3rd slot" equals 0 this field and subsequent binary data fields are not included in the PGN.

---

22 **Binary Data – 4th slot** *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: n *Command Parameter:* Optional

DD142 Binary Bit Field — Binary data bit field.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Application specific data for the 4th slot.
When the "Number of Bits in Binary Data Field - 4th slot" equals 0 this field and subsequent binary data fields are not included in the PGN.

---

23 **Binary Data – 5th slot** *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: n *Command Parameter:* Optional

DD142 Binary Bit Field — Binary data bit field.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Application specific data for the 5th slot.
When the "Number of Bits in Binary Data Field - 5th slot" equals 0 this field is not included in the PGN.

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AIS Long-Range Broadcast Message

PGN: 129813
hex: 1FB15

This parameter group provides data associated with the ITU-R M.1371 Message 27 Long-Range AIS Broadcast Message. This message is primarily intended for long-range detection of AIS Class A and Class B “SO” mobile stations by satellite. This message has similar content to ITU-R M.1371 Messages 1, 2 and 3. Note that future revisions to the ITU-R M.1371 VHF Data Link Messages may result in their spare or reserved bits being defined with a specific meaning, requiring the spare or reserved parameter in this parameter group to have the corresponding new meaning in future revisions of this standard.

Single Frame: N *Priority Default:* 5 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 233

Field #	Field Name						
1	**Sequence ID**		Byte Field Size: 1 Bit Field Size:			Request Parameter: Optional Command Parameter: Optional	
	DD056 Sequence ID			An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	Range: 0 to 252		Resolution: 1 bit		Unit-less number
2	**Message ID**		Byte Field Size: Bit Field Size: 6			Request Parameter: Optional Command Parameter: Optional	
	DD188 AIS Message Identifier			Message Identifier (range of 0 to 63). See the latest version of ITU-R M.1371 for more information.			
	DF52 Bit field	**bit(n)**	Range: Variable		Resolution: 1		Used to construct bit fields
	27 equals Long-Range AIS Broadcast Message						
3	**Repeat Indicator**		Byte Field Size: Bit Field Size: 2			Request Parameter: Optional Command Parameter: Optional	
	DD185 AIS Repeater Indicator			Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3). 0 = Default 1 = First retransmission 2 = Second retransmission 3 = Final retransmission See the latest version of ITU-R M.1371 for more information.			
	DF52 Bit field	**bit(n)**	Range: Variable		Resolution: 1		Used to construct bit fields
4	**User ID**		Byte Field Size: 4 Bit Field Size:			Request Parameter: Optional Command Parameter: Optional	
	DD010 Generic numeric ID, large			Number of route, waypoint, event, mark, etc.			
	DF55 Integer, 32 bit unsigned	**uint32**	Range: 0 to 4,294,967,292		Resolution: 1 bit		Unit-less number
	MMSI Number of mobile station reporting its position.						
5	**Longitude**		Byte Field Size: 4 Bit Field Size:			Request Parameter: Optional Command Parameter: Optional	
	DD023 Longitude, WGS-84			Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10’s of minute.			
	DF25 Longitude	**int32**	Range: +/- 180 deg		Resolution: 1x10E-7 deg		"-" = West, resolution ~1.1 cm

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6 **Latitude**
Byte Field Size: 4 | Request Parameter: Optional
Bit Field Size: | Command Parameter: Optional

DD022 Latitude, WGS-84
Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23 Latitude **int32** Range: +/- 90 deg Resolution: 1x10E-7 deg "-" = South, resolution ~1.1 cm

---

7 **Position Accuracy**
Byte Field Size: | Request Parameter: Optional
Bit Field Size: 1 | Command Parameter: Optional

DD184 AIS Position Accuracy
0 = low accuracy > 10m such as nondifferential GNSS (default),
1 = high accuracy < 10m such as DGNSS

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

8 **Raim Flag**
Byte Field Size: | Request Parameter: Optional
Bit Field Size: 1 | Command Parameter: Optional

DD189 AIS RAIM-flag
0 = RAIM not in use (default),
1 = RAIM in use

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

9 **Navigation Status**
Byte Field Size: | Request Parameter: Optional
Bit Field Size: 4 | Command Parameter: Optional

DD183 AIS Navigational Status
0 = under way using engine,
1 = at anchor,
2 = not under command,
3 = restricted manoeuvrability,
4 = constrained by her draught,
5 = moored,
6 = aground,
7 = engaged in fishing,
8 = under way sailing,
9 = reserved for future amendment of navigational status for ships carrying DG, HS, or MP, or IMO hazard or pollutant category C (HSC),
10 = reserved for future amendment of navigational status for ships carrying DG, HS or MP, or IMO hazard or pollutant category A (WIG),
11-14 = reserved for future use,
15 = not defined (default)

See the latest version of ITU-R M.1371 for more information.

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

---

10 **Position Latency**
Byte Field Size: | Request Parameter: Optional
Bit Field Size: 1 | Command Parameter: Optional

DD388 AIS Position Latency Flag
0 = reported position latency is less than 5 seconds
1 = reported position latency is greater than 5 seconds = default

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

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11 **AIS Spare** *Byte Field Size:* *Bit Field Size:* resv 1 *Request Parameter* *Command Parameter:*

DD311 AIS Spare Field

Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.

DF115 Bit field defaulting to zer **bit0(n)** *Range:* N2KUnitless *Resolution:* 1

Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use

This field mirrors the "spare" bit field found within the corresponding AIS message, such that future definition within the AIS message can also be accommodated within this field.

---

12 **Speed Over Ground** *Byte Field Size:* 2 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD044 Generic Speed

DF35 Speed **uint16** *Range:* 0 to 655.32 m/s *Resolution:* 1x10E-2 m/s 1 Knot = 0.5144 m/s

---

13 **Course Over Ground** *Byte Field Size:* 2 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD165 Course-Over-Ground (COG)

The direction of the path over ground actually followed by a vessel.

DF02 Angle **uint16** *Range:* 0 to 2Pi rad *Resolution:* 1x10E-4 rad Resolution ~0.0057deg, 1 deg = .01745 rad

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Loran-C TD Data

PGN: 130052
hex: 1FC04

This provides Time Difference (TD) lines of position of Loran-C signals relative to a single Group Repetition Interval.

Field #1, Group Repetition Interval (GRI), is identified as a request parameter for this Parameter Group. All providers of this PGN shall accept and process requests based upon the requested value of Field #1. A device receiving an ISO Request (PGN 059904) for this PGN, shall respond by providing as many of these PGNs as necessary for every GRI that has associated data fields. If a Complex Request Group Function (PGN 126208) requesting this PGN is received, the receiving device shall respond in the following manner: If no requested fields have been included with the Complex Request, than the response is to return one or more PGNs, just like responding to the ISO Request (PGN 059904) described above. If the Complex Request (PGN 126208) includes the GRI field, then the response shall be filtered by the field (Field #1) and field value (GRI #) contained within the request.

For example, if the Complex Request for this PGN contained a value of 9960 for field 1, the Group Repetition Interval (GRI), and this was a GRI that the device was operating with or had information about, than the device would respond by providing a single PGN with Time Difference measurement data associated with the 9960 GRI value requested.

If the GRI requested is considered valid by the device, but the device was not operating on that GRI, or had no data associated with the GRI requested, there are two possible responses:
1 - The device responds with the PGN containing the GRI requested and all other fields set to the value indicating "Data not available".
2 - The device responds with the Acknowledge Group PGN (126208) containing the error state of "0x2 = Temporarily unable to comply".

If the request was global no response would be required. If the request was addressed to the device, than either response 1 or 2 would apply.

If the GRI requested is not considered a valid GRI by the receiving device, then the appropriate response would be the Acknowledge Group PGN (126208), containing the error state for the requested GRI field of "0x3 = Request or command parameter out-of-range;".

Single Frame: No *Priority Default:* 3 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 9

Field #	*Field Name*					
1	**Group Repetition Interval (GRI)**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Required *Command Parameter:* Optional	
	DD027 Loran-C GRI			Group Repetition Interval (GRI) in nano-sec. Often cited in units of 10 micro-sec. (i.e., 9960 = 99,600,000 ns)		
	DF45 Time interval, precise	**int32**	*Range:* +/- 2.14 s	*Resolution:* 1x10E-9 s		
2	**Master Range**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD029 Loran - Range (Time)			The actual propagation time of a Loran-C signal from the station to a receiver in nano-sec.		
	DF45 Time interval, precise	**int32**	*Range:* +/- 2.14 s	*Resolution:* 1x10E-9 s		
3	**V Secondary TD**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD028 Loran-C TD			Loran-C Time difference (TD) in nano-sec. The arrival time of a Loran-C secondary station signal minus the arrival time of the master station signal.		
	DF45 Time interval, precise	**int32**	*Range:* +/- 2.14 s	*Resolution:* 1x10E-9 s		

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Loran-C TD Data

PGN: 130052
hex: 1FC04

4	**W Secondary TD**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD028 Loran-C TD		Loran-C Time difference (TD) in nano-sec. The arrival time of a Loran-C secondary station signal minus the arrival time of the master station signal.			
	DF45 Time interval, precise	**int32**	*Range:*	+/- 2.14 s	*Resolution:*	1x10E-9 s
5	**X Secondary TD**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD028 Loran-C TD		Loran-C Time difference (TD) in nano-sec. The arrival time of a Loran-C secondary station signal minus the arrival time of the master station signal.			
	DF45 Time interval, precise	**int32**	*Range:*	+/- 2.14 s	*Resolution:*	1x10E-9 s
6	**Y Secondary TD**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD028 Loran-C TD		Loran-C Time difference (TD) in nano-sec. The arrival time of a Loran-C secondary station signal minus the arrival time of the master station signal.			
	DF45 Time interval, precise	**int32**	*Range:*	+/- 2.14 s	*Resolution:*	1x10E-9 s
7	**Z Secondary TD**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD028 Loran-C TD		Loran-C Time difference (TD) in nano-sec. The arrival time of a Loran-C secondary station signal minus the arrival time of the master station signal.			
	DF45 Time interval, precise	**int32**	*Range:*	+/- 2.14 s	*Resolution:*	1x10E-9 s
8	**Station status: Master**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD030 Loran-C station status		MSB: to LSB: xxx1 = Station in use, xx1x = Low SNR, x1xx = Cycle error, 1xxx = Blink where x = don't care			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
9	**Station status: V**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD030 Loran-C station status		MSB: to LSB: xxx1 = Station in use, xx1x = Low SNR, x1xx = Cycle error, 1xxx = Blink where x = don't care			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields

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10	**Station status: W**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD030 Loran-C station status		MSB: to LSB: xxx1 = Station in use, xx1x = Low SNR, x1xx = Cycle error, 1xxx = Blink where x = don't care			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
11	**Station status: X**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD030 Loran-C station status		MSB: to LSB: xxx1 = Station in use, xx1x = Low SNR, x1xx = Cycle error, 1xxx = Blink where x = don't care			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
12	**Station status: Y**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD030 Loran-C station status		MSB: to LSB: xxx1 = Station in use, xx1x = Low SNR, x1xx = Cycle error, 1xxx = Blink where x = don't care			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
13	**Station status: Z**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD030 Loran-C station status		MSB: to LSB: xxx1 = Station in use, xx1x = Low SNR, x1xx = Cycle error, 1xxx = Blink where x = don't care			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
14	**Mode**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD025 Mode, Data		0x0 = Autonomous mode, 0x1 = Differential, enhanced mode, 0x2 = Estimated mode, 0x3 = Simulator mode, 0x4 = Manual mode, 0x5 to 0xD = Reserved 0xE = Error, 0xF = Data not available			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields

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Loran-C TD Data

PGN: 130052
hex: 1FC04

15	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*		
		Bit Field Size:	resv 4	*Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
	Used to align subsequent data on a byte boundary.					

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Loran-C Range Data

PGN: 130053
hex: 1FC05

This provides Propagation times (Ranges) of Loran-C signals relative to a single Group Repetition Interval.

Field #1, Group Repetition Interval (GRI), is identified as a request parameter for this Parameter Group. All providers of this PGN shall accept and process requests based upon the requested value of Field #1. A device receiving an ISO Request (PGN 059904) for this PGN, shall respond by providing as many of these PGNs as necessary for every GRI that has associated data fields. If a Complex Request Group Function (PGN 126208) requesting this PGN is received, the receiving device shall respond in the following manner: If no requested fields have been included with the Complex Request, than the response is to return one or more PGNs, just like responding to the ISO Request (PGN 059904) described above. If the Complex Request (PGN 126208) includes the GRI field, then the response shall be filtered by the field (Field #1) and field value (GRI #) contained within the request.

For example, if the Complex Request for this PGN contained a value of 9960 for field 1, the Group Repetition Interval (GRI), and this was a GRI that the device was operating with or had information about, than the device would respond by providing a single PGN with Range measurement data associated with the 9960 GRI value requested.

If the GRI requested is considered valid by the device, but the device was not operating on that GRI, or had no data associated with the GRI requested, there are two possible responses:
1 - The device responds with the PGN containing the GRI requested and all other fields set to the value indicating "Data not available".
2 - The device responds with the Acknowledge Group PGN (126208) containing the error state of "0x2 = Temporarily unable to comply".

If the request was global no response would be required. If the request was addressed to the device, than either response 1 or 2 would apply.

If the GRI requested is not considered a valid GRI by the receiving device, then the appropriate response would be the Acknowledge Group PGN (126208), containing the error state for the requested GRI field of "0x3 = Request or command parameter out-of-range;".

Single Frame: No *Priority Default:* 3 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 10

Field #	*Field Name*					
1	**Group Repetition Interval (GRI)**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Required *Command Parameter:* Optional	
	DD027 Loran-C GRI			Group Repetition Interval (GRI) in nano-sec. Often cited in units of 10 micro-sec. (i.e., 9960 = 99,600,000 ns)		
	DF45 Time interval, precise	**int32**	*Range:* +/- 2.14 s	*Resolution:* 1x10E-9 s		
2	**Master Range**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD029 Loran - Range (Time)			The actual propagation time of a Loran-C signal from the station to a receiver in nano-sec.		
	DF45 Time interval, precise	**int32**	*Range:* +/- 2.14 s	*Resolution:* 1x10E-9 s		
3	**V Secondary Range**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD029 Loran - Range (Time)			The actual propagation time of a Loran-C signal from the station to a receiver in nano-sec.		
	DF45 Time interval, precise	**int32**	*Range:* +/- 2.14 s	*Resolution:* 1x10E-9 s		

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Loran-C Range Data

PGN: 130053
hex: 1FC05

4	**W Secondary Range**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD029 Loran - Range (Time)		The actual propagation time of a Loran-C signal from the station to a receiver in nano-sec.			
	DF45 Time interval, precise	**int32**	*Range:* +/- 2.14 s	*Resolution:*	1x10E-9 s	
5	**X Secondary Range**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD029 Loran - Range (Time)		The actual propagation time of a Loran-C signal from the station to a receiver in nano-sec.			
	DF45 Time interval, precise	**int32**	*Range:* +/- 2.14 s	*Resolution:*	1x10E-9 s	
6	**Y Secondary Range**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD029 Loran - Range (Time)		The actual propagation time of a Loran-C signal from the station to a receiver in nano-sec.			
	DF45 Time interval, precise	**int32**	*Range:* +/- 2.14 s	*Resolution:*	1x10E-9 s	
7	**Z Secondary Range**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD029 Loran - Range (Time)		The actual propagation time of a Loran-C signal from the station to a receiver in nano-sec.			
	DF45 Time interval, precise	**int32**	*Range:* +/- 2.14 s	*Resolution:*	1x10E-9 s	
8	**Station status: Master**		*Byte Field Size:* *Bit Field Size:* 4		*Request Parameter* *Command Parameter:*	Optional Optional
	DD030 Loran-C station status		MSB: to LSB: xxx1 = Station in use, xx1x = Low SNR, x1xx = Cycle error, 1xxx = Blink where x = don't care			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:*	1	Used to construct bit fields
9	**Station status: V**		*Byte Field Size:* *Bit Field Size:* 4		*Request Parameter* *Command Parameter:*	Optional Optional
	DD030 Loran-C station status		MSB: to LSB: xxx1 = Station in use, xx1x = Low SNR, x1xx = Cycle error, 1xxx = Blink where x = don't care			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:*	1	Used to construct bit fields

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10	**Station status: W**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD030 Loran-C station status		MSB: to LSB: xxx1 = Station in use, xx1x = Low SNR, x1xx = Cycle error, 1xxx = Blink where x = don't care			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
11	**Station status: X**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD030 Loran-C station status		MSB: to LSB: xxx1 = Station in use, xx1x = Low SNR, x1xx = Cycle error, 1xxx = Blink where x = don't care			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
12	**Station status: Y**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD030 Loran-C station status		MSB: to LSB: xxx1 = Station in use, xx1x = Low SNR, x1xx = Cycle error, 1xxx = Blink where x = don't care			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
13	**Station status: Z**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD030 Loran-C station status		MSB: to LSB: xxx1 = Station in use, xx1x = Low SNR, x1xx = Cycle error, 1xxx = Blink where x = don't care			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields
14	**Mode**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD025 Mode, Data		0x0 = Autonomous mode, 0x1 = Differential, enhanced mode, 0x2 = Estimated mode, 0x3 = Simulator mode, 0x4 = Manual mode, 0x5 to 0xD = Reserved 0xE = Error, 0xF = Data not available			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1 Used to construct bit fields

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Loran-C Range Data

PGN: 130053
hex: 1FC05

15	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 4	*Command Parameter:*	

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

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Loran-C Signal Data

PGN: 130054
hex: 1FC06

SNR, ECD, and ASF values of Loran-C signals

Field #1, Group Repetition Interval (GRI), is identified as a request parameter for this Parameter Group. All providers of this PGN shall accept and process requests based upon the requested value of Field #1. A device receiving an ISO Request (PGN 059904) for this PGN, shall respond by providing as many of these PGNs as necessary for every GRI that has associated data fields. If a Complex Request Group Function (PGN 126208) requesting this PGN is received, the receiving device shall respond in the following manner: If no requested fields have been included with the Complex Request, than the response is to return one or more PGNs, just like responding to the ISO Request (PGN 059904) described above. If the Complex Request (PGN 126208) includes the GRI field, then the response shall be filtered by the field (Field #1) and field value (GRI #) contained within the request.

For example, if the Complex Request for this PGN contained a value of 9960 for field 1, the Group Repetition Interval (GRI), and this was a GRI that the device was operating with or had information about, than the device would respond by providing a single PGN with Range measurement data associated with the 9960 GRI value requested.

If the GRI requested is considered valid by the device, but the device was not operating on that GRI, or had no data associated with the GRI requested, there are two possible responses:
1 - The device responds with the PGN containing the GRI requested and all other fields set to the value indicating "Data not available".
2 - The device responds with the Acknowledge Group PGN (126208) containing the error state of "0x2 = Temporarily unable to comply".

If the request was global no response would be required. If the request was addressed to the device, than either response 1 or 2 would apply.

Single Frame: No | *Priority Default:* 3 | *Default Update Rate:* 1000 milliseconds | *Frequency:* 1. cycles per second
Destination: Global | *Query Support:* Required | *Command Support:* Optional | *ACK Rqmnts:* None

Original Reference ID # 11

Field #	*Field Name*					
1	**Group Repetition Interval (GRI)**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* Required *Command Parameter:* Optional	
	DD027 Loran-C GRI			Group Repetition Interval (GRI) in nano-sec. Often cited in units of 10 micro-sec. (i.e., 9960 = 99,600,000 ns)		
	DF45 Time interval, precise	**int32**	*Range:* +/- 2.14 s	*Resolution:* 1x10E-9 s		
2	**Station identifier**		*Byte Field Size:* char 1 *Bit Field Size:*		*Request Parameter* Required *Command Parameter:* Optional	
	DD031 Loran-C Station ID			1-character code for the Loran-C station: M = master, V, W, X, Y, Z		
	DF63 String, fixed	**char8(n)**	*Range:* 0 to 1,785 characters	*Resolution:* 1 char		0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.
3	**Station SNR**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD026 Loran-C SNR			Signal at standard sampling point / RMS noise in 3kHz BW at 100KHz		
	DF31 dB, relative measure	**int16**	*Range:* +/- 327.64 dB	*Resolution:* 1x10E-2 dB		

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Loran-C Signal Data

PGN: 130054
hex: 1FC06

4	**Station ECD**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD032 Loran-C ECD		Envelope-to-Cycle Discrepancy (ECD) of the Loran-C pulse			
	DF45 Time interval, precise	**int32**	*Range:* +/- 2.14 s		*Resolution:* 1x10E-9 s	
5	**Station ASF**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD033 Loran-C ASF		Additional Secondary Factor (ASF) associated with the propagation of the signal from a Loran-C station			
	DF45 Time interval, precise	**int32**	*Range:* +/- 2.14 s		*Resolution:* 1x10E-9 s	

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Label

PGN: 130060
hex: 1FC0C

The Label PGN is used to set and retrieve a text label assigned to a particular device or a particular hardware resource within a particular device. The Label PGN supports multiple-channel devices. Field programmability of the fields within this PGN over the network is optional, but if implemented shall be performed using the NMEA Command Group Function (126208).

Global ISO Requests (PGN 059904) for this Parameter Group shall be ignored.

Global NMEA Request Group Functions (PGN 126208) for this Parameter Group that do not specify Request Parameters shall be ignored.

Response to a global or an addressed NMEA Request Group Function (PGN 126208) for this Parameter Group that specifies one or more Request Parameters shall be the transmission of this PGN once for each Hardware Channel ID with fields that match the specified parameters.

Response to an addressed ISO Request (PGN 059904) for this Parameter Group shall be the transmission of this PGN once for each defined Hardware Channel ID, and once for the device itself.

Response to an addressed NMEA Request Group Function (PGN 126208) for this Parameter Group that does not specify any request parameters shall be the transmission of this PGN once for each defined Hardware Channel ID, and once for the device itself. Response to a valid NMEA Command Group Function (PGN 126208) in addition to any required acknowledgement shall be the transmission of this PGN once for each defined Hardware Channel ID affected by the command. Additionally, any other device operations that change the values for any Hardware Channel ID configuration shall also result in the transmission of this PGN for each defined Hardware Channel affected.

To retrieve labels from a device, issue an addressed ISO request for this PGN to the device. Devices will respond to the ISO request by transmitting an occurrence of this PGN for each channel that is capable of being labeled, plus one for the device itself.

To change the value of a label for a data source instance within a multiple-channel device, an NMEA Command Group Function (126208) is sent to the device with the “Hardware Channel ID” field set the appropriate Hardware Channel ID and the “Label” field set to the desired new label value.

To change the value of a label for a device taken as a whole, an NMEA Command Group Function (126208) is sent to the device with the “Hardware Channel ID” field set to 252 (0xFC), and with the “Label” field set to the desired label text. The Label message transmitted for Hardware Channel ID 252 (0xFC) will always have Data Source Instance Field Number, Secondary Enumeration Field Number, and Parameter Field Number Fields set to 255 (0xFF).

To change the value of a label for an entire PGN applicable to all fields within that PGN, an NMEA Command Group Function (126208) is sent to the device with the “Hardware Channel ID” field set to the appropriate Hardware Channel ID, and with the “Label” field set to the desired label text. The Label message transmitted in response will always have Data Source Instance Field Number, Secondary Enumeration Field Number, and Parameter Field Number Fields set to 255 (0xFF).

If a text label is shared by more than one Hardware Channel ID, then all occurrences affected by the change shall be transmitted. For example, a single-channel device shall support Hardware Channel ID values of 0, for the single sensor channel, and 252, for the device as a whole. It may share a single text label between these two Hardware Channel ID fields, however; in this case, changing the label with Hardware Channel ID 252 will result in the label with Hardware Channel ID 0 also being changed to the same value, resulting in a transmission for both Hardware Channel ID 0 and 252.

Re-transmission of this PGN any time changes are made to its parameters is done so that all displays in the system will be able to update their stored versions of the labels as they are changed. Any displays or display software in the system that implement the Label PGN are required to monitor the network for occurrences of this PGN and to replace this data with the contents of the received message.

Note 1: Every Command Group Function transmitted for this PGN shall contain a value for Field 1(Hardware Channel ID). The value commanded shall identify the hardware channel that the remaining commanded fields apply to.

Refer to Appendix D for more detailed explanations and application examples.

Single Frame:	N	*Priority Default:*	7	*Default Update Rate:*	Upon Request	milliseconds	*Frequency:*	#Type!	cycles per second

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Label

PGN: 130060
hex: 1FC0C

Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 220

1 **Hardware Channel ID** *Byte Field Size:* 1 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Note 1

DD357 Hardware Channel ID

0 through 251 = Identifier for a single "real" hardware resource, such as a temperature sensor or a single switch contact. Each label in a device will have a unique Hardware Channel ID. Values in this range are to be assigned to hardware resources sequentially beginning with 0 and increasing by one until all resources are assigned.

252 (0xFC) = Refers to the device as a whole, and the label associated with this value is the label for the entire device.

253 = Reserved
254 = Error
255 = Data Not Available

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

Uniquely identifies a hardware resource within a device to which the label applies.
Labels with this field set to 252 (0xFC) shall have the PGN field, Data Source Instance Field Number field, Data Source Instance Value field, Secondary Enumeration Field Number field, Secondary Enumeration Field Value field, and Parameter Field Number field set to their appropriate Data Not Available values.

2 **PGN** *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: 24 *Command Parameter:* Optional

DD009 PGN

24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

PGN of the parameter group to which this label applies

3 **Data Source Instance Field Number** *Byte Field Size:* 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD005 Generic numeric ID, short

Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

For parameter groups that are transmitted multiple times by the same node with differing data source instances, each representing different measurements; this field identifies the field number of that data source instance.

The value 255 means that the Label is not specific to an instance.

4 **Data Source Instance Value** *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: 8 *Command Parameter:* Optional

DD128 Generic instance

0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

The data source instance value to be matched to determine which transmission of the parameter group is to be used as the data source for this channel. If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.

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Label

PGN: 130060
hex: 1FC0C

5 **Secondary Enumeration Field Number** *Byte Field Size:* 1 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

For parameter groups that are transmitted multiple times by the same node with differing data type, each representing different measurements; this field identifies the field number of that data type.

The value 255 means that the Label is not specific to a Secondary Enumeration.

6 **Secondary Enumeration Field Value** *Byte Field Size:* 1 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Optional

DD358 Secondary Enumeration Field Value

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The data type value to be matched to determine which transmission of the parameter group is to be used as the data source for this channel.

7 **Parameter Field Number** *Byte Field Size:* 1 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

This field indicates the field number of the field in the parameter group specified by the PGN Field (Field 2) containing the measurement which this label identifies.

The value 255 means that the Label is not specific to a field.

8 **Label** *Byte Field Size:* 8 or 16 n *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character 2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

Contains the text of the label. If this field is programmable, the device should be able to store a Label with a minimum length of 32 ASCII or 16 Unicode characters. If the field is not programmable, labels should not exceed more than 32 ASCII or 16 Unicode characters.

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Channel Source Configuration

PGN: 130061
hex: 1FC0D

The Channel Source Configuration parameter group is used to identify data sources that a device receives from the NMEA network to satisfy device operational requirements. For example, if a device stores the vessel location any time an event monitored by the device occurs, and there are more than one GPS device aboard the vessel, this parameter group may be used to report and also command which GPS is used/to use by the device. An example may be a MOB sensor.

This parameter group is especially useful for 'black box' devices that have no native GUI for direct configuration, and therefore would require configuration over the network.

The input being reported/configured is identified using a Data Source Channel ID, in similar fashion to the Label parameter group (PGN 130060) Hardware Channel ID. However, the Data Source Channel ID in this case is related to a channel of information being consumed by the device from the network, where the Label parameter group Hardware Channel ID was a channel of information being produced/transmitted to the network. An example could be an autopilot that requires rudder position feedback. The autopilot may assign Data Source Channel ID 0 to rudder position feedback. When the autopilot transmits this parameter group using Data Source Channel ID 0, it is indicating which specific rudder angle adapter is being used for rudder position feedback. In the event that a device supports both this parameter group and the Label parameter group, no assumptions should be made regarding the relationship between source channels and label channels.

Global ISO Requests (PGN 059904) for this Parameter Group shall be ignored.

Global NMEA Request Group Functions (PGN 126208) for this Parameter Group that do not specify Request Parameters shall be ignored.

Response to a global or an addressed NMEA Request Group Function (PGN 126208) for this Parameter Group that specifies one or more Request Parameters shall be the transmission of this PGN once for each Channel Source ID with fields that match the specified parameters.

Response to an addressed ISO Request (PGN 059904) for this Parameter Group shall be the transmission of this PGN once for each defined Channel Source ID.

Response to an addressed NMEA Request Group Function (PGN 126208) for this Parameter Group that does not specify any request parameters shall be the transmission of this PGN once for each defined Channel Source ID.

Response to a valid NMEA Command Group Function (PGN 126208) in addition to any required acknowledgement shall be the transmission of this PGN once for each defined Channel Source ID affected by the command. Additionally, any other device operations that change the values for any Channel Source ID configuration or selection shall also result in the transmission of this PGN for each defined Channel Source ID affected.

The NMEA Command group function can be used to set the values for a device that broadcasts this parameter group. The level of programmability is dependent on the manufacturer. On receipt of a valid command, the response should be to immediately use the new criteria to select a new data source, and once selected this parameter group should be transmitted with the result of that selection. Improperly formatted commands or command value combinations not supported by the manufacturer should be negatively acknowledged in their entirety.

Note 1: When NMEA Command group function support is provided, minimum support shall include a value for field 1. Every command shall include a value for field 1 and shall be used to identify the input channel being configured/commanded.

Single Frame:	N	*Priority Default:*	7	*Default Update Rate:*	Upon Request	milliseconds	*Frequency:*	#Type!	cycles per second
Destination:	Global	*Query Support:*	Required	*Command Support:*	Optional		*ACK Rqmnts:*	None	

Field #	*Field Name*	Original Reference ID # 221

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Channel Source Configuration

PGN: 130061
hex: 1FC0D

1	**Data Source Channel ID**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Required Note 1

DD359 Data Source Channel ID

0 through 252 = Identification of a data values used by the device, such as a value used in a calculation, a value logged when an event occurs, or a value converted/output as a physical stimulus. Data Source Channel IDs in this range are assigned sequentially to data requirements beginning with 0 and increasing until all data requirements are assigned.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

Uniquely identifies a required data input for the node.

2	**Source Selection Status**	*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter* *Command Parameter:*	Optional Optional

DD360 Source Selection Status

0b00 = No source selected for this channel, NAME and mask if provided represent template to match.
0b01 = Source selected for this channel, NAME identifies source, mask if provided identifies template.
0b10 = Reserved.
0b11 = No Data Available.

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3	**NMEA Reserved**	*Byte Field Size:* *Bit Field Size:* resv 2		*Request Parameter* *Command Parameter:*	

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

4	**NAME Selection Criteria Mask**	*Byte Field Size:* *Bit Field Size:* 12		*Request Parameter* *Command Parameter:*	Optional Optional

DD327 NAME Selection Criteria Mask

0xxx xxxx xxx1 = NAME shall match Unique ID field
0xxx xxxx xx1x = NAME shall match Manufacturer Code field
0xxx xxxx x1xx = NAME shall match Device Instance Lower field
0xxx xxxx 1xxx = NAME shall match Device Instance Upper field
0xxx xxx1 xxxx = NAME shall match Device Function field
0xxx xx0x xxxx = Reserved bit (aligns with NAME Reserved field 6)
0xxx x1xx xxxx = NAME shall match Device Class field
0xxx 1xxx xxxx = NAME shall match System Instance field
0xx1 xxxx xxxx = NAME shall match Industry Group field
0x0x xxxx xxxx = Reserved bit (aligns with NAME Reserved field 10)
00xx xxxx xxxx = Reserved bit
1111 1111 1111 = Data not Available, Mask not used
'x' = don't care
Note that multiple matches are possible.
Inactive field selections shall be transmitted as 0.
Reserved bits within the field shall be transmitted as 0, except when transmitting "Data Not Available, Mask not used".

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Identifies which NAME field(s) of the Address Claim Parameter Group (PGN 060928) must be matched when selecting the source for the data channel.

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Channel Source Configuration

PGN: 130061
hex: 1FC0D

5 **Source NAME** — Byte Field Size: 8 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Optional

DD320 Network ID NAME — This field is defined by fields 1- 10 of PGN 60928

DF56 Integer, 64 bit unsigned **uint64** Range: 0 to (2E+64)-4 Resolution: 1 bit Unit-less number

Address Claim NAME of the node used as a source for this data channel

---

6 **PGN** — Byte Field Size: — Bit Field Size: 24 — Request Parameter: Optional — Command Parameter: Optional

DD009 PGN — 24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Parameter group to be received for this data channel.

---

7 **Data Source Instance Field Number** — Byte Field Size: 1 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** Range: 0 to 252 Resolution: 1 bit Unit-less number

For parameter groups that are transmitted multiple times by the same node with differing data source instances, each representing different measurements; this field identifies the field number of that data source instance.

---

8 **Data Source Instance Value** — Byte Field Size: — Bit Field Size: 8 — Request Parameter: Optional — Command Parameter: Optional

DD128 Generic instance

0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

The data source instance value to be matched to determine which transmission of the parameter group is to be used as the data source for this channel. If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.

---

9 **Secondary Enumeration Field Number** — Byte Field Size: 1 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** Range: 0 to 252 Resolution: 1 bit Unit-less number

For parameter groups that are transmitted multiple times by the same node with differing data type, each representing different measurements; this field identifies the field number of that data type.

---

10 **Secondary Enumeration Field Value** — Byte Field Size: 1 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Optional

DD358 Secondary Enumeration Field Value

DF53 Integer, 8 bit unsigned **uint8** Range: 0 to 252 Resolution: 1 bit Unit-less number

The data type value to be matched to determine which transmission of the parameter group is to be used as the data source for this channel.

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Channel Source Configuration

PGN: 130061
hex: 1FC0D

11 **Parameter Field Number**

Byte Field Size: 1
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

For parameter groups that contain multiple fields representing different measurements of the same parameter, this field identifies a specific field within the transmitted parameter group to be used as the data source for this channel.

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Route and WP Service - Database List

PGN: 130064
hex: 1FC10

Complex request for this PGN should return a list of Databases in which a navigation Device organizes its Routes and WPs.

A Database may contain one WP-List and multiple Routes.

A device need not support the concept of multiple Databases. The support of this PGN is then optional and the default Database ID = 0 shall be used in other "Route and Waypoint Service" PGNs. If supporting this PGN the Number of Databases defaults to 1.

The reply should use the same transfer protocol as the request.
ISO request shall, if this PGN is supported, return the 3 first fields with respectively: NA, 0 and the Number of Databases Available.
For a complete description of the Route and WP PGNs, see the application note in Appendix D.

Single Frame: No *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 96

Field #	Field Name					
1	**Start Database ID**	*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Required *Command Parameter:* Optional	
	DD007 Generic numeric ID, medium			Number of route, waypoint, event, mark, etc.		
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number	
	1st Database ID requested/sent. If not specified in the request, the reply shall default to the lowest available Database ID. If the requested Database does not exist, the default response is the next highest Database ID.					
2	**nItems**	*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Required *Command Parameter:* Optional	
	DD007 Generic numeric ID, medium			Number of route, waypoint, event, mark, etc.		
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number	
	n Databases requested/sent. If not specified in the request, the reply shall include all Databases available (or as many as the transport protocol space permits).					
3	**Number of Databases available**	*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD007 Generic numeric ID, medium			Number of route, waypoint, event, mark, etc.		
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number	
4	**Database ID**	*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD007 Generic numeric ID, medium			Number of route, waypoint, event, mark, etc.		
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number	
	Databases shall be included in this PGN in the order of increasing ID.					

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5 **Database Name**

Byte Field Size: 8 or 16 n
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character

2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

---

6 **Database Timestamp**

Byte Field Size: 4
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD158 Generic time of day — 24 hour clock, 0 = midnight, time is in UTC

DF06 Time of day **uint32** *Range:* 0 to 86,401 s *Resolution:* 1x10E-4 s

~24 hours, 0 = midnight, range allows for up to two leap seconds per day

---

7 **Database Datestamp**

Byte Field Size: 2
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD039 Generic date — Days since January 1, 1970, Date is relative to UTC Time.

DF41 Date, day count **uint16** *Range:* 0 to 65,532 days *Resolution:* 1 day

0 = January 1, 1970, max = ~179 years

---

8 **WP Position Resolution**

Byte Field Size:
Bit Field Size: 4

Request Parameter Optional
Command Parameter: Optional

DD238 WP Position Resolution

0= [>0.1min.],
1= [<=0.1& >0.01min.], 2= [<=0.01 & >0.001]
3= [<=0.001 & >0.0001]
4= [<=0.000 1 & >=0.000 001min.] 5-6 =reserved,
7= Not available (not known)
(1min. = 0.01667deg.)

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1

Used to construct bit fields

---

9 **NMEA Reserved**

Byte Field Size:
Bit Field Size: resv 4

Request Parameter
Command Parameter:

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1

Used to construct bit fields

Used to align subsequent data on a byte boundary.

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Route and WP Service - Database List

PGN: 130064
hex: 1FC10

10	**Number of Routes in Database**		*Byte Field Size:*	2	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD007 Generic numeric ID, medium		Number of route, waypoint, event, mark, etc.			
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:*	0 to 65,532	*Resolution:*	1 bit Unit-less number
11	**Number of WPs in Database**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD010 Generic numeric ID, large		Number of route, waypoint, event, mark, etc.			
	DF55 Integer, 32 bit unsigned	**uint32**	*Range:*	0 to 4,294,967,292	*Resolution:*	1 bit Unit-less number
	Includes WPs from the WP-List and all other WPs embedded in Routes					
12	**Number of Bytes in Database**		*Byte Field Size:*	4	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD010 Generic numeric ID, large		Number of route, waypoint, event, mark, etc.			
	DF55 Integer, 32 bit unsigned	**uint32**	*Range:*	0 to 4,294,967,292	*Resolution:*	1 bit Unit-less number
13	**Fields 4 thru 12 repeat as needed**		*Byte Field Size:*	?	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD000 Undefined		Application specific, defined at time of use			
	DF00 Undefined	**Undefined**	*Range:*	undefined	*Resolution:*	undefined Application specific, defined at time of use.

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Route and WP Service - Route List

PGN: 130065
hex: 1FC11

Complex request for this PGN should return a list of Routes in a Database.

A Database may contain Routes identified with Route ID in the range 0-65532. There may be empty gaps (the Route ID does not represent a valid/existent Route) anywhere in this range. This PGN shall include valid Routes only.

The reply should use the same transfer protocol as the request.
ISO request for this PGN shall return 059392 with a positive ACK if the PGN is supported
For a complete description of the Route and WP PGNs, see the application note in Appendix D.

Single Frame: No *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 97

1 **Start Route ID** *Byte Field Size:* 2 *Bit Field Size:* *Request Parameter* Required *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

1st Route ID requested/sent.
If not specified in the request, the reply shall default to the lowest available Route ID.
If the requested Route is not available, the default response is the next highest Route ID.

2 **nItems** *Byte Field Size:* 2 *Bit Field Size:* *Request Parameter* Required *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

n Routes requested/sent.
If not specified in the request, the reply shall include all Routes available in the Database (or as many of them as the transport protocol space permits).

3 **Number of Routes available in Database** *Byte Field Size:* 2 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

4 **Database ID** *Byte Field Size:* 2 *Bit Field Size:* *Request Parameter* Required *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

Should be specified in the request. Is typically gathered from a prior "Database List PGN".
If not specified in the request, the reply may use a default Database.

5 **Route ID** *Byte Field Size:* 2 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

Routes shall be included in this PGN in the order of increasing ID, skipping non-valid Routes.

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Route and WP Service - Route List

PGN: 130065
hex: 1FC11

6	**Route Name**		*Byte Field Size:*	8 or 16 n	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD004 Generic name string, short			Name of place, route, waypoint, destination, vessel, vehicle, etc.			
	DF50 String, variable, short	**ch8or16(n)**	*Range:*	0 to 250 ASCII or 0 to 125 Unicode Characters	*Resolution:*	1 ASCII or 1 Unicode Character	2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.
7	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*		
			Bit Field Size:	resv 2	*Command Parameter:*		
	DD001 Reserved field			Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.						
8	**WP Identification Method**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	2	*Command Parameter:*	Optional	
	DD240 WP Identification Method			0=WP's in WP-List, 1=WP embedded in Rute, 2=Reserved 3=Null (info not available)			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
9	**Route Status**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	4	*Command Parameter:*	Optional	
	DD239 Route Status			0=Active, 1=Inactive, 2=Deleted, 3-13= Reserved, 14=Error, 15= Null			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
10	**Fields 5 thru 9 repeat as needed**		*Byte Field Size:*	?	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD000 Undefined			Application specific, defined at time of use			
	DF00 Undefined	**Undefined**	*Range:*	undefined	*Resolution:*	undefined	Application specific, defined at time of use.

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Route and WP Service - Route/WP-List Attributes

PGN: 130066
hex: 1FC12

Complex request for this PGN should return the attributes of a Route or the WP-List.

WARNING: The Route Track may not be defined by the Waypoint positions only: The Navigation Method (GC/RL) and Radius of Turn at each Waypoint are optional additional parameters defining the Track.
PGN 130066 field #8 "Critical supplementary parameters" shall be used to determine if there are additional parameters to the Waypoint positions.
For a complete description of the Route and WP PGNs, see the application note in Appendix D.

A Database may contain one WP-List and multiple Routes. The Database ID and the Route ID shall be specified in the request/reply. Route ID = 65535 (NA) indicates that the request/reply is addressing the WP-List.

This PGN contains parameters common for the Route or WP-List. The individual Waypoints with positions and other associated parameters must be requested/transferred in other “Route and WP Service” PGNs, such as PGN 130067 “Route - WP Name & Position”.

The reply should use the same transfer protocol as the request.
ISO request for this PGN shall return 059392 with a positive ACK if the PGN is supported.

Single Frame:	No	*Priority Default:*	7	*Default Update Rate:*		milliseconds	*Frequency:*	NA	cycles per second
Destination:	Global	*Query Support:*	Required	*Command Support:*	Optional		*ACK Rqmnts:*	None	

Field # *Field Name* — Original Reference ID # 98

1 Database ID — *Byte Field Size:* 2 — *Bit Field Size:* — *Request Parameter:* Required — *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** — *Range:* 0 to 65,532 — *Resolution:* 1 bit — Unit-less number

2 Route ID — *Byte Field Size:* 2 — *Bit Field Size:* — *Request Parameter:* Required — *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** — *Range:* 0 to 65,532 — *Resolution:* 1 bit — Unit-less number

Set to 65535 (NA) to access the WP-List in the Database.

3 Route/WP-List Name — *Byte Field Size:* 8 or 16 n — *Bit Field Size:* — *Request Parameter:* Optional — *Command Parameter:* Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** — *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters — *Resolution:* 1 ASCII or 1 Unicode Character — 2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

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Route and WP Service - Route/WP-List Attributes

PGN: 130066
hex: 1FC12

4 Route/WP-List Timestamp

Byte Field Size: 4 — Request Parameter: Optional
Bit Field Size: — Command Parameter: Optional

DD158 Generic time of day — 24 hour clock, 0 = midnight, time is in UTC

DF06	Time of day	**uint32**	Range:	0 to 86,401 s	Resolution:	1x10E-4 s	~24 hours, 0 = midnight, range allows for up to two leap seconds per day

5 Route/WP-List Datestamp

Byte Field Size: 2 — Request Parameter: Optional
Bit Field Size: — Command Parameter: Optional

DD039 Generic date — Days since January 1, 1970, Date is relative to UTC Time.

DF41	Date, day count	**uint16**	Range:	0 to 65,532 days	Resolution:	1 day	0 = January 1, 1970, max = ~179 years

6 Change at Last Timestamp

Byte Field Size: — Request Parameter: Optional
Bit Field Size: 8 — Command Parameter: Optional

DD237 Changed at timestamp flag

0000 0000 = No change in data.
0xxx xxx1 = WP: Change in main data (Position, Name)
0xxx xx1x = WP: Change in supplementary parameters (or new added)
0xxx x1xx = Changed no. of WP's in Route/WP-List, and/or name changed/added
0xxx 1xxx = Route: Change supplementary parameters (or new added)
0xx1 xxxx = Reserved
0x1x xxxx = Reserved
01xx xxxx = Other not specifyed change
1111 1111 = This flag is not supported
x = don't care

DF52	Bit field	**bit(n)**	Range:	Variable	Resolution:	1	Used to construct bit fields

7 Number of WPs in the Route/WP-List

Byte Field Size: 2 — Request Parameter: Optional
Bit Field Size: — Command Parameter: Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54	Integer, 16 bit unsigned	**uint16**	Range:	0 to 65,532	Resolution:	1 bit	Unit-less number

8 Critical supplementary parameters

Byte Field Size: — Request Parameter: Optional
Bit Field Size: 8 — Command Parameter: Optional

DD258 Critical supplementary Route parameters

0000 0000 = There are no parameters additional to the Waypoint Positions
xxxx xxx1 = Navigation Method (GC/RL), PGN 130069
xxxx xx1x = XTE Limit, PGN 130069
xxxx x1xx = Spare
xxxx 1xxx = Spare
xxx1 xxxx = Spare
xx1x xxxx = Spare
x1xx xxxx = Spare
1xxx xxxx = Spare

Each of these bitflags represents a parameter. If a flag is '1', the parameter must be downloaded and appended to the Route information. Ignoring any of these parameters will not be safe.

The spare flags shall be transmitted as '0's.

DF52	Bit field	**bit(n)**	Range:	Variable	Resolution:	1	Used to construct bit fields

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9 **Navigation Method** — Byte Field Size: — Bit Field Size: 2 — Request Parameter: Optional — Command Parameter: Optional

DD119 Calculation Type
0 = Great Circle calculations,
1 = Rhumb Line calculations,
2 = Error,
3 = Null

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Not applicable to the WP-List
NOTE: This is the default Navigation Method for the Route. It may be altered for specific legs. Ref. field 8, Critical supplementary parameters.

---

10 **WP Identification Method** — Byte Field Size: — Bit Field Size: 2 — Request Parameter: Optional — Command Parameter: Optional

DD240 WP Identification Method
0=WP's in WP-List,
1=WP embedded in Rute,
2=Reserved
3=Null (info not available)

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Not applicable to the WP-List

---

11 **Route Status** — Byte Field Size: — Bit Field Size: 4 — Request Parameter: Optional — Command Parameter: Optional

DD239 Route Status
0=Active,
1=Inactive,
2=Deleted,
3-13= Reserved,
14=Error,
15= Null

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Not applicable to the WP-List

---

12 **XTE Limit for the Route** — Byte Field Size: 2 — Bit Field Size: — Request Parameter: Optional — Command Parameter: Optional

DD149 Distance ordered
A commanded distance like radius order, off-track limit, etc.

DF74 Distance, rough **int16** Range: +/-32,764 m Resolution: 1 m

No negative values. The limit applies to both sides of the track.
Not applicable to the WP-List.
NOTE: This is the default XTE-Limit for the Route. It may be altered for specific legs. Ref. field 8, Critical supplementary parameters.

---

13 **NMEA Reserved** — Byte Field Size: — Bit Field Size: resv 0 — Request Parameter: — Command Parameter:

DD001 Reserved field
Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

Additional future Route parameters (eg. Radius of Turn, etc). Fields are normally not reserved at the end of a PGN because it is a general rule that new parameters may be appended to an existing (non-single frame) PGN. This is a reminder.

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Route and WP Service - Route - WP Name & Position

PGN: 130067
hex: 1FC13

Complex request of this PGN should return the Waypoints belonging to a Route.

WARNING: The Route Track may not be defined by the Waypoint positions only: The Navigation Method (GC/RL) and Radius of Turn at each Waypoint are optional additional parameters defining the Track.
PGN 130066 field #8 "Critical supplementary parameters" shall be used to determine if there are additional parameters to the Waypoint positions.
For a complete description of the Route and WP PGNs, see the application note in Appendix D.

The Waypoints of a Route are numbered with RPS# continuously from 0 and up in the Forward Direction. RPS# is the sequence number of the Waypoint in the Route. (Route Point Sequence #).

A Database may contain one WP-List and multiple Routes. The WP-List holds Waypoints identified by WPID in the range 0-65532. Each of these may be single or belong to one or multiple Routes in the Database. A Waypoint may be private to a particular Route and not exist in the WP List, its WPID shall then be 65535(NA).

The Database ID and Route ID shall be specified in the request/reply. These will typically be selected from the information received in other ""Route and WP Service"" PGNs previously requested from the same Device.

The Waypoint Name (if available) and Position shall be filled with valid data, even if the Waypoints are from the WP-List which the receiver may already have downloaded.

The reply should use the same transfer protocol as the request.
ISO request for this PGN shall return 059392 with a positive ACK if the PGN is supported.

Single Frame: No *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 99

1 **Start RPS#** *Byte Field Size:* 2 *Bit Field Size:* *Request Parameter* Required *Command Parameter:* Optional

DD007 Generic numeric ID, medium Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

RPS# of the 1st requested/sent Waypoint.
If not specified in the request, the default is the first RPS# (=0) of the Route.

2 **nItems** *Byte Field Size:* 2 *Bit Field Size:* *Request Parameter* Required *Command Parameter:* Optional

DD007 Generic numeric ID, medium Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

n Waypoints requested/sent.
If not specified in the request, the reply shall include all WPs of the Route (or as many thereof as the transport protocol space permits).

3 **Number of WPs in the Route** *Byte Field Size:* 2 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD007 Generic numeric ID, medium Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

The receiver will use this parameter to determine if it has received all WPs of the complete route.

4 **Database ID** *Byte Field Size:* 2 *Bit Field Size:* *Request Parameter* Required *Command Parameter:* Optional

DD007 Generic numeric ID, medium Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

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5	**Route ID**		*Byte Field Size:*	2		*Request Parameter*	Required	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD007 Generic numeric ID, medium			Number of route, waypoint, event, mark, etc.				
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:*	0 to 65,532	*Resolution:*	1 bit	Unit-less number	
6	**WPID**		*Byte Field Size:*	2		*Request Parameter*	Optional	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD007 Generic numeric ID, medium			Number of route, waypoint, event, mark, etc.				
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:*	0 to 65,532	*Resolution:*	1 bit	Unit-less number	
	Shall have valid data if the Waypoint exists in the WP List. The Waypoints shall be included in the order of increasing RPS# (The order of appearance in the Forward Direction of the Route).							
7	**WP Name**		*Byte Field Size:*	8 or 16 n		*Request Parameter*	Optional	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD004 Generic name string, short			Name of place, route, waypoint, destination, vessel, vehicle, etc.				
	DF50 String, variable, short	**ch8or16(n)**	*Range:*	0 to 250 ASCII or 0 to 125 Unicode Characters	*Resolution:*	1 ASCII or 1 Unicode Character	2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.	
	Max. 30 ASCII or Unicode Characters							
8	**WP Latitude**		*Byte Field Size:*	4		*Request Parameter*	Optional	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD022 Latitude, WGS-84			Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.				
	DF23 Latitude	**int32**	*Range:*	+/- 90 deg	*Resolution:*	1x10E-7 deg	"-" = South, resolution ~1.1 cm	

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9	**WP Longitude**		*Byte Field Size:*	4	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD023 Longitude, WGS-84			Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.			
	DF25 Longitude	**int32**	*Range:*	+/- 180 deg	*Resolution:*	1x10E-7 deg	"-" = West, resolution ~1.1 cm
10	**Fields 6 thru 9 repeat as needed**		*Byte Field Size:*	?	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD000 Undefined			Application specific, defined at time of use			
	DF00 Undefined	**Undefined**	*Range:*	undefined	*Resolution:*	undefined	Application specific, defined at time of use.

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Route and WP Service - Route - WP Name

PGN: 130068
hex: 1FC14

Complex request of this PGN should return the Waypoints belonging to a Route.

WARNING: The Route Track may not be defined by the Waypoint positions only: The Navigation Method (GC/RL) and Radius of Turn at each Waypoint are optional additional parameters defining the Track.
PGN 130066 field #8 "Critical supplementary parameters" shall be used to determine if there are additional parameters to the Waypoint positions.
For a complete description of the Route and WP PGNs, see the application note in Appendix D.

The Waypoints of a Route are numbered with RPS# continuously from 0 and up in the Forward Direction. RPS# is the sequence number of the Waypoint in the Route. (Route Point Sequence #).

A Database may contain one WP-List and multiple Routes. The WP-List holds Waypoints identified by WPID in the range 0-65532. Each of these may be single or belong to one or multiple Routes in the Database. A Waypoint may be private to a particular Route and not exist in the WP List, its WPID shall then be 65535(NA).

The Database ID and Route ID shall be specified in the request/reply. These will typically be selected from the information received in other ""Route and WP Service"" PGNs previously requested from the same Device.

The Waypoint Name (if available) and Position shall be filled with valid data, even if the Waypoints are from the WP-List which the receiver may already have downloaded.

The reply should use the same transfer protocol as the request.
ISO request for this PGN shall return 059392 with a positive ACK if the PGN is supported.

Single Frame: No *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 100

1	**Start RPS#**	*Byte Field Size:* 2		*Request Parameter*	Required	
		Bit Field Size:		*Command Parameter:*	Optional	
	DD007 Generic numeric ID, medium		Number of route, waypoint, event, mark, etc.			
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number	
	1st RPS# requested/sent					
2	**nItems**	*Byte Field Size:* 2		*Request Parameter*	Required	
		Bit Field Size:		*Command Parameter:*	Optional	
	DD007 Generic numeric ID, medium		Number of route, waypoint, event, mark, etc.			
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number	
	n RPS# requested/sent					
3	**Number of WPs in the Route**	*Byte Field Size:* 2		*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	
	DD007 Generic numeric ID, medium		Number of route, waypoint, event, mark, etc.			
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number	
4	**Database ID**	*Byte Field Size:* 2		*Request Parameter*	Required	
		Bit Field Size:		*Command Parameter:*	Optional	
	DD007 Generic numeric ID, medium		Number of route, waypoint, event, mark, etc.			
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number	

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Route and WP Service - Route - WP Name

PGN: 130068
hex: 1FC14

5	**Route ID**		*Byte Field Size:*	2		*Request Parameter*	Required	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD007 Generic numeric ID, medium			Number of route, waypoint, event, mark, etc.				
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:*	0 to 65,532	*Resolution:*	1 bit	Unit-less number	
6	**WPID**		*Byte Field Size:*	2		*Request Parameter*	Optional	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD007 Generic numeric ID, medium			Number of route, waypoint, event, mark, etc.				
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:*	0 to 65,532	*Resolution:*	1 bit	Unit-less number	
7	**WP Name**		*Byte Field Size:*	8 or 16 n		*Request Parameter*	Optional	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD004 Generic name string, short			Name of place, route, waypoint, destination, vessel, vehicle, etc.				
	DF50 String, variable, short	**ch8or16(n)**	*Range:*	0 to 250 ASCII or 0 to 125 Unicode Characters	*Resolution:*	1 ASCII or 1 Unicode Character	2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.	
	Max. 30 ASCII or Unicode Characters							
8	**field 6 thru 7 repeat as needed**		*Byte Field Size:*	?		*Request Parameter*	Optional	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD000 Undefined			Application specific, defined at time of use				
	DF00 Undefined	**Undefined**	*Range:*	undefined	*Resolution:*	undefined	Application specific, defined at time of use.	

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Route and WP Service - XTE Limit & Navigation Method

PGN: 130069
hex: 1FC15

Complex request of this PGN will return XTE Limit and/or Navigation Method specific to individual legs of a Route.

The Database ID and Route ID shall be specified in the request/reply.

The parameters apply to the one leg after the Waypoint identified with RPS# in the Forward Direction of the Route and overrides any Route default XTE Limit and Navigation Method.

Waypoints where none of these parameters has valid data shall not be included in this PGN.

The reply should use the same transfer protocol as the request.
ISO request for this PGN shall return 059392 with a positive ACK if the PGN is supported.
For a complete description of the Route and WP PGNs, see the application note in Appendix D.

Single Frame: No *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 101

Field #	Field Name				
1	**Start RPS#**	*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Required Optional
	DD007 Generic numeric ID, medium		Number of route, waypoint, event, mark, etc.		
	DF54 Integer, 16 bit unsigned **uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number	
	1st RPS# requested/sent If not specified in the request, the default is the lowest RPS# with valid data. If the requested RPS# has no valid data, the default response is the next highest RPS#.				
2	**nItems**	*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Required Optional
	DD007 Generic numeric ID, medium		Number of route, waypoint, event, mark, etc.		
	DF54 Integer, 16 bit unsigned **uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number	
	n RPS# requested/sent. If not specified in the request, the reply shall include all RPS# with valid data (or as many the transfer protocol space permits).				
3	**Number of Waypoints with a specific XTE Limit or Nav. Method**	*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD007 Generic numeric ID, medium		Number of route, waypoint, event, mark, etc.		
	DF54 Integer, 16 bit unsigned **uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number	
	Note: This may be less than Number of Waypoints in the Route.				
4	**Database ID**	*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Required Optional
	DD007 Generic numeric ID, medium		Number of route, waypoint, event, mark, etc.		
	DF54 Integer, 16 bit unsigned **uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number	
5	**Route ID**	*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Required Optional
	DD007 Generic numeric ID, medium		Number of route, waypoint, event, mark, etc.		
	DF54 Integer, 16 bit unsigned **uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number	

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Route and WP Service - XTE Limit & Navigation Method

PGN: 130069
hex: 1FC15

6	**RPS#**	*Byte Field Size:*	2	*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

Waypoints without individually specific XTE Limit or Navigation Method shall not be included.

7	**XTE limit in the leg after WP**	*Byte Field Size:*	2	*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	

DD149 Distance ordered — A commanded distance like radius order, off-track limit, etc.

DF74 Distance, rough **int16** *Range:* +/-32,764 m *Resolution:* 1 m

No negative values. The limit applies to both sides of the track.

8	**Nav. Method in the leg after WP**	*Byte Field Size:*		*Request Parameter*	Optional	
		Bit Field Size:	2	*Command Parameter:*	Optional	

DD119 Calculation Type

0 = Great Circle calculations,
1 = Rhumb Line calculations,
2 = Error,
3 = Null

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

9	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*		
		Bit Field Size:	resv 6	*Command Parameter:*		

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

10	**Fields 6 thru 9 repeat as needed**	*Byte Field Size:*	?	*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	

DD000 Undefined — Application specific, defined at time of use

DF00 Undefined **Undefined** *Range:* undefined *Resolution:* undefined Application specific, defined at time of use.

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Route and WP Service - WP Comment

PGN: 130070
hex: 1FC16

Complex request of this PGN should return supplementary Comments attached to Waypoints in a Route or a WP-List.

Waypoints without a Commment shall not be included in this PGN.

If the Route ID is set to 65535 (NA), the Comments will be for the Waypoints in the WP-List.
The Database ID shall be specified in the request/reply.

The reply should use the same transfer protocol as the request.
ISO request for this PGN shall return 059392 with a positive ACK if the PGN is supported.
For a complete description of the Route and WP PGNs, see the application note in Appendix D.

Single Frame: No *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 102

1 **Start ID** *Byte Field Size:* 2 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

1st requested/sent WPID in a WP-List or RPS# in a Route.
If not specified in the request, the default is the lowest WPID/RPS# with a Comment.
If the requested WPID/RPS# does not have a Comment, the default response is the next highest WPID/RPS#.

2 **nItems** *Byte Field Size:* 2 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

n Waypoints requested/sent.
If not specified in the request, the default is the all the Waypoints with a Comment (or as many the transfer protocol permits).

3 **Number of WPs with Comments** *Byte Field Size:* 2 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

4 **Database ID** *Byte Field Size:* 2 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

5 **Route ID** *Byte Field Size:* 2 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

Set to 65535 (NA) to access the WP-List in the Database.

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Route and WP Service - WP Comment

PGN: 130070
hex: 1FC16

6	**WPID / RPS#**	*Byte Field Size:*	2	*Request Parameter*	Optional		
		Bit Field Size:		*Command Parameter:*	Optional		
	DD007 Generic numeric ID, medium		Number of route, waypoint, event, mark, etc.				
	DF54 Integer, 16 bit unsigned **uint16**	*Range:*	0 to 65,532	*Resolution:*	1 bit	Unit-less number	
	WPID shall be used when addressing a WP-List. RPS# shall be used when addressing a Route.						
7	**Comment**	*Byte Field Size:*	8 or 16	*Request Parameter*	Optional		
		Bit Field Size:		*Command Parameter:*	Optional		
	DD198 Generic name string, Medium		Medium size text strings.				
	DF51 String, variable, medium **ch8or16(n)**	*Range:*	0 to 1,782 ASCII or 0 to 891 Unicode Characters	*Resolution:*	1 ASCII or 1 Unicode Character	3 to 1,785 bytes. First and Second bytes in string (unit16) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Third byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 3 bytes, i.e. Count = 3) is a null string.	
	Max 1782 ASCII or 891 Unicode characters						
8	**Fields 6 thru 7 repeat as needed**	*Byte Field Size:*	?	*Request Parameter*	Optional		
		Bit Field Size:		*Command Parameter:*	Optional		
	DD000 Undefined		Application specific, defined at time of use				
	DF00 Undefined **Undefined**	*Range:*	undefined	*Resolution:*	undefined	Application specific, defined at time of use.	

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Route and WP Service - Route Comment

PGN: 130071
hex: 1FC17

Complex request of this PGN should return supplementary Comments attached to Routes.

The Database ID shall be specified in the request/reply.
Routes without a comment shall not be included in this PGN.

The reply should use the same transfer protocol as the request.
ISO request for this PGN shall return 059392 with a positive ACK if the PGN is supported.
For a complete description of the Route and WP PGNs, see the application note in Appendix D.

Single Frame: No *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 103

1 **Start Route ID** *Byte Field Size:* 2 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

1st Route ID requested/sent.
If not specified in the request, the reply shall default to the lowest Route ID with a Comment.
If the requested Route has no Comment, the default response is the next highest Route ID.

2 **nItems** *Byte Field Size:* 2 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

n Routes requested/sent.
If not specified in the request, the reply shall include all Routes with a Comment in the Database (or as many of them as the transport protocol space permits).

3 **Number of Routes with Comments** *Byte Field Size:* 2 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

4 **Database ID** *Byte Field Size:* 2 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

5 **Route ID** *Byte Field Size:* 2 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

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Route and WP Service - Route Comment

PGN: 130071
hex: 1FC17

6	**Comment**		*Byte Field Size:*	8 or 16	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD198 Generic name string, Medium		Medium size text strings.				
	DF51 String, variable, medium	**ch8or16(n)**	*Range:*	0 to 1,782 ASCII or 0 to 891 Unicode Characters	*Resolution:*	1 ASCII or 1 Unicode Character	3 to 1,785 bytes. First and Second bytes in string (unit16) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Third byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 3 bytes, i.e. Count = 3) is a null string.
7	**Fields 5 thru 6 repeat as needed**		*Byte Field Size:*	?	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD000 Undefined		Application specific, defined at time of use				
	DF00 Undefined	**Undefined**	*Range:*	undefined	*Resolution:*	undefined	Application specific, defined at time of use.

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Route and WP Service - Database Comment

PGN: 130072
hex: 1FC18

Complex request of this PGN should return supplementary Comments attached to Databases in the navigation Device.

Databases without a Comment shall not be included in this PGN.

The reply should use the same transfer protocol as the request.
ISO request for this PGN shall return 059392 with a positive ACK if PGN is supported.
For a complete description of the Route and WP PGNs, see the application note in Appendix D.

Single Frame: No *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 104

1 **Start Database ID** *Byte Field Size:* 2 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

1st Database ID requested/sent.
If not specified in the request, the reply shall default to the lowest Database ID with a Comment.
If the requested Database has no Comment, the default response is the next highest Database ID.

2 **nItems** *Byte Field Size:* 2 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

n Databases requested/sent.
If not specified in the request, the reply shall include all Databases with a Comment (or as many of them as the transport protocol space permits).

3 **Number of Databases with comments** *Byte Field Size:* 2 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

4 **Database ID** *Byte Field Size:* 2 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

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Route and WP Service - Database Comment

PGN: 130072
hex: 1FC18

5	**Comment text**	*Byte Field Size:*	8 or 16	*Request Parameter*	Optional		
		Bit Field Size:		*Command Parameter:*	Optional		
	DD198 Generic name string, Medium		Medium size text strings.				
	DF51 String, variable, medium **ch8or16(n)**	*Range:*	0 to 1,782 ASCII or 0 to 891 Unicode Characters	*Resolution:*	1 ASCII or 1 Unicode Character	3 to 1,785 bytes. First and Second bytes in string (unit16) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Third byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 3 bytes, i.e. Count = 3) is a null string.
	Max. 1782 ASCII or 891 Unicode characters						
6	**Fields 4 thru 5 repeat as needed**	*Byte Field Size:*	?	*Request Parameter*	Optional		
		Bit Field Size:		*Command Parameter:*	Optional		
	DD000 Undefined		Application specific, defined at time of use				
	DF00 Undefined **Undefined**	*Range:*	undefined	*Resolution:*	undefined	Application specific, defined at time of use.	

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Route and WP Service - Radius of Turn

PGN: 130073
hex: 1FC19

Complex request of this PGN should return the Radius of Turn at specific Waypoints of a Route.

For a complete description of the Route and WP PGNs, see the application note in Appendix D. The Radius of Turn is described in Appendix D.5.15.6.

The Database ID and Route ID shall be specified in the request/reply.

The Radius overrides any Route default Radius of Turn.
Only those Waypoints with an individually specified Radius of Turn shall be included in this PGN. The Waypoints are identified with RPS#.

The reply should use the same transfer protocol as the request.
ISO request for this PGN shall return 059392 with a positive ACK if the PGN is supported.

Single Frame: No *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 108

1	**Start RPS#**	*Byte Field Size:* 2	*Request Parameter*	Required
		Bit Field Size:	*Command Parameter:*	Optional
	DD007 Generic numeric ID, medium	Number of route, waypoint, event, mark, etc.		
	DF54 Integer, 16 bit unsigned **uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number
	1st RPS# requested/sent If not specified in the request, the default is the first RPS# with valid data. If the requested RPS# has no valid data, the default response is the next highest RPS#.			
2	**nItems**	*Byte Field Size:* 2	*Request Parameter*	Required
		Bit Field Size:	*Command Parameter:*	Optional
	DD007 Generic numeric ID, medium	Number of route, waypoint, event, mark, etc.		
	DF54 Integer, 16 bit unsigned **uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number
	n RPS# requested/sent If not specified in the request, the reply shall include all RPS# with valid data (or as many the transfer protocol space permits).			
3	**Number of Waypoints with a specific Radius of Turn**	*Byte Field Size:* 2	*Request Parameter*	Optional
		Bit Field Size:	*Command Parameter:*	Optional
	DD007 Generic numeric ID, medium	Number of route, waypoint, event, mark, etc.		
	DF54 Integer, 16 bit unsigned **uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number
	Note: This may be less than the number of Waypoints in the Route			
4	**Database ID**	*Byte Field Size:* 2	*Request Parameter*	Required
		Bit Field Size:	*Command Parameter:*	Optional
	DD007 Generic numeric ID, medium	Number of route, waypoint, event, mark, etc.		
	DF54 Integer, 16 bit unsigned **uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number
5	**Route ID**	*Byte Field Size:* 2	*Request Parameter*	Required
		Bit Field Size:	*Command Parameter:*	Optional
	DD007 Generic numeric ID, medium	Number of route, waypoint, event, mark, etc.		
	DF54 Integer, 16 bit unsigned **uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number

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Route and WP Service - Radius of Turn

PGN: 130073
hex: 1FC19

6	**RPS#**	*Byte Field Size:*	2	*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

Waypoints using the Route-default Radius or no Radius shall not be included.

7	**Radius of Turn**	*Byte Field Size:*	2	*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	

DD149 Distance ordered — A commanded distance like radius order, off-track limit, etc.

DF74 Distance, rough **int16** *Range:* +/-32,764 m *Resolution:* 1 m

No negative values.

8	**Fields 6 and 7 repeated as needed**	*Byte Field Size:*	?	*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	

DD000 Undefined — Application specific, defined at time of use

DF00 Undefined **Undefined** *Range:* undefined *Resolution:* undefined Application specific, defined at time of use.

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Route and WP Service - WP List - WP Name & Position

PGN: 130074
hex: 1FC1A

Complex request of this PGN should return the Waypoints of a WP-List.

WARNING: The Route Track may not be defined by the Waypoint positions only: The Navigation Method (GC/RL) and Radius of Turn at each Waypoint are optional additional parameters defining the Track.
PGN 130066 field #8 "Critical supplementary parameters" shall be used to determine if there are additional parameters to the Waypoint positions.
For a complete description of the Route and WP PGNs, see the application note in Appendix D.

The WP-List has Waypoints identified by WPID in the range 0-65532. Each of these may be single or belong to one or multiple Routes in the Database. There may be empty gaps anywhere in the range of WPIDs. These (non-valid WPs) shall not be included in this PGN. A Waypoint is valid when its Position is valid.

A Database may contain one WP-List and multiple Routes. The Database ID shall be specified in the request/reply. The Database ID will typically be selected from the information received in the ""Route and WP Service - Database List"" PGN previously requested from the same Device.

The reply should use the same transfer protocol as the request.
ISO request for this PGN shall return 059392 with a positive ACK if the PGN is supported.

Single Frame: No *Priority Default:* 7 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 107

1 **Start WPID** *Byte Field Size:* 2 *Bit Field Size:* *Request Parameter* Required *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

1st requested/sent WPID.
If not specified in the request, the default is the lowest WPID with a valid Waypoint.
If the requested Waypoint is not valid, the default response is the next highest WPID.

2 **nItems** *Byte Field Size:* 2 *Bit Field Size:* *Request Parameter* Required *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

n Waypoints requested/sent.
Non-valid Waypoints shall be skipped and is not included in this count.
If not specified in the request, the reply shall include all valid WPs of the WP-List (or as many thereof as the transport protocol space permits).

3 **Number of valid WPs in the WP-List** *Byte Field Size:* 2 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

4 **Database ID** *Byte Field Size:* 2 *Bit Field Size:* *Request Parameter* Required *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

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Route and WP Service - WP List - WP Name & Position

PGN: 130074
hex: 1FC1A

5 **NMEA Reserved**

Byte Field Size: *Bit Field Size:* resv | 16

Request Parameter *Command Parameter:*

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

---

6 **WPID**

Byte Field Size: 2 *Bit Field Size:*

Request Parameter Optional *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

The Waypoints shall be included in the order of increasing WPIDs.
Non-valid WPs shall not be included.

---

7 **WP Name**

Byte Field Size: 8 or 16 | n *Bit Field Size:*

Request Parameter Optional *Command Parameter:* Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character

2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

Max. 30 ASCII or Unicode Characters

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8 **WP Latitude**

Byte Field Size: 4 *Bit Field Size:*

Request Parameter Optional *Command Parameter:* Optional

DD022 Latitude, WGS-84 — Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23 Latitude **int32** *Range:* +/- 90 deg *Resolution:* 1x10E-7 deg "-" = South, resolution ~1.1 cm

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Route and WP Service - WP List - WP Name & Position

PGN: 130074
hex: 1FC1A

9	**WP Longitude**		*Byte Field Size:*	4	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD023 Longitude, WGS-84			Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.			
	DF25 Longitude	**int32**	*Range:*	+/- 180 deg	*Resolution:*	1x10E-7 deg	"-" = West, resolution ~1.1 cm
10	**Fields 6 thru 9 repeat as needed**		*Byte Field Size:*	?	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD000 Undefined			Application specific, defined at time of use			
	DF00 Undefined	**Undefined**	*Range:*	undefined	*Resolution:*	undefined	Application specific, defined at time of use.

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Wind Data

PGN: 130306
hex: 1FD02

Direction and speed of Wind. True wind can be referenced to the vessel or to the ground. The Apparent Wind is what is felt standing on the (moving) ship, I.e., the wind measured by the typical mast head instruments.
The boat referenced true wind is given by the vector sum of Apparent wind and vessel's heading and speed though the water. The ground referenced true wind is given by the vector sum of Apparent wind and vessel's heading and speed over ground.

Single Frame: Yes | *Priority Default:* 2 | *Default Update Rate:* 100 milliseconds | *Frequency:* 10. cycles per second
Destination: Global | *Query Support:* Optional | *Command Support:* Optional | *ACK Rqmnts:* None

Field # *Field Name* — Original Reference ID # 81

#	Field Name						
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD056 Sequence ID			An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252		*Resolution:* 1 bit		Unit-less number
2	**Wind Speed**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD044 Generic Speed						
	DF35 Speed	**uint16**	*Range:* 0 to 655.32 m/s		*Resolution:* 1x10E-2 m/s		1 Knot = 0.5144 m/s
3	**Wind Direction**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional		
	DD045 Wind Direction						
	DF02 Angle	**uint16**	*Range:* 0 to 2Pi rad		*Resolution:* 1x10E-4 rad		Resolution ~0.0057deg, 1 deg = .01745 rad
4	**Wind Reference**		*Byte Field Size:* *Bit Field Size:* 3		*Request Parameter* Optional *Command Parameter:* Optional		
	DD205 Wind Reference			0x00 = Theoretical Wind (ground referenced, referenced to True North; calculated using COG/SOG) 0x01 = Theoretical Wind (ground referenced, referenced to Magnetic North; calculated using COG/SOG) 0x02 = Apparent Wind (relative to the vessel centerline) 0x03 = Theoretical (Calculated to Centerline of the vessel, refernced to ground; calculated using COG/SOG) 0x04 = Theoretical (Calculated to Centerline of the vessel, refernced to water; calculated using Heading/Speed through Water) 0x05 = Reserved 0x06 = Error 0x07 = Null			
	DF52 Bit field	**bit(n)**	*Range:* Variable		*Resolution:* 1		Used to construct bit fields
5	**NMEA Reserved**		*Byte Field Size:* *Bit Field Size:* resv 21		*Request Parameter* *Command Parameter:*		
	DD001 Reserved field			Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:* Variable		*Resolution:* 1		Used to construct bit fields
	Used to align subsequent data on a byte boundary.						

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Environmental Parameters - DEPRECATED

PGN: 130310
hex: 1FD06

This PGN has been deprecated (as of version 1.200, PGN 130311 replaced PGN 130310) and is not recommended for new designs. However, support of PGN 130310 may be necessary to ensure compatibility with legacy equipment.

PGN 130311 has also been deprecated as of version 1.210.

The following PGNs are recommended to be used for new designs: 130314-Actual Pressure, 130315-Set Pressure, 130316-Temperature-Extended Range.

The latest definition of PGN 130310 before deprecation was as follows:

Local atmospheric environmental conditions.

Single Frame: Yes *Priority Default:* 5 *Default Update Rate:* 500 milliseconds *Frequency:* 2. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 46

Field #	Field Name					
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD056 Sequence ID			An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.		
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
2	**Water Temp**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD043 Generic Temperature					
	DF39 Temperature, low	**uint16**	*Range:* 0 to 655.32 deg K	*Resolution:* 1x10E-2 deg K		
3	**Outside Ambient Air Temp.**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD043 Generic Temperature					
	DF39 Temperature, low	**uint16**	*Range:* 0 to 655.32 deg K	*Resolution:* 1x10E-2 deg K		
4	**Atmospheric Pressure**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD049 Generic Pressure					
	DF47 Pressure, medium	**uint16**	*Range:* 0 to 6,553,200 Pa	*Resolution:* 1x10E+2 Pa		
5	**NMEA Reserved**		*Byte Field Size:* *Bit Field Size:* resv 8		*Request Parameter* *Command Parameter:*	
	DD001 Reserved field			Variable number of reserved bits, all set to logic "1"		
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
	Used to align subsequent data on a byte boundary.					

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Environmental Parameters- DEPRECATED

PGN: 130311
hex: 1FD07

This PGN has been deprecated (as of version 1.210) and is not recommended for new designs. However, support of PGN 130311 may be necessary to ensure compatibility with legacy equipment.

This PGN 130311 was originally created as a replacement to PGN 130310 in version 1.200.

This PGN 130311 has since been deprecated (as of version 1.210, PGN's 130312, 130313, 130314, and 130315 replaced PGN 130311) and is not recommended for new designs. However, support of PGN 130311 may be necessary to ensure compatibility with legacy equipment.

PGN 130312 has also been deprecated and replaced by PGN 130316 as of version 1.310.

The following PGNs are recommended to be used for new designs: 130314-Actual Pressure, 130315-Set Pressure, 130316-Temperature-Extended Range.

The latest definition of PGN 130311 before deprecation was as follows:

Environmental Conditions contains Temperature, Humidity, and Atmospheric Pressure. This PGN is a rework of PGN 130310.

Single Frame: Yes *Priority Default:* 5 *Default Update Rate:* 500 milliseconds *Frequency:* 2. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 91

Field #	Field Name					
1	**Sequence ID**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Optional	
	DD056 Sequence ID		An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
2	**Temperature Instance**		*Byte Field Size:* *Bit Field Size:* 6		*Request Parameter* Optional *Command Parameter:* Optional	
	DD229 Temperature Instance		0x00 = Sea Temperature, 0x01 = Outside Temperature, 0x02 = Inside Temperature, 0x03 = Engine Room Temperature, 0x04 = Main Cabin Temperature, 0x05 = Reserved, thru 0x3E = Reserved, 0x3F = Data Not Available			
	DF52 Bit field Used to qualify contents of field 4	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
3	**Humidity Instance**		*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter* Optional *Command Parameter:* Optional	
	DD230 Humidity Instance		0x00 = Inside Humidity, 0x01 = Outside Humidity, 0x02 = reserved, 0x03 = Data Not Available			
	DF52 Bit field Used to qualify contents of field 5	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	

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Environmental Parameters- DEPRECATED

PGN: 130311
hex: 1FD07

4	**Temperature**		*Byte Field Size:* 2			*Request Parameter*	Optional
			Bit Field Size:			*Command Parameter:*	Optional
	DD043 Generic Temperature						
	DF39 Temperature, low	**uint16**	*Range:*	0 to 655.32 deg K	*Resolution:*	1x10E-2 deg K	
5	**Humidity**		*Byte Field Size:* 2			*Request Parameter*	Optional
			Bit Field Size:			*Command Parameter:*	Optional
	DD231 Humidity			Relative Humidity			
	DF84 Percent, Relative Measur	**int16**	*Range:*	-131.072% to 131.056%	*Resolution:*	4x10E-3 %	
6	**Atmospheric Pressure**		*Byte Field Size:* 2			*Request Parameter*	Optional
			Bit Field Size:			*Command Parameter:*	Optional
	DD049 Generic Pressure						
	DF47 Pressure, medium	**uint16**	*Range:*	0 to 6,553,200 Pa	*Resolution:*	1x10E+2 Pa	

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Temperature - DEPRECATED

PGN: 130312
hex: 1FD08

This PGN 130312 was originally created as a replacement for Temperature provided by PGN 130311 in version 1.210.

PGN 130312 has also been deprecated and replaced by PGN 130316 as of version 1.310. However, support of PGN 130312 may be necessary to ensure compatibility with legacy equipment.

The following PGN is recommended to be used for new designs: 130316-Temperature-Extended Range.

The latest definition of PGN 130312 before deprecation was as follows:

This PGN contains the Sequence ID, a Temperature Instance, Temperature Source, Temperature Value, and Set Temperature. For example the Temperature might be the temperature of sea water or live well as defined by the Temperature Source (field 3), with a Temperature Instance (field 2) of 1. Using Set Temperature (field 5) this PGN can also be used to control temperature or to report a targeted temperature. This PGN is a rework of PGN 130311 and was introduced in version 1.210 of this standard.

Single Frame: Yes *Priority Default:* 5 *Default Update Rate:* 2000 milliseconds *Frequency:* .5 cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 196

1 **Sequence ID** *Byte Field Size:* 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD056 Sequence ID

An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

2 **Temperature Instance** *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: 8 *Command Parameter:* Optional

DD128 Generic instance

0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.

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Temperature - DEPRECATED

PGN: 130312
hex: 1FD08

3	**Temperature Source**		Byte Field Size:		Request Parameter	Optional
			Bit Field Size:	8	Command Parameter:	Optional
	DD291 Temperature Source		00 = Sea Temperature 01 = Outside Temperature 02 = Inside Temperature 03 = Engine Room Temperature 04 = Main Cabin Temperature 05 = Live Well Temperature 06 = Bait Well Temperature 07 = Refrigeration Temperature 08 = Heating System Temperature 09 = Dew Point Temperature 10 = Wind Chill Temperature, Apparent 11 = Wind Chill Temperature, Theoretical 12 = Heat Index Temperature 13 = Freezer Temperature 14 = Exhaust Gas Temperature 15 through 128 Reserved 129 through 252 Generic Temperature Sources other than those defined 253 = Not Supported 254 = Error 255 = No Change / Data Not Available			
	DF52 Bit field	**bit(n)**	Range:	Variable	Resolution: 1	Used to construct bit fields
4	**Actual Temperature**		Byte Field Size:	2	Request Parameter	Optional
			Bit Field Size:		Command Parameter:	Optional
	DD043 Generic Temperature					
	DF39 Temperature, low	**uint16**	Range:	0 to 655.32 deg K	Resolution:	1x10E-2 deg K
5	**Set Temperature**		Byte Field Size:	2	Request Parameter	Optional
			Bit Field Size:		Command Parameter:	Optional
	DD043 Generic Temperature					
	DF39 Temperature, low	**uint16**	Range:	0 to 655.32 deg K	Resolution:	1x10E-2 deg K
6	**NMEA Reserved**		Byte Field Size:		Request Parameter	
			Bit Field Size:	resv 8	Command Parameter:	
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	Range:	Variable	Resolution: 1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.					

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Humidity

PGN: 130313
hex: 1FD09

This PGN contains the Sequence ID, a Humidity Instance, Humidity Source, the Actual Humidity, and the Set Humidity Value. For example the Humidity might be Inside as defined by the Humidity Source (field 3) with a Humidity Instance (field 2) of 2. Using Set Humidity (field 5) this PGN can also be used to control humidity or to report a targeted humidity. This PGN is a rework of PGN 130311 and was introduced in version 1.210 of this standard, it is to be used for new designs.

Single Frame: Yes *Priority Default:* 5 *Default Update Rate:* 2000 milliseconds *Frequency:* .5 cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 198

1 Sequence ID
Byte Field Size: 1 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD056 Sequence ID
An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

2 Humidity Instance
Byte Field Size: *Bit Field Size:* 8 *Request Parameter* Optional *Command Parameter:* Optional

DD128 Generic instance
0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.

3 Humidity Source
Byte Field Size: *Bit Field Size:* 8 *Request Parameter* Optional *Command Parameter:* Optional

DD292 Humidity Source
00 = Inside Humidity
01 = Outside Humidity
02 through 128 Reserved

129 through 252 Generic Humidity Sources other than those defined
253 = Not Supported
254 = Error
255 = No Change / Data Not Available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

4 Actual Humidity
Byte Field Size: 2 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD231 Humidity
Relative Humidity

DF84 Percent, Relative Measur **int16** *Range:* -131.072% to 131.056% *Resolution:* 4x10E-3 %

5 Set Humidity
Byte Field Size: 2 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD231 Humidity
Relative Humidity

DF84 Percent, Relative Measur **int16** *Range:* -131.072% to 131.056% *Resolution:* 4x10E-3 %

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Humidity

PGN: 130313
hex: 1FD09

6	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*		
		Bit Field Size:	resv 8	*Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
	Used to align subsequent data on a byte boundary.					

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Actual Pressure

PGN: 130314
hex: 1FD0A

This PGN contains a Sequence ID, Pressure Instance, Pressure Source, and the Pressure Value. For example Pressure might be Atmospheric Pressure as defined by the Pressure Source (Field 3) with a Pressure Instance (Field 2) of 1.

Some devices may provide the capability to measure multiple instances of the same measurement type. An example of this might be a device that measures three filter pressures. For such a device, this PGN would be sent three times, once for each filter pressure measurement (Field 4) of a specific pressure source (Field 3) with a unique pressure instance value in Field 2.

PGN 130314 replaces PGN 130311 as of version 1.310 of this standard, and is to be used for new designs. The deprecated PGN 130311 provides compatibility with legacy equipment.

Single Frame: Yes *Priority Default:* 5 *Default Update Rate:* 2000 milliseconds *Frequency:* .5 cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 195

1 Sequence ID
Byte Field Size: 1 *Bit Field Size:*
Request Parameter Optional *Command Parameter:* Optional

DD056 Sequence ID
An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

The Sequence ID field (SID) is used to link this PGN to other related PGN's from the same source address. When no linkage exists, the value of the SID shall be set to 255.

2 Pressure Instance
Byte Field Size: *Bit Field Size:* 8
Request Parameter Optional *Command Parameter:* Optional

DD128 Generic instance
0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.

3 Pressure Source
Byte Field Size: *Bit Field Size:* 8
Request Parameter Optional *Command Parameter:* Optional

DD289 Pressure Source
00 = Atmospheric Pressure
01 = Water Pressure
02 = Steam Pressure
03 = Compressed Air Pressure
04 = Hydraulic Pressure
05 = Filter Pressure
06 = Altimiter Setting
07 = Oil Pressure
08 = Fuel Pressure
09 thru 128 Reserved 129 thru 252
Generic pressure sources other than those defined
253=Reserved
254=Error
255=No Change / Data Not Available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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Actual Pressure

PGN: 130314
hex: 1FD0A

4	**Pressure**		*Byte Field Size:* 4		*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD290 Pressure					
	DF103 Pressure	**int32**	*Range:* +/- 2.1E8	*Resolution:* 1x10E-1 Pa		
5	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*	
			Bit Field Size: resv 8		*Command Parameter:*	
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
	Used to align subsequent data on a byte boundary.					

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Set Pressure

PGN: 130315
hex: 1FD0B

This PGN contains the Sequence ID, a Pressure Instance, Pressure Source, and the Set Pressure Value. This PGN can be sent to a device that controls pressure to change its targeted pressure, or it can be sent out by the control device to indicate its current targeted pressure. This PGN is a rework of PGN 130311 and was introduced in version 1.210 of this standard, it is to be used for new designs.

Single Frame: Yes *Priority Default:* 5 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 197

1 Sequence ID
Byte Field Size: 1 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD056 Sequence ID
An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

2 Pressue Instance
Byte Field Size: *Bit Field Size:* 8 *Request Parameter* Optional *Command Parameter:* Optional

DD128 Generic instance
0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.

3 Pressure Source
Byte Field Size: *Bit Field Size:* 8 *Request Parameter* Optional *Command Parameter:* Optional

DD289 Pressure Source
00 = Atmospheric Pressure
01 = Water Pressure
02 = Steam Pressure
03 = Compressed Air Pressure
04 = Hydraulic Pressure
05 = Filter Pressure
06 = Altimiter Setting
07 = Oil Pressure
08 = Fuel Pressure
09 thru 128 Reserved 129 thru 252
Generic pressure sources other than those defined
253=Reserved
254=Error
255=No Change / Data Not Available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

4 Pressure
Byte Field Size: 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD290 Pressure

DF103 Pressure **int32** *Range:* +/- 2.1E8 *Resolution:* 1x10E-1 Pa

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Set Pressure

PGN: 130315
hex: 1FD0B

5	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 8	*Command Parameter:*	

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on a byte boundary.

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Temperature, Extended Range

PGN: 130316
hex: 1FD0C

This parameter group is used to report Temperature measurements that are not included in parameter groups for a specific equipment type. For example, this parameter group would not be used to report Engine Temperature, which is already included in Engine Parameters, Dynamic (PGN 127489). The Sequence ID field may be used to synchronize the measurement reported by this parameter group within a related group of measurements. The Set Temperature field may be used to report a target temperature, or using the Command Group Function (PGN 126208) to set a target or control temperature.

This PGN performs the same function and replaces the Temperature parameter group Temperature (PGN 130312) for all new designs subsequent to version 1.301, and provides a wider temperature range data type for Actual Temperature to accommodate high temperature applications such as Exhaust Gas Temperature.

Note1: Transmissions of this parameter group using 126208 command group function must include a value for field 2, Temperature Instance, which shall be interpreted as the temperature measurement or channel input that any remaining commanded fields apply to.

Single Frame: Yes *Priority Default:* 5 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 205

1 **Sequence ID** *Byte Field Size:* 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD056 Sequence ID

An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

2 **Temperature Instance** *Byte Field Size:* *Request Parameter* Optional
Bit Field Size: 8 *Command Parameter:* Note 1

DD128 Generic instance

0 = Instance 0
1 = Instance 1
2 = Instance 2
n = Instance n, where n < 253
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

If a Request Parameter or Command Parameter is supported on any other Data Field, you must also support the capability on this Data Field.

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Temperature, Extended Range

PGN: 130316
hex: 1FD0C

3	**Temperature Source**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	8	*Command Parameter:*	Optional

DD291 Temperature Source

00 = Sea Temperature
01 = Outside Temperature
02 = Inside Temperature
03 = Engine Room Temperature
04 = Main Cabin Temperature
05 = Live Well Temperature
06 = Bait Well Temperature
07 = Refrigeration Temperature
08 = Heating System Temperature
09 = Dew Point Temperature
10 = Wind Chill Temperature, Apparent
11 = Wind Chill Temperature, Theoretical
12 = Heat Index Temperature
13 = Freezer Temperature
14 = Exhaust Gas Temperature
15 through 128 Reserved
129 through 252 Generic Temperature Sources other than those defined
253 = Not Supported
254 = Error
255 = No Change / Data Not Available

DF52	Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields

4	**Actual Temperature**	*Byte Field Size:*	3	*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	

DD314 Temperature, extended range and precision

DF105	Temperature, extended ra	**uint24**	*Range:*	0 to 16,777.212 deg K	*Resolution:* 0.001 deg K	

5	**Set Temperature**	*Byte Field Size:*	2	*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	

DD130 Temperature, high

DF38	Temperature, high	**uint16**	*Range:*	0 to 6,553.2 deg K	*Resolution:* 1x10E-1 deg K	

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Tide Station Data

PGN: 130320
hex: 1FD10

Tide station measurement data including station location, numeric identifier, and name.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 12

Field #	Field Name						
1	**Mode**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	4	*Command Parameter:*	Optional	
	DD025 Mode, Data		0x0 = Autonomous mode, 0x1 = Differential, enhanced mode, 0x2 = Estimated mode, 0x3 = Simulator mode, 0x4 = Manual mode, 0x5 to 0xD = Reserved 0xE = Error, 0xF = Data not available				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
2	**Tide Tendency**		*Byte Field Size:*		*Request Parameter*	Optional	
			Bit Field Size:	2	*Command Parameter:*	Optional	
	DD038 Tide direction		msb/lsb: 00 = Falling, 01 = Rising, 10 = Error, 11 = Unavailable, Unknown				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
3	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*		
			Bit Field Size:	resv 2	*Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.						
4	**Measurement date**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD039 Generic date		Days since January 1, 1970, Date is relative to UTC Time.				
	DF41 Date, day count	**uint16**	*Range:*	0 to 65,532 days	*Resolution:*	1 day	0 = January 1, 1970, max = ~179 years
5	**Measurement time**		*Byte Field Size:*	4	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD158 Generic time of day		24 hour clock, 0 = midnight, time is in UTC				
	DF06 Time of day	**uint32**	*Range:*	0 to 86,401 s	*Resolution:*	1x10E-4 s	~24 hours, 0 = midnight, range allows for up to two leap seconds per day

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Tide Station Data

PGN: 130320
hex: 1FD10

6 Station location, latitude

Byte Field Size: 4 | Bit Field Size:
Request Parameter: Optional | Command Parameter: Optional

DD022 Latitude, WGS-84

Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23	Latitude	**int32**	Range: +/- 90 deg	Resolution: 1x10E-7 deg	"-" = South, resolution ~1.1 cm

7 Station location, longitude

Byte Field Size: 4 | Bit Field Size:
Request Parameter: Optional | Command Parameter: Optional

DD023 Longitude, WGS-84

Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF25	Longitude	**int32**	Range: +/- 180 deg	Resolution: 1x10E-7 deg	"-" = West, resolution ~1.1 cm

8 Tide level

Byte Field Size: 2 | Bit Field Size:
Request Parameter: Optional | Command Parameter: Optional

DD041 Tide Level

This value is relative to mean lower low water (MLLW).

DF46	Distance, signed, medium	**int16**	Range: +/- 32.764 m	Resolution: 1x10E-3 m	

9 Tide level standard deviation

Byte Field Size: 2 | Bit Field Size:
Request Parameter: Optional | Command Parameter: Optional

DD040 Standard Deviation for tide level data

The following text is from NOAA and is placed here to convey an understanding of the expected magnitude of the values. "The tide gage processes 181 water level samples, 1 second apart, centered on the 6 minute mark. It then computes the standard deviation of the Samples. Samples more than 3 sigma's away from the average are called "outliers". The "outliers" are removed from the samples and the standard deviation is recomputed. The number of outliers indicates how many samples were discarded. The standard deviation is a measure of noise in the water level environment. In a sheltered location, the typical standard deviation can be as low as 0.001 to 0.010 meters. More open locations could be higher, such as 0.150 meters. Every location is different.

DF13	Distance, short	**uint16**	Range: 0 to 655.32 m	Resolution: 1x10E-2 m	

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Tide Station Data

PGN: 130320
hex: 1FD10

10 **Station ID String**

Byte Field Size: 8 or 16 | n
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character

2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

15 characters maximum.

If this field is not specified in the "Command Request" or an ISO Request is made of this PGN, the response will with the unit's Station ID. Otherwise if this field is specified only the units with a matching Station ID will respond with this PGN.

---

11 **Station Name String**

Byte Field Size: 8 or 16 | n
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character

2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

50 characters maximum.

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Salinity Station Data

PGN: 130321
hex: 1FD11

Salinity station measurement data including station location, numeric identifier, and name.

Single Frame: No *Priority Default:* 6 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 13

1 Mode
Byte Field Size: *Bit Field Size:* 4 *Request Parameter* Optional *Command Parameter:* Optional

DD025 Mode, Data
0x0 = Autonomous mode,
0x1 = Differential, enhanced mode,
0x2 = Estimated mode,
0x3 = Simulator mode,
0x4 = Manual mode,
0x5 to 0xD = Reserved
0xE = Error,
0xF = Data not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

2 NMEA Reserved
Byte Field Size: *Bit Field Size:* resv 4 *Request Parameter* *Command Parameter:*

DD001 Reserved field
Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
Used to align subsequent data on a byte boundary.

3 Measurement Date
Byte Field Size: 2 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD039 Generic date
Days since January 1, 1970, Date is relative to UTC Time.

DF41 Date, day count **uint16** *Range:* 0 to 65,532 days *Resolution:* 1 day 0 = January 1, 1970, max = ~179 years

4 Measurement time
Byte Field Size: 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD158 Generic time of day
24 hour clock, 0 = midnight, time is in UTC

DF06 Time of day **uint32** *Range:* 0 to 86,401 s *Resolution:* 1x10E-4 s ~24 hours, 0 = midnight, range allows for up to two leap seconds per day

5 Station location, latitude
Byte Field Size: 4 *Bit Field Size:* *Request Parameter* Optional *Command Parameter:* Optional

DD022 Latitude, WGS-84
Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23 Latitude **int32** *Range:* +/- 90 deg *Resolution:* 1x10E-7 deg "-" = South, resolution ~1.1 cm

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Salinity Station Data

PGN: 130321
hex: 1FD11

6	**Station location, longitude**		*Byte Field Size:*	4	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD023 Longitude, WGS-84			Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.			
	DF25 Longitude	**int32**	*Range:*	+/- 180 deg	*Resolution:*	1x10E-7 deg	"-" = West, resolution ~1.1 cm
7	**Salinity**		*Byte Field Size:*	4	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD042 Salinity measure			The average Salinity of ocean water is about 35 grams of salts per kilogram of sea water (g/kg), usually written as 35 ppt which is read as 35 parts per thousand.			
	DF49 Ratio, Relative measure	**float32**	*Range:*	Variable	*Resolution:*	Floats	Unit-less number
8	**Water Temperature**		*Byte Field Size:*	2	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD043 Generic Temperature						
	DF39 Temperature, low	**uint16**	*Range:*	0 to 655.32 deg K	*Resolution:*	1x10E-2 deg K	
9	**Station ID String**		*Byte Field Size:*	8 or 16 n	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD004 Generic name string, short			Name of place, route, waypoint, destination, vessel, vehicle, etc.			
	DF50 String, variable, short	**ch8or16(n)**	*Range:*	0 to 250 ASCII or 0 to 125 Unicode Characters	*Resolution:*	1 ASCII or 1 Unicode Character	2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

15 characters maximum.

If this field is not specified in the "Command Request" or an ISO Request is made of this PGN, the response will with the unit's Station ID. Otherwise if this field is specified only the units with a matching Station ID will respond with this PGN.

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Salinity Station Data

PGN: 130321
hex: 1FD11

10	**Station Name String**		*Byte Field Size:*	8 or 16	n	*Request Parameter*	Optional
			Bit Field Size:			*Command Parameter:*	Optional
	DD004 Generic name string, short					Name of place, route, waypoint, destination, vessel, vehicle, etc.	
	DF50 String, variable, short	**ch8or16(n)**	*Range:*	0 to 250 ASCII or 0 to 125 Unicode Characters	*Resolution:*	1 ASCII or 1 Unicode Character	2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

50 characters maximum.

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Current Station Data

PGN: 130322
hex: 1FD12

Current station measurement data including station location, numeric identifier, and name.

Single Frame: No | *Priority Default:* 6 | *Default Update Rate:* 1000 milliseconds | *Frequency:* 1. cycles per second
Destination: Global | *Query Support:* Optional | *Command Support:* Optional | *ACK Rqmnts:* None

Field # *Field Name* — Original Reference ID # 14

#	Field Name							
1	**Mode**		*Byte Field Size:*		*Request Parameter*	Optional		
			Bit Field Size:	4	*Command Parameter:*	Optional		
	DD025 Mode, Data		0x0 = Autonomous mode, 0x1 = Differential, enhanced mode, 0x2 = Estimated mode, 0x3 = Simulator mode, 0x4 = Manual mode, 0x5 to 0xD = Reserved, 0xE = Error, 0xF = Data not available					
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields	
2	**State**		*Byte Field Size:*		*Request Parameter*	Optional		
			Bit Field Size:	3	*Command Parameter:*	Optional		
	DD046 Water Current State		000 = Flood, 001 = Slack, 010 = Ebb, 011 = Reserved, 100 = Reserved, 101 = Reserved, 110 = Error, 111 = Unavailable, Unknown					
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields	
3	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*			
			Bit Field Size:	resv 1	*Command Parameter:*			
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"					
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields	
	Used to align subsequent data on a byte boundary.							
4	**Measurement date**		*Byte Field Size:*	2	*Request Parameter*	Optional		
			Bit Field Size:		*Command Parameter:*	Optional		
	DD039 Generic date		Days since January 1, 1970, Date is relative to UTC Time.					
	DF41 Date, day count	**uint16**	*Range:*	0 to 65,532 days	*Resolution:*	1 day	0 = January 1, 1970, max = ~179 years	
5	**Measurement time**		*Byte Field Size:*	4	*Request Parameter*	Optional		
			Bit Field Size:		*Command Parameter:*	Optional		
	DD158 Generic time of day		24 hour clock, 0 = midnight, time is in UTC					
	DF06 Time of day	**uint32**	*Range:*	0 to 86,401 s	*Resolution:*	1x10E-4 s	~24 hours, 0 = midnight, range allows for up to two leap seconds per day	

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Current Station Data

PGN: 130322
hex: 1FD12

6 **Station location, latitude**

Byte Field Size: 4 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD022 Latitude, WGS-84

Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF23	Latitude	**int32**	*Range:* +/- 90 deg	*Resolution:* 1x10E-7 deg	"-" = South, resolution ~1.1 cm

7 **Station location, longitude**

Byte Field Size: 4 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD023 Longitude, WGS-84

Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.

DF25	Longitude	**int32**	*Range:* +/- 180 deg	*Resolution:* 1x10E-7 deg	"-" = West, resolution ~1.1 cm

8 **Measurement depth**

Byte Field Size: 4 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD047 Water Depth

Water depth measured from the water surface

DF09	Distance	**uint32**	*Range:* 0 to ~4.295x10E+7 m	*Resolution:* 1x10E-2 m	

9 **Current speed**

Byte Field Size: 2 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD044 Generic Speed

DF35	Speed	**uint16**	*Range:* 0 to 655.32 m/s	*Resolution:* 1x10E-2 m/s	1 Knot = 0.5144 m/s

10 **Current flow direction**

Byte Field Size: 2 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD048 Current flow direction

Direction towards which current flows. Degrees relative to True North.

DF02	Angle	**uint16**	*Range:* 0 to 2Pi rad	*Resolution:* 1x10E-4 rad	Resolution ~0.0057deg, 1 deg = .01745 rad

11 **Water Temperature**

Byte Field Size: 2 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD043 Generic Temperature

DF39	Temperature, low	**uint16**	*Range:* 0 to 655.32 deg K	*Resolution:* 1x10E-2 deg K	

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12 **Station ID String**

Byte Field Size: 8 or 16 | n
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character

2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

15 characters maximum.

If this field is not specified in the "Command Request" or an ISO Request is made of this PGN, the response will with the unit's Station ID. Otherwise if this field is specified only the units with a matching Station ID will respond with this PGN.

---

13 **Station Name String**

Byte Field Size: 8 or 16 | n
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character

2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

50 characters maximum

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Meteorological Station Data

PGN: 130323
hex: 1FD13

Meteorological station measurement data including station location, numeric identifier, and name.

Single Frame: No | *Priority Default:* 6 | *Default Update Rate:* 1000 milliseconds | *Frequency:* 1. cycles per second
Destination: Global | *Query Support:* Optional | *Command Support:* Optional | *ACK Rqmnts:* None

Original Reference ID # 15

Field #	Field Name						
1	**Mode**		Byte Field Size:		Request Parameter	Optional	
			Bit Field Size: 4		Command Parameter:	Optional	
	DD025 Mode, Data		0x0 = Autonomous mode, 0x1 = Differential, enhanced mode, 0x2 = Estimated mode, 0x3 = Simulator mode, 0x4 = Manual mode, 0x5 to 0xD = Reserved 0xE = Error, 0xF = Data not available				
	DF52 Bit field	**bit(n)**	Range: Variable	Resolution: 1	Used to construct bit fields		
2	**NMEA Reserved**		Byte Field Size:		Request Parameter		
			Bit Field Size: resv 4		Command Parameter:		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	Range: Variable	Resolution: 1	Used to construct bit fields		
	Used to align subsequent data on a byte boundary.						
3	**Measurement date**		Byte Field Size: 2		Request Parameter	Optional	
			Bit Field Size:		Command Parameter:	Optional	
	DD039 Generic date		Days since January 1, 1970, Date is relative to UTC Time.				
	DF41 Date, day count	**uint16**	Range: 0 to 65,532 days	Resolution: 1 day	0 = January 1, 1970, max = ~179 years		
4	**Measurement time**		Byte Field Size: 4		Request Parameter	Optional	
			Bit Field Size:		Command Parameter:	Optional	
	DD158 Generic time of day		24 hour clock, 0 = midnight, time is in UTC				
	DF06 Time of day	**uint32**	Range: 0 to 86,401 s	Resolution: 1x10E-4 s	~24 hours, 0 = midnight, range allows for up to two leap seconds per day		
5	**Station location, latitude**		Byte Field Size: 4		Request Parameter	Optional	
			Bit Field Size:		Command Parameter:	Optional	
	DD022 Latitude, WGS-84		Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.				
	DF23 Latitude	**int32**	Range: +/- 90 deg	Resolution: 1x10E-7 deg	"-" = South, resolution ~1.1 cm		

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Meteorological Station Data

PGN: 130323
hex: 1FD13

6	**Station location, longitude**		Byte Field Size: 4		Request Parameter	Optional	
			Bit Field Size:		Command Parameter:	Optional	
	DD023 Longitude, WGS-84			Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.			
	DF25 Longitude	**int32**	Range: +/- 180 deg	Resolution: 1x10E-7 deg	"-" = West, resolution ~1.1 cm		
7	**Wind Speed**		Byte Field Size: 2		Request Parameter	Optional	
			Bit Field Size:		Command Parameter:	Optional	
	DD044 Generic Speed						
	DF35 Speed	**uint16**	Range: 0 to 655.32 m/s	Resolution: 1x10E-2 m/s	1 Knot = 0.5144 m/s		
8	**Wind Direction**		Byte Field Size: 2		Request Parameter	Optional	
			Bit Field Size:		Command Parameter:	Optional	
	DD045 Wind Direction						
	DF02 Angle	**uint16**	Range: 0 to 2Pi rad	Resolution: 1x10E-4 rad	Resolution ~0.0057deg, 1 deg = .01745 rad		
9	**Wind Reference**		Byte Field Size:		Request Parameter	Optional	
			Bit Field Size: 3		Command Parameter:	Optional	
	DD205 Wind Reference			0x00 = Theoretical Wind (ground referenced, referenced to True North; calculated using COG/SOG) 0x01 = Theoretical Wind (ground referenced, referenced to Magnetic North; calculated using COG/SOG) 0x02 = Apparent Wind (relative to the vessel centerline) 0x03 = Theoretical (Calculated to Centerline of the vessel, refernced to ground; calculated using COG/SOG) 0x04 = Theoretical (Calculated to Centerline of the vessel, refernced to water; calculated using Heading/Speed through Water) 0x05 = Reserved 0x06 = Error 0x07 = Null			
	DF52 Bit field	**bit(n)**	Range: Variable	Resolution: 1	Used to construct bit fields		
10	**NMEA Reserved**		Byte Field Size:		Request Parameter		
			Bit Field Size: resv 5		Command Parameter:		
	DD001 Reserved field			Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	Range: Variable	Resolution: 1	Used to construct bit fields		
	Used to align subsequent data on a byte boundary.						
11	**Wind Gusts**		Byte Field Size: 2		Request Parameter	Optional	
			Bit Field Size:		Command Parameter:	Optional	
	DD053 Wind gusts			Peak wind gust speed. Sustained wind over an interval of 5 seconds.			
	DF35 Speed	**uint16**	Range: 0 to 655.32 m/s	Resolution: 1x10E-2 m/s	1 Knot = 0.5144 m/s		

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Meteorological Station Data

PGN: 130323
hex: 1FD13

12	**Atmospheric Pressure**		*Byte Field Size:* 2		*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD049 Generic Pressure					
	DF47 Pressure, medium	**uint16**	*Range:* 0 to 6,553,200 Pa	*Resolution:*	1x10E+2 Pa	
13	**Air Temperature**		*Byte Field Size:* 2		*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD043 Generic Temperature					
	DF39 Temperature, low	**uint16**	*Range:* 0 to 655.32 deg K	*Resolution:*	1x10E-2 deg K	
14	**Station ID String**		*Byte Field Size:* 8 or 16 n		*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD004 Generic name string, short		Name of place, route, waypoint, destination, vessel, vehicle, etc.			
	DF50 String, variable, short	**ch8or16(n)**	*Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters	*Resolution:*	1 ASCII or 1 Unicode Character	2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

15 characters maximum.

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15	**Station Name String**		*Byte Field Size:*	8 or 16	n	*Request Parameter*	Optional
			Bit Field Size:			*Command Parameter:*	Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50	String, variable, short	**ch8or16(n)**	*Range:*	0 to 250 ASCII or 0 to 125 Unicode Characters	*Resolution:*	1 ASCII or 1 Unicode Character	2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

50 characters maximum.

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Moored Buoy Station Data

PGN: 130324
hex: 1FD14

Moored buoy measurement data including station location and numeric identifier.

Single Frame: No | *Priority Default:* 6 | *Default Update Rate:* 1000 milliseconds | *Frequency:* 1. cycles per second
Destination: Global | *Query Support:* Optional | *Command Support:* Optional | *ACK Rqmnts:* None

Original Reference ID # 16

Field #	Field Name						
1	**Mode**		Byte Field Size:		Request Parameter	Optional	
			Bit Field Size: 4		Command Parameter:	Optional	
	DD025 Mode, Data		0x0 = Autonomous mode, 0x1 = Differential, enhanced mode, 0x2 = Estimated mode, 0x3 = Simulator mode, 0x4 = Manual mode, 0x5 to 0xD = Reserved 0xE = Error, 0xF = Data not available				
	DF52 Bit field	**bit(n)**	Range: Variable	Resolution: 1	Used to construct bit fields		
2	**NMEA Reserved**		Byte Field Size:		Request Parameter	Optional	
			Bit Field Size: resv 4		Command Parameter:		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"				
	DF52 Bit field	**bit(n)**	Range: Variable	Resolution: 1	Used to construct bit fields		
	Used to align subsequent data on a byte boundary.						
3	**Measurement date**		Byte Field Size: 2		Request Parameter	Optional	
			Bit Field Size:		Command Parameter:	Optional	
	DD039 Generic date		Days since January 1, 1970, Date is relative to UTC Time.				
	DF41 Date, day count	**uint16**	Range: 0 to 65,532 days	Resolution: 1 day	0 = January 1, 1970, max = ~179 years		
4	**Measurement time**		Byte Field Size: 4		Request Parameter	Optional	
			Bit Field Size:		Command Parameter:	Optional	
	DD158 Generic time of day		24 hour clock, 0 = midnight, time is in UTC				
	DF06 Time of day	**uint32**	Range: 0 to 86,401 s	Resolution: 1x10E-4 s	~24 hours, 0 = midnight, range allows for up to two leap seconds per day		
5	**Station location, latitude**		Byte Field Size: 4		Request Parameter	Optional	
			Bit Field Size:		Command Parameter:	Optional	
	DD022 Latitude, WGS-84		Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.				
	DF23 Latitude	**int32**	Range: +/- 90 deg	Resolution: 1x10E-7 deg	"-" = South, resolution ~1.1 cm		

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Moored Buoy Station Data

PGN: 130324
hex: 1FD14

6	**Station location, longitude**		*Byte Field Size:*	4		*Request Parameter*	Optional	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD023 Longitude, WGS-84				Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10's of minute.			
	DF25 Longitude	**int32**	*Range:*	+/- 180 deg		*Resolution:*	1x10E-7 deg	"-" = West, resolution ~1.1 cm
7	**Wind Speed**		*Byte Field Size:*	2		*Request Parameter*	Optional	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD044 Generic Speed							
	DF35 Speed	**uint16**	*Range:*	0 to 655.32 m/s		*Resolution:*	1x10E-2 m/s	1 Knot = 0.5144 m/s
8	**Wind Direction**		*Byte Field Size:*	2		*Request Parameter*	Optional	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD045 Wind Direction							
	DF02 Angle	**uint16**	*Range:*	0 to 2Pi rad		*Resolution:*	1x10E-4 rad	Resolution ~0.0057deg, 1 deg = .01745 rad
9	**Wind Reference**		*Byte Field Size:*			*Request Parameter*	Optional	
			Bit Field Size:	3		*Command Parameter:*	Optional	
	DD205 Wind Reference				0x00 = Theoretical Wind (ground referenced, referenced to True North; calculated using COG/SOG) 0x01 = Theoretical Wind (ground referenced, referenced to Magnetic North; calculated using COG/SOG) 0x02 = Apparent Wind (relative to the vessel centerline) 0x03 = Theoretical (Calculated to Centerline of the vessel, refernced to ground; calculated using COG/SOG) 0x04 = Theoretical (Calculated to Centerline of the vessel, refernced to water; calculated using Heading/Speed through Water) 0x05 = Reserved 0x06 = Error 0x07 = Null			
	DF52 Bit field	**bit(n)**	*Range:*	Variable		*Resolution:*	1	Used to construct bit fields
10	**NMEA Reserved**		*Byte Field Size:*			*Request Parameter*		
			Bit Field Size:	resv 5		*Command Parameter:*		
	DD001 Reserved field				Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:*	Variable		*Resolution:*	1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.							
11	**Wind Gusts**		*Byte Field Size:*	2		*Request Parameter*	Optional	
			Bit Field Size:			*Command Parameter:*	Optional	
	DD053 Wind gusts				Peak wind gust speed. Sustained wind over an interval of 5 seconds.			
	DF35 Speed	**uint16**	*Range:*	0 to 655.32 m/s		*Resolution:*	1x10E-2 m/s	1 Knot = 0.5144 m/s

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Moored Buoy Station Data

PGN: 130324
hex: 1FD14

12	**Wave Height**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD050 Wave Height			Wave height is calculated as the highest one-third of all of the wave heights during a 20-minute sampling period. Note: Accelerometers or inclinometers on board the buoys measure the heave acceleration or the vertical displacement of the buoy hull during the wave acquisition time. A Fast Fourier Transform (FFT) is applied to the data by the processor on board the buoy to transform the data from the temporal domain into the frequency domain. Note that the raw acceleration or displacement measurements are not transmitted shore-side. Response amplitude operator (RAO) processing is then performed on the transformed data to account for both hull and electronic noise. It is from this transformation that non-directional spectral wave measurements (i.e., wave energies with their associated frequencies) are derived. Along with the spectral energies, measurements such as significant wave height (WVHGT), average wave period (AVGPD), and dominant period (DOMPD) are also derived from the transformation.		
	DF13 Distance, short	**uint16**	*Range:* 0 to 655.32 m		*Resolution:* 1x10E-2 m	
13	**Dominate Wave Period**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD051 Wave Period in seconds			Average wave period of all waves during a 20-minute period. Dominant wave period is the period with the maximum wave energy. Note: See DD50 note.		
	DF66 Time interval, .01sec	**uint16**	*Range:* 0 to 655.32s		*Resolution:* 1x10E-2sec	
14	**Atmospheric Pressure**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD049 Generic Pressure					
	DF47 Pressure, medium	**uint16**	*Range:* 0 to 6,553,200 Pa		*Resolution:* 1x10E+2 Pa	
15	**Pressure Tendency Rate**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD052 Pressure Rate			Positive value indicates Rising, Negative value indicates Falling.		
	DF48 Pressure rate	**int16**	*Range:* +/- 327,640 Pa/hr		*Resolution:* 1x10E+1 Pa/hr	+ = increasing rate
16	**Air temperature**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD043 Generic Temperature					
	DF39 Temperature, low	**uint16**	*Range:* 0 to 655.32 deg K		*Resolution:* 1x10E-2 deg K	
17	**Water temperature**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD043 Generic Temperature					
	DF39 Temperature, low	**uint16**	*Range:* 0 to 655.32 deg K		*Resolution:* 1x10E-2 deg K	

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Moored Buoy Station Data

PGN: 130324
hex: 1FD14

18 **Station ID String**

Byte Field Size: 8 or 16 | n
Bit Field Size:

Request Parameter Required
Command Parameter: Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character

2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

15 characters maximum.

If this field is not specified in the "Command Request" or an ISO Request is made of this PGN, the response will with the unit's Station ID. Otherwise if this field is specified only the units with a matching Station ID will respond with this PGN.

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Payload Mass

PGN: 130560
hex: 1FE00

The Payload Mass parameter group is used to transmit the mass associated with vessel payloads. Since the SI units of mass (kilogram) are commonly understood by end users as the weight at standard gravitational acceleration, devices receiving this parameter group may display the transmitted value as either mass or weight.

Measurement status includes a 'Locked' status in addition to the stable status. Some scales may have the ability to filter out small changes to the measurement due to vessel motion or other influences. These scales may use the 'Locked' status to indicate that the transmitted Payload Mass will remain fixed until a significant change in mass is detected. The value for 'significant' may vary by manufacturer, but is usually 10% to 20% or more of the current mass measurement.

Applications for this PGN vary from recreational to commercial use. This may include, but is not limited to weighing fish, or weighing cargo .

Response to Requests:
Field #4, Measurement ID, is identified as a request parameter for this Parameter Group. All providers of this PGN may accept and process requests based upon the requested value of Field # 4. A device receiving an ISO Request (PGN 059904) for this PGN, shall respond by providing as many of these PGNs as necessary for every Measurement ID that has associated data fields. If a Complex Request Group Function (PGN 126208) requesting this PGN is received, the receiving device shall respond in the following manner: If no requested fields have been included with the Complex Request, than the response is to return one or more PGNs, just like responding to the ISO Request (PGN 059904) described above. If the Complex Request (PGN 126208) includes the Measurement ID field, then the response shall be filtered by the field (Field #4) and field value (Measurement ID) contained within the request.

For example, if the Complex Request for this PGN contained a value of 2 for field 4, the Measurement ID, and this was a measurement channel that the device was operating with or had information about, than the device would respond by providing a single PGN with measurement data associated with the Measurement ID channel value requested.

If the Measurement ID requested is considered valid by the device, but the device was not operating on that measurement channel, or had no data associated with the measurement channel requested, there are two possible responses:
1 - The device responds with the PGN containing the measurement channel requested and all other fields set to the value indicating "Data not
available".
2 - The device responds with the Acknowledge Group PGN (126208) containing the error state of "0x2 = Temporarily unable to comply".

If the request was global no response would be required. If the request was addressed to the device, than either response 1 or 2 would apply.

If the Measurement ID requested is not considered a valid measurement channel by the receiving device, then the appropriate response would be the Acknowledge Group PGN (126208), containing the error state for the requested measurement ID field of "0x3 = Request or command parameter out-of-range;".

Examples of equipment that might report Payload Mass include loading/unloading equipment and scales.

Single Frame: Yes *Priority Default:* 5 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 204

1 **Sequence ID** *Byte Field Size:* 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD056 Sequence ID
An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number
Sequence number used to associate Payload Mass transmissions with other parameter groups being transmitted from the same source address.

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Payload Mass

PGN: 130560
hex: 1FE00

2 **Measurement Status**
Byte Field Size: *Bit Field Size:* 3
Request Parameter Optional
Command Parameter: Optional

DD312 Scale Measurement Status

Enumerated value to indicate measurement status
0x00 = Current Changing – the payload acceleration is changing
0x01 = Current Stable – the payload acceleration has reached equilibrium
0x02 = Measurement Locked – the payload measurement is fixed until a significant mass change occurs (see description)
0x04 = Almanac Mass – the payload mass is being transmitted from memory
0x07 = Error or no status

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Indicates if the measurement provided is current, and whether the measurement is stable.

---

3 **NMEA Reserved**
Byte Field Size: *Bit Field Size:* resv 5
Request Parameter *Command Parameter:*

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

---

4 **Measurement ID**
Byte Field Size: 1 *Bit Field Size:*
Request Parameter Optional
Command Parameter: Optional

DD005 Generic numeric ID, short

Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

Unique measurement identifier. May be used to differentiate between multiple channels of information on the same equipment or scale.

---

5 **Payload Mass**
Byte Field Size: 4 *Bit Field Size:*
Request Parameter Optional
Command Parameter: Optional

DD313 Mass

DF104 Mass **uint32** *Range:* 0 - 429,496.7292 kilograms *Resolution:* 1x10E-4 kg

Payload mass in kilograms. Corrected for gravitational acceleration.

---

6 **NMEA Reserved**
Byte Field Size: *Bit Field Size:* resv 8
Request Parameter *Command Parameter:*

DD001 Reserved field

Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary.

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Watermaker Input Setting and Status

PGN: 130567
hex: 1FE07

This PGN may be requested or used to command and configure a number of Watermaker controls. The Command Group Function PGN 126208 is used perform the following: start/stop a production, start/stop rinse or flush operation , start/stop low and high pressure pump and perform an emergency stop. The Request Group Function PGN 126208 or ISO Request PGN 059904 may be used to request this PGN. This PGN also provides Watermaker status and measurement information. The PGN is broadcast periodically.

Single Frame: N *Priority Default:* 6 *Default Update Rate:* 2500 milliseconds *Frequency:* .4 cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Required *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 223

1 Watermaker Operating State
Byte Field Size: *Bit Field Size:* 6 *Request Parameter* Optional *Command Parameter:* Optional

DD363 Watermaker Operating State
0 = Stopped
1 = Starting
2 = Running
3 = Stopping
4 = Flushing
5 = Rinsing
6 = Initiating
7 = Manual Mode
62 = Error
63 = Unavailable

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

2 Production Start/Stop
Byte Field Size: *Bit Field Size:* 2 *Request Parameter* Optional *Command Parameter:* Required

DD002 Generic status pair
MSB/LSB:
00 = [No, Off, Disabled, Reset, "0"],
01 = [Yes, On, Enabled, Set, "1"],
10 = Error,
11= [Unavailable, Unknown]

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

When commanding this field with PGN 126208, 00 Stops the Production and 01 Starts the Production. When this PGN is sent, 01 indicates the Production is ON and 00 indicates the production is off

3 Rinse Start/Stop
Byte Field Size: *Bit Field Size:* 2 *Request Parameter* Optional *Command Parameter:* Required

DD002 Generic status pair
MSB/LSB:
00 = [No, Off, Disabled, Reset, "0"],
01 = [Yes, On, Enabled, Set, "1"],
10 = Error,
11= [Unavailable, Unknown]

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

When commanding this field with PGN 126208, 00 Stops the Rinse/Flush and 01 Starts the Rinse/Flush. When this PGN is sent, 01 indicates the Rinse/Flush function is ON and 00 indicates the Rinse/Flush function is off

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4	**Low Pressure Pump Status**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	2	*Command Parameter:*	Required

DD002 Generic status pair

MSB/LSB:
00 = [No, Off, Disabled, Reset, "0"],
01 = [Yes, On, Enabled, Set, "1"],
10 = Error,
11= [Unavailable, Unknown]

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

When commanding this field with PGN 126208, 00 turns off the Low Pressure Pump and 01 starts the Low Pressure Pump. When this PGN is sent, 00 indicates the Low Pressure Pump is off and 01 indicates the Low Pressure Pump is on.

5	**High Pressure Pump Status**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	2	*Command Parameter:*	Required

DD002 Generic status pair

MSB/LSB:
00 = [No, Off, Disabled, Reset, "0"],
01 = [Yes, On, Enabled, Set, "1"],
10 = Error,
11= [Unavailable, Unknown]

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

When commanding this field with PGN 126208, 00 turns off the High Pressure Pump and 01 starts the High Pressure Pump. When this PGN is sent, 00 indicates the High Pressure Pump is off and 01 indicates the High Pressure Pump is on.

6	**Emergency Stop**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	2	*Command Parameter:*	Required

DD002 Generic status pair

MSB/LSB:
00 = [No, Off, Disabled, Reset, "0"],
01 = [Yes, On, Enabled, Set, "1"],
10 = Error,
11= [Unavailable, Unknown]

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

When commanding this field with PGN 126208, 01 initiates the Emergency Stop function of the Watermaker, all other values are ignored. Emergency Stop can only be activated by the value of 01. When this PGN is sent, normal operation is indicated by the value 00 which means the Emergency Stop is inactive (OFF).

7	**Product Solenoid Valve Status**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	2	*Command Parameter:*	Optional

DD364 Sensor Status

MSB/LSB:
00 = OK
01 = Warning
10 = Error
11= [Unavailable, Unknown]

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

8	**Flush Mode Status**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	2	*Command Parameter:*	Optional

DD002 Generic status pair

MSB/LSB:
00 = [No, Off, Disabled, Reset, "0"],
01 = [Yes, On, Enabled, Set, "1"],
10 = Error,
11= [Unavailable, Unknown]

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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9	**Salinity Status**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD364 Sensor Status		MSB/LSB: 00 = OK 01 = Warning 10 = Error 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
10	**Feed Pressure Status**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD364 Sensor Status		MSB/LSB: 00 = OK 01 = Warning 10 = Error 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
11	**Oil Change Indicator Status**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD364 Sensor Status		MSB/LSB: 00 = OK 01 = Warning 10 = Error 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
12	**Filter Status**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD364 Sensor Status		MSB/LSB: 00 = OK 01 = Warning 10 = Error 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
13	**System Status**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD364 Sensor Status		MSB/LSB: 00 = OK 01 = Warning 10 = Error 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
14	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*	
			Bit Field Size:	resv 2	*Command Parameter:*	
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.					

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Watermaker Input Setting and Status

PGN: 130567
hex: 1FE07

15	**Salinity**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD365 Salinity, Watermaker					
	DF112 Salinity	**uint16**	*Range:* 0 to 65,535 ppm	*Resolution:*	1 ppm	
16	**Product Water Temperature**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD043 Generic Temperature					
	DF39 Temperature, low	**uint16**	*Range:* 0 to 655.32 deg K	*Resolution:*	1x10E-2 deg K	
17	**Pre-filter Pressure**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD049 Generic Pressure					
	DF47 Pressure, medium	**uint16**	*Range:* 0 to 6,553,200 Pa	*Resolution:*	1x10E+2 Pa	
18	**Post-filter Pressure**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD049 Generic Pressure					
	DF47 Pressure, medium	**uint16**	*Range:* 0 to 6,553,200 Pa	*Resolution:*	1x10E+2 Pa	
19	**Feed Pressure**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD366 Pressure, Watermaker					
	DF113 Pressure, Compound	**int16**	*Range:* +/- 32,764 Kpa	*Resolution:*	1 Kpa	
20	**System High Pressure**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD225 Generic Pressure High					
	DF29 Pressure	**uint16**	*Range:* 0 to 65,532,000 Pa	*Resolution:*	1x10E+3 Pa	
21	**Product Water Flow**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD131 Flow rate, low					
	DF18 Flow rate, low	**int16**	*Range:* +/-3.2764 cu-m/hr	*Resolution:*	1x10E-4 cu-m/hr	
22	**Brine Water Flow**		*Byte Field Size:* 2 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD131 Flow rate, low					
	DF18 Flow rate, low	**int16**	*Range:* +/-3.2764 cu-m/hr	*Resolution:*	1x10E-4 cu-m/hr	
23	**Run Time**		*Byte Field Size:* 4 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD132 Run time, Engine					
	DF67 Time interval, large	**uint32**	*Range:* 0 to ~4.295x10E+9 s	*Resolution:*	1 sec	

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Entertainment - Current File and Status

PGN: 130569
hex: 1FE09

Details the current file being played and the current play status by Audio/Video zone.
Sent on start up, upon any change, upon request and periodically while a Audio/Video device is playing a file.
Fields can be commanded by a client using the NMEA Command Group Function (PGN 126208) to change the currently playing audio/video source, current file, adjust volume, change the playing status, save a favorite or tune a radio source.
A device receiving an ISO Request (PGN 059904) for this PGN, shall respond by providing as many of these PGNs as necessary for every Audio/Video zone.
If a Request Group Function (PGN 126208) requesting this PGN is received, the receiving device shall respond in the following manner :
•If no requested fields have been included with the Request Group Function then the response is to return one or more PGNs, just like responding to the ISO Request (PGN 059904).
•If the Request Group Function (PGN 126208) includes the Zone Number, Audio/Video Source Type, Audio/Video Source Number or File ID fields, then the response shall be filtered by these fields contained within this request resulting in one or more responses.
If the fields requested are not valid then the appropriate response would be the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied”. If an individual field is invalid the field parameter error code for that field shall be “0x3 = Requested or command parameter out-of-range”. If the individual fields are valid but the combination of fields together are invalid then just the error state for PGN error code (Field 3) of “0x3 = Access denied” would be in the Acknowledge Group Function (PGN 126208) response.

PGN Transmit Frequency is 1 cycle per second per Audio/Video Zone.

Further details can be found in the NMEA2000 Entertainment Application Notes in Appendix D.

Single Frame:	N	*Priority Default:*	6	*Default Update Rate:*	500	milliseconds	*Frequency:*	2.	cycles per second
Destination:	Global	*Query Support:*	Required	*Command Support:*	Required		*ACK Rqmnts:*	Yes	

Field # *Field Name* — Original Reference ID # 234

1 **Zone Number**
Byte Field Size:
Bit Field Size: 8
Request Parameter Required
Command Parameter: Required

DD396 Entertainment Zones

0 = All Zones
1 = Zone 1
2 = Zone 2
3 = Zone 3
4 - 252 = User Defined

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

An Audio/Video zone is a room or area of the vessel in which sound will be distributed. Each zone can be separately controlled and could play a different source of audio or video.

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Entertainment - Current File and Status

PGN: 130569
hex: 1FE09

2	**Audio/Video Source Type**	*Byte Field Size:*		*Request Parameter*	Required
		Bit Field Size:	8	*Command Parameter:*	Required

DD389 Audio/Video Source Type

0 = Vessel Alarm
1 = AM
2 = FM
3 = Weather
4 = DAB
5 = Aux
6 = USB
7 = CD
8 = MP3
9 = Apple iOS
10 = Android
11 = Bluetooth
12 = Sirius XM
13 = Pandora
14 = Spotify
15 = Slacker
16 = Songza
17 = Apple Radio
18 = Last FM
19 = Ethernet
20 = Video MP4
21 = Video DVD
22 = Video BlueRay
23 = HDMI
24 = Video
25 - 252 = User defind
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3	**Audio/Video Source Number**	*Byte Field Size:*	1	*Request Parameter*	Required
		Bit Field Size:		*Command Parameter:*	Required

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

Audio/Video Source Number per Type. There can be more than one of each type of audio source available on a A/V Device (e.g. 3 USB Sources).

4	**File ID**	*Byte Field Size:*	4	*Request Parameter*	Required
		Bit Field Size:		*Command Parameter:*	Required

DD010 Generic numeric ID, large — Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number

ID of a file, song, station which is unique per Audio/Video source.

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Entertainment - Current File and Status

PGN: 130569
hex: 1FE09

5	**Play Status**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	8	*Command Parameter:*	Required
	DD390 Play Status		0 = Play (Normal functionality) 1 = Pause 2 = Stop 3 = FF (1x) 4 = FF (2x) 5 = FF (3x) 6 = FF (4x) 7 = RW (1x) 8 = RW (2x) 9 = RW (3x) 10 = RW (4x) 11 = Skip Ahead 12 = Skip Back 13 = Jog Ahead 14 = Jog Back 15 = Seek Up 16 = Seek Down 17 = Scan Up 18 = Scan Down 19 = Tune Up 20 = Tune Down 21 = Slow Motion (.75x) 22 = Slow Motion (.5x) 23 = Slow Motion (.25x) 24 = Slow Motion (.125x) 25 - 252 = User Defined 253 = Reserved 254 = Error 255 = Not available			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
6	**Elapsed Track/Chapter Time**		*Byte Field Size:*	2	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Note 1
	DD391 Time, Standard		Time, 1 Second Resolution			
	DF80 Time, 1sec	**uint16**	*Range:* 0 to 65532 seconds	*Resolution:* 1 second		
7	**Track/Chapter Time**		*Byte Field Size:*	2	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Note 1
	DD391 Time, Standard		Time, 1 Second Resolution			
	DF80 Time, 1sec	**uint16**	*Range:* 0 to 65532 seconds	*Resolution:* 1 second		
8	**Repeat Status**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Note 1
	DD392 Repeat		0 = Off 1 = One (Current File) 2 = All (Play Queue) 3 - 14 = Reserved 15 = Data Not Available / Do Not Change			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	

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9 **Shuffle Status**

Byte Field Size:
Bit Field Size: 4

Request Parameter Optional
Command Parameter: Note 1

DD393 Shuffle

0 = Off
1 = Play Queue
2 = All
3 - 14 = Reserved
15 = Data Not Available / Do Not Change

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

If the shuffle status is set to Play Queue the A/V device will play all the files in the Play Queue in random order.

If the shuffle status is set to all the A/V device will play all files from a source in a random order. The A/V Device will no longer play files in the Play Queue.

---

10 **Save Favorite Number (write only)**

Byte Field Size: 1
Bit Field Size:

Request Parameter Optional
Command Parameter: Note 1

DD005 Generic numeric ID, short

Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

Used to save current station as favorite. Tuner based sources only: AM/FM/DAB/SiriusXM.
When this PGN is generated as a status or report, this field is set to not available (0XFF).

---

11 **Play Favorite Number**

Byte Field Size: 1
Bit Field Size:

Request Parameter Optional
Command Parameter: Note 1

DD005 Generic numeric ID, short

Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

Used to command AV to play indicated favorite station

---

12 **Thumbs Up/Thumbs Down**

Byte Field Size:
Bit Field Size: 8

Request Parameter Optional
Command Parameter: Note 1

DD394 Thumbs Up / Thumbs Down

0 = None
1 = Thumbs Up
2 = Thumbs Down
3 = Reserved

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

13 **Signal Strength**

Byte Field Size: 1
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD263 Generic Absolute Percentage 0-252%

DF93 Percent, Absolute **uint8** *Range:* 0 - 252% *Resolution:* 1%

Generic Absolute Percentage 0-252%

---

14 **Radio Frequency**

Byte Field Size: 4
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD016 Radio Tx or Rx Frequency

DF21 Frequency **uint32** *Range:* 0 to ~4.295x10E+10 Hz *Resolution:* 10 Hz

---

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Entertainment - Current File and Status

PGN: 130569
hex: 1FE09

15	**HD Frequency Multicast**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

Digital sub channel used for multicasting digital radio or television stations on a single frequency.

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Entertainment - Library Data File

PGN: 130570
hex: 1FE0A

Details the library data for an individual Audio/Video file.
Fields can be commanded by a client using the NMEA Command Group Function (PGN 126208) to add or remove a file from a play queue. A command would include the Audio/Video Source Type, Audio Video Source Number, File ID, Library Data Type, Play Queue Zone Number and In Play Queue fields.
A device receiving an ISO Request (PGN 059904) for this PGN, shall respond with this PGN with all fields set to invalid.
If a Request Group Function (PGN 126208) requesting this PGN is received, the receiving device shall respond in the following manner :
•If no requested fields have been included with the Request Group Function then the response is to return the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied”.
•If the Request Group Function (PGN 126208) includes the Audio/Video Source Type, Audio/Video Source Number, File ID and Library Data Type fields, then the response shall be filtered by these fields contained within this request resulting in one or more responses.
If the fields requested are not valid then the appropriate response would be the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied”. If an individual field is invalid the field parameter error code for that field shall be “0x3 = Requested or command parameter out-of-range”. If the individual fields are valid but the combination of fields together are invalid then just the error state for PGN error code (Field 3) of “0x3 = Access denied” would be in the Acknowledge Group Function (PGN 126208) response.
Further details can be found in the NMEA2000 Entertainment Application Notes in Appendix D

Single Frame:	N	Priority Default:	6	Default Update Rate:		milliseconds	Frequency:	NA	cycles per second
Destination:	Global	Query Support:	Required	Command Support:	Required	ACK Rqmnts:	Yes		

Original Reference ID # 235

Field #	Field Name				
1	**Audio/Video Source Type**	Byte Field Size:		Request Parameter	Required
		Bit Field Size:	8	Command Parameter:	Required

DD389 Audio/Video Source Type

0 = Vessel Alarm
1 = AM
2 = FM
3 = Weather
4 = DAB
5 = Aux
6 = USB
7 = CD
8 = MP3
9 = Apple iOS
10 = Android
11 = Bluetooth
12 = Sirius XM
13 = Pandora
14 = Spotify
15 = Slacker
16 = Songza
17 = Apple Radio
18 = Last FM
19 = Ethernet
20 = Video MP4
21 = Video DVD
22 = Video BlueRay
23 = HDMI
24 = Video
25 - 252 = User defind
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** Range: Variable Resolution: 1 Used to construct bit fields

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Entertainment - Library Data File

PGN: 130570
hex: 1FE0A

2	**Audio/Video Source Number**	*Byte Field Size:*	1	*Request Parameter*	Required	
		Bit Field Size:		*Command Parameter:*	Required	

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

Audio/Video Source Number per Type

There can be more than one of each type of audio source available on a A/V Device (e.g. 3 USB Sources).

3	**File ID**	*Byte Field Size:*	4	*Request Parameter*	Required	
		Bit Field Size:		*Command Parameter:*	Required	

DD010 Generic numeric ID, large — Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number

ID of a file, song, station which is unique per Audio/Video source.

4	**Library Data Type**	*Byte Field Size:*		*Request Parameter*	Required	
		Bit Field Size:	8	*Command Parameter:*	Optional	

DD395 Library Data Type

0 = File
1 = Playlist Name
2 = Genre Name
3 = Album Name
4 = Artist Name
5 = Track Name
6 = Station Name
7 = Station Number
8 = Favorite Number
9 = Play Queue
10 - 252 = Reserved

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

5	**Library Data Name**	*Byte Field Size:*	8 or 16 n	*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character

2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

0 to 125 Unicode Characters

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6	**Track/Chapter Number**		*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* *Command Parameter:*	Optional Optional
	DD007 Generic numeric ID, medium		Number of route, waypoint, event, mark, etc.				
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number		
	Generic numeric ID, medium						
7	**Station Number**		*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* *Command Parameter:*	Optional Optional
	DD007 Generic numeric ID, medium		Number of route, waypoint, event, mark, etc.				
	DF54 Integer, 16 bit unsigned	**uint16**	*Range:* 0 to 65,532	*Resolution:* 1 bit	Unit-less number		
	Generic numeric ID, medium						
8	**Favorite Number**		*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* *Command Parameter:*	Optional Optional
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
	Used to save current station as favorite Tuner based sources only: AM/FM/DAB/SiriusXM						
9	**Radio Frequency**		*Byte Field Size:* 4 *Bit Field Size:*			*Request Parameter* *Command Parameter:*	Optional Optional
	DD016 Radio Tx or Rx Frequency						
	DF21 Frequency	**uint32**	*Range:* 0 to ~4.295x10E+10 Hz	*Resolution:* 10 Hz			
	Frequency						
10	**HD Frequency Multi-Cast**		*Byte Field Size:* 1 *Bit Field Size:*			*Request Parameter* *Command Parameter:*	Optional Optional
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.				
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number		
	Generic numeric ID, short						
11	**Play Queue Zone Number**		*Byte Field Size:* *Bit Field Size:* 8			*Request Parameter* *Command Parameter:*	Optional Note 1
	DD396 Entertainment Zones		0 = All Zones 1 = Zone 1 2 = Zone 2 3 = Zone 3 4 - 252 = User Defined				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		
12	**In Play Queue**		*Byte Field Size:* *Bit Field Size:* 2			*Request Parameter* *Command Parameter:*	Optional Note 1
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]				
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields		

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Entertainment - Library Data Group

PGN: 130571
hex: 1FE0B

Details the library data for an individual Audio/Video group. A group is a list of Ids of other groups or files.
A device receiving an ISO Request (PGN 059904) for this PGN, shall respond with this PGN with all fields set to invalid.
If a Request Group Function (PGN 126208) requesting this PGN is received, the receiving device shall respond in the following manner :
•If no requested fields have been included with the Request Group Function then the response is to return the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied”.
•If the Request Group Function (PGN 126208) includes the Audio/Video Source Type, Audio/Video Source Number, Group Type, Group ID and Index of first ID in PGN fields, then the response shall be filtered by these fields contained within this request resulting in one response.
If the fields requested are not valid then the appropriate response would be the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied”. If an individual field is invalid the field parameter error code for that field shall be “0x3 = Requested or command parameter out-of-range”. If the individual fields are valid but the combination of fields together are invalid then just the error state for PGN error code (Field 3) of “0x3 = Access denied” would be in the Acknowledge Group Function (PGN 126208) response.
Further details can be found in the NMEA2000 Entertainment Application Notes in Appendix D.

Single Frame:	N	*Priority Default:*	6	*Default Update Rate:*		milliseconds	*Frequency:*	NA	cycles per second
Destination:	Global	*Query Support:*	Required	*Command Support:*	Optional	*ACK Rqmnts:*	Yes		

Field # *Field Name* Original Reference ID # 236

1 **Audio/Video Source Type**

Byte Field Size:
Bit Field Size: 8

Request Parameter Required
Command Parameter: Optional

DD389 Audio/Video Source Type

0 = Vessel Alarm
1 = AM
2 = FM
3 = Weather
4 = DAB
5 = Aux
6 = USB
7 = CD
8 = MP3
9 = Apple iOS
10 = Android
11 = Bluetooth
12 = Sirius XM
13 = Pandora
14 = Spotify
15 = Slacker
16 = Songza
17 = Apple Radio
18 = Last FM
19 = Ethernet
20 = Video MP4
21 = Video DVD
22 = Video BlueRay
23 = HDMI
24 = Video
25 - 252 = User defind
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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Entertainment - Library Data Group

PGN: 130571
hex: 1FE0B

2 **Audio/Video Source Number**
Byte Field Size: 1 *Bit Field Size:*
Request Parameter Required
Command Parameter: Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number
Audio/Video Source Number per Type

There can be more than one of each type of audio source available on a A/V Device (e.g. 3 USB Sources).

---

3 **Group Type**
Byte Field Size: *Bit Field Size:* 8
Request Parameter Required
Command Parameter: Optional

DD395 Library Data Type

0 = File
1 = Playlist Name
2 = Genre Name
3 = Album Name
4 = Artist Name
5 = Track Name
6 = Station Name
7 = Station Number
8 = Favorite Number
9 = Play Queue
10 - 252 = Reserved

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

4 **Play Queue Zone Number**
Byte Field Size: *Bit Field Size:* 8
Request Parameter Required
Command Parameter: Optional

DD396 Entertainment Zones

0 = All Zones
1 = Zone 1
2 = Zone 2
3 = Zone 3
4 - 252 = User Defined

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
Audio Video Zone Number

---

5 **Group ID**
Byte Field Size: 4 *Bit Field Size:*
Request Parameter Required
Command Parameter: Optional

DD010 Generic numeric ID, large — Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number
ID of a group which is unique per Audio/Video source.

---

6 **Index of first ID in PGN**
Byte Field Size: 2 *Bit Field Size:*
Request Parameter Required
Command Parameter: Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number
Generic numeric ID, medium

---

7 **Number of IDs in this PGN**
Byte Field Size: 2 *Bit Field Size:*
Request Parameter Optional
Command Parameter: Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number
Generic numeric ID, medium

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Entertainment - Library Data Group

PGN: 130571
hex: 1FE0B

8 **Total number of IDs available**

Byte Field Size: 2 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD007 Generic numeric ID, medium — Number of route, waypoint, event, mark, etc.

DF54 Integer, 16 bit unsigned **uint16** *Range:* 0 to 65,532 *Resolution:* 1 bit Unit-less number

Generic numeric ID, medium

---

9 **ID Type**

Byte Field Size: — *Request Parameter* Optional
Bit Field Size: 8 — *Command Parameter:* Optional

DD397 ID Type

0 = Group
1 = File
2 = Reserved
3 = Do Not Use / Data Not Available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

10 **ID**

Byte Field Size: 4 — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD010 Generic numeric ID, large — Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number

This field will be either a File ID or Group ID depending on the value of Field 9 ID Type.

---

11 **ID Name**

Byte Field Size: 8 or 16 n — *Request Parameter* Optional
Bit Field Size: — *Command Parameter:* Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character 2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

0 to 125 Unicode Characters

---

12 **Fields 9 – 11 repeat as needed**

Byte Field Size: ? — *Request Parameter*
Bit Field Size: — *Command Parameter:*

DD000 Undefined — Application specific, defined at time of use

DF00 Undefined **Undefined** *Range:* undefined *Resolution:* undefined Application specific, defined at time of use.

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Entertainment - Library Data Search

PGN: 130572
hex: 1FE0C

Details the library data for an individual Audio/Video group. A group is a list of Ids of other groups or files.
A device receiving an ISO Request (PGN 059904) for this PGN, shall respond with this PGN with all fields set to invalid.
If a Request Group Function (PGN 126208) requesting this PGN is received, the receiving device shall respond in the following manner :
•If no requested fields have been included with the Request Group Function then the response is to return the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied”.
•If the Request Group Function (PGN 126208) includes the Audio/Video Source Type, Audio/Video Source Number, Group Type, Group ID and Index of first ID in PGN fields, then the response shall be filtered by these fields contained within this request resulting in one response.
If the fields requested are not valid then the appropriate response would be the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied”. If an individual field is invalid the field parameter error code for that field shall be “0x3 = Requested or command parameter out-of-range”. If the individual fields are valid but the combination of fields together are invalid then just the error state for PGN error code (Field 3) of “0x3 = Access denied” would be in the Acknowledge Group Function (PGN 126208) response.
Further details can be found in the NMEA2000 Entertainment Application Notes in Appendix D.

Single Frame: N *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Optional *ACK Rqmnts:* Yes

Field # *Field Name* Original Reference ID # 237

1 **Audio/Video Source Type** *Byte Field Size:* *Bit Field Size:* 8 *Request Parameter* Required *Command Parameter:* Optional

DD389 Audio/Video Source Type

0 = Vessel Alarm
1 = AM
2 = FM
3 = Weather
4 = DAB
5 = Aux
6 = USB
7 = CD
8 = MP3
9 = Apple iOS
10 = Android
11 = Bluetooth
12 = Sirius XM
13 = Pandora
14 = Spotify
15 = Slacker
16 = Songza
17 = Apple Radio
18 = Last FM
19 = Ethernet
20 = Video MP4
21 = Video DVD
22 = Video BlueRay
23 = HDMI
24 = Video
25 - 252 = User defind
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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Entertainment - Library Data Search

PGN: 130572
hex: 1FE0C

2 **Audio/Video Source Number**

Byte Field Size: 1 *Bit Field Size:*

Request Parameter Required
Command Parameter: Optional

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

Audio/Video Source Number per Type

There can be more than one of each type of audio source available on a A/V Device (e.g. 3 USB Sources).

---

3 **Group ID**

Byte Field Size: 4 *Bit Field Size:*

Request Parameter Required
Command Parameter: Optional

DD010 Generic numeric ID, large — Number of route, waypoint, event, mark, etc.

DF55 Integer, 32 bit unsigned **uint32** *Range:* 0 to 4,294,967,292 *Resolution:* 1 bit Unit-less number

ID of a group which is unique per Audio/Video source.

---

4 **Library Group Type 1**

Byte Field Size: *Bit Field Size:* 8

Request Parameter Required
Command Parameter: Optional

DD395 Library Data Type

0 = File
1 = Playlist Name
2 = Genre Name
3 = Album Name
4 = Artist Name
5 = Track Name
6 = Station Name
7 = Station Number
8 = Favorite Number
9 = Play Queue
10 - 252 = Reserved

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

---

5 **Library Data Name 1**

Byte Field Size: 8 or 16 n *Bit Field Size:*

Request Parameter Required
Command Parameter: Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character

2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

The name string in this field will be of the type in Field 4 – Library Data Type 1.

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6	**Library Group Type 2**		*Byte Field Size:*		*Request Parameter*	Required	
			Bit Field Size:	8	*Command Parameter:*	Optional	
	DD395 Library Data Type		0 = File 1 = Playlist Name 2 = Genre Name 3 = Album Name 4 = Artist Name 5 = Track Name 6 = Station Name 7 = Station Number 8 = Favorite Number 9 = Play Queue 10 - 252 = Reserved				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields
7	**Library Data Name 2**		*Byte Field Size:*	8 or 16 n	*Request Parameter*	Required	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD004 Generic name string, short		Name of place, route, waypoint, destination, vessel, vehicle, etc.				
	DF50 String, variable, short	**ch8or16(n)**	*Range:*	0 to 250 ASCII or 0 to 125 Unicode Characters	*Resolution:*	1 ASCII or 1 Unicode Character	2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.
	The name string in this field will be of the type in Field 6 – Library Data Type 2.						
8	**Library Group Type 3**		*Byte Field Size:*		*Request Parameter*	Required	
			Bit Field Size:	8	*Command Parameter:*	Optional	
	DD395 Library Data Type		0 = File 1 = Playlist Name 2 = Genre Name 3 = Album Name 4 = Artist Name 5 = Track Name 6 = Station Name 7 = Station Number 8 = Favorite Number 9 = Play Queue 10 - 252 = Reserved				
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields

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9	**Library Data Name 3**	*Byte Field Size:*	8 or 16 n	*Request Parameter*	Required	
		Bit Field Size:		*Command Parameter:*	Optional	

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50	String, variable, short	**ch8or16(n)**	*Range:*	0 to 250 ASCII or 0 to 125 Unicode Characters	*Resolution:* 1 ASCII or 1 Unicode Character	2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

The name string in this field will be of the type in Field 8 – Library Data Type 3.

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Entertainment - Supported Source Data

PGN: 130573
hex: 1FE0D

Details the Audio/Videos sources supported by an Audio/Video device.
Sent upon change or in response to a request.
Pre-defined sources can be left unnamed or customized with a name, e.g. “iOs” vs “Steve's iPhone”
Pre-defined sources allow display devices to present customized user interfaces.
User-defined sources allow forward compatibility.
An ISO Request (59904) shall, if this PGN is supported return the first 3 fields with respectively NA, 0 and 0.
If a Request Group Function (PGN 126208) requesting this PGN is received, the receiving device shall respond in the following manner :
•If no requested fields have been included with the Request Group Function then the response is to return the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied”.
•If the Request Group Function (PGN 126208) includes the Index of first Source ID in this PGN field, then the response shall be filtered by these fields contained within this request resulting in one response containing one or more supported source data.
If the fields requested are not valid then the appropriate response would be the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied”. If an individual field is invalid the field parameter error code for that field shall be “0x3 = Requested or command parameter out-of-range”.
A NMEA Command Group Function (PGN 126208) can be used by a client to change the name of a source. The command shall include the Index of first Source ID in this PGN, Source ID, Audio/Video Source Type, Audio/Video Source Number and Audio/Video Source Name fields.
Fields 4 and 5 are repeated as needed.
Further details can be found in the NMEA2000 Entertainment Application Notes in Appendix D.

Single Frame: N *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Required *ACK Rqmnts:* Yes

Field # *Field Name* Original Reference ID # 242

1	**Index of first Source ID in this PGN**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Required Optional
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned Generic numeric ID, short	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
2	**Number of Source IDs in this PGN**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned Generic numeric ID, short	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
3	**Total number of Source IDs available**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Optional
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned Generic numeric ID, short	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
4	**Source ID**		*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Required
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned Generic numeric ID, short	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	

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Entertainment - Supported Source Data

PGN: 130573
hex: 1FE0D

5	**Audio/Video Source Type**	*Byte Field Size:*		*Request Parameter*	Required
		Bit Field Size:	8	*Command Parameter:*	Required

DD389 Audio/Video Source Type

0 = Vessel Alarm
1 = AM
2 = FM
3 = Weather
4 = DAB
5 = Aux
6 = USB
7 = CD
8 = MP3
9 = Apple iOS
10 = Android
11 = Bluetooth
12 = Sirius XM
13 = Pandora
14 = Spotify
15 = Slacker
16 = Songza
17 = Apple Radio
18 = Last FM
19 = Ethernet
20 = Video MP4
21 = Video DVD
22 = Video BlueRay
23 = HDMI
24 = Video
25 - 252 = User defind
253 = Reserved
254 = Error
255 = Not available

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

6	**Audio/Video Source Number**	*Byte Field Size:*	1	*Request Parameter*	Required
		Bit Field Size:		*Command Parameter:*	Required

DD005 Generic numeric ID, short — Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number

Audio/Video Source Number per Type
There can be more than one of each type of audio source available on a A/V Device (e.g. 3 USB Sources).

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Entertainment - Supported Source Data

PGN: 130573
hex: 1FE0D

7 **Audio/Video Source Name**

Byte Field Size: 8 or 16 n
Bit Field Size:

Request Parameter Optional
Command Parameter: Optional

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50 String, variable, short **ch8or16(n)** *Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters *Resolution:* 1 ASCII or 1 Unicode Character

2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16).
Control byte = 0 => Unicode characters
Control byte = 1 => ASCII characters
A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

0 to 125 Unicode Characters

8 **Supported Play Status**

Byte Field Size:
Bit Field Size: 32

Request Parameter Optional
Command Parameter: Required

DD403 Supported Play Status

xxxx xxxx xxx xxxx xxxx xxxx xxxx xxx1 = Play (Normal)
xxxx xxxx xxx xxxx xxxx xxxx xxxx xx1x = Pause
xxxx xxxx xxx xxxx xxxx xxxx xxxx x1xx = Pause
xxxx xxxx xxxx xxx xxxx xxxx xxxx 1xxx = FF (1x)
xxxx xxxx xxxx xxx xxxx xxxx xxx1 xxxx = FF (2x)
xxxx xxxx xxxx xxx xxxx xxxx xx1x xxxx = FF (3x)
xxxx xxxx xxxx xxx xxxx xxxx x1xx xxxx = FF (4x)
xxxx xxxx xxxx xxx xxxx xxxx 1xxx xxxx = RW (1x)
xxxx xxxx xxxx xxx xxxx xxx1 xxxx xxxx = RW (2x)
xxx xxxx xxxx xxxx xxxx xx1x xxxx xxxx = RW (3x)
xxxx xxxx xxxx xxx xxxx x1xx xxxx xxxx = RW (4x)
xxxx xxxx xxx xxxx xxxx 1xxx xxxx xxxx = Skip Ahead
xxxx xxxx xxxx xxx xxx1 xxxx xxxx xxxx = Skip Back
xxxx xxxx xxx xxxx xx1x xxxx xxxx xxxx = Jog Ahead
xxxx xxxx xxx xxxx x1xx xxxx xxxx xxxx = Jog Back
xxxx xxxx xxx xxxx 1xxx xxxx xxxx xxxx = Seek Up
xxxx xxxx xxx xxx1 xxxx xxxx xxxx xxxx = Seek Down
xxxx xxxx xxxx x1x xxxx xxxx xxxx xxxx = Scan Up
xxxx xxxx xxx x1xx xxxx xxxx xxxx xxxx = Scan Down
xxxx xxxx xxx 1xxx xxxx xxxx xxxx xxxx = Tune Up
xxxx xxxx xxx1 xxx xxxx xxxx xxxx xxxx = Tune Down
xxxx xxxx xx1x xxx xxxx xxxx xxxx xxxx = Slow Mot. (.75x)
xxxx xxxx x1xx xxx xxxx xxxx xxxx xxxx = Slow Mot. (.5x)
xxxx xxxx 1xxx xxx xxxx xxxx xxxx xxxx = Slow Mot. (.25x)
xxxx xxx1 xxx xxxx xxxx xxxx xxxx xxxx = Slow Mot. (.125x)
xxxx xx1x xxxx xxx xxxx xxxx xxxx xxxx = Reserved
xxxx x1xx xxxx xxxx xxxx xxxx xxxx xxxx = Reserved
xxx 1xxx xxxx xxxx xxxx xxxx xxxx xxxx = Reserved
xxx1 xxxx xxxx xxx xxxx xxxx xxxx xxxx = Reserved
xx1x xxx xxxx xxxx xxxx xxxx xxxx xxxx = Reserved
x1xx xxxx xxxx xxx xxxx xxxx xxxx xxxx = Reservd
1xxx xxxx xxx xxxx xxxx xxxx xxxx xxxx = Reservd

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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9	**Supported Browsing Methods**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	16	*Command Parameter:*	Optional
	DD404 Supported Browsing Methods		xxxx xxxx xxxx xxx1 = Filename xxxx xxxx xxxx xx1x = Playlist xxxx xxxx xxxx x1xx = Genre xxxx xxxx xxxx 1xxx = Album xxxx xxxx xxx1 xxxx = Artist xxxx xxxx xx1x xxxx = Track xxxx xxxx x1xx xxx = Station Name xxxx xxxx 1xxx xxx = Station Number xxxx xxx1 xxxx xxx = Favorite Number xxxx xx1x xxxx xxx = Play Queue xxxx x1xx xxxx xxxx = Reserved xxxx 1xxx xxxx xxxx = Reserved xxx1 xxxx xxxx xxxx = Reserved xx1x xxxx xxxx xxxx = Reserved x1xx xxxx xxxx xxxx = Reserved 1xxx xxxx xxxx xxxx = Reserved		
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
10	**Thumbs Supported**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
11	**Source Connected**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
12	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*	
			Bit Field Size:	resv 4	*Command Parameter:*	
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.					
13	**Repeat Supported**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD405 Repeat Supported		xxx1 = Song xx1x = Play Queue x1xx = Reserved 1xxx = Reserved			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields

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Entertainment - Supported Source Data

PGN: 130573
hex: 1FE0D

14	**Shuffle Supported**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD409 Shuffle Supported		xxx1 = Play Queue xx1x = All x1xx = Reserved 1xxx = Reserved			
	DF52 Bit field	**bit(n)**	*Range:*	Variable	*Resolution:* 1	Used to construct bit fields
15	**Fields 4 – 12 repeat as needed**		*Byte Field Size:*	?	*Request Parameter*	
			Bit Field Size:		*Command Parameter:*	
	DD000 Undefined		Application specific, defined at time of use			
	DF00 Undefined	**Undefined**	*Range:*	undefined	*Resolution:* undefined	Application specific, defined at time of use.

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Entertainment - Supported Zone Data

PGN: 130574
hex: 1FE0E

Details the Audio/Video Zones supported by an Audio/Video device.
Sent upon change or in response to a request.
An ISO Request (59904) shall, if this PGN is supported return the first 3 fields with respectively NA, 0 and 0.
If a Request Group Function (PGN 126208) requesting this PGN is received, the receiving device shall respond in the following manner :
•If no requested fields have been included with the Request Group Function then the response is to return the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied”.
•If the Request Group Function (PGN 126208) includes the Index of first Zone Number in this PGN field, then the response shall be filtered by these fields contained within this request resulting in one response containing one or more supported zone data.
If the fields requested are not valid then the appropriate response would be the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied”. If an individual field is invalid the field parameter error code for that field shall be “0x3 = Requested or command parameter out-of-range”.
A NMEA Command Group Function (PGN 126208) can be used by a client to change the name of a zone. The command shall include the Index of first Zone Number in this PGN, Zone Number and Zone Name fields.
Fields 4 and 5 are repeated as needed.
Further details can be found in the NMEA2000 Entertainment Application Notes in Appendix D.

Single Frame: N *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Required *ACK Rqmnts:* Yes

Original Reference ID # 241

Field #	*Field Name*					
1	**Index of first Zone Number in this PGN**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Required *Command Parameter:* Required		
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned Generic numeric ID, short	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
2	**Number of Zone Numbers in this PGN**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Required		
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned Generic numeric ID, short	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
3	**Total number of Zone Numbers available**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Optional *Command Parameter:* Required		
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned Generic numeric ID, short	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
4	**Zone Number**	*Byte Field Size:* *Bit Field Size:* 8		*Request Parameter* Optional *Command Parameter:* Required		
	DD396 Entertainment Zones		0 = All Zones 1 = Zone 1 2 = Zone 2 3 = Zone 3 4 - 252 = User Defined			
	DF52 Bit field Generic numeric ID, short	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	

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Entertainment - Supported Zone Data

PGN: 130574
hex: 1FE0E

5	**Zone Name**	*Byte Field Size:*	8 or 16 n	*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Required	

DD004 Generic name string, short — Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50	String, variable, short	**ch8or16(n)**	*Range:* 0 to 250 ASCII or 0 to 125 Unicode Characters	*Resolution:* 1 ASCII or 1 Unicode Character	2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

0 to 125 Unicode Characters

6	**Fields 4 – 5 repeat as needed**	*Byte Field Size:*	?	*Request Parameter*	Optional	
		Bit Field Size:		*Command Parameter:*	Optional	

DD000 Undefined — Application specific, defined at time of use

DF00	Undefined	**Undefined**	*Range:* undefined	*Resolution:* undefined	Application specific, defined at time of use.

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Trim Tab Status

PGN: 130576
hex: 1FE10

Provides data on various small craft control surfaces and speed through the water. Used primarily by display or instrumentation devices. This PGN title was changed from "Small Craft Status" to "Trim Tab Status" as of Database Version 2.00.

Single Frame: Yes *Priority Default:* 2 *Default Update Rate:* 200 milliseconds *Frequency:* 5. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Field # *Field Name* Original Reference ID # 42

1 **Port trim tab** *Byte Field Size:* 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD138 Generic percent of range

DF30 Percent, Relative measure **int8** *Range:* +/- 124% *Resolution:* 1%
Range 0 - 100%, where 0% =Full Up and 100% = Full Down Positions

2 **Starboard trim tab** *Byte Field Size:* 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD138 Generic percent of range

DF30 Percent, Relative measure **int8** *Range:* +/- 124% *Resolution:* 1%
Range 0 - 100%, where 0% =Full Up and 100% = Full Down Positions

3 **NMEA Reserved** *Byte Field Size:* *Request Parameter*
Bit Field Size: resv 48 *Command Parameter:*

DD001 Reserved field Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
Used to align subsequent data on a byte boundary.

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Direction Data

PGN: 130577
hex: 1FE11

The purpose of this PGN is to group three fundamental vectors related to vessel motion,

- Speed and heading referenced to the water
- Speed and course referenced to ground
- Current speed and flow direction

Products that are the primary form of navigation would be appropriate for construction and transmission of this sentence, either by combining PGN's 128259, 129026 or through direct measurement.

The SID (Sequence Identification Number) should only be used to synchronize data when PGN 126992 (Time Stamp) is available, otherwise it should be set to unavailable.

This PGN should only be used when all three elements are available and by a primary navigation product.

Single Frame: No *Priority Default:* 3 *Default Update Rate:* 1000 milliseconds *Frequency:* 1. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 36

Field #	Field Name					
1	**Data Mode**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	4	*Command Parameter:*	Optional
	DD025 Mode, Data		0x0 = Autonomous mode, 0x1 = Differential, enhanced mode, 0x2 = Estimated mode, 0x3 = Simulator mode, 0x4 = Manual mode, 0x5 to 0xD = Reserved 0xE = Error, 0xF = Data not available			
	DF52 Bit field	**bit(n)**	*Range:* Variable		*Resolution:* 1	Used to construct bit fields
2	**Set/COG/Heading Ref.**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Optional
	DD117 Direction reference		0 = True, 1 = Magnetic, 2 = Error, 3 = Null			
	DF52 Bit field	**bit(n)**	*Range:* Variable		*Resolution:* 1	Used to construct bit fields
3	**NMEA Reserved**		*Byte Field Size:*		*Request Parameter*	
			Bit Field Size:	resv 2	*Command Parameter:*	
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field	**bit(n)**	*Range:* Variable		*Resolution:* 1	Used to construct bit fields
	Used to align subsequent data on a byte boundary.					
4	**Sequence ID**		*Byte Field Size:*	1	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD056 Sequence ID		An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.			
	DF53 Integer, 8 bit unsigned	**uint8**	*Range:* 0 to 252		*Resolution:* 1 bit	Unit-less number

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Direction Data

PGN: 130577
hex: 1FE11

5	**Course Over Ground**		*Byte Field Size:*	2	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD165 Course-Over-Ground (COG)		The direction of the path over ground actually followed by a vessel.			
	DF02 Angle	**uint16**	*Range:*	0 to 2Pi rad	*Resolution:*	1x10E-4 rad	Resolution ~0.0057deg, 1 deg = .01745 rad
6	**Speed Over Ground**		*Byte Field Size:*	2	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD044 Generic Speed					
	DF35 Speed	**uint16**	*Range:*	0 to 655.32 m/s	*Resolution:*	1x10E-2 m/s	1 Knot = 0.5144 m/s
7	**Heading**		*Byte Field Size:*	2	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD167 Heading		The horizontal direction in which a ship actually points or heads at any instant, expressed in angular units from a reference direction, usually from 000 at the reference direction clockwise through 359 degrees.			
	DF02 Angle	**uint16**	*Range:*	0 to 2Pi rad	*Resolution:*	1x10E-4 rad	Resolution ~0.0057deg, 1 deg = .01745 rad
8	**Speed through Water**		*Byte Field Size:*	2	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD044 Generic Speed					
	DF35 Speed	**uint16**	*Range:*	0 to 655.32 m/s	*Resolution:*	1x10E-2 m/s	1 Knot = 0.5144 m/s
9	**Set**		*Byte Field Size:*	2	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD048 Current flow direction		Direction towards which current flows. Degrees relative to True North.			
	DF02 Angle	**uint16**	*Range:*	0 to 2Pi rad	*Resolution:*	1x10E-4 rad	Resolution ~0.0057deg, 1 deg = .01745 rad
10	**Drift**		*Byte Field Size:*	2	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Optional
	DD044 Generic Speed					
	DF35 Speed	**uint16**	*Range:*	0 to 655.32 m/s	*Resolution:*	1x10E-2 m/s	1 Knot = 0.5144 m/s

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Vessel Speed Components

PGN: 130578
hex: 1FE12

This PGN provides a single transmission that accurately describes the speed of a vessel by component vectors.

This information is relevant for large vessels and would typically be provided by a product that interfaces to sensors such as dual axis logs. Products that can only measure speed in one direction should not use this PGN.

Single Frame: No *Priority Default:* 2 *Default Update Rate:* 250 milliseconds *Frequency:* 4. cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Optional *ACK Rqmnts:* None

Original Reference ID # 58

Field #	Field Name					
1	**Longitudinal Speed, Water-referenced**	*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD160 Generic speed, signed			Positive values represent ahead or starboard transverse speed and negative values represent astern or port transverse speed.		
	DF36 Speed, signed	**int16**	*Range:* +/-32.764 m/s	*Resolution:* 1x10E-3 m/s		
2	**Transverse Speed, Water-referenced**	*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD160 Generic speed, signed			Positive values represent ahead or starboard transverse speed and negative values represent astern or port transverse speed.		
	DF36 Speed, signed	**int16**	*Range:* +/-32.764 m/s	*Resolution:* 1x10E-3 m/s		
3	**Longitudinal Speed, Ground-referenced**	*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD160 Generic speed, signed			Positive values represent ahead or starboard transverse speed and negative values represent astern or port transverse speed.		
	DF36 Speed, signed	**int16**	*Range:* +/-32.764 m/s	*Resolution:* 1x10E-3 m/s		
4	**Transverse Speed, Ground-referenced**	*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD160 Generic speed, signed			Positive values represent ahead or starboard transverse speed and negative values represent astern or port transverse speed.		
	DF36 Speed, signed	**int16**	*Range:* +/-32.764 m/s	*Resolution:* 1x10E-3 m/s		
5	**Stern Speed, Water-referenced**	*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD160 Generic speed, signed			Positive values represent ahead or starboard transverse speed and negative values represent astern or port transverse speed.		
	DF36 Speed, signed	**int16**	*Range:* +/-32.764 m/s	*Resolution:* 1x10E-3 m/s		
6	**Stern Speed, Ground-referenced**	*Byte Field Size:* 2 *Bit Field Size:*			*Request Parameter* Optional *Command Parameter:* Optional	
	DD160 Generic speed, signed			Positive values represent ahead or starboard transverse speed and negative values represent astern or port transverse speed.		
	DF36 Speed, signed	**int16**	*Range:* +/-32.764 m/s	*Resolution:* 1x10E-3 m/s		

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Entertainment - System Configuration Status

PGN: 130580
hex: 1FE14

Details the configuration status for the entire Audio/Video device.
Sent at startup, upon change or is response to a request.
Fields output as Data Not Available will be understood to be unsupported.
A NMEA Command Group Function (PGN 126208) is used by a client to change the Audio/Video Configuration.
Further details can be found in the NMEA2000 Entertainment Application Notes in Appendix D.

Single Frame: N *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Optional *Command Support:* Required *ACK Rqmnts:* Yes

Field # *Field Name* Original Reference ID # 238

1 Power
Byte Field Size: *Request Parameter* Optional
Bit Field Size: 2 *Command Parameter:* Required

DD002 Generic status pair
MSB/LSB:
00 = [No, Off, Disabled, Reset, "0"],
01 = [Yes, On, Enabled, Set, "1"],
10 = Error,
11= [Unavailable, Unknown]

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

2 Default Settings
Byte Field Size: *Request Parameter* Optional
Bit Field Size: 2 *Command Parameter:* Required

DD398 Default Setting
0 = Save current settings as user default
1 = Load user default (soft reset)
2 = Load manufacturer default (hard reset)
3 = Data Not Available / Do Nothing

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

3 Tuner Regions
Byte Field Size: *Request Parameter* Optional
Bit Field Size: 4 *Command Parameter:* Optional

DD399 Tuner Regions
0 = US
1= Europe
2 = Asia
3 = Middle East
4 = Latin America
5 = Australia
6 = Russia
7 - 15 = Reserved

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

4 Supported Number of Favorites (read only)
Byte Field Size: 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional

DD005 Generic numeric ID, short
Number of route, waypoint, event, mark, etc.

DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number
Used by the A/V device to note the number of favorites it supports

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Entertainment - Zone Configuration Status

PGN: 130581
hex: 1FE15

Details the configuration status for an Audio/Video zone.
Sent at startup, upon change or in response to a request.
An Audio/Video zone is a room or area of the vessel in which sound will be distributed. Each zone can be separately controlled and could play a different source of audio or video.
Fields output as Data Not Available will be understood to be unsupported.
Fields can be commanded by a client using the NMEA Command Group Function (PGN 126208) to change the volume limit, fade, balance, Non-Fader Sub Volume, Equalizer Settings or mute a zone.
A device receiving an ISO Request (PGN 059904) for this PGN, shall respond by providing as many of these PGNs as necessary for every Audio/Video zone.
If a Request Group Function (PGN 126208) requesting this PGN is received, the receiving device shall respond in the following manner :
•If no requested fields have been included with the Request Group Function then the response is to return one or more PGNs, just like responding to the ISO Request (PGN 059904).
•If the Request Group Function (PGN 126208) includes the Zone Number then the response shall be filtered by these fields contained within this request resulting in one response.
If the fields requested are not valid then the appropriate response would be the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied”. If an individual field is invalid the field parameter error code for that field shall be “0x3 = Requested or command parameter out-of-range”.
Further details can be found in the NMEA2000 Entertainment Application Notes in Appendix D.

Single Frame: N *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Required *ACK Rqmnts:* Yes

Field # *Field Name* Original Reference ID # 240

1 Zone Number
Byte Field Size: *Bit Field Size:* 8
Request Parameter Required
Command Parameter: Required

DD396 Entertainment Zones

0 = All Zones
1 = Zone 1
2 = Zone 2
3 = Zone 3
4 - 252 = User Defined

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields
Audio Video Zone Number

2 Volume Limit
Byte Field Size: 1 *Bit Field Size:*
Request Parameter Optional
Command Parameter: Required

DD263 Generic Absolute Percentage 0-252%

DF93 Percent, Absolute **uint8** *Range:* 0 - 252% *Resolution:* 1%
Generic Absolute Percentage 0-252%

3 Fade
Byte Field Size: 1 *Bit Field Size:*
Request Parameter Optional
Command Parameter: Required

DD138 Generic percent of range

DF30 Percent, Relative measure **int8** *Range:* +/- 124% *Resolution:* 1%
Negative = Back
0 = Center
Positive = Front

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Entertainment - Zone Configuration Status

PGN: 130581
hex: 1FE15

4	**Balance**	*Byte Field Size:* 1		*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Required
	DD138 Generic percent of range					
	DF30 Percent, Relative measure **int8**	*Range:* +/- 124%	*Resolution:* 1%		
	Negative = Left 0 = Center Positive = Right				
5	**Non-Fader, Sub Volume**	*Byte Field Size:* 1		*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Required
	DD263 Generic Absolute Percentage 0-252%					
	DF93 Percent, Absolute **uint8**	*Range:* 0 - 252%	*Resolution:* 1%		
	Generic Absolute Percentage 0-252%				
6	**Equalizer – Treble**	*Byte Field Size:* 1		*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Required
	DD138 Generic percent of range					
	DF30 Percent, Relative measure **int8**	*Range:* +/- 124%	*Resolution:* 1%		
	Generic percent of range				
7	**Equalizer – Mid Range**	*Byte Field Size:* 1		*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Required
	DD138 Generic percent of range					
	DF30 Percent, Relative measure **int8**	*Range:* +/- 124%	*Resolution:* 1%		
	Generic percent of range				
8	**Equalizer – Bass**	*Byte Field Size:* 1		*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Required
	DD138 Generic percent of range					
	DF30 Percent, Relative measure **int8**	*Range:* +/- 124%	*Resolution:* 1%		
	Generic percent of range				
9	**EQ Preset Type**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size: 8		*Command Parameter:*	Required
	DD401 EQ Preset Type	0 = Flat 1 = Rock 2 = Hall 3 = Jazz 4 = Pop 5 = Live 6 = Classic 7 = Vocal 8 = Arena 9 = Cinema 10 - 252 = User Defined			
	DF52 Bit field **bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	

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Entertainment - Zone Configuration Status

PGN: 130581
hex: 1FE15

10	**Audio Filter Setting**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	8	*Command Parameter:*	Required

DD402 Audio Filter Setting

0 = Full Range
1 = High Pass
2 = Low Pass
3 = Band Pass
4 = Notch Filter
5 - 15 = Reserved

DF52	Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields

11	**High-Pass Filter Frequency (read only)**	*Byte Field Size:*	4	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Required

DD016 Radio Tx or Rx Frequency

DF21	Frequency	**uint32**	*Range:*	0 to ~4.295x10E+10 Hz	*Resolution:*	10 Hz

Also Notch Frequency

12	**Low-Pass Filter Frequency (read only)**	*Byte Field Size:*	4	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Required

DD016 Radio Tx or Rx Frequency

DF21	Frequency	**uint32**	*Range:*	0 to ~4.295x10E+10 Hz	*Resolution:*	10 Hz

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Entertainment - Zone Volume Status

PGN: 130582
hex: 1FE16

Details the volume status for an Audio/Video Zone.
An Audio/Video zone is a room or area of the vessel in which sound will be distributed. Each zone can be separately controlled and could play a different source of audio or video.
Fields can be commanded by a client using the NMEA Command Group Function (PGN 126208) to change the volume or mute a zone.
A device receiving an ISO Request (PGN 059904) for this PGN, shall respond by providing as many of these PGNs as necessary for every Audio/Video zone.
If a Request Group Function (PGN 126208) requesting this PGN is received, the receiving device shall respond in the following manner :
•If no requested fields have been included with the Request Group Function then the response is to return one or more PGNs, just like responding to the ISO Request (PGN 059904).
•If the Request Group Function (PGN 126208) includes the Zone Number then the response shall be filtered by these fields contained within this request resulting in one response.
If the fields requested are not valid then the appropriate response would be the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied”. If an individual field is invalid the field parameter error code for that field shall be “0x3 = Requested or command parameter out-of-range”.
Further details can be found in the NMEA2000 Entertainment Application Notes in Appendix D.

Single Frame: Yes *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Required *ACK Rqmnts:* Yes

Original Reference ID # 239

Field #	*Field Name*					
1	**Zone Number**		*Byte Field Size:*		*Request Parameter*	Required
			Bit Field Size:	8	*Command Parameter:*	Required
	DD396 Entertainment Zones		0 = All Zones 1 = Zone 1 2 = Zone 2 3 = Zone 3 4 - 252 = User Defined			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
	Audio Video Zone Number					
2	**Zone Volume – Absolute**		*Byte Field Size:*	1	*Request Parameter*	Optional
			Bit Field Size:		*Command Parameter:*	Required
	DD263 Generic Absolute Percentage 0-252%					
	DF93 Percent, Absolute	**uint8**	*Range:* 0 - 252%	*Resolution:* 1%		
	Generic Absolute Percentage 0-252%					
3	**Zone Volume – Relative (Write Only)**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Required
	DD400 Audio Volume – Relative		0 = Volume Up 1 = Volume Down 2 = Reserved 3 = Do Not Change/Data Not Available			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
4	**Mute**		*Byte Field Size:*		*Request Parameter*	Optional
			Bit Field Size:	2	*Command Parameter:*	Note 1
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	

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Entertainment - Zone Volume Status

PGN: 130582
hex: 1FE16

5	**NMEA Reserved**	*Byte Field Size:*		*Request Parameter*	
		Bit Field Size:	resv 4	*Command Parameter:*	

DD001 Reserved field — Variable number of reserved bits, all set to logic "1"

DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

Used to align subsequent data on byte boundary

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Entertainment - Available Audio EQ Presets

PGN: 130583
hex: 1FE17

Details the Audio EQ Presets available on an Audio/Video device.
Sent upon change or in response to a request.
Pre-defined presets can be left unamed or customized with a name, e.g. “Stereo XYZ Rock”
Pre-defined presets allow display devices to present customized user interfaces.
User-defined sources allow forward compatibility.
An ISO Request (59904) shall, if this PGN is supported return the first 3 fields with respectively NA, 0 and 0.
If a Request Group Function (PGN 126208) requesting this PGN is received, the receiving device shall respond in the following manner :
•If no requested fields have been included with the Request Group Function then the response is to return the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied”.
•If the Request Group Function (PGN 126208) includes the Index of first EQ Preset Type in this PGN field, then the response shall be filtered by these fields contained within this request resulting in one response containing one or more EQ Presets.
If the fields requested are not valid then the appropriate response would be the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied”. If an individual field is invalid the field parameter error code for that field shall be “0x3 = Requested or command parameter out-of-range”.
A NMEA Command Group Function (PGN 126208) can be used by a client to change the name of an EQ Preset. The command shall include the Index of first EQ Preset in this PGN, EQ Preset Type and EQ Preset Name fields.
Fields 4 and 5 are repeated as needed.
Further details can be found in the NMEA2000 Entertainment Application Notes in Appendix D.

Single Frame: N *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Required *ACK Rqmnts:* Yes

Field # *Field Name* Original Reference ID # 243

1	**Index of first EQ Preset Type in this PGN**	*Byte Field Size:* 1 *Bit Field Size:*	*Request Parameter*	Required
			Command Parameter:	Optional
	DD005 Generic numeric ID, short	Number of route, waypoint, event, mark, etc.		
	DF53 Integer, 8 bit unsigned **uint8** Generic numeric ID, short	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number
2	**Number of EQ Preset Types in this PGN**	*Byte Field Size:* 1 *Bit Field Size:*	*Request Parameter*	Optional
			Command Parameter:	Optional
	DD005 Generic numeric ID, short	Number of route, waypoint, event, mark, etc.		
	DF53 Integer, 8 bit unsigned **uint8** Generic numeric ID, short	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number
3	**Total number of EQ Preset Types available**	*Byte Field Size:* 1 *Bit Field Size:*	*Request Parameter*	Optional
			Command Parameter:	Optional
	DD005 Generic numeric ID, short	Number of route, waypoint, event, mark, etc.		
	DF53 Integer, 8 bit unsigned **uint8** Generic numeric ID, short	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number

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Entertainment - Available Audio EQ Presets

PGN: 130583
hex: 1FE17

4	**EQ Preset Type**	*Byte Field Size:*		*Request Parameter*	Optional
		Bit Field Size:	8	*Command Parameter:*	Optional

DD401 EQ Preset Type

0 = Flat
1 = Rock
2 = Hall
3 = Jazz
4 = Pop
5 = Live
6 = Classic
7 = Vocal
8 = Arena
9 = Cinema
10 - 252 = User Defined

DF52	Bit field	**bit(n)**	*Range:*	Variable	*Resolution:*	1	Used to construct bit fields

5	**EQ Preset Name**	*Byte Field Size:*	8 or 16 n	*Request Parameter*	Optional
		Bit Field Size:		*Command Parameter:*	Optional

DD004 Generic name string, short

Name of place, route, waypoint, destination, vessel, vehicle, etc.

DF50	String, variable, short	**ch8or16(n)**	*Range:*	0 to 250 ASCII or 0 to 125 Unicode Characters	*Resolution:*	1 ASCII or 1 Unicode Character	2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.

0 to 125 Unicode Characters

6	**Fields 4 – 5 repeat as needed**	*Byte Field Size:*	?	*Request Parameter*	
		Bit Field Size:		*Command Parameter:*	

DD000 Undefined

Application specific, defined at time of use

DF00	Undefined	**Undefined**	*Range:*	undefined	*Resolution:*	undefined	Application specific, defined at time of use.

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Entertainment - Bluetooth Devices

PGN: 130584
hex: 1FE18

Details the Bluetooth Devices known by an Audio/Video device.
Sent in response to a request.
An ISO Request (59904) shall, if this PGN is supported return the first 3 fields with respectively NA, 0 and 0.
If a Request Group Function (PGN 126208) requesting this PGN is received, the receiving device shall respond in the following manner :
•If no requested fields have been included with the Request Group Function then the response is to return the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied”.
•If the Request Group Function (PGN 126208) includes the Index of first Bluetooth Address in the PGN field then the response shall be filtered by these fields contained within this request resulting in one response containing one or more Bluetooth device details.
If the fields requested are not valid then the appropriate response would be the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied”. If an individual field is invalid the field parameter error code for that field shall be “0x3 = Requested or command parameter out-of-range”.
A NMEA Command Group Function (PGN 126208) can be used by a client to forget a Bluetooth device. The command shall include the Bluetooth Source Number and Forget Bluetooth Device fields.
Further details can be found in the NMEA2000 Entertainment Application Notes in Appendix D.

Single Frame: N *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Required *ACK Rqmnts:* Yes

Field # *Field Name* Original Reference ID # 245

1 **Index of first Bluetooth Address in this PGN**
Byte Field Size: 1 *Request Parameter* Required
Bit Field Size: *Command Parameter:* Required
DD005 Generic numeric ID, short Number of route, waypoint, event, mark, etc.
DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number
Generic numeric ID, short

2 **Number of Bluetooth Addresses in this PGN**
Byte Field Size: 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Required
DD005 Generic numeric ID, short Number of route, waypoint, event, mark, etc.
DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number
Generic numeric ID, short

3 **Number of Bluetooth Addresses available**
Byte Field Size: 1 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Optional
DD005 Generic numeric ID, short Number of route, waypoint, event, mark, etc.
DF53 Integer, 8 bit unsigned **uint8** *Range:* 0 to 252 *Resolution:* 1 bit Unit-less number
Generic numeric ID, short

4 **Bluetooth Device Address**
Byte Field Size: 6 *Request Parameter* Optional
Bit Field Size: *Command Parameter:* Required
DD408 Bluetooth Device Address Integer 48bit unsigned, Unit-less number
DF117 Integer 48 bit unsigned **uint48** *Range:* 0 to 281474976710656 *Resolution:* 1 Unit-less number

5 **Bluetooth Device Status**
Byte Field Size: *Request Parameter* Required
Bit Field Size: 8 *Command Parameter:* Required
DD407 Bluetooth Device Status
0 = Connected
1 = Not Connected (In Memory)
2 = Not Paired (Available)
3 - 14 = Reserved
15 = Data Not Available
DF52 Bit field **bit(n)** *Range:* Variable *Resolution:* 1 Used to construct bit fields

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Entertainment - Bluetooth Devices

PGN: 130584
hex: 1FE18

6	**Bluetooth Device Name**		*Byte Field Size:*	8 or 16 n	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD004 Generic name string, short			Name of place, route, waypoint, destination, vessel, vehicle, etc.			
	DF50 String, variable, short	**ch8or16(n)**	*Range:*	0 to 250 ASCII or 0 to 125 Unicode Characters	*Resolution:*	1 ASCII or 1 Unicode Character	2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.
	0 to 125 Unicode Characters						
7	**Bluetooth Signal Strength**		*Byte Field Size:*	1	*Request Parameter*	Optional	
			Bit Field Size:		*Command Parameter:*	Optional	
	DD263 Generic Absolute Percentage 0-252%						
	DF93 Percent, Absolute	**uint8**	*Range:*	0 - 252%	*Resolution:*	1%	
	Generic Absolute Percentage 0-252%						
8	**Fields 4 – 7 repeat as needed**		*Byte Field Size:*	?	*Request Parameter*		
			Bit Field Size:		*Command Parameter:*		
	DD000 Undefined			Application specific, defined at time of use			
	DF00 Undefined	**Undefined**	*Range:*	undefined	*Resolution:*	undefined	Application specific, defined at time of use.

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Entertainment - Bluetooth Source Status

PGN: 130585
hex: 1FE19

Details the status of a Bluetooth Audio/Video source.
Sent upon change or in response to a request.
An ISO Request (59904) shall, if this PGN is supported return the first field set to NA.
If a Request Group Function (PGN 126208) requesting this PGN is received, the receiving device shall respond in the following manner :
•If no requested fields have been included with the Request Group Function then the response is to return the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied”.
•If the Request Group Function (PGN 126208) includes the Bluetooth Source Number field or Bluetooth Pairing Status field then the response shall be filtered by these fields contained within this request resulting in one response.
If the fields requested are not valid then the appropriate response would be the Acknowledge Group Function (PGN 126208), containing the error state for PGN error code (Field 3) of “0x3 = Access denied”. If an individual field is invalid the field parameter error code for that field shall be “0x3 = Requested or command parameter out-of-range”.
A NMEA Command Group Function (PGN 126208) can be used by a client to forget a Bluetooth device. The command shall include the Bluetooth Source Number and Forget Bluetooth Device fields.
Further details can be found in the NMEA2000 Entertainment Application Notes in Appendix D.

Single Frame: Yes *Priority Default:* 6 *Default Update Rate:* milliseconds *Frequency:* NA cycles per second
Destination: Global *Query Support:* Required *Command Support:* Required *ACK Rqmnts:* Yes

Original Reference ID # 244

Field #	*Field Name*					
1	**Bluetooth Source Number**	*Byte Field Size:* 1 *Bit Field Size:*		*Request Parameter* Required *Command Parameter:* Required		
	DD005 Generic numeric ID, short		Number of route, waypoint, event, mark, etc.			
	DF53 Integer, 8 bit unsigned Generic numeric ID, short	**uint8**	*Range:* 0 to 252	*Resolution:* 1 bit	Unit-less number	
2	**Bluetooth Pairing Status**	*Byte Field Size:* *Bit Field Size:* 4		*Request Parameter* Required *Command Parameter:* Required		
	DD406 Bluetooth Pairing Status		0 = Discovering 1 = Connect 2 = Connecting (read only) 3 = Not Connected / Disconnect 4 - 15 = Reserved			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
3	**Forget Bluetooth Device**	*Byte Field Size:* *Bit Field Size:* 2		*Request Parameter* Optional *Command Parameter:* Required		
	DD002 Generic status pair		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]			
	DF52 Bit field	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	
4	**NMEA Reserved**	*Byte Field Size:* *Bit Field Size:* resv 2		*Request Parameter* *Command Parameter:*		
	DD001 Reserved field		Variable number of reserved bits, all set to logic "1"			
	DF52 Bit field Used to align subsequent data on byte boundary.	**bit(n)**	*Range:* Variable	*Resolution:* 1	Used to construct bit fields	

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Entertainment - Bluetooth Source Status

PGN: 130585
hex: 1FE19

5	**Bluetooth Device Address**	*Byte Field Size:* 6 *Bit Field Size:*		*Request Parameter* *Command Parameter:*	Optional Required

DD408 Bluetooth Device Address — Integer 48bit unsigned, Unit-less number

DF117 Integer 48 bit unsigned — **uint48** — *Range:* 0 to 281474976710656 — *Resolution:* 1 Unit-less number

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National Marine Electronics Association



NMEA 2000 ®

Appendix B.2 -- Data Dictionary

STANDARD FOR SERIAL-DATA NETWORKING OF MARINE ELECTRONIC DEVICES

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Version 2.100
February 2015

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD000	Undefined	DF00			Application specific, defined at time of use
DD001	Reserved field	DF52	resv		Variable number of reserved bits, all set to logic "1"
DD002	Generic status pair	DF52	2		MSB/LSB: 00 = [No, Off, Disabled, Reset, "0"], 01 = [Yes, On, Enabled, Set, "1"], 10 = Error, 11= [Unavailable, Unknown]
DD003	Generic command pair	DF52	2		00 = [Turn Off, Disable, Reset, Make "0"], 01 = [Turn On, Enable, Set, Make "1"], 02 = Reserved, 03 = No action
DD004	Generic name string, short	DF50		n	Name of place, route, waypoint, destination, vessel, vehicle, etc.
DD005	Generic numeric ID, short	DF53			Number of route, waypoint, event, mark, etc.
DD006	Generic counter, short	DF53			Numeric count, event counter, sequence counter
DD007	Generic numeric ID, medium	DF54			Number of route, waypoint, event, mark, etc.
DD008	Generic counter, medium	DF54			Numeric count, event counter, sequence counter
DD009	PGN	DF52	24		24-bit Parameter Group Number (PGN) expressed in binary, LSB is transmitted first
DD010	Generic numeric ID, large	DF55			Number of route, waypoint, event, mark, etc.
DD011	DSC Symbol Definitions	DF53			Integer numbers within the range 000 to 127 representing DSC Symbols defined by ITU-R M.493 Table 3 for: Phasing and Unique Functions; Format Specifier; Category; Nature of Distress; First Telecommand; Second Telecommand
DD012	DSC Address or Geographic Area	DF63		5	Individual characters having only decimal values in the range 0 to 127 for the DSC symbols defined by ITU-R M.493 are used to code the address or geographic area as defined in ITU-R M.493-9 Section 5. This may represent an individual MMSI, a group MMSI, or a geographic area.
DD013	DSC Position, Channel, or Frequency	DF63		6	Individual characters having only decimal values in the range 0 to 127 for the DSC symbols defined by ITU-R M.493 are used to code the DSC channel or frequency defined in ITU-R M.493 Table 13.
DD014	DSC Expansion Symbol Definitions	DF53			Integer numbers within the range 000 to 127 representing DSC Symbols defined by ITU-R M.821 Table 1.

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD015	DSC Symbol String	DF50		n	Individual characters having only decimal values in the range 0 to 127 for the DSC symbols defined by ITU-R M.493 are used to code: Telephone number as defined by ITU-R M.493 Section 8.2.3; DSC Expansion Data as defined by ITU-R M.821 Section 2.
DD016	Radio Tx or Rx Frequency	DF21			
DD017	Radio Tx or Rx Channel	DF63		6	MF/HF telephone channels to have first digit 3 followed by ITU channel numbers with leading zeros as required. MF/HF teletype channels to have first digit 4; the second and third digit give the frequency bands; and the fourth to sixth digits ITU channel numbers; each with leading zeros as required. VHF channels to have he first digit 9 followed by zero. The next number is “1” indicating the ship station’s transmit frequency is being used as a simplex channel frequency, or “2” indicating the coast station’s transmit frequency is being used as a simplex channel frequency, “0” otherwise. The remaining three numbers are the VHF channel numbers with leading zeros as required.
DD018	Radio Tx Power	DF28			
DD019	Mode, Radiotelephone	DF52	8		Radiotelephone mode settings: 0 = F3E/G3E simplex, telephone; 1 = F3E/G3E duplex, telephone; 2 = J3E, telephone; 3 = H3E, telephone; 4 = F1B/J2B FEC NBDP, telex/teleprinter; 5 = F1B/J2B ARQ NBDP, telex/teleprinter; 6 = F1B/J2B receive only, teleprinter/DSC; 7 = F1B/J2B, teleprinter/DSC; 8 = A1A Morse, tape recorder; 9 = A1A Morse, Morse key/head set; 10 =F1C/F2C/F3C, FAX-machine; 11-253 = reserved; 254 = error; 255 = unavailable/do not change
DD020	Radio Channel Bandwidth	DF26			
DD021	NMEA 0183 ASCII string	DF50		n	ASCII characters of the number and type specified in NMEA 0183 Standard for Interfacing Marine Electronic Devices
DD022	Latitude, WGS-84	DF23			Latitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10’s of minute.
DD023	Longitude, WGS-84	DF25			Longitude referenced to WGS-84. The resolution of the latitude and longitude fields shall be fixed at 1 decimal place of minutes (1/10 of a minute). If a higher resolution is provided to an AIS unit, the receiving AIS unit shall truncate to 1/10’s of minute.
DD024	Altitude, WGS-84	DF15			Altitude referenced to WGS-84

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD025	Mode, Data	DF52	4		0x0 = Autonomous mode, 0x1 = Differential, enhanced mode, 0x2 = Estimated mode, 0x3 = Simulator mode, 0x4 = Manual mode, 0x5 to 0xD = Reserved 0xE = Error, 0xF = Data not available
DD026	Loran-C SNR	DF31			Signal at standard sampling point / RMS noise in 3kHz BW at 100KHz
DD027	Loran-C GRI	DF45			Group Repetition Interval (GRI) in nano-sec. Often cited in units of 10 micro-sec. (i.e., 9960 = 99,600,000 ns)
DD028	Loran-C TD	DF45			Loran-C Time difference (TD) in nano-sec. The arrival time of a Loran-C secondary station signal minus the arrival time of the master station signal.
DD029	Loran - Range (Time)	DF45			The actual propagation time of a Loran-C signal from the station to a receiver in nano-sec.
DD030	Loran-C station status	DF52	4		MSB: to LSB: xxx1 = Station in use, xx1x = Low SNR, x1xx = Cycle error, 1xxx = Blink where x = don't care
DD031	Loran-C Station ID	DF63		1	1-character code for the Loran-C station: M = master, V, W, X, Y, Z
DD032	Loran-C ECD	DF45			Envelope-to-Cycle Discrepancy (ECD) of the Loran-C pulse
DD033	Loran-C ASF	DF45			Additional Secondary Factor (ASF) associated with the propagation of the signal from a Loran-C station
DD034	Time-elapsed/Time-to-go	DF40			Time interval in milli-sec. "-" = time elapsed since event, "+" = time to go before event.
DD035	Data transmit interval	DF65			Time interval between data transmissions Where: 0x0000 0000 = Turn-off transmission, 0xFFFF FFFE = Restore Default Interval 0xFFFF FFFF = Do not change interval
DD036	Data transmit offset	DF66			Offset in transmit time from time of request command: 0x0 = transmit immediately 0xFFFF = Do not change offset.
DD037	Error codes, Acknowledgement	DF52	4		0x0 = No Error/Acknowledge, 0x1 = PGN not supported, 0x2 = PGN temporarily not available, 0x3 = Access denied, 0x4 = Request or Command is not supported, 0x5 = Definer Tag is not supported, 0x6 = Read or Write is not supported

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD038	Tide direction	DF52	2		msb/lsb: 00 = Falling, 01 = Rising, 10 = Error, 11 = Unavailable, Unknown
DD039	Generic date	DF41			Days since January 1, 1970, Date is relative to UTC Time.
DD040	Standard Deviation for tide level data	DF13			The following text is from NOAA and is placed here to convey an understanding of the expected magnitude of the values. "The tide gage processes 181 water level samples, 1 second apart, centered on the 6 minute mark. It then computes the standard deviation of the Samples. Samples more than 3 sigma's away from the average are called "outliers". The "outliers" are removed from the samples and the standard deviation is recomputed. The number of outliers indicates how many samples were discarded. The standard deviation is a measure of noise in the water level environment. In a sheltered location, the typical standard deviation can be as low as 0.001 to 0.010 meters. More open locations could be higher, such as 0.150 meters. Every location is different.
DD041	Tide Level	DF46			This value is relative to mean lower low water (MLLW).
DD042	Salinity measure	DF49			The average Salinity of ocean water is about 35 grams of salts per kilogram of sea water (g/kg), usually written as 35 ppt which is read as 35 parts per thousand.
DD043	Generic Temperature	DF39			
DD044	Generic Speed	DF35			
DD045	Wind Direction	DF02			
DD046	Water Current State	DF52	3		000 = Flood, 001 = Slack, 010 = Ebb, 011 = Reserved, 100 = Reserved, 101 = Reserved, 110 = Error, 111 = Unavailable, Unknown
DD047	Water Depth	DF09			Water depth measured from the water surface
DD048	Current flow direction	DF02			Direction towards which current flows. Degrees relative to True North.
DD049	Generic Pressure	DF47			

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD050	Wave Height	DF13			Wave height is calculated as the highest one-third of all of the wave heights during a 20-minute sampling period. Note: Accelerometers or inclinometers on board the buoys measure the heave acceleration or the vertical displacement of the buoy hull during the wave acquisition time. A Fast Fourier Transform (FFT) is applied to the data by the processor on board the buoy to transform the data from the temporal domain into the frequency domain. Note that the raw acceleration or displacement measurements are not transmitted shore-side. Response amplitude operator (RAO) processing is then performed on the transformed data to account for both hull and electronic noise. It is from this transformation that non-directional spectral wave measurements (i.e., wave energies with their associated frequencies) are derived. Along with the spectral energies, measurements such as significant wave height (WVHGT), average wave period (AVGPD), and dominant period (DOMPD) are also derived from the transformation.
DD051	Wave Period in seconds	DF66			Average wave period of all waves during a 20-minute period. Dominant wave period is the period with the maximum wave energy. Note: See DD50 note.
DD052	Pressure Rate	DF48			Positive value indicates Rising, Negative value indicates Falling.
DD053	Wind gusts	DF35			Peak wind gust speed. Sustained wind over an interval of 5 seconds.
DD054	Elevation	DF04			Angle above or below the horizon. -90° to +90°; negative below the horizon
DD055	DOP	DF69			Dilution of Precision (DOP) indicates the contribution of satellite configuration geometry to positioning error. A lower DOP value is preferred because less error is being introduced. Reported as components: HDOP (Horizontal), VDOP (Vertical), TDOP (Time). Minimum DOP value is 1.0 (no error introduced).
DD056	Sequence ID	DF53			An upward counting number used to tie related information together between different PGNs . For example, the SID would be used to tie together the COG, SOG and RAIM values to a given position. 255=no valid position fix to tie it to. Range 0 to 252 for valid position fixes.
DD057	SNR Value	DF31			SNR expressed in C/No
DD058	Mode, GNSS	DF52	3		0 = 1D, 1 = 2D, 2 = 3D, 3 = Auto, 4-5 = Reserved, 6 = Error, 7 = Null

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD059	Mode, DGNSS	DF52	3		0 = Off, 1 = Auto, 2 = RTCM SC104 Pseudorange Corrections, 3 = RTCA SC159 Network Corrections, 4 = RTK Differential Corrections, 5 = Reserved, 6 = Error; 7 = Null
DD060	Differential Age	DF66			Age of Differential corrections
DD061	Roll	DF04			Oscillation of ship about it's longitudinal axis. Roll to the starboard is positive.
DD062	Pitch	DF04			Oscillation of ship about it's latitudinal axis. Bow moving up is positive.
DD063	Yaw	DF04			Oscillation of ship about it's vertical axis. Bow moving to starboard is positive.
DD064	Latitude, Local	DF23			Datum is included in package or via Datum Packet
DD065	Longitude, Local	DF25			
DD066	Time Source	DF52	4		0x0 = GPS, 0x1 =GLONASS, 0x2 = WWV or equivalent Radio Station Time Sync, 0x3 = Local Cesium clock, 0x4 = Local Rubidium clock, 0x5 = Local Crystal clock, 0x6 - 0xE = reserved, 0xF = Null
DD067	Quality, GNSS	DF52	4		0 = no GPS, 1 = GNSS fix, 2 = DGNSS fix, 3 = Precise GNSS*, 4 = RTK Fixed Integer, 5 = RTK Float, 6 = Estimated (DR) mode, 7 = Manual Input, 8 = Simulate mode, 9-13 = Reserved, 14 = Error, 15 = Null. *Precise GNSS means no deliberate degradation (such as SA) and higher resolution code (P-code), and 2 frequencies are used to correct atmospheric delays.
DD068	Datum	DF63		4	4-character code for the datum currently being output for the position solution. The datum's are defined in the IHO Publication S-60, Appendices B and C. The first three characters are the datum ID as per the IHO tables. The fourth character is the local datum subdivision code. A null character indicates the datum or subdivision code is unknown or not

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD069	Geoidal Separation	DF15			The difference between the earth ellipsoid and mean-sea-level (geoid) defined by the reference datum used in the position solution, "-" = mean-sea-level below ellipsoid. Reference datum is defined in this packet.
DD070	Ref Station Type	DF52	4		Reference Station Type. 0x0=GPS; 0x1=GLONASS; 0x2 to 0xD=Reserved; 0XE=Error; 0XF=Null
DD071	Ref Station	DF52	12		Reference Station ID. Reference Station number as provided by the Service Provider.[Reference document required]
DD072	Range Residual Mode	DF52	2		Range Residual used in position calculation or range residuals were calculated after the position. 0=range residuals were used to calculate data; 1=range residuals were calculated after the position., 2=Error, 3=Null
DD073	Range Residuals	DF79			Range Residual value in meters
DD074	Satellite ID Number	DF53			0 = value not used, 1-32 = GPS, 33-64 = SBAS, Satellite, Based Augmentation System (ie WAAS) 65-96 = GLONASS. For GLONASS, satellites are identified by 64+satellite slot number. The slot numbers are 1 through 24 for the full GLONASS constellation of 24 satellites, this gives a range of 65 through 88. The numbers 89 through 96 are available if slot numbers above 24 are allocated to on-orbit spares.
DD075	Error Distances	DF13			Error distances expressed in meters.
DD076	Receiver channel number	DF53			The channel number of the correction receiver. If the receiver only has one beacon input channel, this value shall be 1; 0 is undefined.
DD077	Differential Correction Receiver frequency	DF21			This is the input frequency of the correction receiver.

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD078	Differential Correction Broadcast Bit Rate	DF52	5		This is the bit rate of the correction receiver. 0 = 25bps, 1 = 50bps, 2 = 100bps, 3 = 200bps, 4 = 300bps, 5 = 500bps, 6 = 1200bps, 7 = 2400bps, 8 = 4800bps, 9 = 9600bps 10 = 19200bps, 11 = 38400bps, 12 = 57600bps, 13-29 = Reserved, 30 = Error, 31 = Null
DD079	Mode, Bit Rate	DF52	3		This is the mode of operation for the correction receiver. 0 = Auto bit rate set, 1 = Manual bit rate set, 2-5 = Reserved 6 = Error, 7 = Null.
DD080	Correction Receiver Signal Strength	DF16			This is the signal strength expressed in dB with respect to 1uV/m.
DD081	SNR Value	DF31			SNR expressed in dB.
DD082	GPS Week Number	DF54			GPS week number. Starting on 6 Jan., 1980. The GPS week number roll over will not affect this value, i.e., it will continue to count up 1022, 1023, 1024, 1025.
DD083	SV Health Bits	DF52	8		SV health, bits 17-24 of each almanac page. Reference ICD-GPS-200 paragraph 20.3.3.5.1.3, Table 20-VII and Table 20-VIII
DD084	Eccentricity, e	DF52	16		Eccentricity, e. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.
DD085	Almanac Reference Time, toa	DF52	8		Almanac reference time. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.
DD086	Almanac parameter, (sigma)I	DF52	16		Inclination angle. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.
DD087	Almanac parameter, OMEGADOT	DF52	16		Rate of right ascension, OMEGADOT. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.
DD088	Almanac parameter, (A)1/2	DF52	24		Root of semi-major axis. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.
DD089	Almanac parameter, (omega)	DF52	24		Argument of Perigee. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD090	Almanac parameter, (omega)0	DF52	24		Longitude of ascension node. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.
DD091	Almanac parameter, M0	DF52	24		Mean anomaly. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.
DD092	Almanac parameter, af0	DF52	11		Clock Parameter 1. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.
DD093	Almanac parameter, af1	DF52	11		Clock Parameter 2. Reference ICD-GPS-200 Table 20-VI for scaling factors and units.
DD094	Almanac parameter, NA	DF54			Calendar day count within the four year period beginning with the previous leap year
DD095	Almanac parameter, CnA	DF52	1		Generalized health of the Satellite, reference GLONASS ICD.
DD096	Almanac parameter, HnA	DF52	5		Carrier frequency number respectively, reference GLONASS ICD.
DD097	Almanac parameter, (epsilon)nA	DF52	16		Eccentricity, reference GLONASS ICD Section 4.5 Table 4.3 (fill unused bits with zeros).
DD098	Almanac parameter, (deltaTnA)DOT	DF52	8		Rate of change of the draconitic circling time, reference GLONASS ICD Section 4.5 Table 4.3 (fill unused bits with zeros).
DD099	Almanac parameter, (omega)nA	DF52	16		Argument of Perigee, reference GLONASS ICD Section 4.5 Table 4.3 (fill unused bits with zeros).
DD100	Almanac parameter, (delta)TnA	DF52	24		Correction to the average value of the draconitic circling time, reference GLONASS ICD Section 4.5 Table 4.3 (fill unused bits with zeros).
DD101	Almanac parameter, tnA	DF52	24		Time of the ascension node, almanac reference time, reference GLONASS ICD Section 4.5 Table 4.3 (fill unused bits with zeros).
DD102	Almanac parameter, (lambda)nA	DF52	24		Greenwich longitude of the ascension node, reference GLONASS ICD Section 4.5 Table 4.3 (fill unused bits with zeros).
DD103	Almanac parameter, (delta)inA	DF52	24		Correction to the average value of the inclination angle, reference GLONASS ICD Section 4.5 Table 4.3 (fill unused bits with zeros).
DD104	Almanac parameter, (tau)cA	DF52	28		System time scale correction, reference GLONASS ICD Section 4.5 Table 4.3 (fill unused bits with zeros).
DD105	Almanac parameter, (tau)nA	DF52	12		Course value of the time scale shift, reference GLONASS ICD Section 4.5 Table 4.3 (fill unused bits with zeros).

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD106	Latitude, Offset	DF23			Offsets change with position; position in the local datum is offset from the position in the reference datum in the directions indicated: Plocal datum = Pref datum + offset.
DD107	Longitude, Offset	DF25			
DD108	Axis Delta shift	DF15			Delta Shift in X, Y, or Z axis from WGS 84.
DD109	Axis Rotational shift	DF70			Rotational shift in X, Y, or Z axis from WGS 84. Rotations presented use the geodetic sign convention. When looking along the positive axis towards the origin, counter-clockwise rotations are positive.
DD110	Scale	DF49			Scale factor expressed in parts-per-million
DD111	Ellipsoid Semi-major Axis	DF15			Semi-major axis (a) of the User Datum ellipsoid.
DD112	Ellipsoid Flattening Inverse	DF49			Flattening (1/f) of the user Datum ellipsoid.
DD113	Course	DF02			The horizontal direction in which a vessel is steered or intended to be steered, expressed as angular distance 000 north, clockwise through 359 degrees. Not to be confused with Course-over-ground, Track, or Heading.
DD114	XTE	DF15			Cross-track-error of a route. "-" = Left of track, need to steer right
DD115	Distance	DF15			
DD117	Direction reference	DF52	2		0 = True, 1 = Magnetic, 2 = Error, 3 = Null
DD118	Heading Sensor Reading	DF02			Primary output of heading as indicated by the heading sensor.
DD119	Calculation Type	DF52	2		0 = Great Circle calculations, 1 = Rhumb Line calculations, 2 = Error, 3 = Null
DD120	Distance, voyage	DF11			
DD121	Time, Local Offset	DF71			Local offset from UTC to obtain Local Time. This value includes Time Zone, daylight Savings Time, etc.
DD122	Mark Type	DF52	4		0 = Collision, 1 = Turning Point, 2 = Reference (general), 3 = Wheelover, 4 = Waypoint, 5-13 = Reserved, 14 = Error, 15 = Null

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD123	To / From Indicator	DF52	3		0 = time from mark, 2 = time to go to mark, 3 = estimated time of arrival (ETA) at mark, 4-5=Reserved, 6=Error, 7=Null
DD124	PRN Usage Status	DF52	4		0 = Not Tracked, 1 = Tracked but not used in solution, 2 = Used in solution without Differential corrections, 3 = Differential Corrections available, 4 = Tracked with Differential Corrections, 5 = used with Differential Corrections, 6-13 =Reserved, 14 = Error, 15 = No Selection
DD125	Differential Source	DF52	4		0 = Auto Select, 1 = Loran Communications 2 = MSK Beacon, 3 = FM Subcarrier, 4 = AIS (Automatic Identification System), 5 = Other Ground-based Radio, 6 = SBAS (Satellite Based Augmentation System, 7 = Other Satellite, 8-13 = Reserved, 14 = Error, 15 = No Selection
DD126	Differential Operating Mode	DF52	4		0 = Manual, 1 = Auto Power, 2 = Auto Range, 3-13 = Reserved, 14 = Error, 15 = No Selection
DD127	Generic Direction -True	DF02			Degrees clockwise relative to True North.
DD128	Generic instance	DF52	8		0 = Instance 0 1 = Instance 1 2 = Instance 2 n = Instance n, where n < 253 253 = Reserved 254 = Error 255 = Not available
DD129	Rate of rotation	DF72			
DD130	Temperature, high	DF38			
DD131	Flow rate, low	DF18			
DD132	Run time, Engine	DF67			
DD133	Rudder deflection	DF30			Percent rudder deflection from the Dead Ahead position. Negative is to port, positive is to starboard. +100% is hard to starboard, -100% is hard to port.

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD134	Run time, Trip	DF65			
DD135	Volume	DF44			
DD136	Voltage, DC	DF42			
DD137	Generic variable bit field	DF52	n		
DD138	Generic percent of range	DF30			
DD139	Error codesTransmit interval acknowledgement	DF52	4		0x0 = No Error/Acknowledge, 0x1 = Transmit Interval /Priority not supported, 0x2 = Transmit interval is less than measurement/calculation interval, 0x3 = Access denied. 0x4 = Request is not supported.
DD140	Current, Electric	DF07			+ = current consumed, - = current supplied
DD141	Error Codes Command Acknowledgement	DF52	4		0x0 = No Error/Acknowledge; 0x1 = Invalid request or command parameter field; 0x2 = Temporarily unable to comply; 0x3 = Request or command parameter out-of-range; 0x4 = Access denied; 0x5 = Request or Command is not supported; 0x6 = Read or Write is not supported;
DD142	Binary Bit Field	DF52	n		Binary data bit field.
DD143	Satellite Service ID	DF52	16		Satellite Service ID number as provided by the Service Provider.[Reference document required]
DD144	Group Function, Request/Command/Acknowledge	DF53			0 = Request Message, 1 = Command Message, 2 = Acknowledge Message, 3 = Read Fields, 4 = Read Fields Reply, 5 = Write Fields, 6 = Write Fields Reply
DD145	Group Function, Transmit and Receive PGN List	DF53			0 = Transmit PGN List Message, 1 = Receive PGN List Message
DD146	Rudder Angle Signed	DF04			Rudder angle where positive values are starboard and negative values are port
DD147	Directional Command	DF52	3		MSB/LSB: 000 = No Order, 001 = Move to starboard, 010 = Move to port.
DD148	Angular Limit	DF02			
DD149	Distance ordered	DF74			A commanded distance like radius order, off-track limit, etc.

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD150	Rate of Turn	DF73			+ = Bow turning to starboard, 1 deg/min = .00029 rad/sec
DD151	Magnetic Heading Correction	DF04			Positive values are Easterly and negative values are Westerly.
DD152	Turn Method	DF52	3		MSB/LSB: 000 = Rudder Limit controlled, 001 = turn rate controlled, 010 = radius controlled. Definitions: Rudder – An operator set rudder limit determines the maximum turning capability available, in which case the system outputs rudder orders up to and limited by the rudder limit. Turn Rate – An operator set turn rate limit determines the maximum turning capability available, in which case the system controls the rate of turn using whatever value of rudder order is necessary to control the rate. Radius – An operator set turn radius determines the maximum turning capability available which in this case is described as the minimum turn radius, in which case the system controls the rate of turn corresponding to the set turn radius.
DD153	Steering Mode	DF52	3		MSB/LSB: 000 = Main Steering, 001 = Non-Follow-up Device, 010 = Follow-up Device, 011 = Heading Control Standalone, 100 = Heading Control, 101 = Track Control. Definitions: Main Steering/Outside System – The main steering system is in use. Non-Follow-up Device – The system provides non-follow-up control. Rudder is moved in the commanded direction but a specific angle is not maintained. Follow-up – The system provides follow-up control. Rudder is moved to the commanded angle and maintained at that angle. Heading Control Standalone – The system works as a standalone heading controller. Heading Control – The system works as a remotely controlled heading controller with the commanded course input from an external device. Track Control – The system works as a track controller by correcting a commanded course. Corrections are based on additionally received track errors.
DD154	Manufacturer's Code	DF52	11		Company identifier as specified by NMEA
DD155	Identity	DF52	21		Unique identifier for use in the NAME field in ISO11783-5.
DD156	Functional Division	DF52	4		MSB/LSB: 0000 = Reserved, 0001 = Steering, 0010 = Voyage Management, 0100 = Propulsion.
DD157	Control Method	DF52	4		When Steering Function. MSB/LSB: 0000 = Main Steering/Outside of System, 0001 = Non-Follow-up Device, 0010 = Follow-up Device, 0011 = Heading Control Standalone, 0100 = Heading Control, 0101 = Track Control. [When Voyage Management Function:]. [When Propulsion Function:].

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD158	Generic time of day	DF06			24 hour clock, 0 = midnight, time is in UTC
DD159	Station Control Message Type	DF52	4		MSB/LSB: 0000 = 3D Current Control, 0001 = 3D Assign Control, 0010 = 3D Offer Control, 0011 = 3D Request Control, 0100 = 3D Accept Control, 0101 = 3D Release Control, 0110 = 3D Refuse Control. Definitions:Current/Assign Control - Method for notifying all devices of the current control station or to notify a specific device that it shall become the controlling station. Offer Control - Method for notifying a specific device that it may become the controlling station or the controlling device's method for notifying the system that it is willing to relinquish control Request Control - Method for notifying the station manager that a device wishes to become the controlling location or a device desires the controlling location be set to a specific device. Accept Control - Method for a device to acknowledge when control is assigned to the device. Release Control - Method for notifying the controlling station that it is no longer the controlling station. Refuse Control - Method for a device to refuse control when assigned by the steering manager or for the steering manager to refuse request from a device seeking to change the control location.
DD160	Generic speed, signed	DF36			Positive values represent ahead or starboard transverse speed and negative values represent astern or port transverse speed.
DD161	Transducer Offset	DF46			Positive values represent distance from transducer to water line and negative values represent distance from the transducer to the keel.
DD162	Water Depth At Transducer	DF09			Depth relative to the transducer location. Range of value specified in "Maximum Depth Range" (field 4).
DD163	Autopilot Override	DF52	2		1 = Yes, 0 = No. Yes means a temporary interruption of the selected steering mode. As long as this field is Yes, Steering Mode and Turn Mode shall be ignored by the heading/track controller and its computing parts shall operate as if manual steering was selected.
DD164	Bearing	DF02			The horizontal direction of one terrestrial point from another, expressed as the angular distance from a reference direction, measured from 000 at the reference direction clockwise through 359 degrees.
DD165	Course-Over-Ground (COG)	DF02			The direction of the path over ground actually followed by a vessel.

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD166	Track	DF02			The intended or desired horizontal direction of travel with respect to the earth. The track expressed in degrees of the compass may differ from the course due to allowances made in the course for such factors as sea and weather conditions in order to achieve the desired track. This field represents the course line (leg) between two waypoints. It may be altered dynamically in a track-controlled turn along a pre-planned radius.
DD167	Heading	DF02			The horizontal direction in which a ship actually points or heads at any instant, expressed in angular units from a reference direction, usually from 000 at the reference direction clockwise through 359 degrees.
DD168	Industry Group	DF52	3		0 = Global; 1 = On-Highway; 2 = Agricultural and Forestry; 3 = Construction; 4 = Marine; 5 = Industrial - Process Control - Stationary (Gen-Sets) 6 = Reserved for SAE 7 = Reserved for SAE
DD169	Generic instance (4-bit)	DF52	4		0x0 to 0xF = Instance number 0 to 15;
DD170	NMEA Device Class	DF52	7		Dependent on Industry Group DD 168, reference NMEA Class & Function Codes, Appendix B6.
DD171	NMEA Function Code	DF52	8		Dependent on NMEA Device Class DD170, reference NMEA Class & Function Codes, Appendix B6.
DD172	NMEA Manufacturer Code	DF52	11		Assigned by NMEA Committee
DD173	NMEA Unique Number	DF52	21		Binary number assigned by manufacturer to ensure that the NAME field for each manufactured device is unique, reference NMEA Network Management Section 8.
DD174	Generic instance (5-bit)	DF52	5		0x00 to 0x1F = Instance 0 to 31;
DD175	Dominant Bit	DF52	1		Set = 0
DD176	Network Address	DF52	8		0x00 to 0xFB (0 to 251) = Claimable NMEA network address space; 0xFC (252) = Reserved; 0xFD (253) = Reserved; 0xFE (254) = Null address; 0xFF (255) = Global address
DD177	ISO 11783 ACK Status	DF52	8		0x00 = Positive Acknowledgment; 0x01 = Negative Acknowledgment; 0x02 = PGN supported but access denied; 0x03 to 0xFF = Reserved

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD178	Group Function Value	DF52	8		Group Function of PGN being acknowledged. This field identifies for a device the specific group function of a PGN being acknowledged or declined. This field is not used if the PGN being acknowledged or declined is not a group function PGN.
DD179	Group Function, Connection Management	DF52	8		0x10 = Request to Send; 0x11 = Clear to Send; 0x13 = End of Message; 0x20 = Broadcast Announce Message; 0xFF = Abort; 0x00 to 0xF, 0x12, 0x14 to 0x1F, 0x21 to 0xFE = Reserved
DD180	Multi-packet frame counter	DF52	8		Valid range 0x01 to 0xFF
DD181	Multi-packet packetized data	DF52	56		56 bits organized as seven 8-bit bytes, each of the bytes is transmitted in the same order as it would in a standard (i.e., single frame) message. Unused bytes shall be filled with 0xFF.
DD182	Priority, Set	DF52	4		0x0 to 0x7 = commanded priority value; 0x8 = do not change priority; 0x9 = return priority to default; 0xA to 0xF = reserved
DD183	AIS Navigational Status	DF52	4		0 = under way using engine, 1 = at anchor, 2 = not under command, 3 = restricted manoeuvrability, 4 = constrained by her draught, 5 = moored, 6 = aground, 7 = engaged in fishing, 8 = under way sailing, 9 = reserved for future amendment of navigational status for ships carrying DG, HS, or MP, or IMO hazard or pollutant category C (HSC), 10 = reserved for future amendment of navigational status for ships carrying DG, HS or MP, or IMO hazard or pollutant category A (WIG), 11-14 = reserved for future use, 15 = not defined (default) See the latest version of ITU-R M.1371 for more information.
DD184	AIS Position Accuracy	DF52	1		0 = low accuracy > 10m such as nondifferential GNSS (default), 1 = high accuracy < 10m such as DGNSS See the latest version of ITU-R M.1371 for more information.
DD185	AIS Repeater Indicator	DF52	2		Used by the repeater to indicate how many times a message has been repeated (range of 0 to 3). 0 = Default 1 = First retransmission 2 = Second retransmission 3 = Final retransmission See the latest version of ITU-R M.1371 for more information.

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD186	AIS Time Stamp	DF52	6		0-59 = UTC second when the report was generated, 60 = time stamp not available (default), 61 = positioning system is in manual input mode, 62 = Electronic position fixing system operates in estimated (dead reckoning) mode, 63 = positioning system is inoperative See the latest version of ITU-R M.1371 for more information.
DD187	AIS Communication State	DF52	19		The Communication State contains information used by the various TDMA slot allocation algorithms and synchronization information See the latest version of ITU-R M.1371 for more information.
DD188	AIS Message Identifier	DF52	6		Message Identifier (range of 0 to 63). See the latest version of ITU-R M.1371 for more information.
DD189	AIS RAIM-flag	DF52	1		0 = RAIM not in use (default), 1 = RAIM in use See the latest version of ITU-R M.1371 for more information.
DD190	Heading Source	DF52	4		0=Gyro, 1=Magnetic Compass, 2=Hall Effect Compass, 3=Flux Gate Compass, 4=GPS Attitude Determination System, 5-13=reserved, 14=error, 15=null
DD191	AIS Electronic Positioning Device Type	DF52	4		0 =Undefined (default) 1 = GPS 2 = GLONASS 3 = Combined GPS/GLONASS 4 = Loran-C 5 = Chayka 6 = Integrated Navigation System 7 = Surveyed; For fixed AtoN and virtual AtoN, the charted position should be used. The accurate position enhances its function as a radar reference target 8 = Galilieo 9-15 = Reserved for future use. See the latest version of ITU-R M.1371 for more information.
DD192	Generic String, ASCII, Fixed length	DF63		n	Length specified by PGN field definition.
DD193	Ship/Cargo Type	DF52	8		0=Not Available or no ship (default), 1-99= (See the latest version of ITU-R M.1371 Section 3.3.8.2.3.2 Table 18), 100-199=Reserved for Regional (See the latest version of ITU-R M.1371), 200-255=Reserved for future (See the latest version of ITU-R M.1371).
DD194	Distance, medium	DF75			Dependent upon PG Field definition.

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD195	Distance, short	DF13			Dependent upon PG Field definition.
DD196	Draft	DF13			The depth of a ship in the water. The vertical distance between the waterline and the keel.
DD197	U.AIS Communications State Indicator	DF52	2		0=SOTDMA, 1=ITDMA, 2=Error, 3=Null
DD198	Generic name string, Medium	DF51			Medium size text strings.
DD199	Distance, Unsigned	DF09			
DD200	Water Depth, Maximum Range	DF74			
DD201	Generic instance 2 (3-bit)	DF52	3		0x0 to 0x7 = instance 0 to 7
DD202	Latitude (Extended Resolution)	DF76			Latitude referenced to WGS-84
DD203	Longitude (Extended Resolution)	DF77			Longitude referenced to WGS-84
DD204	Altitude (Extended Resolution)	DF78			Altitude referenced to WGS-84
DD205	Wind Reference	DF52	3		0x00 = Theoretical Wind (ground referenced, referenced to True North; calculated using COG/SOG) 0x01 = Theoretical Wind (ground referenced, referenced to Magnetic North; calculated using COG/SOG) 0x02 = Apparent Wind (relative to the vessel centerline) 0x03 = Theoretical (Calculated to Centerline of the vessel, refernced to ground; calculated using COG/SOG) 0x04 = Theoretical (Calculated to Centerline of the vessel, refernced to water; calculated using Heading/Speed through Water) 0x05 = Reserved 0x06 = Error 0x07 = Null

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD206	Engine Discrete Warning Status	DF52	16		xxxx xxxx xxxx xxx1 = Check Engine, xxxx xxxx xxxx xx1x = Over Temperature, xxxx xxxx xxxx x1xx = Low Oil Pressure, xxxx xxxx xxxx 1xxx = Low Oil Level, xxxx xxxx xxx1 xxxx = Low Fuel Pressure, xxxx xxxx xx1x xxxx = Low System Voltage, xxxx xxxx x1xx xxxx = Low Coolant Level, xxxx xxxx 1xxx xxxx = Water Flow, xxxx xxx1 xxxx xxxx = Water in Fuel, xxxx xx1x xxxx xxxx = Charge Indicator, xxxx x1xx xxxx xxxx = Preheat Indicator, xxxx 1xxx xxxx xxxx = High Boost Pressure, xxx1 xxxx xxxx xxxx = Rev Limit Exceeded, xx1x xxxx xxxx xxxx = EGR System, x1xx xxxx xxxx xxxx = Throttle Position Sensor, 1xxx xxxx xxxx xxxx = Engine Emergency Stop Mode where x = don't care Inactive states shall be transmitted as 0. Reserved and unsupported bits within the field shall be transmitted as 0.
DD207	Type of System	DF52	4		0x0 = GPS 0x1 = GLONASS 0x2 = GPS+GLONASS 0x3 = GPS+SBAS(WAAS) 0x4 = GPS+SBAS+GLONASS 0x5 - 0xF = Reserved for future combinations
DD208	Type of Fluid	DF52	4		0x00 = Fuel, 0x01 = Fresh Water, 0x02 = Waste Water, 0x03 = Live Well, 0x04 = Oil, 0x05 = Black Water (Sewage) 0x06 = Fuel (Gasoline) 0x07 = Reserved, thru 0x0D = Reserved, 0x0E = Error, 0x0F = Data not available
DD209	GNSS Integrity	DF52	2		0 = No Integrity checking,* 1 = Safe, 2 = Caution, 3 = Unsafe * means the receiver does not have this capability
DD210	Time Value, resolution 1 sec	DF80			Time in seconds
DD211	Time Value, resolution 0.1 sec	DF81			
DD212	Station Heath	DF52	4		0x00 = Not Working, 0x01 = Unmonitored, 0x02 = Healthy & Operational, 0x03 = Healthy & in Test Mode, 0x04 = In Test Mode - DO NOT USE, 0x05 - 0x15 = Reserved
DD213	Distance, int 32 4dp	DF83			

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD214	Generic Speed	DF82			
DD215	Generic Percent of Range, Medium	DF84			
DD216	Track Status	DF52	4		xx00 = Cancelled or Not Available, xx01 = Initial Acquisition Target, xx10 = Tracking Target, xx11 = Lost Target, x0xx = Reported Target No, x1xx = Reported Target Yes, 0xxx = Acquision Manual, 1xxx = Acquision Auto, where x = don't care
DD217	Unstable/Reference Target	DF52	2		xx00 = Unreferenced Target, xx01 = Referenced Target, xx10 = Unstable Target, xx11 = Stable Target, x0xx = Reported Target No, x1xx = Reported Target Yes, 0xxx = Acquision Manual, 1xxx = Acquision Auto, where x = don't care
DD218	Direction reference for target	DF52	2		0 = True, 1 = Magnetic, 2 = Error, 3 = Releative
DD219	Standard Deviation	DF13			
DD220	Measure	DF14			
DD221	Transmission Discrete Warning Status	DF52	8		xxxx xxx1 = Check Tranmission, xxxx xx1x = Over Temperature, xxxx x1xx = Low Oil Pressure, xxxx 1xxx = Low Oil Level, xxx1 xxxx = Sail Drive, xx1x xxxx = reserved, x1xx xxxx = reserved, 1xxx xxxx = reserved, where x = don't care
DD222	Transmission Gear	DF52	2		0 = Forward. 1 = Neutral, 2 = Reverse, 3 = [Unavailable, Unknown]

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD223	Engine Discrete Warning Status	DF52	16		xxxx xxxx xxxx xxx1 = Warning Level 1, xxxx xxxx xxxx xx1x = Warning Level 2, xxxx xxxx xxxx x1xx = Power Reduction, xxxx xxxx xxxx 1xxx = Maintenance Needed, xxxx xxxx xxx1 xxxx = Engine Comm Error, xxxx xxxx xx1x xxxx = Sub or Secondary Throttle, xxxx xxxx x1xx xxxx = Neutral Start Protect, xxxx xxxx 1xxx xxxx = Engine Shutting Down, xxxx xxx1 xxxx xxxx = reserved, xxxx xx1x xxxx xxxx = reserved, xxxx x1xx xxxx xxxx = reserved, xxxx 1xxx xxxx xxxx = reserved, xxx1 xxxx xxxx xxxx = reserved, xx1x xxxx xxxx xxxx = reserved, x1xx xxxx xxxx xxxx = reserved, 1xxx xxxx xxxx xxxx = reserved, where x = don't care Inactive states shall be transmitted as 0. Reserved and unsupported bits within the field shall be transmitted as 0.
DD224	Rate of Turn	DF85			+ = Bow turning to starboard, 1 deg/min = .00029 rad/sec
DD225	Generic Pressure High	DF29			
DD226	NMEA Certification Level	DF53			0 = Level A 1 = Level B 2 = Not Applicable
DD227	Volume	DF86			
DD228	Generic speed, signed - large	DF87			Positive values represent ahead or starboard transverse speed and negative values represent astern or port transverse speed.
DD229	Temperature Instance	DF52	6		0x00 = Sea Temperature, 0x01 = Outside Temperature, 0x02 = Inside Temperature, 0x03 = Engine Room Temperature, 0x04 = Main Cabin Temperature, 0x05 = Reserved, thru 0x3E = Reserved, 0x3F = Data Not Available
DD230	Humidity Instance	DF52	2		0x00 = Inside Humidity, 0x01 = Outside Humidity, 0x02 = reserved, 0x03 = Data Not Available
DD231	Humidity	DF84			Relative Humidity

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD232	Variation Source	DF52	4		0x00 = Manual Entry, "Variation has been entered via key entry" 0x01 = Automatic - Chart, "Variation is derived from cartograhy present in system" 0x02 = Automatic Table, "Variation has been computed from tabular based system" 0x03 = Automatic Calculation, "Variation has been derived via calculation" 0x04 = WMM 2000, "Variation is calc via World Magnetic Model for 2000" 0x05 = WMM 2005, "Variation is calc via World Magnetic Model for 2005" 0x06 = WMM 2010, "Variation is calc via World Magnetic Model for 2010" 0x07 = WMM 2015, "Variation is calc via World Magnetic Model for 2015" 0x08 = WMM 2020, "Variation is calc via World Magnetic Model for 2020" 0x09 = Reserved, thru 0x0E = Reserved, 0x0F = Data Not Available
DD233	Time Value (Short resolution 5 msec)	DF88			
DD234	Latitude	DF89			
DD235	Longitude	DF90			
DD236	Altitude	DF91			
DD237	Changed at timestamp flag	DF52	8		0000 0000 = No change in data. 0xxx xxx1 = WP: Change in main data (Position, Name) 0xxx xx1x = WP: Change in supplementary parameters (or new added) 0xxx x1xx = Changed no. of WP's in Route/WP-List, and/or name changed/added 0xxx 1xxx = Route: Change supplementary parameters (or new added) 0xx1 xxxx = Reserved 0x1x xxxx = Reserved 01xx xxxx = Other not specifyed change 1111 1111 = This flag is not supported x = don't care
DD238	WP Position Resolution	DF52	4		0= [>0.1min.], 1= [<=0.1& >0.01min.], 2= [<=0.01 & >0.001] 3= [<=0.001 & >0.0001] 4= [<=0.000 1 & >=0.000 001min.] 5-6 =reserved, 7= Not available (not known) (1min. = 0.01667deg.)
DD239	Route Status	DF52	4		0=Active, 1=Inactive, 2=Deleted, 3-13= Reserved, 14=Error, 15= Null

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD240	WP Identification Method	DF52	2		0=WP's in WP-List, 1=WP embedded in Rute, 2=Reserved 3=Null (info not available)
DD241	Navigation Direction	DF52	3		0=Forward, 1=Reverse, 2-5 Reserved, 6= Error, 7=Null (info not available)
DD242	Data Terminal Equipment (DTE)	DF52	1		0=Available, 1=not available. See the latest version of ITU-R M.1371 for more information.
DD243	AIS Sequence Number	DF52	2		Range 0-3 See the latest version of ITU-R M.1371 for more information.
DD244	AIS Retransmit Flag	DF52	1		0=No retransmission, 1=retransmitted. See the latest version of ITU-R M.1371 for more information.
DD245	AIS Communication State Selctor Flag	DF52	1		0=SOTDMA communication state, 1=ITDMA communication state follows. See the latest version of ITU-R M.1371 for more information.
DD246	AIS Transceiver Information	DF52	5		0 = Channel A VDL reception, 1 = Channel B VDL reception, 2 = Channel A VDL transmission, 3 = Channel B VDL transmission, 4 = Own information not broadcast, 5-31 = Reserved.
DD250	Types of Aids-to-Navigation	DF52	5		0=Not available (default), 1-15=fixed aid-to-navigation, 16-31=floating aid-to-navigation. See the latest version of ITU-R M.1371 for more information.
DD251	AIS Off Position Indicator	DF52	1		For floating aids-to-navigation only. 0=On position, 1=off position. See ITU-R M.1371-1 for more information.
DD252	AIS Power	DF52	1		0=High (default), 1=low. See the latest version of ITU-R M.1371 for more information.

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD253	AIS Tx/Rx Mode	DF52	4		0=Tx A/Tx B, Rx A/Rx B (default), 1=Tx A, Rx A/Rx B, 2=Tx B, Rx A/Rx B, 3-15=not used. See the latest version of ITU-R M.1371 for more information.
DD254	AIS Addressed or Broadcast Message Indicator	DF52	1		0=Broadcast geographical area message (default), 1=addressed message (to individual station(s)). See the latest version of ITU-R M.1371 for more information.
DD255	AIS Channel Bandwidth	DF52	1		0=default (as specified by channel number), 1=12.5 kHz bandwidth. See the latest version of ITU-R M.1371 for more information.
DD256	AIS Transitional Zone Size	DF52	3		See the latest version of ITU-R M.1371 for more information.
DD257	Load Equivalency Number	DF53			The "Load Equivalency Number" is defined in the NMEA 2000 Standard's Main Document in section 2.4.7 Interface Power. Please consult this for details.
DD258	Critical supplementary Route parameters	DF52	8		0000 0000 = There are no parameters additional to the Waypoint Positions xxxx xxx1 = Navigation Method (GC/RL), PGN 130069 xxxx xx1x = XTE Limit, PGN 130069 xxxx x1xx = Spare xxxx 1xxx = Spare xxx1 xxxx = Spare xx1x xxxx = Spare x1xx xxxx = Spare 1xxx xxxx = Spare Each of these bitflags represents a parameter. If a flag is '1', the parameter must be downloaded and appended to the Route information. Ignoring any of these parameters will not be safe.
DD259	AC Acceptability	DF52	2		0x00 = Bad Level, 0x01 = Bad Frequency, 0x02 = Being Qualified, 0x03 = Good
DD260	Voltage, AC RMS	DF96			
DD261	Power (watts)	DF94			
DD262	Volt Amps Reactive Power (VAR)	DF92			
DD263	Generic Absolute Percentage 0-252%	DF93			

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD264	Charger Operating State	DF52	4		0x00 = Not Charging, 0x01 = Bulk, 0x02 = Absorption, 0x03 = Overcharge, 0x04 = Equalize, 0x05 = Float, 0x06 = No Float 0x07 = Constant VI, 0x08 = Disabled, 0x09 = Fault, thru 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available
DD265	Charger Mode	DF52	4		0x00 = Standalone, 0x01 = Primary, 0x02 = Secondary, 0x03 = Echo 0x04 = Reserved thru 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available
DD266	Invertor Operating State	DF52	4		0x00 = Invert, 0x01 = AC Passthru, 0x02 = Load Sense, 0x03 = Fault, 0x04 = Disabled 0x05 = Reserved thru 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available
DD267	Frequency	DF22			
DD268	Time	DF98			
DD269	Current, Electric, Unsigned	DF95			
DD270	AC Line	DF52	2		0x00 = Line 1, 0x01 = Line 2, 0x02 = Line 3 0x03 = Reserved
DD271	Power Factor	DF97			Cosine of the angle between the AC voltage and current
DD272	Charging Algorithm	DF52	4		0x00 = Trickle 0x01 = CVCC (Constant Voltage Constant Current) 0x02 = 2 Stage (No float) 0x03 = 3 Stage 0x04 = Reserved thru 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD273	Waveform	DF52	3		0x00 = Sine Wave 0x01 = Modified Sine Wave 0x02 = Reserved thru 0x05 = Reserved 0x06 = Error 0x07 = Data Not Available
DD274	Batt Temp - No Sensor	DF52	4		0x00 = Cold 0x01 = Warm 0x02 = Hot 0x03 = Reserved thru 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available
DD275	Inverter Mode	DF52	4		0x00 = Standalone 0x01 = Series Master 0x02 = Series Slave 0x03 = Parallel Master 0x04 = Parallel Slave 0x05 = Reserved thru 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available
DD276	Power, medium (watts)	DF28			
DD277	AGS Mode	DF52	4		0x00 = Off 0x01 = On 0x02 = Automatic 0x03 = Reserved thru 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available
DD278	AGS Operating State	DF52	4		0x00 = Quiet Time 0x01 = Auto On 0x02 = Auto Off 0x03 = Manual On 0x04 = Manual Off 0x05 = Generator Shutdown 0x06 = External Shutdown 0x07 = Fault 0x08 = Suspend 0x09 = Not Operating 0x0A = Reserved thru 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD279	Generator State	DF52	4		0x00 = Preheating 0x01 = Start Delay 0x02 = Cranking 0x03 = Starter Cooling 0x04 = Warming Up 0x05 = Cooling Down 0x06 = Spinning Down 0x07 = Shutdown Bypass 0x08 = Stopping 0x09 = Running 0x0A = Stopped 0x0B = Crank Delay 0x0C = Reserved 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available
DD280	Generator On Reason	DF52	8		0x00 = Not On 0x01 = DC Voltage Low 0x02 = Battery State Of Charge Low 0x03 = AC Current High 0x04 = Contact Closed 0x05 = Manual On 0x06 = Exercise 0x07 = Non Quiet Time 0x08 = External On Via AGS 0x09 = External On Via Generator 0x0A = Unable to Stop 0x0B = Reserved thru 0xFD = Reserved 0xFE = Error 0xFF = Data Not Available
DD281	Generator Off Reason	DF52	8		0x00 = Not Off 0x01 = DC Voltage High 0x02 = Battery State Of Charge High 0x03 = AC Current Low 0x04 = Contact Opened 0x05 = Reached Absorption 0x06 = Reached Float 0x07 = Manual Off 0x08 = Max Run Time 0x09 = Max Auto Cycle 0x0A = Exercise Done 0x0B = Quiet Time 0x0C = External Off Via AGS 0x0D = Safe Mode 0x0E = External Off Via Generator 0x0F = External Shutdown 0x10 = Auto Off 0x11 = Fault 0x12 = Unable to Start 0x13 = Reserved thru 0xFD = Reserved 0xFE = Error 0xFF = Data Not Available

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD282	Battery Type	DF52	4		0x00 = Flooded 0x01 = GEL 0x02 = AGM 0x03 = Reserved thru 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available
DD283	Battery Capacity (Coulombs)	DF100			
DD284	Nominal Voltage	DF52	4		0x00 = 6 Volts 0x01 = 12 Volts 0x02 = 24 Volts 0x03 = 32 Volts 0x04 = 36 Volts 0x05 = 42 Volts 0x06 = 48 Volts 0x07 = Reserved thru 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available
DD285	Battery Chemistry	DF52	4		0x00 = Lead Acid 0x01 = LiIon 0x02 = NiCad 0x03 = ZnO 0x04 = NiMH 0x05 = Reserved thru 0x0D = Reserved 0x0E = Error 0x0F = Data Not Available
DD286	Peukert Exponent	DF101			
DD287	AC Ripple	DF102			
DD288	DC Type	DF52	8		0x00 = Battery, 0x01 = Alternator, 0x02 = Convertor, 0x03 = Solar Cell, 0x04 = Wind Generator, 0x05 = Reserved, thru 0xFD = Reserved 0xFE = Error 0xFF = Data Not Available

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD289	Pressure Source	DF52	8		00 = Atmospheric Pressure 01 = Water Pressure 02 = Steam Pressure 03 = Compressed Air Pressure 04 = Hydraulic Pressure 05 = Filter Pressure 06 = Altimiter Setting 07 = Oil Pressure 08 = Fuel Pressure 09 thru 128 Reserved 129 thru 252 Generic pressure sources other than those defined 253=Reserved 254=Error 255=No Change / Data Not Available
DD290	Pressure	DF103			
DD291	Temperature Source	DF52	8		00 = Sea Temperature 01 = Outside Temperature 02 = Inside Temperature 03 = Engine Room Temperature 04 = Main Cabin Temperature 05 = Live Well Temperature 06 = Bait Well Temperature 07 = Refrigeration Temperature 08 = Heating System Temperature 09 = Dew Point Temperature 10 = Wind Chill Temperature, Apparent 11 = Wind Chill Temperature, Theoretical 12 = Heat Index Temperature 13 = Freezer Temperature 14 = Exhaust Gas Temperature 15 through 128 Reserved 129 through 252 Generic Temperature Sources other than those defined 253 = Not Supported 254 = Error 255 = No Change / Data Not Available
DD292	Humidity Source	DF52	8		00 = Inside Humidity 01 = Outside Humidity 02 through 128 Reserved 129 through 252 Generic Humidity Sources other than those defined 253 = Not Supported 254 = Error 255 = No Change / Data Not Available
DD293	Speed Water Reference Type	DF52	8		00 = Paddle Wheel 01 = Pitot Tube 02 = Doppler Log 03 = Correlation Log (Ultra-Sound) 04 = EM Log (Electro - Magnetic) 05 through 128 Reserved 129 through 252 Generic Speed Sources other than those defined 253 = Not Supported 254 = Error 255 = Do Not Change/Not Available

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD294	AIS ClassB Unit Flag	DF52	1		0 = Class B SOTDMA unit 1 = Class B “CS” unit See the latest version of ITU-R M.1371 for more information.
DD295	AIS Class B Display Flag	DF52	1		0 = No display available; not capable of displaying ITU-R M.1371 Messages12 and 14 1 = Equipped with integrated display displaying ITU-R M.1371 Messages 12 and 14 See the latest version of ITU-R M.1371 for more information.
DD296	AIS Class B DSC Flag	DF52	1		0 = Not equipped with DSC function 1 = Equipped with DSC function (dedicated or time-shared) See the latest version of ITU-R M.1371 for more information.
DD297	AIS Class B Band Flag	DF52	1		0 = Capable of operating over the upper 525 kHz band of the marine band 1 = Capable of operating over the whole marine band (irrelevant if “Class B Message 22 flag” is 0) See the latest version of ITU-R M.1371 for more information.
DD298	AIS Class B Msg 22 Flag	DF52	1		0 = No frequency management via Message 22 , operating on AIS1 and AIS2 only 1 = Frequency management via Message 22 See the latest version of ITU-R M.1371 for more information.
DD299	AIS Mode Flag	DF52	1		0 = Station operating in autonomous mode (default) 1 = Station operating in assigned mode See the latest version of ITU-R M.1371 for more information.
DD300	Ship/Cargo Filter	DF52	8		0 = all types (default) 1 - 99 - See ITU-R M.1371 Table "Identifiers to be used by ships to report their type" 100 - 199 = Reserved for regional use 200 - 255 = Reserved for future use See the latest version of ITU-R M.1371 for more information.
DD301	AIS Station Type	DF52	4		0 = All types of mobiles (default) 1 = Reserved for future use 2 = All types of Class B mobile stations 3 = SAR airborne mobile station 4 = AtoN station 5 = Class B “CS” shipborne mobile station only 6 = Inland waterways 7 to 9 = Reserved for regional use 10 to 15 = Reserved for future use See the latest version of ITU-R M.1371 for more information.

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD302	AIS Reporting Interval for Class B	DF52	4		0 = As given by the autonomous mode 1 = 10 min 2 = 6 min 3 = 3 min 4 = 1 min 5 = 30 sec 6 = 15 sec 7 = 10 sec 8 = 5 sec 9 = Next shorter reporting interval 10 = Next longer reporting interval 11 = 2 sec (not applicable to the Class B “CS”) 12 - 15 = Reserved for future use See the latest version of ITU-R M.1371 for more information.
DD303	AIS Quiet Time	DF52	4		0 = No quiet time commanded 1-15 = Quiet time of 1 to 15 min See the latest version of ITU-R M.1371 for more information.
DD304	AIS Version Indicator	DF52	2		0 = Station compliant with AIS edition 0 1-3 = Station compliant with AIS editions 1, 2, and 3 See the latest version of ITU-R M.1371 for more information.

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD305	AIS Aids to Navigation Type	DF52	5		0 = Type of AtoN not specified (default) 1 = Reference point 2 = RACON 3 = Fixed structures off-shore, such as oil platforms, wind farms; This code identifies an obstruction that is fitted with an AtoN AIS station 4 = Spare, Reserved for future use Fixed AtoN: 5 = Light, without sectors 6 = Light, with sectors 7 = Leading Light Front 8 = Leading Light Rear 9 = Beacon, Cardinal N 10 = Beacon, Cardinal E 11 = Beacon, Cardinal S 12 = Beacon, Cardinal W 13 = Beacon, Port hand 14 = Beacon, Starboard hand 15 = Beacon, Preferred Channel port hand 16 = Beacon, Preferred Channel starboard hand 17 = Beacon, Isolated danger 18 = Beacon, Safe water 19 = Beacon, Special mark Floating AtoN: 20 = Cardinal Mark N 21 = Cardinal Mark E 22 = Cardinal Mark S 23 = Cardinal Mark W 24 = Port hand Mark 25 = Starboard hand Mark 26 = Preferred Channel Port hand 27 = Preferred Channel Starboard hand 28 = Isolated danger 29 = Safe Water 30 = Special Mark 31 = Light Vessel/LANBY/Rigs NOTE 1: The types of aids to navigation listed above are based on the IALA Maritime Buoyage System, where applicable. NOTE 2: There is potential for confusion when deciding whether an aid is lighted or unlighted. Competent authorities may wish to use the regional/local section of the message to indicate this. See Latest version of ITU-R M.1371 for more information.
DD306	Off position indicator	DF52	1		For floating AtoN, only: 0 = on position 1 = off position NOTE 1: This flag should only be considered valid by receiving station, if the AtoN is a floating aid, and if time stamp is equal to or below 59. For floating AtoN the guard zone parameters should be set on installation.
DD307	Virtual AtoN Flag	DF52	1		0 = real AtoN at indicated position (default) 1 = virtual AtoN, does not physically exist See note 2 of MSG 21 in ITU-R M.1371 for more information.

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD308	AIS Assigned Mode Flag	DF52	1		0 = Station operating in autonomous and continuous mode (default) 1 = Station operating in assigned mode
DD309	AtoN Status	DF52	8		Reserved for indicating AtoN Status; Only defined value is 00000000 (default).
DD310	Special Maneuver Indicator	DF52	2		0 = not available (default) 1 = not engaged in special Maneuver 2 = engaged in special Maneuver (i.e.: regional passing arrangement on Inland Waterway) 3 = reserved
DD311	AIS Spare Field	DF115	resv		Variable number of reserved bits, all set to logic "0". Normally, spare or reserved bits in NMEA network messages are encoded with logic 1's, however for AIS PGNs the unused or reserved bits are to be encoded as logic 0's.
DD312	Scale Measurement Status	DF52	3		Enumerated value to indicate measurement status 0x00 = Current Changing – the payload acceleration is changing 0x01 = Current Stable – the payload acceleration has reached equilibrium 0x02 = Measurement Locked – the payload measurement is fixed until a significant mass change occurs (see description) 0x04 = Almanac Mass – the payload mass is being transmitted from memory 0x07 = Error or no status
DD313	Mass	DF104			
DD314	Temperature, extended range and precision	DF105			
DD315	Alert Type	DF52	4		0 = Reserved 1 = Emergency Alarm 2 = Alarm 3 = Reserved 4 = Reserved 5 = Warning 6 = Reserved 7 = Reserved 8 = Caution 13 = Reserved 14 = Data out of range 15 = Data not available
DD316	Alert Category	DF52	4		0 = Navigational 1 = Technical 13 = Reserved 14 = Data out of range 15 = Data not available
DD317	Alert System	DF53			Values defined by the Alert Codes in Appendix B
DD318	Alert Sub-System	DF53			
DD319	Alert ID	DF54			

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD320	Network ID NAME	DF56			This field is defined by fields 1- 10 of PGN 60928
DD321	Temporary Silence Status	DF52	1		0 = Not Temporary Silence 1 = Temporary Silence
DD322	Acknowledge Status	DF52	1		0 = Not Acknowledged 1 = Acknowledged
DD323	Escalation Status	DF52	1		0 = Not Escalated 1 = Escalated
DD324	Temporary Silence Support	DF52	1		0 = Not Supported 1 = Supported
DD325	Acknowledge Support	DF52	1		
DD326	Escalation Support	DF52	1		
DD327	NAME Selection Criteria Mask	DF52	12		0xxx xxxx xxx1 = NAME shall match Unique ID field 0xxx xxxx xx1x = NAME shall match Manufacturer Code field 0xxx xxxx x1xx = NAME shall match Device Instance Lower field 0xxx xxxx 1xxx = NAME shall match Device Instance Upper field 0xxx xxx1 xxxx = NAME shall match Device Function field 0xxx xx0x xxxx = Reserved bit (aligns with NAME Reserved field 6) 0xxx x1xx xxxx = NAME shall match Device Class field 0xxx 1xxx xxxx = NAME shall match System Instance field 0xx1 xxxx xxxx = NAME shall match Industry Group field 0x0x xxxx xxxx = Reserved bit (aligns with NAME Reserved field 10) 00xx xxxx xxxx = Reserved bit 1111 1111 1111 = Data not Available, Mask not used 'x' = don't care Note that multiple matches are possible. Inactive field selections shall be transmitted as 0. Reserved bits within the field shall be transmitted as 0, except when transmitting "Data Not Available, Mask not used".
DD328	Trigger Condition	DF52	4		0 = Manual 1 = Auto 2 = Test 3 = Disabled 13 = Reserved 14 = Data out of range 15 = Data not available
DD329	Threshold Status	DF52	4		0 = Normal 1 = Threshold Exceeded 2 = Extreme Threshold Exceeded, field added to support 0183 3 = Low Threshold Exceeded, field added to support 0183 4 = Extreme Low Threshold Exceeded, field added to support 0183 13 = Reserved 14 = Data out of range 15 = Data not available

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD330	Alert State	DF53			0 = Disabled 1 = Normal 2 = Active 3 = Silenced 4 = Acknowledged 5 = Awaiting Acknowledge 253 = Reserved 254 = Data out of range 255 = Data not available
DD332	Response Command	DF52	2		0 = Acknowledge 1 = Temporary Silence 2 = Test Command off 3 = Test Command on
DD333	Alert Control	DF52	2		0 = Disabled 1 = Enabled
DD334	Reactivation Period	DF67			Range: 0 to ~4.295x10E+9 s
DD335	Temporary Silence Period	DF67			
DD336	Escalation Period	DF67			
DD337	Trigger Method	DF52	8		0 = Manual 1 = Automatic 2 = Threshold Reached Ascending 3 = Threshold Reached Descending 253 = Reserved 254 = Data out of range 255 = Data not available
DD338	Threshold Data Format	DF53			As an example to reference an unsigned high voltage (DF96), the Threshold Data format would be decimal 96. Range 0 to 252.
DD339	Threshold Level	DF00			This parameter is defined by the Threshold Data Format and is a preset value used to trigger an Alert.
DD340	Value Data Format	DF53			As an example to reference an unsigned high voltage (DF96), the Threshold Data format would be decimal 96. Range 0 to 252.
DD341	Value Data	DF00			The value data range is defined by the value data format. The Value data is the dynamic parameter used to trigger the alert.

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD342	Converter Operating State	DF52	8		0x0=Off, 0x1=Low Power Mode, 0x2=Fault, 0x3=Bulk, 0x4=Absorption, 0x5=Float, 0x6=Storage, 0x7=Equalize, 0x8=Pass thru, 0x9=Inverting, 0xA=Assisting, 0xB-0xFE=Reserved, 0xFF=Not Available
DD343	Converter Temperature State	DF52	2		0x0=Ok, 0x1=Warning, 0x2=Over Temperature, 0x3=Not Available
DD344	Converter Overload State	DF52	2		0x0=Ok, 0x1=Warning, 0x2=Overload, 0x3=Not Available
DD345	Converter Low DC Voltage State	DF52	2		0x0=Ok, 0x1=Warning, 0x2=DC voltage too low, 0x3=Not Available
DD346	Converter Ripple State	DF52	2		0x0=Ok, 0x1=Ripple Warning, 0x2=Ripple Too High, 0x3=Not Available
DD347	Voltage, AC, High Range	DF106			6553.3 = Reserved 6553.4 = Out of Range 6553.5 = Not Available or Do Not Change
DD348	Frequency-Medium	DF107			
DD349	Power- Extended Range	DF108			+2,147,483,645 = Reserved +2,147,483,646 = Out of Range +2,147,483,647 = Data not available or Do not change
DD350	Maximum Depth Range	DF109			Device classification of the Maximum Range over which water depth can be measured. 253 = Deeper than 2,520 meters 254 = Error 255 = Data Not Available

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD351	Connection State	DF52	8		Commandable States- Codes that may be reported as states and commanded using PGN 126208 0 = OFF Connection is OFF - Hi Impedance 1 = LOW Connected to Ground 2 = HIGH Connected to Supply 3 = TRIPPED Max. Device capabilities Exceeded - Connection is OFF 4 = PWM Connection is Pulse Width Modulated - Dimmer 5 = TIMED_OFF Connection is OFF for TimeOFF, then HIGH 6 = TIMED_ON Connection is HIGH for TimeON, then OFF 7 = CYCLE_OFF Connection is OFF for TimeOFF, then CYCLE_ON 8 = CYCLE_ON Connection is HIGH for TimeON, then CYCLE_OFF 9-127 = Reserved State Only- Codes that can only be reported as states 128 = FAULT Internal Failure - Connection is OFF 129-191 = Reserved Command Only- Codes that may be commanded using PGN 126208, but are never reported as states 192 = RESET Reset TRIPPED to OFF 193 = DESTROY Command to Destroy Fuse - Disable Output - set FAULT state 194-253 = Reserved 254 = Error 255 = Data not Available
DD352	Connection Status	DF52	16		x xxxx xxxx xxx1 = SENSED_HIGH-HIGH voltage level detected x xxxx xxxx xx1x = SENSED_LOW- voltage level detected x xxxx xxxx x1xx = SHORT- Current draw Exceeds device capability x xxxx xxxx 1xxx = NO_LOAD- When HIGH, No Load detected - Amps=0 x xxxx xxx1 xxxx = FUSE_FAIL- Blown or removed x xxxx xx1x xxxx = FUSE_DESTRUCT - Fuse was Commanded to destroy x xxxx x1xx xxxx = GFCI_FAULT- Ground fault leakage detected x xxxx 1xxx xxxx = OVER_VOLTAGE- Voltage Greater than Over Voltage Value x xxx1 xxxx xxxx = OVER_CURRENT- Current Greater than Over Current Value x xx1x xxxx xxxx =UNDER_VOLTAGE- Voltage Less than Under Voltage Value x x1xx xxxx xxxx = UNDER_CURRENT- Current Less than Under Current Value x 0xxx xxxx xxxx = Reserved 0 xxxx xxxx xxxx = Reserved where x = don't care Inactive states shall be transmitted as 0. Reserved and
DD353	AIS Channel Management Source	DF52	3		Source identifiers: 0 = When this PGN is sent to an AIS unit 1 = ITU-R M.1371 message 22: Channel Management addressed message 2 = ITU-R M.1371 message 22: Channel Management broadcast geographical area message 3 = NMEA 0183 ACA Sentence or NMEA Network PGN # 129806 4 = DSC Channel 70 Telecommand 5 = Operator manual input 6 thru 7 = Reserved for future use

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD354	AIS Channel Management Parameters Usage	DF52	2		0 = “Not in use” when this PGN is sent from an AIS unit or when commanded with PGN 126208 to a AIS Base Station 1 = “In use” when this PGN is sent from an AIS unit or when commanded with PGN 126208 to an AIS Base Station 2 - 3 = Reserved for future use AIS Mobile units shall not accept commands on this field with PGN 126208.
DD355	Language ID	DF53			0 = English (US) 1 = English (UK) 2 = Arabic 3 = Chinese (simplified) 4 = Croatian 5 = Danish 6 = Dutch 7 = Finnish 8 = French 9 = German 10 = Greek 11 = Italian 12 = Japanese 13 = Korean 14 = Norwegian 15 = Polish 16 = Portuguese(Brazilian) 17 = Russian 18 = Spanish 19 = Sweedish
DD356	Speed Direction	DF52	4		00 = Forward (Ahead) 01 = Astern (Backward) 02 through 14 Reserved 15 = Data Not Available
DD357	Hardware Channel ID	DF53			0 through 251 = Identifier for a single “real” hardware resource, such as a temperature sensor or a single switch contact. Each label in a device will have a unique Hardware Channel ID. Values in this range are to be assigned to hardware resources sequentially beginning with 0 and increasing by one until all resources are assigned. 252 (0xFC) = Refers to the device as a whole, and the label associated with this value is the label for the entire device. 253 = Reserved 254 = Error 255 = Data Not Available
DD358	Secondary Enumeration Field Value	DF53			
DD359	Data Source Channel ID	DF53			0 through 252 = Identification of a data values used by the device, such as a value used in a calculation, a value logged when an event occurs, or a value converted/output as a physical stimulus. Data Source Channel IDs in this range are assigned sequentially to data requirements beginning with 0 and increasing until all data requirements are assigned.

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD360	Source Selection Status	DF52	2		0b00 = No source selected for this channel, NAME and mask if provided represent template to match. 0b01 = Source selected for this channel, NAME identifies source, mask if provided identifies template. 0b10 = Reserved. 0b11 = No Data Available.
DD361	Event Time- Short	DF111			Interval of time remaining before event change
DD362	Operational Status & Control	DF52	3		xx1 = Locked- Disable/Enable ability to Change State x1x = Security-Disable/Enable ability to Lock/Unlock 0xx = Reserved where x = don't care Inactive states shall be transmitted as 0. Reserved and unsupported bits within the field shall be transmitted as 0.
DD363	Watermaker Operating State	DF52	6		0 = Stopped 1 = Starting 2 = Running 3 = Stopping 4 = Flushing 5 = Rinsing 6 = Initiating 7 = Manual Mode 62 = Error 63 = Unavailable
DD364	Sensor Status	DF52	2		MSB/LSB: 00 = OK 01 = Warning 10 = Error 11= [Unavailable, Unknown]
DD365	Salinity, Watermaker	DF112			
DD366	Pressure, Watermaker	DF113			
DD367	Voltage, DC, High Range/Resolution	DF106			6553.3 = Reserved 6553.4 = Out of Range 6553.5 = Not Available or Do Not Change
DD368	Current, DC, High Range/Resolution	DF114			83886.06 = Reserved 83886.07 = Out of Range 83886.08 = Not Available or Do Not Change
DD369	Man Over Board (MOB) Status	DF52	3		
DD370	Position Source	DF52	3		Identifies the source of the position information 0=MOB Position estimated by the Vessel 1=MOB position reported by MOB emitter 2-5= Reserved 6=Error 7=Unknown / Unavailable

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD371	Battery Status	DF52	3		0=Good 1=Low 2–5=Reserved 6=Error 7=Unavailable
DD372	Heave	DF14			Vertical displacement perpendicular to (smooth, wave-free water on) the earth's surface. The height that a vessel is lifted by the crest of a wave (-), or the depth that a vessel drops in the trough of a wave (+). This convention conforms to the North-East-Down(NED) coordinate system, which is also referred to as positive down.
DD373	Delay	DF66			Delay added by calculations. When commanded: 0x0000 - 0xfff9 = Value within valid uint16 range 0xfffa = Set minimum value 0xfffb = Set maximum value 0xfffc = Restore Factory Defaults
DD374	Measurement Delay Source	DF52	4		0 = User Defined 1 = Minimum Supported Value 2 = Factory Default Value 3 = Maximum Supported Value 4 thru 13 = Reserved 14 = Data out of range 15 = Data not available
DD375	AES Encryption Key	DF51			This is an ASCII string containing Hex characters. A 128-bit Key would require 32 ASCII Hex characters for a total string size of 35 characters including the two count bytes and control byte. A 192-bit key would require 48 ASCII Hex characters for a total string size of 51 characters. A 256-bit key would require 64 ASCII Hex characters for a total string size of 67 characters. A 512-bit key would require 128 ASCII Hex characters for a total string size of 131 characters. A 1024-bit key would require 256 ASCII Hex characters for a total string size of 259 characters.
DD376	Command / Report	DF52	3		0 = Command 1 = Report 3 – 5 = Reserved 6 = Error 7 = Data not available

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD378	Transmit Mode of Operation	DF52	3		0 = Normal: When in this mode of operation the AIS transponder defaults to operate in Autonomous & Continuous mode, unless switched to Assigned or Polled mode. The transponder sends and receives AIS messages and eAIS messages. 1 = Receive-Only: When in this mode of operation the AIS transponder does not transmit any AIS messages, regardless of any commands received from base stations or interrogations from other vessels. The transponder receives AIS messages and eAIS messages, but transmits nothing (maintaining radio silence). 2 = Restricted: When in this mode of operation the AIS transponder only transmits eAIS messages, while no non-eAIS messages are sent. The transponder continues to receive AIS messages and eAIS messages. 3 – 5 = Reserved for future assignments 6 = Error 7 = Data not available
DD379	Encrypted Message Selection	DF52	3		0 = ITU-R M.1371 Messages 25 and 26 1 = ITU-R M.1371 Messages 6 and 8 2 – 5 = Reserved for future assignments 6 = Error 7 = Data not available
DD380	Non Volatile Persistence	DF52	3		0 = Transponder will default to last known Mode of Operation when power is cycled. 1 = Transponder will default to Restricted Mode of Operation when power is cycled. 2 -5 = Reserved for future assignments 6 = Error 7 = Data Not available
DD381	Type of AIS Asset	DF52	3		0 = Vessel 1 = Aircraft 2 -5 = Reserved for future assignment 6 = Error 7 = Data not available
DD382	AES Encryption Key Size	DF52	3		0 = 128 bits 1 = 192 bits 2 = 256 bits 3 – 5 = Reserved for future assignment 6 = Error 7 = Data not available
DD383	Data Interval	DF116			Reporting interval in seconds. 0xFFFD=Reserved 0xFFFE=Error 0xFFFF=Data not available
DD384	CAN State	DF52	2		00= error active 01= error passive 10= bus off 11= not available Represents the worst state detected since the last transmission of this information.

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD385	Device Status	DF52	2		00= operational 01= fault 10= reserved 11= not available Operational = device has no active error conditions Fault = active error conditions exist
DD386	AIS Binary Structure	DF52	1		0 = unstructured binary data (no Application Identifier bits used) 1 = binary data coded as defined by using the 16-bit Application Identifier
DD387	AIS Destination Indicator	DF52	1		0 = Broadcast (no Destination ID field used) 1 = Addressed (Destination ID field uses 30 data bits for MMSI
DD388	AIS Position Latency Flag	DF52	1		0 = reported position latency is less than 5 seconds 1 = reported position latency is greater than 5 seconds = default
DD389	Audio/Video Source Type	DF52	8		0 = Vessel Alarm 1 = AM 2 = FM 3 = Weather 4 = DAB 5 = Aux 6 = USB 7 = CD 8 = MP3 9 = Apple iOS 10 = Android 11 = Bluetooth 12 = Sirius XM 13 = Pandora 14 = Spotify 15 = Slacker 16 = Songza 17 = Apple Radio 18 = Last FM 19 = Ethernet 20 = Video MP4 21 = Video DVD 22 = Video BlueRay 23 = HDMI 24 = Video 25 - 252 = User defind 253 = Reserved 254 = Error 255 = Not available

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD390	Play Status	DF52	8		0 = Play (Normal functionality) 1 = Pause 2 = Stop 3 = FF (1x) 4 = FF (2x) 5 = FF (3x) 6 = FF (4x) 7 = RW (1x) 8 = RW (2x) 9 = RW (3x) 10 = RW (4x) 11 = Skip Ahead 12 = Skip Back 13 = Jog Ahead 14 = Jog Back 15 = Seek Up 16 = Seek Down 17 = Scan Up 18 = Scan Down 19 = Tune Up 20 = Tune Down 21 = Slow Motion (.75x) 22 = Slow Motion (.5x) 23 = Slow Motion (.25x) 24 = Slow Motion (.125x) 25 - 252 = User Defined 253 = Reserved 254 = Error 255 = Not available
DD391	Time, Standard	DF80			Time, 1 Second Resolution
DD392	Repeat	DF52	4		0 = Off 1 = One (Current File) 2 = All (Play Queue) 3 - 14 = Reserved 15 = Data Not Available / Do Not Change
DD393	Shuffle	DF52	4		0 = Off 1 = Play Queue 2 = All 3 - 14 = Reserved 15 = Data Not Available / Do Not Change
DD394	Thumbs Up / Thumbs Down	DF52	8		0 = None 1 = Thumbs Up 2 = Thumbs Down 3 = Reserved
DD395	Library Data Type	DF52	8		0 = File 1 = Playlist Name 2 = Genre Name 3 = Album Name 4 = Artist Name 5 = Track Name 6 = Station Name 7 = Station Number 8 = Favorite Number 9 = Play Queue 10 - 252 = Reserved

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD396	Entertainment Zones	DF52	8		0 = All Zones 1 = Zone 1 2 = Zone 2 3 = Zone 3 4 - 252 = User Defined
DD397	ID Type	DF52	8		0 = Group 1 = File 2 = Reserved 3 = Do Not Use / Data Not Available
DD398	Default Setting	DF52	2		0 = Save current settings as user default 1 = Load user default (soft reset) 2 = Load manufacturer default (hard reset) 3 = Data Not Available / Do Nothing
DD399	Tuner Regions	DF52	4		0 = US 1= Europe 2 = Asia 3 = Middle East 4 = Latin America 5 = Australia 6 = Russia 7 - 15 = Reserved
DD400	Audio Volume – Relative	DF52	2		0 = Volume Up 1 = Volume Down 2 = Reserved 3 = Do Not Change/Data Not Available
DD401	EQ Preset Type	DF52	8		0 = Flat 1 = Rock 2 = Hall 3 = Jazz 4 = Pop 5 = Live 6 = Classic 7 = Vocal 8 = Arena 9 = Cinema 10 - 252 = User Defined
DD402	Audio Filter Setting	DF52	8		0 = Full Range 1 = High Pass 2 = Low Pass 3 = Band Pass 4 = Notch Filter 5 - 15 = Reserved

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD403	Supported Play Status	DF52	32		x̄xxx xxxx xxx xxxx xxxx xxxx xxxx xxx1 = Play (Normal) x̄xxx xxxx xxx xxxx xxxx xxxx xxxx xx1x = Pause x̄xxx xxxx xxx xxxx xxxx xxxx xxxx x1xx = Pause x̄xxx xxxx xxxx xxx xxxx xxxx xxxx 1xxx = FF (1x) x̄xxx xxxx xxxx xxx xxxx xxxx xxx1 xxxx = FF (2x) x̄xxx xxxx xxxx xxx xxxx xxxx xx1x xxxx = FF (3x) x̄xxx xxxx xxxx xxx xxxx xxxx x1xx xxxx = FF (4x) x̄xxx xxxx xxxx xxx xxxx xxxx 1xxx xxxx = RW (1x) x̄xxx xxxx xxxx xxx xxxx xxx1 xxxx xxxx = RW (2x) x̄xx xxxx xxxx xxxx xxxx xx1x xxxx xxxx = RW (3x) x̄xxx xxxx xxxx xxx xxxx x1xx xxxx xxxx = RW (4x) x̄xxx xxxx xxx xxxx xxxx 1xxx xxxx xxxx = Skip Ahead x̄xxx xxxx xxxx xxx xxx1 xxxx xxxx xxxx = Skip Back x̄xxx xxxx xxx xxxx xx1x xxxx xxxx xxxx = Jog Ahead x̄xxx xxxx xxx xxxx x1xx xxxx xxxx xxxx = Jog Back x̄xxx xxxx xxx xxxx 1xxx xxxx xxxx xxxx = Seek Up x̄xxx xxxx xxx xxx1 xxxx xxxx xxxx xxxx = Seek Down x̄xxx xxxx xxxx x1x xxxx xxxx xxxx xxxx = Scan Up x̄xxx xxxx xxx x1xx xxxx xxxx xxxx xxxx = Scan Down x̄xxx xxxx xxx 1xxx xxxx xxxx xxxx xxxx = Tune Up x̄xxx xxxx xxx1 xxx xxxx xxxx xxxx xxxx = Tune Down x̄xxx xxxx xx1x xxx xxxx xxxx xxxx xxxx = Slow Mot. (.75x) x̄xxx xxxx x1xx xxx xxxx xxxx xxxx xxxx = Slow Mot. (.5x) x̄xxx xxxx 1xxx xxx xxxx xxxx xxxx xxxx = Slow Mot. (.25x) x̄xxx xxx1 xxx xxxx xxxx xxxx xxxx xxxx = Slow Mot. (.125x) x̄xxx xx1x xxxx xxx xxxx xxxx xxxx xxxx = Reserved x̄x̄xx x1xx xxxx xxxx xxxx xxxx xxxx xxxx = Reserved x̄xx 1xxx xxxx xxxx xxxx xxxx xxxx xxxx = Reserved x̄x̄x1 xxxx xxxx xxx xxxx xxxx xxxx xxxx = Reserved x̄x̄1x xxx xxxx xxxx xxxx xxxx xxxx xxxx = Reserved x1x̄xx xxxx xxxx xxx xxxx xxxx xxxx xxxx = Reservd 1x̄xx xxxx xxx xxxx xxxx xxxx xxxx xxxx = Reservd
DD404	Supported Browsing Methods	DF52	16		xxxx xxxx xxxx xxx1 = Filename xxxx xxxx xxxx xx1x = Playlist xxxx xxxx xxxx x1xx = Genre xxxx xxxx xxxx 1xxx = Album xxxx xxxx xxx1 xxxx = Artist xxxx xxxx xx1x xxxx = Track x̄xxx xxxx x1xx xxx = Station Name x̄xxx xxxx 1xxx xxx = Station Number x̄xxx xxx1 xxxx xxx = Favorite Number x̄xxx xx1x xxxx xxx = Play Queue xxxx x1xx xxxx xxxx = Reserved xxxx 1xxx xxxx xxxx = Reserved xxx1 xxxx xxxx xxxx = Reserved xx1x xxxx xxxx xxxx = Reserved x1xx xxxx xxxx xxxx = Reserved 1xxx xxxx xxxx xxxx = Reserved
DD405	Repeat Supported	DF52	4		xxx1 = Song xx1x = Play Queue x1xx = Reserved 1xxx = Reserved
DD406	Bluetooth Pairing Status	DF52	4		0 = Discovering 1 = Connect 2 = Connecting (read only) 3 = Not Connected / Disconnect 4 - 15 = Reserved

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DD #	DD Name	DF #	Bits	Chars	Data Dictionary Definition
DD407	Bluetooth Device Status	DF52	8		0 = Connected 1 = Not Connected (In Memory) 2 = Not Paired (Available) 3 - 14 = Reserved 15 = Data Not Available
DD408	Bluetooth Device Address	DF117			Integer 48bit unsigned, Unit-less number
DD409	Shuffle Supported	DF52	4		xxx1 = Play Queue xx1x = All x1xx = Reserved 1xxx = Reserved
DD410	Engine / Transmission Instance	DF52	8		This data instance field shall be enumerated on a globally unique basis beginning on the Port side with instance number 0 and incrementing towards Starboard. Devices in Propulsion Class 50 and Function Code 140 shall be the first enumeration sequence (i.e. 0, 1, 2, 3, ...n) prior to moving to a different Class and Function Code. Devices of the next Class and Function Code shall continue the enumeration sequence (i.e. n+1, n+2,...).
DD411	User Defined Alert Assignment	DF52	2		0 = Instantiate (new) User Defined Alert 1 = Remove (existing) User Defined Alert 2 = Null – not applicable

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National Marine Electronics Association



NMEA 2000 ®

Appendix B.3 -- Data Format

STANDARD FOR SERIAL-DATA NETWORKING OF MARINE ELECTRONIC DEVICES

NMEA 2000® -- is a Registered Trademark of the National Marine Electronics Association, Inc.

Version 2.100
February 2015

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DF #	DF Description	Range	Resolution	Type	Data Format Notes
DF00	Undefined	undefined	undefined	Undefined	Application specific, defined at time of use.
DF01	Acceleration	+/- 327.64 m/s*s	1x10E-2 m/s*s	int16	1G = 9.80665 m/s*s
DF02	Angle	0 to 2Pi rad	1x10E-4 rad	uint16	Resolution ~0.0057deg, 1 deg = .01745 rad
DF03	Angle, precise	Variable radians	Floats, radian	float32	Resolution ~1x10E-6 rad at range of 2 Pi
DF04	Angle, signed	+/-Pi rad	1x10E-4 rad	int16	Resolution ~0.0057deg
DF05	Angular acceleration	+/- 3,276.4 rad/s*s	1x10E-1 rad/s*s	int16	
DF06	Time of day	0 to 86,401 s	1x10E-4 s	uint32	~24 hours, 0 = midnight, range allows for up to two leap seconds per day
DF07	Current, electric, high	+/- 3,276.4 A	1x10E-1 A	int16	
DF08	Current, electric, low	+/- 3.2764 A	1x10E-4 A	int16	
DF09	Distance	0 to ~4.295x10E+7 m	1x10E-2 m	uint32	
DF10	Distance, length, signed	+/- 3.2764 m	1x10E-4 m	int16	
DF100	Battery Capacity	0 - 235,915,200 Coulombs	3600 C	uint16	This maps directly into Ampere Hours (AH) where 3600C = 1AH.
DF101	Peukert Exponent	1 - 1.5	2x10E-3	uint8	
DF102	AC Vrms	0 - 65.532	1x10E-3 V	uint16	
DF103	Pressure	+/- 2.1E8	1x10E-1 Pa	int32	

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DF #	DF Description	Range	Resolution	Type	Data Format Notes
DF104	Mass	0 - 429,496.7292 kilograms	1x10E-4 kg	uint32	
DF105	Temperature, extended range and precision	0 to 16,777.212 deg K	0.001 deg K	uint24	
DF106	Voltage AC- High Range	0 - 6553.2 V	0.1V	uint16	
DF107	Frequency-Medium	0-6553.2 Hz	0.1 Hz	uint16	
DF108	Power- Extended Range	-2,147,483,648 to +2,147,483,644 Watts	1W	int32	
DF109	Distance, Rough Approximation	0 - 2,520 meters	10 meters	uint8	
DF11	Distance, long	0 to ~4.295x10E+9 m	1 m	uint32	
DF110	PWM Duty Cycle	0-100%	1%	uint8	
DF111	Time Interval Medium-Short	0.00 - 2.52 seconds	0.01 seconds	uint8	
DF112	Salinity	0 to 65,535 ppm	1 ppm	uint16	
DF113	Pressure, Compound	+/- 32,764 Kpa	1 Kpa	int16	
DF114	Current, DC, high range/resolution	-83886.08 A to 83886.05 A	0.01A	int24	
DF115	Bit field defaulting to zero when NULL	N2KUnitless	1	bit0(n)	Special data format applicable to AIS field that must be set to zero for a NULL data condition or when unused - applies to AIS spare fields and AIS fields that are for future use
DF116	Time Interval, .001 sec.	0 - 65.532s	1x10E-3 sec.	uint16	

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DF #	DF Description	Range	Resolution	Type	Data Format Notes
DF117	Integer 48 bit unsigned	0 to 281474976710656	1	uint48	Unit-less number
DF12	Distance, long, precise	Variable meters	Floats, meters	float64	Resolution ~0.1 u-meter at 99,000 km
DF13	Distance, short	0 to 655.32 m	1x10E-2 m	uint16	
DF14	Distance, short, signed	+/-327.64 m	1x10E-2 m	int16	
DF15	Distance, signed	+/-~2.147x10E+7 m	1x10E-2 m	int32	
DF16	Electric field	+/-327.64 dB re: uV/m	1x10E-2 dB re: uV/m	int32	
DF17	Flow rate, high	+/-32.764 cu-m/hr	1x10E-3 cu-m/hr	int16	
DF18	Flow rate, low	+/-3.2764 cu-m/hr	1x10E-4 cu-m/hr	int16	
DF19	Force, high	+/- 327,640 N	10 N	int16	"-" = compression
DF20	Force, low	+/- 3,276.4 N	1x10E-1 N	int16	"-" = compression
DF21	Frequency	0 to ~4.295x10E+10 Hz	10 Hz	uint32	
DF22	Frequency, low	0 to 655.32 Hz	1x10E-2 Hz	uint16	
DF23	Latitude	+/- 90 deg	1x10E-7 deg	int32	"-" = South, resolution ~1.1 cm
DF25	Longitude	+/- 180 deg	1x10E-7 deg	int32	"-" = West, resolution ~1.1 cm
DF26	Frequency, mid	0 to 65,532 Hz	1 Hz	uint16	
DF27	Magnetic flux density	0 to 6.5532x10E-2T	1x10E-6 T	uint16	1 Tesla = 10,000 gauss

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DF #	DF Description	Range	Resolution	Type	Data Format Notes
DF28	Power	0 to 65,532 W	1 W	uint16	
DF29	Pressure	0 to 65,532,000 Pa	1x10E+3 Pa	uint16	
DF30	Percent, Relative measure	+/- 124%	1%	int8	
DF31	dB, relative measure	+/- 327.64 dB	1x10E-2 dB	int16	
DF32	Resistance, electric, high	0 to ~4.295x10E+9 Ohm	1 Ohm	uint32	
DF33	Resistance, electric, low	0 to 655.32 Ohm	1x10E-2 Ohm	uint16	
DF34	Rotational rate	+/- 32,764 RPM	1 RPM	int16	
DF35	Speed	0 to 655.32 m/s	1x10E-2 m/s	uint16	1 Knot = 0.5144 m/s
DF36	Speed, signed	+/-32.764 m/s	1x10E-3 m/s	int16	
DF38	Temperature, high	0 to 6,553.2 deg K	1x10E-1 deg K	uint16	
DF39	Temperature, low	0 to 655.32 deg K	1x10E-2 deg K	uint16	
DF40	Time interval, signed, standard	+/- ~2.148x10E+6 s	1x10E-3 s	int32	
DF41	Date, day count	0 to 65,532 days	1 day	uint16	0 = January 1, 1970, max = ~179 years
DF42	Voltage, high	+/- 327.64 V	1x10E-2 V	int16	
DF43	Voltage, low	+/- 3.2764 V	1x10E-4 V	int16	
DF44	Volume	0 to 65.532 cu m	1x10E-3 cu m	uint16	

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DF #	DF Description	Range	Resolution	Type	Data Format Notes
DF45	Time interval, precise	+/- 2.14 s	1x10E-9 s	int32	
DF46	Distance, signed, medium	+/- 32.764 m	1x10E-3 m	int16	
DF47	Pressure, medium	0 to 6,553,200 Pa	1x10E+2 Pa	uint16	
DF48	Pressure rate	+/- 327,640 Pa/hr	1x10E+1 Pa/hr	int16	+ = increasing rate
DF49	Ratio, Relative measure	Variable	Floats	float32	Unit-less number
DF50	String, variable, short	0 to 250 ASCII or 0 to 125 Unicode Characters	1 ASCII or 1 Unicode Character	ch8or16(n)	2 to 252 bytes. First byte in string (uint8) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Second byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 2 bytes, i.e. Count = 2) is a null string.
DF51	String, variable, medium	0 to 1,782 ASCII or 0 to 891 Unicode Characters	1 ASCII or 1 Unicode Character	ch8or16(n)	3 to 1,785 bytes. First and Second bytes in string (unit16) is the Count byte indicating the number of bytes in the string, including the Count and Control bytes. Third byte in string is the Control byte. The Control byte indicates if the string consists of ASCII characters (Char8) or Unicode characters (Char16). Control byte = 0 => Unicode characters Control byte = 1 => ASCII characters A string with no characters (total length of 3 bytes, i.e. Count = 3) is a null string.

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DF #	DF Description	Range	Resolution	Type	Data Format Notes
DF52	Bit field	Variable	1	bit(n)	Used to construct bit fields
DF53	Integer, 8 bit unsigned	0 to 252	1 bit	uint8	Unit-less number
DF54	Integer, 16 bit unsigned	0 to 65,532	1 bit	uint16	Unit-less number
DF55	Integer, 32 bit unsigned	0 to 4,294,967,292	1 bit	uint32	Unit-less number
DF56	Integer, 64 bit unsigned	0 to (2E+64)-4	1 bit	uint64	Unit-less number
DF57	Integer, 8 bit 2's complement	-128 to +124	1 bit	int8	Unit-less number
DF58	Integer, 16 bit 2's complement	-32,768 to +32,764	1 bit	int16	Unit-less number
DF59	Integer, 32 bit 2's complement	-2,147,483,648 to +2,147,483,644	1 bit	int32	Unit-less number
DF60	Integer, 64 bit 2's complement	-(2E+63) to +(2E+63)-4	1 bit	int64	Unit-less number
DF61	Floating point, IEEE 754 single precision	Variable	Floats	float32	Unit-less number
DF62	Floating point, IEEE 754 double precision	Variable	Floats	float64	Unit-less number
DF63	String, fixed	0 to 1,785 characters	1 char	char8(n)	0 to 1,785 bytes. Character count not included, length is specified by application in Data Dictionary. Unused characters shall be treated as not available, and filled with [0xFF] = Data not available.

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DF #	DF Description	Range	Resolution	Type	Data Format Notes
DF64	Floating point, IEEE 754 expanded	Variable	Floats	float80	Unit-less number
DF65	Time interval, standard	0 to ~4.295x10E+6 s	1x10E-3 s	uint32	
DF66	Time interval, .01sec	0 to 655.32s	1x10E-2sec	uint16	
DF67	Time interval, large	0 to ~4.295x10E+9 s	1 sec	uint32	
DF68	Power, low	0 to 65.532 W	1x10E-3 W	uint16	
DF69	Ratio, Relative measure, small	+/-327.64	1x10E-2	int16	Unit-less number
DF70	Angle, tiny	Variable radians	Floats, radian	float32	
DF71	Time interval, medium	+/-32,764 minutes	1.0 minute	int16	
DF72	Rotational rate, unsigned	0-16,383 RPM	1/4 RPM	uint16	
DF73	Angular rate, signed	+/-1.0 rad/s	1/32 x 10E-3 rad/s	int16	Resolution 0.1 deg/min
DF74	Distance, rough	+/-32,764 m	1 m	int16	
DF75	Distance, Medium	0 to 6553.2 m	1x10E-1 m	uint16	
DF76	Latitude (Extended)	+/- 90 deg	1x10E-16 deg	int64	"-" = South, resolution ~.01 nanometer
DF77	Longitude (Extended)	+/- 180 deg	1x10E-16 deg	int64	"-" = West, resolution ~.01 nanometer
DF78	Distance (Extended)	+/-~9.223x10E+12 m	1x10E-6 m	int64	

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DF #	DF Description	Range	Resolution	Type	Data Format Notes
DF79	Distance signed fine	+/-~2.147x10E+4 m	1x10E-5 m	int32	
DF80	Time, 1sec	0 to 65532 seconds	1 second	uint16	
DF81	Time, 0.1sec	0 to 6553.2 seconds	1x10E-1 sec	uint16	
DF82	Speed, signed 4dpt	+/-3.2764 m/s	1x10E-4 m/s	int16	
DF83	Distance, signed 4dp	+/-~2.147x10E+5 m	1x10E-4 m	int32	
DF84	Percent, Relative Measure, Medium	-131.072% to 131.056%	4x10E-3 %	int16	
DF85	Angular rate, signed - Precise	+/-67.0 rad/s (approx. 230703 deg/min	1/32 x 10E-6 rad/s	int32	
DF86	Volume. Large	0 to~4.296x10E+5 cu m	1x10E-4 cu m	uint32	
DF87	Speed, signed - large	+/- 327.66 m/s (+/- 636 knots)	1x10E-2 m/s	int16	
DF88	Time Interval, short	0 to 1.26 sec	5x10E-3 sec	uint8	
DF89	Latitude, 24 bit	+/-83 sec (")	1x10E-5 sec(")	int24	
DF90	Longitude, 24 bit	+/-83 sec (")	1x10E-5 sec(")	int24	
DF91	Altitude, 24 bit	+/-8,388m	1x10E-3m	int24	
DF92	Power - VAR	0 - 4,294,967,292 VAR	1 VAR	uint32	
DF93	Percent, Absolute	0 - 252%	1%	uint8	
DF94	Power	0 - 4,294,967,292 W	1 W	uint32	

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DF #	DF Description	Range	Resolution	Type	Data Format Notes
DF95	Current, electric, high	0 - 6553.2 A	1x10E-1 A	uint16	
DF96	Voltage, high, unsigned	+/- 655.32 V	1x10E-2 V	uint16	
DF97	Power Factor	+/- 1.00	1x10E-2	int8	
DF98	Time interval, medium, unsigned	0 - 65,532 minutes	1 minute	uint16	
DF99	Power	0 - 65,532 watts	1 W	uint16	

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National Marine Electronics Association



NMEA 2000 ®

Appendix B.4 -- Data Type

STANDARD FOR SERIAL-DATA NETWORKING OF MARINE ELECTRONIC DEVICES

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Version 2.100
February 2015

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Type	Size	Description	Range	Notes
int16	2	16 bit 2's complement integer	-32,768 to +32,767	Extreme positive range of numbers reserved: +32,767 = Data not available or Do Not Change; +32,766 = Out of range; +32,765 = Reserved
char16(n)	2char	16 bit characters, Unicode16 Standard (ISO 10646)	character set	Reserved characters: 0x009F = Not Available or Null; 0x009E = Reserved
uint16	2	16 bit unsigned integer	0 to 65,535	Extreme positive range of numbers reserved: 65,535 = Data not available or Do Not Change; 65,534 = Out of range; 65,533 = Reserved
int24	3	24 bit signed number	-8,388,608 to 8,388,607	Extreme positive range of numbers reserved: +8,388,607 = Data not available or Do Not Change; +8,388,606 = Out of range; +8,388,605 = Reserved
uint24	3	24 bit unsigned integer	0 to 16,777,215	Extreme positive range of numbers reserved: 16,777,215 = Data not available or Do Not Change; 16,777,214 = Out of range; 16,777,213 = Reserved
int32	4	32 bit 2's complement integer	-2,147,483,648 to +2,147,483,647	Extreme positive range of numbers reserved: +2,147,483,647 = Data not available or Do Not Change; +2,147,483,646 = Out of range; +2,147,483,645 = Reserved
uint32	4	32 bit unsigned integer	0 to 4,294,967,295	Extreme positive range of numbers reserved: 4,294,967,295 = Data not available or Do Not Change; 4,294,967,294 = Out of range; 4,294,967,293 = Reserved
uint48	6	48 bit unsigned integer	0 to 281474976710656	Extreme positive range of numbers reserved: 281474976710656 = Data not available or Do Not Change; 281474976710655 = Out of range; 281474976710654 = Reserved
int64	8	64 bit 2's complement integer	-(2E+63) to +(2E+63)-1	Extreme positive range of numbers reserved: Largest positive number = Data not available or Do Not Change; Next lowest positive value = Out of range; Second lowest positive value = Reserved
uint64	8	64 bit unsigned integer	0 to (2E+64)-1	Extreme positive range of numbers reserved: Largest positive number = Data not available or Do Not Change; Next lowest positive value = Out of range; Second lowest positive value = Reserved
int8	1	8 bit 2's complement integer	-128 to +127	Extreme positive range of numbers reserved: +127 = Data not available or Do Not Change; +126 = Out of range; +125 = Reserved
char8(n)	char	8 bit characters, ISO 8859-1 (not limited to ASCII	character set	Reserved characters: [0xFF] = Data not available or Null; [0x00] = Reserved
uint8	1	8 bit unsigned integer	0 to 255	Extreme positive range of numbers reserved: 255 = Data not available or Do Not Change; 254 = Out of range; 253 = Reserved
bit(n)		bit field	0 to 1	When available, high-order multi-bit field values are reserved: Highest value = Data not available or Do Not Change; Next lowest value = Reserved. Bit fields may be used to represent numerical values requiring the full range (e.g., the integer values 253 to 255 not available in uint8) of a number, additional bits may be added to the field for status (e.g., 9th and 10th bits to an 8-bit number) if necessary.
bit0(n)		bit field	0 to 1	Used in PGNs to defined a bit field whose "Null data" value is zero. This relevant to AIS fields.
ch8or16(n)	8 or 16	char8(n) or char16(n)	character set	
float64	8	IEEE 754 double precision floating point		Extreme range of "Not a Number" reserved: 0x7FFF FFFF FFFF FFFF = Data not available or Do Not Change; 0x7FFF FFFF FFFF FFFE = Out of range; 0x7FFF FFFF FFFF FFFD = Reserved
float80	10	IEEE 754 expanded floating point		Extreme range of "Not a Number" reserved: 0x7FFF FFFF FFFF FFFF FFFF = Data not available or Do Not Change; 0x7FFF FFFF FFFF FFFF FFFE = Out of range; 0x7FFF FFFF FFFF FFFF FFFD = Reserved

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Type	Size	Description	Range	Notes
float32	4	IEEE 754 single precision floating point		Extreme range of “Not a Number” reserved: 0x7FFF FFFF = Data not available or Do Not Change; 0x7FFF FFFE = Out of range; 0x7FFF FFFD = Reserved
Undefined	?	Undefined	Application specific, defined when used	Application specific. Defined at time of use

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NMEA 2000 Appendix B-5

Alert Codes

Version 1.000

July 2013

STANDARD FOR SERIAL DATA NETWORKING OF MARINE ELECTRONIC DEVICES

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
1	Navigation	0	Safety Systems	0	Man Overboard	0	Man Overboard Active	0	000000000
2	Navigation	0		0			Man Over Board System unavailable	1	000000001
3	Technical	1		0	Watertight Door Controller	1	Hydraulic fluid reservoir level low	0	100010000
4	Technical	1		0			Gas pressure low	1	100010001
5	Technical	1		0			Electrical power loss	2	100010002
6	Technical	1		0	Hull (shell) door controller	2	Door open or locking device not secured (representative)	0	100020000

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
7	Technical	1		0			Power fail	1	100020001
8	Technical	1		0	Fire door Controller	3	System abnormal	0	100030000
9	Technical	1		0			Power fail	1	100030001
10	Technical	1		0	Fire Heat detection type	4	System fail	0	100040000
11	Technical	1		0			Power fail	1	100040001
12	Technical	1		0	Fire Smoke detection type	5	System fail	0	100050000
13	Technical	1		0			Power fail	1	100050001

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
14	Technical	1	Electrical Generation	1	Electric power generator plant	0	Voltage – Low	0	101000000
15	Technical	1		1			Voltage - High	1	101000001
16	Technical	1		1			Voltage - Unavailable	2	101000002
17	Technical	1		1			Current – Low	3	101000003
18	Technical	1		1			Current - High	4	101000004
19	Technical	1		1			Current - Unavailable	5	101000005
20	Technical	1		1			Frequency - Low	6	101000006

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
21	Technical	1		1			Frequency - High	7	101000007
22	Technical	1		1			Frequency - Unavailable	8	101000008
23	Technical	1		1			Failure of online generator	9	101000009
24	Technical	1		1			Failure of online generator - Unavailable	10	10100000A
25	Technical	1		1			Bearing lub. oil inlet pressure – Low	11	10100000B
26	Technical	1		1			Bearing lub. oil inlet pressure – High	12	10100000C
27	Technical	1		1			Bearing lub. oil inlet pressure – Unavailable	13	10100000D

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
28	Technical	1		1			Generator cooling inlet pump or fan motor fail	14	10100000E
29	Technical	1		1			Generator cooling inlet pump or fan motor fail - Unavailable	15	10100000F
30	Technical	1		1			Generator cooling medium temperature – Low	16	101000010
31	Technical	1		1			Generator cooling medium temperature – High	17	101000011
32	Technical	1		1			Generator cooling medium temperature – Unavailable	18	101000012
33	Technical	1	Electrical Generation	1	Electric power generator plant - Coolant	1	Coolant Level - Low	0	101010000
34	Technical	1		1			Coolant Level - High	1	101010001

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
35	Technical	1		1			Coolant Level - Unavailable	2	101010002
36	Technical	1		1			Coolant Flow - High	3	101010003
37	Technical	1		1			Coolant Flow - Low	4	101010004
38	Technical	1		1			Coolant Flow - Unavailable	5	101010005
39	Technical	1		1			Coolant Pressure - Low	6	101010006
40	Technical	1		1			Coolant Pressure - High	7	101010007
41	Technical	1		1			Coolant Pressure - Unavailable	8	101010008

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
42	Technical	1		1			Coolant Temperature - Low	9	101010009
43	Technical	1		1			Coolant Temperature - High	10	10101000A
44	Technical	1		1			Coolant Temperature - Unavailable	11	10101000B
45	Technical	1		1			Oily contamination of Coolant	12	10101000C
46	Technical	1	Electrical Generation	1	Electric power generator plant - Seawater Cooling	2	Seawater Level - High	0	101020000
47	Technical	1		1			Seawater Level - Low	1	101020001
48	Technical	1		1			Seawater Level Sensor - Unavailable	2	101020002

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
49	Technical	1		1			Seawater Flow - Low	3	101020003
50	Technical	1		1			Seawater Flow - High	4	101020004
51	Technical	1		1			Seawater Flow Sensor - Unavailable	5	101020005
52	Technical	1		1			Seawater Pressure - Low	6	101020006
53	Technical	1		1			Seawater Pressure - High	7	101020007
54	Technical	1		1			Seawater Pressure Sensor - Unavailable	8	101020008
55	Technical	1		1			Seawater Temperature - Low	9	101020009

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
56	Technical	1		1			Seawater Temperature - High	10	10102000A
57	Technical	1		1			Seawater Temperature Sensor - Unavailable	11	10102000B
58	Technical	1		1			Oily contamination of Cooling Seawater	12	10102000C
59	Technical	1	Electrical Generation	1	AC Power Source - Bus	3	Average Current Low	0	101030000
60	Technical	1		1			Average Current High	1	101030001
61	Technical	1		1			Average Current Unavailable	2	101030002
62	Technical	1		1			Average Frequency Low	3	101030003

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
63	Technical	1		1			Average Frequency High	4	101030004
64	Technical	1		1			Average Frequency Unavailable	5	101030005
65	Technical	1		1			Average Line-Line Voltage Low	6	101030006
66	Technical	1		1			Average Line-Line Voltage High	7	101030007
67	Technical	1		1			Average Line-Line Voltage Unavailable	8	101030008
68	Technical	1		1			Average Line-Neutral Voltage Low	9	101030009
69	Technical	1		1			Average Line-Neutral Voltage High	10	10103000A

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
70	Technical	1		1			Average Line-Neutral Voltage Unavailable	11	10103000B
71	Technical	1		1			Phase A - Frequency Low	12	10103000C
72	Technical	1		1			Phase A - Frequency High	13	10103000D
73	Technical	1		1			Phase A - Frequency Unavailable	14	10103000E
74	Technical	1		1			Phase A - Phase B Voltage Low	15	10103000F
75	Technical	1		1			Phase A - Phase B Voltage High	16	101030010
76	Technical	1		1			Phase A - Phase B Voltage Unavailable	17	101030011

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
77	Technical	1		1			Phase A - Neutral Voltage Low	18	101030012
78	Technical	1		1			Phase A - Neutral Voltage High	19	101030013
79	Technical	1		1			Phase A - Neutral Voltage Unavailable	20	101030014
80	Technical	1		1			Phase B - Frequency Low	21	101030015
81	Technical	1		1			Phase B - Frequency High	22	101030016
82	Technical	1		1			Phase B - Frequency Unavailable	23	101030017
83	Technical	1		1			Phase B - Phase C Voltage Low	24	101030018

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
84	Technical	1		1			Phase B - Phase C Voltage High	25	101030019
85	Technical	1		1			Phase B - Phase C Voltage Unavailable	26	10103001A
86	Technical	1		1			Phase B - Neutral Voltage Low	27	10103001B
87	Technical	1		1			Phase B - Neutral Voltage High	28	10103001C
88	Technical	1		1			Phase B - Neutral Voltage Unavailable	29	10103001D
89	Technical	1		1			Phase C - Frequency Low	30	10103001E
90	Technical	1		1			Phase C - Frequency High	31	10103001F

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
91	Technical	1		1			Phase C - Frequency Unavailable	32	101030020
92	Technical	1		1			Phase C - Phase A Voltage Low	33	101030021
93	Technical	1		1			Phase C - Phase A Voltage High	34	101030022
94	Technical	1		1			Phase C – Phase A Voltage Unavailable	35	101030023
95	Technical	1		1			Phase C - Neutral Voltage Low	36	101030024
96	Technical	1		1			Phase C - Neutral Voltage High	37	101030025
97	Technical	1	Electrical Generation	1	AC Power Source - Generator	4	Average Current - Low	0	101040000

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
98	Technical	1		1			Average Current - High	1	101040001
99	Technical	1		1			Average Current - Unavailable	2	101040002
100	Technical	1		1			Average Frequency - Low	3	101040003
101	Technical	1		1			Average Frequency - High	4	101040004
102	Technical	1		1			Average Frequency - Unavailable	5	101040005
103	Technical	1		1			Average Line-Line - Low	6	101040006
104	Technical	1		1			Average Line-Line - High	7	101040007

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
105	Technical	1		1			Average Line-Line - Unavailable	8	101040008
106	Technical	1		1			Average Line-Neutral - Low	9	101040009
107	Technical	1		1			Average Line-Neutral - High	10	10104000A
108	Technical	1		1			Average Line-Neutral - Unavailable	11	10104000B
109	Technical	1		1			Phase A Apparent Power - Low	12	10104000C
110	Technical	1		1			Phase A Apparent Power - High	13	10104000D
111	Technical	1		1			Phase A Apparent Power - Unavailable	14	10104000E

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
112	Technical	1		1			Phase A Current - Low	15	10104000F
113	Technical	1		1			Phase A Current - High	16	101040010
114	Technical	1		1			Phase A Current - Unavailable	17	101040011
115	Technical	1		1			Phase A Frequency - Low	18	101040012
116	Technical	1		1			Phase A Frequency - High	19	101040013
117	Technical	1		1			Phase A Frequency - Unavailable	20	101040014
118	Technical	1		1			Phase A - Phase B Voltage - Low	21	101040015

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
119	Technical	1		1			Phase A - Phase B Voltage - High	22	101040016
120	Technical	1		1			Phase A - Phase B Voltage - Unavailable	23	101040017
121	Technical	1		1			Phase A - Neutral Voltage - Low	24	101040018
122	Technical	1		1			Phase A - Neutral Voltage - High	25	101040019
123	Technical	1		1			Phase A - Neutral Voltage - Unavailable	26	10104001A
124	Technical	1		1			Phase A - Power Factor - Low	27	10104001B
125	Technical	1		1			Phase A - Power Factor - High	28	10104001C

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
126	Technical	1		1			Phase A - Power Factor - Unavailable	29	10104001D
127	Technical	1		1			Phase A - Reactive Power - Low	30	10104001E
128	Technical	1		1			Phase A - Reactive Power - High	31	10104001F
129	Technical	1		1			Phase A - Reactive Power - Unavailable	32	101040020
130	Technical	1		1			Phase A - Real Power - Low	33	101040021
131	Technical	1		1			Phase A - Real Power - High	34	101040022
132	Technical	1		1			Phase A - Real Power - Unavailable	35	101040023

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
133	Technical	1		1			Phase B Apparent Power - Low	36	101040024
134	Technical	1		1			Phase B Apparent Power - High	37	101040025
135	Technical	1		1			Phase B Apparent Power - Unavailable	38	101040026
136	Technical	1		1			Phase B Current - Low	39	101040027
137	Technical	1		1			Phase B Current - High	40	101040028
138	Technical	1		1			Phase B Current - Unavailable	41	101040029
139	Technical	1		1			Phase B Frequency - Low	42	10104002A

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
140	Technical	1		1			Phase B Frequency - High	43	10104002B
141	Technical	1		1			Phase B Frequency - Unavailable	44	10104002C
142	Technical	1		1			Phase B - Phase C Voltage - Low	45	10104002D
143	Technical	1		1			Phase B - Phase C Voltage - High	46	10104002E
144	Technical	1		1			Phase B - Phase C Voltage - Unavailable	47	10104002F
145	Technical	1		1			Phase B - Neutral Voltage - Low	48	101040030
146	Technical	1		1			Phase B - Neutral Voltage - High	49	101040031

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
147	Technical	1		1			Phase B - Neutral Voltage - Unavailable	50	101040032
148	Technical	1		1			Phase B - Power Factor - Low	51	101040033
149	Technical	1		1			Phase B - Power Factor - High	52	101040034
150	Technical	1		1			Phase B - Power Factor - Unavailable	53	101040035
151	Technical	1		1			Phase B - Reactive Power - Low	54	101040036
152	Technical	1		1			Phase B - Reactive Power - High	55	101040037
153	Technical	1		1			Phase B - Reactive Power - Unavailable	56	101040038

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
154	Technical	1		1			Phase B - Real Power - Low	57	101040039
155	Technical	1		1			Phase B - Real Power - High	58	10104003A
156	Technical	1		1			Phase B - Real Power - Unavailable	59	10104003B
157	Technical	1		1			Phase C Apparent Power - Low	60	10104003C
158	Technical	1		1			Phase C Apparent Power - High	61	10104003D
159	Technical	1		1			Phase C Apparent Power - Unavailable	62	10104003E
160	Technical	1		1			Phase C Current - Low	63	10104003F

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
161	Technical	1		1			Phase C Current - High	64	101040040
162	Technical	1		1			Phase C Current - Unavailable	65	101040041
163	Technical	1		1			Phase C Frequency - Low	66	101040042
164	Technical	1		1			Phase C Frequency - High	67	101040043
165	Technical	1		1			Phase C Frequency - Unavailable	68	101040044
166	Technical	1		1			Phase C - Phase A Voltage - Low	69	101040045
167	Technical	1		1			Phase C - Phase A Voltage - High	70	101040046

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
168	Technical	1		1			Phase C - Phase A Voltage - Unavailable	71	101040047
169	Technical	1		1			Phase C - Neutral Voltage - Low	72	101040048
170	Technical	1		1			Phase C - Neutral Voltage - High	73	101040049
171	Technical	1		1			Phase C - Neutral Voltage - Unavailable	74	10104004A
172	Technical	1		1			Phase C - Power Factor - Low	75	10104004B
173	Technical	1		1			Phase C - Power Factor - High	76	10104004C
174	Technical	1		1			Phase C - Power Factor - Unavailable	77	10104004D

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
175	Technical	1		1			Phase C - Reactive Power - Low	78	10104004E
176	Technical	1		1			Phase C - Reactive Power - High	79	10104004F
177	Technical	1		1			Phase C - Reactive Power - Unavailable	80	101040050
178	Technical	1		1			Phase C - Real Power - Low	81	101040051
179	Technical	1		1			Phase C - Real Power - High	82	101040052
180	Technical	1		1			Phase C - Real Power - Unavailable	83	101040053
181	Technical	1		1			Total Apparent Power - Low	84	101040054

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
182	Technical	1		1			Total Apparent Power - High	85	101040055
183	Technical	1		1			Total Apparent Power - Unavailable	86	101040056
184	Technical	1		1			Total Power Factor - Low	87	101040057
185	Technical	1		1			Total Power Factor - High	88	101040058
186	Technical	1		1			Total Power Factor - Unavailable	89	101040059
187	Technical	1		1			Total Reactive Power - Low	90	10104005A
188	Technical	1		1			Total Reactive Power - High	91	10104005B

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
189	Technical	1		1			Total Reactive Power - Unavailable	92	10104005C
190	Technical	1		1			Total Real Power - Low	93	10104005D
191	Technical	1		1			Total Real Power - High	94	10104005E
192	Technical	1		1			Total Real Power - Unavailable	95	10104005F
193	Technical	1		1			Total kWh Export - Low	96	101040060
194	Technical	1		1			Total kWh Export - High	97	101040061
195	Technical	1		1			Total kWh Export - Unavailable	98	101040062

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
196	Technical	1		1			Total kWh Import - Low	99	101040063
197	Technical	1		1			Total kWh Import - High	100	101040064
198	Technical	1		1			Total kWh Import - Unavailable	101	101040065
199	Technical	1	Electrical Generation	1	AC Power Source - Utility (Shore Power)	5	Average Current - Low	0	101050000
200	Technical	1		1			Average Current - High	1	101050001
201	Technical	1		1			Average Current - Unavailable	2	101050002
202	Technical	1		1			Average Frequency - Low	3	101050003

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
203	Technical	1		1			Average Frequency - High	4	101050004
204	Technical	1		1			Average Frequency - Unavailable	5	101050005
205	Technical	1		1			Average Line-Line - Low	6	101050006
206	Technical	1		1			Average Line-Line - High	7	101050007
207	Technical	1		1			Average Line-Line - Unavailable	8	101050008
208	Technical	1		1			Average Line-Neutral - Low	9	101050009
209	Technical	1		1			Average Line-Neutral - High	10	10105000A

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
210	Technical	1		1			Average Line-Neutral - Unavailable	11	10105000B
211	Technical	1		1			Phase A Apparent Power - Low	12	10105000C
212	Technical	1		1			Phase A Apparent Power - High	13	10105000D
213	Technical	1		1			Phase A Apparent Power - Unavailable	14	10105000E
214	Technical	1		1			Phase A Current - Low	15	10105000F
215	Technical	1		1			Phase A Current - High	16	101050010
216	Technical	1		1			Phase A Current - Unavailable	17	101050011

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
217	Technical	1		1			Phase A Frequency - Low	18	101050012
218	Technical	1		1			Phase A Frequency - High	19	101050013
219	Technical	1		1			Phase A Frequency - Unavailable	20	101050014
220	Technical	1		1			Phase A - Phase B Voltage - Low	21	101050015
221	Technical	1		1			Phase A - Phase B Voltage - High	22	101050016
222	Technical	1		1			Phase A - Phase B Voltage - Unavailable	23	101050017
223	Technical	1		1			Phase A - Neutral Voltage - Low	24	101050018

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
224	Technical	1		1			Phase A - Neutral Voltage - High	25	101050019
225	Technical	1		1			Phase A - Neutral Voltage - Unavailable	26	10105001A
226	Technical	1		1			Phase A - Power Factor - Low	27	10105001B
227	Technical	1		1			Phase A - Power Factor - High	28	10105001C
228	Technical	1		1			Phase A - Power Factor - Unavailable	29	10105001D
229	Technical	1		1			Phase A - Reactive Power - Low	30	10105001E
230	Technical	1		1			Phase A - Reactive Power - High	31	10105001F

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
231	Technical	1		1			Phase A - Reactive Power - Unavailable	32	101050020
232	Technical	1		1			Phase A - Real Power - Low	33	101050021
233	Technical	1		1			Phase A - Real Power - High	34	101050022
234	Technical	1		1			Phase A - Real Power - Unavailable	35	101050023
235	Technical	1		1			Phase B Apparent Power - Low	36	101050024
236	Technical	1		1			Phase B Apparent Power - High	37	101050025
237	Technical	1		1			Phase B Apparent Power - Unavailable	38	101050026

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
238	Technical	1		1			Phase B Current - Low	39	101050027
239	Technical	1		1			Phase B Current - High	40	101050028
240	Technical	1		1			Phase B Current - Unavailable	41	101050029
241	Technical	1		1			Phase B Frequency - Low	42	10105002A
242	Technical	1		1			Phase B Frequency - High	43	10105002B
243	Technical	1		1			Phase B Frequency - Unavailable	44	10105002C
244	Technical	1		1			Phase B - Phase C Voltage - Low	45	10105002D

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
245	Technical	1		1			Phase B - Phase C Voltage - High	46	10105002E
246	Technical	1		1			Phase B - Phase C Voltage - Unavailable	47	10105002F
247	Technical	1		1			Phase B - Neutral Voltage - Low	48	101050030
248	Technical	1		1			Phase B - Neutral Voltage - High	49	101050031
249	Technical	1		1			Phase B - Neutral Voltage - Unavailable	50	101050032
250	Technical	1		1			Phase B - Power Factor - Low	51	101050033
251	Technical	1		1			Phase B - Power Factor - High	52	101050034

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
252	Technical	1		1			Phase B - Power Factor - Unavailable	53	101050035
253	Technical	1		1			Phase B - Reactive Power - Low	54	101050036
254	Technical	1		1			Phase B - Reactive Power - High	55	101050037
255	Technical	1		1			Phase B - Reactive Power - Unavailable	56	101050038
256	Technical	1		1			Phase B - Real Power - Low	57	101050039
257	Technical	1		1			Phase B - Real Power - High	58	10105003A
258	Technical	1		1			Phase B - Real Power - Unavailable	59	10105003B

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
259	Technical	1		1			Phase C Apparent Power - Low	60	10105003C
260	Technical	1		1			Phase C Apparent Power - High	61	10105003D
261	Technical	1		1			Phase C Apparent Power - Unavailable	62	10105003E
262	Technical	1		1			Phase C Current - Low	63	10105003F
263	Technical	1		1			Phase C Current - High	64	101050040
264	Technical	1		1			Phase C Current - Unavailable	65	101050041
265	Technical	1		1			Phase C Frequency - Low	66	101050042

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
266	Technical	1		1			Phase C Frequency - High	67	101050043
267	Technical	1		1			Phase C Frequency - Unavailable	68	101050044
268	Technical	1		1			Phase C - Phase A Voltage - Low	69	101050045
269	Technical	1		1			Phase C - Phase A Voltage - High	70	101050046
270	Technical	1		1			Phase C - Phase A Voltage - Unavailable	71	101050047
271	Technical	1		1			Phase C - Neutral Voltage - Low	72	101050048
272	Technical	1		1			Phase C - Neutral Voltage - High	73	101050049

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
273	Technical	1		1			Phase C - Neutral Voltage - Unavailable	74	10105004A
274	Technical	1		1			Phase C - Power Factor - Low	75	10105004B
275	Technical	1		1			Phase C - Power Factor - High	76	10105004C
276	Technical	1		1			Phase C - Power Factor - Unavailable	77	10105004D
277	Technical	1		1			Phase C - Reactive Power - Low	78	10105004E
278	Technical	1		1			Phase C - Reactive Power - High	79	10105004F
279	Technical	1		1			Phase C - Reactive Power - Unavailable	80	101050050

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
280	Technical	1		1			Phase C - Real Power - Low	81	101050051
281	Technical	1		1			Phase C - Real Power - High	82	101050052
282	Technical	1		1			Phase C - Real Power - Unavailable	83	101050053
283	Technical	1		1			Total Apparent Power - Low	84	101050054
284	Technical	1		1			Total Apparent Power - High	85	101050055
285	Technical	1		1			Total Apparent Power - Unavailable	86	101050056
286	Technical	1		1			Total Power Factor - Low	87	101050057

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
287	Technical	1		1			Total Power Factor - High	88	101050058
288	Technical	1		1			Total Power Factor - Unavailable	89	101050059
289	Technical	1		1			Total Reactive Power - Low	90	10105005A
290	Technical	1		1			Total Reactive Power - High	91	10105005B
291	Technical	1		1			Total Reactive Power - Unavailable	92	10105005C
292	Technical	1		1			Total Real Power - Low	93	10105005D
293	Technical	1		1			Total Real Power - High	94	10105005E

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
294	Technical	1		1			Total Real Power - Unavailable	95	10105005F
295	Technical	1		1			Total kWh Export - Low	96	101050060
296	Technical	1		1			Total kWh Export - High	97	101050061
297	Technical	1		1			Total kWh Export - Unavailable	98	101050062
298	Technical	1		1			Total kWh Import - Low	99	101050063
299	Technical	1		1			Total kWh Import - High	100	101050064
300	Technical	1		1			Total kWh Import - Unavailable	101	101050065

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
301	Technical	1	Electrical Generation	1	Battery Monitoring	6	State of Charge - Low	0	101060000
302	Technical	1		1			State of Charge -High	1	101060001
303	Technical	1		1			State of Charge - Unavailable	2	101060002
304	Technical	1		1			Temperature - Low	3	101060003
305	Technical	1		1			Temperature - High	4	101060004
306	Technical	1		1			Temperature - Unavailable	5	101060005
307	Technical	1		1			Time Remaining - Low	6	101060006

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
308	Technical	1		1			Time Remaining - High	7	101060007
309	Technical	1		1			Time Remaining - Unavailable	8	101060008
310	Technical	1		1			DC Discharge Current - Low	9	101060009
311	Technical	1		1			DC Discharge Current - High	10	10106000A
312	Technical	1		1			DC Discharge Current - Unavailable	11	10106000B
313	Technical	1		1			DC Power - Low	12	10106000C
314	Technical	1		1			DC Power - High	13	10106000D

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
315	Technical	1		1			DC Power - Unavailable	14	10106000E
316	Technical	1		1			DC Voltage - Low	15	10106000F
317	Technical	1		1			DC Voltage - High	16	101060010
318	Technical	1		1			DC Voltage - Unavailable	17	101060011
319	Technical	1		1			DC Charge Current - Low	18	101060012
320	Technical	1		1			DC Charge Current - High	19	101060013
321	Technical	1		1			DC Charge Current - Unavailable	20	101060014

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
322	Technical	1	Electrical Distribution	2	Circuit Breaker	0	On	0	102000000
323	Technical	1		2			Off	1	102000001
324	Technical	1		2			Tripped	2	102000002
325	Technical	1		2			Data Unavailable	3	102000003
326	Technical	1	Electrical Distribution	2	Load Controller	1	Input Voltage - Low	0	102010000
327	Technical	1		2			Input Voltage - High	1	102010001
328	Technical	1		2			Input Voltage - Unavailable	2	102010002

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
329	Technical	1		2			Input Current - Low	3	102010003
330	Technical	1		2			Input Current - High	4	102010004
331	Technical	1		2			Input Current - Unavailable	5	102010005
332	Technical	1		2			Power Fail	6	102010006
333	Technical	1		2			System Fail	7	102010007
334	Technical	1		2			Faulted Circuit Bilge Pump	8	102010008
335	Technical	1		2			Faulted Circuit Blower	9	102010009

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
336	Technical	1		2			Faulted Circuit NAV Lights	10	10201000A
337	Technical	1		2			Faulted Circuit Anchor Lights	11	10201000B
338	Technical	1	Electrical Distribution	2	Command Source	2	Faulted	0	102020000
339	Technical	1		2			System Fail	1	102020001
340	Technical	1		2			Data Unavailable	2	102020002
341	Technical	1	Electrical Distribution	2	Event Source	3	Faulted	0	102030000
342	Technical	1		2			System Fail	1	102030001

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
343	Technical	1		2			Data Unavailable	2	102030002
344	Technical	1	Electrical Distribution	2	Chargers / inverter / converters	4	Faulted	0	102040000
345	Technical	1		2			Overload	1	102040001
346	Technical	1		2			DC Voltage – low	2	102040002
347	Technical	1		2			DC Voltage – high	3	102040003
348	Technical	1		2			Internal temperature - high	4	102040004
349	Technical	1		2			Battery temperature – high	5	102040005

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
350	Technical	1		2			DC Ripple – high	6	102040006
351	Technical	1		2			AC input failure	7	102040007
352	Technical	1		2			Data Unavailable	8	102040008
353	Technical	1	Steering and Control Surfaces	3	Power unit	0	Stop	0	103000000
354	Technical	1		3			Power fail	1	103000001
355	Technical	1		3			Overload	2	103000002
356	Technical	1		3			Phase fail	3	103000003

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
357	Technical	1		3			Hydraulic fluid level low	4	103000004
358	Technical	1		3			Run	5	103000005
359	Technical	1	Steering and Control Surfaces	3	Control system (actuator or drive unit for steering signal)	1	Power fail	0	103010000
360	Technical	1	Steering and Control Surfaces	3	Trim tabs	2	Low	0	103020000
361	Technical	1		3			High	1	103020001
362	Technical	1		3			Unavailable	2	103020002
363	Technical	1	Propulsion Control	4	Propulsion control	0	Inhibition of starting of propulsion engine	0	104000000

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
364	Technical	1		4			Automatic shutdown	1	104000001
365	Technical	1		4			Automatic slowdown	2	104000002
366	Technical	1		4			Safety system override	3	104000003
367	Technical	1		4			Operating in barred speed range	4	104000004
368	Technical	1		4			System power supply main and emergency feeders – failure	5	104000005
369	Technical	1		4			CPP hydraulic oil pressure – low and high	6	104000006
370	Technical	1		4			CPP hydraulic oil temperature – high and low	7	104000007

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
371	Technical	1		4			Control, alarm or safety system, power supply failure	8	104000008
372	Technical	1	Propulsion Control	4	Remote control system	1	Power fail	0	104010000
373	Technical	1		4			System abnormal	1	104010001
374	Technical	1		4			Governor control abnormal	2	104010002
375	Technical	1		4			Propeller pitch control abnormal	3	104010003
376	Technical	1	Propulsion Control	4	Monitoring system	2	Normal power source – fail	0	104020000
377	Technical	1		4			Individual power supply to control, monitoring and safety systems – fail	1	104020001

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
378	Technical	1		4			Integrated computerized system: data highway abnormal	2	104020002
379	Technical	1		4			Integrated computerized system: duplicated data link – failure	3	104020003
380	Technical	1	Propulsion Control	4	Group alarm system	3	Power fail	0	104030000
381	Technical	1		4			Personnel alarm	1	104030001
382	Technical	1		4			Dead man alarm	2	104030002
383	Technical	1		4			Request backup OOW	3	104030003
384	Technical	1	Propulsion Control	4	System power source	4	Main feeder – fail	0	104040000

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385	Technical	1		4			Emergency feeder – fail	1	104040001
386	Technical	1	Propulsion	5	Engine	0	Exhaust Gas temperature - Low	0	105000000
387	Technical	1		5			Exhaust Gas temperature - High	1	105000001
388	Technical	1		5			Exhaust Gas temperature - Unavailable	2	105000002
389	Technical	1		5			Check Engine	3	105000003
390	Technical	1		5			Check Engine Sensor - Unavailable	4	105000004
391	Technical	1		5			Over temperature	5	105000005

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392	Technical	1		5			Over temperature Sensor - Unavailable	6	105000006
393	Technical	1		5			Oil Pressure - Low	7	105000007
394	Technical	1		5			Oil Pressure - High	8	105000008
395	Technical	1		5			Oil Pressure - Unavailable	9	105000009
396	Technical	1		5			Oil Level - Low	10	10500000A
397	Technical	1		5			Oil Level - High	11	10500000B
398	Technical	1		5			Oil Level - Unavailable	12	10500000C

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
399	Technical	1		5			Fuel Rate - Low	13	10500000D
400	Technical	1		5			Fuel Rate - High	14	10500000E
401	Technical	1		5			Fuel Rate - Unavailable	15	10500000F
402	Technical	1		5			Fuel Pressure - Low	16	105000010
403	Technical	1		5			Fuel Pressure - High	17	105000011
404	Technical	1		5			Fuel Pressure - Unavailable	18	105000012
405	Technical	1		5			System Voltage - Low	19	105000013

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406	Technical	1		5			System Voltage - High	20	105000014
407	Technical	1		5			System Voltage - Low	21	105000015
408	Technical	1		5			Water In Fuel	22	105000016
409	Technical	1		5			Water In Fuel Sensor - Unavailable	23	105000017
410	Technical	1		5			Charge Indicator	24	105000018
411	Technical	1		5			Charge Indicator - Unavailable	25	105000019
412	Technical	1		5			Preheat Indicator	26	10500001A

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
413	Technical	1		5			Preheat Indicator - Unavailable	27	10500001B
414	Technical	1		5			Boost Pressure - Low	28	10500001C
415	Technical	1		5			Boost Pressure - High	29	10500001D
416	Technical	1		5			Boost Pressure - Unavailable	30	10500001E
417	Technical	1		5			Rev Limit Exceeded	31	10500001F
418	Technical	1		5			Rev Limit Exceeded Sensor - Unavailable	32	105000020
419	Technical	1		5			Exhaust Gas Recirculation System	33	105000021

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420	Technical	1		5			Exhaust Gas Recirculation System - Unavailable	34	105000022
421	Technical	1		5			Throttle Position Sensor	35	105000023
422	Technical	1		5			Throttle Position Sensor - Unavailable	36	105000024
423	Technical	1		5			Emergency Stop Mode	37	105000025
424	Technical	1		5			Emergency Stop Mode - Unavailable	38	105000026
425	Technical	1		5			Warning Level 1	39	105000027
426	Technical	1		5			Warning Level 1 Sensor - Unavailable	40	105000028

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
427	Technical	1		5			Warning Level 2	41	105000029
428	Technical	1		5			Warning Level 2 Sensor - Unavailable	42	10500002A
429	Technical	1		5			Power Reduction	43	10500002B
430	Technical	1		5			Power Reduction - Unavailable	44	10500002C
431	Technical	1		5			Maintenance Needed	45	10500002D
432	Technical	1		5			Maintenance Needed Sensor - Unavailable	46	10500002E
433	Technical	1		5			Communication Error	47	10500002F

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434	Technical	1		5			Communication Error Sensor - Unavailable	48	105000030
435	Technical	1		5			Sub or Secondary Throttle	49	105000031
436	Technical	1		5			Sub or Secondary Throttle Sensor - Unavailable	50	105000032
437	Technical	1		5			Neutral Start Protect	51	105000033
438	Technical	1		5			Neutral Start Protect Sensor - Unavailable	52	105000034
439	Technical	1		5			Shutting Down	53	105000035
440	Technical	1		5			Shutting Down Sensor - Unavailable	54	105000036

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441	Technical	1		5			Engine Starter	55	105000037
442	Technical	1		5			Engine Starter Sensor - Unavailable	56	105000038
443	Technical	1		5			Alternator Potential - Low	57	105000039
444	Technical	1		5			Alternator Potential - High	58	10500003A
445	Technical	1		5			Alternator Potential - Unavailable	59	10500003B
446	Technical	1		5			Hours - High	60	10500003C
447	Technical	1		5			Hours - Unavailable	61	10500003D

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
448	Technical	1		5			Tilt/Trim - Low	62	10500003E
449	Technical	1		5			Tilt/Trim -High	63	10500003F
450	Technical	1		5			Tilt/Trim - Unavailable	64	105000040
451	Technical	1		5			Engine Speed - Low	65	105000041
452	Technical	1		5			Engine Speed - High	66	105000042
453	Technical	1		5			Engine Speed - Unavailable	67	105000043
454	Technical	1		5			Fuel Economy - High	68	105000044

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455	Technical	1		5			Fuel Economy - Unavailable	69	105000045
456	Technical	1		5			Fuel Economy - Low	70	105000046
457	Technical	1		5			Percent Engine Load - Low	71	105000047
458	Technical	1		5			Percent Engine Load - High	72	105000048
459	Technical	1		5			Percent Engine Load - Unavailable	73	105000049
460	Technical	1		5			Percent Engine Torque - Low	74	10500004A
461	Technical	1		5			Percent Engine Torque - High	75	10500004B

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462	Technical	1		5			Percent Engine Torque - Unavailable	76	10500004C
463	Technical	1		5			High Boost Pressure	77	10500004D
464	Technical	1		5			High Boost Pressure - Unavailable	78	10500004E
465	Technical	1	Propulsion Engine	5	Engine Coolant	1	Coolant Level - Low	0	105010000
466	Technical	1		5			Coolant Level - High	1	105010001
467	Technical	1		5			Coolant Level - Unavailable	2	105010002
468	Technical	1		5			Coolant Flow - High	3	105010003

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469	Technical	1		5			Coolant Flow - Low	4	105010004
470	Technical	1		5			Coolant Flow - Unavailable	5	105010005
471	Technical	1		5			Coolant Pressure - Low	6	105010006
472	Technical	1		5			Coolant Pressure - High	7	105010007
473	Technical	1		5			Coolant Pressure - Unavailable	8	105010008
474	Technical	1		5			Coolant Temperature - Low	9	105010009
475	Technical	1		5			Coolant Temperature - High	10	10501000A

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476	Technical	1		5			Coolant Temperature - Unavailable	11	10501000B
477	Technical	1		5			Oily contamination of Coolant	12	10501000C
478	Technical	1	Propulsion	5	Engine Seawater cooling	2	Seawater Level - High	0	105020000
479	Technical	1		5			Seawater Level - Low	1	105020001
480	Technical	1		5			Seawater Level Sensor - Unavailable	2	105020002
481	Technical	1		5			Seawater Flow - Low	3	105020003
482	Technical	1		5			Seawater Flow - High	4	105020004

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483	Technical	1		5			Seawater Flow Sensor - Unavailable	5	105020005
484	Technical	1		5			Seawater Pressure - Low	6	105020006
485	Technical	1		5			Seawater Pressure - High	7	105020007
486	Technical	1		5			Seawater Pressure Sensor - Unavailable	8	105020008
487	Technical	1		5			Seawater Temperature - Low	9	105020009
488	Technical	1		5			Seawater Temperature - High	10	10502000A
489	Technical	1		5			Seawater Temperature Sensor - Unavailable	11	10502000B

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490	Technical	1		5			Oily contamination of Cooling Seawater	12	10502000C
491	Technical	1	Propulsion	5	Transmission	3	Check Transmission	0	105030000
492	Technical	1		5			Check Transmission Sensor - Unavailable	1	105030001
493	Technical	1		5			Over temperature	2	105030002
494	Technical	1		5			Over temperature - Unavailable	3	105030003
495	Technical	1		5			Low Oil Pressure	4	105030004
496	Technical	1		5			Low Oil Pressure - Unavailable	5	105030005

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497	Technical	1		5			Low Oil Level	6	105030006
498	Technical	1		5			Low Oil Level - Unavailable	7	105030007
499	Technical	1		5			Sail Drive	8	105030008
500	Technical	1		5			Sail Drive - Unavailable	9	105030009
501	Technical	1	Propulsion	5	Fuel Management	4	Distance to Empty - Low	0	105040000
502	Technical	1		5			Distance to Empty - High	1	105040001
503	Technical	1		5			Distance to Empty - Unavailable	2	105040002

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504	Technical	1		5			Total Fuel Rate - Low	3	105040003
505	Technical	1		5			Total Fuel Rate - High	4	105040004
506	Technical	1		5			Total Fuel Rate - Unavailable	5	105040005
507	Technical	1		5			Time to Empty - Low	6	105040006
508	Technical	1		5			Time to Empty - High	7	105040007
509	Technical	1		5			Time to Empty - Unavailable	8	105040008
510	Technical	1		5			Total Fuel Economy - Low	9	105040009

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511	Technical	1		5			Total Fuel Economy - High	10	10504000A
512	Technical	1		5			Total Fuel Economy - Unavailable	11	10504000B
513	Technical	1		5			Total Fuel Remaining- Low	12	10504000C
514	Technical	1		5			Total Fuel Remaining- High	13	10504000D
515	Technical	1		5			Total Fuel Remaining- Unavailable	14	10504000E
516	Technical	1	Propulsion Diesel Plant	6	Fuel oil	0	Fuel oil tank heating control and temp. display and alarm – high	0	106000000
517	Technical	1		6			Fuel oil engine inlet pressure – low	1	106000001

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518	Technical	1		6			Fuel oil before injunction pump temp – high and low	2	106000002
519	Technical	1		6			Leakage from high pressure pipe	3	106000003
520	Technical	1	Propulsion Diesel Plant	6	Lubricating oil	1	Lub. oil to main bearing pressure – low	0	106010000
521	Technical	1		6			Lub. oil to thrust bearing pressure – low	1	106010001
522	Technical	1		6			Lub. oil to crosshead bearing pressure – low	2	106010002
523	Technical	1		6			Lub. oil to camshaft pressure – low	3	106010003
524	Technical	1		6			Lub oil to camshaft temp – high	4	106010004

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525	Technical	1		6			Lub oil inlet temp – high	5	106010005
526	Technical	1		6			Thrust bearing pads temp temp – high	6	106010006
527	Technical	1		6			Main, crank, crosshead bearing oil outlet temp – high	7	106010007
528	Technical	1		6			Cylinder lubricator, flow rate – low	8	106010008
529	Technical	1		6			Lub oil tanks, level – low	9	106010009
530	Technical	1	Propulsion Diesel Plant	6	Turbo-charger	2	Lub oil inlet, pressure – low	0	106020000
531	Technical	1		6			Lub oil outlet, temp – high	1	106020001

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532	Technical	1	Propulsion Diesel Plant	6	Piston cooling	3	Coolant inlet, pressure – low	0	106030000
533	Technical	1		6			Coolant outlet, temp. – high	1	106030001
534	Technical	1		6			Coolant outlet, flow – low	2	106030002
535	Technical	1		6			Coolant expansion tank, level – low	3	106030003
536	Technical	1	Propulsion Diesel Plant	6	Compressed air	4	Starting air before main shut-off valve, pressure – low	0	106040000
537	Technical	1		6			Control air, pressure – low	1	106040001
538	Technical	1		6			Safety air, pressure – low	2	106040002

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539	Technical	1	Propulsion Diesel Plant	6	Scavenge air	5	Scavenge air box, temp – high	0	106050000
540	Technical	1		6			Scavenge air receiver water, level – high	1	106050001
541	Technical	1	Propulsion Diesel Plant	6	Exhaust gas	6	Exhaust gas, temp – high	2	106060002
542	Technical	1		6			Exhaust gas deviation from average, temp – high	3	106060003
543	Technical	1		6			Exhaust gas before turbo-charger, temp – high	4	106060004
544	Technical	1		6			Exhaust gas after turbo-charger, temp – high	5	106060005
545	Technical	1	Propulsion Diesel Plant	6	Fuel valve coolant	7	Coolant, pressure low	0	106070000

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546	Technical	1		6			Coolant, temp – high	1	106070001
547	Technical	1		6			Coolant expansion tank, level – low	2	106070002
548	Technical	1	Propulsion Diesel Plant	6	Engine	8	Rotation – wrong way	0	106080000
549	Technical	1		6			Engine, over speed	1	106080001
550	Technical	1	Propulsion Steam Turbines Plant	7	Lubrication oil	0	Pressure at bearing inlet – high and low	0	107000000
551	Technical	1		7			Temp at bearing outlet – high	1	107000001
552	Technical	1		7			Filter differential pressure – high	2	107000002

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553	Technical	1		7			Gravity tank level – low	3	107000003
554	Technical	1	Propulsion Steam Turbines Plant	7	Lubricating oil cooling system	1	Pressure – low	0	107010000
555	Technical	1		7			Temp at outlet – high	1	107010001
556	Technical	1		7			Expansion tank level – low	2	107010002
557	Technical	1	Propulsion Steam Turbines Plant	7	Seawater	2	Pressure – low	0	107020000
558	Technical	1	Propulsion Steam Turbines Plant	7	Steam	3	Pressure at throttle – low	0	107030000
559	Technical	1		7			Gland seal exhaust fan – failure	1	107030001

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560	Technical	1		7			Astern guardian valve – fail to open	2	107030002
561	Technical	1	Propulsion Steam Turbines Plant	7	Condensate	4	Condenser level – high and low	0	107040000
562	Technical	1		7			Condensate pump pressure – low	1	107040001
563	Technical	1		7			Condenser vacuum – low	2	107040002
564	Technical	1		7			Salinity – high	3	107040003
565	Technical	1	Propulsion Steam Turbines Plant	7	Rotor	5	Vibration level – high	0	107050000
566	Technical	1		7			Axial displacement – large	1	107050001

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567	Technical	1		7			Overspeed	2	107050002
568	Technical	1	Propulsion Steam Turbines Plant	7			Shaft stopped – excess of set period	3	107050003
569	Technical	1	Propulsion Steam Turbines Plant	7	Power	6	Throttle control system power failure	0	107060000
570	Technical	1	Propulsion Gas Turbine Plant	8	Fuel oil	0	Pressure – low	0	108000000
571	Technical	1		8			Temp – low and high	1	108000001
572	Technical	1	Propulsion Gas Turbine Plant	8	Lubricating oil	1	Inlet pressure – low	0	108010000
573	Technical	1		8			Inlet temp – high	1	108010001

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574	Technical	1		8			Main bearing oil outlet temp – high	2	108010002
575	Technical	1		8			Filter differential pressure – high	3	108010003
576	Technical	1		8			Tank level – low	4	108010004
577	Technical	1	Propulsion Gas Turbine Plant	8	Cooling medium	2	Pressure – low	0	108020000
578	Technical	1		8			Temp – high	1	108020001
579	Technical	1	Propulsion Gas Turbine Plant	8	Starting	3	Stored starting energy level – low	0	108030000
580	Technical	1		8			Automatic starting failure	1	108030001

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
581	Technical	1	Propulsion Gas Turbine Plant	8	Combustion	4	Flame failure	0	108040000
582	Technical	1	Propulsion Gas Turbine Plant	8	Exhaust gas	5	Temp – high	0	108050000
583	Technical	1	Propulsion Gas Turbine Plant	8	Turbine	6	Vibration level – high	0	108060000
584	Technical	1		8			Rotor axial displacement – large	1	108060001
585	Technical	1		8			Overspeed	2	108060002
586	Technical	1		8			Vacuum at compressor inlet – high	3	108060003
587	Technical	1	Propulsion Electric Plant	9	Propulsion generator	0	Bearing lub oil inlet pressure – low	0	109000000

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
588	Technical	1		9			Voltage – off-limit	1	109000001
589	Technical	1		9			Frequency – off-limit	2	109000002
590	Technical	1		9			Stationary windings temperature – high	3	109000003
591	Technical	1		9			Failure of online generator	4	109000004
592	Technical	1		9			Transfer of standby generator	5	109000005
593	Technical	1		9			Generator cooling medium temperature – high	6	109000006
594	Technical	1		9			Generator cooling pump – failure	7	109000007

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
595	Technical	1		9			Inter-pole windings temperature – high	8	109000008
596	Technical	1	Propulsion Electric Plant	9	Propulsion motor – AC	1	Bearing lub oil inlet pressure – low	0	109010000
597	Technical	1		9			Armature voltage – off-limit	1	109010001
598	Technical	1		9			Frequency – off-limit	2	109010002
599	Technical	1		9			Stationary windings temperature – high	3	109010003
600	Technical	1		9			Failure of online generator	4	109010004
601	Technical	1		9			Transfer of standby generator	5	109010005

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
602	Technical	1		9			Motor cooling medium temperature – high	6	109010006
603	Technical	1		9			Motor cooling pump – failure	7	109010007
604	Technical	1	Propulsion Electric Plant	9	Propulsion motor – DC	2	Bearing lub oil inlet pressure – low	0	109020000
605	Technical	1		9			Armature voltage – off-limit	1	109020001
606	Technical	1		9			Motor overspeed	2	109020002
607	Technical	1		9			Failure of online generator	3	109020003
608	Technical	1		9			Transfer of standby generator	4	109020004

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
609	Technical	1		9			Motor cooling medium temperature – high	5	109020005
610	Technical	1		9			Motor cooling pump – failure	6	109020006
611	Technical	1	Propulsion Electric Plant	9	Propulsion SCR	3	Overload (high current)	0	109030000
612	Technical	1		9			SCR cooling medium temperature – high	1	109030001
613	Technical	1		9			SCR cooling pump – failure	2	109030002
614	Technical	1	Propulsion Electric Plant	9	Transformer	4	Transformer winding temp – high	0	109040000
615	Technical	1	Propulsion Boiler	10	Feed water	0	Atmospheric drain tank level – high and low	0	10A000000

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
616	Technical	1		10			Dearator level – high and low	1	10A000001
617	Technical	1		10			Dearator pressure – high and low	2	10A000002
618	Technical	1		10			Feed water pump pressure – low	3	10A000003
619	Technical	1		10			Feed water temp – high	4	10A000004
620	Technical	1		10			Feed water outlet salinity – high	5	10A000005
621	Technical	1	Propulsion Boiler	10	Boiler drum	1	Water level – high and low	0	10A010000
622	Technical	1		10			Water level – low-low	1	10A010001

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
623	Technical	1	Propulsion Boiler	10	Steam	2	Pressure – high and low	0	10A020000
624	Technical	1		10			Superheater outlet temp – high	1	10A020001
625	Technical	1	Propulsion Boiler	10	Air	3	Forced draft fan – failure	0	10A030000
626	Technical	1		10			Rotating air heater motor – failure	1	10A030001
627	Technical	1		10			Fire in boiler casing	2	10A030002
628	Technical	1	Propulsion Boiler	10	Fuel oil	4	Pump pressure at outlet – low	0	10A040000
629	Technical	1		10			Fuel oil temp – high and low	1	10A040001

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
630	Technical	1	Propulsion Boiler	10	Burner	5	Atomizing medium pressure – off-limit	0	10A050000
631	Technical	1		10			Flame of burner – fail	1	10A050001
632	Technical	1		10			Flame sensor – fail	2	10A050002
633	Technical	1		10			Untake gas temp – high	3	10A050003
634	Technical	1	Propulsion Boiler	10	Power	6	Control system power failure	0	10A060000
635	Technical	1	Propulsion Auxiliary Boiler	11	Feed water	0	Feed water outlet salinity – high	0	10B000000
636	Technical	1	Propulsion Auxiliary Boiler	11	Boiler drum	1	Water level – high and low	0	10B010000

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
637	Technical	1	Propulsion Auxiliary Boiler	11	Steam	2	Pressure – high and low	0	10B020000
638	Technical	1		11			Superheater outlet temp – high	1	10B020001
639	Technical	1	Propulsion Auxiliary Boiler	11	Air	3	Supply air pressure – fail	0	10B030000
640	Technical	1		11			Fire in boiler casing	1	10B030001
641	Technical	1	Propulsion Auxiliary Boiler	11	Fuel oil	4	Pump pressure at outlet – low	0	10B040000
642	Technical	1		11			Fuel oil temp – high and low	1	10B040001
643	Technical	1	Propulsion Auxiliary Boiler	11	Burner	5	Flame of burner – fail	0	10B050000

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
644	Technical	1		11			Flame sensor – fail	1	10B050001
645	Technical	1		11			Untake gas temp – high	2	10B050002
646	Technical	1	Propulsion Auxiliary Boiler	11	Power	6	Control system power failure	0	10B060000
647	Technical	1	Propulsion Auxiliary Diesel Engine	12	Fuel oil	0	Fuel oil leakage from injunction pipe	0	10C000000
648	Technical	1		12			Fuel oil temp – high and low	1	10C000001
649	Technical	1		12			Service tank level – low	2	10C000002
650	Technical	1	Propulsion Auxiliary Diesel Engine	12	Lubricating oil	1	Bearing oil inlet pressure – low	0	10C010000

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
651	Technical	1		12			Bearing oil inlet temp – high	1	10C010001
652	Technical	1		12			Crankcase oil mist concentration – high	2	10C010002
653	Technical	1	Propulsion Auxiliary Diesel Engine	12	Cooling medium	2	Pressure – low	0	10C020000
654	Technical	1		12			Temp. – high	1	10C020001
655	Technical	1		12			Expansion tank, level – low	2	10C020002
656	Technical	1	Propulsion Auxiliary Diesel Engine	12	Starting medium	3	Energy level – low	0	10C030000
657	Technical	1	Propulsion Auxiliary Diesel Engine	12	Exhaust gas	4	Exhaust gas, temp – high	0	10C040000

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
658	Technical	1	Propulsion Auxiliary Diesel Engine	12	Engine	5	Engine, overspeed	0	10C050000
659	Technical	1	Propulsion Auxiliary Turbine	13	Lubrication oil	0	Pressure at bearing inlet – low	0	10D000000
660	Technical	1		13			Temp at bearing inlet – high	1	10D000001
661	Technical	1		13			Temp at bearing outlet – high	2	10D000002
662	Technical	1	Propulsion Auxiliary Turbine	13	Lubricating oil cooling system	1	Pressure – low	0	10D010000
663	Technical	1		13			Temp at outlet – high	1	10D010001
664	Technical	1		13			Expansion tank level – low	2	10D010002

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
665	Technical	1	Propulsion Auxiliary Turbine	13	Seawater	2	Pressure – low	0	10D020000
666	Technical	1	Propulsion Auxiliary Turbine	13	Steam	3	Pressure at inlet – low	0	10D030000
667	Technical	1	Propulsion Auxiliary Turbine	13	Condensate	4	Condensate pump pressure – low	0	10D040000
668	Technical	1		13			Condenser vacuum – low	1	10D040001
669	Technical	1	Propulsion Auxiliary Turbine	13	Rotor	5	Overspeed	0	10D050000
670	Technical	1		13			Axial displacement – large	1	10D050001
671	Technical	1	Propulsion Auxiliary Gas Turbine	14	Fuel oil	0	Pressure – low	0	10E000000

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
672	Technical	1		14			Temp – high and low	1	10E000001
673	Technical	1	Propulsion Auxiliary Gas Turbine	14	Lubricating oil	1	Inlet pressure – low	0	10E010000
674	Technical	1		14			Inlet temp – high	1	10E010001
675	Technical	1		14			Bearing oil outlet temp – high	2	10E010002
676	Technical	1		14			Filter differential pressure – high	3	10E010003
677	Technical	1	Propulsion Auxiliary Gas Turbine	14	Cooling medium	2	Pressure – low	0	10E020000
678	Technical	1		14			Temp – high	1	10E020001

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
679	Technical	1	Propulsion Auxiliary Gas Turbine	14	Starting	3	Stored starting energy level – low	0	10E030000
680	Technical	1		14			Ignition failure	1	10E030001
681	Technical	1	Propulsion Auxiliary Gas Turbine	14	Combustion	4	Flame failure	0	10E040000
682	Technical	1	Propulsion Auxiliary Gas Turbine	14	Exhaust gas	5	Temp – high	0	10E050000
683	Technical	1	Propulsion Auxiliary Gas Turbine	14	Rotor	6	Vibration level – high	0	10E060000
684	Navigation	0		14			Rotor axial displacement – large	1	00E060001
685	Navigation	0		14			Overspeed	2	00E060002

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
686	Navigation	0		14			Vacuum at compressor inlet – high	3	00E060003
687	Navigation	0	Sensor	15	Depth	0	Depth Alarm - Low	0	00F000000
688	Navigation	0		15			Depth Alarm - High	1	00F000001
689	Navigation	0		15			Depth Alarm Sensor - Unavailable	2	00F000002
690	Navigation	0		15			Anchor Watch - Low	3	00F000003
691	Navigation	0		15			Anchor Watch - High	4	00F000004
692	Navigation	0		15			Anchor Watch Sesnor - Unavailable	5	00F000005

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
693	Navigation	0	Sensor	15	Wind	1	Apparent Wind Angle - High	0	00F010000
694	Navigation	0		15			Apparent Wind Angle - Low	1	00F010001
695	Navigation	0		15			Apparent Wind Angle Sensor - Unavailable	2	00F010002
696	Navigation	0		15			Apparent Wind Speed - Low	3	00F010003
697	Navigation	0		15			Apparent Wind Speed- High	4	00F010004
698	Navigation	0		15			Apparent Wind Speed Sensor - Unavailable	5	00F010005
699	Navigation	0		15			True Wind Angle - Low	6	00F010006

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
700	Navigation	0		15			True Wind Angle - High	7	00F010007
701	Navigation	0		15			True Wind Angle - Unavailable	8	00F010008
702	Navigation	0		15			True Wind Speed - Low	9	00F010009
703	Navigation	0		15			True Wind Speed - High	10	00F01000A
704	Navigation	0		15			True Wind Speed Sensor / Calculation - Unavailable	11	00F01000B
705	Navigation	0	Sensor	15	Speed Through Water	2	Low	0	00F020000
706	Navigation	0		15			High	1	00F020001

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
707	Navigation	0		15			Speed Through Water Sensor - Unavailable	2	00F020002
708	Navigation	0	Sensor	15	Log	3	High	0	00F030000
709	Navigation	0		15			Log Sensor - Unavailable	1	00F030001
710	Navigation	0	Sensor	15	Trip	4	High	0	00F040000
711	Navigation	0		15			Trip Sensor - Unavailable	1	00F040001
712	Navigation	0	Sensor	15	Heading	5	Course - Low	0	00F050000
713	Navigation	0		15			Course - High	1	00F050001

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
714	Navigation	0		15			Course Sensor - Unavailable	2	00F050002
715	Navigation	0	Sensor	15	Rate of Turn	6	Low	0	00F060000
716	Navigation	0		15			High	1	00F060001
717	Navigation	0		15			Rate of Turn Sensor - Unavailable	2	00F060002
718	Navigation	0	Sensor	15	Pitch	7	Low	0	00F070000
719	Navigation	0		15			High	1	00F070001
720	Navigation	0		15			Pitch Sensor - Unavailable	2	00F070002

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
721	Navigation	0	Sensor	15	Roll	8	Low	0	00F080000
722	Navigation	0		15			High	1	00F080001
723	Navigation	0		15			Roll Sensor - Unavailable	2	00F080002
724	Navigation	0	Sensor	15	Variation	9	Variation Unavailable	0	00F090000
725	Navigation	0	Sensor	15	Rudder Angle	10	High	0	00F0A0000
726	Navigation	0		15			Rudder Angle Sensor - Unavailable	1	00F0A0001
727	Navigation	0	Electronic Equipment	16	Radar	0	CPA	0	010000000

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
728	Navigation	0		16			CPA Function - Unavailable	1	010000001
729	Navigation	0		16			TCPA	2	010000002
730	Navigation	0		16			TCPA Function - Unavailable	3	010000003
731	Navigation	0	Electronic Equipment	16	GPS	1	Quality Low	0	010010000
732	Navigation	0		16			Quality High	1	010010001
733	Navigation	0		16			Quality Function - Unavailable	2	010010002
734	Navigation	0		16			Loss of Signal	3	010010003

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
735	Navigation	0		16			Loss of Signal Sensor - Unavailable	4	010010004
736	Navigation	0		16			COG - Course Alert	5	010010005
737	Navigation	0		16			COG Function - Unavailable	6	010010006
738	Navigation	0		16			SOG - Low	7	010010007
739	Navigation	0		16			SOG - High	8	010010008
740	Navigation	0		16			SOG Function - Unavailable	9	010010009
741	Navigation	0		16			Lat / Long - Inside Radius	10	01001000A

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
742	Navigation	0		16			Lat / Long - Outside Radius	11	01001000B
743	Navigation	0		16			Lat / Long Function - Unavailable	12	01001000C
744	Navigation	0	Electronic Equipment	16	Multi-Functional Equipment	2	Off Course	0	010020000
745	Navigation	0		16			Off Course Function - Unavailable	1	010020001
746	Navigation	0		16			Large XTE	2	010020002
747	Navigation	0		16			Large XTE Function - Unavailable	3	010020003
748	Navigation	0		16			Bearing Origin to Destination Alert	4	010020004

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
749	Navigation	0		16			Bearing Origin to Destination Function - Unavailable	5	010020005
750	Navigation	0		16			Bearing to Waypoint Alert	6	010020006
751	Navigation	0		16			Bearing To Waypoint Function - Unavailable	7	010020007
752	Navigation	0		16			Time to Go - Low	8	010020008
753	Navigation	0		16			Time to Go - High	9	010020009
754	Navigation	0		16			Time to Go Function - Unavailable	10	01002000A
755	Navigation	0		16			VMG to Waypoint - Low	11	01002000B

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
756	Navigation	0		16			VMG to Waypoint - High	12	01002000C
757	Navigation	0		16			VMG to Waypoint Function - Unavailable	13	01002000D
758	Navigation	0	Electronic Equipment	16	AIS	3	CPA	0	010030000
759	Technical	1		16			CPA Function - Unavailable	1	110030001
760	Technical	1		16			TCPA	2	110030002
761	Technical	1		16			TCPA Function - Unavailable	3	110030003
762	Technical	1		16			Tx Malfunction	4	110030004

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
763	Technical	1		16			Antenna VSWR exceeds limit	5	110030005
764	Technical	1		16			Rx Channel 1 Malfunction	6	110030006
765	Technical	1		16			Rx Channel 2 Malfunction	7	110030007
766	Technical	1		16			Rx Channel 70 Malfunction	8	110030008
767	Navigation	0		16			General Equipment Failure	9	010030009
768	Navigation	0		16			MKD Connection Lost	10	01003000A
769	Navigation	0		16			External EPFS Lost	11	01003000B

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770	Navigation	0		16			No Position Sensor in use	12	01003000C
771	Navigation	0		16			Speed Over the Ground (SOG) invalid	13	01003000D
772	Navigation	0		16			Course Over the Ground (COG) Invalid	14	01003000E
773	Navigation	0		16			Heading lost/ Invalid	15	01003000F
774	Navigation	0		16			Rate of Turn (ROT) Invalid	16	010030010
775	Navigation	0	Electronic Equipment	16	Fish Finder	4	Fish Limit - Low	0	010040000
776	Navigation	0		16			Fish Limit - High	1	010040001

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
777	Navigation	0		16			Fish Limit Sensor - Unavailable	2	010040002
778	Navigation	0	Systems	17	Clock	0	Timer	0	011000000
779	Navigation	0		17			Timer Function - Unavailable	1	011000001
780	Navigation	0		17			Countdown Timer	2	011000002
781	Navigation	0		17			Countdown Timer Function - Unavailable	3	011000003
782	Navigation	0		17			Alarm Clock	4	011000004
783	Navigation	0		17			Alarm Clock Function - Unavailable	5	011000005

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
784	Navigation	0	External Environment	18	Atmospheric	0	Barometer - Low	0	012000000
785	Navigation	0		18			Barometer - High	1	012000001
786	Navigation	0		18			Barometer Sensor - Unavailable	2	012000002
787	Navigation	0		18			Dew Point - Low	3	012000003
788	Navigation	0		18			Dew Point - High	4	012000004
789	Navigation	0		18			Dew Point Sensor - Unavailable	5	012000005
790	Navigation	0		18			Heat Index - Low	6	012000006

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
791	Navigation	0		18			Heat Index - High	7	012000007
792	Navigation	0		18			Heat Index Sensor - Unavailable	8	012000008
793	Navigation	0		18			Humidity - Low	9	012000009
794	Navigation	0		18			Humidity - High	10	01200000A
795	Navigation	0		18			Humidity Sensor - Unavailable	11	01200000B
796	Navigation	0		18			Wind Chill, Apparent - Low	12	01200000C
797	Navigation	0		18			Wind Chill, Apparent - High	13	01200000D

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
798	Navigation	0		18			Wind Chill, Apparent Sensor - Unavailable	14	01200000E
799	Navigation	0		18			Wind Chill, Theoretcical - Low	15	01200000F
800	Navigation	0		18			Wind Chill, Theoretcical - High	16	012000010
801	Navigation	0		18			Wind Chill, Theoretical Sensor - Unavailable	17	012000011
802	Technical	1	External Environment	18	Aquatic	1	Sea Temperature - Low	0	112010000
803	Technical	1		18			Sea Temperature - High	1	112010001
804	Technical	1		18			Sea Temperature Sensor - Unavailable	2	112010002

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
805	Technical	1	Internal Environment	19	HVAC	0	Engine Room Temperature - Low	0	113000000
806	Technical	1		19			Engine Room Temperature - High	1	113000001
807	Technical	1		19			Engine Room Temperature Sensor - Unavailable	2	113000002
808	Technical	1		19			Main Cabin Temperature - Low	3	113000003
809	Technical	1		19			Main Cabin Temperature - High	4	113000004
810	Technical	1		19			Main Cabin Temperature Sesnor - Unavailable	5	113000005
811	Technical	1		19			Live Well - Low	6	113000006

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
812	Technical	1		19			Live Well - High	7	113000007
813	Technical	1		19			Live Well Sensor - Unavailable	8	113000008
814	Technical	1		19			Bait Well - Low	9	113000009
815	Technical	1		19			Bait Well - High	10	11300000A
816	Technical	1		19			Bait Well Sesnor - Unavailable	11	11300000B
817	Technical	1		19			Refrigeration - Low	12	11300000C
818	Technical	1		19			Refrigeration - High	13	11300000D

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
819	Technical	1		19			Refrigeration Sensor - Unavailable	14	11300000E
820	Technical	1		19			Heating System - Low	15	11300000F
821	Technical	1		19			Heating System - High	16	113000010
822	Technical	1		19			Heating System Sensor - Unavailable	17	113000011
823	Technical	1		19			Freezer - Low	18	113000012
824	Technical	1		19			Freezer - High	19	113000013
825	Technical	1		19			Freezer Sensor - Unavailable	20	113000014

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
826	Technical	1	Internal Environment	19	Refrigeration Seawater cooling	1	Seawater Level - High	0	113010000
827	Technical	1		19			Seawater Level - Low	1	113010001
828	Technical	1		19			Seawater Level Sensor - Unavailable	2	113010002
829	Technical	1		19			Seawater Flow - Low	3	113010003
830	Technical	1		19			Seawater Flow - High	4	113010004
831	Technical	1		19			Seawater Flow Sensor - Unavailable	5	113010005
832	Technical	1		19			Seawater Pressure - Low	6	113010006

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
833	Technical	1		19			Seawater Pressure - High	7	113010007
834	Technical	1		19			Seawater Pressure Sensor - Unavailable	8	113010008
835	Technical	1		19			Seawater Temperature - Low	9	113010009
836	Technical	1		19			Seawater Temperature - High	10	11301000A
837	Technical	1		19			Seawater Temperature Sensor - Unavailable	11	11301000B
838	Technical	1		19			Oily contamination of Cooling Seawater	12	11301000C
839	Technical	1	Internal Environment	19	Air Conditioning Seawater cooling	2	Seawater Level - High	0	113020000

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
840	Technical	1		19			Seawater Level - Low	1	113020001
841	Technical	1		19			Seawater Level Sensor - Unavailable	2	113020002
842	Technical	1		19			Seawater Flow - Low	3	113020003
843	Technical	1		19			Seawater Flow - High	4	113020004
844	Technical	1		19			Seawater Flow Sensor - Unavailable	5	113020005
845	Technical	1		19			Seawater Pressure - Low	6	113020006
846	Technical	1		19			Seawater Pressure - High	7	113020007

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
847	Technical	1		19			Seawater Pressure Sensor - Unavailable	8	113020008
848	Technical	1		19			Seawater Temperature - Low	9	113020009
849	Technical	1		19			Seawater Temperature - High	10	11302000A
850	Technical	1		19			Seawater Temperature Sensor - Unavailable	11	11302000B
851	Technical	1		19			Oily contamination of Cooling Seawater	12	11302000C
852	Technical	1	Deck Cargo and Fishing equipment systems	20	High voltage rotating machine	0	Stationary windings temperature – high	0	114000000
853	Technical	1	Deck Cargo and Fishing equipment systems	20	Fuel oil system	1	Settling and service tank level – high and low	0	114010000

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
854	Technical	1		20			Overflow tank and drain tank level – high	1	114010001
855	Technical	1	Deck Cargo and Fishing equipment systems	20	Stern tube lub. Oil	2	Tank level – low	0	114020000
856	Technical	1	Deck Cargo and Fishing equipment systems	20	Boiler	3	Automatic shutdown	0	114030000
857	Technical	1	Deck Cargo and Fishing equipment systems	20	Propulsion machinery space	4	Bilge level – high	0	114040000
858	Technical	1		20			Air condition system – fail	1	114040001
859	Technical	1		20			Fire detected	2	114040002
860	Technical	1	Deck Cargo and Fishing equipment systems	20	Chemical cargo system	5	High and low temp of cargo	0	114050000

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
861	Technical	1		20			High temp in tank	1	114050001
862	Technical	1		20			Oxygen concentration in void space	2	114050002
863	Technical	1		20			Malfunctioning of temp controls of cooling system	3	114050003
864	Technical	1		20			Failure of mechanical ventilation of cargo tank	4	114050004
865	Technical	1		20			Low temp in in cargo tanks	5	114050005
866	Technical	1	Deck Cargo and Fishing equipment systems	20	LPG/LNG cargo system	6	High and low temp in cargo tank	0	114060000
867	Technical	1		20			Gas detection	1	114060001

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
868	Technical	1		20			Hull or insulation temp – high	2	114060002
869	Technical	1		20			Cargo high pressure	3	114060003
870	Technical	1		20			Chlorine concentration	4	114060004
871	Technical	1		20			High pressure in chlorine cargo tank	5	114060005
872	Technical	1		20			Liquid cargo in ventilation system – failure	6	114060006
873	Technical	1		20			Vacuum protection of cargo tank – failure	7	114060007
874	Technical	1		20			Inert gas pressure – high	8	114060008

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
875	Technical	1		20			Gas detection equipment – failure	9	114060009
876	Technical	1		20			Gas detection after bursting disk for chlorine – failure	10	11406000A
877	Technical	1	Deck Cargo and Fishing equipment systems	20	Inert gas system	7	Low water pressure	0	114070000
878	Technical	1		20			High water level in scrubber	1	114070001
879	Technical	1		20			Gas temp – high	2	114070002
880	Technical	1		20			IG blower – failure	3	114070003
881	Technical	1		20			Oxygen content volume – high	4	114070004

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
882	Technical	1		20			Power supply of automatic control system – failure	5	114070005
883	Technical	1		20			Low water level in water seal	6	114070006
884	Technical	1		20			High and Low pressure of gas	7	114070007
885	Technical	1		20			Insufficient fuel oil supply	8	114070008
886	Technical	1		20			Power supply – failure	9	114070009
887	Technical	1	Deck Cargo and Fishing equipment systems	20	Bilge Maintenance System	8	Bilge Pump Running	0	114080000
888	Technical	1		20			Bilge Pump Running Sensor Fail	1	114080001

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Alert Number	Alert Category(AC	AC I	Alert System(AS)	AS I	Alert Sub System(AS-S)	AS-S ID	Alert Description(AD)	AD-I	Full Code (Hex)
889	Technical	1		20			Bilge Pump Running Duration	2	114080002
890	Technical	1		20			Bilge Pump Running Duration Sensor Fail	3	114080003
891	Technical	1	Deck Cargo and Fishing equipment systems	20	Bathing Platform System	9	Bathing Platform Up	0	114090000
892	Technical	1		20			Bathing Platform Up Sensor Failure	1	114090001
893	Technical	1	Deck Cargo and Fishing equipment systems	20	Shore Water Supply System	10	Shore Water Supply On	0	1140A0000
894	Technical	1		20			Shore Water Supply Sensor Fail	1	1140A0001

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NMEA 2000 Appendix B-6

Class & Function Codes

Version 2.200

February 2015

STANDARD FOR SERIAL DATA NETWORKING OF MARINE ELECTRONIC DEVICES

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Appendix B.6: NMEA 2000 Class and Function Codes

Version: **2.200** ***Effective Date:*** **2/18/2015**

Class Code	Funct Code	Class / Function Name	Description

Class Code	Funct Code	Class / Function Name	Description
TBD		Class Name	Enter class description

Class Code	Funct Code	Class / Function Name	Description
10		System Tools	Equipment that queries and measures NMEA 2000 bus traffic and may be used to configure, troubleshoot and/or test. System Tools shall not be permanently connected to the NMEA 2000 bus.
	120	NMEA 2000 Certification Tool	This equipment is the NMEA 2000 Certification Tool exclusively. This tool only certifies NMEA 2000 products.
	121	NMEA OneNet Certification Tool	This equipment is the NMEA OneNet Certification Tool exclusively. This tool only certifies NMEA OneNet products.
	122	NMEA Network Gateway Cert. Tool-2000 Interface	This equipment is the NMEA Network Gateway Certification Tool exclusively. This tool only certifies NMEA Network Gateway products on the NMEA 2000 interface.
	123	NMEA Network Gateway Cert. Tool-OneNet Interface	This equipment is the NMEA Network Gateway Certification Tool exclusively. This tool only certifies NMEA Network Gateway products on the NMEA OneNet interface.
	130	Diagnostic	Devices that stress the system for diagnostic purposes
	140	Bus Traffic Logger	Devices that may be connected to a NMEA 2000 backbone to record bus traffic; as used in this class, these devices promiscuously capture and save CAN frames without regard for source or content

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Class Code	Funct Code	Class / Function Name	Description

Class Code	Funct Code	Class / Function Name	Description
20		Safety Systems	Equipment that is inheritantly designed to ensure personnel and/or vessel safety by recording, detecting, and/or alerting on safety/security related occurances.
	110	Alarm Enunciator ***DEPRECATED - Function not for use in new designs!***	Alarm Enunciator {Deprecated - See Display/Alarm Enunciator - Class Code 120, Function Code 140 }
	130	Emergency Position Indicating Radio Beacon (EPIRB)	Devices that transmit ownship position to aid in vessel location for distress and search and rescue
	135	Man Overboard	Devices that detect/report events where personnel are absent from the vessel
	140	Voyage Data Recorder	Devices that collect data from various vessel sensors and stores the information for later retrieval; as used in this class, these devices are typically configured to record specific parameters from specific sources, such as vessel position from a GNSS receiver.
	150	Camera	Devices that record ownship compartment, equipment and personnel video images for the purpose of ensuring ownship security and/or safety.

Class Code	Funct Code	Class / Function Name	Description
25		Inter/Intranetwork Device	NMEA 2000 equipment used as nodes to interconnect communication paths, including both inter- and intra- network communication
	130	PC Gateway	Device that brings both NMEA 2000 data onto a computer, and computer data onto an NMEA 2000 network.
	131	NMEA 2000 to Analog Gateway	Device that brings NMEA 2000 data to an analog system or display
	132	Analog to NMEA 2000 Gateway	Device that converts analog device data and sends this onto the NMEA 2000 network
	133	NMEA 2000 to Serial Gateway	Device that brings NMEA 2000 data to a serial port.
	135	NMEA 0183 Gateway	Device that joins an NMEA 2000 network to NMEA 0183 circuit(s) and provides translation between NMEA 2000 and 0183 data formats.

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Class Code	Funct Code	Class / Function Name	Description
	136	NMEA Network Gateway	This is a bi-directional device that connects an NMEA OneNet Network to an NMEA 2000 Network and provides translation between NMEA 2000 and NMEA OneNet data
	137	NMEA 2000 Wireless Gateway	Transports NMEA 2000 data onto a wireless network and wireless data onto an NMEA 20000 network
	140	Router	Device that joins network segments with the same network protocol. On each side of a router address space, data rate and physical media may differ.
	150	Bridge	Device that joins network segments using the same network protocol and address space. Data rate and physical media may differ on each side to the bridge. A bridge may perform message filtering.
	160	Repeater	Device that receives a signal and retransmits it at a higher level or higher power.

Class Code	Funct Code	Class / Function Name	Description
30		Electrical Distribution	Equipment that monitors and/or controls non-propulsion electrical power use aboard a vessel.
	130	Binary Event Monitor	Provides status and/or notification of binary conditions on the vessel. For example, binary conditions include switch positions (On/Off) and float levels (High Bilge Detected/Not Detected).
	140	Load Controller	Manages (enables/disables or is otherwise in control of) loads under command from a triggering mechanism or control function.
	141	AC/DC Input	Reports electrical properties of AC and/or DC connections feeding power to products/loads.
	150	Function Controller	Logical controller that interprets trigger events to determine functional state transitions. For example, a function controller might be responsible for turning on all bilge pumps in the event of a high bilge condition signaled from a binary event monitor by sending commands to one or more load controllers.

Class Code	Funct Code	Class / Function Name	Description
35		Electrical Generation	Equipment that outputs electrical "power", not used primarily to propel the vessel

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Class Code	Funct Code	Class / Function Name	Description
	140	Engine	Devices that convert chemical energy in a fuel into kinetic energy; Primarily under this class, the engine acts as a kinetic energy source for a generator. This function is normally only used for an Engine ECU separate from the Generator ECU.
	141	DC Generator/Alternator	Converts kinetic energy to DC power; under this function, the kinetic energy source is relatively stable and can respond to load variations. This function is used both by ECUs that report/control only the DC generation parameters, and by ECUs that report/control both DC generation parameters and the associated engine parameters.
	142	Solar Panel (Solar Array)	Converts solar energy to DC power
	143	Wind Generator (DC)	Converts kinetic wind energy to DC power, under this function the kinetic energy source is unstable, and may not sustain a given load for any length of time.
	144	Fuel Cell	Produces DC power by electrochemical conversion.
	145	Network Power Supply	Provides power to an NMEA 2000 network.
	151	AC Generator	Converts kinetic energy to AC power; under this function, the kinetic energy source is relatively stable and can respond to load variations. This function is used both by ECUs that report/control only the DC generation parameters, and by ECUs that report/control both AC generation parameters and the associated engine parameters.
	152	AC Bus	Reports and/or controls AC electrical properties of an AC electrical bus.
	153	AC Mains (Utility/Shore)	Reports and/or controls AC electrical properties of power originating off the vessel.
	154	AC Output	Reports and/or controls AC electrical properties of an AC connection feeding power to a bus or another group of products.
	160	Power Converter - Battery Charger	Device capable of charging a battery/batteries; This function applies to both AC and DC sourced chargers.
	161	Power Converter - Battery Charger+Inverter	Devices that convert AC power to DC power to charge a battery AND convert DC power to AC.
	162	Power Converter - Inverter	Converts DC power to AC power

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Class Code	Funct Code	Class / Function Name	Description
	163	Power Converter - DC	Converts one voltage of DC power to another DC power voltage
	170	Battery	Reports battery status
	180	Engine Gateway	Device that brings information from an engine used for electrical generation onto the NMEA 2000 network.

Class Code	Funct Code	Class / Function Name	Description
40		Steering and Control Surfaces	Equipment used to change the direction and/or attitude of the vessel {Formerly 'Steering Systems'}
	130	Follow-up Controller	Follow-up controller (Helm/JoyStick/etc.)
	140	Mode Controller	Mode controller
	150	Autopilot	Devices that employ a mechanical, electrical or hydraulic system to guide the ship without assistance from a human being (also known as self-steering gear) {Formerly 'Automatic Steering Controller'}
	155	Rudder	Devices that monitor and/or control a control surface used to steer vessel by re-directing the flow of water past the hull (i.e. rudders)
	160	Heading Sensors ***DEPRECATED - Function not for use in new designs!***	Heading sensors {Deprecated - See Navigation/Own ship Attitude - Class Code 60, Function Code 140}
	170	Trim (Tabs)/Interceptors	Devices that monitor and/or control a control surface used to provide lift to compensate for changes in speed, weight distribution and water conditions
	180	Attitude (Pitch, Roll, Yaw) Control	Devices that employ a mechanical, electrical or hydraulic system to maintain a given vessel attitude without assistance from a human being (also known as stabilizer)

Class Code	Funct Code	Class / Function Name	Description
50		Propulsion	Equipment used to move the vessel (ex. Trolling, Thruster, Kicker, and Propulsion devices) {Formerly Propulsion Systems}
	130	Engineroom Monitoring ***DEPRECATED - Function not for use in new designs!***	Engineroom monitoring {Deprecated - See Internal Environment - Class Code 90, Function Code 130}

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Class Code	Funct Code	Class / Function Name	Description
	140	Engine	Devices that convert chemical energy in a fuel into kinetic energy; In this class the engine acts primarily as a propulsion device (Combustion). This function is used both by component level ECUs that report control only propulsion engine parameters and also by system level ECUs that report/control parameters related to an entire propulsion train (for esample an engine and transmission). {Formerly Engine interface}
	141	DC Generator/Alternator	Devices that convert kinetic energy to DC power used to drive propulsion devices (generator). These devices shall be tied to the engine.
	145	Thruster Engine	Includes Engines that power a Thruster
	150	Engine Controller ***DEPRECATED - Function not for use in new designs!***	Engine controller {Deprecated - See Propulsion/Engine - Class Code 50, Function Code 140}
	151	AC Generator	Devices that convert kinetic energy to AC power used to drive propulsion devices (generator). These devices shall be tied to the engine
	155	Motor	Devices that convert electrical energy into kinetic energy; In this class, the motor acts primarily as a propulsion device (Electrical)
	160	Engine Gateway	Device that brings information from an engine used for propulsion onto the NMEA 2000 network
	165	Transmission	Gear systems providing speed-power conversion
	170	Throttle/Shift Control	Local to the propulsion system, may be the actuator or fuel injection system, etc. { Previously "Control Head" }
	180	Actuator ***DEPRECATED - Function not for use in new designs!***	Actuator {Deprecated - unused prior to version 2.00}
	190	Gauge Interface ***DEPRECATED - Function not for use in new designs!***	Gauge interface {Deprecated - unused prior to version 2.00}
	200	Gauge Large ***DEPRECATED - Function not for use in new designs!***	Gauge, large {Deprecated - See Display/Display - Class Code 120, Function Code 130}
	210	Gauge Small ***DEPRECATED - Function not for use in new designs!***	Gauge, small {Deprecated - See Display/Display - Class Code 120, Function Code 130}

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Class Code	Funct Code	Class / Function Name	Description
60		Navigation	Equipment that provide information related to the passage of the vessel and potental obstructions/hazards {Formerly Navigation Systems}
	130	Bottom Depth	Devices that report distance to bottom {Fomerly Sounder, Depth}
	135	Bottom Depth/Speed	Devices that report distance to bottom AND speed through water. This allows combination devices of this type to provide both functions with one address. {Formerly Unnamed}
	140	Ownship Attitude	Devices that report heading, pitch, roll, yaw, angular rates
	145	Ownship Position (GNSS)	Devices that report vessel Latitude, Longitude, ground speed, etc. using satellites (ex. GPS, GLONAS, GALILEO, etc.) {Formerly Global Navigation Satellite System (GNSS)}
	150	Ownship Position (Loran C)	Devices that report vessel Latitude, Longitude, speed (water/ground), etc. using terrestrial instrumentation (i.e. LORAN C)
	155	Speed	Devices that report water/ground speed. {Formerly Speed Sensors}
	160	Turn Rate Indicator ***DEPRECATED - Function not for use in new designs!***	{Deprecated - See Navigation/Own ship Attitude - Class Code 60, Function Code 140}
	170	Integrated Navigation ***DEPRECATED - Function not for use in new designs!***	{Deprecated - See Navigation/Own ship Position - Class Code 60, Function Code 145 OR Navigation/Integrated Navigation System - Class Code 60, Function Code 175}
	175	Integrated Navigation System	Devices that receive navigation data from NMEA 2000 bus and/or internal sources. These devices report back to bus the data selected/used in vessel navigation. The data reported back to the bus should NOT be interpreted as source data for other data consumers.
	190	Navigation Management	Devices that report general information necessary to manage navigational equipment as opposed to measured/sensed parameters. Such information includes route/waypoints, distance to waypoint, cross track error, estimated arrival times, waypoint position, datum to use, variation, navigation events such as waypoint arrival, etc.

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Class Code	Funct Code	Class / Function Name	Description
	195	Automatic Identification System (AIS)	Devices that provides dynamic and static information about other vessels in the waterway for the purpose of collision avoidance. These devices may broadcast ownship dynamic and static information to other vessels.
	200	Radar	Devices that report position of potential obstructions. These devices typically use electromagnetic waves to identify the range, direction and speed of both moving and fixed objects. {Formerly Radar and/or Radar Plotting}
	201	Infrared Imaging	Devices that report position of potential obstructions. These devices typically use infrared radiation generated by both moving and fixed objects to identify their range, direction and speed.
	205	ECDIS ***DEPRECATED - Function not for use in new designs!***	Electronic Chart Display and Information System (ECDIS) {Deprecated - See Display/Display - Class Code 120, Function Code 130}
	210	ECS ***DEPRECATED - Function not for use in new designs!***	Electronic Chart System (ECS) {Deprecated - See Display/Display - Class Code 120, Function Code 130}
	220	Direction Finder ***DEPRECATED - Function not for use in new designs!***	Direction finder {Deprecated - unused prior to version 2.00}
	230	Voyage Status	Devices that report time aggregated data, such as average speed, or total distance traveled.

Class Code	Funct Code	Class / Function Name	Description
70		Communication	Equipment used to communicate ship to ship, ship to shore, or intra ship, such as VHF, SSB, Intercom(s) and SATCOMs, etc. {Formerly Communications Systems}
	130	EPIRB ***DEPRECATED - Function not for use in new designs!***	Emergency Position Indicating Radio Beacon (EPIRB) {Deprecated - See Safety/EPIRB - Class Code 20, Function Code 130}
	140	AIS ***DEPRECATED - Function not for use in new designs!***	Automatic Identification System (AIS) {Deprecated - See Navigation/AIS - Class Code 60, Function Code 195}
	150	DSC ***DEPRECATED - Function not for use in new designs!***	Digital Selective Calling (DSC) {Deprecated - See Communication/Radiotelephone - Class Code 70, Function Code 190}

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Class Code	Funct Code	Class / Function Name	Description
	160	Data Receiver/Transceiver	One or two way communication devices intended primarily for digital data (ex. NavTex, WeatherFax) {Formerly Data Receiver}
	170	Satellite	Satellite Communications
	171	Cellular Data Network	One or two way communications via a cellular data network, including SMS and/or IP traffic
	180	Radio-telephone (MF/HF) ***DEPRECATED - Function not for use in new designs!***	Radio-telephone (MF/HF) {Deprecated - See Communication/Radiotelephone - Class Code 70, Function Code 190}
	190	Radiotelephone	One or two way communication devices intended primarily for voice. This includes equipment with DSC hailing capabilities. {Formerly Radio-telephone (VHF)}

Class Code	Funct Code	Class / Function Name	Description
75		Sensor Communication Interface	Equipment that measures one or more of the same parameter per function code for general purpose use. Devices under this class code may require configuration, such as temperature source. For example a device that reports data from two (2) temperature sensors would require one address (Class 75, Function 130). A device that reports data from one (1) temperature sensor and one (1) pressure sensor would require two (2) addresses, one to report the temperature (Class 75, Function 130) and one to report the pressure (Class 75, Function 140)
	130	Temperature	Devices that measure/report temperature
	140	Pressure	Devices that measure/report pressure
	150	Fluid Level	Devices that measure/report fluid level
	160	Flow	Devices that measure/report flow
	170	Humidity	Devices that measure/report humidity

Class Code	Funct Code	Class / Function Name	Description
80		Instrumentation/General Systems ***DEPRECATED - Class not for use in new designs!***	Instrumentation/General systems {Deprecated class and assigned functions - see specific functions for recommendations for future use}

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Class Code	Funct Code	Class / Function Name	Description
	130	Time/Date Systems ***DEPRECATED - Function not for use in new designs!***	Time/Date systems {Deprecated - unused prior to verision 2.00}
	140	VDR ***DEPRECATED - Function not for use in new designs!***	Voyage Data Recorder (VDR) {Deprecated - See Communication/Radiotelephone - Class Code 70, Function Code 190}
	150	Integrated Instrumentation ***DEPRECATED - Function not for use in new designs!***	Integrated instrumentation {Deprecated - See Sensor Communication Interface - Class Code 75}
	160	General Purpose Displays ***DEPRECATED - Function not for use in new designs!***	General purpose displays {Deprecated - See Display/Display-Class Code 120, Function Code 130}
	170	General Sensor Box ***DEPRECATED - Function not for use in new designs!***	General sensor box {Deprecated - See Sensor Communication Interface - Class Code 75}
	180	Weather Instruments ***DEPRECATED - Function not for use in new designs!***	Weather instruments {Deprecated - See External Environment/Atmospheric - Class Code 85, Function Code 130}
	190	Transducer/General ***DEPRECATED - Function not for use in new designs!***	Transducer/general {Deprecated - See Sensor Communication Interface - Class Code 75}
	200	NMEA 0183 Converter ***DEPRECATED - Function not for use in new designs!***	NMEA 0183 converter {Deprecated - See Inter/Intranetwork Device/ NMEA 0183 Gateway - Class Code 25, Function Code 135}

Class Code	Funct Code	Class / Function Name	Description
85		External Environment	Equipment monitoring the meteorological conditions on the exterior of the vessel.
	130	Atmospheric	Devices that measure/report weather conditions such as Wind, Barometric Pressure, Temperature, Humidity, Dew point, Wind Chill, Heat Index, etc.
	160	Aquatic	Devices that measure/report water conditions such as Temperature, Salinity, Current, Tide, Wave Frequency/Height, etc.

Class Code	Funct Code	Class / Function Name	Description
90		Internal Environment	Equipment monitoring the conditions of interior spaces such as cabins, compartments and engine rooms. {Formerly Environmental (HVAC) Systems}

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Class Code	Funct Code	Class / Function Name	Description
	130	HVAC	Systems that report and/or control climate conditions such as a heating system, an airconditioning system, etc. For example a heating system that measures cabin temperatures. Where the cabin temperature sensor is not on the bus, would use this class/function code. A stand-alone device that measures and reports cabin temperature and is directly on the bus should use class 75, function 130.

Class Code	Funct Code	Class / Function Name	Description
100		Deck, Cargo and Fishing Equipment	Deck, cargo and fishing equipment systems
	TBD	Function Name	Enter function description
	130	Scale (Catch)	Equipment used to measure/weigh aquatic species caught in commercial or recreational fishing activities.

Class Code	Funct Code	Class / Function Name	Description
110		Human Interface Device	Devices that provide an input interface with little or no feedback to the user. Example Systems could include: Devices which have keys, buttons, knobs, sliders or other input mechanical components which are digitized by micro processor and communicated to other network devices such as displays. Example PGNs could include: Binary Status Report (127501), Switch Bank Control (127502), Proprietary PGNs
	130	Button Interface	Devices that communicate momentary switch interactions to the network. Example Products could include: Keyboards, keypads, remote control devices Most likely using Proprietary PGNs designed specifically for a product's specific set of keys.
	135	Switch Interface	Devices that communicate non-momentary switch interactions to the network. Example Products could include: Light switches, relay control switches, pump switches. Example PGNs could include Binary Status Report (127501), Switch Bank Control (127502), Proprietary PGNs
	140	Analog Interface	Devices that communicate analog input interactions to the network. Example Products could include: Light Dimmer Switches & Trim Tab Adjustment most likely using Proprietary PGNs designed specifically for a product's specific purpose.

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Class Code	Funct Code	Class / Function Name	Description

Class Code	Funct Code	Class / Function Name	Description
120		Display	Equipment that provides visual or audible indication/reporting of data parameters
	130	Display	Devices that provide active or passive user interface. A display may also include alarm enunicator functions.
	140	Alarm Enunciator	Stand-alone devices that provide passive visual and/or audible alert indication(s).

Class Code	Funct Code	Class / Function Name	Description
125		Entertainment	Multimedia or other communication equipment not impacting safe vessel navigation.
	130	Multimedia Player	Equipment that provides visual or audible playback of recorded media for entertainment purposes.
	140	Multimedia Controller	Equipment that transmits multimedia commands in response to man-machine input or other external events.

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NMEA 2000 Appendix B - Revision History

1	Sep 2001 1.001	Initial Release
2	Oct 2001 1.001	This Revision History Log Added
3	Oct 2001 1.001	DF09 Range Correction from 10E+5 to 10E+7
4	Oct 2001 1.001	DF21 Range Correction from 10E+10 to 10E+8
5	Dec 2001 1.001	DD176 Network Addresses updated the allocations. Corrections made to agree with specifications
6	Jan 2002 1.001	PGN 127489 - added a second Discrete Status Field 12
7	Jan 2002 1.001	DD223 Dictionary item added for 127489
8	Jan 2002 1.001	PGN 129545 inserted field 3 (Reserve of 6bits) for alignment and corrected Latitude expected error's Data Dictionary reference from DD001 to DD220
9	Sep 2002 1.002	PGNs 060160 & 060416 (Transport Protocol) chgd from priority 7 to 6 as defined by ISO
10	Sep 2002 1.002	PGN 060416 correct "ABORT" description labeling
11	Nov 2002 1.002	PGN 126996 corrected Single Frame "Yes" to "No", field 2 corrected grammer from "manufactures" to "manufacturer's"
12	Nov 2002 1.002	PGN 130321 Field 6 DD022 corrected to DD023
13	Nov 2002 1.002	PGN 129540 Added field 3 Reserve to pad previous field 2 - Mode to Byte, this new field insertion shifted all fields above 2 up 1
14	Nov 2002 1.002	PGN 129033 In description corrected incorrect pgn reference of 128012 to 126992
15	Nov 2002 1.002	PGN 126998 changed note field from 50 char to 70 ASCII or 35 Unicode
16	Nov 2002 1.002	PGN 130577 in description corrected pgn reference from 128006 & 128007 to 128259 & 129026. Also removed no longer appropriate reference to 128008.
17	Nov 2002 1.002	PGN 130577 in description corrected pgn reference from 128012 to 126992

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18	May 2003 1.003	Changed Ack Rqmnts Field in PGNs 059904, 060416, 060928, 065240,126208
19	May 2003 1.003	PGN 128520 - removed field 15 Reserve Bits is not needed at end.
20	May 2003 1.003	PGN 128520 - changed fields14 & 13 fromDD007 to DD050 var length short ASCII
21	May 2003 1.003	DD045 - Removed description with reference to ground, mag or true
22	May 2003 1.003	PGN 059392 - added clarification to description that message will always be sent with a destination address of 255. (Per agreement with SAE/ISO)
23	May 2003 1.003	PGN 127488 in description corrected pgn reference from 127509 to 127489
24	May 2003 1.003	PGN 129033 in description corrected pgn reference from 128012 to 126992
25	May 2003 1.003	PGN 129029 in description corrected pgn reference from 128005 to 129025
26	Jul 2003 1.004	PGN 126996 field 1 clarified version to "NMEA 2000 Database Version"
27	Jul 2003 1.004	PGN 127251 Rate of turn added along with DD125, DF85
28	Jul 2003 1.004	PGN 128520 fields 13 & 14 changed to ASCII String Fields
29	Jul 2003 1.004	DD217 correct upper 2 bit field definitions
30	Jul 2003 1.004	PGN 129540 GNSS Sats in View. Corrected field references, added missing filed 16 Range Residuals "n", Clarified Description.
31	Sep 2003 1.004	Moved Eng Trim from PGN 127489 to PGN 127488
32	Oct 2003 1.005	PGN 127251 Rate of Turn, deleted Vessel Heading
33	Nov 2003 1.005	Moved "The message will always be sent with a destination address of 255" from ISO Request (pgn 059904) to ISO Acknowledge (pgn 059392)
34	Nov 2003 1.005	Time & Date PGN 129033 corrected old PGN reference 128012 to 126992
35	Feb 2004 1.100	PGN 127489 changed Fuel Pressure Range (field 9); required adding DD225 with DF29 range
36	Feb 2004 1.100	PGN 126208 Command Group, Description clarification made

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37	Feb 2004 1.100	PGN 129283 Cross Track Error, removed field 3, increase Reserve field size to adjust
38	Feb 2004 1.100	PGN 129550 chgs fields 3&4 & PGN 129551 chgs fields 8,9
39	Mar 2004 1.100	PGN 127505 Fluid Level, added field 4 - Tank Capacity
40	Mar 2004 1.100	PGN 129284 Navigation Data - updated Description
41	Mar 2004 1.100	PGN 129291 Set & Drift, Rapid Update - updated Description
42	Mar 2004 1.100	PGN 130577 Direction Data - field 8 added "through water" to Speed name
43	Apr 2004 1.101	PGN 127505 Fluid Level, revised field 4 size and resolution added DF86 & DD227 to support the change
44	Apr 2004 1.101	PGN 129283 Cross Track Error, added field 4 "Navigation Terminated" to previously "reserve" field
45	Apr 2004 1.101	PGN 129808 DSC Call Info, fields 8, 22, 24 use DD015, no other reference - DD015 was changed from DF63 to DF50. Notes added to each field defining maximum size
46	Apr 2004 1.101	PGN 126208 Command Group, Ack group 2 field 3 PGN error code state 0x4 added , 0xF removed
47	Apr 2004 1.101	PGN 127489 Eng Parms, Dynamic - add % Torque and % Load
48	Apr 2004 1.101	PGN 127489 Eng Parms, Status 2 - added Maintenance Required and Comm error Alarms
49	Apr 2004 1.101	PGN 130311 Environmental Parameters added to be used insterad of PGN 130310 in new designs. Allows for instance of temp and humidity. More flexible. DD129, DD130, and DD131added to support this.
50	May 2004 1.111	PGN 129027 Position Delta & PGN 129028 Altitude Delta addded
51	May 2004 1.111	DD233 thru DD235 & DF88 thru DF90 added to support PGN129027&129028
52	May 2004 1.111	PGN 127496 Trip Parameters, Vessel
53	May 2004 1.111	PGN 127497 Trip Parameters, Engine - revised per discussions last meeting
54	May 2004 1.111	Nav & WP PGN's 129285, 130064 thru 130072 added
55	May 2004 1.111 +	PGN 127258 new PGN Name chgd from Headding variation to Magnetic Variation

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56	Sep 2004 1.111 B	Engine Parms Dynamic, added to Field 12, Engine discrete Status 2, the following status bits: Sub or Secondary Throttle, Neutral Start Protect, and Engine Shutting Down.
57	Sep 2004 1.111 B	Added description to Titles on first cover page on all reports
58	Sep 2004 1.111 B	PGN 126996 Product Information - added field 8 - Load Equivalency Number, DD242 added in support of this, other text clarifications
59	Oct 2004 1.111 B	Way Point and Route PGNs and supporting items updated.
60	Oct 2004 1.111 B	AIS PGNss and supporting items added and/or updated
61	Nov 2004 1.111 B	PGN 127508 - Field 5 SID added, replaced spare
62	Nov 2004 1.200	DD237 - bit field typo corrections
63	Nov 2004 1.200	PGN 130065 fields 7,8,9 realignment
64	Nov 2004 1.200	PGN 130066 fields 9,1,11 realignment
65	Nov 2004 1.200	WP & Route PGNs 130066, 130067, 130068, 130073, 130074 Descriptions updated
66	May 2009 1.200 A	Fluid Level Corrected DF84 range and resolution (used in PGN 127505)
67	May 2009 1.200 A	Added to PGN Reports the display of the PGN number in hexidecimal format. Located below original Decimal PGN #
68	May 2009 1.200 A	PGN 060928 ISO Address Claim - field 11 note field changed from "Only values less than 254 shall be used" to "Only values less than 252 shall be used"
69	Jul 2005 1.200 A	Added extensions to Command Group PGN 126208
70	Aug 2005 1.200 A	Added Proposed Power PGNs (PGN #s not yet assigned)
71	Aug 2005 1.200 A	PGN 126208 Request group Function, field 3 Transmission Interval added state 0xFFFF FFFE Restore default Time Interval
72	Aug 2005 1.200 A	PGN 059392 ISO Ack remove last line of description "This message will always be sent with a destination address of 255"
73	Sep 2005 1.200 B	PGN 059392 ISO Ack add last line of description "The destination address of this PGN shall always contain a destination specific address" and note about version 1.000
74	Sep 2005 1.200 B	PGN 129796 AIS Acknowledge correct field 10 from DD001 to DD010

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75	Jul 2006 1.200 C	Power PGN Support added DF92 - DF102
76	Jul 2006 1.200 C	Power PGN Support added DD258 - DD288
77	Jul 2006 1.200 C	Power PGNs added: 127503 AC Input Status 127504 AC Output Status 127506 DC Detailed Status 127507 Charger Status 127509 Inverter Status 127510 Charger Configuration Status 127511 Inverter Configuration Status 127512 Automatic Generator Start Status 127513 Battery Configuration Status 127514 Automatic Generator Start Configuration Status
78	Aug 2006 1.200 D	Meeting Updates PGN 060928,127250, int24,129039,129538,130323, DD025, DD070, DD232
79	Sep 2006 1.210	PGNs added 130312,130313,130314,130315 to replace 130311 for future designs
80	Sep 2006 1.210	PGN 128259 designate field 3 not for new designs refer to 129026 for value
81	May 2009 1.300	Revised and added AIS Class B PGNs in accordance with AIS Class B working group report dated 24 April 2006, and as modified July 2008. - Revised: 129039, 129040 - Restored from vers 1.111: 129806 - New: 129807, 129809, 129810 - Data Definitions added: DD294 - DD304
82	May 2009 1.300	Implemented structural changes to database to make database maintenance easier: - Added VersionHistory table and changed all reports to list version identified for this file when printing - Changed "Pick_PGN_Form" to also refer to version identified for this file - Added RevisionHistory table, imported revision history data from "PGNs by Number" report footer, removed revision history data from report footer - Added "RevisionHistory-Edit" form for entering/editing revision history data - Added new "RevisionHistoryReport" and Revision History button on "Pick_PGN_Form" to launch it
84	May 2009 1.300	Revised AIS Class A PGNs to reflect changes and corrections noted during Update of AIS Class B PGNs; PGNs, affected: 129794, 129795, 129797, 129798, 129801, 129802.
85	Jun 2009 1.300 B	Corrected PGN 130315 and 130313 destination; changed from no value to Global.
86	Mar 2010 1.300 B	Revised PGNs 127503 AC Input Status, and 127504 AC Output Status. Marked as not for use in new designs per Standards Committee Meeting minutes of Feb 2008 and referred manufacturers to J1939-75.
87	Apr 2010 1.301	Structural change to create PGN sub-selection groups to allow printing Parameter Groups either by the full list or by the following selections: AIS, Environmental, Navigation, Power, Propulsion, Steering. All selections include the subselections Mandatory and Other. New selection group reports include electronic licensing agreement.

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88	May 2010 1.301	Structural/formatting changes: - Modified reports used for standard publishing to include to include electronic licensing agreement, and cleaned up formatting so all reports are similar in formation and have consistent use of headers and footers. Reports modified: - Parameter_Group_ByNumber_Report - Data_Formats_ByNumber_Report - Data_Dictionary_Report - Type_ByType_Report - Created PGN Field List report.
89	Jul 2010 1.301 A	Added PG Short Name and PG Short Description to IDRef_Tbl for recording public unlicensed data dumps.
90	Sep 2010 1.301 A	Revise PGN 126208 to clarify that the Command group function always requires an acknowledgement either positively or negatively, and Request group functions only require an acknowledgement if the request cannot be completed. [August 2005 Standards Committee Meeting, Agenda Item 17b]
91	Sep 2010 1.301 A	(For Confirmation) Per request from Jacop Lie Simonsen, change PGN 127258 from priority 7 to priority 6 in recognitiion of the fact the the PGN contains real-time data. [October 2005 Standards Committee, Agenda Item 16]
92	Sep 2010 1.301 A	(For Confirmation) Change description for System Time (PGN 126992) and rename and update Time and Date (PGN 129033) in accordance with Standards Committee decision. Also removed/reworded Application Notes as necessary to ensure there is no conflict with the revised description [June 2007 Standards Committee Meeting, Agenda Item 7]
93	Sep 2010 1.301 A	Add to description for Engine Parameters, Static (PGN 127498) to require that no bit fields in Engine Discrete Status 1 and 2 may be set to unused/unimplemented. Also added requirement to DD206 and DD223. [May 2008 Standards Committee Meeting, Agenda Item 11]
94	Sep 2010 1.301 A	Add value ??? "Atmospheric Pressure, Corrected for Altitude" to Actual Pressure (PGN 130314), Field 3, DD289. [October 2007 Standards Committee Meeting, Agenda Item 6]
95	Sep 2010 1.301 A	Revise DF63, String Fixed, to require that unused characters be filled with 0xFF = Data not available. [July 2008 Standards Committee Meeting, Agenda Item 2]
96	Sep 2010 1.301 A	Corrected typographical error in Int24 definition: Range was mis-stated, should be -8,388608 to 8,388,607. Also impacted definition of reserved values. [Reported by Actisense 2010-02-17]
97	Sep 2010 1.301 A	Corrected DF102 by revising resolution to match the specified range. Also revised all resulutions with fractional values to conform to standard "1x10E-n units". Affected DFs are: DF66, DF81, DF84, DF97, DF101, DF102, DF103. [Reported by Actisense 2010-02-17]
98	Sep 2010 1.301 A	(For Confirmation) Add requirement that Product Information (PGN 126996) parameter groups may only be requested from one device at a time. [July 2008 Standards Committee Meeting, Agenda Item 3]
99	Sep 2010 1.301 A	Corrected typo in GNSS Differential Correction Receiver Signal, Field 9, Differential Signal Detection Mode; removed invalid designation of 3 reserved bits. [Reported by Actisense 2010-02-17]
100	Sep 2010 1.301 A	Corrected field designations for GNSS Differential Correction Receiver Interface (PGN 129550), resolved missing field 6 by changing fields 7,8 to 6,7. [Reported by Actisense 2010-02-17]
101	Oct 2010 1.301 A	Added new parameter group AIS Aids to Navigation (AtoN) Report (PGN 129041); Also created new Data Definitions DD305, DD306, DD307, DD308, DD309; Also updated DD191 to define meaning for a value of 7. (Incorporated minor typographical fixes to new parameter group and DD's 10/6 & 10/19.)

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103	Jan 2011 1.301 B	Revised SystemTime (126992) to state that when transmitted it always represents current system time; Revised Local Time Offset (PGN 129033) to remove the phrase "with the SID set to 255". (reverses previous revision 92) [September 2010 Standards Committee Meeting, Agenda Item 12]
104	Jan 2011 1.301 B	Modified Sequence ID (DD 056) to show a valid range of from 0 to 252, replacing a range from 0 to 250. [September 2010 Standards Committee Meeting, Agenda Item 12]
105	Feb 2011 1.301 B	Modified DD302 in accordance with AIS 1371-4 Updates dated 15 February 2011 by L. Luft. [February 2011 Standards Committee Meeting, Agenda Item 11] Net change is to revise order of enumerations 9, 10, and 11.
106	Feb 2011 1.301 B	Revised AIS Class A Position Report (PGN 129038) in accordance with AIS 1371-4 Updates dated 15 February by L. Luft. Also Created new DD310 & DD311. [February 2011 Standards Committee Meeting, Agenda Item 11]
107	Feb 2011 1.301 B	Revised AIS Group Assignement (PGN 129807) to remove 'Class B' from its title; Added 'AIS Transceiver Information field to PGNs 129807, 129809, and 129810. [February 2011 Standards Committee Meeting, Agenda Item 11]
108	Mar 2011 1.301 B	Additional changes and confirmation of revisions 105-108 in accordance with L.Luft markup. [nmea-2000-corrigendum-1-2011_LAL_Mar_7_1]
109	Mar 2011 2.000	Added Payload Mass parameter group (PGN 130560) in accordance with approvde PGN Template, including required DD312 Scale Measurement Status, DD313 Mass, and DF104 Mass; Added Command Support Field to IDRef_Tbl and Command Parameter Field to IDRef_Detail_Tbl [February 2011 Standards Committee Meeting, Agenda Item 12]
110	Jul 2011 2.000	Revised Address Claim (PGN 60928) to provide additional clarification of requirements and add a reference to address mode and request frequency limits established in Section 8.3.2; Revised PGN List - Received PGNs group function and PGN List - Transmitted PGNs group function to provide typographical corrections and to establish limitations on when requests for them may be made globally; Revised Product information (PGN 126996) and Configuration Information (PGN 126998) both to establish limitations on when requests for them may be made globally. [February 2011 Standards Committee Meeting, Agenda Item 15]
111	Jul 2011 2.000	Revised Binary Switch Bank Status (PGN127501) and Switch Bank Control (PGN127502 descriptions in accordance with descriptions developed at the meeting of the Power Distribution Group with the PGN committee in August 2010. Also added Command Support and Command Parameter field values according to the intent of the description and meeting notes. [February 2011 Standards Comittee Meeting, Agenda Item 4]
112	Jul 2011 2.000	Added visability for Command Support and Command parameter fields added in revision 109. Includes changes to Queries "Test_Report_Query," "New_Report_Query," and "PG_Selection_Report_Query," and changes to reports "Test_Report," "Parameter_Group_Report," "Parameter_Group_By_Number_Report," and "Parameter_Group_Selection_Report_By_Number."
113	Jul 2011 2.000	Added new parameter group: Temperature, Extended Range (PGN 130316), and deprecated parameter group Temperature (PGN 130312). Added new Data Dictionary item Temperature, extended range and precision (DD314). Added new Data Format item Temperature, extended range and precision (DF105). Added new type uint24. [February 2011 Standards Committee Meeting, Agenda Item 9]
114	Jan 2012 2.000	Added new parameter group: Payload Mass (PGN 130560)

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116	Feb 2012 2.000	Revised the following AIS PGN's to include Sequence ID (SID) as the last field,so these PGN's can be linked to other PGN's: 129038, 129039, 129040, 129793, 129794, 129796, 129798, 129800, 129801, 129802, 129803, 129804, 129805, 129806, 129807, 129810 129810
117	Apr 2012 2.000	Edited AIS ACA Channel Assignment Message (PGN 129806). Field 9 has become the Source ID field. Fields 11 thru 14 have Lat Lon Resolution specified to 1/10 min. Addition of In Use Flag (Field 24), Time of in-use Flag Cahnge (Field 26), SID (Field 27).
118	Mar 2012 2.000	Edited Output rate of PGN 130313
119	Mar 2012 2.000	Edited Bit Field size of PGN 129285
121	Mar 2012 2.000	Added Exhaust Gas Temperature as a state to Field 3 of PGN 130316.
122	Apr 2012 2.000	Deprecated PGN's 130310, 130311, 130312. Within the PGN descriptions lists all details and suggested PGN to use as replacements
123	Mar 2012 2.000	Added the following states to Filed 3 (DD289) to PGN 130314 - Filter Pressure, Altimiter Setting, Oil Pressure, and Fuel Pressure.
124	May 2012 2.000	Added "Maximum Depth Range" (Field 4) to PGN 128267
125	May 2012 2.000	Added "Speed Direction" (Field 5) to PGN 128259
126	May 2012 2.000	Added new ALERT Parameter Groups: Alert (PGN 126983), Alert Response (PGN 126984), Alert Text (PGN 126985), Alert Configuration (PGN 126986), Alert Threshold (PGN 126987), Alert Value (PGN 126988). Added 900 Alert codes to be used with Alert PGN's Added Alert document Appendix which details Alert PGN's and code implementation.
127	May 2012 2.000	Added Default update rate (2000ms) to PGN 130313 Humidity
128	May 2012 2.000	Added Field 8 (Consumed Amp Hours) to PGN 127506
129	Aug 2012 2.000	Added new Parameter Groups: Label (PGN 130060) and Channel Source Configuration (PGN 130061)
130	2.000	Added new Power Parameter Groups: Load Controller Connection State and Control (PGN 127500), AC Power / Current, Phase A (PGN 127744), AC Power / Current, Phase B (PGN 127745), AC Power / Current, Phase C (PGN 127746), AC Voltage/Frequency, Phase A (PGN 127747) , AC Voltage/Frequency, Phase B (PGN 127748), AC Voltage/Frequency, Phase C (PGN 127749), Converter (Inverter/Charger) Status (PGN 127750)
135	Sep 2012 2.000	Changed state options of Query Support box for all PGN headers New state options are: Required(formerly YES), Optional, Prohibited(formerly NO) States are described in Appendix A, Version 2.00
136	Sep 2012 2.000	Changed state options of Request Parameter box for ALL Fields within ALL PGN's New state options are: Required(formerly YES), Optional, Note(which is described in the PGN header) States are described in Appendix A, Version 2.00

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137	Sep 2012 2.000	PGN 060416 (CTS Group Function) Added PGN description.
138	Sep 2012 2.000	PGN 060416 (EOM Group Function) Added PGN description.
139	Sep 2012 2.000	PGN 060416 (RTS Group Function) Added PGN description.
143	Sep 2012 2.000	PGN 126208 (Command Group Function) Added to description Note 1: When sending a complex command to a device, when accepted if the PGN referenced has command capability, then these fields shall contain the commanded values.
144	Sep 2012 2.000	PGN 126208 (Request Group Function) Edited description and added the following: NMEA encourages manufacturers to provide the capability of altering the values of transmission time for NMEA certified products on the network. When the values of transmission time or delay before the first transmission are altered, they should be maintained after power cycling.
145	Sep 2012 2.000	Edited fields within all PGN's to reference new Class & Function Codes Appendix B6 (where applicable)
146	Sep 2012 2.000	All PGN's- removed "NMEA 2000" specifics in PGN descriptions in preparation for NMEA OneNet. These are now referred to as "NMEA Network Messages"
147	Sep 2012 2.000	126996 Field 1 added the following note: NMEA encourages manufacturers to provide the capability to support multiple NMEA Network Message database versions within their product. This provides a means for manufacturers products to retain backwards compatability with previous versions of the database and existing legacy products.
148	Sep 2012 2.000	Added NEW Command Support header box to ALL PGN's Command Support state options are: Required, Optional, Prohibited States are described with detail in Appendix A, Version 2.00
149	Sep 2012 2.000	Added NEW Command Parameter box to ALL Fields within ALL PGN's Command Parameter state options are: Required, Optional, Prohibited, Note(which is described in the PGN header) States are described with detail in Appendix A, Version 2.00

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150	Sep 2012 2.000	Changed the Query Support header box value on listed PGN's below to the following values: 126998 Optional to Prohibited 126208 Optional to Required 065240 Blank to Required 060416 Blank to Required 127245 Optional to Required 127250 Optional 127251 Optional 127527 Optional 127258 No to Optional 127488 Required 127489 Optional to Required 127493 Optional to Required 127496 No to Optional 127497 Optional to Required 127498 Optional to Required 127505 Optional to Required 127507 Required 127508 Optional to Required 129027 No to Optional 129028 No to optional 129038 No to optional 129285 No to optional 129547 No to optional 129039 No to optional 129040 No to optional 129401 No to optional 129792 No to optional 129793 No to optional 129794 No to optional 129795 No to optional 129796 No to optional 129797 No to optional 129798 No to optional 129799 No to optional 129800 No to optional 129802 No to optional 129803 No to optional 129804 No to optional 129805 No to optional 129806 No to optional 129807 No to optional

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151	Sep 2012 2.000	Changed the Request Parameter box value on the following PGN's and Field Numbers within the PGN's 060416 Field 1 Optional to Required 126208 Field 1 Optional to Required 126208 Field 2 Optional to Required 126998 Field 1,2,3 Optional to Prohibited 127237 Field 1,2,3,4,5,6,7,9,10,11,12,13,14,15,16,17,18 No to Optional 127245 Field 1,2,4,5 Optional to Required 127250 Field 1,2,3,4,5 Optional 127251 Field 1,2 Optional 127527 Field 1,2,3,4 Optional 127258 Field 1,2,3,4,5 No to Optional 127489 Field 1,2,3,4,5,6,7,8,9,11,12,13,14 Optional to Required 127493 Field 1 Optional to Required 127496 Field 1,2,3,4 No to Optional 127497 Field 1 Optional to Required 127498 Field 1 Optional to Required 127500 Field 2 Required 127501 Field 1 Required 127505 Field 1 Required 127507 Field 1,2 Required 127508 Field 1 Required 127509 Field 1,2 Required 127510 Field 1,2 Required 127511 Field 1,2 Required 127512 Field 1,2 Required 127513 Field 1 Required 127514 Field 1,2 Required 129793 Field 1 Required 129796 Field 1 Required
152	Sep 2012 2.000	Added Sequence ID (SID) Sentence to PGN 129794 header description.
153	Sep 2012 2.000	PGN 060416 (Abort Group Function) Added PGN description.
155	Sep 2012 2.000	PGN 060416 (BAM Group Function) Added PGN description.
156	Nov 2012 2.000	Discussion regarding Deprecation of PGN 127502- Temporary decision was to keep 127502 until NMEA consults with SAE regarding obtaining more addressable PGN's. NMEA will then create a new addressable PGN based on 127502. Date is TBD.
157	Jan 2013 2.000	Re-named references to "Table 8 Class & Function Codes" to "Appendix B.6 Class & Function Codes"
158	Jan 2013 2.000	Deprecated PGN's As of January 2013: 127503 127504 130310 130311 130312 127507 127509

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160	Mar 2013 2.100	Released Power PGN Corrigendum containing the following PGNs 127744 AC Power / Current Phase A 127745 AC Power / Current Phase B 127746 AC Power / Current Phase C 127747 AC Voltage / Frequency Phase A 127748 AC Voltage / Frequency Phase B 127749 AC Voltage / Frequency Phase C 127750 Converter (Inverter / Charger) Status 127751 DC Voltage and Current 127500 Load Controller Connection State and Control
161	Mar 2013 2.100	Appendix B.6 Class & Function Codes Added to Class 10- Functions 120, 121, 122, 123. Added to Class 25- Functions 136, 137. Changed Solar Panel to Function 142 in Class 35. Removed NMEA 2000 Wireless Gateway from Class 90.
162	Dec 2013 2.100	PGN 127509, changed Field 7 DD item to Ripple Voltage DD287. Previously Field 7 was DD001 and this was incorrectly entered in draft version 1.4. This PGN was deprecated in V2.000. DD item was changed to align correct number of bits. Garmin identified this issue.
163	Jan 2014 2.100	Heartbeat PGN 126993 . Changed reserved Field 3 bit boundary error. Was incorrect at 24 bits, corrected to 40 bits.
164	Jan 2014 2.100	PGN 126208 Group Function PGN. Moved ACK Requirements to a Note at the end of the PGN text description for Command and Request Group function. This makes the page look cleaner and read easier as the note was too big for the ACK Requirements box which usually is filled with "none".
165	Jan 2014 2.100	PGN 126208 Read & Write Fields Group Function Clarifications: The clarification of the Read Fields and Write Fields group functions (and their associated Reply group functions) was necessary to clarify their use to interrogate and configure PGNs that have at least one instance field.
166	Jan 2014 2.100	PGN 126208 Text Descriptions clarification:The text descriptions for each specific group function have now been placed within the header of each group function. Previously, all text describing each of the group functions were only listed in the Request Group Function PGN description.
167	Dec 2014 2.100	(PDW) In Table "DD#_Tbl", DD351 was missing Bit field size - should be 8 bits. Fixed. This fix is also related to a fix at RevisionID 168 below
168	Dec 2014 2.100	(PDW) In Table "IDRef_Detail_Tbl", entry 1741 ("State"), has a value set for "Reserved bits". This should be blank - This relates to Revision ID 167 above - because the reserved bit size was set, the PGN report worked, but the definition was incorrect, this is not a reserved field.
169	Dec 2014 2.100	(PDW) In Table "DD#_Tbl", DD352 was set to a bit field size of 13. Should be 16 bits. Although the three bits at the end are "reserved", the whole field size is actually 16 bits. Fixed. This fix is also related to a fix at RevisionID 170 below.
170	Dec 2014 2.100	(PDW) In Table "IDRef_Detail_Tbl", entry 1742 ("Status") has a value of 16 set for "Reserved bits". This should be blank - This relates to Revision ID 169 above - because the reserved bit size was set, the PGN report worked, but the definition was incorrect, this is not a reserved field.
171	Dec 2014 2.100	(PDW) In Table "DD#_Tbl", DD353 ("AIS Channel Management Source") had no bit field size. Should be 3 bits. Fixed. This fix is also related to a fix at RevisionID 172 below.

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172	Dec 2014 2.100	(PDW) In Table "IDRef_Detail_Tbl", entry 1566 ("Source Identifier") has a value of 3 set for "Reserved bits". This should be blank - This relates to Revision ID 171 above - because the reserved bit size was set, the PGN report worked, but the definition was incorrect, this is not a reserved field.
173	Dec 2014 2.100	(PDW) In Table "DD#_Tbl", DD330 ("Alert State") uses DF53, which is an integer byte field (8 bits), but also has a bit field size set of 8. The bit field size is not required, it has been removed.
174	Dec 2014 2.100	(PDW) In Table "Type_Tbl", bit0(n) has been added - makes clear a bit type that defaults to a value of "0" when NULL or unused - clarifies this requirement for AIS fields. Linked to Revision ID 175 below.
175	Dec 2014 2.100	(PDW) In Table "DF#_Tbl",DF118 has been added - makes clear a data format that defaults to a value of "0" when NULL or unused - clarifies this requirement for AIS fields. Linked to Revision ID 174 above.
176	Dec 2014 2.100	(PDW) In Table "IDRef_Detail_Tbl", The following fields marked "AIS Spare" currently set to "DD001" need to change to "DD311" Where "RefID_Detail_ID" = 1038, 1056,1068, 1075, 1087, 1106, 1117, 1119, 1111, 1130, 1151, 1137, 1142, 1146, 1164, 1157, 1172, 1176, 1218, 1224, 1241, 1252, 1267, 1283, 1557, 1539, 1545, 1531, 1563, 1579, 1598 (All are definitions of the AIS Spare bits, which should use DD311 as this definition ensures that the description shows that all bit default to zero for null data)
177	Dec 2014 2.100	(PDW) In Table "IDRef_Detail_Tbl", The following fields marked "Reserved for Regional Applications" used in AIS MEssages currently set to "DD001" have been changed to "DD311" Where "RefID_Detail_ID" = 1104, 1191, 1192, 1205, 1206, 1253 (All are definitions of the AIS Spare bits, which should use DD311 as this definition ensures that the description shows that all bit default to zero for null data)
178	Dec 2014 2.100	(PDW) In Table "IDRef_Detail_Tbl", entry 1524 ("Message ID") has a value of 6 set for "Reserved bits". This should be blank - this is not a reserved field, the bit field is defined in the DD type DD188 "Bit field size" of 6 bits
179	Dec 2014 2.100	(PDW) In Table "IDRef_Detail_Tbl", entry 1525 ("Repeat Indicator") has a value of 2 set for "Reserved bits". This should be blank - this is not a reserved field, the bit field is defined in the DD type DD185 "Bit field size" of 2 bits
180	Dec 2014 2.100	(PDW) In Table "IDRef_Detail_Tbl", entry 1547 ("Repeat Indicator") has a value of 2 set for "Reserved bits". This should be blank - this is not a reserved field, the bit field is defined in the DD type DD185 "Bit field size" of 2 bits
181	Dec 2014 2.100	(PDW) In Table "IDRef_Detail_Tbl", entry 1549 ("Type of Ship and Cargo") has a value of 8 set for "Reserved bits". This should be blank - this is not a reserved field, the bit field is defined in the DD type DD193 "Bit field size" of 8 bits
182	Dec 2014 2.100	(PDW) In Table "IDRef_Detail_Tbl", entry 1582 ("Message ID") has a value of 6 set for "Reserved bits". This should be blank - this is not a reserved field, the bit field is defined in the DD type DD188 "Bit field size" of 6 bits
183	Dec 2014 2.100	(PDW) In Table "IDRef_Detail_Tbl", entry 1583 ("Repeat Indicator") has a value of 2 set for "Reserved bits". This should be blank - this is not a reserved field, the bit field is defined in the DD type DD185 "Bit field size" of 2 bits
184	Dec 2014 2.100	(PDW) In Table "IDRef_Detail_Tbl", entry 1782 ("In-Use flag") has a value of 2 set for "Reserved bits". This should be blank - this is not a reserved field, the bit field is defined in the DD type DD354 "Bit field size" of 2 bits. Linked to Revision ID 185 below.
185	Dec 2014 2.100	(PDW) In Table "DD#_Tbl", DD354 ("AIS Channel Management Parameters Usage") had no bit field size. Should be 2 bits. Fixed. This fix is related to the fix at RevisionID 184 above.

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186	Dec 2014 2.100	(PDW) In Table "DD#_Tbl", DD391 ("Time, Standard") uses DF65, which is an unsigned integer 32 bit field (32 bits), but also has a bit field size set of 32. The bit field size is not required, it has been removed, as it is already defined in the DF table
187	Dec 2014 2.100	(PDW) In Table "IDRef_Detail_Tbl" PGN 129550 [IDRef ID 27] has bad field sequence - field "7" should be field "6" and field "8" should be field "7"
189	Dec 2014 2.100	(PDW) In Table "IDRef_Detail_Tbl" ID= 1517 was not being used - it had no ID ref, name or other details. It has been re-assigned to PGN 127512 (AGS status), as this PGN had 44 unassigned bits at the end. In the default single packet PGNS, it should be made clear that remaining bits must be set to one. Adding this field 4 DD01 reserved bit definition of 44 bits makes this clear.
190	Dec 2014 2.100	(PDW) PGN 129538 has 6 bits remaining at the end of the message that should be set to "reserved" Added field 11 to this PGN with DD001 and reserved size of 6 bits to show these bits must always be set to "1"
191	Dec 2014 2.100	Removed DF115 as this was not associated with any specific Data Dictionary Item or PGN. DF115 Time Interval was the same as DF116 Time interval. DF116 has more detail and is used with the Heartbeat PGN, Field #1. DF115 is now available for future use of data dictionary items.
192	Dec 2014 2.100	ID Reference 119, PGN 129803 Field 21 has been changed to 6 bits instead of 4 bits so it is aligned on the correct byte boundary. Versions beyond this version, this field has been corrected to 6 bits, as it was 4.
193	Dec 2014 2.100	DD311 now links to DF115, instead of DF118 which was available for future use based on the 12/9/2014 change, see RevisionID 191. DF118 is now open for future use
194	Dec 2014 2.100	Added "Status" field at the far right of the IDRef_Tbl to indicate if the PGN is "Current", "Draft", or "Deprecated"
195	Dec 2014 2.100	Removed "orphan" ID References 230, 222, 224, 225 as these were not used
196	Jan 2015 2.100	Added new Data Dictionary Item specific to Engine / Transmission Instance. DD410. Updated engine and transmission PGNs to reflect using the DD410 instead of DD128. These specific PGNs are 127493, 127488, 127489, 127497, 127498
197	2.100	Added 12 new Entertainment / Audio PGNs: Entertainment Current File & Status 130569, Entertainment Library Data File 130570, Entertainment Library Data Group 130571, Entertainment Library Data Search 130572, Entertainment System Configuration Status 130580, Entertainment Zone Volume Status 130582, Entertainment Zone Configuration Status 130581, Entertainment Supported Zone Data 130574, Entertainment Supported Source Data 130573, Entertainment Available Audio EQ Presets 130583, Entertainment Bluetooth Source Status 130585, Entertainment Bluetooth Devices 130584
198	2.100	Updated Alert Configuration PGN 126986 to include instructions as to how to create custom user defined alerts. Added a new 2 bit Field 11 called User defined alert assignment along with new Data Dictionary item DD411. Moved all proceeding fields ahead by one and reduced NMEA Reserved Field 12 from 6 bits down to 4 bits.
199	Feb 2015 2.100	Increased the recommended update rate of PGN 127500 to 15,000 per request of manufacturer member Maretron.
200	Feb 2015 2.100	Combined Reserved fields in AIS PGN ID 232 starting at Field 12 and adjusted remaining fields accordingly.

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Rev ID	Date Version	Description
201	Feb 2015 2.100	Combined Reserved fields in AIS PGN ID 231 starting at Field 7 and adjusted remaining fields accordingly.
202	2.100	Added 3 new AIS PGNs Single Slot 129811, Multi Slot 129812, Long Range Binary 129813
203	2.100	Added Watermaker PGN 130567
204	2.100	Added Heave PGN 127252
205	2.100	Added Heartbeat PGN 126993
206	2.100	Added MOB PGN 127233

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